

Indigenous languages of the Americas and their structures

Sounds

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The Saguaro Group

Textbooks in Language Sciences 16



Textbooks in Language Sciences

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Indigenous languages of the Americas and their structures

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Edited by

The Saguaro Group

The Saguaro Group (ed.). 2026. *Indigenous languages of the Americas and their structures: Sounds* (Textbooks in Language Sciences 16). Berlin: Language Science Press.

This title can be downloaded at:

<http://langsci-press.org/catalog/book/563>

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ISBN: 978-3-96110-559-5 (Digital)

978-3-98554-179-9 (Softcover color)

978-3-98554-188-1 (Softcover b/w)

ISSN: 2364-6209

DOI: 10.5281/zenodo.20542602

Source code available from www.github.com/langsci/563

Errata: paperhive.org/documents/remote?type=langsci&id=563

Cover and concept of design: Ulrike Harbort

Typesetting: Diana Archangeli, Sebastian Nordhoff

Proofreading: Adriana Rosalina Galván Torres, Aidan Wilson, Aviva

Shimelman, Cesar Perez, Elliott Pearl, Frederic Blum, Harold Somers, Hella

Olbertz, Janina Rado, Jean Nitzke, Kate Lynn Lindsey, Lauren Schneider, Ludger

Paschen, Maria Milla, Mary Ann Walter, Matthew Korte, Nicoletta Romeo,

Philip Duncan, Silvie Strauß

Fonts: Libertinus, Arimo, DejaVu Sans Mono

Typesetting software: Xe_{La}T_EX

Language Science Press

Scharnweberstraße 10

10247 Berlin, Germany

<http://langsci-press.org>

support@langsci-press.org

Storage and cataloguing done by FU Berlin

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Preface

Samantha Prins

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Linguistics & community language work

Teaching and learning, policy and planning, advocacy and visibility, research and documentation, archiving and access are just some of the interdependent and interrelated aspects of community language work. Many different skills, methods, and activities make up a community language worker's toolkit. Projects of all sizes and scopes play important roles in supporting these efforts.

Linguistics, a field concerned with language documentation, description, theory, and — increasingly — revitalization, is also a tool in this work. Linguistics provides a framework for understanding structures and patterns in language. This framework can support a variety of language efforts, including:

- accessing, understanding, and creating language documentation,
- creating language resources,
- developing teaching methods and materials,
- teaching and learning languages.

Knowing a bit about phonetics and phonology (the sounds and sound patterns of a language), for example, is useful for things like developing a practical writing system or teaching pronunciation, as well as making dictionaries and creating new words. (An understanding of morphology — the internal structures of words — is also helpful in making dictionaries and adding new words to the vocabulary of a language.) These applications highlight just a few of the junctions of linguistics and community language work; linguistic knowledge can support any activity that involves a practical understanding of language.

While there are clear connections between linguistics and community language work, these connections are not always made in useful or transparent

ways. For instance, resources that focus on Indigenous languages and their structures (for example, journal articles, dissertations, and reference grammars) tend to be written for audiences with advanced training in linguistics. On the other hand, beginner-friendly linguistic resources (for example, introductory linguistics courses, textbooks and workbooks, and pop linguistics articles), tend to feature more widely-studied, “large” languages and the structures and patterns common to them. Further, linguistic literature is generally more focused on analysis and theory with minimal attention to the practical applications that are critical in community language work.

On a more foundational level, the motivations, goals, and approaches of linguistics and community language work can be quite different. In light of these differences, along with the checkered and often problematic history of linguistic research in Indigenous communities, linguistic frameworks for working with language can be viewed as irrelevant, misguided, or even harmful to languages and communities. Contemporary movements in linguistics have sought to address these issues by, for example, developing ethical research practices, placing higher value on consultation and collaboration with speakers and communities, and amplifying Indigenous voices from within in the field. While these and other paradigm shifts have effected positive changes, the work is still ongoing.

With all this in mind, the *Indigenous Languages of the Americas & their Structures project* is grounded in these foundational questions: Where do linguistics and community language work intersect and where do they diverge? Which tools from linguistics are most useful in community language work? How can we share these tools in grounded, engaging, and accessible ways? These questions shaped this volume, and shape the on-going development of future *ILAS* volumes. These remain open questions, but the hope is that *ILAS* will contribute to this conversation, and in doing so, help support language workers and communities in achieving their goals.

What is *ILAS*?

Indigenous Languages of the Americas & their Structures (ILAS, for short) is envisioned as a collection of digital, open-access resources about linguistics as it relates to Indigenous languages and community language work. Drawing from the Indigenous Languages of North, Central, and South America, *ILAS* examines language patterns and structures, introduces linguistic concepts and analysis for understanding them, and explores applications in community language work. The goal of this collection is to support language workers of all kinds by teaching linguistics as a practical tool for understanding and sustaining languages.

This volume, *ILAS: Sounds*, focuses on phonology (sound patterns in language), as well as topics in phonetics (language sounds and gestures) and morphology (words and word-building).

Making *ILAS*

The idea for *ILAS* first came about during a graduate phonology course at the University of Arizona. We had an especially large cohort of students engaged in community language work in some way, and so the class was organized around sound patterns in the Indigenous languages of the Americas. Over the course of that semester, a recurring theme was the difficulty in sourcing viable data and descriptions for studying sound patterns — resources which are ethically sourced, well-supported with examples, and not overly bound-up in a particular theory. This problem, needless to say, is not specific to phonology. Across the sub-fields in linguistics, coverage of this kind tends to be somewhat limited for Indigenous languages.

Driven by our experiences in that class and in linguistics as a whole, a working group was formed to discuss linguistic resources for and about Indigenous languages, and how these resources might be leveraged to support community language efforts. These discussions led to the idea to create such a resource, and in turn to the establishment of the *ILAS* project and the Saguaro Group to author and edit it.

The Saguaro Group includes Indigenous and non-Indigenous faculty and students with various backgrounds and expertise in linguistics and community language work. The structure of the group was intended to be non-hierarchical; the idea being that its members, with the support of external advisors, consultants, and contributors, would collaborate in all aspects of development of the *ILAS* project, and decision-making would be done by consensus. The framing, composition, review and revision of this book were done in that ethos.

Even with a diverse team working together on the project, we had to grapple with the fact that both linguistics and community language work are large and complex topics. To keep the project cohesive (and moreover, feasible), we decided to take a local approach in both content and scope. The languages featured in the series are all rooted in the places and contexts we, our colleagues, and our neighbors come from and work in — hence, *Indigenous Languages of the Americas*. In the same vein, we draw on our collective knowledge and experiences to ground linguistics as a practical tool within the context of community language work.

This book, *ILAS: Sounds*, is the pilot volume of the series. In documenting, describing, teaching, or learning a language, a common place to start is with the

smallest building block — that is, the sounds that make up that language. To mirror that process, and as a nod to our beginnings in a phonology class, we made that the starting point for this series as well.

Note to readers

A companion to this text series is the website at <https://sites.google.com/email.arizona.edu/ilas-ua>. It houses general information about the project, including news and updates, guiding principles, the book proposal, Saguaro Group membership, a map of languages featured in the project, and our contact information. We invite you to get in touch to share about your own language work, your experiences with linguistics, and to get involved in future volumes in the *ILAS* project if you're interested. Use the form at <https://sites.google.com/email.arizona.edu/ilas-ua/contact?authuser=0>.

We also welcome any feedback you might have to share, particularly as it relates to the representation of languages and language data. As a part of our editorial process, we were very selective and intentional about the kinds of language examples and sources we included in this book (more information about the vetting process can be found on the website, linked above). Despite these efforts, we understand that certain language examples or descriptions may still be deemed inappropriate for wide circulation. If something looks inaccurate, misrepresented, or like it shouldn't be publicly accessible, please let us know — digital publishing helps to ensure that anything in need of revision can be quickly addressed. Corrections or omissions can be made at the request of communities and stakeholders.

Above all, we hope that you enjoy this book, and that it is helpful in achieving your language goals, whatever they may be.

User guide

Diana Archangeli^a, Samantha Prins^a & Corey Roberts^b

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This book is an introduction to the sounds and sound patterns of languages. It explains concepts using examples, it has exercises (called Comprehension Checks) to test your comprehension, there are guides for applying material to your own language, the Language Explorations, and you will find concrete suggestions for ways this knowledge is helpful to language workers in the final part of each chapter.

Not everyone has the benefit of other people — friends, classmates, teachers — to share the learning process with. This book is written to be useful for learning even when there is no one else to answer questions and explain concepts. We do our best to provide clear explanations and examples. We do not assume prior knowledge of linguistic concepts and terms.

This User Guide gives our suggestions on getting the most out of the book on your own. We address four topics:

1. how to decide what to read
2. how to read the book
3. how to use the Comprehension Checks and Answers
4. how to approach the Language Explorations

1 Deciding what to read

You may choose to read only part of this book. You may find that certain parts are more useful and relevant for you while other parts are less so. All of this may be very new to you, and you may want to read only part of each chapter the first time through. You may already have some experience with linguistic concepts and terms, so that the chapter sequence we have chosen may not fit your situation. You may want to devise your own path through the text. You

may have a limited amount of time and wish to be very selective in what you read. There may be other reasons why you will choose to read only part of this book. You can always come back later for more.

To help you decide which parts to read, we give an overview of all chapters in Chapter 1 Section 1.5. We also begin each chapter and major section by laying out the topics covered in that part. We encourage you to read these introductions to determine whether the content of that part is useful to you. We hope you find and follow your own path.

If you are a little unsure about your ability with a particular topic, you could try the Comprehension Checks in that section to test whether you have control of the material. If you are quite sure you know the material, you could jump directly to the Language Explorations to use the concepts in the context of your language — and to build your language description. You could also skip ahead to the last section of the chapter to get ideas for ways to use the skills and concepts in working with your language.

If you have no familiarity at all with this material, we recommend that you start with one of the first three chapters, Chapter 2 on language sounds, Chapter 3 on making words, and Chapter 4 on how the same language sometimes sounds very different, depending on the speaker. These three chapters can be read in any order. We chose to begin with sounds, since this book is about sounds. The chapter on words (Chapter 3) may be the most accessible of the three so you may choose to start there. Chapter 4 addresses variation — the observation that different people pronounce some words in the same language quite differently from each other, an issue that is very important in many language communities, so that may be where you want to start.

2 Reading this book

Our strongest recommendations in reading this book are, first, to take your time and, second, to focus on what is useful to you. It helps to reflect after reading each section to make sure you've understood the concepts and explanations, before moving on to a new section. The structure of this book is designed to make it easy to take breaks and to skip sections that aren't relevant for you.

2.1 Short sections

Throughout this book, subsections are short, at most just a few pages. Each section contains a complete discussion of some concept — a single, bite-sized piece.

This means that you can read this book in small doses, just a few pages at a time. Even when you are short on time, you can read a few pages and still accomplish something. When you have time but find a particular section more difficult to understand, you can stop and reflect or re-read after just a few pages, before progressing to the next section.

2.2 Examples

Examples in this volume differ from those that you may find in introductory linguistic textbooks because in this book, examples are typically very short. A big part of our exposition style is to show that words can differ in some way, and then to discuss that difference. Usually we can illustrate a difference with just two words. For example, in Chapter 3, we discuss two words from Zenzontepec Chatino, *ndetiyaa* ‘PROGRESSIVE.arrive’ and *ngutiyaa* ‘COMPLETIVE.arrive’. We see from the translations that part of each word means ‘arrive’ and the other part has a different meaning (PROGRESSIVE — ‘is arriving’ — and COMPLETIVE — ‘arrived’). Two words are sufficient to show these differences.

Many times when we present examples, we use font color to direct your attention to certain parts of a word or its meaning. Black font shows the focused part of the example while gray font shows the backgrounded part of the example. This is illustrated in (1).

- (1) Use of font color to focus attention
 - a. Focus on what is the same in two words

ndetiyaa ‘PRG.arrive’
ngutiyaa ‘CPL.arrive’
 - a. Focus on what is different between two words

ndetiyaa ‘PRG.arrive’
ngutiyaa ‘CPL.arrive’

The method of giving examples using a very small number of words shows that you can begin to identify language structure even when you don’t have a large number of words in a language. You don’t need a long list of words or a whole dictionary. It is important, though, to remind yourself regularly that identifying a pattern is preliminary — you will want to test your patterns using more words in your language, to increase your certainty that you’ve found a general property of your language. Sometimes there is a pattern that works with some words while another pattern is found with other words — languages are not always internally consistent. You may also need to watch for variation, where different speakers use different patterns (see Chapter 4).

2.3 Glossary

We include an extensive glossary of the many terms that are introduced in this volume, terms that linguists use. The number of terms is a bit daunting, so this glossary can be very useful. We introduce and use linguistic terms so that if you read work produced by linguists about your language, you will have an edge for understanding the terms used by the authors.

Glossary terms are blue in the text; you can click on them and be taken to the glossary entry where you will find a short definition as well as other page numbers where the word is also found. For example, in Chapter 2, you will find the sentence “The [velum](#) can be raised and lowered.” Because the word *velum* is in blue text, you know that it is a glossary entry. If you are reading online, you can click on the word and you will be taken to the glossary entry for *velum*. In the glossary entry, you can click on the page number of the page where you were in the reading and you will be returned to the appropriate page in the text (in this case, p. xx).

If you want more context in understanding a particular term, you can click on the page number that is bolded in the glossary entry. This takes you to the page in the volume where the concept is most fully discussed (the term is both italicized and blue on that page; click on it to get back to the glossary).

2.4 Re-reading

For many people, it will work best to read one section at a time, to reflect on that section, and then, when you are sure you are ready, to go on to the next section.

We sometimes use this same strategy in our presentation. We circle back and re-visit a concept, reinforcing the initial presentation and then going a little further with the idea. Re-reading is a way for you to do the same — to read sections and chapters more than once. The first time gives you a general overview and the second time lets you really focus on the parts of most relevance to you and your language.

2.5 If you have a reading partner...

Reading and discussing the material with friends is highly recommended. If you know someone else who is reading this book, we strongly encourage you to discuss the concepts with them, as well as your answers to the Comprehension Checks and to the Language Explorations. Talking through language puzzles with another person often helps figure out whether you understand, and each person gains clarity about the ideas through discussion.

3 Using the Comprehension Checks

Many sections end with a Comprehension Check. A Comprehension Check is a question that asks you to use the concepts you've just learned about to come up with an answer.



Comprehension Check 1

A Comprehension Check is in a box like this one.

Some of these Comprehension Checks may be quite easy for you; some will be difficult and will require focus and thinking to work out the puzzle. Even when they are difficult, we urge you to try your best to come up with an answer before looking at the answer we came to. A box with our answer follows each Comprehension Check.



Here's our answer

The answer to a Comprehension Check is in a box like this one.

Completing a Comprehension Check before looking at the answer will help you know whether you've really understood the concepts of the section. If you go straight to our answer (or if you skip reading our answer) you may not realize that there is something you didn't understand completely. The Comprehension Checks are a very important part of the design we use to help you to understand and control the knowledge, even in the absence of a teacher.

When you compare your answer and the one that we came up with, sometimes you will find that your answer is essentially the same answer that we have. In this case, you can be confident that you are understanding the material in the chapter.

It is not uncommon for people to come up with different answers to questions about language. One of the difficult properties of studying language is that there

are often many different ways to understand the same structure. How, then, can you learn from having a different answer from us?

We recommend that you re-read the explanation we give of our answer to see whether you understand our reasoning. Once you are pretty sure you understand why we came up with our answer, think about your answer again. Do you think your answer is a viable alternative answer or do you think you missed something as you worked things out? By working through the differences, you will come to a better understanding. Realizing that you got something wrong and then thinking it through is part of learning. Realizing that you got it right, but in a different way, shows your creative thinking. You may even come up with an answer that is better than the one we offer.

The important thing is to think more deeply about both answers. Remind yourself that your answer may not be wrong — it may simply be different.

4 Approaching the Language Explorations

In addition to Comprehension Checks, there are invitations to explore your own language, scattered throughout the book. Language Explorations invite you to participate in the discovery and description of the phonology of your own language — they provide a real-world test of your increased understanding. The Language Explorations also help you to identify the concepts that are most relevant to your language situation, helping guide your reading choices throughout the volume.

Language Explorations are enclosed in boxes.



Exploring my language

A Language Exploration is in a box like this one.

Each Language Exploration takes concepts that you have read about and played with (through Comprehension Checks), and asks you to apply the concepts to your language. Sometimes, these will take time and creative thinking. If you have the opportunity, it can help to discuss your thoughts with someone else.

There are a number of benefits to taking the Language Explorations seriously. First, as you complete each one, you are reinforcing what you have just learned. More tangibly, as you complete each Language Exploration, you are completing the basic elements of a description of the sounds and sound patterns of your language. You are revealing the phonology of your language. This description will be one that you can use in many different ways as a language worker — in writing system design, dictionary organization, language classes, and other revitalization efforts.

5 Conclusion

In reading through this User Guide, and again as you go through Chapter 1, think about what parts of this book sound the most useful to you. Chart your path through the chapters so that you are learning what is most relevant to you as you understand your language. When you begin a chapter, read the introduction and select the parts that address questions that you have about your language. Be ready to read only part of each chapter. You can always come back and read another section later on, if you discover you want or need to know more on some topic.

One reaction you may have after reading this User Guide is thinking *It's going to take a long time to read this book*. It is true that it takes time to stop and reflect and re-read, and it takes time to complete the Comprehension Checks and Language Explorations in each chapter. However, if you select a route through the book that addresses your biggest concerns, you won't be reading every page, and you will move through the material more quickly. The payoff is that you will have honed many new skills and you will have created a sketch of the phonology of your language — both the skills and the sketch contribute to further work with your language, fostering its health and sharing its beauty with others.

Acknowledgments

A project of this size cannot be created without the help of many people.

We greatly appreciate the hard work done by Martin Saniga (Sappony) and especially Vickie Jeffries (Occaneechi Band of the Saponi Nation) and Shaheen A. Hedgepeth (Saponi Nation of Ohio), who read earlier versions of several chapters. Their suggestions, comments, and encouragement helped make this a better book.

Thanks to those who contributed to the works cited, especially those who shared their languages. Where names are missing, it is because we tried but could not reach the authors. If you can help us get in touch with one of those people, please reach out so that we can improve future iterations of this volume. You can contact us at <https://sites.google.com/email.arizona.edu/ilas-ua/contact?authuser=0>.

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Rosalia Badhorse, Northern Cheyenne

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Mosiah Bluecloud, Kickapoo

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Angela Hughes, O'otham

Kevan Joe, Navajo

Jeremy Kuhn, Sign language

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Acknowledgments

Hunter Thompson Lockwood, Potawatomi
Anna Mai, Ja'a Kumiai
Steve Marlett, Seri
Teresa McFarland, Filomena Mata Totonac
Melanie McKay-Cody, Plains Indian Sign Language
Gessiane Picanço, Mundurukú
Samantha Prins, Sign language and Niisti'powahsin (Blackfoot)
Eduardo Ribeiro, Karajá
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Chapter 1

Introduction

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Here's a puzzle. Suppose you love words. And you love your language. You think that it would be wonderful to have a dictionary of the words in your language. Your friends start to get tired of you talking about this and finally one says “Why don't you just make a dictionary for us?”

It stops you cold. What would it take to make a dictionary?

Maybe your language does not yet have a writing system, so to make a dictionary requires developing one. How do you go about creating a way of writing words that is helpful to speakers (or would-be speakers) of your language?

Maybe your language is written, but it is written in different ways by different people, and you hear that it is because some people use a phonetic writing system while others use a phonemic writing system. *What do phonetic and phonemic mean?* you wonder. How do you use that information in making a useful dictionary?

Maybe there are lists of words and meanings, but the lists alone aren't very helpful for people trying to learn the language. The word lists don't show relationships between words, so they provide no help when you are deciding what form of a word to use in different kinds of sentences or questions.

Maybe the word lists include symbols you don't understand, such as [ʒɑ:di:] (which means ‘arrowhead’ in Potawatomi) — what are [ʒ] and [ɑ] and [:] and why are there square brackets around them? And how are they pronounced? All of these symbols are explained in Chapter 2.

Maybe it has linguistic terms as seen in *ndetiyaa* ‘PRG.arrive’ and *ngutiyaa* ‘CPL.arrive’ (Zenzontepec Chatino) — what are PRG and CPL? To explain these abbreviations: PRG is shorthand for PROGRESSIVE, meaning the action is on-going.

So *ndetiyaa* translates roughly as ‘is arriving’. CPL means COMPLETIVE, meaning the action has been completed. So *ngutiyaa* translates roughly as ‘arrived’ or ‘has arrived’. If you can’t wait to learn about this, you can jump ahead to Chapter 3, Section 3.2.3.

For that matter, why are these in small caps, what does the dot (.) mean, and why are the words written in italics? These conventions are also explained in Chapter 3.¹

As you face the challenge of making a dictionary for your language, you realize that these examples just scratch the surface of what you wonder about. To get started, you need a description of the sounds of your language and how those sounds are put together in words. That description will answer these questions and more as you go about creating a dictionary for your language.

These same challenges, as well as many other questions, might come up if you are wanting to teach your language to someone else, or to write a description of your language, or to read something written by a linguist about your language. If you have very little background in language study, you may be asking yourself:

How can I get answers to questions about my language when I don’t even know what questions to ask?

If you have already managed some language study, your question might be a little different:

How do I create a description of the phonology — the sounds and sound patterns — of my language?

A common answer to either question might be a recommendation to go to a university or college and study linguistics. You may even have looked into such a program. And you learned that university and college degrees require several years, partly because they require you to learn about some subjects that may not be all that useful to you in reaching your goals. And they are very expensive.

Maybe you are fortunate enough to enroll in such a program, but you find the way that you are taught about language sound patterns doesn’t address your needs and you want another resource.

¹Potawatomi is an Algonquian language of the Algic family. The community is concentrated around the Great Lakes, as well as in Kansas and Oklahoma. Our source is Lockwood (2017); we thank Hunter Lockwood for improving our discussion of Potawatomi here and in other parts of the volume. Zenzontepec Chatino is spoken in Oaxaca, Mexico, and is an Otomanguean language; our source is Campbell (2014). We thank Eric Campbell for making sure that we represented the language appropriately by thoroughly checking the examples and discussion. You will see more of both languages in this book.

Maybe you have found some short programs — just one or two months at a university, programs that are designed for language workers. You are able to manage those costs — the time away from family, community, and work, and the money for tuition, travel, and living expenses. And then you find that in one or two months, you have learned enough about language sounds to start asking your questions, but not enough to create a description of the phonology of your language.

You need something more.

You may decide to find an introductory textbook in linguistics so you can study on your own. This can be frustrating as well. Textbooks are typically written for people who have the luxury of time for reading and studying the material, and of having a teacher to help explain the difficult parts. Introductory textbooks are also typically written for a generic audience, leaving it to the teacher to show the class how the material can be used in different contexts. They typically rely mostly on examples from major world languages rather than from Indigenous languages.

You really don't want to give up your efforts with your language, but you might be at a loss as to how you can get the knowledge and skills that you need to progress.

This book is written for you.

Here's our promise. Each language has a unique combination of sounds and sound patterns used in constructing words. Even if you haven't thought about language this way, you already know this — you've heard people speaking a language and just from the sounds you can confidently say *that's not a language I know* or *I know that language*. The term *phonology* is used to refer to the distinct sound patterns that make each language unique.²

If you read this book, by the end you will know what is meant by the *sound patterns* that make up the phonology of a language. You will have found sound patterns in your language and you will know how to talk about them with others. You will have tools to use when you read writings by linguists about the sound patterns in your language. Perhaps best yet, you will have a lot of ideas about ways to use this knowledge in your efforts as a community language worker.

Here's our plan. We have just made you a pretty big promise. It is a lot to get from reading a single volume, even one as long as this one. Can we deliver?

²The blue font of the word *phonology* tells you the term is in the glossary. You can click on the word to see the glossary entry — the term and a brief explanation. Then click on this page's number in the glossary entry (p. 1) to get back to this page. See p. xx for more on how to use glossary entries.

Each chapter in the book begins with a puzzle, a promise, and a plan. The basic plan in this chapter, Chapter 1, is to lay out what is coming in the remainder of the book, both in terms of what each chapter holds and also in terms of strategies used throughout the book to realize our promise. The list below sketches how the content of each section in this chapter fits into the goal of the chapter.

1. Our goal in creating this book is for it to be accessible even if you don't have a teacher to ask for clarifications along the way. Our most significant inspiration to achieve this end is laid out in Section 1.1.
2. Section 1.2 explores other ways that we hope will help make this book useful to people who work with Indigenous languages in their communities.
3. There are some things that this book does not attempt to do, addressed in Section 1.3.
4. Section 1.4 explains what we are trying to achieve with the volume — to create a practical introductory guide to be used in understanding the sound patterns of a language.
5. Section 1.5 may be extremely useful to you as you decide what to read. The book is divided into three conceptual parts, each with a few chapters in it. We now introduce each of the parts of the book as well as the leading concepts and skills in each chapter. The parts are:
 - Background and general information, Chapters 2, 3, and 4
 - Sounds of languages, Chapters 5, 6, and 7
 - Sounds in action, Chapters 8 and 9
6. We have used the same template for each chapter, laid out in Section 1.6. Importantly, each chapter ends with a discussion of ways that the material introduced in the chapter might be of use in language work.
7. Section 1.8 summarizes our hopes for the ways that this book will benefit you.

We begin with our inspiration.

1.1 Lessons in Indigenous spaces

We have taken several measures to accomplish our goals, based on the discussions that led to this volume, which are reviewed in the Preface (beginning on p. xiii). A great source of inspiration throughout the volume comes from Mosiah Bluecloud's thoughts about how knowledge gets shared between people. Mosiah earned both a Master's degree in Native American Linguistics and a Doctorate of Philosophy in Linguistics, both at the University of Arizona. He is an experienced language instructor and language advocate who believes that language connects

past, present, and future culture caretakers to hold communities together, as well as providing access to one's identity and to ways of being.



Lessons in Indigenous spaces

Mosiah Bluecloud, Kickapoo

I've built my career around the two concepts of *teaching in bite-sized pieces* and *comprehension checks*. I've found that with these two concepts mastered I've been able to foster language acquisition in even the most "unreachable" students — from students with "behavioral issues" to students that simply thought they were beyond help in acquiring a language due to age, perceived intelligence, or prior experience.

I've been trained in several language teaching methodologies and I've studied many others. What I've found in my experience, study, and conversations with students and teachers is that all the successful language acquisition methods are the same at their core. Successful teaching happens when the complex is broken down into simple bite-sized pieces and given to students in a structured and sometimes reordered way. The second major key is that no bite-sized piece is ever given until the previous bite-sized piece has been successfully received by the student. We ascertain a student's understanding of bite-sized pieces of information by using comprehension checks.

I feel that this method of teaching is the oldest in the world and one doesn't need to look far to see that it is still operating in Indigenous spaces. Let's begin with a question. How do we pass on ceremony? One might think that it must surely be passed on by watching and observing, but most ceremonies are so hugely intricate in ways and whys of being. Growing up Kickapoo, I would be told little bits and pieces at every one ceremony. Just one thing and one thing only. "Notice the way the sticks are put into the fire. They are placed carefully like for a controlled burn," my uncle once said. At the next ceremony I would be told to join in, in helping build the fires inside the wikiyup. Wikiyup is a traditional Kickapoo structure for living and also for ceremonies. My uncle would watch me lay the sticks in the fire the correct way, with the end that grows

out and away from the tree towards the fire. If I had done well, I would be given another bite-sized bit of info and another comprehension check. And this is the way the ceremony is passed along to the next generation.

We follow these strategies for passing knowledge to another person. We present information in short sections, our bite-sized pieces. We ask questions that allow you to check your understanding (Comprehension Checks). We provide opportunities to extend your knowledge (Language Explorations).

Bite-sized pieces mean that sections are short, examples demonstrate one point at a time, and that we sometimes circle back to a familiar topic to extend it further.

Each Comprehension Check is in a box marked with a question mark. You are asked a question that tests how well you are understanding the material just covered.



Comprehension Check 1

A Comprehension Check is in a box like this one.

It is followed by an answer in a box marked with a check mark.



Here's our answer

The answer to a Comprehension Check is in a box like this one.

The answer usually follows the question immediately. There may be times that you are tempted to just read our answer without doing the Comprehension Check for yourself. If you give in to temptation, you will lose an opportunity to learn or to confirm your learning.

Language Explorations are presented in boxes marked with a pencil. The Language Explorations have two goals. First, they consolidate the knowledge you are gaining by practicing the skills and concepts with your own language. Second, as you accumulate answers to the different Language Explorations, you are building a basic description of the phonology of your language. This description can then serve as a foundation for continued language work.



Exploring my language

A Language Exploration is in a box like this one.

1.2 Other recognition of language work in Indigenous spaces

There are other measures we have taken because this book is written especially for people who work with Indigenous languages in their communities.

1.2.1 Examples from Indigenous languages

Our examples are almost exclusively drawn from Indigenous languages of the Americas. This is one way to honor and respect these languages, and to make this book more relevant to those who work with Indigenous languages. We hope at least one of our examples is from a language that you know or that is related to one you know.

Most examples are from works that were published recently. The reason for selecting from recent publications is that we have greater trust that good ethics and research practice were followed when the work was done, so there is a greater likelihood that the community is comfortable with sharing information about their language. Where a language is from an older work, we explain why we think the community would not object to our using their language in this book.

1.2.2 Open access

We have chosen to publish this book with an open-access press, through Language Science Press (<https://langsci-press.org/>). There are two reasons for this.

First, people can download the book at no charge. If someone wants a hard-copy of the volume, it is possible to buy a copy through the press for the cost of printing and binding the book. The cost of a physical version of the book is as low as we could make it.

Second, it is easy to change the content of the book as needed. This is important in two ways.

- A reader may find the presentation is lacking in some way — it isn't clear, or a critical concept is omitted, or there is something that is just plain wrong.
- A language community may find that there is an error in the way that we have presented words from their language, or a community may really prefer that information about their language not be shared in a publication.

If we are informed of either kind of problem, we can quickly change the text, so that future downloads have the updated version. Please send us a note through the form at <https://sites.google.com/email.arizona.edu/ilas-ua/contact?authuser=0> so that we can make appropriate changes.

1.2.3 Encouragement to explore your language

Some Indigenous languages already have a grammar and a dictionary, or have been written about extensively by anthropologists or linguists. Yours may be one of these languages; you may be wondering whether there is anything more to be said.

It is important to remember that many published works are not focused on the language knowledge and experience of speakers, rememberers, knowledge keepers and learners — the people who have what is known as *passive knowledge*. People with passive knowledge are experts with valuable language expertise.

Passive knowledge about language is important. It is extremely difficult to write down every property of the sound system of a language, the knowledge that speakers have of the language, that learners gain from the experience of learning the language, and so on. Someone else may have already written an article, a description, a dissertation, or a book about your language, and you may feel that there is little room for additional work to be done by a community linguist. However, passive knowledge of your language contains information that can determine whether a publication on your language is accurate and complete, or whether it is incomplete, has errors, or reports on a different dialect. The knowledge you will gain from this book will help you to access and share that passive knowledge — held by you or by members of your community who are willing to work with you — to enrich and improve the existing written records.

Through the Language Explorations, this volume invites you to tap into both passive knowledge and active knowledge as you explore your language and develop a deeper active awareness.

1.3 What this book is not

The book is not a complete introduction to linguistics. The focus is on the sound patterns of languages, and how to describe the phonology of a language. But there is a lot more to languages than the sounds and sound patterns. We don't address those other aspects in any depth — there is just one chapter about words and almost nothing about sentences and meanings. These are all important aspects of a language. However, this book is not an introduction to all aspects of linguistics.

This book is not short. We have done our best to divide the concepts into bite-sized chunks, and to build from one concept to the next. This makes each concept more accessible. It also makes the book longer.

This book is not (necessarily) easy to read. The material is complex. We encourage you to take time for reflection to make connections, to do the Comprehension Checks, to explore your language with the Language Explorations. We also encourage you to jump around to the parts that are of most value to you, going back to fill in knowledge gaps when necessary.

1.4 What this book is

This book is a guide for community language workers striving to know more about the sounds of their language. As such, the book is both explanatory and practical. On the explanatory side, we introduce and explain concepts. On the practical side, we introduce strategies for discovering the way a concept is realized in a specific language and we offer suggestions on the way this knowledge can be put to use by a language worker.

This book is an introduction. We do not expect prior knowledge of linguistic terms or linguistic analysis skills. (If you already have some background, you might find it useful to go through the introductory sections of each chapter to figure out whether there is anything new for you in our discussions.)

This book is about language and phonology generally. We introduce key concepts and explain key strategies for understanding the phonology of any language using examples from a wide range of Indigenous languages. The Language Explorations give guidance on applying these concepts to your language.

This book is practical. We explain concepts and provide tools and strategies for using those concepts when examining your language. This book is not a treatise about a particular theoretical approach to language sounds. At the same time, we take pains to introduce key concepts used by theoretical linguists in their writings about Indigenous languages, to help you make sense of linguistic work that has been published about your language.

This book is a means to an end. Our promise to you about the concepts in the book is that they will help you understand the phonology of your language. Our promise to you about the Comprehension Checks is that they will cement your understanding. Our promise to you about the Language Explorations is that your answers together can form the basis of a complete, grounded, and relevant description of the sounds of your language. Together, the knowledge and skills give you tools to use in further work with your language.

1.5 What to expect in the book

The book is divided into three parts. The first section gives background information, the second part focuses on the sounds we find in words, and the third part looks at the way sounds interact with each other inside words.

1.5.1 Background and general information, Chapters 2, 3, 4

The book begins with background information about sounds used in languages (Chapter 2) and about how words can be built from smaller parts (Chapter 3). The third of these chapters (Chapter 4) provides general information about variation within a language, to understand better why speakers don't all say the same words the same way.

Sounds, Chapter 2. Chapter 2 is the first background chapter. Chapter 2 examines the sounds used in languages — how to make the sounds, how to identify them, the symbols that are used to represent the sounds, and the terms used in describing sounds. The Language Explorations begin the process of describing the sounds that are found in your language; having such a description is useful in developing an **orthography** (writing system) and when teaching someone how to pronounce words in your language.

The study of sounds is called **phonetics**. We try to provide enough information about phonetics for this volume to make sense and so that you can ask questions about the sounds of your own language. We do not attempt to provide a complete

discussion of the production and perception of language sounds — that is the topic of a different book.

If you are not familiar with symbols and terms for sounds (symbols like [ʃ], [ɔ], [dʒ], [ɲ] and so on; terms like *fricative*, *continuant*, *aspiration*, and so on), we urge you to begin with Chapter 2. We also urge you to take your time with this chapter, reading it in small chunks so you have time to absorb and think about what you've read — and to figure out which parts of the chapter you can safely skip. There is a lot in this chapter, and it may feel overwhelming — especially if you don't give yourself time.

If the symbols and terms are familiar concepts for you, you may want to skip this chapter. You can test yourself with the Comprehension Checks. Even if you skip the chapter, we suggest you work through the Language Explorations, so that you develop as complete a sketch of your language as possible. The Language Explorations discovering the sounds of your language form the foundation for describing the phonology of your language.

Words, Chapter 3. The second background chapter, Chapter 3, shows how words can be put together from smaller parts, called *morphemes*. This chapter shows how to look at the words of a language and find morphemes in those words, the meaningful sub-parts that contribute to whole words. The Language Explorations encourage you to find words in your language that have more than one morpheme in them. Understanding the sub-parts of words is particularly valuable in using a language, teaching a language, and developing a dictionary.

The study of morphemes is called *morphology*. Our goal is to provide enough background in morphology for this volume; we leave a more complete explanation for a different book.

If you are not familiar with finding roots and affixes in words or with *three-line glosses*, you will find this chapter useful — and necessary before reading Chapter 9. If you think you know the material, again you might try the Comprehension Checks to be sure. And again, doing the Language Explorations will contribute to your language sketch.

Variation, Chapter 4. The next chapter, Chapter 4, explores something that every language faces: There is *variation* — sometimes, different people pronounce the same word in different ways.

This chapter explains different types of variation. It gives you tools for identifying the details of different variants so that you can precisely describe particular patterns of variation. This is particularly useful in cases where differences in language develop when a community is split geographically, or by age, or by some other factor. It is also useful when there are few speakers left and they say certain

words differently from each other. The Language Explorations give insight into the properties of variation in your language.

Interestingly, the tools used to identify properties of variation are the same tools that are used to identify properties of other types of sound patterns; you will see them repeated in later chapters. However, this is not a volume about variation; we discuss it in a single chapter, leaving full discussion for a different book.

If you are interested in variation, read this chapter. You can read the rest of the book without reading this chapter, although variation is occasionally mentioned elsewhere in the book. You might also want to read this chapter before reading Chapters 7 and 8, to gain more fluency with the analytic strategies we introduce. The Language Explorations in this chapter might be of great interest to you if your community is wrestling with problems that stem from variation in pronunciation.

1.5.2 Sounds of languages

Through the Language Explorations in Chapter 2, you will develop a list of sounds in your language, a description of those sounds, and example words illustrating each sound. This gives the basic elements from which words are formed. However, sounds play different roles in a language. Chapters 5, 6, and 7 take a closer look at the sounds of languages and the different ways that sounds are used in languages — specific properties that make the phonology of each language unique to that language. With each chapter, if you think you might skip it, we urge you to work through the Comprehension Checks to be sure you know the material. We also encourage you to complete the Language Explorations so that you end up with a fairly complete language sketch.

Contrast, Chapter 5. Chapter 5 explains how some sounds are *contrastive*, and others are not contrastive. It shows how to determine whether a pair of sounds is contrastive. The Language Explorations in this chapter give you practice with those skills as you begin to make a list of the contrastive sounds in your language. If you are already familiar with contrast and with using *minimal pairs*, you can be sure by doing the Comprehension Checks and Language Explorations. If those go well, skip this chapter and go on to the next one.

Syllables, Chapter 6. The second chapter in this part of the book, Chapter 6, looks at the distribution of sounds in words through the concept of the *syllable*. The basic ideas are (i) each word is made up of one or more syllables in sequence, and (ii) each syllable is typically made up of a vowel, with the possibility of some

consonants before and/or after the vowel. This means that words are generally made up of strings of sounds that alternate between consonants and vowels.

Chapter 6 introduces terms used to describe syllables and strategies used to identify the syllables of a language. The Language Explorations help you define the syllables of your language. If you already have experience with syllables and their subparts (like *onsets*, *nuclei*, and *codas*) you might check your knowledge with the Comprehension Checks before jumping ahead to the next chapter. As always, we strongly recommend that you try the Language Explorations, even if you don't read the entire chapter, so that your final description of your language is as complete as possible when you finish.

Distribution, Chapter 7. The next chapter, Chapter 7, shows how some sounds are found only in special places in words, while other sounds occur more freely in the words of a language: Some sounds have *limited distribution* while others have more free *distribution*. You will learn what this means and how to determine whether a sound has limited or free distribution. The Language Explorations help you determine the distribution of different sounds in your language. If you think that you know how to determine the distribution of a sound in a language, you can test yourself with the Comprehension Checks and the Language Explorations.

1.5.3 Sounds in action

In the preceding chapters, the book focused on inspecting words and learning about sound patterns from that examination. If you've kept up with the Language Explorations, you have a fairly complete basic understanding of the phonology of your language — what makes it sound different from all other languages. This section adds the finishing touches. Here, we examine sound patterns by comparing words to each other. In Chapter 8 we compare unrelated words while in Chapter 9 we look at words that are related to each other because they share some morpheme. These comparisons reveal how sounds interact dynamically in language.³

Complementary distribution, Chapter 8. Chapter 8 addresses a very special type of distribution, called *complementary distribution*. If you've been confused when you read or heard the terms *phoneme* and *allophone* and *phonemic*, this chapter is for you: This is where the terms are explained. Complementary distribution goes a long way in helping to understand why differences show up in writing systems

³You may remember that we mentioned earlier that Chapter 3 is critical for understanding Chapter 9. Chapter 3 introduces the morpheme; Chapter 9 relies on this concept.

for the same language, in understanding linguistic writing, and in making sense of the pronunciation of words with internal structure. Even if you've skipped other chapters, you may not want to skip this one. If you are working to develop a useful language sketch, the Language Explorations in this chapter are very important for you.

Alternation, Chapter 9. Chapter 9 examines the pronunciation of specific morphemes when they are combined with other morphemes. In some cases, we simply put the morphemes together and we are done. In other cases, when morphemes are put together, some sound is not what we expected; the difference between what we expect and what is said is called an *alternation*. Alternations are important in learning to pronounce words appropriately; they are also important in building dictionaries and language lessons. Alternations are the crowning part of a language description, drawing on all the parts that make up the phonology of a language.

1.6 What to expect in each chapter

Each chapter begins with a puzzle, a promise, and a plan. The puzzle raises a type of question you may have faced as a language worker, or simply as someone who thinks about language. There is a brief answer to the puzzle at the beginning of the chapter. At the end of each chapter, we return to the puzzle and connect the contents of the chapter to answer the puzzle.

The promise tells you what the chapter delivers. This gives you an idea of the contents of the chapter, so that you can decide whether it covers material of use to you.

The plan tells you what is in each part of the chapter and how those parts contribute to the whole. It also gives suggestions on different ways to read the chapter, depending on your personal goals.

A short introduction briefly explains the key concept of the chapter and why it is important. The introduction also sketches the topics of the major sections.

There are several major sections. These sections introduce concepts, strategies, and tools and show how to use the material in the context of your language. Each major section begins with an introduction about the content, to help you decide whether it offers material you would like to understand.

Each chapter ends with the same two sections. The first of these provides a more complete answer to the puzzle, connecting the puzzle to the material you have just studied, and then explores a related issue to give more depth to your knowledge. It ends with suggestions on ways that a language worker can use

the material of the chapter in their work. We offer suggestions in five general categories for using your hard-earned knowledge.

1. Language learning (for both teachers and students)
2. Creating language resources (e.g. grammars, dictionaries, writing systems)
3. Language documentation and description
4. Language revitalization
5. Reading work by linguists about your language

The final section concludes the chapter, reviewing what you know as a result of working through the chapter.

1.7 What do I do if I'm brand new to this?

In some ways, this book is an instruction manual. Each chapter focuses on a particular phonological phenomenon, the concepts related to understanding that phenomenon, and the analytic techniques for analyzing it in your own language. The chapters are structured so that the basic concepts and skills are introduced in the first part of each chapter, with refinements and further examples coming in later sections. Finally, each chapter includes Language Explorations to help you develop aspects of a phonological sketch of your own language.

If this is all new material for you, we recommend that you consider alternatives to reading the book from start to finish. Here's a path that may work better for you.

- (1) Suggested chapter sections for newcomers to phonology

Chapter 1 *Introduction*

This chapter tells what you can expect from this book. Our recommendation is to read the whole chapter.

Chapter 2 *Sounds: the foundation of oral language*

Oral language is communicated through sounds produced with the mouth. This chapter explains how sounds are made, described, and characterized.

Our recommendation is to start each major section to choose the best parts for understanding the sounds of your language. We also recommend that you read the discussion in Section 2.5.

Chapter 3 *Morphemes: The foundation of word-building*

Some words are put together from smaller pieces, like bracelets being made from beads. This chapter introduces the parts of words, called morphemes, and how

to find them. Our recommendation is to read through Section 3.3, and to read the discussion in 3.7.

Chapter 4 Variation: Different but the same

Communities or individuals may use different pronunciations for the same word; this is called variation. This chapter explains different kinds of variation and strategies for figuring out patterns of variation. These same skills are used in subsequent chapters to figure out other types of patterns. Our recommendation is to read through Section 4.2, and to read the discussion in Section 4.5.

Chapter 5 Contrast: Where sounds meet language

Contrast defines the sounds that are most critical in your language. This chapter tells what contrast is and how to find the contrastive sounds of your language. Our recommendation is to read through Section 5.2, and the discussion in Section 5.6.

Chapter 6 Syllables: Putting segments into words

Syllables are made up of sounds, and help organize the sounds of words. In this chapter we look at different types of syllables and show how to identify them. Our recommendation is to read through Section 6.4 and the discussion in Section 6.5.

Chapter 7 Distribution: Where sounds are found in words

The sounds of a language do not appear just anywhere. Rather, each language has its own patterns for sounds — their distribution. We use the skills introduced in Chapter 4 to help understanding patterns of distribution in a language. Our recommendation is to read through Section 7.2 as well as the discussion in Section 7.8.

Chapter 8 Complementary distribution: The same but different

Complementary distribution is a specific kind of distribution involving two or more sounds. Complementary distribution is central to the linguistic concepts of phonemes and allophones, and of underlying representations. We show how to discover patterns of complementary distribution using skills from Chapters 4 and 7. Our recommendation is to read through Section 8.3 and the discussion in Section 8.7.

Chapter 9 Alternations: Tuning into a language

Alternations are phonological patterns that arise when morphemes are put together to make words. Alternations are explained using knowledge about morphemes from Chapter 3 and knowledge about phonological analysis from Chapters 4, 7, and 8. Our recommendation is to read through Section 9.2, and the discussion in Section 9.7.

Chapter 10 *Conclusion: Looking back and going forward*

We close the book with a review of what you can learn from the book and why it is important. The discussion focuses on ways to use your phonological sketch in your role as a language worker. Our recommendation is to read the whole chapter.

Other than with Chapter 2, we propose that you read the first two or three sections of each chapter, and the discussion section, then move on. The discussion sections explore how a language worker might use some of the knowledge gained in the chapter. We recommend reading each of those sections, to help put the knowledge in context. Chapter 2 provides background on the sounds of languages, and we expect that you will want to find the parts that are relevant for the sounds of your language.

1.8 What we hope for you, with this volume

This book has been written to resolve these initial puzzles:

How can I get answers to questions about my language when I don't even know what questions to ask?

How do I create a description of the phonology – the sounds and sound patterns – of my language?

Each chapter will give you tools and skills, and concepts and terms to use when working with your language, reading about your language, and trying to explain your language to other people.

At the very least, reading this book will give you the knowledge you need to ask your questions. In many cases, this book will give you the knowledge you need to answer the questions yourself, and even to explain your answers to others. And you will find that the exercises along the way give you the foundation of a description of the phonology of your language.

We have four hopes for when you finish reading this book.

1. Our first hope is that you have learned concepts and strategies that are useful to you in understanding the phonology of your own language.
2. Our second hope is that you can use this book as a reference, to return to when you need a reminder or to review some concept.

3. Our third hope is that you will be inspired by some of our suggestions about ways to use this knowledge in your efforts as a language worker — either to use these suggestions directly, or to change them in a way that fits you, your language, and your community.
4. Our fourth hope is that you will share with us your experiences with the book. Because we can modify the book easily (as noted above, by making this an open-access book that users can download), there is a lot of flexibility for making corrections and improvements for future readers. There are many things we would like to know about. You can contact us at <https://sites.google.com/email.arizona.edu/ilas-ua/contact>.
 - We want the text to be very clear, so please let us know if you are confused by parts of the presentation or directions (such as in the Comprehension Checks and Language Explorations).
 - We respect language communities and languages, and are doing our best to represent words from each language appropriately. If we have not done so, please let us know so that we can make corrections.
 - We want this text to be practical, applicable, and useful. If you've found this book to be helpful, we would really like to hear how you used the knowledge you gained.



Coahuiltecan

Miguel Acosta

“I am a lineal descendant of the Coahuiltecan people who inhabited Central South Texas and going South into the Mexican States of Coahuila, Nuevo Leon and Tamaulipas. I am a member of the Coahuiltecan Tāp Pilam (People of this land) Nation and the Coahuilteco Language Reclamation Program... [A]t the AILDI⁴ conference ... [o]ne of the most important things I learned... was that other communities had started their language reclamation program by studying documents available on their language. One of the first speakers at that conference was Daryl Baldwin... of the Miami Nation who introduced himself in his native language. It is hard to describe the feelings I experienced when he stated, “I learned my language from the documents.” This experience, as well as the rest of the

conference, caused so much excitement, hope and inspiration for me and later the community to continue our efforts of reclaiming our heritage language.”

Acosta (2022: 10-16)



AILDI

AILDI stands for the American Indian Language Development Institute. Founded in 1978, AILDI’s mission is to “provide critical training to strengthen efforts to revitalize and promote the use of Indigenous languages across generations. This is accomplished by engaging educators, schools, Indigenous communities and policy makers nationally and internationally through outreach, transformative teaching, purposeful research and collaborative partnerships. As a result of our work, we envision that the larger society will know that language revitalization is critical to sustain and reinforce Indigenous linguistic, cultural, and spiritual health and identity.” The AILDI website is <https://aildi.arizona.edu/>.

1.9 An invitation to proceed

We invite you to select your first chapter — Chapter 2 for fundamentals about the sounds used in languages, Chapter 3 to look at words, or some other chapter. (If you know you are a beginner, the best place to start is with either Chapter 2 or Chapter 3.) As you progress through sections and chapters, you may want to circle back and read (or reread) an earlier section or chapter. Or you may decide to jump ahead to a later section.

Read each chapter’s introduction to make sure that it will deliver what you are expecting or needing. Do the same as you come to each major section, determining whether to keep reading or whether to skip some section(s).

Stop along the way to think through the Comprehension Checks. Test your understanding, and learn from coming up with right answers, or from reflection on answers that are different from ours.

Diana Archangeli, Rosalia Badhorse & Samantha Prins

Enjoy exploring your own language through the Language Explorations, learning how to create each piece of a description of the phonology of your language.

Chapter 2

Sounds: the foundation of oral language

Diana Archangeli

University of Arizona

Here's a puzzle. The big day has finally arrived — you receive a copy of a dissertation that describes your language. You've been waiting for it; finally it is here. At last, you will be able to pronounce words in your language as they once were said. You open the pages to get started — and you feel like you can't read a word.

This could happen to you if your language is Tutelo-Saponi, a Siouan language spoken primarily in Virginia and North Carolina in the southeastern United States, and you've just received a copy of Oliverio (1997). You flip it open randomly and come across the word meaning 'dog' written in different ways, including *čʰɔki* and [tsʰɤkʰ]. Your heart sinks. How on earth can you make sense of this? (Our primary source for Tutelo-Saponi is Roberts (2020). Corey Roberts also checked all Tutelo-Saponi words in this chapter.)

How do the weird symbols used by linguists relate to the way a language sounds?

To decode Tutelo-Saponi pronunciation, there are several things to figure out. Why are some words in italics and others in square brackets? What is the little raised *ʰ*? What is the tail on the vowel? What are the two dots after the vowel? Does *č* relate to [ts]? Finally, why does one word end with *i* while the other word ends with [kʰ], yet they both mean 'dog'?

That's a lot to decode. You cannot necessarily tell how a language sounds by looking at the written form of the language — unless you also know what the symbols mean, as well as how to move your tongue and lips to make the relevant sounds.

There are a few things that can help with decoding a description of a language. There are conventions about the use of square brackets and of italics. Square brackets indicate that these symbols are really how the word is pronounced, so if you are focusing on speaking the language, you want to focus on [ts^hʊk^h] rather than *č^hoki*. Italics are conventionally used for how a word is written for reading, not for pronunciation. If your interest is in reading and writing, forms in italics may be more relevant for you.

As for the symbols themselves, each one is designed to tell how the sound is pronounced. For example, the raised ^h means that when the preceding consonant is made, it is followed by a little puff of air, called *aspiration*; the tail on the vowel tells us that the vowel is a *nasal vowel*, made with air coming through the nose as well as through the mouth. Aspiration and nasality, as well as other symbols and pronunciations, are explained in this chapter.

An additional challenge is that not all linguistic writings make use of the same conventions. Fortunately, more recent publications are becoming more consistent in the use of symbols. Unfortunately, many writings about Indigenous languages in the Americas are dated well before the 1950s.

Here's our promise. The goal of this chapter is to give you knowledge to help with the decoding. By working through this chapter, you will learn about the different parts of the [vocal tract](#) and how they move in order to produce different kinds of sounds. You will also learn tips for pronouncing some of the sounds.

You will learn the [features](#) that are commonly used to identify language sounds so that when you encounter these different features as you read about your language, you will have some background to put those terms in context. For example, in Tutelo-Saponi, the feature [nasal] is important for vowels; if a vowel is described by the feature [nasal], we know that air should be coming out of the nose when the vowel is pronounced. Notice that the feature is placed in square brackets: [nasal]. There is a list of features with their definitions in the back of the book, along with a chart showing which features are typical for different sounds.

You will learn the [symbols](#) used to indicate the pronunciation of a word. When symbols are used to indicate the pronunciation of a word, they are customarily placed in square brackets, as we saw with [ts^hʊk^h]. The symbols we use in this volume are those found in the [International Phonetic Alphabet](#), or *IPA* for short, advocated by the International Phonetic Association. Appendix A gives a list of IPA symbols (at the back of the book).



Americanist Phonetic Notation (APN)

In some cases, you will find sources that use non-IPA symbols; we convert these to IPA symbols for consistency. A common alternative symbol system, especially in older works on the Indigenous languages of the Americas, is called the *Americanist Phonetic Notation* (APN). The Americanist Phonetic Notation is discussed in Section 2.5.2. (Some people use the name *NAPA*, or *North American Phonetic Alphabet*, instead of APN.)

In this volume, we start our discussion with symbols that have a straightforward relation to sounds, such as [p] and [b], and use those to launch our discussion into unusual symbols, for example [ɸ] and [β].

It is very important to understand the difference between the symbols used to show how something is pronounced and the way words are written for other people to read. In some cases, there is a high degree of similarity between the two, but in other cases, the IPA symbols and the written word can look quite different. Other chapters in this volume (especially Chapters 8 and 9) give some guidance on understanding why the written word can look very different from the way a word is pronounced.

Here's our plan. In this chapter, we focus on sounds. The goal is to help readers understand how to identify, make, and represent (with symbols) the sounds of their language. It is not written with a specific language in mind. As you will see in reading this chapter, there are a lot of different ways that humans can create sounds, a whole lot of different ways. This means that there are a whole lot of different sounds — many of which will be irrelevant to your language. We have tried to write this chapter so that you can select sections to read that are relevant for your language, and skip other sections. The problem, of course, is that you may not yet know which sections you can safely skip.

We have also tried to write this chapter so that it can be a resource for you to come back to at a later time, either to remind yourself of something, or to learn about a new type of sound that you think you might need for your language after all. Because this is a resource chapter, it is quite long, and you may find it most effective to read it in small chunks instead of all at once. We've tried to write each section — and each subsection — as a self-contained description of some

component of the way sounds are produced, to make it easy to complete reading about each concept in a short amount of time.

Finally, we do not include every sound found in every language in this chapter — that is the topic of another volume, delving more deeply into how language sounds are made. Our goal is to provide a tool that you can start using, and to give you a foundation to build on as you learn more about sounds, particularly the sounds of your language. We make some mention of signed language, but our main focus is on spoken language. Signed languages are very important in some Indigenous communities in the Americas, and deserves their own volume.

The list below tells what is in each section. It also gives guidance on whether you might want to skip a particular section.

1. Section 2.1 explains the basics of how to make language sounds, focusing on the major difference in the production of consonants and vowels. This section also introduces the articulators within the mouth. If the images and labels in Figure 2 and Figure 3 are familiar to you, you may want to skip Section 2.1 and start with Section 2.2.
2. We begin with two major distinctions in sounds, due to two articulators that are not part of the mouth itself — the nasal cavity (nose) and the larynx (voicebox). Section 2.2 explains two very common articulation effects used in both consonants and vowels — voicing (due to the larynx) and nasality (due to the position of the velum). Voicing and/or nasality are used in most languages, so they may well be important in your language. If the brief explanation of voicing and nasality above reminds you of knowledge you already have, you might want to skip this section.
3. Section 2.3 explores how consonant sounds are produced. In general, consonants are made by constricting the stream of air coming out of the mouth. Section 2.3 looks at the two key properties of this constriction, the type of constriction (complete, partial) in Section 2.3.1, and the location of the constriction in the mouth in Section 2.3.2. Overall, Section 2.3 is a very long section. We encourage you to make use of the small sections by giving yourself time to absorb the material from one subsection before moving on to the next.
4. Section 2.4 is about producing vowels. The first part (Section 2.4.1) addresses some very common vowels, starting with the three most common ones. This may be all you need for your language. Some languages have some of the less common vowels. Some of the less common vowels make use of the same articulatory gestures that we find in the most common vowels, but those gestures are combined in uncommon ways, discussed in

Section 2.4.2. Other vowels are made using additional articulatory properties, discussed in Section 2.4.3.

5. Section 2.5 does three things. It shows how you now have a full answer to the puzzle that introduced the chapter. It discusses the question of why we use the IPA symbols to represent a language instead of using a practical writing system. Finally, there is a discussion of ways that a community language worker might use the knowledge gained in this chapter. Some readers may want to jump ahead to that section first, so that they are primed to think about how to make use of the knowledge as they read the chapter.
6. Section 2.6 recaps what you have learned from this chapter.

Throughout the chapter, there are exercises to ensure that you understand the material being presented, the Comprehension Checks. If you think that your background may allow you to skip some of the sections of this chapter, you might try the Comprehension Checks to make sure. There are also Language Explorations, encouraging you to apply the material that you are learning to your own language. We urge you to do these explorations even if you are skipping major parts of the chapter, so that you build a more complete description of the sounds and sound patterns of your language.



What is a word?

Have you ever wondered how we know something is a word? Some people say that if it is in the dictionary, it is a word. But what if your language has no dictionary? That does not mean that there are no words.

It can be very difficult to determine whether something is a word, in particular if you are listening to a recording of fluent speakers talking and you are not very familiar with the language. Speech is a continuous flow: When we speak, we do not put pauses between words (try talking to someone that way and see the curious looks you get).

Figuring out what counts as a word in a language involves the interaction between sounds, morphemes, and sentence structure. Since this book focuses on sounds, we won't go into the question of what makes something a word.

2.1 Introduction

When a person is using a language, their body is involved. Whether speaking or signing, language is produced through body movements that allow the transfer of an idea from one person's head into another person's head.

In this chapter, we introduce the parts of the body used in producing spoken language, the *articulators*. We explain what each articulator is and the different ways each one is used in spoken language. We also give hints on how to produce the different sounds. We find videos can be helpful for this, and there are videos online to help you learn to pronounce sounds. We consulted <https://enunciate.arts.ubc.ca/linguistics/world-sounds/>, which has explanatory videos of both consonants and vowels along with audio recordings. These videos show the movements of both the lips and the tongue, two of the articulators.

We use the term *feature* for the ways that the articulators can be manipulated to produce language sounds. We write features inside square brackets, like [labial] — the feature that indicates lips make some kind of constriction. There is a list of features in Appendix B.

Multiple features are used simultaneously to produce a single sound. The different feature combinations enable people to produce a myriad of sounds (or signs) — far more than are used in a single language. We can determine the *similarity* of two sounds by finding how many features they have in common. The feature differences tell us very precisely how the two sounds differ from each other.

Features classify groups of sounds. For instance, the feature [consonantal] classifies the group of all consonants. Groups of sounds classified by a feature — or by a set of features — is called a *natural class*. The significance of features and natural classes will become more apparent, and be very useful, when you read more of this book, especially Chapters 5, 8, and 9.

In addition, understanding articulators and features helps you pronounce new sounds. It helps as you listen to identify sounds that are new to you. And it makes it easier to teach others to pronounce new types of sounds. We explore these and other ways that articulators and features contribute to the job of a language worker at the end of this chapter, in Section 2.5.3. For now, let's start with an overview of how language sounds are produced.

2.1.1 The spoken word: A fancy whistle

Think about how you can get sound out of a whistle. You blow air into it, and if you send the air through with the right force, you get a noise. If you pick up a

whistle with a different shape, the sound is likely to be different. The mechanics remain the same — send air through the whistle’s tube and you produce noise.

Roughly speaking, this is exactly how we produce our language sounds. We send air out of our lungs and through a tube that ends with our mouth and/or nose. If we do it right, we produce noise. We can change the shape of the tube by manipulating different articulators. These manipulations produce different noises. If we do it all exactly right for the language we are speaking, the noises we produce are the sounds of the language.

2.1.2 The lungs

The lungs start everything off, sending air out of the body. Using more force makes the sounds louder; less force makes the sounds softer. (We learn to control the pressure — and so the volume.) But the important thing is that air is coming out from the lungs through the mouth and/or nose, as shown by the blue arrows in Figure 1.

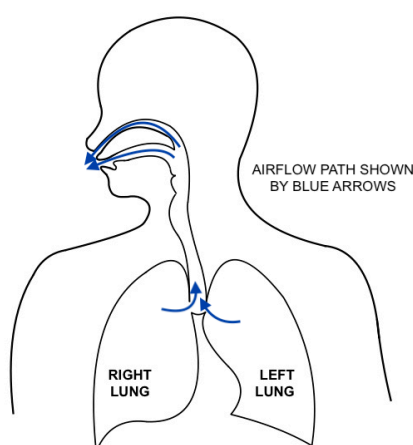


Figure 1: The lungs

Try this. If you are skeptical, try speaking out loud while breathing in. It is not easy.

2.1.3 Articulation: Making consonants and vowels

Blowing air out of your mouth isn't enough to produce the sounds of language. We need to change the shape of our mouths in very precise ways to create the sounds used in the words of our language.

Try this. Exhale slowly. Then inhale and, on the next slow exhale, open and close your mouth repeatedly, making sure your lips come together when you close. Do this for a few breaths, just to get the hang of it.

Do you hear yourself saying something like [ababababa] or [mamamamama]? It isn't speech, because we don't have a meaning attached to this sequence, but it is the beginning of speech. And doing this reveals a very fundamental property of spoken language: Most words are composed of sequences of sounds that involve opening and closing our mouths.

Roughly, we have two types of sounds — we call the ones with a more open mouth *vowels* and the ones with a more closed mouth *consonants*. In the exercise above, you produced vowels when your lips were open. When your lips were closed, you produced consonants. Words are composed of series of vowels and/or consonants.

Try this. To test whether something is a consonant, say the sound you are interested in. Holding your mouth very still in the position to make the sound, say it and then, without moving any part of your mouth, breathe in and try to say the sound again. Try this with the consonant [t]. It is impossible to make two [t]-sounds in sequence without moving the tongue. This contrasts with what happens when you say the vowel [a]. It is quite easy to make two [a]-sounds in a row without changing the position of the tongue.

There are some consonants that are made without moving the tongue or lips (such as [m] or [l]), so this trick doesn't work for all consonants. But any sound where you change the position of the tongue or lips while making the sound is a consonant — or you've accidentally chosen two sounds instead of one to test.

Features are used to characterize vowel and consonant sounds. The feature [+vocalic] refers to a sound that is a vowel or is functioning as a vowel and not as a consonant (a sound that is [−vocalic]). The feature [+consonantal] refers to something that is a consonant, not a vowel — vowels are [−consonantal]. Vowels

are members of the natural class of [+vocalic] sounds; consonants are members of the natural class of [+consonantal] sounds.

Take a moment to check how well you are understanding consonants and vowels, with the first Comprehension Check in this chapter.

In this chapter, most comprehension checks give an exercise using words from South Bolivian Quechua language (SBQ). SBQ forms part of the Quechua language family, spoken in South America. (If you are a speaker of South Bolivian Quechua, please do not panic. There is still much you can do with the Language Explorations. You can find other words to illustrate the sounds. And our sources may not have the same sounds that you find.)



Comprehension Check 1

Gaby G. Vargas Melgarejo, South Bolivian Quechua

If you are unsure about finding vowels and consonants, here's a chance to test yourself. Below are a few words from South Bolivian Quechua. (The one symbol that may be unfamiliar to you is [ɾ]. It is a tap and sounds a bit like a very fast [d]. Taps are discussed later in this chapter, in Section 2.3.1.4.3.)

[manu]	'debtor'
[tura]	'brother'
[pupu]	'belly button'
[haku]	'let's go'

Your job is to identify which sounds are consonants and which are vowels. We suggest you start by saying these words as best you can. Think about how your mouth opens and closes — which sounds are likely to be consonants and which are likely to be vowels? You can make decisions based on how far open your mouth gets. You can also use the trick on p. 28 to help you determine whether a sound is a consonant or a vowel.

Make a table with three lists — consonants, vowels, unknown. (We include the “unknown” category for sounds that you aren't sure about.) Make columns for *sound*, *example*, and *gloss*. (Linguists typically use the word *gloss* instead of “meaning” or “translation”, to indicate that nuances

of meaning are not included.) We give a partial table below, with one of the vowels to give you an example to follow. The sound, [a], is in the first column. The second column shows an example word that illustrates [a] — [manu]. The final column tells what [manu] means ('debtor').

South Bolivian Quechua sounds

<i>sound</i>	<i>example</i>	<i>gloss</i>
--------------	----------------	--------------

Vowels: [+vocalic]		
--------------------	--	--

[a]	[manu]	'debtor'
-----	--------	----------

...

Consonants: [+consonantal]		
----------------------------	--	--

...

Unknown: [+vocalic] or [+consonantal]?		
--	--	--

...

As a reminder before you look at our answer: With this and all Comprehension Checks, you will learn the most if you write down your answer before looking at our answer. Be aware that if your answer disagrees with ours, it may still be a good answer. Think about our answer so that you understand it, then think about your answer and decide whether you missed something, or whether your answer is a reasonable alternative to ours.



Here's our answer

Our table is below; here are a few explanatory comments. First, notice that each word appears in both the Vowels list and the Consonants list, since each of these words has both vowels and consonants in it. A word like [tura] 'brother' is found four times because it has four distinct sounds ([t], [u], [r], and [a]), but [pupu] 'belly button' occurs only twice because it has only two different sounds ([p] and [u]).

Second, you will have noticed that we included not just the sounds, like [a], but also the words where we found the sounds, including the meanings for the words. This is a good practice to keep track of why you think particular sounds are found in your language.

These nine sounds — two vowels, five consonants, and two unknowns — are not the extent of sounds in South Bolivian Quechua. They are just the sounds found in the four words that we started with. A person working with this language would continue to seek all sounds in the language by considering more words.

South Bolivian Quechua sounds

<i>sound</i>	<i>example</i>	<i>gloss</i>
Vowels: [+vocalic]		
[a]	[manu]	‘debtor’
	[haku]	‘let’s go’
	[tura]	‘brother’
[u]	[pupu]	‘belly button’
	[tura]	‘brother’
	[haku]	‘let’s go’
Consonants: [+consonantal]		
[m]	[manu]	‘debtor’
[n]	[manu]	‘debtor’
[p]	[pupu]	‘belly button’
[t]	[tura]	‘brother’
[k]	[haku]	‘let’s go’
Unknown: [+vocalic] or [+consonantal]?		
[ɾ]	[tura]	‘brother’
[h]	[haku]	‘let’s go’



Exploring my language

It is useful and important to have a list of the sounds of your language, and to know which ones are vowels and which are consonants. In this Exploration, you begin to make a list of vowels and consonants of your language. There are three steps.

1. *Choose a few short words from your language and write them down.* Depending on how long the words are, you might only need one or two words. You want 8 to 10 different sounds to work with to begin your list. (As you write down words, you may be a bit puzzled about what symbols to use to represent the sounds. At this point, do the best you can. This chapter is designed to give you confidence in symbol choice, and we are still early in the chapter.)
2. *Prepare a list with three section headings (Vowels: [+vocalic], Consonants: [+consonantal], and Unknown: [+vocalic] or [+consonantal]?) and three column headings (sounds, example words, and glosses).* There is a table template in the Comprehension Check on p. 29; the answer to that Comprehension Check shows what the filled-in table looks like. Be sure to use the name of your language in your table's title.
3. *Determine the category for each sound (Consonant, Vowel, Unknown) and put them in the appropriate place in your table.* To do this, say each word aloud carefully, identifying each sound in the word. Now say each sound alone, focusing on how open your mouth is. Use this information to help you decide which sounds are likely to be vowels and which are likely to be consonants – and which are still ambiguous. Use the trick on p. 28 to double-check your decisions. Write the example word and its meaning next to the sound symbol. Be aware that any word with more than one sound in it will appear more than once in your lists.

Congratulations! You have started a list of the sounds of your language. The Language Explorations in this chapter continue to examine the consonants and vowels of your language, so keep track of the lists you create in these Explorations.

As you work with your language, feel free to add to your lists. Each time you add a sound, be sure to write down the word where you found the sound, so you have a point of reference when you come back to your list – especially if you are having trouble figuring out which symbols to use for some of the sounds. Also, it is okay to have more than one example for a sound, so go ahead and put down extra examples if you like. This can be especially useful with sounds that you are unsure about.

2.1.4 The articulators

In order to have words with more variety in them than sequences of [ma] and [ba], we need to “use a whistle of a different shape”. We do this by changing the shape of our *vocal tract*. The vocal tract holds the *articulators* — the parts of the body involved in making distinct speech sounds, from the lips and nose to the larynx.

There is a challenge in really understanding the vocal tract and articulators. When we look at someone speaking, we can’t see most of what is going on. Lip movement is visible, and sometimes we can see a bit of the tongue. But the rest is hidden from view. Because of this, we have work to do in order to understand how the articulators move to produce the sounds of a language. A necessary first step is to become familiar with the different articulators. You may want to refer back to this section frequently until the articulators (and their names) are truly familiar to you.

Try this. Sit very still. Now that you are still, make a short sentence — speak it aloud or, if you are a signer, sign the sentence. Do it again in slow motion, paying close attention to the parts of your body that are moving. Those parts that must move in order to produce the sentence are articulators in the language (whether spoken or signed). Since we can’t see most of the articulators in spoken language, this exercise is to help you become more aware of your articulators by feeling them move.

What articulators did you notice? We hope you noticed that your tongue moves a lot, and that your lips also are important; if you were signing, you probably noticed that not only do your fingers and hands move, but so do your arms, lips, and even other body parts.

This kind of awareness correlates with what is easy to see. Did you notice any other movements as you produced your sentence? As you read this chapter, keep thinking about the sounds or signs in your sentence — and other sentences in your language — to try to figure out what features are involved in producing your language.

To help locate the different vocal tract articulators as we introduce them, we use images. The images show what is called a *midsagittal* view — a profile view sliced right through the middle of the head. The first such image is in Figure 2. Both Figure 2 and other images in this chapter are the result of magnetic resonance imaging (MRI) technology (done by the author). MRI images are created by measuring movement of iron in the blood, represented with white pixels. Teeth and bone show up as black, as does air.

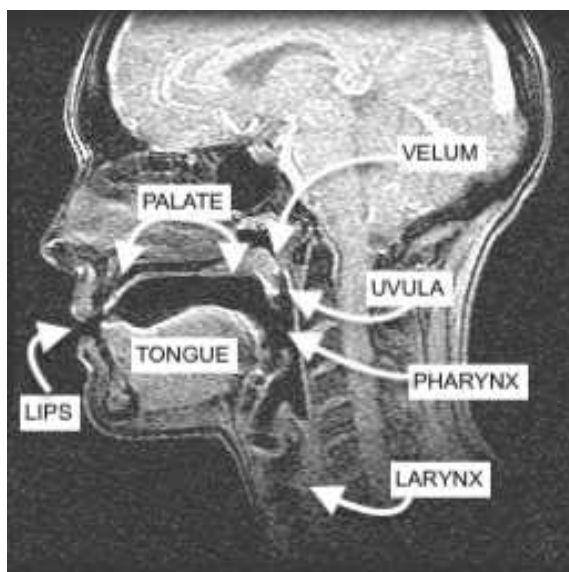


Figure 2: The midsagittal view of the vocal tract

Look at the position of the tongue. We see that the tip is curved up towards the teeth (not visible in the MRI image). The body of the tongue is somewhat low, so there is a good-sized air pocket in the mouth (remember: Air is black in the MRI image in Figure 2. The lips and tongue are light colored (because of the blood in them).

Now compare Figure 2 and Figure 3. The image in Figure 3 is a line drawing based on Figure 2 (with teeth added), in case it is helpful to see both images. You can see the profile of the nose and lips: The person is facing to the left. Can you find the tongue shape in Figure 3, with the air pocket above it? (Hint: The air pocket is white in this image.)

We control movement of the lips and tongue so we call them the *active articulators*. Active articulators move towards or even touch *passive articulators*, the parts of the vocal tract whose movement is not critical when producing language sounds, such as the teeth. *Features* typically refer to the active articulators, while descriptions of sounds are often made in terms of the passive articulators. For example, the sound [k] is made by raising the back of the tongue to the velum. You will see [k] described as a velar sound, referencing the velum, the relevant passive articulator. You will also see [k] classified by the feature [dorsal], referencing the active articulator, the tongue dorsum (the back part of the tongue). (This may be

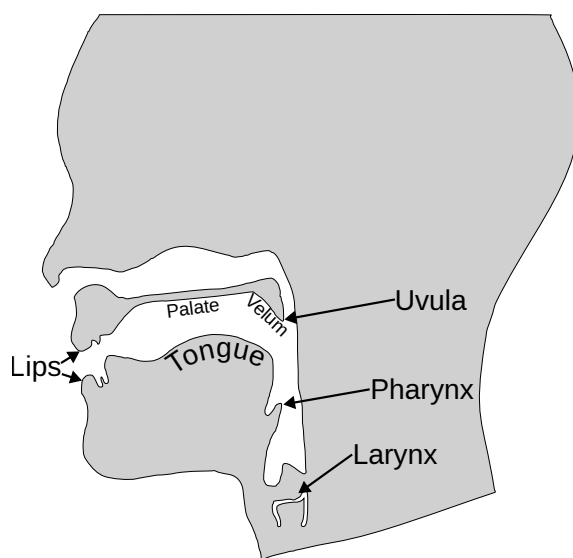


Figure 3: The midsagittal view of the vocal tract

a bit confusing. As we continue to explain the articulators, it will become easier to remember the connections.)

The roof of the mouth, or *palate*, is among the passive articulators. The palate is divided into the hard palate, in the front (between the two arrows labeled “palate” in Figure 2 and Figure 3, and the soft palate, in the back. In linguistic writing, the soft palate is usually called the *velum*. The velum is an active articulator, because we can raise it and lower it. Interestingly, the velum is also a passive articulator, because the back of the tongue can be pulled close to the velum, or even make contact with it. (Our ability to move the velum is very important outside of language — when raised, the velum prevents food or liquid from getting into the nose.)

Try this. Go to a mirror. Open your mouth very wide, stick your tongue out and say “aaaahhhh” while looking in the mirror. You will likely see something hanging down in the back of your mouth. This is made up of the velum, at the top, and the *uvula* at the bottom. The uvula is a bit of flesh that hangs off the velum. (The name comes from a Latin word referring to a grape, which the uvula resembles. Touching the uvula provokes a gag reflex, which is useful to prevent choking.)

The uvula is a passive articulator — the tongue can be pulled back close to the uvula. The uvula is not an active articulator. The uvula moves only when the velum moves: The uvula goes along for the ride.

The rear pharyngeal wall, on the back side of the *pharynx*, is our last passive articulator. Some languages use a sound made by pulling the base of the tongue back close to that wall. These are called pharyngeal sounds; we don't discuss pharyngeal sounds further in this book.

In addition to the lips and tongue, the *larynx* (or *voice box*) is made up of active articulators. (The front wall of the larynx is the Adam's apple.) The larynx houses the *vocal folds* (or *vocal cords*), thin sheets of muscle that we can move in different ways. These muscles play an important role, in speech. (They are also very important in preventing choking.)



Articulators in Sign Language

Samantha Prins and Sybil Vachaud

Sign languages, like spoken languages, can be described in terms of their *phonetics* and *phonology*. When we talk about the sounds of spoken languages, we refer to *articulators* like the lips, teeth, and tongue. For Sign languages, the relevant articulators can include the hands, arms, and even the face, head, and body. Just like using place and manner of articulation to describe speech sounds, signs are conventionally described with parameters like location, handshape, orientation, movement, and non-manual markers (note that depending on the language, non-manual markers may or may not be specified for a given sign).

- *Location*: where the sign is produced on the body or in space
- *Handshape*: the configuration of the hand(s)
- *Orientation*: directional orientation of the palm(s)
- *Movement*: how the articulators move during the production of the sign
- *Non-manual markers*: can include postures of the mouth, face, head, and body

In Plains Indian Sign Language (PISL, a variety of North American In-

dian/Indigenous Sign Language also known as Hand Talk), for example, the Northern Cheyenne sign for HUNGRY can be described in the following way:

‘Hungry’ in PISL (McKay-Cody 2019)

Signed by James Wooden Legs, recorded by Robert Cody Jr.



- Location: torso and elbow in motion
- Handshape: flat hand
- Orientation: palm facing upward
- Movement: moving sideways

Describing signs in this way allows us to better understand them for both linguistic analysis and language teaching, which together can support the maintenance and revitalization of Sign languages.

2.1.5 Summary

We have introduced several terms in this section, to label different parts of the vocal tract. The figures in Figure 2 and Figure 3 include most of these terms. It is a good figure to come back to if you need a refresher as you go forward. You can also find terms in the glossary, for a quick reminder of what they mean.

The next sections in this chapter talk about how the different articulators can be manipulated, to create different kinds of sounds, the consonants and vowels used in languages. These discussions review the articulators. By the time you get to the end of this chapter, the many terms introduced in this section should be much more familiar to you.

2.2 Effects due to larynx and velum

Recall the whistle analogy we used earlier in this chapter, and how speech is rather like a very fancy whistle. We push air through the vocal tract to get sound; we change the nature of the sound by changing the nature of the vocal tract.

We can make these changes in two general ways. We can change the kind of resonance by altering the flow of air (by narrowing the vocal folds to create voicing) and by adding a resonating chamber (by lowering the velum). We can also change the very shape of the vocal tract by moving the tongue and the lips. These movements can be done independently — a quick look back at Figure 2 shows that the tongue, lips, larynx, and velum are quite independent of each other.

We start our discussion of the effects due to different articulators by considering the larynx and the velum. (In case you need a reminder, the website <https://enunciate.arts.ubc.ca/linguistics/world-sounds/> gives a way for you to listen to the sounds discussed here, as well as some notes on how they are produced.)

2.2.1 Larynx

As air moves out of the lungs, it passes through the **vocal folds** inside the **larynx**. We can change the position and tension of the vocal folds for different effects. For example, the vocal folds can be held wide open so that air passes freely through them, or they can be held so close to each other that, as air passes, the airflow is changed, becoming turbulent. (If you have trouble imagining this, think about the flow of water in a stream. Where the valley is wide, the stream flows smoothly. Where the canyon walls close in and narrow the stream, we get turbulent water. The same kind of effects happen with airflow depending on the amount of space between the vocal folds.)

Try this. Put your hands over your ears. (Yes, do it.) Now, keeping your hands over your ears, try saying a very long [s] out loud — don't put a vowel in front of it, just make the hissing noise, just [sssss]. Keep your hands in position and try saying a very long [z] out loud — again, just the buzzing noise, only [zzzz]. Do you feel a difference? If you didn't, try again. Or place your fingers on your Adam's apple and say the two sounds. Typically, the [z] sounds much louder, and you can feel your fingers vibrate a little, while the [s] is a softer sound (though still hissy) and there is no vibration to be felt. The difference between the soft [s] and the buzzy [z] is due to the vocal folds — when they are close to each other we get turbulence that results in the louder, more buzzy sound.

Speak aloud: If you whisper the [zzz], you won't feel the difference. Whispering is achieved by preventing the vocal folds from vibrating.

Noisiness and vibration are a consequence of the vocal folds vibrating. The **voiced** sounds are those made with the vocal folds held close to each other and loosely enough so that they vibrate as air passes between them. Sounds made with the vocal folds pulled apart (so they don't vibrate) are called **voiceless** sounds. The vibrating [z] is voiced, while the quieter [s] is voiceless. The relevant feature is [±voice]. The “±” notation reminds us that there are two possibilities: [+voice] classifies voiced sounds and [–voice] classifies voiceless sounds.

Note: Some sounds, like [s] or [m], are easy to hold for a long time. Others, like [p] and [k], are very quick — because of the way they are made, you cannot hold them for a long time. Usually, when we say these we add a vowel before and/or after the consonant. However, because vowels are voiced, if you have vowels on either side of a consonant it can be hard to tell whether a consonant is voiced or not, so you need to listen closely. (If you want to feel this, try saying [apa] and [aba] several times to see whether you can feel or hear a difference — [p] is voiceless and [b] is voiced, so there is a difference to notice.)

Incorporating the larynx gives us two more natural classes — the voiced sounds and the voiceless sounds. Vowels are typically voiced, but some consonants are voiced and others are voiceless, so we will encounter this distinction throughout this book.



Comprehension Check 2

Gaby G. Vargas Melgarejo, South Bolivian Quechua

Test yourself on distinguishing voiced and voiceless consonants. Use one of the methods from p. 39 to determine whether the sound is voiced ([+voice]) or voiceless ([−voice]). You might want to make a table with two columns, one for the [+voice] and one for the [−voice] sounds.

Consider each of the South Bolivian Quechua consonants that you identified in the previous Comprehension Check and repeated here, as well as the unknown sounds. (Since vowels are typically voiced, we ignore them for this exercise).

For fun, we add one new word, [suju] ‘fetus’. (The symbol [j] is used for the sound often spelled with a y; [j] is discussed in this chapter in Section 2.3.1.4.1.) You will need to figure out which sounds are vowels in [suju], which are consonants, and which you are undecided about.

[manu]	‘debtor’
[tura]	‘brother’
[pupu]	‘belly button’
[haku]	‘let’s go’
[suju]	‘fetus’

We hope that you identified [u] as a vowel. The other two sounds, [s] and [j], are more tricky — they don’t seem like vowels, but the mouth does not close completely to produce the sounds. We put them in the “unknown” category for the moment. Now, before you read on, be sure to make your voicing table for the South Bolivian Quechua consonants.



Here’s our answer

Here’s the table we came up with for voicing in South Bolivian Quechua consonants.

South Bolivian Quechua sounds: [±voice]

[+voice] (voiced)

Vocal folds vibrate

Consonants

[m] [manu] 'debtor'

[n] [manu] 'debtor'

[−voice] (voiceless)

Vocal folds still

[p] [pupu] 'belly button'

[t] [tura] 'brother'

[k] [haku] 'let's go'

Unknown

[ɾ] [tura] 'brother' [h] [haku] 'let's go'

[j] [suju] 'fetus' [s] [suju] 'fetus'

There is a bonus to checking voicing for the unknown sounds: Two of these sounds are voiceless, [h] and [s]. Vowels are typically voiced. This means that [h] and [s] are likely to be consonants, so we can move both [h] and [s] out of the “unknown” set and into the “consonants” set, resolving two of our “unknown” questions.

We repeat the list of consonants below, to show our new understanding. We also reorganize the list from above to put all voiced consonants together and all voiceless consonants together.

South Bolivian Quechua consonants: [±voice]

voiced	{	[m]	[manu]	'debtor'
		[n]	[manu]	'debtor'

voiceless	{	[p]	[pupu]	'belly button'
		[t]	[tura]	'brother'
		[k]	[haku]	'let's go'
		[h]	[haku]	'let's go'
		[s]	[suju]	'fetus'

Unknown

voiced	{	[ɾ]	[tura]	'brother'
		[j]	[suju]	'fetus'

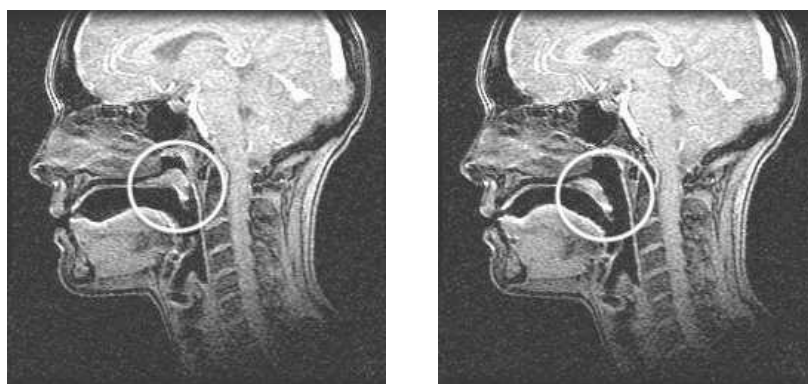
We intentionally have the columns in a different order in the two tables — there are many different ways to organize information, and this gives you two to look at. Choose a system that makes sense to you and makes it easy for you to explain the information to other people.

2.2.2 Velum

Once air has passed through the larynx, it comes towards the mouth past the pharynx, to the velum and uvula. The **velum** can be raised and lowered.

In speech, a raised velum directs air out through the mouth, not through the nose. The result is an **oral** sound. When the velum is lowered, it allows air to pass through the nasal tract (as well as through the mouth, since the lowered velum does not block off the mouth). When this happens, the sound produced is **nasal**. The feature that separates oral from nasal sounds is the feature $[\pm\text{nasal}]$: $[-\text{nasal}]$ sounds are oral and $[\text{+nasal}]$ sounds are nasal.

Look at the two images in Figure 4. Find the velum inside the circle (go back to Figure 2 if you need a reminder on where the velum is). Is the velum raised or lowered in Figure 4a? How about in Figure 4b?



a. velum position for oral airflow b. velum position for nasal airflow

Figure 4: Oral and nasal sounds

Do you see, in Figure 4a,b, how the position of the velum determines whether air through the lungs can flow through the nose? The raised velum in Figure 4a directs air into the oral cavity, producing an oral sound. Air can't escape through the nose because the velum has closed off that channel. If the velum is lowered, as in Figure 4b, air can flow out through the nose as well as through the mouth.

Try this. In Figure 4b, we see that when the velum is lowered, air can flow out of the nose. This is true even if the lips are closed, preventing air from flowing out of the mouth. If you want to convince yourself of this, make an $[m]$ — a $[\text{nasal}]$ sound — and hold it (as if you were humming or singing).

While that is going on, plug your nose. Why does everything stop? (Think for a minute before reading on.)

Since the lips are closed for the [nasal] sound [m], there is nowhere for the air coming through your nose to go. It is this nasal airflow that gives the nasal [m] its special sound.

Lowering the velum creates dual resonating chambers, the mouth (oral cavity) and the nose (nasal cavity). Even if we keep the tongue and lips in exactly the same position, we hear very different sounds depending on whether the velum is raised or lowered.

Try this. Repeat aloud the sequence [ama...aba]. The vowels are the same. The consonants [m] and [b] are very **similar** to each other, both made with closed lips and both voiced. The difference between [m] and [b] is that [m] is made with a lowered velum and so is a nasal sound, while [b] is made with a raised velum and so is an oral sound. Listen to yourself – or ask a friend to say the sequence aloud for you to listen to. The sound difference that tells you [m] is [m] and [b] is [b] is the difference between a nasal sound and an oral sound. You can try the same thing with the sequences [ana] and [ada]: [n] is nasal while [d] is oral.

Test yourself – did you figure out the two additional natural classes that were introduced in this section?

We hope you said that this section introduces the natural class of [–nasal] sounds and the natural class of [+nasal] sounds.

2.2.3 Summary

Changing what we do with the larynx results in voiced or voiceless sounds. Changing what we do with the velum results in oral or nasal sounds. Most languages have both voiced and voiceless consonants and oral and nasal consonants. Voicelessness and nasality are occasionally found with vowels as well, though voiceless vowels and nasal vowels are far less common than voiced vowels and oral vowels. The mechanisms for producing voicelessness and nasality in vowels are the same as the ones used for producing these effects in consonants.

Notice that these properties create oppositions – voiced or voiceless, nasal or oral. The action needed to create voicing means the sound cannot be simultaneously voiceless and voiced; the action needed to create a nasal sound means

that the sound cannot be simultaneously oral and nasal. The properties describe the sounds; features classify sounds for phonological analysis, and show that sounds are either [+voice] or [–voice], not both, and either [+nasal] or [–nasal], not both.

Linguists use an array to show the features of sounds. For example, the sounds [t], [s] are voiceless and oral while [d], [z] are voiced and oral. The sound [n] is voiced and nasal. The array in Table 1 shows one way of representing how these features classify the sounds.

Table 1: Feature array classifying sounds

	[±voice]	[±nasal]
[t], [s]	–	–
[d], [z]	+	–
[n]	+	+

The array in Table 1 is a dense way to present information. To use this to find the features for a particular sound, you identify the sound in the leftmost column. Suppose you want to find the features for [d], in the second row. As you scan across the row for [d] (and also for [z]), you come to a cell with “+”. You check the heading for the column with the “+” and see that the feature there is [±voice]. Since the [d] row has “+” in the [±voice] column, you know that [d] is classified as [+voice].

To test how well you’ve understood this, keep going across the [d] row until you come to a “–”. What does this “–” tell you? Think for a moment before you read on.

We hope you saw how that the “–” in the [d] row is in the column headed [±nasal]. The value “–” in this column tells you that [d] is classified as [–nasal], so [d] is an oral sound.

These features can combine with each other – a sound might be oral and voiced like [d], or oral and voiceless like [t]; a sound might be nasal and voiced like [m], or nasal and voiceless, symbolized [ṁ]. (The small circle under the [m] in [ṁ] tells us that the sound is like [m], but voiceless.) What a sound is – [t] or [k], [a] or [u], etc. – is determined by movements of the tongue and lips, the topic of the next sections. We begin with consonants – sounds that involve complete or near blockage of the airflow through the vocal tract. We then turn to vowels – sounds that allow the air to flow freely through the oral tract.



Comprehension Check 3

Gaby G. Vargas Melgarejo, South Bolivian Quechua

Test your ability to identify nasal and oral consonants, again with words from South Bolivian Quechua, repeated here.

[manu]	‘debtor’
[tura]	‘brother’
[pupu]	‘belly button’
[haku]	‘let’s go’
[suju]	‘fetus’

Take all the consonants and sort them into the two categories, [–nasal] (oral) and [+nasal] (nasal). As always, start by saying these words aloud. Use the plug-your-nose strategy (see p. 42) to determine oral vs. nasal articulation.

Since nasal sounds are typically voiced, you can ignore all voiceless consonants in the South Bolivian Quechua words – this leaves [m], [n], and the two unknowns, [r] and [j]. (These sound a bit like sounds that are often spelled with *r* and *y* respectively.) Since we are looking at oral and nasal consonants, you can also safely ignore all the vowels.

Be sure to make your own table before looking at ours, or else this won’t count as a Comprehension Check.



Here's our answer

We made this table.

South Bolivian Quechua voiced consonants: [$\pm$ nasal]

[+nasal] (nasal)			[−nasal] (oral)		
Velum lowered			Velum raised		
[m]	[manu]	'debtor'	[ɾ]	[tura]	'brother'
[n]	[manu]	'debtor'	[j]	[suju]	'fetus'

The two [+nasal] sounds are from the consonants category in the previous Comprehension Check while the two [−nasal] sounds are from the “unknown” category. Being [−nasal] does not tell whether the unknown sounds are consonants or vowels.



Exploring my language

It is likely that your language has both voiceless and voiced consonants as well as both oral and nasal consonants — most languages do.

Start refining your list of consonants by first identifying which are voiced and which are voiceless. To do this, you can use the hands-over-the-ears method from p. 39. (Be sure to check the sounds that you aren't sure about (your “unknown” list), since they could be either consonants or vowels.)

Once you've sorted your consonants into voiced and voiceless sounds, you can sort the voiced consonants into those that are oral and those that are nasal, using the plug-your-nose test (see p. 42). If needed, add a few more consonants to make sure you have both nasal and oral consonants on your list — or to convince yourself that there are no nasal consonants in your language.

Modify your table to include the voiced/voiceless and the nasal/oral categories. Be sure to include example words and their meanings to illustrate each consonant.

Finally, make use of the “unknown” category if needed.

2.3 Making consonants

We now know how we make voiced and voiceless sounds, and how we make nasal and oral sounds. But knowing this does not tell us the difference between two nasals, such as [m] and [n], or two voiceless consonants, such as [t] and [s]. We need to know more.

These differences are created by movement of the tongue and lips. The tongue in particular is a very agile articulator, able to move in a variety of ways to produce very different effects (like the difference between [t] and [s]). The tongue’s agility is because the tongue is attached only at one end, the root of the tongue. The other end is free to move in different ways in your mouth. (We hope you are moving your tongue around your mouth right now to test this out.) The lips also are able to produce a variety of effects, depending on whether they are tightly held together (stopping airflow), or are held gently close to each other allowing a bit of air to escape, or are quite open (whether rounded or not), allowing air to flow freely. (You are trying these too, aren’t you?)

There are two general categories of articulator movement in making consonants. One category defines the degree of constriction in the vocal tract by how close the active articulator gets to a passive articulator (ranging from some degree of openness to completely touching). The degree-of-constriction features define the *manner of articulation* for a sound. The other category defines where the constriction is found in the vocal tract — does the active articulator approach a passive articulator in the front, middle, or back of the mouth? The location of constriction features define the *place of articulation* for a sound.

This section is divided into two subsections: Section 2.3.1 explores manner of articulation and Section 2.3.2 examines place of articulation. Each begins with a general description, followed by discussion of some of the more common sounds illustrating manner or place properties. Features combine in many different ways to create language sounds; this chapter aims at providing a basic understanding of these properties, leaving a more complete discussion for a volume focused on phonetics.

2.3.1 Manner of articulation

We structure our discussion of the manner of articulation of consonants as a series of properties with two options for each. These properties identify different aspects of the manner of articulation; collectively, they define the manner of articulation of a consonant.

The first difference in distinguishing sounds by manner of articulation is air-flow. Is there some obstruction or impediment to airflow, or is air allowed to flow freely from the lungs, without restriction? This property divides sounds into two categories. A sound that has some degree of obstruction is called an *obstruent*. A sound without any impediment to the airflow is called a *sonorant*. Obstruction to the airflow causes turbulence, so the sounds with obstruction, the obstruents, have *turbulent air flow*. The sonorants, produced without obstruction, have smoothly flowing or *resonant airflow*. Most sources use the feature $[\pm\text{sonorant}]$ to distinguish $[\text{+sonorant}]$ sonorants from $[\text{−sonorant}]$ obstruents. If you are thinking “Aha — two more natural classes, $[\text{+sonorant}]$ sonorants and $[\text{−sonorant}]$ obstruents”, you are right! (You may also see the term *resonant* used instead of *sonorant*.)

Try this. Say a vowel out loud. Decide whether you think it is an obstruent or a sonorant. Do the same thing with a couple of other vowels. What do you think — obstruent or sonorant?

We hope you decided that the airflow was not impeded in any way while you made the vowels, and concluded that the vowels you produced are sonorants. In fact, all vowels are sonorants.

Try this. Say the consonant $[p]$. Is the air impeded? What about if you say $[s]$ or $[z]$? What do you think — obstruent or sonorant?

We hope you decided that the airflow was impeded with each of these sounds, making them obstruents. With the $[p]$, the lips close and completely stop the air, which is definitely an obstruction. With $[s]$ and $[z]$, the tongue almost touches the front of the palate — it is so close that the air does not flow smoothly. Rather, it is turbulent, and this turbulence makes a hissy or buzzy noise, called *frication*, as the air comes out of your mouth.

Our first difference, $[\text{−sonorant}]$ obstruents vs. $[\text{+sonorant}]$ sonorants, is visualized in Figure 5. Features are shown in square brackets, like $[\text{+sonorant}]$, following the standard convention for features.

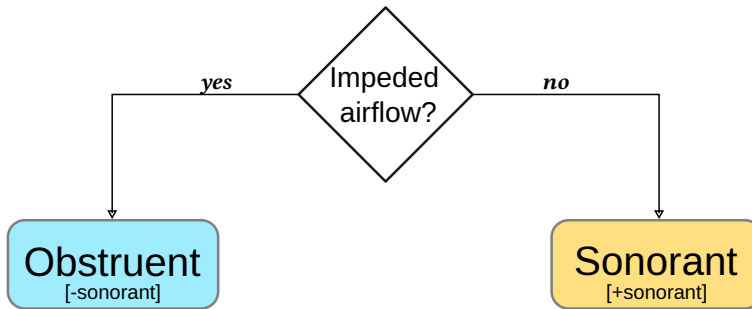


Figure 5: Classifying obstruents and sonorants

Do you think all consonants are obstruents? They are not. Consider, for example, the consonant [m]. You can see that your lips are closed, blocking air out of the mouth. But you also know that the velum is lowered, allowing air to pass freely through the nose. This is why nasal consonants, like [m] and [n], are sonorants. Other consonants are also sonorants. Obstruents are discussed in the next three sections, stops in Section 2.3.1.1, fricatives in Section 2.3.1.2, and affricates in Section 2.3.1.3, while the sonorant consonants are discussed in Section 2.3.1.4.

As we discuss the different consonant classes, we will comment on whether each is a [–sonorant] obstruent or a [+sonorant] sonorant.

2.3.1.1 Stops and continuants

Oral stops and continuants are two types of obstruent consonants. The difference between a *stop* and a *continuant* is whether the air is stopped completely coming out of the mouth. These two types of sounds are distinguished by the feature [±continuant]. If the air is stopped in the mouth, we have a [–continuant] sound, also called a stop. (Another term for stop is *plosive*.) If the air is not stopped in the mouth, we have a [+continuant] sound. The stop/continuant distinction makes a difference with consonants only. This is because air flows freely out of the mouth when making a vowel, so all vowels are continuants.

Try this. Say [p] and [f]. Which involves the complete stopping of airflow and which allows the airflow to continue?

We hope you noted that with [p], the lips stop the airflow completely, so [p] is a [–continuant] stop. Other stops include [b], [t], [d], [k], and [g], sounds we explore more in Section 2.3.2. With [f], there is continual airflow, making

[f] a [+continuant] sound. The airflow is turbulent, making a whooshy sound. The lip opening — and in particular the teeth touching the lip — is enough of an obstruction that [f] is not a sonorant. The whooshy sound is due to the friction of the air flowing through a narrow space; these sounds are described as *fricatives*.

What about [w], another consonant made with a lot of lip action — stop or continuant? Airflow continues throughout the sound [w], so [w] is also [+continuant]. But the sound is not whooshy or hissy or buzzy — [w] is a sonorant. Some sonorants are [+continuant], like [w]. We address sonorants in Section 2.3.1.4.

Try this. Say [sa], but hold the [s] for as long as you can before saying the vowel: [ssssssa]. You can do that because [s] is a continuant. Now try doing the same thing with [ta], holding the [t] for as long as you can before saying the vowel. It is very different — you have silence until the [t] is released into the vowel. That is because [t] is a stop.

We show the classification of obstruent stops (with the features [–sonorant, –continuant]) and fricatives (with the features [–sonorant, +continuant]) in Figure 6. Features are shown in square brackets as are example sounds. We show only stop sounds at this point because that’s where our focus is in this section. (We show a figure with fricative sounds in Figure 7, when we explore fricatives.)

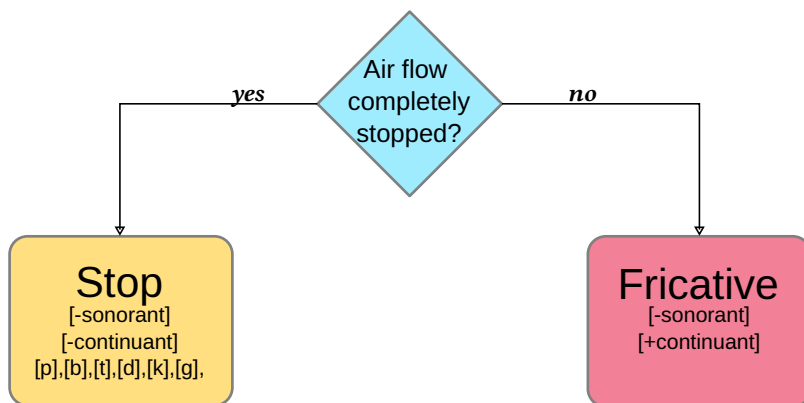


Figure 6: Classifying the [–sonorant] obstruents as stops or fricatives

Before moving on to fricatives, let’s look a little more deeply at stops. Stops have two parts, the *closure* and the *release*. Closure happens when the lips or the

tongue are used rather suddenly to block the passage of air. At some point, the closure that stops the airflow is let go, or released.

We can release an obstruent in different ways. Release is when the closure is let loose — the lips are opened or the tongue drops and air can flow again. There are also stops that are *unreleased*, where the closure smoothly transitions into the next sound or into normal breathing, without an audible release (an audible release is also known as *burst*) when the closure ends. For discussion of other types of release, see Sections 2.3.1.3 and 2.3.2.5.

Importantly, the airflow deciding whether a sound is a stop or a continuant is relative to the mouth. Air coming out of the nose “doesn’t count” for deciding whether a sound is a continuant. Because of this, nasal consonants like [m] and [n] are [–continuant] stops. (They are also sonorants, as already established). Nasal consonants have an oral release along with the nasal airflow. The stops, which are [–continuant], and the continuants, which are [+continuant], add two more natural classes.

You may have noticed that every feature introduces another *natural class*. Importantly, natural classes are defined by one or more features. For instance, the features [–voice, +continuant], or “voiceless continuants”, describe a natural class defined by two features; [+voice, –sonorant, –continuant], or “voiced obstruent stops”, is a natural class defined by three features.



Comprehension Check 4

Gaby G. Vargas Melgarejo, South Bolivian Quechua

It’s time to test yourself on identifying the [–continuant] stops and the [+continuant] continuants, again using the words from South Bolivian Quechua, and adding one more: [tʃuri] ‘son’. Focus on the oral consonants and unknown sounds only. To review, nasal consonants are [–continuant] (stops) because air is stopped in the mouth by the tongue touching some passive articulator. We know they are stops, so we don’t need to check them. Similarly, air is not stopped when making a vowel, so vowels are all [+continuant] (continuant). The full list of words is given below, including the new word, [tʃuri] ‘son’. See discussion in Section 2.3.1.3 for the sound [tʃ]; it is spelled *ch* in some languages (and sounds rather like the sound at the beginning of the English word *choose*).

[manu]	‘debtor’	[haku]	‘let’s go’
[tura]	‘brother’	[suju]	‘fetus’
[pupu]	‘belly button’	[tʃuri]	‘son’

Use the tests you’ve tried already for determining whether sounds are stops or continuants, and modify your table accordingly.



Here’s our answer

Here is the categorization we came up with for stops and continuants among the oral sounds we have from South Bolivian Quechua.

South Bolivian Quechua consonants: [±continuant]

[–continuant] (stops)

Stopped airflow

Consonants

[p] [pupu] ‘belly button’

[t] [tura] ‘brother’

[k] [haku] ‘let’s go’

Unknown

[tʃ] [tʃuri] ‘son’

[+continuant] (continuants)

Continuous airflow

[s] [suju] ‘fetus’

[h] [haku] ‘let’s go’

[r] [tura] ‘brother’

[j] [suju] ‘fetus’

2.3.1.2 Fricatives

We’ve seen that some continuant consonants are obstruents, like [p] and [f], and some are sonorants, like [w]. In this section, we explore one type of obstruent, the **fricative**. There is no special “fricative” feature. Instead, two familiar features together identify fricatives: Fricatives are [+continuant, –sonorant].

Fricatives are divided into two classes, those that are noisier and those that are softer. The noisy fricatives are called **strident** sounds while the soft ones are **non-strident** sounds. The feature [±strident] distinguishes the two. Stridents are noisy because air is being forced through a very narrow opening across a fairly hard surface. The sounds [f], [v], [s], [z], [ʃ], and [ʒ] are [+strident] fricatives, with air

forced across the teeth or the **alveolar ridge**. The alveolar ridge is right behind the upper teeth. (If you can't wait, jump forward to Figure 12 to see where it is.) The term **sibilant** is sometimes used to refer to a subset of the strident fricatives, [s], [z], [ʃ], and [ʒ]. If any of these sounds or symbols are unfamiliar to you, you can go to the website at <https://enunciate.arts.ubc.ca/linguistics/world-sounds/>. You can hear each of these different sounds, finding them by their symbols.)



Frication in Cheyenne

Rosalia Badhorse, Northern Cheyenne

Northern Cheyenne is a Plains Algonquian language (in the Algic family) spoken in Montana, in the United States. Badhorse discusses a class of consonants called fricatives that are found in the Northern dialect of the Cheyenne language as follows:

“Think of this as friction, or the way your mouth revs up to pronounce all of those [fricative] sounds... For example, there is a friction-like occurrence between your lips and teeth that is almost closed or narrow when pronouncing the “v” sound.”

Badhorse (2023: 48)

Other fricatives are non-strident (or [–strident]) because they are made with constriction involving soft tissue rather than the hard teeth or bony alveolar ridge. The symbols for non-strident fricatives may be unusual to you: [ɸ], [β], [θ], [ð], [x] and [ɣ] are non-strident fricatives. (At this point, those symbols may mean nothing to you because we have yet to address place of articulation; we do that in Section 2.3.2.)

In Table 2, we give an array showing how the manner features introduced up to this point classify sounds.

Again, we can find out how a sound is classified by scanning its row and matching the “+” and “–” values with the feature heading the column. We can quickly see that all of these sounds are not sonorants, by scanning the first feature column. The second feature column separates the stops [t] and [d] from the fricatives. The third feature column, for [±strident], only has values for the two fricatives. This is because stridency is not relevant for stops. It can only classify fricatives.

Table 2: Manner feature array classifying sounds

		[±sonorant]	[±continuant]	[±strident]
stops	[p], [b], [t], [d], [k], [g]	–	–	
sibilants	[f], [v], [s], [z], [ʃ], [ʒ]	–	+	+
other fricatives	[ɸ], [β], [θ], [ð], [x], [ɣ]	–	+	–

Another way of showing how to use the features [±continuant] and [±strident] to classify fricatives is shown in Figure 7.

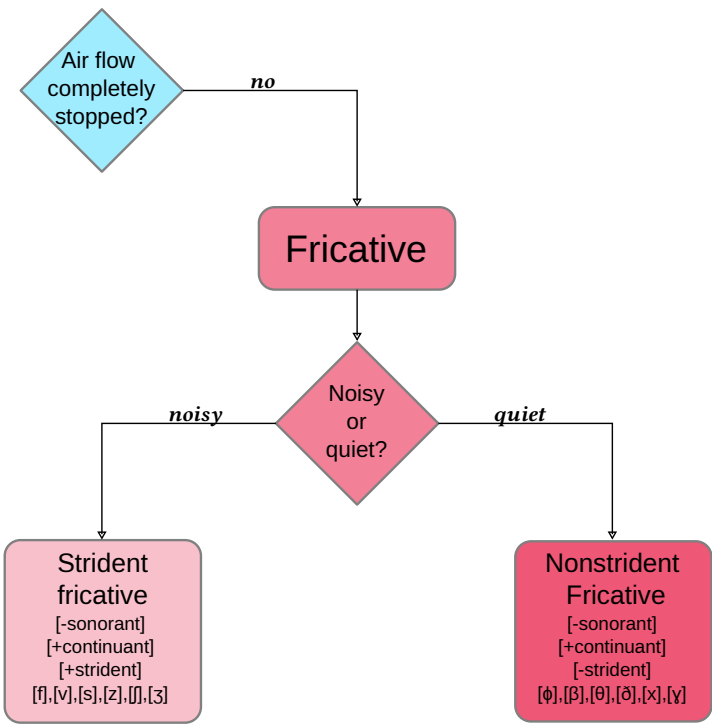


Figure 7: Classifying the [+continuant] fricatives

2.3.1.3 Affricates

In some languages, there is a type of sound that combines a stop and a fricative. Since these sounds involve a stop closure (obstructing the airflow), they are classed as [–continuant]. Yet they have a fricative component. Usually, the feature [±delayed release] is used to indicate that the stop’s release is very slow ([+delayed release]), so that the sound starts as a stop but ends as a fricative. A stop-with-fricative-release is called an *affricate*.

The stop part of an affricate typically comes first, and the fricative part is typically second. We write affricate sounds with a *tiebar* above the two symbols, showing that they are one sound. For example, $\widehat{[ts]}$ represents a single sound, an affricate, that begins with a stop closure – [t] – and ends with a fricative release – [s]. The sound $\widehat{[ts]}$ contrasts with the sounds symbolized as [ts], a sequence of [t] followed by [s].¹

The feature [delayed release] helps classify [–continuant] sounds into true stops ([–delayed release]) and affricates ([+delayed release]). Often, people will only talk about [delayed release], meaning [+delayed release], assuming that the usual situation is for consonants to be [–delayed release]. This is noted in the stop oval in Figure 8.

When we look at different sources on languages, we see that affricates are represented in a variety of ways, both in terms of their features and in terms of the symbols used. The simplest feature representation is to say that these are both stops and continuants. Here are some of the more common symbols.

- $\widehat{[ts]}$. This uses a tiebar joining two symbols, one for the stop portion and one for the fricative portion. This is the method we use in this volume.
- [ts̩]. The stop and fricative parts are smooshed together to make one symbol. We don’t like this method because it can be hard to distinguish [ts̩] from [ts].
- [ts̩]. The stop and fricative parts are written independently of each other. We don’t like this method because it is impossible to distinguish the affricate from a sequence of [t] followed by [s].
- [c]. The IPA symbol [c] denotes a voiceless palatal plosive. In languages without this sound [c] is sometimes used for the affricate $\widehat{[ts]}$.

The symbols in this list are all for coronal affricates. Another common type of affricate is made further back in the mouth. The corresponding symbols are $\widehat{[tʃ]}$ (the one we use), [tʃ̩], [tʃ̩], and [ç].

¹You may be wondering how to tell whether something is a sequence of two sounds or is a single sound. We address this issue in Chapter 7, Section 7.7.1.

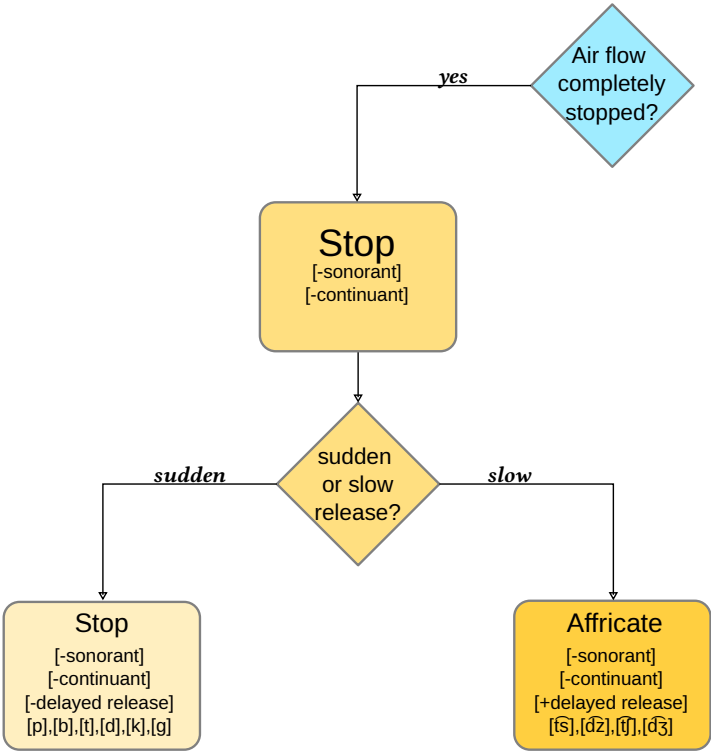


Figure 8: Classifying the [-continuant] stops and affricates

As you read about your language, be alert to how the authors represent affricates, if your language has any.



Comprehension Check 5

Gaby G. Vargas Melgarejo, South Bolivian Quechua

Test your comprehension by determining whether any of the South Bolivian Quechua oral consonants are affricates, and whether any of the continuants are strident continuants.

Here are the words.

[manu]	‘debtor’	[haku]	‘let’s go’
[tura]	‘brother’	[suju]	‘fetus’
[pupu]	‘belly button’	[tʃuri]	‘son’



Here’s our answer

We found two strident sounds, one of which is a fricative [s], and one affricate [tʃ]. (You may recall that [tʃ] sounds much like the initial sound in the English word *choose*.) There is also a non-strident fricative, [h]. South Bolivian Quechua fricatives and affricates: [±strident] and [±delayed release]

Fricatives	$\left[\begin{array}{l} +\text{continuant} \\ -\text{delayed release} \\ +\text{strident} \end{array} \right]$	[s]	[suju]	‘fetus’
	$\left[\begin{array}{l} -\text{continuant} \\ -\text{delayed release} \\ +\text{strident} \end{array} \right]$	[h]	[haku]	‘let’s go’
Affricates	$\left[\begin{array}{l} -\text{continuant} \\ -\text{delayed release} \\ +\text{strident} \end{array} \right]$	[tʃ]	[tʃuri]	‘son’

Reminder: Fricatives have continuous airflow so are [+continuant] while affricates are [–continuant, +delayed release] because they have stopped airflow with a slow release.

In the next Language Exploration, your task is to identify the manner of articulation of the obstruents you’ve identified in your language. To help with this, we first summarize our discussion of the manner of articulation of obstruents.

This sounds like a big job. It helps to break it into smaller steps, as illustrated in the visuals in Figures 5–8. We’ve put these images together into a single visual, in Figure 9 – doing so shows you that there are essentially four steps, the four diamonds. The squares hold possible answers.

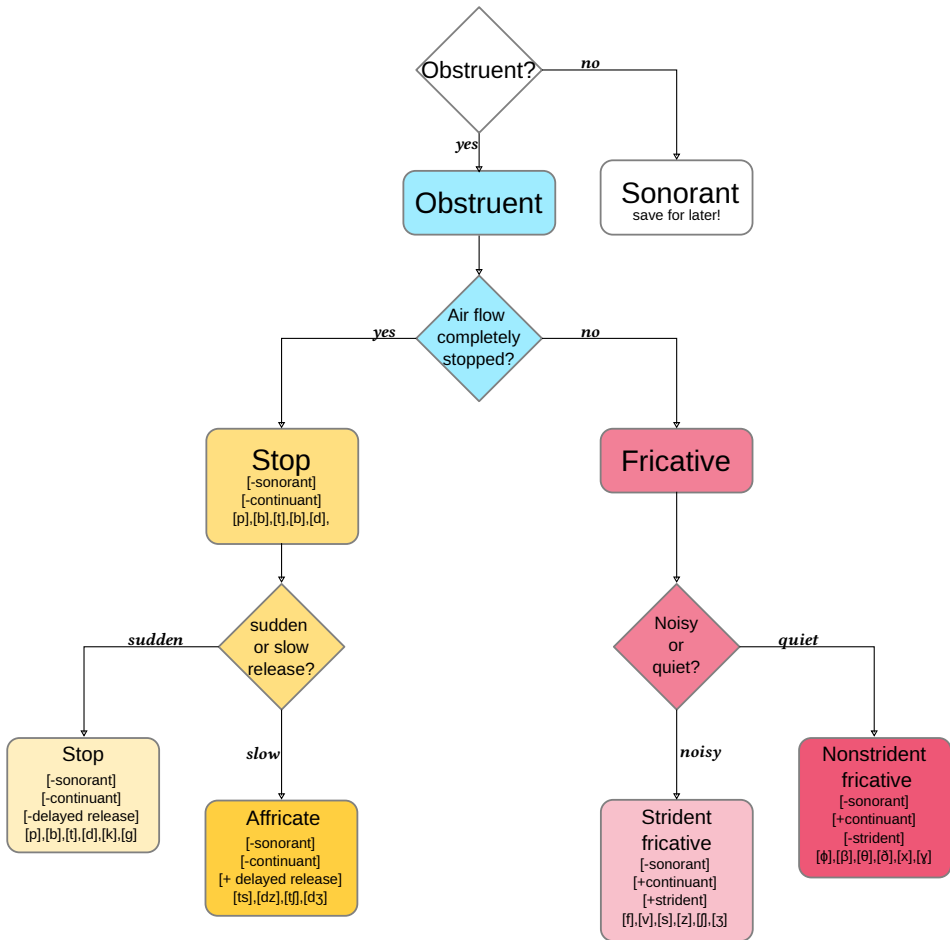


Figure 9: Classifying types of obstruents

1. Is it an **Obstruent?** Identify $[-\text{sonorant}]$ sounds (the obstruents, made with obstructed airflow).
2. Is the **Airflow completely stopped?**
 - a) Any obstruent made with complete obstruction of the oral airflow is a $[-\text{continuant}]$ sound.
 - Is there **Sudden or slow release?**
 - If the release is sudden, the sound is a stop.
 - If the release is slow, the sound is an affricate.

b) Any obstruent made without completely stopping the air is a [+continuant] sound. Sounds that are [–sonorant, +continuant] are called fricatives.

- Are the fricatives Noisy or quiet?
 - Noisy fricatives are [+strident].
 - Quiet fricatives are [–strident].



Exploring my language

Revisit the list of consonants you have been working with in your Language Explorations. Your task is to sort the obstruents into four categories:

1. stops [–continuant, –delayed release]
2. affricates [–continuant, +delayed release]
3. strident fricatives (sibilants) [+continuant, +strident]
4. non-strident fricatives [+continuant, –strident]

Use the steps summarized in Figure 9 to figure them out.

2.3.1.4 Sonorant consonants: Nasals, liquids, approximants

Sonorant consonants are consonants that have resonating airflow, through either the nose, the *nasal* consonants like [m] and [n], or the mouth, the oral *approximant* consonants like [j] and [w].

Sonorant nasal consonant features are [+sonorant, +nasal, –continuant] and sonorant approximant consonant features are [+sonorant, –nasal, +continuant]. Sonorant consonants are typically voiced, and nasals, glides, and liquids are all typically voiced sounds. We have already discussed nasal consonants, in Section 2.2.2 of this chapter. In this section, we consider the approximants, the oral sonorant consonants.

Approximants get their name because the active articulator nears, or approximates, the passive articulator, but not enough to cause turbulence (so they aren't fricatives). There are two classes of approximants — the *glides* and the *liquids*. Both types of sounds may be familiar to you already, but if you have questions about what the different symbols represent, please go to <https://enunciate.arts.ubc.ca/linguistics/world-sounds/> and listen to the relevant recordings.

2.3.1.4.1 Glides Generally, glides are round [w] and palatal [j] (also written [y] in some sources). There is no special “glide” feature. Glides are sonorants because the mouth is open enough for the airflow to resonate. Glides are continuants because the airflow through the mouth does not get impeded. Glides are oral because air flows out the mouth, not the nose. These three features together, [+sonorant, +continuant, –nasal], identify glides, without resorting to an additional feature. However, it is often very convenient to simply call these sounds “glides”.

2.3.1.4.2 Liquid approximants: Laterals There are also two types of liquids, the *l*-like sounds and *r*-like sounds. First, we address the *l*-like sounds, which are called *laterals*, a term that refers to the sides of the tongue.

The feature for these sounds is named [±lateral] because, regardless of what else is happening with the tongue, there is always space for air to flow along one side of the tongue or the other. The tongue touches the roof of the mouth somewhere for a lateral, yet airflow still occurs along the side of the tongue.

Try this. Put your tongue in position to make an *l*-sound, and breathe in. If it is a cool day or the air conditioning is strong, you will feel coolness along one side of your tongue, but not along the center of your tongue. Now do the same with an *s*-sound — practice a few *s*-sounds, then hold your tongue in the *s* position and breathe in. You are likely to feel coolness down the center of your tongue but not along the sides.

The difference is due to the way we position our tongues. With [s], we make a groove down the middle of our tongues that air shoots out through, and this is where we feel the coolness when we breathe in. With [l], the tongue creates closure with the roof of the mouth, but the side of the tongue drops and allows air to flow freely — and when we breathe in, that’s where we feel the coolness.

In addition to lateral sonorants, there are lateral obstruents, the lateral fricative and the lateral affricate. The *lateral fricative* is exactly what the name sounds like — the tongue is in position to make a lateral, but is close enough to the edges of the mouth that the friction of air coming through the narrow passage causes turbulence, the hallmark of fricative sounds. The symbol for the lateral fricative is [ɬ], which some people call a “belted *l*”.

Laterals are chameleon-like in that sometimes they fall into the same class as stops, the [–continuant] sounds, and sometimes they pattern with the other approximants, the [+continuant] sounds. When you think about how they are

made, this makes sense: The tip of the tongue stops the air flow, as is done with [–continuant] sounds, while the side of the tongue allows air to flow smoothly through the mouth, as with [+continuant] sounds. In many languages, it doesn't really matter which classification laterals are given, [–continuant] or [+continuant].

Try this. Read the description of the lateral fricative (or belted *l*) above and try making it. If it sounds like a plain [l], you'll need to widen your tongue so that the space available for airflow is narrower.

The *lateral affricate* is made with a normal stop closure at the start (just like how we stop the air with our tongue when making a [t] or [d] sound), followed by a lateral fricative release. As with the other affricates, the symbol for a lateral affricate is made with a tiebar joining the [t] and the [ʃ]: [tʃ̟].

2.3.1.4.3 The rhotics, or *r*-sounds. There are several *rhotics*, made in different ways. Rhotics are [+sonorant, +voice, –nasal, +continuant]. (They are continuant because they are made with the front of the tongue free, not touching any part of the roof of the mouth.) Many resources do not give much information about what kind of *r*-sound is used in a language, but you may be lucky and find a description with good detail. Because there are so many different types of rhotics, we do not go into detail here, beyond mentioning the trill (or “rolled *r*”) and the tap.

The trill, [r]. A *trill* is made by putting the tongue in position to make a [d] or an [l]. With your tongue in that position, relax it so that when air flows past the tongue, it causes the tongue to vibrate a little. The best way to learn to make this sound is to practice a lot, putting it between two vowels. It is a bit of a challenge to learn how much to relax the tongue, but practice does pay off. Note that the tongue is relaxed, not an active participant in the “rolling” process. Some people use a feature [trill] for trilled rhotics.

The tap, [ɾ]. A *tap* is a very quick sound, just like the name implies. To make a tap, you essentially make a [d] sound just as quickly as you can. Again, practice helps. It might also help to practice [ada] and [ara] (with the tap) in succession so it is easier to focus on making the [d] and the [ɾ] different from each other. Some sources use the word *flap* instead of tap.



Comprehension Check 6

Gaby G. Vargas Melgarejo, South Bolivian Quechua

The job now is to identify approximants in South Bolivian Quechua.

You know that approximants are [+sonorant], so you can safely ignore obstruents since they are [–sonorant] (the obstruents in these words are [t], [b], [k], [s], and [tʃ]).

Approximants are consonants, so you can ignore vowels since they are [–consonantal]. (The vowels in these words are [a], [u], [i]).

Approximants are continuants, so you can ignore stops since they are [–continuant]. In addition to the oral obstruent stops [p], [t], [k], and [tʃ], there are the two [+nasal] stops [m] and [n].

Not many sounds are left. Your job is to classify the approximants as glides, laterals, or rhotics. Here are the words again.

[manu]	‘debtor’	[haku]	‘let’s go’
[tura]	‘brother’	[suju]	‘fetus’
[pupu]	‘belly button’	[tʃuri]	‘son’



Here’s our answer

Here’s our table of approximants in our words from South Bolivian Quechua.

South Bolivian Quechua approximants

[+sonorant, consonantal, +continuant]

Glides			Rhotics		
[j]	[suju]	‘fetus’	[r]	[tura]	‘brother’
(No laterals found.)					



Exploring my language

Does your language have any approximants? Identify any on your list of [+sonorant] sounds, dividing them into liquids (laterals and rhotics) and glides. Be sure to include example words and their glosses.

2.3.1.5 Reviewing manner of articulation

The key to distinguishing the consonant classes discussed in this section revolves around the type of closure and release made in the vocal tract — is the closure complete or almost complete? Does the degree of closure/release cause turbulence in the airflow? There is so much variability in the nature of the (almost) closure and the release that we have identified several dimensions of variation for consonants.

There are **obstruents**, consonants made by obstructing the vocal tract in some way. Obstruents are in opposition to **sonorants**, made with no vocal tract obstruction. Sonorants are typically voiced; obstruents can be either voiceless or voiced; usually a language with voiced obstruents also has comparable voiceless obstruents, though the converse is not necessarily the case (languages with voiceless obstruents do not always have the voiced counterparts). You may find it helps to refer to the visual summary in Figure 9 as we discuss manner of articulation for obstruents.

There are **stops** and **continuant**s; stops are made with complete blockage of the oral airflow. Continuant s are made with different degrees of narrowing, but no complete closure. If the narrowing is sufficient to cause airflow turbulence, the consonant is an obstruent. If the narrowing does not cause airflow turbulence, the sound is a sonorant.

Obstruents with a continuant component include both **fricatives** and **affricates**. Fricatives have only a continuant component while affricates are a stop with a continuant release, making them a sequence of a stop and a fricative. Fricatives — and the fricative part of an affricate — are either **strident** or **non-strident**, depending on whether the air is forced through a very narrow channel (strident sounds are made this way — they are noisier).

Sonorant consonants include nasals and approximants. **Nasal** consonants are typically stops made with the velum lowered so air passes through the vocal tract

and out the nose while making the sound. Oral sonorants are the sounds called *approximants*, the *glides* [w] and [j] and the *liquids* (*rhotics*, or *r*-sounds, and *laterals*, or *l*-sounds).

In discussing these different types of sounds — sonorants and obstruents (indicated by [±sonorant]), nasal and oral sounds ([±nasal]), voiced and voiceless sounds ([±voice]), stops and continuants ([±continuant]) — we have talked about how they are made, how they sound, and the features that are used to classify them. The features uniquely identify each class of sounds, as shown by Table 3, a figure that is repeated in Appendix B for convenience.

Table 3: Manner of articulation features distinguishing consonant classes

description	[±sonorant]	[±nasal]	[±voice]	[±continuant]
nasal stop	+	+	+	–
approximant	+	–	+	+
voiced stop	–	–	+	–
voiced fricative	–	–	+	+
voiceless stop	–	–	–	–
voiceless fricative	–	–	–	+

Sonorant sounds are typically voiced, as shown in Table 3. We do sometimes find voiceless sonorants (nasals, approximants, liquids, as well as vowels). Keep an ear out with your language to see whether you hear voiceless sonorants. Additional features — such as [strident], [delayed release], and [trill] — make further distinctions in the consonant categories.

With this array of consonant classes, it may seem like we ought to be done. But there is another set of properties that is needed to uniquely classify specific sounds. We need to know where the narrowest constriction is found in the vocal tract. This property is called *place of articulation*. Place of articulation makes the difference between stops in individual categories, like the two nasal stops [m] and [n], or the two voiceless stops [t] and [k]. Place of articulation is the topic of the next section.



Exploring my language

Review the chart(s) you have made for the consonants in your language. Check the manner of articulation features for each consonant. If needed, tidy it up – you have created a resource that you can use in all chapters in the rest of this book – including this one, when we examine place of articulation.

2.3.2 Place of articulation

Place of articulation refers to where, in the vocal tract, any narrowing or constriction takes place when making a consonant.

It really helps to have a good feel for the inside of your mouth when trying to understand [place of articulation](#). Most of us don't spend much time feeling around inside our mouths so we aren't always aware of where different parts are. A labeled image like the one in Figure 2, repeated below in Figure 10 for ease of reference, is a starting point.

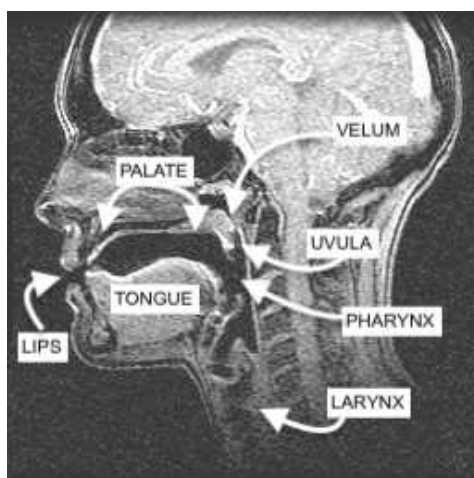


Figure 10: Midsagittal view of the vocal tract, repeated from Figure 2

In this section, we really work at becoming aware of the active and passive articulators so we become adept at identifying where the narrowest constriction occurs when making a consonant. We begin at the lips and work our way back in the vocal tract to the larynx. As we go, we introduce the symbols for the different sounds that we encounter, as well as the place of articulation features. Combining manner of articulation features with place of articulation features allows us to classify the consonants of a language.

2.3.2.1 Constrictions with the lips and the feature [labial]

We have a fair amount of control over our lips. We can press them together completely to produce a stop sound like [p]; we can hold them close to each other but relaxed to produce a non-strident fricative, like [ɸ]; we can catch the lower lip lightly with our upper teeth to produce a strident fricative, like [f] (the teeth create a narrow enough opening that the airflow is turbulent). We can even put our lips in a round shape, to create consonants like [w] and vowels like [o]. Sounds using the lips in any of these ways have the feature [labial].

Try this. If you didn't do it while you read that last paragraph, go back and read it again. This time, try making sounds in each of the ways described. (We'll give a little more detail below, but it helps to get a rough idea of how the articulators work as we start the discussion.)

Labial stops, oral and nasal. To quickly review, when we completely stop the airflow through the oral tract, we create a **stop** sound. If the velum is lowered while we stop the air in the oral tract, we make a **nasal** stop. Now we can add place of articulation: If the air is blocked with the lips, we produced a [labial stop], either oral or nasal. The [labial] stops are voiceless [p], voiced [b], and nasal [m].

Labial fricatives — strident and non-strident. When airflow is not blocked completely, but escapes through a narrow opening, the sound produced is a fricative. When the opening is extremely narrow so that air is forced over the teeth, the fricative is noisier. Noisy fricatives are strident fricatives, voiceless [f] and voiced [v]. A slightly less narrow opening produces non-strident fricatives, voiceless [ɸ] and voiced [β].²

²Some resources use more specific descriptive terms to refer to labial sounds, separating *bilabial* sounds (those made with both lips, [p], [b], [m], [ɸ], and [β]) and *labiodental* sounds (those made with the teeth and tongue together, [f] and [v]).

Try this. To learn to make a non-strident labial fricative, it helps to start with a labial stop. Let's start with a voiced one between two vowels: [aba]. Say [aba] aloud a few times. Get the feel of the lips coming all the way together. Now say it again, except lighten up the pressure of the lips closing. Keep repeating the sequence, but with less and less pressure until you realize that there is a consonant there, but the lips are not closing completely. (If you back off on the closure too much, you won't get an obstruent — you'll end up with something more like [awa] even though your lips aren't rounded.)

The sound you are producing by relieving the pressure only slightly is a voiced labial fricative, [β]. (We can also make a voiceless labial fricative, [ɸ]. Start exactly the same way as you did to say [aβa], except with a voiceless labial stop, not a voiced labial stop: [apa], not [aba].)

Labial approximant. If we round our lips to make a consonant, the opening is so large that we cannot make an obstruent — we make a sonorant approximant instead. A consonant made simply by lip rounding is the glide [w].

Labials: Sounds made with the lips. The eight consonants introduced here are the most common consonants made with the lips. Each is classified by the feature [labial] so they are all members of the class of [labial] consonants, even though some might be described as bilabial and some might be described as labiodental. The image in Figure 11 shows the variety of labial consonants. We use three conventions in this and other images exploring place of articulation.

1. Following the standard convention, feature names are given in square brackets, like [labial], as are IPA symbols, like [p] and [b].
2. We use the convention of plain text for the place of articulation descriptive terms, like *Bilabial* and *Labiodental*.
3. We use a single color to show sounds that share a key place of articulation feature. In this example, the color red is used for bilabials because they involve both lips. The shaded red-to-yellow indicates that labiodentals involve more than the lips.

We have already seen that there are a number of different manners of articulation (see Table 3 for a review). As visualized in Figure 11 and detailed in Table 4, we can combine manner of articulation features with the feature [labial] to distinguish the eight different labial consonants from each other.³ (Recall the web-

³Close comparison with Table 3 shows that we have omitted the feature [±nasal] in Table 4. We have done this because the figure fits the page as long as we don't add a column for the feature [±nasal]. Why did we choose to omit [±nasal] instead of some other feature? Because there is only one [+nasal] labial sound, [m] — all other sounds in the figure are [−nasal]. This makes it easy to remember how the feature [±nasal] fits into the table.

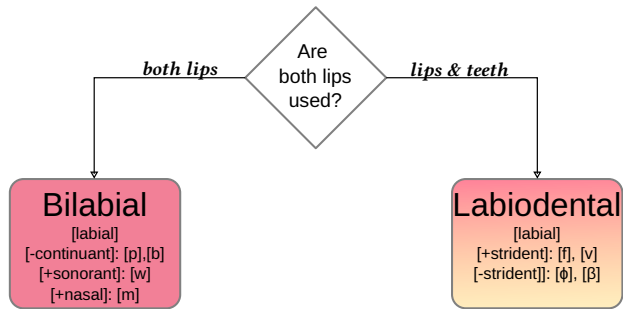


Figure 11: Identifying [labial] consonants

site <https://enunciate.arts.ubc.ca/linguistics/world-sounds/> where you can hear recordings of sounds whose symbols are not familiar, like [β] and [ɸ].)

Table 4: Manner of articulation features distinguishing [labial] consonants

labials	description	[±sonorant]	[±voice]	[±continuant]
[m]	nasal stop	+	+	–
[w]	approximant	+	+	+
[b]	voiced stop	–	+	–
[β], [v]	voiced fricative	–	+	+
[p]	voiceless stop	–	–	–
[ɸ], [f]	voiceless fricative	–	–	+

The place features distinguishing [labial] consonants do not distinguish between labial and labiodental sounds – the latter are distinguished by manner features. All of the labial sounds have the feature [labial]. As with other features, some researchers classify sounds with [±labial]. Others simply use [labial] for the class, and don’t have a “[–labial]” class. We follow the [labial]-or-nothing system pattern.

You may be wondering why we are talking about both features and descriptive terms. We do this because this is what is found when you read about languages in other sources. A sound can be referred to by either its features or its description, or by a combination of the two. For example, [p] is a “voiceless bilabial stop”, a “[labial, –voice, –continuant]”, a “[labial] voiceless stop”, and so on. Descriptions of languages typically combine these methods of identifying sounds, so it is helpful to know that both refer to the same sound, [p]. Features are more common in writings by linguists, especially those describing the sound patterns of a language. For these reasons, we include both methods of talking about sounds

in this chapter, even though this can be a bit overwhelming when it is new. It may help to remind yourself that features typically refer to the active articulator and descriptions typically refer to the passive articulator. (One exception to this general trend is the descriptive term “nasal” and feature [nasal], which both refer to air passing through the nose, the consequence of raising the velum.)



Comprehension Check 7

Gaby G. Vargas Melgarejo, South Bolivian Quechua

We are starting to make a new set of lists. In this set, we will be identifying the South Bolivian Quechua consonants by place of articulation, beginning at the lips, the front of the mouth. Your master list can be headed *South Bolivian Quechua consonants by place of articulation* and your first section can be labeled *Lips: [labial]*.

Here are the ten consonants from the previous Comprehension Checks, in one table. The consonant identified in each group is in black text; other consonants and vowels are in gray text. For example, in [p] [pupu] ‘belly button’, only the sound [p] is in black text, illustrating that the sound [p] is in South Bolivian Quechua.

South Bolivian Quechua consonants [+consonantal]

Voiceless stops			Voiceless fricatives		
[p]	[pupu]	‘belly button’	[s]	[suju]	‘fetus’
[t]	[tura]	‘brother’	[h]	[haku]	‘let’s go’
[k]	[haku]	‘let’s go’			
Voiceless affricate					
[tʃ]	[tʃuri]	‘son’			
Sonorant nasals					
[m]	[manu]	‘debtor’			
[n]	[manu]	‘debtor’			
Sonorant glide			Sonorant rhotic		
[j]	[suju]	‘fetus’	[r]	[tura]	‘brother’

Your task now is to review these consonants and find the labial ones. Create a table of labial consonants in South Bolivian Quechua, following the model in Table 4. Only use consonants that you find in the words

you have from South Bolivian Quechua — don't simply copy all the consonants from Table 4. You are making a chart for the labial consonants of one language, not a chart of all labial consonants. Be sure to include example words.



Here's our answer

Here's how we made our chart, using the consonants identified in earlier Comprehension Checks.

South Bolivian Quechua consonants with [labial] place of articulation

labials	description	[±sonorant]	[±voice]	[±continuant]
[m]	nasal stop	+	+	—
[p]	voiceless stop	—	—	—

Examples

[m]	[manu]	'debtor'
[p]	[pupu]	'belly button'



Exploring my language

Have you found any [labial] consonants in your language? If not, think a bit and see if you can find any words with a [labial] consonant. (Not all languages have labial consonants.)

If you've identified some [labial] consonants, place them in a table like the one in Table 4. Once again, be sure to include the example words (as modeled in the previous Comprehension Check). These help you and help someone else who is reading your work.

2.3.2.2 Constrictions with the front of the tongue and the feature [coronal]

Suppose you have identified a consonant that is made with constriction in the mouth, but the constriction is not made with the lips — the constriction is made with the tongue. The next question to ask is whether you are using the front of the tongue or the back (or body) of the tongue.

The front part of our tongue is very agile and can be moved in many different ways. Before discussing some of the sounds made with the front of the tongue (the tongue tip and **tongue blade** — the broad part right behind the tip), let's get familiar with moving the tongue tip against the relevant passive articulators.

Try this. Press your tongue firmly against the roof of your mouth. Move your tongue so that you feel around the roof of your mouth. Imagine that there is something stuck to the roof of your mouth and you are trying to find it with your tongue and to push it out of your mouth. Now turn the tip of your tongue back then push it forward firmly along the roof of your mouth towards your teeth. Now do the whole bit of tongue gymnastics again, while looking at Figure 12. Can you feel the curve of the length of the palate? Do you feel the hard bump right behind your teeth?

You may notice that, as you slide your tongue towards your teeth, there comes a point where there is a bit of a bulge or hill that your tongue has to get past in order to reach the teeth. It is right behind the teeth; it is called the **alveolar ridge**. In Figure 12, an arrow points up to the alveolar ridge. (One meaning for *alveolus* is the bony socket containing a tooth. The alveolar ridge contains the sockets for the upper teeth.) The **palate** is the roof of the mouth, between the two arrows pointing down.

Now that you know about the alveolar ridge and palate, try running your tongue along the roof of your mouth again. Do you feel both your alveolar ridge and your palate?

Try this. Move your tongue around your mouth, changing the pressure as you touch the teeth, the alveolar ridge, the post-alveolar region, and the palate with your tongue. Use more pressure; use less pressure. Maybe even get as close as you can without actually touching.

Try doing this exercise — and the preceding one — while vocalizing. Can you identify different sounds as you do this?

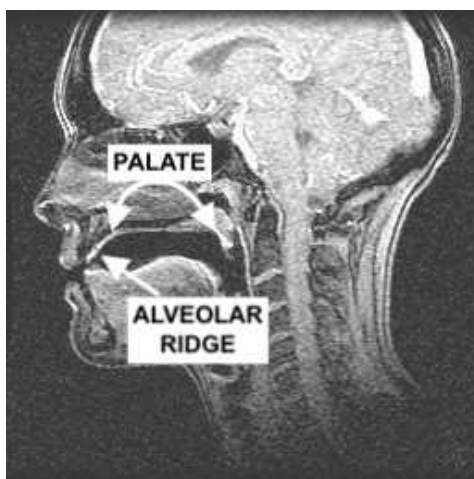


Figure 12: The alveolar ridge

Sounds made with the front of our tongues have the feature [coronal]. These activities give you a sense of the dexterity we have with the front of our tongues – where we can place it, whether it is pointed or flat, how close we can get to a passive articulator without touching it. This agility allows us to make a variety of different coronal sounds, by making different types of constriction (see Section 2.3.1) at different passive articulators that we can reach with the front of our tongues.

Most languages have coronal sounds; the most common coronal sounds are made with the tongue tip against the alveolar ridge or against the teeth. We can make both coronal stops and coronal fricatives.

Try this. To make a coronal stop, press the front of your tongue firmly against your alveolar ridge. When you release your tongue, you should hear something like a [t] or a [d]. It might help to put the stop between two vowels to start with ([ata]), then try putting the stop before a vowel ([ta]) and after a vowel ([at]), to get a feel for how the sound works at the beginning and end of a word.

Once you feel confident with those sounds, try the same thing, except this time put your tongue against the back side of your upper teeth instead of against the alveolar ridge. These will be dental sounds.

Now try both places of articulation, but this time don't make a complete closure — just get the tongue tip really close to the alveolar ridge, then really close to the teeth, so you make fricatives, such as [s] or [z].

Touching the alveolar ridge makes an **alveolar** stop; the feature for such stops is [coronal]. We can make voiceless, voiced, and nasal coronal (alveolar) stops. The symbols for these sounds are [t], [d], and [n] respectively. Narrow constriction makes a fricative; alveolar fricatives include voiceless [s] and voiced [z].

Touching the teeth makes a **dental** stop, whether voiceless, voiced, or nasal. The feature that differentiates dentals from plain coronals is [±distributed] — referring to using the broad **tongue blade**, not the narrow tip. (The tongue blade is the area right behind the tip.) Dentals are [+distributed].

The symbols for the three dental stops are exactly the same as the symbols used for alveolar stops except dentals have a **diacritic** called a **sub-bridge** beneath: voiceless [t̪], voiced [d̪], and nasal [n̪]. A diacritic is an extra mark added to a symbol, that changes the meaning of the symbol slightly. In this case, the sub-bridge diacritic tells us that the sound is modified to include dental articulation — that is, touching the teeth when the sound is produced. The sub-bridge looks a bit like a tooth, which may help you remember that a sound with the sub-bridge is a dental sound, made with the tongue blade against the teeth. Distributed fricatives include the non-strident voiceless [θ] and voiced [ð] as well as the strident voiceless [ʃ] and voiced [ʒ].

Alveolar and dental sounds are both made in the front of the mouth and so are classed together as [+anterior] sounds. There are also coronal sounds that are [−anterior], made with constriction further back in the mouth.

Try this. Try touching the following with the tip of your tongue — the teeth, the alveolar ridge, the palate. Notice how your tongue curls back a bit when it gets to the palate — at least, it will if you are using the tongue tip to touch the palate. The curling gesture is used in making **retroflex** sounds. Symbols for retroflexes are like symbols for alveolar consonants with a long tail curling to the right underneath: voiceless [ɬ], voiced [ɖ], nasal [ɳ], fricatives [ʂ] and [ʐ].

Now try touching the palate with the front of your tongue, *without* curling the tip back. Sounds made with constriction here are called **post-alveolar** sounds. Post-alveolar fricatives (voiceless [ç] and voiced [ʝ]) and affricates (voiceless [tʃ] and voiced [dʒ]) are fairly common. Sometimes these are called **palatal**.

We do not discuss retroflex or post-alveolar sounds further in this chapter, but you can go to <https://enunciate.arts.ubc.ca/linguistics/world-sounds/> to hear recordings of retroflex and post-alveolar sounds.

Finally, most laterals, like [l], and most rhotics, like [ɾ] and [r], are typically made with some contribution from the front of the tongue, so laterals and rhotics are typically [coronal] sounds as well.

Coronals: Sounds made with the front of the tongue. The coronal consonants introduced here are the most common ones to find. From front to back, we have the [+anterior] coronals, dental and alveolar, then the [−anterior] coronals, retroflex and post-alveolar, visually laid out in Figure 13. The yellow path identifies the most common coronal sounds, the [coronal, +anterior, −distributed] alveolar consonants.

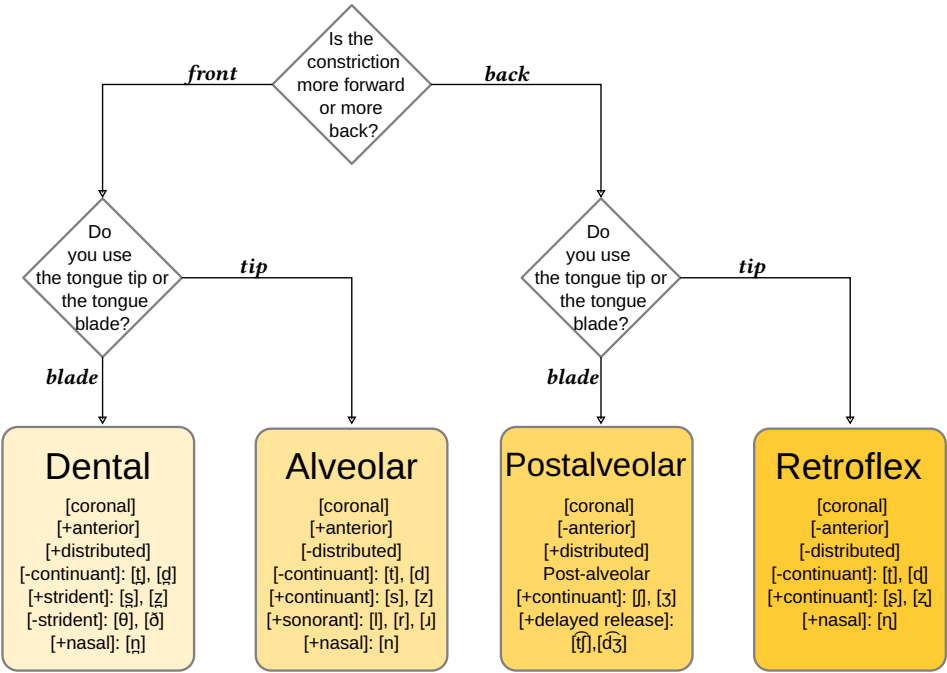


Figure 13: Identifying [coronal] consonants

The details of the place of articulation features distinguishing these four types of coronals from each other are given in Table 5. As we move from the front of the mouth to the back of the mouth, the yellow color becomes more intense. The

most common of the [coronal] sounds is possibly the alveolar sounds, with the features [coronal, +anterior, –distributed]. (See p. 68 about the use of [coronal] instead of [±coronal].)

Table 5: Place of articulation features distinguishing [coronal] consonants

coronals	description	[coronal]	[±anterior]	[distributed]
[t̪], [d̪] [ɲ]	dental	[coronal]	+	+
[t], [d], [n]	alveolar	[coronal]	+	–
[t̠], [d̠], [ɳ]	retroflex	[coronal]	–	–
[t̠], [d̠], [ɳ]	post-alveolar	[coronal]	–	+

Place of articulation is more complex for coronal sounds because the tip of the tongue is so flexible and so mobile. Each of the four place categories noted in Table 5 can combine with the different manner of articulation categories found in Table 3, to characterize the different kinds of coronal consonants – stops, fricatives, nasals, and voiced or voiceless sounds.

Many languages have only one series of coronal stops, either dentals or alveolars. Resources about such languages handle the situation in different ways. Some will describe the articulation carefully, telling why the sounds are classed the way they are. Others will not, so you don’t actually know whether the sounds are dental or alveolar until you do your own analysis. Of those that do carefully describe the way the sounds are pronounced, most use the alveolar symbols [t], [d], and [n], even when the dental symbols [t̪], [d̪], and [ɲ] are more accurate, because the first set of symbols is easier to write – even though the symbols are a bit misleading.



Comprehension Check 8

Gaby G. Vargas Melgarejo, South Bolivian Quechua

Now you will add a section to your master list of South Bolivian Quechua consonants by place of articulation; the section can be called *Tongue tip: [Coronal]*.

Review the consonant chart below to identify the coronal consonants. Again, black text emphasizes the consonant illustrated by each word while gray text de-emphasizes other sounds. For instance, [m] [manu] ‘debtor’ illustrates the consonant [m] while [n] [manu] ‘debtor’ emphasizes the consonant [n].

South Bolivian Quechua consonants [+consonantal]

Voiceless stops			Voiceless fricatives		
[p]	[pupu]	‘belly button’	[s]	[suju]	‘fetus’
[t]	[tura]	‘brother’	[h]	[haku]	‘let’s go’
[k]	[haku]	‘let’s go’			
Voiceless affricate					
[tʃ]	[tʃuri]	‘son’			
Sonorant nasals					
[m]	[manu]	‘debtor’			
[n]	[manu]	‘debtor’			
Sonorant glide			Sonorant rhotic		
[j]	[suju]	‘fetus’	[r]	[tura]	‘brother’

Now create a table of the coronal consonants, following the model in Table 4 (but changed to show coronals).

Again, use only the coronal consonants that you find in the words in South Bolivian Quechua. You are making a chart for the coronal consonants of one language, not a chart of all coronal consonants. Be sure to include example words.



Here’s our answer

This may have been more challenging than the labial chart, because there are so many different place of articulation features for coronals. Here’s how we made our chart, using the consonants identified in earlier Comprehension Checks. The feature [distributed] is abbreviated as [dist] here.

The affricate [tʃ] is also [+delayed release], but we've left this feature out of the table to make the table fit the page.

South Bolivian Quechua consonants by place of articulation

Tongue tip: [coronal]

coronals	description	[±continuant]	[±anterior]	[dist]
[n]	nasal stop	–	+	
[t]	oral stop	–	+	
[s]	fricative	+	+	
[tʃ]	affricate	–	–	+
[r]	tap	+	+	

Examples

[n]	[manu]	‘debtor’
[t]	[tura]	‘brother’
[s]	[suju]	‘fetus’
[tʃ]	[tʃuri]	‘son’
[r]	[tura]	‘brother’

We've taken a shortcut with the feature array, by omitting two columns, [±sonorant] and [±voice], to save space in the width of the table. Our move relies on two properties of sounds: (i) nasals and taps are sonorants ([+sonorant]) while fricatives and affricates are obstruents ([–sonorant]) so we know which sounds are sonorants by other properties of those sounds, and (ii) sonorants are typically voiced and all the obstruents we have seen so far in South Bolivian Quechua are voiceless.



Exploring my language

The job now is to find the [coronal] consonants in your language and create a table showing the consonants, their descriptive terms, and their features. Use the model from the preceding Comprehension Check in creating the table for your language.

Be sure to include example words for each coronal sound.

2.3.2.3 Constriction with the tongue body and the feature [dorsal]

In figuring out the [labial] and [coronal] sounds, we made use of the fact that we tend to be somewhat aware of our lips and tongue tip. We can easily see the lip gestures when a person speaks, and we sometimes can see the front of the tongue. When we pay close attention to the front of the tongue, we are generally aware of its location. It is very sensitive; we can feel where it is in the mouth.

What if you know the constriction is in your mouth, but it isn't made with the lips or with the front of the tongue? To answer this question, we move farther back in the mouth to sounds that involve constriction caused by movement of the body of the tongue. These sounds are called [dorsal] sounds, because they are made with the tongue body, a part of the tongue that is also called the *dorsum*. Take a minute to get some awareness of your *tongue body*.

Try this. We have two suggestions for feeling your tongue body. You can swallow, paying special attention to the contact between the body of your tongue — the part behind the agile front — and the back of your mouth. The part of your tongue that contacts the back of your mouth is the tongue body.

Alternatively, put your tongue in position to say a word beginning with [k] and hold it there, in that “k position”, without saying the word. Again, notice where the contact is made, both on your tongue and on the roof of your mouth. This tongue part is the tongue body.

The images in Figure 14 may also help. See how in both images, the tongue makes a lopsided ball in the mouth. We've circled the balls in Figure 14 to help you focus. The tongue body is circled in Figure 14b, where the tip of the tongue is excluded from the circle.

The image in Figure 14a shows a *post-alveolar* sound, which is a type of coronal sound. The highest part of the tongue ball is close to the palate, towards (but behind) the alveolar ridge, so it is called “post-alveolar”. We see that the tongue blade is very close to the roof of the mouth, making this a coronal sound, not a dorsal sound.

For the dorsal sound in Figure 14b, the highest part of the tongue ball is at the very back of the tongue, putting it right up against the velum, making this a *velar* sound. The descriptive term “velar” comes from this location. The distinctive feature is named [dorsal] because the tongue *dorsum* (another word for tongue body) makes the constriction. (You may recall that distinctive features typically refer to active articulators while phonetic descriptions typically refer to passive articulators.)



a. post-alveolar constriction



b. velar constriction

Figure 14: Tongue position for post-alveolar and velar constriction

There are dorsal sounds that are not velar. A **palatal** sound such as voiceless [c] has constriction made by the tongue body a bit in front of the velum. A **uvular** sound such as voiceless [q] has its constriction lower, at the uvula, not the velum.



Comprehension Check 9

As a quick review, which sound in Figure 14 is a stop and which is a continuant? Review your manner of articulation knowledge as well: Which of the two is oral and which is nasal? Give yourself an extra challenge and write down your reasons for your answers.



Here's our answer

The images are repeated here for convenience, along with our answers. Our explanations follow.



a. a post-alveolar sound



b. a velar sound

Stop/continuant. The post-alveolar sound is a continuant and the velar sound is a stop.

Looking at the post-alveolar sound, we see black between the tongue and the roof of the mouth all the way from the back of the mouth to the front of the tongue, allowing air to flow without complete obstruction. This means the post-alveolar sound is a continuant.

On the other hand, with the velar sound, we can see the bulge at the back of the tongue is pressed firmly against the back part of the palate and the velum, a long closure that completely stops airflow through the mouth.

Oral/nasal. The post-alveolar sound is oral and the velar sound is nasal.

Looking at the post-alveolar sound, we see the velum raised to close off airflow through the nose. There is no clear passage for air to get out that way. The post-alveolar is oral.

For the velar sound, we see that the velum is lowered, giving a wide opening for air to flow out through the nose, making this a nasal sound.

The sounds shown by these MRIs are a post-alveolar fricative [ʃ] and a velar nasal stop [ŋ].

Dorsal consonants. Dorsal stops, whether oral or nasal, are made with complete oral closure with the tongue dorsum (or back). It is most common to find velar consonants — consonants that are [+high, dorsal].

You likely have a velar stop in your language, voiceless [k] or voiced [g] or both. Say [k] or [g] a few times, feeling the back of your tongue make contact. Some languages also have a velar nasal, whose symbol is [ŋ].

Try this. Start with a velar stop, either [k] or [g]. Now try making the corresponding palatal stop — move the high part of your tongue forward a little as you make the closure. The IPA symbols for palatal stops are voiceless [c], voiced [ɟ], and nasal [ɲ].

The sound [j] is a palatal sonorant. Palatal stops often sound a little like they have a [j] release; some sources write palatal stops as voiceless *ty*, voiced *dy*, and nasal *ny*; the sound [j] is sometimes written as *y*.

The fricatives are made just as expected, with narrowing (not closure). Constriction narrows the space between the tongue (the active articulator) and the palate or velum (the passive articulator), but it does not close it completely. You can practice this by making the velar stop, [k] or [g], between vowels several times, then gradually making less contact until you produce a fricative. The symbols for velar fricatives are voiceless [x] and voiced [ɣ]. Because dorsal fricatives are made in the back half of the mouth, the articulators making the sounds are all very soft, so there is no contrast between strident and non-strident dorsal fricatives.

The most common dorsal approximant is [w] which is also a labial sound. It is called a *labio-velar* approximant, because it has constriction in two parts of the mouth.

Dorsals: Sounds made with the body of the tongue. Dorsal sounds involve the tongue body, raised for palatals and velars and pulled back and low for uvulars, summarized visually in Figure 15. The most common [dorsal] place of articulation is [dorsal, +back], which are the velar sounds. The color becomes more intense as we move from the front of the mouth to the back of the mouth.

Both palatals and velars are classed as [dorsal] and [+high]; velar sounds are also [+back]. Uvulars are [dorsal] and [+back] and [−high] because, while the dorsum is pulled back, it is also lower in the mouth.

Place features for dorsal consonants are shown in Table 6. The example symbols show voiceless stops, voiced stops, and nasal stops for each place of articulation. (See p. 68 about the use of [dorsal] instead of [±dorsal].)

Table 6: Place of articulation features distinguishing [dorsal] consonants

	description	[dorsal]	[±high]	[±back]
[c, ɟ, ɲ]	palatal	[dorsal]	+	−
[k, g, ŋ]	velar	[dorsal]	+	+
[q, ɢ, ɴ]	uvular	[dorsal]	−	+

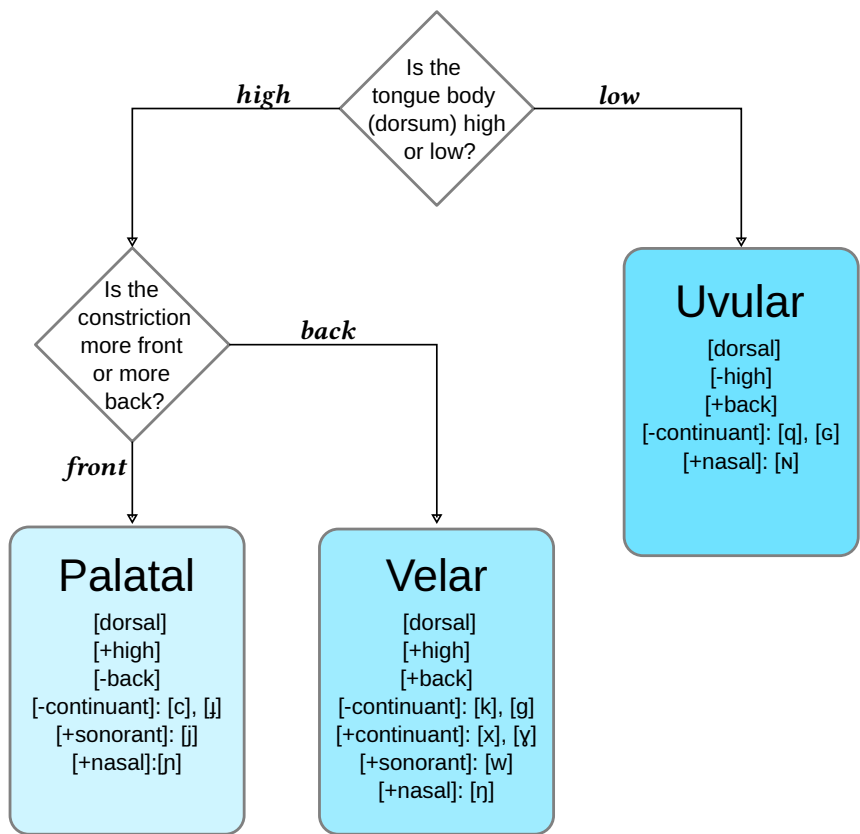


Figure 15: Identifying [dorsal] consonants



Comprehension Check 10

Here is a review of both manner of articulation features and place of articulation features. There are three images. For each, identify the manner and place of articulation. For bonus points, see if you can identify each sound being made with its proper symbol.



Here's our answer

The manner of articulation is [+sonorant, +nasal, –continuant] for all three sounds. In all three images, the velum is lowered, allowing free passage of air through the nose. This makes each sound [nasal]. Nasal airflow is resonant, so each sound is [+sonorant]. Finally, as we see in our discussion of manner of articulation, each sound involves complete closure somewhere in the mouth, so each sound is [–continuant]. These images show three nasal stops. Place of articulation is explained below the images.



[m]

[n]

[ŋ]

The images show three different places of articulation. In the leftmost image, the tongue is in a neutral position, while the lips are pressed together. Sounds made with constriction at the lips are [labial] sounds. The symbol for a labial nasal stop is [m].

In the center image, the front of the tongue rests at the teeth-end of the alveolar ridge. Sounds made with the front of the tongue have the feature [coronal]. The symbol for a coronal nasal stop is [n].

In the rightmost image, the back of the tongue is raised against the velum. Such sounds have the feature [dorsal] since they are made with the tongue dorsum. The symbol for a dorsal nasal stop is [ŋ].

And now, we turn to the larynx as a place of articulation.

2.3.2.4 Laryngeal constriction – or not

What if you know something is a consonant, but there is no constriction in the mouth – not with the lips and not with the tongue? These are laryngeal consonants.

In addition to all the places of articulation in the vocal tract, and all the ways that the active articulators (lips and tongue) make constrictions by approaching or contacting a passive articulator (teeth, palate, velum, uvula, and pharyngeal wall), we also have sounds made by manipulating the larynx, while the rest of the vocal tract is without constriction. We do this in two ways – by closing the vocal folds tightly to make a stop and by opening the vocal folds completely to make a fricative.

A laryngeal stop is commonly called a *glottal stop*; its symbol is [ʔ]. This sound is found in many languages, and you will find it represented in a variety of ways as you read different sources. Two common alternative representations are ['] and [ʔ]. We use the IPA symbol [ʔ] but you may see one of the others as you explore other resources.

Try this. If you aren't familiar with what a glottal stop sounds like, try saying two vowels in a row, [a...a]. Do you feel a little catch between them, way in the back of your throat? Try again, saying the two vowels a little more quickly if you didn't feel it the first time. That little catch is your glottis closing between the two vowels, making a glottal stop. You aren't saying [a:] where the vowel configuration is held for a longer time. If you feel that catch, you are saying [aʔa]. Because this sound requires the glottis to constrict, it is given the feature [+constricted glottis].

The glottal fricative [h] is made by opening the entire vocal tract, from glottis to lips, with no constriction anywhere. Try it. Try putting that sound between vowels: [aha]. In order to produce [h], the glottis must be wide open. This is called [+spread glottis]; sounds with a more closed glottis are [–spread glottis]. There is a fair amount of discussion in the literature about whether the laryngeal consonants [ʔ] and [h] are sonorants or obstruents. Much of the time it doesn't matter which they are so we don't commit to either feature for these two sounds.

To summarize, when you're sure you have a consonant but there is no obvious oral place of articulation, you may have a consonant with a laryngeal place of articulation, shown visually in Figure 16. The features [constricted glottis] and [spread glottis] are used to identify the primary laryngeal consonants, [ʔ] and [h].

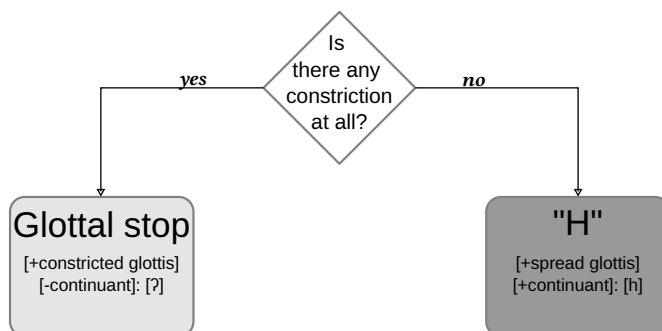


Figure 16: Identifying laryngeal consonants



Comprehension Check 11

Gaby G. Vargas Melgarejo, South Bolivian Quechua

Now it is time to revisit the South Bolivian Quechua word list, add to it, and then make a table showing both the [dorsal] and the [laryngeal] consonants. The two sections are *Tongue body: Dorsal* and *Laryngeal: Glottis*. Here are two new words to add: [tʃʰaqe] ‘thick crushed grain soup’ and [qowi] ‘guinea pig’. These words add two consonants to our list, [tʃʰ] and

[q]. Here are the words, with the two new words added.

[manu]	‘debtor’	[haku]	‘let’s go’
[tura]	‘brother’	[suju]	‘fetus’
[pupu]	‘belly button’	[tʃuri]	‘son’
[qowi]	‘guinea pig’	[tʃʰaqe]	‘thick crushed grain soup’

Use the models from the previous Comprehension Checks, modifying the table for the dorsal and laryngeal consonants, and including [tʃʰ] and/or [q] if appropriate.



Here’s our answer

Here’s how we made our chart, using the consonants identified in earlier Comprehension Checks.

South Bolivian Quechua consonants with [dorsal] or laryngeal place of articulation

dorsals	description	[±continuant]	[±back]	[±high]
[k]	velar stop	–	+	+
[q]	uvular stop	–	+	–

Larynx

laryngeals	description	[±continuant]	[±spread glottis]
[h]	‘h’	+	+

Examples

[k]	[haku]	‘let’s go’
[q]	[qowi]	‘guinea pig’
	[tʃʰaqe]	‘thick crushed grain soup’
[h]	[haku]	‘let’s go’



Exploring my language

Are there any dorsal sounds in your language? Does your language have [h] or [ʔ] (or both)? Now is the time to add these consonants to your charts showing places of articulation of consonants in your language.

Continue to use the model from previous Language Explorations in this chapter, including example words.

Review your collection of charts. You have a good start on a list of the sounds of your language — something you can keep adding to as you find new sounds, and something that can serve as the basis for building a dictionary or a writing system ([orthography](#)) if needed.

2.3.2.5 Secondary articulation

We take a brief moment to discuss a phenomenon called [secondary articulation](#): Sometimes producing a sound involves movement of two articulators — tongue and lips, tongue and glottis, lips and glottis. There are two general types of secondary articulation, those where the primary articulation is in the mouth and the secondary articulation is in the glottis, and those where the primary articulation is an obstruent or a stop and the secondary articulation is a sonorant.

Laryngeal secondary articulation. In our discussion of laryngeal sounds in Section 2.3.2.4, we talked about a laryngeal continuant [h] and a laryngeal stop [ʔ] (the glottal stop). Each of these gestures (spread glottis and constricted glottis respectively) can be added to a sound with a primary oral articulation.

Sounds with secondary articulation involving a spread glottis are called [aspirated](#) sounds; there is a tiny puff of air that accompanies the basic sound. Aspiration is annotated by a raised [h]. If the [h] comes first, as in [h p], the sound is pre-aspirated, and if the [h] comes second as in [p^h], the sound is post-aspirated.

Try this. To tell whether a sound is aspirated, you can hold your hand (or even a piece of paper) in front of your mouth when you say the sound. If you feel a strong puff of air — strong enough to move the paper — the sound is likely aspirated. (We do not recommend trying this with a lit match or lit birthday candle. While aspiration will cause the flame to waver, you might burn your lips.)

Glottal constriction can be achieved in a variety of different ways, leading to a variety of different types of secondary articulation involving constricted glottis — *ejectives*, *implosives*, and *glottalized sonorants*, all of which are found in some of the Indigenous languages of the Americas (Maddieson 2013). These are less common than a lot of other sounds, however, so we don't discuss them in this volume.

Sonorant secondary articulation. We have identified three active articulators in the mouth — the lips, the tongue, and the velum. Each of these can contribute a secondary articulation to a sound. They are called labialization, palatalization, and nasalization, respectively.

Labialization adds rounding to the basic sound, almost like you are making a “w” when you make the sound. The symbol for this is a raised “w”: Labialized labials are [p^w], [b^w], and [m^w]. The feature for labialization is [+round].

Palatalization adds constriction with the tongue blade toward the palate, designated by the feature [+high]. Palatalization adds a [j]-like quality to the basic sound, like you are saying a “y” at the same time that you are saying the basic sound. The palatalization symbol is a raised “j”, for example [k^j], [g^j], and [ŋ^j].

Finally, *nasalization* results when the velum is lowered for part of a sound. Nasalization can occur either before the basic sound (pre-nasalization) or after the basic sound (post-nasalization). The symbol is a raised symbol for a nasal, typically of the same place of articulation as the basic sound: [n^t] is a pre-nasalized [t], [n^d] is a pre-nasalized [d]. If these sounds are post-nasalized, the raised nasal follows the basic sound: The representation [dⁿ] indicates a post-nasalized [d].



Exploring my language

Review the consonants in your language. Do you think any have secondary articulation? If so, add a *Secondary articulation* category to your master list. Try using the correct symbols and be sure to include the example words.

The various types of secondary articulation are summarized visually in Figure 17. The colors in this image correspond to colors used in other images. The laryngeal features have the same gray used for [h] and for [ʔ] in Figure 16, the red

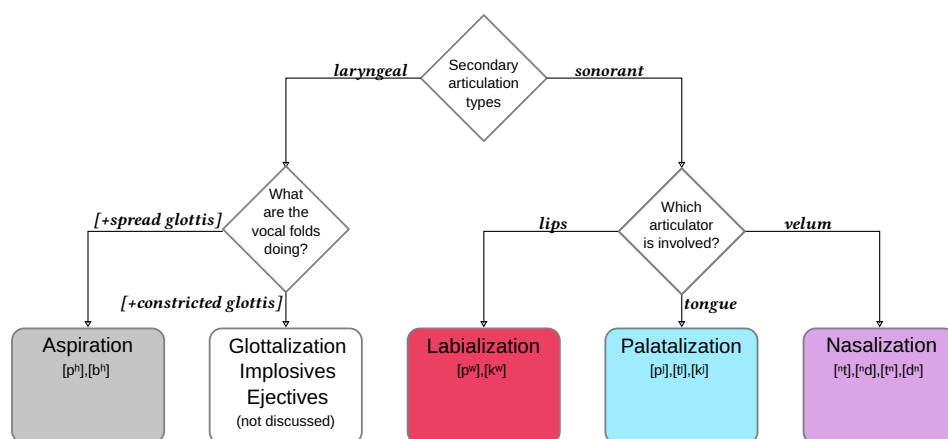


Figure 17: Identifying secondary articulation

for labialization corresponds to that used for [labial] sounds in Figure 11. Palatalization is shown with the blue used for dorsal sounds in Figure 15. (The color chosen for nasalization, purple, is random since this is the only image referring to nasal, and purple isn't used elsewhere for any sound class.)

2.3.3 Conclusion

This concludes our discussion of consonants. In the big picture, you want to remember that producing speech is a bit like blowing a whistle — the air goes out through a tube and that makes a sound. As the shape of the tube changes, the sound changes. In our discussion of consonants, we looked at a number of ways that we use our vocal tracts — from larynx to lips and nose — to change the nature of the tube and so change the sound that comes out.

These changes are of two types, those involving the manner of articulation and those involving the place of articulation. The **manner of articulation** determines the way that air passes through the vocal tract along two different dimensions. First, what is the basic type of airflow? It can be resonant or turbulent. Sounds with resonant airflow are called **sonorants** and are classified as [+sonorant]. Turbulent airflow, due to some obstruction, gives rise to sounds called **obstruents** and are classified as [−sonorant].

Airflow can be nasal or oral, depending on whether it passes in part through the nose or only through the mouth. Nasal sounds are classed as [+nasal]; oral sounds are classed as [−nasal].

Airflow can also be modified in the larynx, with vibration of the vocal folds producing the [+voice] voiced sounds and the lack of vibration resulting in the [−voice] voiceless sounds.

Second, does the airflow pass through a constriction or not? If there is constriction, is it complete or partial? Complete constriction gives us stops; narrowing gives us fricatives.

The **place of articulation** tells us where in the mouth constriction is found. Constriction can be made with the lips, the [labial] sounds. It can be made with the front part of the tongue, either the tip or the blade, the [coronal] sounds. Constriction can also be made with the body of the tongue, the [dorsal] sounds. Finally, constriction can even be made in the larynx, sounds with [constricted glottis] or, with no laryngeal constriction at all, with [spread glottis]. Although we list only four “moving parts”, there are far more than four places of articulation because each of these articulators is quite agile, especially the front part of the tongue.

In Figure 18, we compile all the primary articulation images from this section into one figure, to review the information presented. Questions are in diamonds – the initial question is indicated by a heavy border around the diamond. Connecting arrows are labeled with the articulations that answer those questions, and the oval boxes give the descriptive term for the class of sounds, the relevant features and IPA symbols.

In Figure 18, the labial section contains the red boxes, and comes from Figure 11; the coronal section (with the yellow boxes) comes from Table 13. The dorsal section (with the blue boxes) is found in Figure 15, and the laryngeal part (with the gray boxes) was seen in Figure 16. Remember that features, like [coronal], are in square brackets. Descriptive terms, like “alveolar” are in bolded text.

You are prepared now with the knowledge necessary to figure out how to pronounce the consonants of your language. Also, you likely have a better understanding of why people can really struggle to learn how to pronounce sounds and words in another language – there are so many different moving parts to learn about, and a small difference in location can produce a big difference in sound.

Even though we’ve worked hard to understand consonants, we aren’t done yet. Words are not made exclusively of consonants – there are also vowels in words. In the next section, we address vowels.

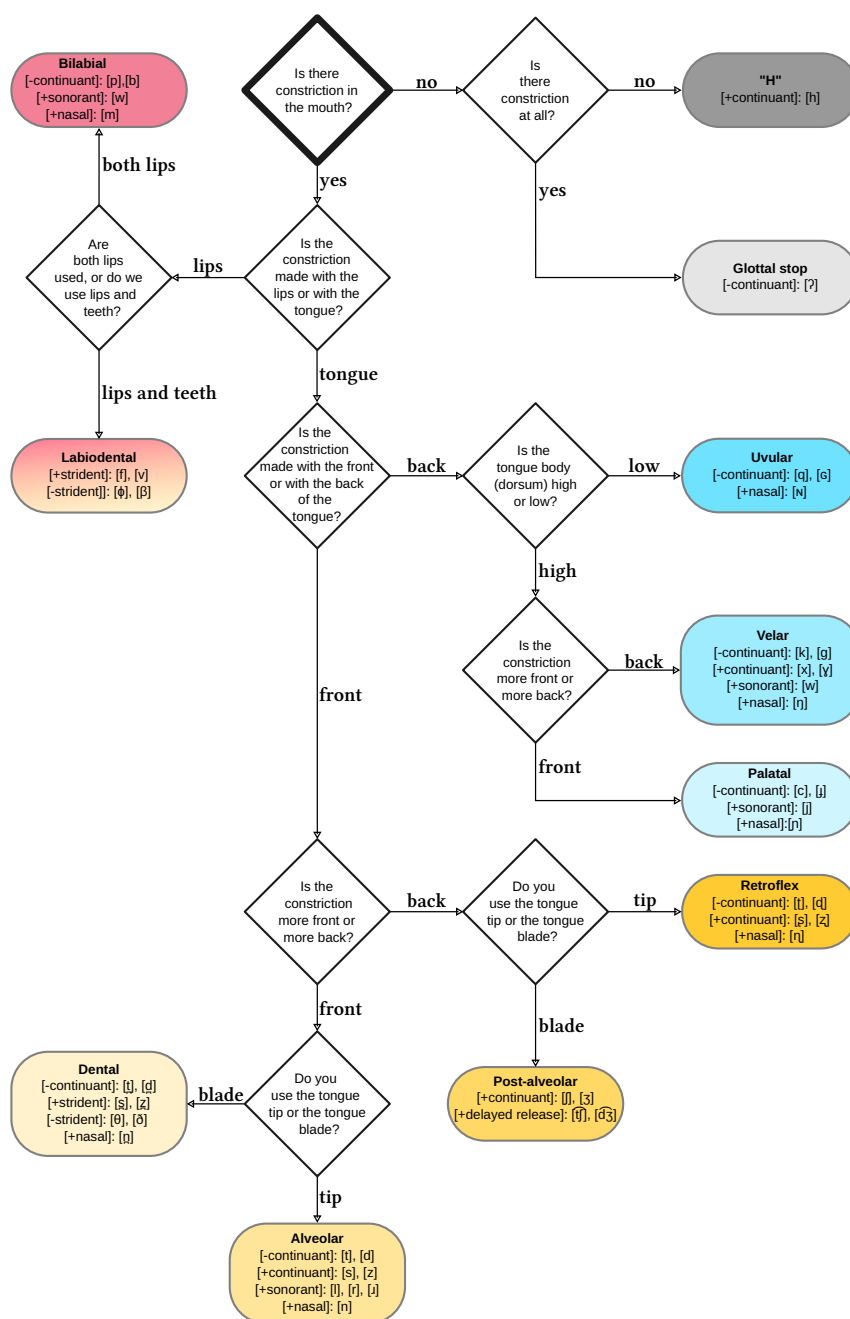


Figure 18: Identifying the place of articulation

2.4 Making vowels

The first part of our discussion of vowels, Section 2.4.1, covers properties of the most commonly found vowels. The second and third parts briefly address some vowel types that are not found as commonly. We recommend that you read the introduction (Section 2.4.1), and then determine whether you have enough to work with the vowels of your language. If you need more information, at that time you are likely to be able to decide whether Section 2.4.2 (which tells about additional “plain” vowels) is what you need or whether you need information about “enhanced” vowels, covered in Section 2.4.3.

We already know quite a bit about vowels. We know they are [+sonorant] because the vocal tract is open enough that there is no turbulent air when making a vowel. As sonorant sounds, vowels are also typically [+voice], not [–voice]. Because vowel sounds are made with a fairly open mouth, the tongue and lips do not get very close to the passive articulators. The absence of a passive articulator anchor makes it a bit more difficult to figure out how to make vowels. All the exercises we did before to become aware of our tongues in our mouths are very important now, as we try to be aware of the position of our tongues without the tongue actually touching anything. Descriptions and features match more directly with vowels — they are typically described by the positions of the active articulators, the tongue and the lips (and sometimes the velum and larynx). (This follows logically, since the active articulators don’t connect with or approach passive articulators.)

2.4.1 The most common vowels

People who look at languages from all over the world say that the most common vowels are [i], [a], and [u]. Before talking about how these sounds are produced, described, and classified, let’s look briefly at how they sound. We use English examples to illustrate each. Black text is used to emphasize the sounds we are talking about in (1).

(1) The three most common vowels

[i]	[bit]	‘beet’
[a]	[bat]	‘bought’, ‘bot’
[u]	[but]	‘boot’



English examples in this book

We apologize for using English examples in this introduction. Our reason for doing so in this case is because the lack of tongue contact with vowels makes it very hard to know what sound a vowel symbol refers to without actually hearing a word. We assume all our readers are familiar with English pronunciation and so rely on that as a way to begin our discussion of vowel sounds. We will return to examples from Indigenous language as soon as the basics of vowel production have been described. There are also web resources where you can click on IPA symbols and hear the sound that each symbol represents, for example the site <https://enunciate.arts.ubc.ca/linguistics/world-sounds/>. For those interested in English, in most American English dialects, “bought” and “bot” sound the same. This is not true in all of the world’s Englishes — you might also hear [bɒt] ‘bought’.

In this section, we explain how these vowels are produced, described, and classified by features. We also bring up a few other fairly common vowels, [e], [o], and [æ], (as in *bait*, *boat*, and *bat* respectively). Many readers may find that this section provides what they want to know about vowels and can skip both Section 2.4.2 and 2.4.3. Others may need more information and decide to read one or both of those sections. In Section 2.4.2, we show how the features introduced in this section combine to describe vowels beyond the ones introduced in Section 2.4.1. In Section 2.4.3, we address other vowel features that are found in some languages. In case it is useful, there is a definition of each feature in Appendix B.

2.4.1.1 The lips can round

Lip position is a characteristic of all vowels. The lips have two basic shapes for vowels — they can be either rounded or not (also called **spread lips**). The vowel [u] is rounded and is classified by the feature [+round]. (Refer to example (1) for how to say the sound [u].)

Try this. Say the vowel [u] out loud while noticing the shape of your lips.

Now we turn to some non-round vowels, the vowels [i] and [a]. Since the vowels [i] and [a] are not rounded, they are classified by the feature [–round]. (Refer to example (1) for how [i] and [a] sound.)⁴

Try this. Say the vowels [i] and [a] aloud and notice the shape of your lips while you make the vowels.

Did you notice that the corners of your mouth are pulled back and your lips are close together when you make the sound [i], and that the corners come in a little as your mouth opens wider to make the sound [a]? Importantly, there is no pursing of the lips when making these sounds; the lips are not rounded.

Try this. Again, we are saying vowels out loud. This time, switch between round and non-round vowels – [u]-[i]-[u]-[i]. This may make it easier for you to feel how your lips change position as you make these vowels.

The feature [±round] is used to state whether the lips are rounded when making a vowel: [–round] means the lips are spread and [+round] means the lips are rounded. There are two classes of vowels, the [–round] vowels and the [+round] vowels, shown in Table 7.

Table 7: [±round] vowel classes

	[i]	[a]	[u]
[±round]	–	–	+

You may have noticed that the array in Table 7 is “flipped” in comparison to the consonant arrays we presented earlier: Here, sound symbols head the columns while feature names label the rows. There are two reasons for doing this. First, it saves space on the page – the information in Table 7 requires three rows if each row is headed by a vowel, but only one row if rows are headed by features. Second, both ways of orienting feature arrays are used in linguistic writings, so it is important to become familiar with both if you read more broadly about your language.

2.4.1.2 The tongue can raise and lower

The tongue moves along two dimensions – it can move up and down, and it can move forward and backward. Tongue body height is a characteristic of all vowels.

⁴Some sources classify vowels like [u] and [o] as [round], and simply have no [–round] feature for vowels like [i] and [a].

The up-and-down movement gives multiple dimensions; the two extremes are high and low.

Try this. Go to a mirror and say the vowels [i] (as in *beet*) and [a] (as in *bought* and *bot*) out loud. Can you see that your lips are almost touching with [i] and are wide open for [a]?

The openness of the mouth is a clue to the position of your tongue. To get space in your mouth to draw the tongue down for a low vowel like [a], we need to open our mouths more. Conversely, for a high vowel like [i], our mouths are almost shut, because the tongue is very high in the mouth when we make the vowel [i].

Try this. Say the two vowels again, focusing on the movement of your tongue without using the mirror. Try to feel how the tongue gets lower in your mouth as you progress from [i] to [a].

Try this. Now say the vowels [i] and [a] alternating each with [u] (as in *boot*): [i]-[u]-[i]-[u] and [a]-[u]-[a]-[u]. Notice the height of your tongue. Is the height of [u] more like that of [i] or of [a]?

We hope you found the height of [u] to be more like that of [i] than that of [a]. If you didn't sense this, try again, thinking about how your mouth opens for a lower vowel and closes for a higher vowel.

We show images of the two vowels [i] and [a] in Figure 19. The arrow in the two images is in exactly the same place in each image. The tip of the arrow touches the high point of the tongue for the vowel [i]. The space between the arrow tip and the tongue surface is quite large for [a], illustrating why we class [a] as a low vowel.

The same is true for the round vowel [u] when compared with [a]. Try the same two exercises, this time using the vowels [u] and [a] (and trying to ignore the fact that your lips are rounding for [u] but not for [a]). We hope you can both see and feel how your tongue moves higher and lower in your mouth, depending on which vowel you are making.

Vowel height is described with different terms. Perhaps the most common are high and low: [i] and [u] are high vowels, [a] is a low vowel. You will also see descriptions of [i] and [u] as *close* vowels (because the mouth is more closed) and [a] as an *open* vowel (because the mouth is more open). Shortly, we will introduce vowels with an in-between height, often called mid vowels.

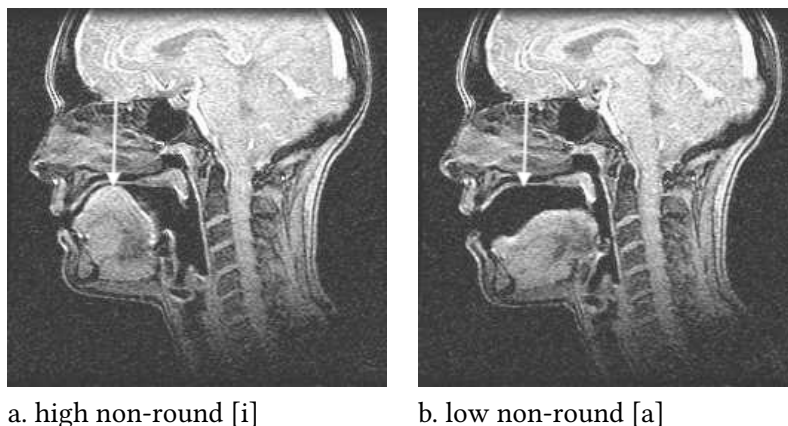


Figure 19: Non-round vowels with different tongue height: [i] and [a]

The features used to classify vowel height reflect the features used for consonants. High vowels are classified as [+high], using the feature [±high]. Low vowels are classified as [+low], using the feature [±low].

Here is a chart showing the height classifications for our three vowels, in Table 8.



Is the low back vowel represented with [a] or [ɑ]?

To be precise, [ɑ] is the correct IPA symbol for the back low vowel, not [a]. The symbol [a] is most appropriate for a front vowel that is even lower than [æ]. However, most languages do not use both [a] and [ɑ] as different vowels, and researchers frequently use the symbol [a] for the low vowel. This is because [a] is found easily on a keyboard while [ɑ] is found only on specialized keyboards (or, more recently, with special typesetting software). If your sources don't describe the low vowels adequately, you won't know whether [a] or [ɑ] is appropriate unless you do your own analysis. We follow the same convention, using [a] generally, and [ɑ] only when our source used [ɑ], for example in the box about Niisti'powahsin on p. 117.

Table 8: [±high] and [±low] vowel classes

	[i]	[a]	[u]
[±high]	+	–	+
[±low]	–	+	–

2.4.1.3 The tongue can go forward and backward

Forward and backward movement of the tongue can take place with the body of the tongue.⁵ Tongue body movement is a characteristic of all vowels. The primary feature is [±back], but some sources use [front] for [–back] and [back] for [+back].

Try this. Say the vowels [i] and [u] out loud, one after the other: [i]-[u]-[i]-[u]. You may immediately notice that your lips go from spread to rounded position, and you know why – [i] is a [–round] vowel and [u] is a [+round] vowel. The challenge here is to focus on your tongue: Feel it move forward and backward. Can you feel which vowel is the [–back] vowel and which one is the [+back] vowel?

We hope you felt the tongue moving towards the front of the mouth in the vowel [i] and toward the back of the mouth in [u]. If you didn't, it may help to look at the MRI images for these vowels, given in Figure 20. The arrow pointing to the middle of the palate is in exactly the same place in the two images, to give a reference point for comparison.

Notice where the highest part of the tongue is in each image. For [i], the highest part of the tongue is in front of the arrow – way up at the front part of the mouth, close to the alveolar ridge. The vowel [i] is a [–back] vowel. For [u], the highest part of the tongue is in the back of the mouth, behind the arrow. The vowel [u] is a [+back] vowel. (Here and elsewhere, we continue to use the convention of gray text to minimize the part of the word that is of less interest to our discussion.)

Try this. You can try feeling the front/back movement with low vowels. Use [a], the vowel that is already familiar to you, and [æ], found in English [bæt].

We hope you felt the tongue moving towards the front of the mouth with [æ] (as it did with [i]), and toward the back of the mouth with [a] (as it did with [u]).

⁵The root of the tongue, where it attaches to the floor of the mouth, can also go back and forth. Tongue root movement is discussed in Section 2.4.3.1.

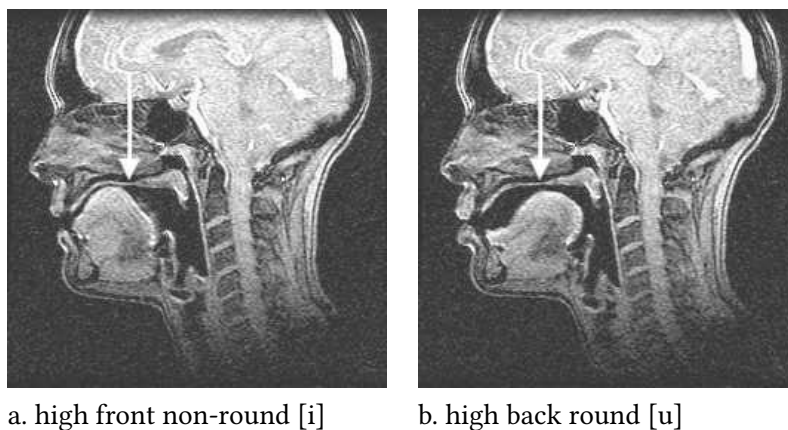


Figure 20: Two [+high] vowels, [i] and [u]

MRIs for [−back] [æ] and [+back] [a], the two low vowels, are given in Figure 21. The front-back distinction with [æ] and [a] is less dramatic because the tongue is always a little further back in the mouth with lower vowels.

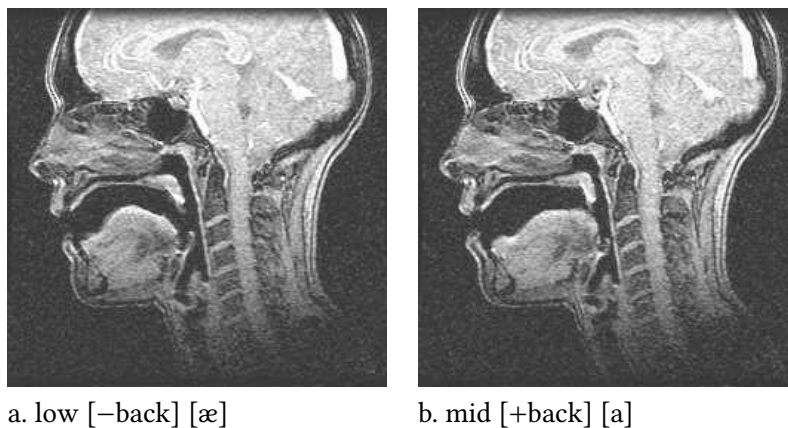


Figure 21: Two low vowels, [æ] and [a]

The features [−back] and [+back] are typically used to classify the front and back classes of vowels respectively, shown in Table 9.

2.4.1.4 Two additional common vowels

In addition to the high vowels [i] and [u] and the low vowels [æ] and [a], there are two fairly common mid vowels, [e] and [o]. These vowels sound a lot like the

Table 9: [±back] vowel classes

	[i]	[æ]	[a]	[u]
[±back]	–	–	+	+

vowels in the English words *bait* and *boat*. However, typically fluent speakers of English raise their tongues at the very end of these mid vowels, saying [beit] and [bout], not [bet] and [bot]. (Fluent speakers of Spanish, among many other languages, do not have this raising at the end of the mid vowels.)

Try this, if you are an English speaker. Say the word *lay* very slowly. Can you feel your tongue raise at the end? You can check the movement of your jaw in the mirror if the movement is too subtle to feel it at first.

Now say the word *low* very slowly. Again, do you feel your mouth closing slightly at the end of the vowel? This is the tongue raising that is typically of English, but not of most languages with the mid vowels [e] and [o].

We started this discussion by stating that [e] and [o] are mid vowels. We also need to determine the position of the lips. Can you do that?

Try this. Say [e]-[o]-[e]-[o]-[e]-[o] while watching yourself in the mirror. Is either vowel [–round] (because it involves spread lips)? Is either vowel [+round] (because your lips are rounded when you say the vowel)?

If you didn't figure out that [e] is [–round] and [o] is [+round], try the exercise again now that you know what to look for. The change in lip position should be noticeable visually and easy to feel as you change between the two vowels.

Now let's turn to whether the tongue body is forward or back in the mouth.

Try this. Say [e]-[o]-[e]-[o]-[e]-[o], focusing on your tongue and not your lips. Is either vowel [–back] (because the tongue is further forward in the mouth)? Is either vowel [+back] (because your tongue is further back when you say the vowel)?

We hope you figured out that [e] is [–back] and [o] is [+back]. You may not have done so: Tongue body movement is more subtle. It may help to try the exercise again now that you know what to look for. It may also help to start the sequence with [o] rather than with [e], in case moving the tongue forward is more noticeable for you.

At this point we have established that [e] is [–round, –back] and [o] is [+round, +back]. We have also asserted that both vowels are not high nor low, but mid – classified as [–high, –low]. Can you think of a way to test this assertion?

Recall that tongue height has to do with how high the body of the tongue is in your mouth. So you can test this by seeing where [e] and [o] fall with respect to the high vowels [i] and [u] and the low vowels [æ] and [a].

Try this. Go to a mirror and watch yourself say the vowels [i]-[æ]. Can you see that your lips are almost touching with [i] and are wide open for [æ]? Now add [e] to the vowels – [i]-[e]-[æ]. Where is [e] in the mouth-open continuum?

Try the same thing with [u] and [a], then with [u]-[o]-[a].

If you are really enthusiastic, try the same thing without the mirror, to see whether you can feel your tongue body position for the three vowels in each sequence.

We hope that you decided that the tongue body position for [e] and [o] falls between that for the high vowels [i] and [u] and that for the low vowels [æ] and [a]. As such, the height of [e] and [o] is between high and low – a commonly used term is mid.

The three tongue heights are illustrated in Figure 22, which adds the image for the mid vowel [e] to the images for [i] and [a] that we showed in Figure 19. Again, the arrow is in exactly the same place in each image. You can see how the distance between the tip of the arrow and the tongue surface increases as the vowels change from high [i] to mid [e] to low [a].

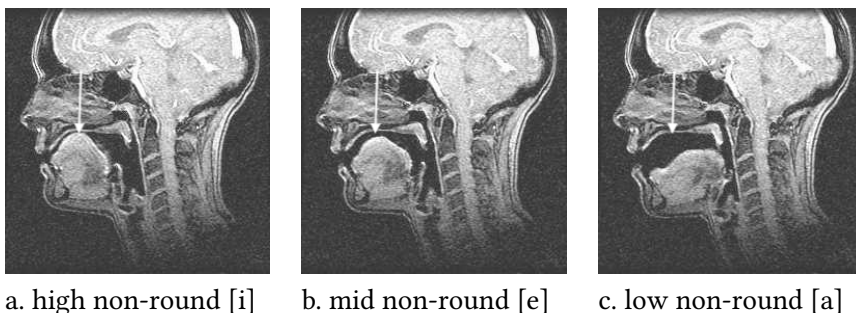


Figure 22: Non-round vowels of different heights: [i], [e], and [a]

2.4.1.5 Summary

We have explored six common vowels in this section. The first three, [i], [u], and [a], are the vowels most commonly found in languages. They may be the most common because each is maximally different from the others — the highest and lowest vowels, the most front and the most back vowels. Not all languages have all three of these vowels, but it is not surprising to find one, two, or all three in a language.

To these three vowels, we added another low vowel, the front [æ], and two mid vowels, [e] and [o]. Again, not all languages have these vowels, but it is not surprising to find any of them in a language.

To characterize the vowels, we introduced four features, [±round], [±high], [±low], and [±back]. We showed how each of the six vowels can be classified by a particular certain combination of these features. (At the end of this section is a Comprehension Check for you to create a feature chart for the six vowels discussed in this section.)

For describing the vowels of many languages, the information in this section is enough. The term vowel *quality* refers to the sound of a vowel that can be classified by these features — such as [i], [a], [u], etc.⁶ That means we can say that many languages use these five vowel qualities, or use some of them. However, in some languages, we find other vowel qualities, ones that are classified by other combinations of these same features.

If you found all the vowels of your language described in this section, you might decide to jump ahead all the way to the discussion at the end of this chapter, in Section 2.5. On the other hand, your language may have some other vowel qualities that we haven't touched on yet — for example, front round vowels or back non-round vowels. In that case, you will want to read the next section, 2.4.2. Or your language may have some additional vowel properties, such as tongue root position, nasality, laryngeal properties, or length. These are discussed in Section 2.4.3, so you may want to go there next. But first, regardless of what part you read next, please try the Comprehension Check to reaffirm your understanding of this section and do the Language Exploration, to consolidate what you know about the vowels of your language.

⁶In some languages, tongue root position also contributes to vowel quality. See Section 2.4.3.1.



Comprehension Check 12

Try making a table for the six vowels, [i], [e], [æ], [a], [o], and [u]. We give the features for [i] to start you off. It is common with vowels to present height features first, followed by the back and round features, so that is how we present it here.

	[i]	[e]	[æ]	[a]	[o]	[u]
[±high]	+					
[±low]	–					
[±back]	–					
[±round]	–					



Here's our answer

Here's the table we came up with. We placed [æ] between [e] and [a], so that the front vowels are on the left of the table and the back vowels are on the right. You will find different sources order vowels differently in comparable tables.

	[i]	[e]	[æ]	[a]	[o]	[u]
[±high]	+	–	–	–	–	+
[±low]	–	–	+	+	–	–
[±back]	–	–	–	+	+	+
[±round]	–	–	–	–	+	+



Comprehension Check 13

Gaby G. Vargas Melgarejo, South Bolivian Quechua

Let's go back to our words from South Bolivian Quechua, repeated here for convenience.

[manu]	'debtor'	[haku]	'let's go'
[tura]	'brother'	[suju]	'fetus'
[pupu]	'belly button'	[tʃuri]	'son'
[qowi]	'guinea pig'	[tʃʰaqe]	'thick crushed grain soup'

What vowels do you find? Make a table to show the features of the vowels of South Bolivian Quechua. Be sure to include an example word to illustrate each vowel.



Here's our answer

We made an array showing the South Bolivian Quechua vowel features, illustrated with example words.

Five South Bolivian Quechua vowels

	[i]	[e]	[a]	[o]	[u]
[±high]	+	–	–	–	+
[±low]	–	–	+	–	
[±back]	–	–	+	+	+
[±round]	–	–	–	+	+

Examples

[i]	[tʃuri]	‘son’
[e]	[tʃʰaqe]	‘thick crushed grain soup’
[a]	[manu]	‘debtor’
[o]	[qowi]	‘guinea pig’
[u]	[pupu]	‘belly’

Every South Bolivian Quechua word that we’ve seen has at least one vowel, so there are lots of examples of some of these vowels. Since it is easy to find examples, we include only one word to illustrate each vowel.

There are other ways to show the features and vowels in a table. For example, sometimes vowels are listed down the side of the table instead of across the top. We show this layout here.

An alternative vowel layout

	[±high]	[±low]	[±back]	[±round]
[i]	+	–	–	–
[e]	–	–	–	–
[a]	–	+	+	–
[o]	–	–	+	+
[u]	+	–	+	+

Choose a presentation method that makes sense to you and makes it easy for you to explain to others.



Exploring my language

The focus now is to find out whether your language has the five most common vowels. Review the words that you have already collected, looking for the five vowels, [i], [e], [a], [o], and [u].

Make a table of the vowels that you find and include example words to illustrate each vowel. Did you find all five? Did you find other vowels besides these five? Make a note of the words, and which vowel in the word

is confusing. We hope that, by the end of our discussion of vowels, you will be able to resolve all of the “unknown” vowels.

A challenge that we sometimes face with vowel sounds is that their spelling uses letters that are different than the IPA symbols used for the sounds. For example, you might see something written with *ee* which sounds like the high front vowel [i] (remember that the IPA symbol [e] is a mid vowel, not the high vowel [i]). To address the complication that is introduced by spelling systems, be sure to say each word aloud a few times, trying hard to feel the location of your tongue with each vowel.

2.4.2 Additional vowel qualities

In Section 2.4.1 we introduced four vowel features, [±high], [±low], [±back], and [±round], and showed how these features combine to classify the five most common vowels.

We have seen examples of all possible combinations of [±high] and [±low] and how they classify high, mid, and low vowels. High vowels are classified by the features [+high, −low], mid vowels by [−high, −low], and low vowels by [−high, +low]. Logically, there are no [+high, +low] vowels, because these features are contradictory.

We have not seen all possible combinations of [±back] and [±round]. The five most common vowels include the [−back, −round] vowels [i] and [e], and the [+back, +round] vowels [u] and [o], and one [+back, −round] vowel, the low [a], but that doesn’t mean other combinations are impossible. After all, lip position and tongue position are independent of each other. The front round and back non-round vowels are the focus of this section. We start with front round vowels, vowels classified by [−back, +round].

It can be hard to imagine these vowels without knowing what they sound like. You can visit the website at <https://enunciate.arts.ubc.ca/linguistics/world-sounds/> to listen to recordings of people saying these vowels.

2.4.2.1 Front round vowels

In this section, we explore combining rounded lips with a forward tongue position, [−back, +round] vowels. We start our discussion with high vowels, although there are front round mid vowels and front round low vowels. We will get to them too.

It may not surprise you to hear that lip-rounding makes the difference between a [–back, –round] vowel like [i] and a [–back, +round] vowel like [y]. How do we produce such front round vowels?

Try this. First say [i]. Now say [i] again, holding it for a long time. As you hold the [i], gradually round your lips as you do so, but *do not move your tongue*. Can you hear a difference between the [i] you started with and the sound you end with? That sound is likely to be [y], if you’ve rounded your lips without moving your tongue.

Practice gradually rounding your lips while saying [i] a few times before trying just the end points – [i] and [y] (or “[i]-with-rounded-lips”).

Be careful with the symbol [y] – it is easily confused with the letter y (whose symbol is [j]). The sound [y] is a [+high, –back, +round] vowel.

There is also a mid vowel that is front and rounded, the [–high, –low, –back, +round] vowel [ø]. By now, we hope you know how to produce this vowel and are already trying to say [ø] out loud. If not, we’ve got an exercise for you.

Try this. Start with [e], the mid, front, non-round vowel. Say the vowel very slowly, gradually rounding your lips as you say it. As with the [i]-[y] exercise, *do not move your tongue* – just add lip rounding. Do you hear the difference between the starting [e] and the ending [ø]?

Practice the slow slide from [e] to [ø] a few times before trying the two end points, [e] and [ø].

Finally, it is also possible to have a front round low vowel. This vowel, the [+low, –back, +round] vowel [œ], is made by producing a front low vowel with rounded lips.

Try this. Start with [æ], the low, front, non-round vowel, the vowel of the English word *bat*. Say the [æ] very slowly, and gradually rounding your lips as you say it. As with the [i]-[y] and the [e]-[ø] exercises, *do not move your tongue* – just add lip rounding. (Some people find it a little difficult to round the lips when making a low vowel because the mouth is quite open. Try to keep it open as you round your lips.) Do you hear the difference between the starting [æ] and the ending [œ]?

Practice gradually changing non-round [æ] to round [œ] a few times before trying the two end points, [æ] and [œ].

The features for the [–back] vowels are given in Table 10. The last row, showing values for [±round], distinguishes between the front non-round vowels [i], [e], and [æ], and the front round vowels, [y], [ø], and [œ].

Table 10: Features for [–back] vowels

	[i]	[e]	[æ]	[y]	[ø]	[œ]
[±high]	+	–	–	+	–	–
[±low]	–	–	+	–	–	+
[±back]	–	–	–	–	–	–
[±round]	–	–	–	+	+	+

The feature combinations in Table 10 classify the primary vowel qualities for front vowels.

2.4.2.2 Back non-round vowels

You have already encountered vowels that are [+back, +round] and a vowel that is [+back, –round]: The high vowel [u] and the mid vowel [o] are both [+back] and [+round] while the low vowel [a] is [+back, –round].

In this section, we look at high and mid vowels that are [+back] and [–round], and we look at a low vowel that is [+back, +round]. We start with the [+high, +back, –round] vowel [ʉ].

Try this. Say [u] out loud. Now say [u] again, slowly. As you are saying [u], gradually pull the corners of your lips back, so that your lips are no longer round. (Use facial muscles to do this – you don’t need to use your fingers.) Be careful to keep your tongue in the same place and only manipulate your lip position. Can you hear a difference between the starting [u] and the ending [ʉ]?

Now try making just the end-point sounds, [u] and [ʉ].

(If you want to really challenge your tongue and lips, make all four high vowels: [i], [y], [u], and [ʉ].)

You know where we are going now – to the mid back vowels, round [o] and non-round [ɤ].

Try this. Start by saying [o] out loud. Now say [o] again, slowly. As you are saying [o], gradually pull the corners of your lips back, so that your lips are no longer round. (You know to keep your tongue in the same place, right?)

Do you hear a difference between the starting [o] and the ending [ʏ]?
Once you master the gradual change from [o] to [ʏ], try making just the end-point sounds, [o] and [ʏ].
(Take the mid-vowel challenge and say all four – [e], [ø], [o], and [ʏ].)

The last vowel to try is the low, back, round vowel [ɒ]. This is the round counterpart to the low, back, non-round vowel [a].

Try this. Start by saying [a] out loud, first with a normal production and then more slowly. When you are saying [a] slowly, start rounding your lips. Of course you are keeping your tongue in the same place. Do you hear a difference between the starting [a] and the ending [ɒ]?
Once you master the gradual change from [a] to [ɒ], try making just the end-point sounds, [a] and [ɒ].
(Take the low-vowel challenge and say all four – [æ], [ɛ], [ɒ], and [a].)

The features for the [+back] vowels are given in Table 11. The last row, showing values for the feature [±round], classifies the difference between the more commonly found [u], [o], [a], and the less common [ʉ], [ɤ], and [ɔ].

Table 11: Features for [+back] vowels

	[u]	[o]	[ɒ]	[ʉ]	[ɤ]	[a]
[±high]	+	–	–	+	–	–
[±low]	–	–	+	–	–	+
[±back]	+	+	+	+	+	+
[±round]	+	+	+	–	–	–

2.4.2.3 Central vowels

Some sources describe vowels as *central*. This term is used when the highest part of the tongue seems to be in the middle of the mouth, between front vowels and back vowels.

The classifying features for central vowels are tricky, because many sources try to avoid adding the feature “central”. In languages that have central vowels but no back, unround vowels, it is common to find the features [+back, –round] used to classify the central vowels. You might also find a feature [±front] used, with front vowels classified by [+front, –back], central vowels as [–front, –back], and back vowels as [–front, +back]. If a language has round central vowels and

round back vowels, the $[\pm\text{front}]$ system can classify all vowels uniquely while the system using only the features $[\pm\text{back}]$ and $[\pm\text{round}]$ cannot do so. Features for central vowels vary by sources, so if you are reading something that talks about central vowels, watch for the features being used.

The high, mid, and low central vowel symbols are given in Table 12 for both the non-round and the round central vowels.

Table 12: Central vowel classifications, using $[\pm\text{back}]$ and $[\pm\text{front}]$

		$[\pm\text{high}]$	$[\pm\text{low}]$	$[\pm\text{round}]$	$[\pm\text{back}]$	$[\pm\text{front}]$
high	$[\text{i}]$	+	–	–	–	–
	$[\text{u}]$	+	–	+	–	–
mid	$[\text{ə}]$	–	–	–	–	–
	$[\text{e}]$	–	–	+	–	–
low	$[\text{e}]$	–	+	–	–	–

Visit the website at <https://enunciate.arts.ubc.ca/linguistics/world-sounds/> to find recordings where you can hear someone saying the central vowels.

2.4.2.4 Summary

This section explored vowels that are classified by combinations of the features $[\pm\text{back}]$ and $[\pm\text{round}]$. We've added a lot of vowels by doing this. The most common of this extended set of vowels — our original five plus the front round vowels and the back non-round vowels — are shown with their classifying features in Table 13. Close inspection of the array in Table 13 will show you that each vowel is uniquely classified by its features. (We abbreviate feature names to fit the array on the page, Hi, Lo, Bk, and Rd instead of $[\pm\text{high}]$, $[\pm\text{low}]$, $[\pm\text{back}]$, and $[\pm\text{round}]$ respectively; central vowels are omitted.)

Table 13: Vowels and their features

	$[\text{i}]$	$[\text{y}]$	$[\text{u}]$	$[\text{u}]$	$[\text{e}]$	$[\text{ø}]$	$[\text{ɣ}]$	$[\text{o}]$	$[\text{æ}]$	$[\text{œ}]$	$[\text{a}]$	$[\text{ɒ}]$
Hi	+	+	+	+	–	–	–	–	–	–	–	–
Lo	–	–	–	–	–	–	–	–	+	+	+	+
Bk	–	–	+	+	–	–	+	+	–	–	+	+
Rd	–	+	–	+	–	+	–	+	–	+	–	+



Exploring my language

Does your language have vowels that are not among the five most common vowels ([i], [e], [a], [o], and [u])? If so, say words with the additional vowels out loud and try to identify the vowel quality for each vowel. Be very careful that you do not get fooled by the way vowel sounds are spelled.

Does your language have any [–back, +round] vowels – the ones that are front and round? Are there [+back, –round] vowels other than [a]? Or is there a low back and round vowel?

As you find examples of words showing any of these vowels, add those vowels and words to a table like the one in Table 13 to illustrate the vowels in your language.

This may resolve the remaining questions you had about vowels in your language. If not, just make a note of remaining vowels and keep going. Language is complex. If you try to answer every question in detail before moving on, you may never get anywhere.

2.4.2.5 Summary

This section explored combinations of the features [±back] and [±round] with different vowel heights. This led to a lot of different vowel qualities – we show 12 different combinations in Table 13, which does not include the central vowels shown in Table 12.

The vowels which are classified by these features are called *modal vowels*. We have identified 12 modal vowels (ignoring the central vowels). Most languages use at most five or six modal vowels.

With just a few vowels, it is easy to put them all in one table. In Tables 14–17, we show three different ways of representing the South Bolivian Quechua vowels in a table.

Table 14 focuses on the classifying features by listing the feature values (like [+high] or [–high]) in the cells of the table. Each row shows the features for a specific vowel. When you see vowels presented in this way, it is likely that the work is focusing on phonological patterns found in the language.

Table 14: South Bolivian Quechua vowels: Focus on features

	[±high]	[±low]	[±round]	[±back]
[i]	+	–	–	–
[e]	–	–	–	–
[a]	–	+	–	+
[o]	–	–	+	+
[u]	+	–	+	+

Table 15 uses descriptive terms for the labels, rather than features. The arrangement of vowels also focuses on the descriptions, by putting the high vowels in the top row and the low vowels in the bottom row. Traditionally, the front of the mouth is to the left, so front vowels are at the left and the back vowels are at the right. This type of array is more useful for describing the sounds where there is no discussion of phonological patterns.

Table 15: South Bolivian Quechua vowels: Focus on phonetic properties

	front	central	back
high	[i]		[u]
mid	[e]		[o]
low		[a]	

Occasionally, you might see a hybrid arrangement of the sort in Table 16, which arrays the vowels in the table by the height and frontness dimensions, and uses feature labels instead of descriptive labels.

Table 16: South Bolivian Quechua vowels: Hybrid focus

	[–round] [–back]	[–round] [+back]	[+round] [+back]
[+high]	[i]		[u]
[–high, –low]	[e]		[o]
[+low]		[a]	

In Tables 15 and 16, the cells are filled by the vowel symbols. The rows show vowel height and columns show the front-back dimension, with or without rounding. Often the feature or description labels are omitted completely in this kind of layout; the author assumes the reader will fill them in – the top row shows the highest vowels and the bottom row shows the lowest vowel(s), the

front of the mouth is to the left and the back of the mouth is to the right. This kind of vowel chart is illustrated in Table 17.

Table 17: Vowel chart for South Bolivian Quechua

[i]	[u]
[e]	[o]
[a]	

This ends our discussion of modal vowels. Often, when linguists talk about how many vowels a language has, they are really talking about the number of modal vowels in the language. Modal vowels can be enhanced by other properties – in a sense, secondary articulation for vowels. These properties increase the total number of vowels in the language. If you see a list of 10 vowels and a statement that the language “has five vowels”, the modal/non-modal distinction is probably the source of the discrepancy, even if the author doesn’t say that explicitly.

We turn to the properties that characterize non-modal vowels in the next section. If you have already identified all the vowels of your language, you might want to read the first paragraph of Section 2.4.3, so you know what is in this section in case you discover later that you need it. After that quick preview, you may prefer to jump ahead to Section 2.5, to see how a language worker might make use of the information about sounds that is presented in this chapter.



Exploring my language

Create a single vowel table for your language, along the lines of your preferred way of representing vowels – see Tables 14–17 for models. Include a list of example words after the table. Be sure to title the table appropriately, including the name of your language.

2.4.3 Enhancing properties and non-modal vowels

Just as consonants have secondary articulations, it is possible to modify the quality of a modal vowel. We briefly mention six different types of modifications

here, the position of the tongue root in Section 2.4.3.1, nasality in Section 2.4.3.2, laryngeal features in Section 2.4.3.3, length in Section 2.4.3.4, tone in Section 2.4.3.5, and stress in Section 2.4.3.6. These discussions are intentionally brief: A deeper discussion belongs in a book focusing on phonetic properties, but it is worth being aware of these enhancements just in case vowels in your language are enhanced in one (or more) of these ways. If you think your language makes use of one or more of these modifications, you might want to jump directly to the relevant section(s).

2.4.3.1 Tongue root

Tongue root position (advanced or retracted) may enhance some or all of the modal vowels in a language. We use a feature [$\pm$ advanced tongue root] to categorize these vowels. The feature [+advanced tongue root] is sometimes abbreviated [+atr] or even [atr] while the feature [–advanced tongue root] is abbreviated [–atr] or [rtr] for “retracted tongue root”. We use [+atr] and [–atr] when we need to shorten the terms.

We show the difference between advanced and retracted high back vowels in Figure 23. Importantly, in the pair of images, the position of the arrow is fixed with respect to the head. In the lefthand image, the tip of the arrow touches where the tongue root advances to. In the righthand image, the arrow tip doesn’t reach the edge of the tongue root, because the tongue root is retracted, while the arrow has stayed in the same place.

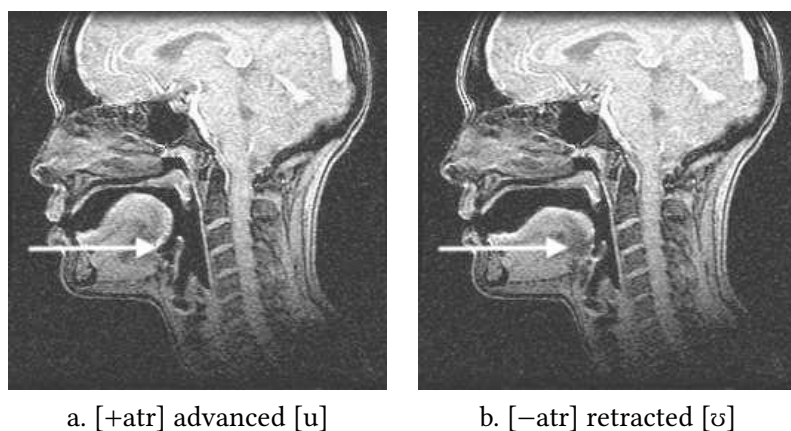


Figure 23: Tongue root position with high back vowels

Note how, in Table 23b, the tip of the arrow falls short of the tongue root. The tongue root is pulled quite far back in the mouth, towards the pharyngeal wall.

Try this. Try saying high advanced and retracted vowels aloud. The high front advanced vowel is [i] — sometimes [i] is spelled *ee* as in English *beet*. The high front retracted vowel is [ɪ], sometimes spelled *i* (as in English *bit*). Say each vowel a few times to get the feel of the tongue root moving forwards in the advanced vowels and backwards in the retracted vowels.

Now do the same with the high back advanced and retracted vowels. The high back advanced vowel is [u], spelled *oo* in the English word *boot*. The high back retracted vowel is [ʊ], which can also be spelled *oo*, as in the English word *book*.

You’ve seen the symbols for the advanced [u], [i] and retracted [ʊ], [ɪ] above. In Table 18, we show the advanced and retracted symbols for high and mid vowels, the most common advanced/retracted pairs. Central vowels and low vowels do not often have an advanced/retracted opposition, so they are left out of this table. The feature used is [±advanced tongue root], with advanced vowels being [+advanced tongue root] and retracted vowels being [−advanced tongue root].

Table 18: Advanced and retracted symbols

		[+high]			[−high, −low]		
[+advanced tongue root]	[+atr]	i	y	u	e	ø	o
[−advanced tongue root]	[−atr]	ɪ	ʏ	ʊ	ɛ	œ	ɔ

In resources, sometimes you will find both advanced and retracted vowels in a single table; other times, advanced vowels are in one table and retracted vowels are in another table. An example showing the two-table model is given in Table 19. We use the symbol [ə] for an advanced low vowel; some sources use the symbol [æ].

Table 19: One way to show advanced and retracted vowels in a single language

advanced tongue root			retracted tongue root		
high	i	u	high	ɪ	ʊ
mid	e	o	mid	ɛ	ɔ
low	ə		low	a	

2.4.3.2 Nasality

To check whether a vowel is oral or nasal is easy — just plug your nose. If you are making an oral vowel, there will be no difference. If you are making a nasal

vowel, when you plug your nose it will feel a bit congested through your nasal tract as air that was flowing smoothly through your nose is suddenly blocked. Nasal vowels have the feature [+nasal].

If your language has nasal vowels, you will want to be able to make them.

Try this. If you are unfamiliar with making nasal vowels, here’s an exercise for you. First make an [m]. From the [m] open your mouth to make a vowel but concentrate on only opening your lips, and not allowing the velum to flip up to close off the nasal passage. It can be tricky to learn to independently manipulate the different articulators, so if this is tough, try a few times each day until you are comfortable with it and can easily make nasalized vowels.

You can test your proficiency by alternating your nasal vowels with oral consonants so instead of saying [mãmõ] (for example) try saying [bãbõ] and, for a bigger challenge, [pãpõ].

There are two common diacritics used to show that a vowel is nasal. The IPA system puts a wavy line, or *tilde*, over the vowel: [a] is oral and [ã] is nasal. The Americanist Phonetic Notation (APN) method is to put a curved tail, or *ogonek*, under the vowel: [a] is oral and [a̰] is nasal. People typically use only one of these systems when writing about a language. The ogonek (also called “nasal hook”) is often used in languages that also mark stress or tone, to prevent overcrowding in the space above the vowel (since tone and stress diacritics are both put above the vowel). To show that a vowel is nasal, just add the relevant diacritic, as shown in Table 20. Do pay attention because some transcription systems use other diacritics to show nasal vowels. It is important to read carefully to know what the diacritics mean.

Table 20: Symbols for nasal vowels

IPA	ã	tilde over vowel
APN	a̰	tail (or “nasal hook”) under vowel

Nasal vowels may be shown in a table separate from the oral vowels, similar to the two tables shown in Table 19 for tongue root position.

2.4.3.3 Glottalization, creak, and aspiration

We saw in Section 2.3.2.4 of this chapter that we can add laryngeal features to consonants. We can do the same with vowels. The vocal folds in the glottis

are typically a bit relaxed with vowels, as they are with other sonorants — so vowels (and other sonorants) are typically voiced. Here are some other variants. In general, languages which use one or more of these variants also have several modal vowels.

Table 21: Vowels with laryngeal features

	symbol	name & description
a.	[ḁ]	<i>Voiceless</i> . The vocal folds are opened far enough that they don't vibrate. The effect is comparable to whispering a vowel. Voiceless vowels are often at the ends of words and/or next to voiceless consonants.
b.	[aʔ], [a'a]	<i>Glottalized</i> . Closing the vocal folds briefly during the production of a vowel adds a kind of sharp interruption to the vowel. Read your resources carefully if you suspect glottalized vowels to make sure you understand what is being talked about.
c.	[a̰]	<i>Creaky</i> . According to Picanço (2005: 16), <i>creaky voice</i> describes a mode of vocal fold vibration that involves harshness accompanied by lowered pitch (Ladefoged 1971). ⁹ You may associate this with a gravelly quality to the sound, or slowed vocal fold vibration so that you can hear each beat as they hit each other.
d.	[a̤]	<i>Breathy</i> . A vowel can be halfway between voiced and voiceless. The vocal folds still vibrate, so the vowel is voiced, but they are also pulled apart so a lot of air comes through, sounding breathy. We call this <i>breathy voice</i> .

Vowels with laryngeal features may be shown in a table that is separate from the table with modal vowels, as shown in Table 19 for tongue root position.



Voiceless vowels in Niisti'powahsin

Samantha Prins

Niisti'powahsin (Blackfoot, a Plains Algonquian language spoken in Montana and Canada) has three primary vowels, the high front /i/, the low central /a/, and the mid back /o/. Sometimes these vowels are *voiceless*, meaning they are produced like a whisper, without any vibration of the vocal folds. Speakers often describe these vowels as soundless, silent sounds, or puffs of air (Bliss & Gick 2009).

Where do we find voiceless vowels in Niisti'powahsin? Vowels in Niisti'powahsin are often voiceless when they are in a final position. A common place to find a voiceless vowel is at the end of an utterance. In connected speech, this will often correlate to the end of a phrase or sentence. We can see this below, where the word-final vowels are voiced in the middle of the sentence, but voiceless at the end as in this example from Prins (2019).

Tsáániwa áó'ohkoitapiiyiwa isóómonii'pawa.

[tsa:niwa o:xkoitapi:w iso:moni:pq]

'Johnny received a gift.'

When a word is uttered in isolation (like the answer you might give if someone asks you "What is the word for *moon* in Niisti'powahsin?"), the final vowel of that word is usually voiceless. The next example (from Bliss & Gick 2009) shows a word-final voiceless vowel.

ki'somma [ki?som:q] 'moon'

How do we know a vowel is there if we can't hear it? Voiceless vowels can be tricky to identify – in the same way it can be harder to hear a whisper than a shout. But if you listen closely, you can sometimes hear a whispered puff of air where the vowel would be.

Sometimes, voiceless vowels are so quiet that they disappear completely. These "missing" vowels are still there though—we know this because they leave a visual trace. Speakers can tell not only where a missing vowel would go, they can tell exactly which vowel it would be if it were fully pronounced. By looking at the shape of the lips during the produc-

tion of a voiceless vowel, speakers can distinguish an [q̥] from an [i̥], for example, even when there is no perceptible sound to rely upon (Bliss & Gick 2009).

Why pay attention to sounds we can't really hear? Voiceless vowels are an important part of the sound signature of Niisti'powahsin. Paying attention to which vowels are voiced and which are voiceless is one way for learners to sound more like their elders. Furthermore, in certain contexts, which voiceless vowel is used can even change the meaning of what you're saying. We can see this in the example from Frantz & Russell (2017) below.

<i>óóhкотoki</i>	<i>óóhкотoka</i>
[ox ^w котoki̥]	[ox ^w котokḁ]
'small stone, rock, pebble'	'large stone, rock'

2.4.3.4 Length

Sometimes, we hear a vowel that seems extra *long*. This can be done for effect (see Chapter 4 Section 4.3.2 for a case of this type). Sometimes it is done because in the language, a long vowel and a short vowel are two different sounds. To make a long vowel, you simply hold it a little longer than you hold a short vowel. The *length* difference between long and short vowels depends on both the language and the speaker; it is highly variable. The better you know a language that uses a length difference, the more proficient you will become at making the right length distinctions for that language.

Vowel tables often separate short vowels in one table and long vowels in another table.

The IPA symbol to mark length is to put two triangles after the sound, as in short [a] and long [aː]. As you might imagine, many people take a typographical short cut and use a regular colon (made of two dots, short [a] and long [aː]); the colon is used in the Americanist notation (APN). There is a very subtle difference between the symbols, the IPA [ː] and the APN [ː]. If you notice [aː] instead of [aː], do not be surprised.

You may also see a repeated symbol to indicate length, long [aa] vs. short [a]. This method can be confusing. The disadvantage with doubling the symbol is that the long sound is written exactly the same way we would write two short sounds in a row. If we use the “ː” symbol, there is a visual difference between [aː] and [aa]. We lose this visual difference if we use double letters: The written form

showing two short sounds in succession is [aa], which is exactly the same as the written form of a single long sound, [a:], under the double-letter convention.

Length can also be observed with consonants; long consonants are represented the same ways. We see short [t]; a long version may be represented as [t:], [t:], or [tt] (which is, again, confusing).

2.4.3.5 Tone

Some languages use *pitch* to change the meaning of an utterance. Pitch changes are made by changing the tension in the vocal folds. Higher tension correlates to higher pitch. Languages that use pitch in this way are called *tone* languages. We don't use the term pitch because pitch refers to a measurable entity and in language, the pitches used are all relative to the speaker and to the other pitches in the utterance. We use the term *tone* to step back from an absolute pitch value and remind ourselves that, as with so many things in language, we are talking about something that gets its value in relation to the sounds around it.

Making toned vowels is a little like singing, except that there is no musical score telling you exactly what note to sing — you just have indicators of whether a vowel is higher or lower than the previous one. If you are learning a language that has tone, don't try to match the pitch used by the instructor because that will be confusing — what is important in language is whether one vowel is relatively higher or lower pitched than another and not the musical notes of the vowels.

Some languages differentiate two tones, a higher tone and a lower tone. Others differentiate three tones, higher, mid, and lower. Some even have super high tones or super low tones. The symbol for a high tone is an accent mark that angles up (an *acute accent*), an untuned [a] and a high-toned [á]. A low tone is indicated by an accent mark that angles down (a *grave accent*), an untuned [a] and a low-toned [à]. A mid tone is a flat line above the vowel, an untuned [a] and a mid-toned [ā]. However, as you have already discovered, there are a lot of typographical conventions in linguistic descriptions. Quite often, if a language has only two tones, one is marked and the other is not. It is usually the high tone that is marked, so we have high-toned [á] and low-toned [a]. If there is also a mid tone, it is the mid tone that usually is not shown, so we have high-toned [á], low-toned [à], and mid-toned [a]. It is important to read descriptions, if the author has provided one, to know for sure what the tone marks mean.

Languages may have level tones, where the pitch is relatively constant throughout the duration of the vowel. Languages may also have contour tones where the tone on a single vowel changes as the vowel is said.

A vowel with a falling tone starts with a higher pitch and goes to a lower pitch. It is shown by a sequence of high and low tones together, a symbol called a *circumflex*, a high-toned [á], a low-toned [à], and a falling-toned [â].

A rising tone on a vowel starts with a lower pitch and ends with a higher pitch. A rising tone is just the opposite of a falling tone, with the low tone part first combining into a symbol called a *hacek* (pronounced [hætʃɛk]), a low-toned [à], a high-toned [á], and a rising-toned [ǎ].

These common tone symbols are given in Table 22.

Table 22: Common symbols for tones

high tone	acute accent over vowel
mid tone	flat line over vowel
low tone	grave accent over vowel
falling tone	circumflex over vowel
rising tone	hacek over vowel

Strategies for understanding tone and tone patterns are similar to the strategies introduced in this text. However, tone can be very complex, in particular because of its relative nature. If your language seems to have tone and you are having trouble figuring it out, we suggest trying to find a person who will help you. Tone is not necessarily easy.

A further complication is that speakers often use *intonation* as well. Linguistic intonation can be thought of as tone for sentences rather than words. In some languages, for example, a rise at the end of a sentence changes that sentence from a statement to a question. Intonation can also be used to express a speaker’s attitude towards something. Such sentence-level tone patterns can make it difficult to figure out tone patterns for words.



The ambiguity of “high” and “low”

Another problem can arise when reading about a tone language. You might see a statement about “high vowels” or “low vowels”. Typically, this would mean the features [±high] and [±low], but sometimes writers really mean “high-toned vowels” and “low-toned vowels”, and they don’t realize that they’ve been ambiguous. If you get confused about the use of the terms “high”, “mid”, and “low”, ask yourself whether the writer means vowel features or tone levels. That may help.

2.4.3.6 Stress

Some languages give greater prominence to one or more vowels in each word, as opposed to other vowels. Prominence can be shown by length, pitch, amplitude (loudness), or a combination of some or all of the three. Linguistic prominence is called *stress*; you may also see it called *accent*. Usually (but not always), people use the term *accent* when the location of the prominence has to be learned with each word and use the term *stress* when there is a general pattern of which vowel(s) receive stress in a word (a rule might be something like “the next to last vowel is stressed”). Stress is sometimes shown in the same way as high tone — [á] is a stressed vowel in such a system. Sometimes it is shown with a small mark in front of the stressed vowel, [ˈa]. Often, stress is described but not transcribed, so stressed and unstressed vowels look exactly the same in the IPA transcription.

A language that (primarily) uses pitch to indicate an accented vowel is called a *pitch accent* language.



Exploring my language

We have just looked at tongue root position, length, nasality, laryngeal features, tone, and stress. If you identified any of these enhancements as properties of vowels in your language, create category headings (like *Nasal vowels* or *Tone*) in your master list and add appropriate examples.

2.4.3.7 Summary: Suprasegmental properties

We talk about the sounds of languages as *segments*; but tone, stress, and length are a somewhat different category. In each case the quality of the basic sound remains the same, but there is some additional element — the relative pitch, relative strength, or relative duration can vary for linguistic purposes. Collectively, these properties are called *suprasegmental* properties, because they go beyond the segment.

2.4.4 Conclusion

In looking at how vowels are produced, we see the importance of tongue position up and down in the mouth as well as forward and backward in the mouth.

Whether the lips are rounded or not are additional options. Those two basic articulators, tongue and lips, account for most of what is going on with vowels. The basic vowel sounds can be modified by adding other dimensions — position of the tongue root, velum, or glottis, as well as whether the language incorporates tone or stress.

2.5 Discussion

You've learned a lot in this chapter, and you've started to put all this knowledge to use. You've learned about the articulators and how they work together to make language sounds; you've learned how to move those articulators to make language sounds that are new to you; you've learned how to identify how a sound is being articulated so that you can identify it by its features. And you've started making a list of all the sounds in your language, complete with descriptions of the sounds and with words illustrating the sounds.

2.5.1 Resolving the puzzle

You are also in position to address the puzzle we confronted at the beginning of this chapter, a question inspired by finding words written in different ways in language resources, such as the word meaning 'dog' written as both *č^hɔki* and [ts^hʊk^h].

How do the weird symbols used by linguists relate to the way a language sounds?

With your new understanding of IPA symbols and writing systems, you have a new appreciation of the similarities and differences between the two ways that 'dog' is written.

From Section 2.3.1.3, you know about affricates, sounds that start with a stop and have a delayed release. You know that affricates are written in many different ways — [ts] and [č] are two such ways (*č^hɔki* and [ts^hʊk^h]). From Section 2.3.2.4, you know that the little raised ^h tells you the preceding sound is aspirated, or followed by a tiny puff of air (*č^hɔki* and [ts^hʊk^h]). From Section 2.4.1, you know that [o] and [u] are both back round vowels, differing only because the tongue is higher for [u] than it is for [o]. From Section 2.4.3.2, you know that a little tail under a vowel is called an ogonek, and that it is a diacritic telling you that air comes out the nose when you make the vowel (*č^hɔki* and [ts^hʊk^h]). Using the IPA

symbols introduced in this chapter, we might represent this word as [ts^höki] or [ts^hük^h].

There are three differences between č^hŋki (or [ts^höki]) and [ts^hük^h] (or [ts^hük^h]) that we have yet to discuss. First, why does one word end with a CV *ki* while the other word ends with an aspirated consonant and no vowel, č^hŋki vs. [ts^hük^h]? Second, why does one word have ŋ while the other has [u]? Both of these questions have the same types of possible answers. Either difference could be variation of some sort, the topic of Chapter 4. Either difference could be an alternation, the topic of Chapter 9. Either difference could be due to mishearing the speaker, a topic we do not discuss.

The third difference between the two forms is the format for each: č^hŋki is in italics while [ts^hük^h] is in square brackets. Typically, italics are reserved for the way words are written for the purpose of reading a language while square brackets indicate that you are looking at a [transcription](#), a way of writing meant to show how a word actually sounds when pronounced.

You may have found some discrepancies between the writing system used for your language (if there is one) and the symbols used to show how the words sound. Most written languages have such discrepancies. Why do we use a separate system for writing and for sounds? Why not use one system only? We address these questions first, before turning to ways you might use your new knowledge.

2.5.2 Using symbols to represent sounds

We introduced the [International Phonetic Alphabet \(IPA\)](#) right at the beginning of this chapter, a set of symbols used internationally to represent the sounds of language. As we introduced new types of sounds, we discussed how they are pronounced and we showed the IPA symbols for those sounds. When you represent the sounds of a word with IPA symbols, you [transcribe](#) the word. The resulting representation is a [transcription](#) of the word.

Transcribing a word can be difficult. It is a lot of work to learn and remember all those different symbols. Why do we bother? Why don't we just use the writing system to represent words, if a language has one?

A writing system, also called an [orthography](#), is an important way to represent the words of a language. (In some languages, there is no standard, publicly accepted writing system. Some people may develop a writing system to use until there is an official writing system for the community — this is called a [practical orthography](#).) The purpose of a writing system is to present words for reading.

The only way to use a writing system to correctly pronounce words is if you already know how the words are pronounced. This is because in writing systems some words are written in ways that do not match how they sound.

This kind of discrepancy happens for a variety of reasons; we discuss a few here. (i) The way a word sounds may have details (like aspiration, nasalization, tone, or stress) that are not represented in the orthography, so the orthography doesn't tell us precisely how each word is pronounced. (ii) Often, a writing system is created by a person who is familiar with writing another language, and adopts ways of writing sounds that work well for that language, but don't always fit the new situation. (iii) If a writing system has been around for a long time, it is likely that the language no longer sounds like it did when the orthography was created. So the letters used for writing no longer match the sounds of the language (if in fact they did match at the time the system was created). (iv) In many cases, a language has multiple dialects and some words are pronounced differently depending on the dialect spoken. (See Chapter 4 on dialects.) In this case, a shared orthography helps with written communication between dialects, but it also means that the sounds in some of the dialects do not match the letters used for writing the words. Whatever the cause of the discrepancies, when letters and sounds don't match, we need to know the spelling rules to know how to pronounce words. The spelling rules will be individual to each language or dialect.

Transcription using the IPA uses internationally accepted sound-symbol correspondences. When we are trying to learn how to pronounce the words of a language, having a transcription that shows how each word sounds is more helpful than knowing how it is spelled in the writing system.

There can be frustrations in working with transcriptions because not every resource uses the IPA. There are many different ways that languages have been transcribed, especially in older resources (the IPA was invented in the late 1800s but was not widely adopted in the Americas until much later). An alternative that is often used to transcribe Indigenous languages of the Americas is called the *Americanist Phonetic Notation*, (also *Americanist transcription*) or *APN* for short.⁷ The APN makes heavy use of diacritics, rather than having new symbols for distinct sounds. For example, the APN uses [š] — an [s] with a diacritic on top — where the IPA has a separate symbol, [ʃ]. Diacritics are challenging, in part because they can be hard to see, which means there can be a lot of errors. Stacking multiple diacritics on a single symbol can also make them all hard to read.

⁷The Americanist Phonetic Notation is also sometimes called the *North American Phonetic Alphabet*, or *NAPA*.

Another problem is that there is no single APN. Many users of Americanist transcription introduce unique symbols. This is a challenge for anyone using materials that have been recorded using the APN — especially if the author is unclear about what specific symbols are meant to represent, or if there are multiple sources with different authors and so (perhaps) different systems. If you are working with an APN resource, it is really important to find the description of what each symbol represents.

The IPA is maintained by the International Phonetic Association. Every few years, the alphabet is revisited and new symbols or diacritics may be added. You can get a copy of the most recent version of the IPA on their website.⁸

At the same time, writing systems (orthographies) serve an important purpose. They are a valuable tool for language communication, which makes them important in learning and maintaining a language. When we include words from different languages in this volume, if we have access to an orthography we show both the orthographic representation, in italics, and the IPA transcription, in square brackets. Sometimes this means that we need to convert an Americanist transcription into the IPA in order both to be consistent within the volume and also so that we do not need to explain every non-IPA symbol that we use. At other times this means we need to convert an orthographic representation into an IPA representation, following conventions presented in the source. We have checked our conversions where possible with the original authors.

To illustrate, here is an example from Kickapoo, an Algonquian branch) spoken in Oklahoma; the word in Table 23 is from Bluecloud (2020). Kickapoo has a writing system, and the *graphemes* (or letters) are a little different from the way words actually sound (as is usually the case). Because Bluecloud shows the sounds of words in IPA as well as using the writing system, someone using Bluecloud (2020) can learn about how Kickapoo is written and read, and can also learn how to pronounce the Kickapoo words.

Table 23: Kickapoo: Transcription and writing system (orthography)

orthography	transcription	meaning
<i>thiithapaakwi</i>	[ðiðəp ^h ak ^{wi}]	‘sugar’

There are several differences between the orthography and the IPA transcription in Table 23. Starting at the beginning of the word, we see the sequence

⁸We accessed the chart at <https://www.internationalphoneticassociation.org/content/ipa-chart> on August 18, 2023.

spelled *th*. In fact, it occurs twice in this word. Is this [th], [t^h], [ð], [θ], or something else? The IPA transcription tells us the sequence spelled *th* is pronounced [ð]. Knowing this, we can sound more like a Kickapoo speaker when we ask for ‘sugar’.

There are two vowels written with double letters as well, *ii* and *aa*. The IPA transcription of the word tells us that these are not pronounced as long vowels, but as short vowels, [i] and [a] respectively. Additionally, the vowel written *a* is pronounced as a mid vowel [ə], and the *kw* sequence looks like two separate consonants but is transcribed as a single consonant with secondary articulation, [k^w].

In some languages, the transcription is not in IPA. For example, in the puzzle at the beginning of this chapter, the Tutelo-Saponi word meaning ‘dog’ is written in Americanist transcription, not IPA. We convert the Americanist transcription to IPA in this volume so that the reader does not need to keep changing transcription systems with each example. In Table 24 we see how the word meaning ‘dog’ is represented in IPA, as well as in three other systems. The example is taken from Roberts (2020), who compares the different ways that words are transcribed in different sources. (Tutelo-Saponi is a Siouan language.)

Table 24: Tutelo-Saponi transcriptions for ‘dog’

source	transcription
IPA (used in this volume)	[t ^h õ:ki]
Americanist (Oliverio 1997)	[č ^h ó:ki]
Hale (1883)	[tçõŋg]
Yesáh intertribal community orthography	č ^h qg or č ^h qki

The standard convention is to present the writing system, or orthography, in italics, transcription in square brackets, and the meaning (or *gloss*) between single quotation marks. We follow these conventions in this volume.

2.5.3 Using my hard-earned knowledge

This chapter has introduced you to the articulators, both passive and active, and how we move and position the active articulators to produce different sounds. The chapter also introduced you to the various features that are commonly used to characterize the position and movement of the articulators. And the chapter showed the symbols from the International Phonetic Alphabet (IPA) that are used to show what each sound actually sounds like.

This is a lot of information. What can a language worker do with this knowledge?

Language learning. When teaching a language, one big challenge can be to teach the students how to pronounce unfamiliar sounds. With the information in this chapter, a teacher can figure out precisely what the components of an unfamiliar sound are as well as what familiar sounds are most similar to the unfamiliar ones (because they share many features). This allows the teacher to design pronunciation exercises that target the unfamiliar articulation, and other exercises that use the familiar sounds as a scaffold to build to the unfamiliar sounds. For example, if a language has front round vowels like [y] and [ø], the teacher can scaffold the pronunciation of the unfamiliar sound on the pronunciation of front non-round vowels, like [i] and [e]: Say [i], hold the tongue in that position, and round your lips — the result is [y]. Similarly with [e] and [ø].

Students who understand what the articulators are and how they move are in a great position to learn how to produce and recognize unfamiliar sounds that they encounter in the language classroom.

Language resources — dictionaries, grammars, orthographies. Grammars and dictionaries often include a section talking about how the sounds of the language are pronounced. This chapter gives a common vocabulary for talking about how language sounds are articulated. Someone creating a description of sounds for a dictionary or grammar may not want to use all of the technical terms introduced here, but it is easier to come up with descriptions that will work for a community once you know what the technical terms refer to and what sounds they create. Examples of this kind of restating are found in Bluecloud (2020: 19, 21) and Roberts (2020: 27–28). (We have taken a minor liberty with both sources, using square brackets around IPA symbols and italics for graphemes, to be consistent with the rest of this volume.)



[k] and [j] in Kickapoo

Mosiah Bluecloud, Kickapoo

[k] is a voiceless velar stop. It is produced with the fat part of the tongue touching the soft palate (velum).

[j] is a voiced palatal approximant. It is produced with the fat part of the tongue close enough to the roof of the mouth (hard palate), without touching it. The grapheme *y* is used in the Kickapoo orthography to represent the palatal consonant sound.

Bluecloud (2020: 19, 21)



[ʔ] and [x] in Tutelo-Saponi

Corey Roberts, *Occaneechi Saponi*

...the glottal stop, marked as [ʔ] in the IPA and as ' in the Occaneechi orthography. This sound is characterized by a closing in the throat followed by a release of air...

...a velar fricative... marked as [x] in the IPA and as *q* in the Occaneechi orthography. This sound is similar to what we associate with [h], however with more vibration at the back of the throat as if attempting to produce phlegm.

Roberts (2020: 27–28)

Language documentation and description. The documentation or description of a language can never be complete — language is too varied and complex. But we can make choices about what is included in the effort. Including some discussion of how a language sounds is critical if one purpose is to facilitate speaking the language. To this end, identifying the sounds of a language and describing and documenting the articulation of those sounds is essential, even if nothing else can be done to describe the sound patterns of a language. If you did the Explorations, you have already documented a partial list of the sounds of your language.

Language revitalization. One challenge in revitalizing a language is to increase the number of speakers (or signers); another challenge is encouraging learners to use the language more. Being confident of one's pronunciation helps some people feel more comfortable using even a few words of a language outside the learning situation.

Yet another challenge with revitalization is that in some languages, new words

are needed. One reason for this is that sometimes words are “borrowed” from another language, especially from languages that are not Indigenous to the Americas. Once you know the sounds of your language, you might be able to recognize some of these words because they include a sound from the source language that is not found in your language.

In other cases, there is a concept that speakers would like to be able to express, but resources on the language pre-date the introduction of the concept, so there is no word available — for example, the resources for a language dating from the 19th century would include no words for *internet*, *cellphone*, or *chocolate chip cookies*.

In such cases, the best solution may be to simply create a new word for the language (a neologism), making it as similar in sound to other words of the language. To this end, knowing the sounds used in words in your language means that you know what sounds to choose from when creating new words.

Reading linguists’ resources about your language. If you’ve found a text about your language that was written by a linguist, chances are that you found it very difficult to read. One reason is because there are many words that linguists use with very specific meanings. Here is another word of that sort — the word *segment*. For a linguist, the term *segment* is used to refer to a discrete unit of speech — we’ve used the term *sound* for this concept here, relying on an informal understanding that the term *sound* can encompass both discrete sounds and a continuous stream of sound. You might see reference to the *segments of a language*, rather than the sounds of a language.

When linguists write about the segments of a language, they typically include descriptions that use terms introduced in this chapter. If you come across a paragraph like the one about Chimariko (once spoken in Northern California, language family under dispute) in (2), there may be parts that are still confusing (partly because the paragraph is very dense), but our hope is that you now understand the basic point of such a description. (The only word in the description of Chimariko obstruents that is not covered in this chapter is the word *phonemic*. We address the concept of *phonemes* in Chapter 8 Section 8.7.2.)

- (2) Chimariko obstruent description, Jany (2009: 15)

“Stops occur in three series: plain, ejective, and aspirated. They distinguish five places of articulation: bilabial, alveolar, post-alveolar, velar, and uvular. There is also a phonemic glottal stop. Fricatives are found in five places of articulation: alveolar, palato-alveolar, velar, uvular, and glottal. Affricates equally occur in three series: plain, ejective, and aspirated in two places of articulation: alveolar and palato-alveolar.”

Another challenge that you may face in working with resources is the problem of authors using different transcription systems. This can happen if your sources use different versions of the Americanist notation, for instance, or if the sources are from very different time periods so that some of them are written prior to any standardization of transcription symbols or of terms for the articulators. Understanding the details of articulator position and movement gives you tools to use in reading texts that use different terms for the articulators, so that you can make an educated guess about what the author is trying to describe. We don't want to ignore such texts: Accessing words from older sources can add to the known words for a language.

2.6 Conclusion

In this chapter, you learned about the passive and active articulators found in the vocal tract. We looked at how consonant sounds are formed with different kinds of constriction — the manner of articulation — made at different places in the vocal tract — the place of articulation. We also looked at how different tongue positions, coupled with rounded or non-rounded lips, create different vowel sounds. In both instances, consonant sounds and vowel sounds, we also looked at added properties that make the basic sound a little bit different.

There are many more ways that humans use their articulators to produce sounds, so we may not have included all the sounds you encounter in your language in this discussion. At the same time, this chapter has a lot of information in it; as we said at the outset, it is more than is needed for any one language. It is meant to serve as an introduction to most of the sounds of the language spoken by any reader. It is also meant to serve as a reference as you look more deeply into the sound patterns of your language in case you need a refresher. We think you now have the tools to figure out how to ask your remaining questions about sounds in your language.

Through the Language Explorations, you were encouraged to explore the sounds in your language to come up with descriptions for both the consonants

and the vowels. If you did this, you now have listed many of the sounds of your language. Are you finished with figuring out how sounds work in your language? Not quite yet — in two different ways.

First, your lists may not be complete yet; you may decide to return to them later to modify them in some ways — by changing the features or the descriptions, or by adding more sounds, or even by removing some sound(s). Identifying the sounds of a language may sound like an easy task, but there are a lot of complex details to keep track of. It can take a while to become confident in the list you propose. But for now, you have made a solid start and have a list to work with.

Second, not just any handful of sounds makes a word. You can't just grab a bunch of sounds and put them together and have a new possible word. Regardless of the language, only some sequences of sounds make reasonable words for that language. Chapters 5–9 address different aspects of the rules that determine whether sounds fit together to make a reasonable word for a language.

Chapter 3

Morphemes: The foundation of word-building

Diana Archangeli

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Here's a puzzle. Zenzontepec Chatino is an Otomanguean language spoken in Oaxaca. Suppose that you're learning Zenzontepec Chatino, and you're excited to finally be able to follow along when your teacher tells a story.

You hear the sentence *lē' ngutiyaa tsaka kuijnyä* [lēʔ ŋgutija: tsaka kʷihnʲäʔ] and happily recognize most of the words — *tsaka* [tsaka] means 'one' and *kuijnyä* [kʷihnʲäʔ] means 'mouse'... what about that mouse? *ngutiyaa* [ŋgutija:] — something to do with 'arriving' you think, because that's what other words ending with *...tiyaa* [...tija:] mean. Will the mouse arrive? Did the mouse arrive? Is it arriving? How can you tell?¹

A hand shoots up and a student announces that the mouse has arrived. The teacher nods, then challenges the class to change the sentence to mean that the mouse is arriving — someone offers the word *ndetiyaa* [ndetija:] — like *ngutiyaa* [ŋgutija:] it ends with *...tiyaa* [...tija:], but the start is different. The teacher smiles approvingly.

We use the convention of gray text to minimize the part of the word that is of less interest to our discussion. Where necessary, we have created transcriptions in the IPA, in consultation with Eric Campbell. We owe him a great debt of gratitude for the care he took in checking every Zenzontepec Chatino word in

¹Our source for most of the Zenzontepec Chatino information is Campbell (2014). The sentence about the mouse, however, is from Campbell (2017). Text in *italics* is a representation in practical [orthography](#); symbols in square brackets show the pronunciation using the [International Phonetic Alphabet](#) (IPA). Appendix A lists IPA symbols if you need a review.

this book. He tells us that the speakers call their language *tsá' jnyá* [tsáʔ hnʲá], while it is called Chatino de Zenzontepec in Spanish.

How do some words help us to understand other words?

The answer starts with the observation that words can be made up of smaller parts, like a bracelet can be made up of beads on a string. These smaller parts combine in different ways to make different words, just as beads combine to make different bracelets. In the Zenzontepec Chatino words *ngutiyaa* [ŋgutija:] and *ndetiyaa* [ndetija:], we can think of *tiyaa* [tija:] as the “bead” meaning ‘arrive’ and other parts as the beads that tell us about whether the action is completed (*nku* [ŋgu]) or on-going (*nte* [nde]).

These smaller parts, these beads on the string, are called *morphemes*. In words with multiple morphemes — *polymorphemic* words — each morpheme contributes meaning to the whole word. Understanding morphemes and how to combine them in words is an important part of understanding and creating vocabulary for a language. If an ‘arrive’ morpheme (*tiyaa* [tija:]) is combined in a word, we get an ‘arrive’ meaning; if the word includes the morpheme *ngu* [ŋgu], it means the action has been completed, and so on.

The organization of morphemes within a word is not directly a major component of a language’s phonological patterns, its sound signature. However, knowing a bit about *morphology* — the study of words and morphemes — is helpful as we go forward with understanding the sound patterns that words follow, especially Chapters 6–9.

Here’s our promise. After reading this chapter, you will be able to look at words in a language to determine whether there are smaller parts that contribute to their meaning. Your skills will enable you to identify the morphemes themselves and to use that knowledge to build likely words in the same language. Language workers can use this knowledge in many ways, for example when designing language lessons, developing dictionaries, and helping community members create missing words to revitalize a language in line with the language itself. You will also be in position to examine how sounds and morphemes interact within words, the topic of Chapter 9.

Here’s our plan. In this chapter, we first show how to use *subtractive analysis*, a method to identify the morphemes in a word, and then we see how to test an analysis. We use one language, Zenzontepec Chatino, for our examples. We do this because the analysis builds on itself: We see in this chapter how discovery

of one morphological pattern leads to other discoveries about the language. Finally, we play with our knowledge of Zenzontepec Chatino morphology. With the Language Explorations, you are invited to do the same with your language.

1. Section 3.1 expands on the concept of related and unrelated words. It is the related words that we explore with morphological analysis, starting with Section 3.3.
2. Section 3.2 introduces a number of conventions used by linguists to show meaning and morphological structure. The primary notation is known as the *three-line-gloss*. The three-line-gloss is both conventional and useful. Because it is conventional, you will find it in many works written by linguists, often without explanation. It is useful because three-line-glosses are a concise way to encode a number of morphological concepts. Those concepts help organize the description of your language.
3. Section 3.3 presents subtractive analysis, a common method used for determining linguistic subparts of words. This is a very handy tool for understanding the complexity of words in your language.

If you are reading this chapter in order to carry out a very basic analysis of your language, you might stop after Section 3.3 and jump ahead to the discussion in Section 3.7. On the other hand, you may also want to read Section 3.4, which is about checking whether your pattern is general in the language.

4. Section 3.4 shows how to make sure that a morphological analysis is general in a language, rather than a pattern for just a few words. After reading this section, you will have learned how to create and test a morphological analysis of your language. Depending on your situation, this may be where you jump to the discussion in Section 3.7, to learn about ways to make use of your new skills.
5. Section 3.5 shows ways that you can use the morphological analysis of just a few words to expand your analysis of related words of other types. This is really useful for extending a learner's usable vocabulary.

If you stop after this section, you will have learned how to carry out and test a morphological analysis of your language (from Sections 3.3 and 3.4); you will also have learned how to scaffold on that analysis to understand other morphological patterns. What you will miss is our discussion of how to make sense out of morphological surprises — that is, what to do when something doesn't follow an expected pattern.

6. Section 3.6 shows ways that we can use words that don't follow an established pattern to learn more about a language.

7. The chapter ends with Section 3.7, where we resolve the chapter puzzle more completely, sketch a few ways that morphology goes beyond the scope of this chapter (it merits a comparable volume of its own), and provides some ideas about using the material presented here in language work.

As with other chapters, there are exercises to help you check on how well you understand the concepts. If you think that you don't need to read this chapter, you can try a few of the Comprehension Checks to make sure. The Language Explorations guide you in beginning a morphological analysis of your language. We urge you to complete these even if you skip reading the chapter so that you continue building the phonological sketch of your language.



Diacritics in practical orthography

Zenzontepec Chatino has both nasal vowels and tone; both are marked in the practical orthography. We saw how nasal vowels and toned vowels are marked in IPA in our discussion in Chapter 2. Here is an explanation of how these properties are indicated in the Zenzontepec Chatino practical orthography. We see Zenzontepec Chatino nasal vowels in *kuë'ë* [k^wẽʔẽ] 'air', for example. Nasal vowels (shown in IPA with a wavy line over the vowel, e.g. [ẽ]) are introduced in Chapter 2 Section 2.4.3.2. A nasal vowel is indicated in the Zenzontepec Chatino [practical orthography](#) with two dots above the rightmost nasal vowel, as in *kuë'ë*. The two dots are called diaeresis.

Zenzontepec Chatino has three tones, high, mid, and low. The practical orthography uses diacritics that are like IPA diacritics, with the convention of leaving low-toned vowels unmarked. Thus, vowels marked with an acute accent, like *á* [á] and *é* [é], are high toned. Those with a macron, like *ā* [ā] and *ē* [ē], are mid toned. Those with no mark, like *a* [a] and *e* [e], are low toned.

In the Zenzontepec Chatino word *telā* [telā] 'night', the first vowel is low toned and the second is mid toned. The low toned vowel is shown with no tone mark at all. In some publications, low tone is marked by a grave accent, like [à] and [è]; in some, mid tone is not marked. Yes, this

can be confusing. Ideally, authors are very clear about the marks being used to indicate tone or other properties. (Eric Campbell is one of these careful authors.) See Chapter 2, Section 2.4.3.5 for more on tone. In the rest of this chapter, you may find some unexpected tones. Please do not let these distract you from the main points in the text.

Tone plays a significant role in Section 3.6.5, so you will have a chance to become more familiar with the tone diacritics in that section.

3.1 Introduction: What are morphemes?

Morphemes are famously described as the smallest meaningful units in a language. Some morphemes can appear as words by themselves — such as the in Zenzontepec Chatino word *kuijnyä* [k^wiɦn^ˈä?] ‘mouse’. (Remember the convention: Italics show the way words are spelled in the practical orthography while square brackets show their transcription in the IPA.) Morphemes which can stand alone on their own as a word are called *free morpheme*.

Other words are composed of more than one morpheme. A *bound morpheme* cannot stand alone — a bound morpheme must be attached to some other morpheme, which in some cases might be free, but in other cases might be another bound morpheme.

As we learned above, the Zenzontepec Chatino words *ngutiyaa* [ŋgutija:] ‘arrived’ and *ndetiyaa* [ndetija:] ‘arriving’ each have a similar part — *tiyaa* [tija:] — and a similar meaning — ‘arrive’. This suggests that *tiyaa* [tija:] is a morpheme meaning ‘arrive’. The other parts of *ngutiyaa* [ŋgutija:] and *ndetiyaa* [ndetija:] — *ngu-* [ŋgu-] and *nde-* [nde-] — are also morphemes. They each contribute a different type of meaning, telling us about the nature of the ‘act of arriving’ (whether it is completed, *ngu-* [ŋgu-], or whether it is on-going, *nde-* [nde-]). They are affixes: An *affix* is always a bound morpheme.

We start by revisiting *ngutiyaa* [ŋgutija:] and *ndetiyaa* [ndetija:], to understand the steps that led us to examining this word pair.

We work with pairs of words to help focus our attention on minimal differences. This is important as a way to get started.



Pairwise comparison in analysis

Comparing pairs is a strategy we repeat throughout this book. For example, in Chapter 2, we saw that two sounds can differ by a single feature (such as [t] and [d] differing only by the feature [voice]). The concept of **minimal pairs** (introduced in Chapter 5) are pairs of words that differ by only one sound – and sometimes by only one feature of one sound. (See Chapter 2 for an explanation of features, the glossary for feature definitions, and Appendix B for a list of segments and their features.)

The word pair *ngutiya* [ŋgutija:] and *ndetiya* [ndetija:] is interesting because the sounds and the meaning are in part the same and so are related, and the sounds and the meaning are in part different and so are not related. Let's look at a few more word pairs to fully understand this idea.

Generally, if we randomly grab two words from a language, they will sound different from each other and they will mean different things. The two Zenzontepec Chatino words *kuë'ë* [k^wẽʔẽ] 'air' and *telā* [telā] 'night' are an example. There is nothing similar in the sounds or in the meanings. They are completely unrelated words. Zenzontepec Chatino also has high, mid, and low tones.

It is possible to find words with related meanings and words with related sounds. Here are two words with related meanings: *nānē'* [nānẽʔ] 'belly of' and *jlākū* [hlākū] 'stomach of'. However, the sounds of these words are completely different from each other. These are unrelated words that accidentally have closely related meanings: These are different words with similar meanings.

We can also find words with related sounds but unrelated meanings. For instance, in the word pair *ti'i* [tiʔi] 'poor' and *ti'ī* [tiʔi] 'breath of', the sounds are almost completely the same, differing only by the tone (low or mid) of the final vowel. However, the meanings are quite different from each other. These are unrelated words that happen to sound almost alike: They are different words with similar sounds.

The three distinct types of unrelated words are summarized in (1).

- (1) Three types of unrelated word pairs (Zenzontepec Chatino)
- | | sounds | meanings | word 1 | | word 2 |
|----|-----------|-----------|--------------------------------------|-------|-------------------------------|
| a. | different | different | <i>kuë'ë</i>
[k ^w ẽʔẽ] | 'air' | <i>telā</i>
[telā] 'night' |

b.	different	similar	<i>nănē</i> [nănēʔ]	‘belly of’	<i>jlākū</i> [hlākū]	‘stomach of’
c.	similar	different	<i>ti’i</i> [tiʔi]	‘poor’	<i>ti’i</i> [tiʔi]	‘breath of’

Laid out this way, we can see that one type of word pair is missing — a word pair with both similar sound and similar meaning. We find that in *ngutijaa* [ŋgu-tija:] ‘arrived’ and *ndetiyaa* [ndetija:] ‘arriving’. There is similarity in sound — the sequence *tiyaa* [tija:] — and there is similarity in meaning — ‘arrive’. These similarities indicate that the words are related in both ways. But there are also differences in sound and meaning so we know that these are not the same word — one word starts with *ngu* [ŋgu] and means that someone has already arrived while the other starts with *nde* [nde] and means that the arriving is on-going.

(2) A related word pair (Zenzontepec Chatino)

sounds	meanings	word 1	word 2
different	different	<i>ngutiyaa</i> ‘arrived’ [ŋgutija:]	<i>ndetiyaa</i> ‘is arriving’ [ndetija:]
same	same	<i>ngutiyaa</i> ‘arrived’ [ŋgutija:]	<i>ndetiyaa</i> ‘is arriving’ [ndetija:]

Pairs with comparable similarity and difference in both sound and meaning are the kind of pairs we are most interested in when we are trying to understand the morphemes of a language. Sorting out the role of the similarities and differences in making words is the heart of morphological analysis.



Comprehension Check 1

Here are some words in Hopi, a Uto-Aztecan language that we use for the Comprehension Checks in this chapter. Make a note of your answers because they may be useful in answering later Comprehension Checks.

Which of these words might be *morphologically related*? Morphologically related words share a common sequence of sounds and a common meaning while the rest of the word has different sounds and different meaning. Be sure to inspect the entire word. Similarities may be found at

the beginning of a word or at the end of a word. (Review the discussion of (2) if you are struggling with this question.)

- | | | | |
|----|---------------------|----------------|---|
| a. | <i>tokiltoyna</i> | [tokiltojna] | ‘set a date’ |
| b. | <i>toko’ma</i> | [tokoʔma] | ‘putting on weight’ |
| c. | <i>tokiltoynani</i> | [tokiltojnani] | ‘will set a date’ |
| d. | <i>sakiwta</i> | [sakiwta] | ‘be lucky with a girl, have a date with a girl’ |

What is it about the words that leads you to your answers?



Here’s our answer

The words *tokiltoyna* and *tokiltoynani* (in a. and c. respectively) are likely to be morphologically related; the other two words are not. Let’s look a little more closely.

The words [tokiltojna] and [tokiltojnani] are almost completely the same, both beginning [tokiltojna...]. And the meanings are very similar – both refer to setting a date. There is similarity in sound and in meaning.

The other two words are different. While [tokoʔma] starts the same way, with [tok...], the meaning (‘to put on weight’) is quite different from ‘to set a date’. The sound similarity seems to be accidental.

It isn’t likely that the Hopi words [sakiwta] and [tokiltojna] are related either. They do not share sounds, and the meanings are quite different too. (With [sakiwta], the ‘date’ has to do with going out with a girl, while [tokiltojna] means something like ‘schedule an event’.)

3.2 Representing words and morphemes

Before we jump into how to carry out a morphological analysis in Section 3.3, we provide some background and useful vocabulary for talking about morphemes and their meanings.

Part of what we want to show in language classes, dictionaries, etc., is the sounds and meaning for each word of the language. This sounds trivial, but sometimes it can be challenging. One type of challenge is that there can be a mismatch between concepts in the language and concepts in the language we are using for translations (like English is used in this book).

In this section, we introduce approaches that are helpful in representing meanings despite the mismatches. These approaches, or conventions, are useful in understanding some writings by linguists because they also use these same conventions. We follow these conventions in this volume as well, so becoming familiar with them will make the rest of the book easier to follow – and you will get a lot of practice with them.

3.2.1 Meaning mismatches and the dot convention

The meanings mismatch arises when the language being described uses one morpheme to express a concept that does not match up with any single morpheme or word in the language being used for describing. We see this in several Zenzontepec Chatino morphemes. For example, the morpheme *lakuā* means ‘get counted’, a concept that English expresses with two words, not one.

There are two general ways that this kind of *gloss* (definition or meaning) is represented in linguistic writings. One way is intuitive: We just write out the translation – ‘get counted’.

The other way is a little less intuitive, but it focuses our attention on the fact that the language being described is different from the language being used to make the description. Instead of separating words in the translation with a space, we separate them with a dot (‘.’) – ‘get.counted’.



The Leipzig Glossing Rules

The dot convention and others like it presented in this chapter are laid out in Comrie et al. (2015) (available online at <https://www.eva.mpg.de/lingua/pdf/Glossing-Rules.pdf>). Comrie et al. (2015) provide a complete set of conventions for language glossing, called the *Leipzig Glossing Rules*, or the *Leipzig Conventions*. We do our best to follow these conventions in our glosses.

An advantage of using dots to separate meanings in the translation of a single word is that we can then use spaces to separate words in the utterance being described and align each word with its single (dot-separated) gloss. This is illustrated in (3).

Note how the left edge of each Zenzontepec Chatino word is aligned with a gloss — for instance, there is a big white space before *tsaka* [tsaka] so that *tsaka* [tsaka] can align with the word ‘one’. This shows that *tsaka* [tsaka] means ‘one’. Note also that the gloss aligned with each word is either a single word or is a group of words connected by dots “.” between the words. White space is used specifically to separate words or their corresponding glosses.²

- (3) Dot convention illustrated

<i>lē'</i>	<i>ngutiya</i>	<i>tsaka</i>	<i>kuijnyä'</i>
then	completive.arrive.there	one	mouse

3.2.2 Three-line glosses

The dot convention and spacing convention illustrated in (3) are used in making a **three-line gloss**. A three-line gloss has (you guessed it!) three lines. The top line shows the words of the language being described, typically in the practical orthography if there is one. (In our examples here, the language is Zenzontepec Chatino). The second line shows the meanings of morphemes, using dots to separate information when a single word is inadequate for translation. The third line shows an intuitive translation.

We give two examples of three-line glosses here, one with the single word *lakua* [lak^wa] ‘how.many’ and one with the sentence in (3), also used in the introduction to this chapter. In each case, the first line shows the words in Zenzontepec Chatino **practical orthography**; the second line shows the word-by-word gloss. The third line shows the intuitive translation. Notice that there is no line showing the IPA transcription in a three-line gloss.

When there is only one word, as in (4), all three rows are aligned at the left.

- (4) Three-line gloss example — single word

<i>lakua</i>
how.many
‘how many’

²‘Completive’ indicates that the action of ‘arrive.there’ has been completed. A loose translation is ‘arrived there’. We return to this point in Section 3.2.3.

When the three-line gloss contains more than one word, as with the sentence in (5), the words in the first line and the glosses in the second line are aligned so that the left edge of each item — word and meaning — match up. In some cases, there can be quite a bit of white space to ensure the alignment. However, the third line, with the intuitive translation, aligns only at the left. There is no extra spacing between words because the words in the intuitive translation do not necessarily match the words or the order of words in the language being described.

- (5) Three-line gloss example — sentence
lē' ngutiyaa tsaka kuijnyä'
 then completive.arrive.there one mouse
 'Then a mouse arrived there.'

In addition to using the dot convention in three-line glossing, another advantage is that it emphasizes differences between the language being described and the language used for writing the description. When we write about a language, we give meanings in the language that we use for writing — in this book, we are writing in English and we use English for the glosses of our words. For example, we give English glosses for the Zenzontepec Chatino words, *ndetiyaa* [ndetija:] 'is arriving', and *ngutiyaa* [ŋgutija:] 'arrived'.

However, English translations reflect the way the English language is structured. Just a little familiarity with another language can show that English translations do not necessarily work well. For example, English requires different forms depending on who is arriving — 'am arriving', 'are arriving', 'is arriving' — but many languages do not require different forms to indicate who is doing the action. (We call this person the *actor*.) Still other languages might require more forms than English has: English uses 'are arriving' for three actors, for 'we', 'you', and 'they', while other languages might have different forms for each of the different actors. Using English translations can imply that a language functions like English when it doesn't.

This situation leads to a second convention, one that makes use of the difference between *content* and *function* morphemes.

3.2.3 Content and function and the SMALL CAPS convention

The meanings discussed above focus on the basic meaning of a word — 'arrive', 'mouse'. This type of morpheme is called the *content* morpheme because it provides the core meaning. But as we have just seen, some words carry more information than just their content. There is also information about who is carrying

out an action, whether the action is on-going or completed, and so on. The words – and morphemes – that provide this information are called *function* words and function morphemes.

Small capital letters (‘small caps’) are used to indicate that a word or morpheme is providing function information. Plain text is used for content words and content morphemes. We revisit our three-line gloss sentence example with the “small caps” convention in mind. Note how *COMPLETIVE* is now written in small caps, showing the function part of the meaning of *ngutiyaa* [ɲgutija:].

- (6) Three-line gloss example – sentence

lē’ ngutiyaa tsaka kuijnyä’
then *COMPLETIVE*.arrive.there one mouse
‘Then a mouse arrived there.’

We know that both *ngutiyaa* [ɲgutija:] and *ndetiyaa* [ndetija:] have a base meaning of ‘arrive’ in Zenzontepec Chatino, which is carried by the morpheme *tiyaa*. We call this morpheme a content morpheme because it carries the core meaning of the word; it is also the *root* or *stem* of the word.

Function morphemes carry another kind of meaning, such as tense (the time of an action, such as *PAST* for completed actions and *FUTURE* for actions that are yet to be undertaken) or case (which gives information about nouns and their role in a sentence, like subject or object). This is called “grammatical meaning”. The function morphemes add this grammatical information to the core meaning that a content morpheme has. For instance, *nde* [nde] and *ngu* [ɲgu] are function morphemes in Zenzontepec Chatino that provide tense information for *tiyaa* [tija:]. The core meaning of *ngutiyaa* [ɲgutija:] and *ndetiyaa* [ndetija:] is ‘arrive’, but these two words are different in whether the action is completed or not.

There are many standard terms used by linguists for different functions. For instance, linguists use the label *PROGRESSIVE* for on-going actions. Using this term allows us to gloss *ndetiyaa* [ndetija:] as ‘*PROGRESSIVE*.arrive’ instead of ‘is.arriving’. Other functions have other conventional names, such as *COMPLETIVE* for an action that is finished, and *PREGRESSIVE* for an action that is going to take place but hasn’t happened yet.

While standard terms are listed in Comrie et al. (2015), there are many writings that predate Comrie et al. (2015), so there is a lot of variation in the names and definitions given to function morphemes. Additionally, some languages make use of function concepts that do not align well with established terms. Ideally, authors are very careful about describing the use of the various function morphemes that appear in their work.

These terms can be quite long and so are often abbreviated to just a few letters. For example, *PROGRESSIVE* might be abbreviated *PRG* or *PROG* and *COMPLE-*

TIVE might be abbreviated **COMPL** or **CPL**. There are some standard abbreviations introduced in the Leipzig Glossing Rules (Comrie et al. 2015; <https://www.eva.mpg.de/lingua/pdf/Glossing-Rules.pdf>), but some authors create their own abbreviations. Unfortunately, sometimes abbreviations are used without explanation, which can be very frustrating for the reader.



Glossing Conventions for Sign Languages

Samantha Prins and Sybil Vachaudéz

We've just looked at the conventions for glossing spoken languages. But what about signed languages? Due to differences in modality, Sign languages have their own unique set of glossing conventions. Some of the common shorthand linguists use for glossing signs include:

- UPPERCASE: refers to the entirety of the manual sign (handshape, orientation, location, movement, but not the non-manual marker)
- lowercase: non-manual markers (produced using the mouth, face, head, or body) — these are often produced simultaneously with manual signs, and can be stacked vertically with a line underneath to mark the duration and scope of the overlap
- CL: classifier sign (refers to a class of objects or entities and their properties, such as size and shape, how they are handled, or how they move through space)
- IX: pointing or index sign (a designated point in space that can refer to entities, locations, etc.)
- ++: indicates repetition of a sign
- ^ or +: indicates the joining of two signs in a compound

For example, the words 'sun' and 'shark' in Highland Mayan Sign Language (HMSL, used by Maya communities in the Guatemalan highlands), can be glossed like this.

‘Sun’ in HSML

Kaqchikel signer recorded by Samantha Prins and Sybil Vachaudéz



upward gaze

SUN

‘sun’

The complete gloss for ‘sun’ tells us that the signer produces ‘sun’ by combining the manual sign SUN (curved palm with splayed fingers) with a non-manual marker (upward gaze).

‘Shark’ in HMSL

Kaqchikel signer recorded by Samantha Prins and Sybil Vachaudéz



FISH+BITE

‘shark’

The gloss tells us that ‘shark’ is signed as a compound of the signs for FISH and BITE.

Detailed description of this kind is useful for documenting and describing Sign languages, especially in writing when photos or videos can't always be included. Knowing how these conventions work can also help you understand what others have written about a particular Sign language, learn more about how words and phrases are formed in that language, and apply that knowledge to teaching or learning signs in your own community.



Compound words

A **compound** is a type of word that is created by putting together two words to make a single word. For example, in Hiráaca, two nouns can combine to make another noun: *naxbíccaadi* ‘bear’s den’ is made by combining *naxbicci* ‘bear’ and *caadi* ‘den’, according to Boyle (2007: 88).

3.2.4 Summary

Why do we start this chapter with the conventions used in presenting words in another language?

Using these standard terms, rather than simply using the intuitive English equivalent, helps separate properties of the language being described from properties of English. We are aligning the meanings of words to their use in the described language, Zenzontepec Chatino in our examples here, rather than in the language used for description, like the English we use in this volume.

Another value to using function terms rather than intuitive glosses is that it helps show meaning similarities and differences in word pairs. With *ndetiyya* [ndetija:] ‘PRG.arrive’ and *ngutiyya* [ŋgutija:] ‘CPL.arrive’, we can easily see the similarity — ‘arrive’ — and the difference — PRG and CPL.

Knowing the standard conventions for presenting words in a linguistic analysis is useful in reading works about your language because some authors don’t explain the conventions.

With these conventions, we are ready to start exploring words that are made up of more than one morpheme — polymorphemic words.

3.3 Relations in a word pair

In this section, we walk through our two words, *ndetiyaa* [ndetija:] ‘PRG.arrive’ (‘is arriving’), and *ngutiyaa* [ŋgutija:] ‘CPL.arrive’ (‘arrived’), identifying the steps we use to find similarities and differences. These are the basic steps used in any morphological analysis. We call the method **subtractive analysis**.

When we examine the sounds of these two words, we see that the end of the word is the same, as shown in (7). (We continue to use gray text to minimize attention to the gray parts of a word.)

- (7) Same: *-tiyaa* ‘arrive’
- a. *ndetiyaa* [ndetija:] ‘PRG.arrive’ (‘arriving’)
 - b. *ngutiyaa* [ŋgutija:] ‘CPL.arrive’ (‘arrived’)

By isolating the part that is the same in both sound and meaning in the two words, we identify one of the morphemes — *tiyaa* [tija:] ‘arrive’.



Comprehension Check 2

Remember Hopi from our first Comprehension Check, on p. 139? Here is another pair of words in Hopi to build on your knowledge of the language. Find the part that has the *same sound and meaning* in both words.

- a. *pakima* [pakima] ‘enter.COMPLETIVE’ ~ ‘entered’
- b. *pakito* [pakito] ‘enter.PREGRESSIVE’ ~ ‘going to enter’



Here's our answer

The part that is the same is *paki* 'enter'.

To see this, we can array the two Hopi words to show what is the same, following the model of (7).

[pakima]	'enter.COMPLETIVE'
[pakito]	'enter.PREGRESSIVE'

We have identified one of the morphemes: [paki] 'enter'.

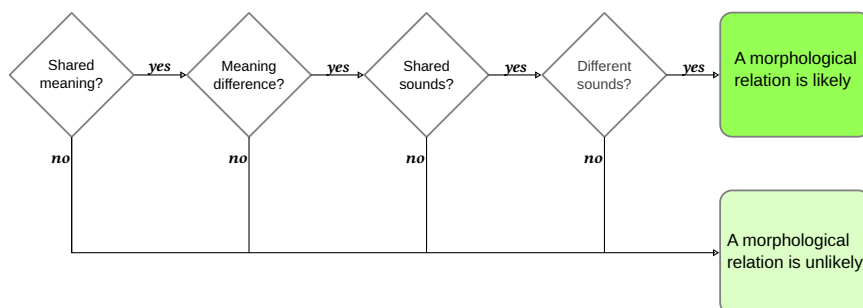


Exploring my language

Find two words in your language that you think are morphologically related. Also find two words that you think are unrelated. Test your pairs.

1. Is there a shared meaning?
2. Is there a difference in meaning?
3. Is there a shared sound sequence?
4. Is there a difference in sound sequences?

You may find it easier to use the decision tree below — it poses the same four questions.



If you answer “yes” to all four questions — that is, there is both common meaning and common sound, but also there is a difference in meaning and a difference in sound — you have identified a pair of words that are likely to be morphologically related. If you answered “no” to any of the questions, then the words are not likely to be morphologically related to each other (although there may be some other type of relation such as variation, discussed in Chapter 4).

This practice can be a first step in understanding relationships among words in your language.

3.3.1 Subtractive analysis

Once we have isolated what we think is a morpheme shared in a word pair, we can “subtract” its sounds and meaning to find out what other morphemes are present in each word. In the Zenzontepec Chatino case, we subtract *tiyaa* [tija:] ‘arrive’ from our two words and are left with two more sound-meaning pairings. In (8), we show subtraction by putting the subtracted sounds and meaning in gray font, and leaving the remainder in black font. We see a pairing between *nde* [nde] and PRG and another pairing between *ngu* [ngu] and CPL.

- (8) Different: *nde*- ‘PROGRESSIVE’ vs. *ngu*- ‘COMPLETIVE’
- a. *ndetiyaa* [ndetija:] ‘PRG.arrive’ (‘is arriving’)
 - b. *ngutiyaa* [ngutija:] ‘CPL.arrive’ (‘arrived’)

From this *subtractive analysis*, we have isolated the sound and meaning of three morphemes: *tiyaa* [tija:] ‘arrive’, *nde* [nde] PROGRESSIVE, and *ngu* [ngu] COMPLETIVE. We also know that *tiyaa* [tija:] comes at the end of the word.

With this information, we can make two general statements about Zenzontepec Chatino word structure, one about words that start with *nde* [nde] and one about words that start with *ngu* [ngu], given in (9).

- (9) Generalizations about word parts in Zenzontepec Chatino
- a. A verb that starts with *nde* [nde] shows that the action is on-going, called PROGRESSIVE.
 - b. A verb that starts with *ngu* [ngu] shows that the action has been completed, called COMPLETIVE.



Comprehension Check 3

Here is the pair of Hopi words familiar from the Comprehension Check on p. 148, where you identified the morpheme *paki* [paki] ‘enter’.

Now use subtractive analysis to find the sound-meaning pairs that make up the other morpheme in each word. Give a statement telling how these morphemes combine to make words.

- a. *pakima* [pakima] ‘enter.COMPLETIVE’ ~ ‘entered’
- b. *pakito* [pakito] ‘enter.PREGRESSIVE’ ~ ‘going to enter’



Here’s our answer

Using subtractive analysis, we remove [paki] from these words. What is left is the affixes.

- a. [pakima] ‘enter.COMPLETIVE’
- b. [pakito] ‘enter.PREGRESSIVE’

In [pakima], we see that the final part, [-ma], corresponds to the completive meaning, ‘entered’. With [pakito], the final part, [-to], corresponds to the pregressive meaning, which gets translated as ‘going to...’: [pakito] ‘going to enter’.

We have identified two more morphemes, the affixes [-ma] COMPLETIVE and [-to] PREGRESSIVE. Both attach at the end of the root: [pakima], [pakito] (and not [mapaki] or [topaki]). This is different from our Zenzontepec Chatino example, where the morphemes attach at the beginning of the word.



Exploring my language

Take the word pair in your language that you think is morphologically related — the pair you found in the exploration on p. 149.

1. Find a morpheme that is common between the two members of your pair by finding the part that has the same meaning and the same sounds.
2. Use the subtractive analysis method to figure out the other morphemes in your pair.
3. List your three morphemes, giving both sound and meaning. You may need to be a little creative when defining the function morphemes in your language. That is ok — aim to find ones that resonate with your language and with speakers of your language. (See the discussion in Section 3.2.3 if you need a reminder about *function* morphemes.)
4. Make a statement about how to combine your three morphemes.

Use statements like the one you just made to help community members identify morphemes in your language. You could also use these statements to structure a language lesson and even to sketch out parts of a grammar or dictionary of your language.

3.3.2 The parts of polymorphemic words

Languages often have words that are made up of single morphemes. These can be words of different types. In our puzzle, we used the sentence *lē' ngutiya tsaka kuijnyä* [lēʔ ŋgutija: tsaka kʷihnʲäʔ] ‘then a mouse arrived’. This sentence has three Zenzontepec Chatino words that have only one morpheme: *lē'* [lēʔ] ‘then’, *tsaka* [tsaka] ‘one’, and *kuijnyä* [kʷihnʲäʔ] ‘mouse’. Words with only one morpheme are called *monomorphemic* words. We also determined that the word *ngutiya* [ŋgutija:] ‘arrived’ has two morphemes — *ngu* [ŋgu] and *tiya* [tija:], making it a *polymorphemic* word.

We can see that the polymorphemic word *ngutiya* [ŋgutija:] is made up of a content morpheme, *tiya* [tija:] ‘arrive’, and a function morpheme, *ngu* [ŋgu] COMPLETIVE. The terms *content* and *function* tell us the type of meaning of each

morpheme, but they don't tell us much else. We have some other terms that provide more information: Generally, morphemes can be categorized as **root** or **affix**. You may also see the term **stem**.

A morpheme that forms the sound and content base for a word is called the **root** — *tiyaa* [tija:] 'arrive' is a root: It carries the basic meaning of the word and it cannot be divided into smaller meaningful units. Some roots have a specific meaning, but some don't have a clear meaning until they are combined with an affix.

An **affix** must attach to a root or a stem. Affixes usually have a functional or grammatical meaning, and they do not form a word by themselves. For example, *nde* [nde] and *ngu* [ŋgu] need to attach to *tiyaa* [tija:], and cannot be used alone. If you want to show the parts of a word, you put a hyphen between the morphemes in the gloss and in the IPA representation: *ngutiyaa* [ŋgu-tija:] 'CPL-arrive'. You will also sometimes see hyphens in the orthographic representation: *ngu-tiyaa* 'CPL-arrive'.

In some cases, roots do not have a clear meaning by themselves, and only get a clear meaning when an affix is attached. We use the term **stem** for a unit with a clear meaning, whether it is made of a root alone, or is made of a combination of a root and affix(es).

There are special terms for different types of affixes, terms that you may already be familiar with. An affix preceding a root or stem is called a **prefix**, while an affix following a root or stem is called a **suffix**. We have seen examples of these in the text. A hyphen is used after a prefix to show that it attaches at the beginning of a form: *ngu-* [ŋgu-]. With a suffix, the hyphen comes before the morpheme.

In some languages, an affix can even occur inside a root, which is called an **infix**. Infixes are marked with a hyphen on both sides of the morpheme.

3.3.3 Refining three-line glosses: Whole-word and word-structure representations

Now that we understand about morphemes and how words can be made up of more than one morpheme (the polymorphemic words), we can write words in two different ways — as whole words (10a) or as a sequences of morphemes, showing the word structure (10b) — that is, as bracelets or as beads on a string, if you liked that analogy. The morpheme-beads are separated from each other with hyphens, as shown in (10b).

(10) Conventions for representing polymorphemic words

a. ‘whole word’

ndetiyaa [ndetija:] ‘PRG.arrive’ (‘is arriving’)

ngutiyaa [ŋgutija:] ‘CPL.arrive’ (‘arrived’)

b. ‘word structure’

nde-tiyaa [ndetija:] ‘PRG-arrive’ (‘is arriving’)

ngu-tiyaa [ŋgutija:] ‘CPL-arrive’ (‘arrived’)

Notice that in the whole-word glosses in (10a), the meanings are separated by a dot “.” while in the word-structure glosses in (10b), the meanings are separated by a hyphen “-”. The dot “.” shows a meaning for the word being glossed without showing any of the structure inside the word – even when that structure exists, as it does with ‘PRG.arrive’.

The hyphen “-” shows structure inside words, and is used in both words and glosses so that we can see how meanings and sounds correspond. A hyphen “-” is used to separate morphemes in a word and to separate meanings in its gloss, and the order of the meanings and the morphemes correspond to each other. These conventions may also be combined, as in *ndē-kītyé* [ndē-kiṭʰéʔ] ‘PRG-get.snapped’, shown in (17). Be aware: The dot “.” is only used in glosses, not in the words being glossed. When a dot is used, the order of the meanings is irrelevant – any order can be used.)



Unfortunate inconsistencies in linguistic writings

Some sources use some other symbol instead of a dot “.”, such as a colon “:”. Again, this can be confusing. You may also see a plus sign “+” or an equals sign “=” separating morphs, instead of the hyphen “-”. The use of these symbols is less standard, and ideally the author will clearly state what is meant by the “:” or “+” or “=” symbol. We hope that authors are very clear about their conventions. We also hope that the material presented here helps you to puzzle out when people are using symbols in a non-standard way, and maybe even to figure out any non-standard system you encounter.

Both the whole-word and the word-structure conventions are used in writings about languages, often in the same work. Including a hyphen “-” to separate morphemes can help learners understand the structure of words but the hyphens can also lead the unsuspecting to think that the sequence of sounds is really two words, not one. On the other hand, when learners are not aware of word-internal structure, it can take longer to understand a language and to see the relations among words. The hyphens can help reveal the structure inside words. Each system has advantages and disadvantages to be aware of.



Comprehension Check 4

Here are our two words from Hopi from earlier Comprehension Checks (on p. 148 and p. 151): [pakima] and [pakito], the COMPLETIVE and PREGRESSIVE forms of ‘enter’, respectively (‘entered’ and ‘going to enter’). Represent each word and its meaning in two ways using the conventions described here – the “whole word” conventions and the “word structure” conventions.



Here’s our answer

In the “whole word” column, there is no internal structure shown in the Hopi words and individual meanings in the glosses are separated by “.” to show that we make no claims about how the meanings align to morphemes.

whole word	word structure
[pakima] ‘enter.COMPLETIVE’	[paki-ma] ‘enter-COMPLETIVE’
[pakito] ‘enter.PREGRESSIVE’	[paki-to] ‘enter-PREGRESSIVE’

In the “word structure” column, we separate the root and the affix with a hyphen “-”, both in the phonological string and in the meaning. The meanings are sequenced so meaning order matches the order of morphemes in the word.



Exploring my language

Take the two words that you identified when exploring your language on p. 152 and represent each in terms of the “whole word” conventions and the “word structure” conventions.

Comparing the two representations can start a discussion about what information to include in a dictionary.

3.3.4 Summary

We have spent more space in this section on conventions than we have on subtractive analysis. That is because the conventions require a lot more explanation. However, if your goal is to better understand a language, subtractive analysis is truly critical.

Subtractive analysis allows us to identify morphemes and their meanings in complex words. The result of subtractive analysis allows us to make statements about how morphemes combine to make words. Once we have an analysis, we can test the statements to determine their validity by looking at more words. When such statements are valid, they can help in interpreting words that we hear and in creating new words.

With this focus on finding the roots and affixes, it may appear that all words are composed of more than one morpheme. But there are morphemes that are words all by themselves, the free morphemes.

You may be satisfied with what you have learned about word structure at this point, and decide to jump ahead to the discussion in Section 3.7. On the other hand, you may want a little more foundation in morphological analysis: We turn next to how to test an analysis. But first we pause to identify morphemes in your language as root, prefix, or suffix.



Exploring my language

You have already identified several morphemes in your language, in the previous Explorations. Now label each of the morphemes that you found as root, prefix, or suffix.

This kind of morpheme identification is often used in a grammar sketch as well as in structuring lesson plans and in brainstorming ideas for organizing dictionary entries.

3.4 Testing our morphological analysis

This section shows how to determine whether a pattern you have identified with one pair of words extends to other words in the language.

If the pattern that we identified by subtractive analysis is a general pattern in Zenzontepec Chatino, we should find more verbs with the same pattern. (If we can't find additional verbs that behave the same way, then we need to look for other morphologically-related word pairs to figure out what is going on.)

To test the analysis, we look for other verbs that start with *nde-* [nde-] and have PROGRESSIVE meaning, and we look for verbs that start with *ngu-* [ngu-] and have COMPLETIVE meaning. There are several verbs that fit these criteria; some are shown in (11). (These two prefixes sometimes have a mid tone instead of a low tone — *ndē-* [ndē-] instead of *nde-* [nde-] and *ngū-* [ngū-] instead of *ngu-* [ngu-]. Some of the verbs in (11) use the mid-toned prefix.)

This is the same prefix pattern that we described in (9), but using different verbs: Here's how we can tell. The words in (11) have the prefixes in bold font. Using the glosses and the phonological forms, we can see that the words in (11a) start with the prefix *nde-* [nde-] 'PROGRESSIVE' while the words in (11b) start with the prefix *ngu-* [ngu-] 'COMPLETIVE'.



Transitive and intransitive verbs

Recall that the PROGRESSIVE prefix translates roughly as ‘is ...ing’ and the COMPLETIVE prefix as ‘...ed’, as in [nden¹ǎfẽʔ] ‘is.getting.angry’ and [ŋgun¹ǎfẽʔ] ‘got.angry’, respectively. The abbreviation **ITRN** stands for **IN-TRANSITIVE**. An INTRANSITIVE verb has an actor only: *ndēlákui* [ndēlák^wi] ‘PROGRESSIVE.ITRN.boil’ or ‘(it) is boiling’ — there is no person referred to who is making the water boil. This contrasts with **TRANSITIVE** verbs, which have both an actor and something/someone acted upon — *nguēkūlákuii* [ŋg^wēkulák^wiūʔ] ‘COMPLETIVE.boil.3rd.person.plural’ or ‘they boiled (it)’. The transitive/intransitive contrast is more significant in a discussion of morphology and **syntax** than it is in a discussion of **phonology**, so we do not expand on it here.

(11) *nde*-stem and *ngu*-stem are common in Zenzontepec Chatino

- | | | | |
|----|------------------|--------------------------|-----------------------------------|
| a. | <i>ndēlákui</i> | [ndēlák ^w i] | ‘PROGRESSIVE.ITRN.boil’ |
| | <i>ndēnǎǎ</i> | [ndēnǎǎ] | ‘PROGRESSIVE.get.cleared.(field)’ |
| | <i>ndelakuā</i> | [ndelak ^w ā] | ‘PROGRESSIVE.get.counted’ |
| | <i>ndenyāshē</i> | [nden ¹ ǎfẽʔ] | ‘PROGRESSIVE.get.angry’ |
| b. | <i>ngūlákui</i> | [ŋgūlák ^w i] | ‘COMPLETIVE.ITRN.boil’ |
| | <i>ngūnǎǎ</i> | [ŋgūnǎǎ] | ‘COMPLETIVE.get.cleared.(field)’ |
| | <i>ngulakuā</i> | [ŋgulak ^w ā] | ‘COMPLETIVE.get.counted’ |
| | <i>ngunyāshē</i> | [ŋgun ¹ ǎfẽʔ] | ‘COMPLETIVE.get.angry’ |

Once the prefixes are identified, the stems are much easier to find — do subtractive analysis and strip off the prefixes to reveal the stems. We show this in (12) by putting the prefixes and their translations into gray font. Each pair shows the same root with different prefixes.

- (12) Subtracting *nde-* and *ngu-* in Zenzontepec Chatino words
- | | | | |
|----|------------------|--------------------------|-----------------------------------|
| a. | <i>ndēlákui</i> | [ndēlák ^w i] | ‘PROGRESSIVE.ITRN.boil’ |
| | <i>ngūlákui</i> | [ŋgūlák ^w i] | ‘COMPLETIVE.ITRN.boil’ |
| b. | <i>ndēnāá</i> | [ndēnāá] | ‘PROGRESSIVE.get.cleared.(field)’ |
| | <i>ngūnāá</i> | [ŋgūnāá] | ‘COMPLETIVE.get.cleared.(field)’ |
| c. | <i>ndelakuā</i> | [ndelak ^w ā] | ‘PROGRESSIVE.get.counted’ |
| | <i>ngulakuā</i> | [ŋgulak ^w ā] | ‘COMPLETIVE.get.counted’ |
| d. | <i>ndenyāshē</i> | [nden ^j āfēʔ] | ‘PROGRESSIVE.get.angry’ |
| | <i>ngunyāshē</i> | [ŋgun ^j āfēʔ] | ‘COMPLETIVE.get.angry’ |

Looking closely at each pair in (12) shows similarity of sound and meaning at the end of each word. Together, this tells us that the words in each pair have the same stem — the sounds that are left when prefixes are removed (or grayed out) and the meaning that is left when the prefix translation is removed.

For example, in the pair in (12a), we see *ndēlákui* [ndēlák^wi] and *ngūlákui* [ŋgūlák^wi]. The part that is in black text in both words is the same sound sequence *lákui* [lák^wi]. When we look at the glosses, we see the same meaning in black text when the prefix meaning is subtracted: ‘ITRN.boil’ is common in ‘PROGRESSIVE.ITRN.boil’ and ‘COMPLETIVE.ITRN.boil’.

These words have the sound-meaning relations that we identified in (9): Words starting with *nde-* [nde-] show PROGRESSIVE meaning while those starting with *ngu-* [ngu-] have COMPLETIVE meaning. We conclude that we have successfully identified a pattern that extends beyond the pair of words that we started with. Subtractive analysis doesn’t always work this neatly, but when it does, you can celebrate — you have identified morphemes in your language and sorted out the appropriate pattern.

While searching for words to find out whether the pattern extends in the language, you might come across other words that puzzle you — words that appear to share a pattern with the words you’ve already identified, but that don’t quite fit. You might find only one of the two members of a pair because the forms you have don’t include the other member. Or you might find a form with a different beginning of the word, such as the Zenzontepec Chatino pair *ndīlákui* [ndīl^ják^wi] ‘(ITRN).boils’ and *kīlyákui* [kīl^ják^wi] ‘(ITRN).will.boil’ (compare those prefixes to the prefixes *ndēlákui* [ndēlák^wi] and *ngūlákui* [ŋgūlák^wi] from (11a)). The next section explores what you can do with these kinds of discoveries.



Comprehension Check 5

In the Comprehension Check on p. 151, you identified two Hopi suffixes, and you worked with them in the Comprehension Check on p. 155. Here, we see how form alone can lead us to suspect a morphological relation.

Based on the forms of these Hopi words, which ones form a pair that you suspect might share the same Hopi suffix pattern that you've been working with?

- a. *hawto* [hawto]
- b. *aaku* [ʔaaku]
- c. *hàwqa* [hàwqa]
- d. *hawma* [hawma]

Give a reason for your answer before you look at our answer.



Here's our answer

The pair a/d shares the pattern from p. 151. They both begin with [haw] and one ends with [-to] while the other ends with [-ma], just like [pakito] and [pakima].

You might have puzzled about [hàwqa] because the word starts almost the same way, leading to the suspicion that maybe [-qa] is another suffix of interest. This is something to check on with other words.

The word in b, [ʔaaku], is quite different from the other three. So this one is the least likely to be morphologically related to the others.

In case you are fond of knowing the meanings too, here are the same words with their glosses and their word structure shown. You can see from the meanings that three of these words are related, sharing the meaning 'descend, go down'.

- | | | | |
|----|--------------|----------|--|
| a. | <i>hawto</i> | [haw-to] | ‘be going down’ or ‘descend-
PREGRESSIVE’ |
| b. | <i>aaku</i> | [ʔaaku] | ‘spoon, ladle’ |
| c. | <i>hàwqa</i> | [hàw-qa] | ‘one who has gone down’ or
‘descend- <i>RELATIVIZER</i> ’ |
| d. | <i>hawma</i> | [haw-ma] | ‘to have gone down’ or ‘descend-
COMPLETIVE’ |



Exploring my language

Get the list of morphemes in your language that you used in the exploration on p. 157.

Now you will test your analysis of these morphemes in your language. Find two other similar word pairs in your language, using the affix or affixes that you identified already. Write them down.

Sets like the one you just created, using the same affixes but different roots, are useful in many ways, including when teaching and learning a language and when setting up a dictionary. Sets of morphologically related words are called *paradigms*.

3.5 Having fun with the analysis

Now that we have a morphological analysis that we are pretty sure is on the right track, we can have some fun discovering the language in new ways. We can uncover other affixes when we encounter them, we can identify stems that we haven't learned yet, and we can create new words that we've never heard before. (If you think you've learned enough for your purposes, this is a place where you can skip ahead to the concluding discussion, in Section 3.7.)

3.5.1 Uncovering other affixes

We know two verbal prefixes in Zenzontepec Chatino. We can use that knowledge to identify other verbal prefixes. For example, suppose we know the words *ndēshikā* [ndēʃikā] ‘PRG.choose’ and *ngūshikā* [ŋgūʃikā] ‘CPL.choose’. First, pause for a moment to reflect: Using subtractive morphology, what is the stem for ‘choose’?

If you thought that the stem meaning ‘choose’ is *shikā* [ʃikā], you are correct. And now imagine that you hear someone say *kishikā* [kiʃikā]. From the context, you think they are talking about choosing sometime in the future. And from the sounds, you suspect that this is a verb root that you know, *shikā* [ʃikā] ‘choose’.

You examine the words and see that the beginning is different — one way you might do this is laid out in (13). Using subtractive morphology, you remove *shikā* [ʃikā] ‘choose’, and you are left with the prefix *ki-* [ki-]. Context again can help sort this out — *ki-* [ki-] is used when an action hasn’t started yet. (Campbell 2014 calls this *POTENTIAL*, abbreviated *POT*. It is intuitively translated as ‘will’.)

(13) Subtractive morphology in action

- a. familiar

<i>ndeshikā</i>	[ndēʃikā]	‘PRG.choose’	‘(is choosing)’
<i>ngushikā</i>	[ŋgūʃikā]	‘CPL.choose’	‘(chose)’
- b. discovery

<i>kishikā</i>	[kiʃikā]	‘POT.choose’	‘(will choose)’
----------------	----------	--------------	-----------------

In (14), we add hyphens to the forms from (13) to make the morphological structure clear.

(14) Illustrating the *ki-* prefix

- a. familiar

<i>nde-shikā</i>	[ndēʃikā]	‘PRG-choose’	‘(is choosing)’
<i>ngu-shikā</i>	[ŋguʃikā]	‘CPL-choose’	‘(chose)’
- b. discovery

<i>ki-shikā</i>	[kiʃikā]	‘POT-choose’	‘(will choose)’
-----------------	----------	--------------	-----------------

At this point, if you really were working with Zenzontepec Chatino, you would want to ensure that this is a pattern and not just the only verb that uses *ki-* [ki-] to mark *POTENTIAL*, so you would look for more words that begin with *ki-* [ki-].

In fact, there are a lot of *ki-*initial words with *POTENTIAL* meaning. Some are given in (15) (these words are from Campbell 2016). We also show the *PROGRESSIVE* form of the same verbs, for comparison. (Recall that a *PROGRESSIVE* form

can be glossed as ‘is doing’, e.g. *nde-shikā* [nde-ʃikā] ‘PROGRESSIVE-choose’ or ‘is choosing’, and that a POTENTIAL form can be glossed as ‘will do’, e.g. *ki-shikā* [ki-ʃikā] ‘POTENTIAL-choose’ or ‘will choose’.)

(15) Support for *ki-* POTENTIAL in Zenzontepec Chatino

verb	PROGRESSIVE	POTENTIAL
a. ‘come.off’	<i>nde-sū</i> [nde-sū]	<i>ki-su</i> [ki-su]
b. ‘get.toasted’	<i>nde-ki’i</i> [nde-kiʔi]	<i>ki-ki’i</i> [ki-kiʔi]
c. ‘fall.over’	<i>ndē-sāá</i> [ndē-sāá]	<i>kī-sāá</i> [kī-sāá]
d. ‘get.beaten’	<i>nde-ká’nē</i> [nde-kǎʔnē]	<i>ki-ka’nē</i> [ki-kǎʔnē]
e. ‘get.snapped’	<i>ndē-kityě</i> [ndē-kitěʔ]	<i>ki-kityě</i> [ki-kitěʔ]



Comprehension Check 6

Mosiah Bluecloud, Kickapoo

We have used two Hopi words in several Comprehension Checks in this chapter, the words [pakima] ‘enter.COMPLETIVE’ or ‘entered’ and [pakito] ‘enter.PREGRESSIVE’ or ‘go to enter’. Here are two questions about other Hopi words. Can you answer these questions, even though you don’t know what the Hopi words mean?

- What tells us that *tsáwikma* [tsáwikma] is COMPLETIVE?
- What tells us that *tsáwikto* [tsáwikto] is PREGRESSIVE?
- Bonus question: What is the verb stem?



Here’s our answer

- [-ma] shows that [tsáwikma] is COMPLETIVE: [tsáwik-ma].
- [-to] shows that [tsáwikto] is PREGRESSIVE: [tsáwik-to].
- Bonus: The verb stem in these words is [tsáwik]. You can figure this

out by stripping off the suffixes: [tsáwik-ma], [tsáwik-to]. ([tsáwik] means ‘come undone’, but there is no way to figure that out without a semantic context.)

3.5.2 Identifying other stems

Once you’ve figured out the verbal prefixes *nde-* [nde-], *ngu-* [ngu-] and *ki-* [ki-], you can subtract these from new words that you encounter to discover new verb stems. Suppose, for example, that you’re in a conversation and someone uses the word *ndelítyi*’ [ndelítʰiʔ]. Context tells you that it likely means ‘is.sinking’. And you have the tool to identify the stem for ‘sink’ – subtraction. As illustrated in (16), you can subtract *nde-* [nde-] from the beginning of the word and simultaneously subtract PROGRESSIVE from the meaning of the word. You’re left with a sound string and its meaning, in this case *lítyi*’ [lítʰiʔ] ‘sink’. (The “raised j”, ^ʰ, is a symbol for secondary palatalization. See Chapter 2 Section 2.3.2.5 on secondary palatalization.)

(16) Subtractive morphology in action!

- | | | | | |
|----|-----------------------------|--------------------|-------------|--------------------------------------|
| a. | word | <i>ndelítyi</i> ? | [ndelítʰiʔ] | ‘PROGRESSIVE.sink’
('is sinking') |
| b. | known morpheme | <i>ndelítyi</i> ? | [ndelítʰiʔ] | ‘PROGRESSIVE.sink’ |
| c. | subtract <i>nde-</i> [nde-] | <i>nde-lítyi</i> ? | [ndelítʰiʔ] | ‘PROGRESSIVE-sink’ |
| d. | conclusion | <i>lítyi</i> ? | [lítʰiʔ] | ‘sink’ |



Comprehension Check 7

Try subtractive morphology yourself. Here are two new words from Hopi. What are the stems in these words? Use subtractive analysis. (See the Comprehension Checks on p. 148 and p. 163 to remind yourself of the suffixes for the COMPLETIVE and PROGRESSIVE.)

- | | | | |
|----|-----------------|------------|--|
| a. | <i>kwúrukma</i> | [kʷuʔʉkma] | ‘collapse.in.a.heap.COMPLETIVE’
~ ‘collapsed.in.a.heap’ |
| b. | <i>ùutato</i> | [ʔùùutato] | ‘close.PROGRESSIVE’
~ ‘going.to.close’ |



Here's our answer

Following the model of familiar meanings, we can guess that 'collapsed.in.a.heap' is 'collapse.in.a.heap.COMPLETIVE'. We know that [-ma] is a COMPLETIVE suffix. Subtracting [-ma] from *kwúrukma* [kʷuʔukma] leaves [kʷuʔuk], which we guess means 'collapse.in.a.heap'. (A dictionary search shows we are right.)

For *ùutato* [ʔùutato] 'going.to.close', we know that the 'going.to' meaning, called PREGRESSIVE, is added by the suffix [-to]. The word [ʔùutato] ends with [-to]. We subtract [-to] from [ʔùutato] and are left with [ʔùuta], which we guess means 'close'. Again a dictionary search confirms this analysis.

Note that when you work with your own language, you may not have a dictionary to check your analysis. Your analysis will be a journey of discovery.



Exploring my language

Subtractive analysis can help language students learn how to understand more about the language they are learning. Using the affixes that you have identified in the exploration on p. 152, create a list of four words that students of your language might use to practice subtractive analysis in order to learn how to build their vocabulary.

3.5.3 Creating new words

Once you have started to understand the morphology of a language, you can put affixes together with stems to create likely words to express meanings that you want to share. For example, you already know that *ngūlákui* [ŋgūlákʷi] means that something has already boiled. But what do you say if the soup is still boiling? We do this by adding a known affix to a known stem. From *ngūlákui* [ŋgūlákʷi],

we can identify *lákui* [lák^wi] as the stem meaning ‘boil’. And we know that both *nde-* [nde-] and *ndē-* [ndē-] are forms of the prefix for the PROGRESSIVE. So we can add these two morphemes together and produce *ndē-lákui* [ndēlák^wi] ‘PROGRESSIVE.boil’ to convey that boiling is on-going. (Speakers of Zenzontepec Chatino learn when to use the low-toned form *nde-* [nde-] and when to use the mid-toned form *ndē-* [ndē-], but this is not something that we cover in this book.)

Basically, if we know stems and prefixes, we can subtract one prefix and add another to create new words to suit new situations.

(17) Subtracting and adding prefixes

- a. *change* ‘is getting snapped’ to ‘got snapped’
- | | | | |
|----------------------|------------------|---------------------------|--------------------|
| known | <i>ndē-kītyé</i> | [ndē-kit ⁱ éʔ] | ‘PRG-get.snapped’ |
| subtract <i>ndē-</i> | <i>ndē-kītyé</i> | [ndē-kit ⁱ éʔ] | ‘PRG- get.snapped’ |
| add <i>ngū-</i> | <i>ngū-kītyé</i> | [ngū-kit ⁱ éʔ] | ‘CPL-get.snapped’ |
- b. *change* ‘got beaten’ to ‘is getting beaten’
- | | | | |
|----------------------|------------------|-------------|------------------|
| known | <i>ngu-kā’né</i> | [ngu-káʔnē] | ‘CPL-get.beaten’ |
| subtract <i>ngu-</i> | <i>ngu-kā’né</i> | [ngu-káʔnē] | ‘CPL-get.beaten’ |
| add <i>nde-</i> | <i>nde-kā’né</i> | [nde-káʔnē] | ‘PRG-get.beaten’ |

It is possible that we might use a stem that follows a different pattern from the one we’ve been working with here. For example, suppose we heard *nde-tēē* [ndetēē] ‘is carrying’. Subtraction gives us the stem *tēē* [-tēē] ‘carry’; we might then cheerfully add *ngu-* [ngu-] to create *ngu-tēē* [ngutēē] ‘CPL-carry’. And a more fluent speaker might laugh a little at us because fluent speakers say *ngui-tyēē* [ng^witēē], not *ngu-tēē* [ngutēē]. This can be discouraging, but it is also very important – unexpected forms, that don’t “follow the rules” are forms that teach us more about the intricacies of a language. If you don’t attempt new words, you may never discover other properties of the language, so have courage to explore. We look at some ramifications of unexpected forms in the next section.



Exploring my language

In doing either of these explorations, it may help to review your answers to the explorations that you have already done in this chapter.

If you are a language teacher, create an exercise using a known polymorphemic word from your language with morphemes that you already

know. You can use your questions to discover a new word or to teach someone else how to expand their vocabulary.

If you are wondering about making a dictionary of your language, try creating a dictionary entry for a polymorphemic word, showing the word and its meaning but also the morphemes of the word so that others can use the entry to relate your form to other words in the language.

3.6 Four surprises we can learn from

We look at four kinds of surprises that arise when playing with the morphology of a language. (This is another place where you might choose to jump to the concluding discussion in Section 3.7. You can always come back and read this later if you decide you need the information.)

3.6.1 Patterns aren't necessarily general!

In our discussion of Zenzontepec Chatino, we focused on verbs that use the prefixes *nde-* [nde-] PROGRESSIVE and *ngu-* [ngu-] COMPLETIVE. However, in Zenzontepec Chatino, some verbs use different prefixes for these same meanings. You may have noticed that we slipped in a different COMPLETIVE form in Section 3.5.3 with the word *ngui-tyěě* [ŋg^witⁱěě] ‘CPL-carry’. Looking at more Zenzontepec Chatino forms shows that there are, in fact, several different prefixes to express COMPLETIVE. Three ways are shown here. Each set of examples includes both a PROGRESSIVE form and a COMPLETIVE form. We see that in each set of examples, the PROGRESSIVE always has the prefix *nde-* [nde-] (in the lefthand column). The COMPLETIVE prefix, in the righthand column, is different in each set. In (18), we see the familiar *ngu-*. (In the COMPLETIVE column, only the prefix is in black, for emphasis.)

(18) Examples with *ngu-* COMPLETIVE

PROGRESSIVE			COMPLETIVE		
<i>nde-jnīī</i>	[ndehnīī]	‘is growing up’	<i>ngu-jnīī</i>	[ŋguhnīī]	‘grew up’
<i>nde-tōō</i>	[ndetōō]	‘is standing’	<i>ngu-tōō</i>	[ŋgutōō]	‘stood’

In (19), we see *nde-* [nde-] for the PROGRESSIVE just as we saw in (18). We also see the briefly mentioned *ngui-* [ŋg^wi-] for the COMPLETIVE, not the familiar *ngu-* [ngu-].

(19) Examples with *ngui-* [ŋg^wi] COMPLETIVE

PROGRESSIVE			COMPLETIVE		
<i>nde-tëë</i>	[nde-tëë]	‘is carrying’	<i>ngui-tyëë</i>	[ŋg ^w i-t ^ʰ ëë]	‘carried’
<i>nde-taa</i>	[nde-taa]	‘is finishing’	<i>ngui-tyaa</i>	[ŋg ^w i-t ^ʰ aa]	‘finished’

Finally, in (20), the PROGRESSIVE again uses the prefix *nde-* [nde-]. The COMPLETIVE, however, has a new form, *nga-* [ŋga-].

(20) Examples with *nga-* [ŋga-] COMPLETIVE

PROGRESSIVE			COMPLETIVE		
<i>nde-nëë</i>	[nde-nëë]	‘is demanding’	<i>nga-nëë</i>	[ŋga-nëë]	‘demanded’
<i>ndē-tāá</i>	[ndē-tāá]	‘is giving’	<i>ngā-tāá</i>	[ŋgā-tāá]	‘gave’

In Zenzontepec Chatino, when you learn a verb, you also have to learn which COMPLETIVE prefix is used with that verb. We can say that verbs are divided into groups based on the completive prefixes. Linguists call groups like this *lexical classes*.

The term lexical class relates to the *lexicon*, the term used by linguists for the list of idiosyncratic information about a language’s words and morphemes. If you read about a relation being *lexical* it usually means that it is not a general pattern, but rather has to be memorized (because it is idiosyncratic, a property of one or a few morphemes). (Sometimes, the words *lexicon* and *dictionary* are used synonymously, to mean a list of words of a language. Yes, this is confusing.)

Returning to Zenzontepec Chatino, each lexical class has its particular form of the COMPLETIVE prefix. One lexical class is illustrated in (18). This class uses the COMPLETIVE prefix *ngu-* [ŋgu-]; examples include the verbs *jnīi* [hnīi] ‘grow.up’ and *töö* [töö] ‘stand’. There is another lexical class, illustrated in (19), that uses *ngui-* [ŋg^wi-], including the verbs *tëë* [tëë] ‘carry’ and *taa* [taa] ‘finish’. And there is a third lexical class, in (20), that requires *nga-* [ŋga-]; examples are *nëë* [nëë] ‘demand’ and *tāá* [tāá] ‘give’.

In learning or teaching a language, it can help to identify properties of lexical classes. Sometimes such classes have properties that help unify them — and make them easier to learn; in other cases, we simply have to memorize which words belong in which class. The COMPLETIVE classes in Zenzontepec Chatino must be learned — there is no unifying property. (In Section 3.6.5, we give an example of phonologically-defined lexical classes.)

3.6.2 One type of word can pair with several different affixes: Paradigms

You may have noticed that the Zenzontepec Chatino examples that we have been exploring, for example the words in (18), (19), and (20), show verb roots with different prefixes that change the verb's tense (the timing of when the action takes place).

We use the term *paradigm* to refer to the set of different affixes that can attach to roots of a particular type, usually verbs or nouns; sometimes adjectives and adverbs also have paradigms. We've been looking at part of the Zenzontepec Chatino verb tense paradigm. In Section 3.6.3.1, we see part of the Zenzontepec Chatino "person-number" paradigm (and we explain "person-number" at that time).

We don't examine different types of paradigms in this book — that is a topic for a volume that focuses on morphology. However, the concept is useful here: When trying to understand the sound patterns in words of a language, knowing paradigms of a language means you understand the sequence of sounds and how those sounds relate to meaning for words in each paradigm.



Exploring my language

Does your language have any paradigms?

Your goal is to find at least two forms for the same root — either two words made from the root with different affixes, or two words, one that is the root alone and one that is the root plus an affix. (It is ok to have more than two forms for the same root, but you don't have a paradigm if you have only one form.) A good starting point is the list you used in the exploration on p. 161.

Once you find your pair, think about similar words in your language — can you find another root that fits the same paradigm?

Make a table with the roots in rows and the affixes in columns, similar to the tables in (18)–(20).

The paradigms you find here will be useful in Chapter 9, especially for the language exploration on p. 495.

Note: There are some languages with very few affixes, so do not be discouraged if you don't find any. There may not be many to find. On the other hand, if you don't find any and you are lucky enough to have resources (people, websites, books) to consult about your language, you could consult those resources to see what they think.

3.6.3 There can be more than one affix in a word!

By now, we are quite comfortable with saying that Zenzontepec Chatino verbs have the structure *prefix-stem*, and the prefix tells us something about how the action takes place. Finding a word like *nde'nē* [ndeʔnē], we suspect the first part means PROGRESSIVE, and we are right — this is the word for 'PROGRESSIVE.do' (or 'is doing'), so by subtraction we assume that 'nē [ʔnē] means 'do'. All is well.

But then we encounter more words with the sequence *nde'nē*, but there's more in these words — they surprise us by having something at the *end* of the word after *nde'nē* [ndeʔnē]: *nde'nēē* [ndeʔnēēʔ] and *nde'nēū* [ndeʔnēūʔ].

(21) Something at the end of the word

- a. *nde'nēē* [ndeʔnēēʔ] 'is doing' followed by *ē* [ēʔ]
- b. *nde'nēū* [ndeʔnēūʔ] 'is doing' followed by *ū* [ūʔ]

The final vowel and [ʔ] are new to our investigation. You may remember what the symbols represent. The symbol [ʔ] indicates a glottal stop; see Chapter 2, Section 2.3.2.4 to learn about glottal stops. You may also remember that the tilde over the vowel, as in [ē], tells us that it is a nasal vowel; see Chapter 2, Section 2.4.3.2, if you need a review of nasal vowels. In our investigation, these sounds are unexpected, so we consider subtractive analysis. A first step is knowing the meanings: They are translated as *nde'nēē* [ndeʔnēēʔ] 'I am doing' and *nde'nēū* [ndeʔnēūʔ] 'they are doing'.

(22) Translations of the words in (21)

- a. *nde'nēē* [ndeʔnēēʔ] 'I am doing'
- b. *nde'nēū* [ndeʔnēūʔ] 'they are doing'

Putting these meanings into a technical translation, we see that both involve PROGRESSIVE and 'do', but the **actor** in each case is different — in one case the actor is also the speaker ('I', or '1.SINGULAR') and in the other case the actor is some group of people not being addressed by the speaker ('they', or '3.PLURAL').

Before turning to our next topic, we pause briefly to explain the use of terms like 1.SINGULAR and 3.PLURAL to describe who is performing an action. (The terms **SINGULAR** and **PLURAL** are abbreviated **SG** and **PL** respectively.) These terms help us to express the meanings of a language directly, rather than forcing intuitive English translations. If this is familiar to you, or unnecessary, skip ahead to Section 3.6.3.2.

3.6.3.1 Person and number

When talking about unnamed actors in language, linguists use the following terms, generally referred to as *person* and *number*, rather than giving a translation in some other language (like English, as shown here).

(23) Person/number terms

1st person singular	1.SG	person speaking	‘I’
1st person plural	1.PL	people including speaker	‘we’
2nd person singular	2.SG	person being spoken to	‘you’
2nd person plural	2.PL	people being spoken to	‘you all’
3rd person singular	3.SG	person being spoken of	‘she/he/it’
3rd person plural	3.PL	people being spoken of	‘they’

These are also sometimes written without the dot: 1SG and 1PL, and so on. So 1SG is the same as 1.SG, which is the same as 1.SINGULAR, and so on. Authors do things differently, and are not always consistent.

There is a lot of variation in how languages refer to these different groups. Some languages have different forms depending on the gender of the person being referred to. Some languages differentiate the 1.PLURAL (‘we’) category depending on whether the speaker is including the people being spoken to, called 1.PLURAL.INCLUSIVE, or is not including those people, called 1.PLURAL.EXCLUSIVE. (The terms *INCLUSIVE* and *EXCLUSIVE* are abbreviated **INCL** and **EXCL** respectively.) English does not distinguish INCLUSIVE and EXCLUSIVE with one word — English speakers have to add a bunch of words to clarify. Other languages, including Zenzontepec Chatino, do distinguish ‘we.INCLUSIVE’ from ‘we.EXCLUSIVE’ (or 1.PLURAL.INCLUSIVE vs. 1.PLURAL.EXCLUSIVE). Here’s an example; the two 1.PLURAL suffixes are bolded in (24), following the analysis in Campbell (2014). You can see that they involve different sounds.

- (24) 1.PL.INCLUSIVE vs. 1.PL.EXCLUSIVE in Zenzontepec Chatino
- a. *ngajyãã* [ŋgahjãã] ‘we (incl.) played’ ‘1.INCL.play’
 - b. *ndakuya* [ndakuja] ‘we (excl.) eat’ ‘1.EXCL.eat’

The INCLUSIVE and EXCLUSIVE glosses are sometimes abbreviated as 1.INCL and 1.EXCL as is done here, since the PLURAL meaning is implied by the meaning of INCLUSIVE and EXCLUSIVE. You may also see these without the dot: 1INCL and 1EXCL.

Another difference you may find is that some languages distinguish between pairs and multiples, having a difference between 1.DUAL (*DUAL* may be abbreviated *DU*) and 1.PLURAL (or 1.PL) and/or between 2.DUAL (or 2.DU) and 2.PLURAL (or 2.PL). Zenzontepec Chatino does not have this difference.



Singular, dual, plural in Navajo

Kevan Joe, Navajo

Navajo makes a distinction between singular, dual, and plural: Note how the beginning of each word is different and so the meaning is also different. (Navajo is an Athabaskan language spoken primarily in Arizona and New Mexico, in the United States of America.)

2.SINGULAR, 2.DUAL, 2.PLURAL in Navajo

- a. *’ilhosh* ‘you (singular) are going to sleep’
- b. *’othosh* ‘you two are going to sleep’ (exactly two)
- c. *da’othosh* ‘you all are going to sleep’ (more than two)

Yet another person/number category you might find is ‘4th person’ — a category that is different from 1st, 2nd, and 3rd person. If you believe you are encountering a 4th person category in your language, please carefully examine how it is used. These tend to not have a single definition that works for all languages, unlike 1st.person, 2nd.person, and 3rd.person.



Fourth person in Kickapoo

Mosiah Bluecloud, Kickapoo

Kickapoo is one language with a 4th person category. It is used to differentiate between the main character in a narrative and everyone else in the narrative. Imagine hearing someone talk about how Superman had lunch with Batman and they say “Superman had lunch with Batman. He tried to pay for it but he wouldn’t allow it”. In English we use the word order of the previous sentence to understand that it is Superman who tried to pay, and Batman is the “he” that wouldn’t allow Superman to pay. However, since word order is free in Kickapoo, Kickapoo has created the 4th person category to keep track of who is doing what regardless of the word order of the previous sentence. (In these examples, we use subscripts to help keep track — ‘he_S’ or ‘him_S’ means Superman, and ‘he_B’ or ‘him_B’ means Batman.)

Example 1

Supermana wiipomeea batmanani.

superman-a wiipom-eea batman-ani
superman-ANIMATE.3RD eat.with-3SG→4SG batman-4TH

‘Superman eats with Batman.’

kochitepahamwa seh chaahi aanomekwa.

kochi-tepaham-wa seh chaahi aanom-ekwa
tries-pay.for.it-3RD but ?? gives.refusal-4TH→3RD

‘He_S tried to pay for it but he_B refused him_S.’

Example 2

batmanani wiipomeea Supermana.

batman-ani wiipom-eea superman-a
batman-4TH eat.with-3SG→4SG superman-animate.3RD

‘Superman eats with Batman.’

kochitepahamwa seh chaahi aanomekwa.

kochi-tepaham-wa seh chaahi aanom-ekwa
 tries-pay.for.it-3RD but EMPHASIS gives.refusal-4TH→3RD
 ‘He_S tried to pay for it but he_B refused him_S.’

In the first sentence of example 2 we can switch the word order and still get the same meaning. We still know who is doing what because of the 3RD and 4TH person suffixes.

3.6.3.2 Person and number suffixes in Zenzontepec Chatino

Returning to our words *nde'něē* [ndeʔněēʔ] ‘I am doing’ and *nde'něū* [ndeʔněūʔ] ‘they are doing’, we can use our subtractive strategy to isolate the additional material and determine its meaning.

(25) Subtractive strategy with more than one affix

a.	word	<i>nde'něē</i>	[ndeʔněēʔ]	‘PRG.do.1.SG’
	subtract <i>nde</i>	<i>nde'-něē</i>	[nde-ʔněēʔ]	‘PRG-do.1.SG’
	subtract <i>'ně</i>	<i>nde'-ně-ē</i>	[nde-ʔně-ēʔ]	‘PRG-do-1.SG’
	morphemes	<i>nde'-ně-ē</i>	[nde-ʔně-ēʔ]	‘PRG-do-1.SG’
b.	word	<i>nde'něū</i>	[ndeʔněūʔ]	‘PRG.do.3.PL’
	subtract <i>nde</i>	<i>nde'-něū</i>	[nde-ʔněūʔ]	‘PRG-do.3.PL’
	subtract <i>'ne</i>	<i>nde'-ně-ū</i>	[nde-ʔně-ūʔ]	‘PRG-do-3.PL’
	morphemes	<i>nde'-ně-ū</i>	[nde-ʔně-ūʔ]	‘PRG-do-3.PL’

By this strategy, we’ve isolated two suffixes, the suffix *-ē* [–ēʔ] for 1.SINGULAR and *-ū* [–ūʔ] for 3.PLURAL. We find these same suffixes with the same meaning in other words, like *ngu-juī-ē* [ŋgu-hʔē-ēʔ] ‘CPL-get-1.SG’ (‘I got’) and *nakuē-ūʔ* [nak^wē-ūʔ] ‘said-3.PL’ (‘they said’), which suggests that this analysis is likely to be correct.

This example also shows how starting small, with a pair of words that are composed of a stem and one affix, can give us the basics for a more complex analysis, where words are made up of three – or even more – morphemes. Sometimes, prefixes stack up at the beginning of a word or suffixes stack up at the end of a word; watch for it in your language.



Exploring my language

Can you find words with three (or even more) morphemes in your language?

One way to do this is to take a word that you have already identified as having a root and an affix and try to add another affix to the word. You may have to make a few attempts before you actually find a combination that really makes a word in your language.

Finding such words — and identifying the morphemes in those words — is a necessary step in understanding some of the complexity of your language — an understanding that is useful to anyone working with the words of a language.

3.6.4 There may not be a pattern!

Sometimes you will find forms in a language which simply do not fit into any pattern. In Zenzontepec Chatino, the verb meaning ‘say’ has two forms only; both forms can be used to mean either PROGRESSIVE and COMPLETIVE, (26).

(26) Two verbs that do their own thing

	PROGRESSIVE		COMPLETIVE	
a.	<i>něě</i>	[něě]	<i>něě</i>	[něě] ‘say, tell’
b.	<i>naküě</i>	[nak ^w ě]	<i>naküě</i>	[nak ^w ě] ‘say, tell’

In general, it is the common, frequently used words that behave differently, as is the case with ‘say’, while less commonly used words tend to follow regular patterns.

3.6.5 Subtraction doesn’t always remove segments!

We’ve seen that suffixes can mark person in Zenzontepec Chatino — segments are added at the end of a word, *-ě* [-ěʔ] for 1.SINGULAR and *-ũ* [-ũʔ] for 3.PLURAL. (More familiarity with Zenzontepec Chatino shows that there are other forms that can be used to mark the person and number, for example some verbs use *-ã* [-ã] for 1.SINGULAR. You can also find a form for 3.PLURAL with an oral vowel, *-ū* [-ūʔ].) The 2.SINGULAR marker is different: 2.SINGULAR is indicated by tone.

Linguists use the term *melody* to talk about tone and other features that are used as morphemes.

Compare the two words in (27). As a reminder about the representation of tones, the tones of the three vowels in *ngulíjī* [ŋgulihi] ‘CPL.get.lost’ are low-toned [u] with no marking, high-toned [i] with an acute accent, and mid-toned [ī] with a macron.³

(27) A “tone melody” morpheme

- | | | | | |
|----|-----------------|-----------|---------------------|------------------|
| a. | <i>ngu-líjī</i> | [ŋgulihi] | ‘CPL-get.lost’ | (‘got lost’) |
| b. | <i>ngū-lījī</i> | [ŋgūlihi] | ‘CPL-get.lost.2.SG’ | (‘you got lost’) |

When we compare these words, we see that the segments are identical. This makes it a little tricky to perform subtractive analysis because there are no segments to subtract. But we do see a difference between the words. (We highlight the tonal differences between the two forms by using black vs. gray letters.) Did you notice that *ngu-líjī* [ŋgu-líhi] ‘CPL-get.lost’ has a low-tone...high-tone...mid-tone sequence while we find mid-tones on every vowel in *ngū-lījī* [ŋgu-lihi] ‘CPL-get.lost.2.SG’? We suspect that mid-tone melodies mark 2.SINGULAR. Finding more verbs for 2.SINGULAR supports this suspicion.

(28) Mid-toned 2.SINGULAR verbs in Zenzontepec Chatino

- | | | | | |
|----|------------------|-------------|------------------|-----------------|
| a. | <i>ngā-tūsú?</i> | [ŋgā-tūsú?] | ‘CPL-grab’ | (‘grabbed’) |
| | <i>ngā-tūsū?</i> | [ŋgā-tūsū?] | ‘CPL-grab.2.SG’ | (‘you grabbed’) |
| b. | <i>ngū-tú?u</i> | [ŋgū-tú?u] | ‘CPL-leave’ | (‘left’) |
| | <i>ngū-tū?ū</i> | [ŋgū-tū?ū] | ‘CPL-leave.2.SG’ | (‘you left’) |

The 2.SINGULAR tonal morpheme results in mid tones on every vowel of the word, with no other changes to the segments in the word.

This example shows that Zenzontepec Chatino has a tone melody to indicate 2.SINGULAR. It is very useful to have the dot “.” notation for representing the meanings of words that include tonal morphemes — *ngātūsū?* [ŋgātūsū?] ‘CPL-grab.2.SG’ (‘you grabbed’) — since the tone is neither a prefix nor a suffix and so cannot be separated in the gloss by a hyphen.

Other features can be used in a melody as well — for example, if the vowels in a word appear with nasality to mark a particular actor, we might talk of that morpheme having a *nasal melody*.

Whether focusing on tones, nasality, or some other property, trying to pronounce the words aloud can really help you to pay very close attention to sound

³Recall the way that tones are marked in the orthography from the box on p. 136. See also Chapter 2 Section 2.4.3.5 on tones.

differences coupled with meaning differences — which is critical in understanding the morphology of a language.

3.6.6 Summary

In this section, we identified a few ways that morphemes can be surprising. There are many more ways that morphemes can surprise us. But in every case finding pairs of words with both same and different parts and isolating both the sameness and the difference gives us insight to test with subtractive analysis.



Comprehension Check 8

Diana Archangeli & Nelson Santiago

Suppose you learn a new Hopi word, *na'uywupma* [naʔuywupma], which means someone 'climbed all the way up secretly' or they 'sneaked right up'. What would you say if someone were 'getting ready to sneak up'? (It may help to review Section 3.3.1 and the Comprehension Checks on p. 148 and p. 151 before you start on this exercise.)

Follow these steps to find out.

1. The known word is [naʔuywupma] 'sneaked right up'. The target word means 'getting ready to sneak up'.
2. What is the Hopi suffix meaning that an action is completed (the COMPLETIVE)? _____
3. What are the sounds of the two morphemes? [naʔuywupma] = [_____ + _____]
4. What is the morpheme that means 'sneak up'? _____
5. What is added to a verb to make the word mean 'is about to happen' (that is, the PREGRESSIVE suffix)? _____
6. Add this to the 'sneak up' part of the word: [_____ + _____]
7. This gives us [_____] 'is about to sneak up'

Once you understand the answer to this Comprehension Check, you will be in good position when you get to Chapter 9, on *alternations*. We use Hopi as our main example in that chapter, and the Comprehension Checks in this chapter have already introduced you to some of the morphology

of that language. You may want to keep track of your answers from this chapter so you are ready to use them when you get to Chapter 9.



Here's our answer

The short answer is that the word you want is [naʔuywupto]. Here are the answers to each question that show how to figure this out.

1. The known word is [naʔuywupma] ‘sneaked right up’. This is a completed action, so it is likely to have the COMPLETIVE suffix. The target word means ‘getting ready to sneak up’, an action which hasn’t taken place yet but is about to take place — which is a PREGRESSIVE meaning. So the target word needs a PREGRESSIVE suffix.
2. [-ma] means COMPLETIVE.
3. The sounds of the two morphemes are [naʔuywup] and [ma].
4. [naʔuywup] means ‘sneak up’.
5. [-to] means PREGRESSIVE.
6. Add [naʔuywup] ‘sneak up’ to [-to] PREGRESSIVE morpheme: [naʔuywup] + [to].
7. This gives us [naʔuywupto] ‘is about to sneak up’.

3.7 Discussion

In this chapter, we showed how words can be made up of subparts, called morphemes. We looked at how to identify those morphemes by using subtractive analysis in pairs of words that are related in both sound and meaning.

With that useful tool, we then played with Zenzontepec Chatino — finding new stems and new affixes, making new words, finding patterns that apply to some words but are not shared throughout the language, as well as noting words that don’t fit any pattern at all. We also discovered that words don’t have to have any affixes at all, and that affixes don’t have to be segments.

Have we learned everything of importance about morphology? And how can we put what we know to good use?

3.7.1 Resolving the puzzle

We started the chapter with a puzzling question; after reading this chapter, we hope you will have a good idea about the answer.

How do some words help us to understand other words?

As we have shown in this chapter, words can be made up of smaller parts, morphemes. When two words make use of the same morpheme, their meanings are related. Each morpheme in a word contributes to its meaning. Understanding the morphological structure of one word gives us insight into the meaning of other words, if we can identify related morphemes in that other word.

3.7.2 There's more to morphology!

Morphology examines the internal structure of words. Since words are made of sounds or signs, there is an interface between phonology and morphology. In this chapter, we have explored adding prefixes and suffixes to words (*prefixation* and *suffixation*), using the strategy of finding word pairs and carrying out subtractive analysis. The critical word pairs have similarities and differences in both sound and meaning, indicating a potential morphological relation. Subtractive analysis involves removing the part that is the same in both words and leaving the parts that are different, in order to find morphemes within the word — the sound-meaning pairing of a subpart of the word.

There are other morphological patterns, including adding a *circumfix*, where a prefix and a suffix must co-occur surrounding a root, an *infix*, where an affix is found inside of another morpheme, and *zero affixation*, where there is a meaning difference but no phonological change. This gives only a small taste of the variety we find in word structure, both in the sounds and in the meanings. And not all morphology involves some kind of affix. For example, a *compound* is formed when two words are put together to make a new word.

Since this book is focused on the sounds of language, we do not go further into morphological structure — we just add a warning that there is much more to the morphological story than has been sketched here.

There is clearly a phonological component to morphological analysis — subtractive analysis involves finding the sound similarities and differences that identify different morphemes within a word pair. In some cases, sound sequences that really look like they are the same morpheme are not completely identical. This

is called an *alternation*; alternations are the topic of Chapter 9. The chapters between this one and Chapter 9 provide information critical for understanding that chapter.



Compounds in Highland Mayan Sign Language

Sybil Vachaudéz and Samantha Prins

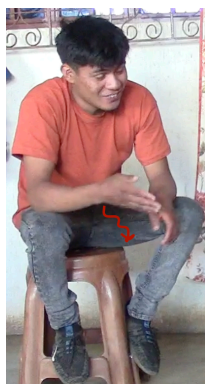
We've just mentioned some of the different ways languages combine existing words to form new ones via compounding. Sign languages also have **compounds** in their word-formation toolkit. In Sign languages, compounds can be sequential or simultaneous.

In sequential compounds, the component signs are produced one after the other. We find sequential compounds in Highland Mayan Sign Language (HMSL, used by Maya communities in the Guatemalan highlands), where the word 'whale' is a compound made up of the signs for FISH and JUMP, signed one after the other.

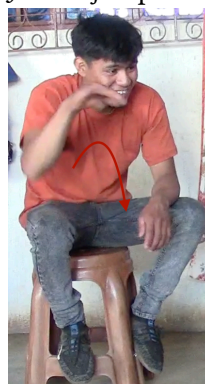
'Whale' in HMSL

Kaqchikel signer recorded by Samantha Prins and Sybil Vachaudéz

FISH 'fish'



JUMP 'jump'

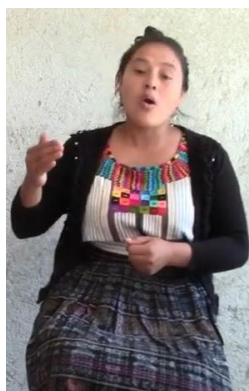


FISH+JUMP
'whale'

HMSL also makes use of simultaneous compounds. Simultaneous compounds are formed by using multiple articulators (e.g., left hand, right hand, face, head, etc.) to produce different pieces of the word at the same time. For example, ‘dog’ is a compound consisting of the classifier sign for MAMMAL (produced with the right hand) and a non-manual ‘barking’ demonstration (produced with the mouth), produced simultaneously.

‘Dog’ in HMSL

Kaqchikel signer recorded by Samantha Prins and Sybil Vachaudes



bark

CL:MAMMAL

‘dog’

While sign and spoken language share many things in common, there are some interesting differences that arise from their unique modalities (visual-spatial vs. audio-oral). Simultaneous compounds highlight one such difference. In Sign languages, using multiple articulators simultaneously is a bit like playing a chord on an instrument. In spoken languages, because only one speech apparatus (the vocal tract) is available, notes are played one at a time.

It’s important to note that in both kinds of compounds, component signs are often reduced (simplified in terms of [movement](#), [handshape](#), etc.) when compared to their independent, non-compounded counter-

parts. This doesn't necessarily change the meaning of the signs, but it can make it easier to articulate them quickly. Knowing about the different strategies for forming compounds, as well as the phonological variation that can accompany them, can help to identify them in your own Sign language, or even to create new ones when signing yourself.

3.7.3 Using my hard-earned knowledge

With this chapter under your belt, you know about morphemes — the sound-meaning pairs that together make up some of our words. And you have developed the skills to isolate morphemes in a language, identifying what their sounds are, what their meanings are, and what order they follow in a word. You've seen how this can extend a person's use of a language, both in discovering stems and affixes that were previously unknown and in using known forms in more contexts. Are there other ways that these skills can help a language worker?

Language learning. Teachers and students alike can make great use of understanding the internal structure of words. The more a teacher understands about the morphological complexity of a language, the more that teacher is able to prepare lessons that help students build from simple to more complex morphological patterns. A teacher can also create drills and conversation structures around different types of morphological patterns to help students become fluent in the whole pattern, not just in the items used in the classroom.

As students get a better grasp of the morphology of the language they are learning, they open themselves to all of the discovery we undertook in this chapter. They learn to identify new stems and new affixes, and are able to use stems and affixes in new (for the students) constructions. This would allow students to increase their ability to communicate fluently. Morphology is an amazing tool for increasing the number of words that students can use and understand.

Language resources — dictionaries, grammars, orthographies. In developing a dictionary, communities make decisions about how to represent words that are morphologically complex — or whether to include them at all. In some cases, the decision might be to include all the words. For Zenzontepec Chatino, this would mean including both *ndetiyaa* [ndetija:] 'PRG.arrive' and *ngutiyaa* [ŋgutija:] 'CPL.arrive', etc. Another decision might be to list stems without prefixes (*tiyaa* [tija:] 'arrive', but not *ndetiyaa* [ndetija:] or *ngutiyaa* [ŋgutija:]); a third choice would be to list stems and put the prefixes as part of a sub-entry — such

as *tiyaa* [tija:] ‘arrive’; *nte-* [nde-], *nku-* [ŋgu-] — with a comment about how to use the prefixes in the preamble to the dictionary. The last is a strategy that is useful when a language has **lexical classes** with different morphology, as we have seen in Zenzontepec Chatino — the dictionary entry shows which class a verb belongs to by listing the prefixes to use. There are other imaginable decisions one might make, depending on what is most useful for speakers of the language. The better the morphology is understood, the more thought can go into these decisions.

Grammars can be structured to start with simple morphology and build to more complex morphology, as we did in our discovery process in this chapter. There is nothing absolute about these decisions, though. For example, in some languages the very same person markers are used to mark the actor with a verb and to indicate ownership of some object. When constructing a grammar or dictionary, the team must decide how to represent such “dual function” morphemes. The choices will depend on what works for the intended audience.

Language documentation and description. A good language description is not simply a list of observations about a language. The most useful descriptions explain and clarify the structure of the language, including the structure of words (the language’s morphology). A good description of the morphology of a language is not simply a list of the morphemes and how they attach to words — it is organized in a way that makes the structure of words accessible to readers. The strategies in this chapter can be used to identify morphemes in a language that form the foundation of a clear and concise description.

Language revitalization. Morphology provides several means of growing the vocabulary, which can be critical in revitalization efforts. As noted above, understanding the basic morphology of a language expands a learner’s vocabulary and fluency dramatically: A learner no longer needs to learn every single word, but can learn a root morpheme and then use morphology to create additional words to fit the desired meaning. Conversely, understanding morphology also adds significantly to a learner’s comprehension, with each part of a word giving the learner a clue about the meaning of the whole. Such rapid vocabulary expansion can give a learner a needed lift in the somewhat daunting task of trying to revitalize a language.

In cases where earlier attempts at description and documentation are incomplete, a grasp of morphology boosts revitalization efforts: Once the sounds and meaning of a root are identified, natural-sounding words can be hypothesized by adding prefixes and suffixes to fill in gaps when the records fall short. Where a

root is missing, one might be hypothesized (perhaps based on roots in a related language or dialect; see Chapter 4) and affixes then added appropriately.

Sometimes records include apparent monomorphemic words. Once the morphology is understood, a researcher might find that an apparent monomorphemic word actually has a prefix or suffix that is familiar, allowing the researcher to suggest a previously unidentified root — which can then be affixed to create new words, again growing the vocabulary.

Finally, a neologism (a word that is created to fill a need, for example is the historical records fall short) sounds more natural in the language when we know enough to add the appropriate affixes.

Reading linguists' resources about your language. Now that you understand three-line glosses and the Leipzig Glossing Rules (Comrie et al. 2015; <https://www.eva.mpg.de/lingua/pdf/Glossing-Rules.pdf>), you may find them extremely helpful. Fortunately, they are used by many (but not all) authors. In general, the earlier a work was created, the less likely it is to follow these conventions. However, even the earlier authors make use of the concepts in this chapter — and often assume that the reader has an understanding of roots, stems, and affixes, and how to identify them. If you come across something a linguist has written about your language, watch for whether the author has their own convention for marking morpheme boundaries within words. You can even build your own three-line glosses, translating the earlier work to make it more accessible going forward.



Glottolog

The Glottolog website, <https://glottolog.org/> is a well-maintained resource for finding materials written about many of the world's languages. Check it for your language.

3.8 Conclusion

Roots, stems, and affixes figure prominently in many language descriptions: You now have an understanding of these concepts and can use them in talking about your language.

You know that morphologically related words have sound and meaning differences due to different morphemes in each, but also have similarity of sound and meaning — a shared morpheme. And you know how to identify such word pairs.

You also know how to perform subtractive analysis to determine whether there is a systematic prefix-root or root-suffix relation to identify in your word pair. You might even find a circumfix-root-circumfix in your language, or a root-and-infix pattern. And you know the steps to take to verify whether a pattern you have found is pervasive in the language.

Finally, with the ideas in Section 3.7.3 as inspiration, you are ready to use this knowledge as you work with your language.

Chapter 4

Variation: Different but the same

Diana Archangeli

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Here's a puzzle. Mamaindê and Negarotê are both spoken in southern Amazonia in the Mato Grosso state of Brazil. They are both Northern Nambiquaran languages, and for the most part, speakers of one language understand speakers of the other. Suppose that two Northern Nambiquara speakers meet and, in the course of their conversation, one speaker uses the word [hiuriru] 'tree'. The other might conclude "Aha – a person from Negarotê land".

Or maybe that conversation took place further to the east, in the state of Tocantins, between speakers of the Macro-Je language Karajá. Someone nearby overhears one of the speakers say [ɔwɔɾɔ] 'tree', and immediately decides "that's a man".

How does a single word tell you about home town or gender?

The answer is that a specific sequence of sounds can do more than convey the meaning of a word – sounds may differ depending on where a person is from, their gender, their age, and other factors. In the Northern Nambiquaran languages, Negarotê speakers use a **diphthong** [iu] where Mamaindê speakers use a long vowel [i:], so Negarotê [hiuriru] and Mamaindê [hi:riru] both mean 'tree'. In Karajá, women and men pronounce some words differently. One difference is that women use the sound [k]; men rarely do: To mean 'tree', a woman might say either [kɔwɔɾɔ] or [ɔwɔɾɔ], while a man says only [ɔwɔɾɔ] and never [kɔwɔɾɔ]. Here and elsewhere, key sounds are in black text to make them more

obvious, and the other sounds are in gray text, to reduce them as a distraction from the point being made.¹

The examples from the Northern Nambiquaran languages and from Karajá illustrate that language forms can be different without having a difference in meaning — the differences provide social information, not information about meaning. The general phenomenon is called **variation**. Variation occurs in all aspects of language — forms (sounds or signs) differ, words differ, sentences differ. Our focus is on variation in form.

Variation is the first major component of a language's unique **phonology**. In this chapter, we explore three types of variation — dialectal, idiolectal, and free. We discover the beauty of variety within language and learn the tools used to determine the phonological nature of each example.

Here's our promise. After reading this chapter, you will be able to identify and analyze language variation when you come across it, determine whether it is a general pattern and if so, determine the properties. These skills are valuable — we use the same tools in Chapters 6–9. Additionally, language workers can use their analysis to better explain variation in their language to students and other community members and to inform decisions about ways to present their language in written form.

Here's our plan. We have three goals with this chapter. The first is to show that variation is everywhere and that it takes many different forms. Next is to explain analytic tools for understanding patterns of variation in the sounds of a language. The last is to demonstrate that the analytic skills introduced here can be used to identify the phonological and phonetic properties of variation, regardless of the type of variation. These are very useful skills — we use the same basic steps throughout the book to understand other types of sound patterns.

Here are more details about each section to help you choose which ones are most important to you right now.

1. We begin with **dialectal variation**, in Section 4.2. Dialectal variation distinguishes groups within a language community: Distinct groups speak different **dialects**. This section introduces the basic analytic steps for understanding a phonological pattern, making an explicit statement about

¹The example from the Northern Nambiquaran languages is from Eberhard (2009: 123); we thank Dave Eberhard for checking and improving our discussion of Northern Nambiquara. The example from Karajá is from Ribeiro (2012: 131); we thank Eduardo Ribeiro for his help improving this example. Neither source includes a practical orthography (that is, a writing system used by a community for reading and writing), so we give forms only in **International Phonetic Alphabet** (IPA) symbols, showing their pronunciation. See Appendix A for descriptions of IPA symbols, as well as Chapter 2 for full discussion.

that pattern, and testing that statement to make sure you've got everything right.

The skills introduced in Section 4.2 are invoked throughout the remaining chapters in this book. We encourage you to read through the end of Section 4.2, if you want to understand the analytic methods used to determine phonological patterns, whether they are the patterns of dialectal variation, alternation, or something else.

2. We next turn to **idiolectal variation**, in Section 4.3. Idiolectal variation distinguishes individuals from each other; we each have our own **idiolect**.
3. The final type of variation is **free variation**, in Section 4.4. There may be two different productions for the same form, used pretty much interchangeably. Free variation may mark a dialect or an idiolect.
4. The chapter concludes in Section 4.5. We expand on our answer to the chapter puzzle, including a summary of the analytic steps in Figure 2. We also answer the question of where the many varieties come from, and we explore several different ways that knowledge about language variation is valuable for language workers.



How speakers refer to Mamaindê and Negarotê

Speakers of the language known as Mamaindê call themselves [mamãintʃiru]. If they refer to their language (which is not that often) they might say either [mamãintʃiru nahaiʔgiru] ‘the Mamaindê, their speech’ or simply and more commonly [nūsa haiʔgiru] ‘our speech’. The name Mamaindê is used by outsiders for both the people and their language; it is a pronunciation based on the word [mamãintʃiru].

Similarly, the language known by outsiders as Negarotê gets its name from the term [nagadoʔtʰu], which refers to the ethnic group. Reference to the language is either [nagadoʔtʰu nahaiʔgiru] ‘the Negarotê, their speech’, or the more commonly used [nūsa haiʔgiru] ‘our speech’.

This information was given to us by Dave Eberhard.

4.1 Introduction: What is variation?

Variation in a linguistic context simply refers to different ways of producing language to mean the same thing. While differences can be found in the words used and in sentence structures, in this chapter we focus on differences in the sounds themselves. Variation shows up both within the speech of a community of speakers and in the speech of a single speaker.

Variation (more specifically *language variation*) is a characteristic of human language. It describes the situation in which multiple different forms express the same meaning in different contexts according to factors outside of language directly, such as geography, social class, gender, etc.

Variety is a general term for any distinctive form of a language or linguistic expression. For instance, a **dialect** is any variety of a language spoken by a group of people that is different from other varieties of the same language. An **idiolect** is the variety of a language that is spoken by an individual.

A **variant** is one of many different ways to say the same word. It can be a single sound, or a word, and so on. For example, in Yauyos Quechua (Section 4.2.2), there are two liquid sounds, [l] and [r], that are dialectal variants. Because [l] and [r] are variants in this language, the forms [walmi] and [warmi] are variants of the same word, meaning ‘woman’.



Variation in different O’otham communities

Angela Hughes, O’otham

O’otham is a Uto-Aztecan language.

“I was born and raised on the Gila River Indian Community (GRIC) in the Village of Santa Cruz. Our village is at the foothills of Sierra Estrella Mountains approximately fifteen miles south of Phoenix. For as far back as I can recall I have heard my family refer to us, our relatives, friends and any other persons who came to visit us from many surrounding O’otham areas as “O’otham”. I remember relatives visiting or people coming to trade with my father. My father was a farmer and always had dried goods, hay, seasonal vegetables, desert harvest goods, and also livestock to sell or trade. When the visitors would leave he would mention who they were and where they came from. When these visits occurred I learned that there are O’otham but from a specific area and location. Hughes (2016: 8)

The speech may differ from district to district or from village to village and in some cases neighbor to neighbor.

This came to my attention at a very early age when I heard my father talk about the sound of our neighbors' pronunciations. My father called it the *O'oks chuchul ñeok* — the 'old hens' talk. Our neighbors' talk is with sounds coming from a rounded mouth form with tongue resting making a nasal 'pohk, pohk' similar to the sound of hens clucking. I heard the speech of relatives who came to visit from different village on Gila River, all with their own distinct sounds such as elongated words or speech that was very slow... As time progressed I heard other variations, from the (S-)Onk, Ak-chin and Tohono O'otham tribes.

The variations of the way we speak are prominent in the O'otham language but there is also different ways of saying words. The words below are examples of certain words are pronounced in my village and how they are said... When I first heard the words as a child, I thought that the person was saying these words incorrectly.

Kwiva Ki:kam	other (village)	
chipk	chikp	'work'
hekapsh	hekachp	'quickly'
vevkam	vevgam or vevam	'seven'

As time evolved I heard the variations from the Tribes of (S-)Onk, Ak-chin and Tohono O'otham. The Tohono O'odham ñeok is considerably difficult to understand because of differences in phonetics. Sometimes the words were completely different from Akimel O'otham especially with Ak Chin and Tohono O'odham."

Hughes (2016: 19-20)

4.2 Dialectal variation

There are times when we meet a person and, from the moment the conversation begins, we have a good idea about some aspect of that person's life — perhaps where they grew up or something about their social community, based simply on the way they produce their language. We know the same language, yet there are differences, and those differences identify each of us as a member of a particular community. These differences stem from *dialectal variation*, showing that we

come from different communities despite knowing “the same” language. We say that these groups speak different *dialects* of the same language.

Dialect membership is reflected in our choice of words and grammatical structures, and it is also reflected in the way certain words are produced.

Non-linguistic use of the word *dialect* can lead to confusion. For example, to emphasize shared history or shared culture between two communities, we may talk about the languages as “dialects of one language” even though the people cannot understand each other’s speech. On the other hand, to emphasize differences between communities, we may say that the communities “speak different languages”, even though there is no difficulty in understanding each other’s speech.

The term dialect has also been used to refer disparagingly to some people’s language, as if the language that they speak is not really a language.

In this work, we use *dialect* as linguists use the term, with no negative connotations — meaning simply the collection of words and grammar rules shared by a group of speakers of a particular language community. Given this definition, everyone who speaks a language also speaks a dialect.

4.2.1 ‘Standard language’

When people talk about dialects, they may say something like “those people speak a dialect of the standard language”. Typically, a language has different varieties; sometimes one of these is thought of as the “standard language”. This raises the question — what is a *standard language*?

We start answering this by addressing what the standard language is not. It is not “the most linguistically advanced form of the language”. Each variety of a language has linguistic legitimacy — viewing the standard language as the one and only “legitimate” form of the language is a social or political statement, not a linguistic evaluation. “Non-standard variants” are not linguistically defective; they simply follow different rules.

Functionally, in many societies there is a variant of the language that is taught in community schools and to non-native speakers in language classes. This variant may be called the “standard language”.

Dialectally, with many languages there is a regional variety that is thought of as the prototype for that language, the “standard language”.

Politically, the variety spoken by the more privileged and powerful groups or communities is viewed as the “standard language”.

Intentionally, the “standard language” can be the result of deliberate language engineering, often carried out by a governmental organization. The process is

known as *standardization*. Standardization supports language revitalization by enabling the development of shared written materials, such as books and dictionaries, rather than requiring separate resources for each linguistic variety. It also encourages the use of a consistent written form for distributing information in the language.

As an example of standardization, Dirección Nacional de Educación Indígena Intercultural Bilingüe (DINEIB, the National Directorate of Bilingual Indigenous Intercultural Education) in Ecuador is responsible for standardizing Ecuadorian Quechua. The language planners identify words of Spanish origin that are used in Quechua and replace them with a word from some regional Quechua variety or with a new Quechua-like word (a *neologism*). (A word in one language that originated in another language is called a *loanword*. The act of doing this is called *borrowing*, so you will also see loan words referred to as borrowed words.) The goal is to facilitate the development of Ecuadorian Quechua and promote bilingual education for Indigenous people. However, standardized Quechua differs from the language used by many Quechua speakers on an everyday basis (King & Haboud (2007)) because each speaker speaks the dialect of their community.



Standardization and variation in Otomi

Gabriela De la Cruz Sánchez, Otomi

Standardization can influence linguistic attitudes and impact strategies aimed at revitalizing and promoting language use. That's the case of Otomi, an Indigenous language spoken across eight states in Mexico, with multiple regional varieties. These varieties differ at all linguistic levels, and some are better documented than others.

A few years ago, Otomi-language textbooks were distributed in Indigenous primary schools in the State of Mexico. Teachers—either speakers of Otomi or Indigenous educators—were assigned to promote the language in these schools. However, the program was eventually discontinued, largely because students and teachers struggled with dialectal differences. In many cases, the materials were not understandable, and the teachers were unprepared to address dialectal differences. As a result, efforts to promote the language and culture often devolved into singing

songs that didn't always resonate with local elders, or performing dances in traditional clothing without deeper engagement.

The dispersion of Otomi across varied environments—from arid deserts to humid regions—limits contact among speakers of different varieties. Additionally, the unequal availability of resources among different groups has hindered the understanding of the various language varieties, making the standardization process even more difficult. Discussions with scholars and politicians have proven insufficient without the direct involvement of speakers themselves. To achieve meaningful progress, it is essential to include local surveys, scholars from the region, and genuine community participation to foster a sense of ownership and pride in the language.

In the remainder of this section, we present three examples to illustrate some of the ways that the production of words mark different dialects. We use specific analytic steps to determine the general patterns of variation identifying each dialect. These analytic steps can be used with any language, to clarify our understanding of dialectal differences (and other types of sound differences). Our first example shows variation in the [liquids](#), [l] and [r], in different dialects of Yauyos Quechua, a member of the Quechua family.

4.2.2 Yauyos Quechua [l]~[r]

Suppose you were trying to learn Quechua in the Yauyos province of Peru. You ask one person how to say 'woman' and you're told to say [walmi]. Another person says you should say [warmi]. Even though these words sound different, both are correct: The different ways of saying these words are used by speakers from different parts of the province.



A note on Quechua

Quechua is spoken in many parts of South America. Yauyos Quechua is spoken in the Peruvian Andes. Our source is Shimelman (2017). Shimelman's book describes the language spoken in the southern districts of the Yauyos province (hence the name Yauyos Quechua), which divides into

five dialect groups, Apurí-Madeán-Viñac, Azángaro-Chocos-Huangáscar, Caca-Hongos, Lincha-Tana and Liscay-San Pedro.

Our main discussion contrasts the speech from Caca-Hongos with that from other regions. Shimelman (2017: 28) states “I have chosen an orthography...” — apparently, the writing system found in Shimelman’s work is not community based, but rather is used as a convenience for the author. We convert Shimelman’s system to IPA in accordance with the system Shimelman has set up. We are grateful to Aviva Shimelman for looking over our discussion of Yayos Quechua, and confirming that our conversion is on target. Aviva also told us that speakers will sometimes call their language *shimi* ‘tongue, language’ or *Quechua* or *lengue* when speaking Spanish.

4.2.2.1 Two regional dialects in Yayos Quechua

Speakers from the Caca-Hongos area are distinguished from speakers of the other Yayos Quechua dialects in part because in Caca-Hongos, speakers use the **lateral** [l], where speakers of the other dialects use the **tap** [ɾ], e.g. *kala*: [kala:] vs. *kara*: [kara:] ‘I was’; *luna* [luna] vs. *runa* [runa] ‘person’; the second word in each pair is from the Apurí-Madeán-Viñac dialect.

The sounds that vary between these dialects are [l] and [ɾ]; both are in the **natural class** called **liquids**, introduced in Chapter 2 Section 2.3.1.4 which explains that the two liquids are distinguished from each other by tongue movement.

As is often the case with languages, the [l]~[ɾ] dialect variation is not absolute: There are some words where even speakers of the Caca-Hongos dialect use [ɾ]. Examples include *uyari:chu* [ujari:cu] ‘I don’t hear’, *uyarila*: [ujarila:] ‘I heard it’, and *rigakuq* [rigakoq] ‘those who water/irrigate’. There are also words with [l] in all dialects, such as *laqatu* [laqatu] ‘slug’, *lashta* [laʃta] ‘snow’, and *alpaka* [alpaka] ‘alpaca’. There is no regular pattern: Usually, Caca-Hongos [l] corresponds to [ɾ] in the other dialects, but there are some words where [l] corresponds to [l] in all dialects and others where [ɾ] corresponds to [ɾ] in all dialects.



Writing in Yauyos Quechua

You may remember that we use italics to show the spelling of words, like *warmi* ‘woman’, and we put words in square brackets, like [warmi] to show the pronunciation in symbols from the IPA. Shimelman’s way of writing these Quechua dialects is a *phonetic writing system* because the letter used for writing either the [l] or the [ɾ] reflects the sound that is used by speakers. For the Cakra-Hongos dialect, Shimelman uses the letter *l* in these words, where for the other dialects, she uses the letter *r*. When there is a direct relation between the sound and its written form, the system is called a phonetic writing system.

Phonetic writing systems make it easier for speakers to learn to read and write a language. The cost of this strategy is that writing in one dialect looks different from writing in another dialect: *warmi* ‘woman’ does not look like *walmi* ‘woman’. With phonetic writing systems, as the sound differences between two dialects increase, so too do the differences between the written word in one dialect and its pronunciation in the other.



Comprehension Check 1

Suppose you are listening to two women in Yauyos speaking together. Here are some words that each one says:

one woman says...		the other woman says...		gloss
<i>limpu</i>	[limpu]	<i>limpu</i>	[limpu]	‘all’
<i>paqarin</i>	[paqarin]	<i>paqarin</i>	[paqarin]	‘tomorrow’
<i>kilu</i>	[kilu]	<i>kiru</i>	[kiru]	‘tooth’

Why is it more likely that the first woman is from Cakra-Hongos?



Here's our answer

The first woman is more likely to be from Cacara-Hongos because she says [kilu] while the second woman says [kiru] for 'tooth'. As we have seen, in Cacara-Hongos, [l] typically corresponds to [r] in other dialects.

The other words are all said the same way by both women so they do not give any clues. Remember that [r] is sometimes found in the Cacara-Hongos dialect, even though [l] is often used in the dialect of Cacara-Hongos where [r] is used in other dialects.

4.2.3 Two dialects in Kashibo-Kakataibo

In our next example of dialectal variation, the pattern is quite interesting because there are two sets of variant sounds and the variation is consistent, but it occurs only in the middle of words. The example comes from dialects of Kashibo-Kakataibo, spoken in Peru; Kashibo-Kakataibo is sometimes identified as Cashibo. With Kashibo-Kakataibo, we slowly work through the analytic steps to take when you suspect there may be dialectal variation and you want to be sure that you understand the full pattern.²

4.2.3.1 The Kashibo-Kakataibo puzzle

Imagine yourself in the situation where you hear two people pronounce the same word differently. For instance, one speaker of Kashibo-Kakataibo, might say [pókʊ] while the other says [pógʊ]. You repeat these words aloud (yes, do it!) because you expect people to use the same word to mean the same thing, so you are a little surprised.

²Kashibo-Kakataibo is in the Pano-Tacanan language family, spoken in the Ucayali province in Peru. We used the freely-available copy of the dissertation (Zariquiey 2011) as our source. The dissertation was later published as Zariquiey (2018). We thank Roberto Zariquiey for taking the time to check our understanding of the Kashibo-Kakataibo languages. This checking was very important because some of the IPA forms shown here are created from the practical orthography based on the correspondences shown in Zariquiey (2011); we did this with the words that were not given in IPA in the published works.

Try this. Say the two words aloud, [pókʊ] and [pógʊ]. Listen for the difference; feel the difference as you say the two words.

Saying the words aloud may help you realize that these speakers almost use the same word: [pókʊ] and [pógʊ] differ only by one sound, [k] or [g]. And these two sounds differ only by one feature, [voice]. Otherwise, both are in the sound class of **dorsal stop**: [k] is a **voiceless** dorsal stop and [g] is a **voiced** dorsal stop. (As a reminder, we discussed features in Chapter 2, the glossary has a definition of [voice] as well as of other features, and Appendix B gives a list of segments and their features.)

Imagine that you also know that the person who used [k] comes from Yamina, along the Lower Aguaytía River, and the person who used [g] from Sinchi Roca, along the San Alejandro River.

The meeting of a linguistic difference (sound) and a societal difference (place) raises the question of whether the differences are connected: Is using [k] or [g] an indicator of the community you are from? It also raises the question of whether the choice between [k] and [g] is the only dialectal difference. In this section, we go through the analytic steps that will answer these questions.

4.2.3.2 Is there a dialectal difference?

The [k]~[g] question identifies a *potential* dialectal pattern.

(1) Notice a difference!

location difference	Lower Aguaytía	San Alejandro	gloss
linguistic difference	púku [pókʊ]	púku [pógʊ]	‘belly’

We want to determine whether this is a single odd word, so we ask *Are there more words with the [k]~[g] relation?* We also want to know whether this is limited to one of the individuals we consulted, so we ask *Does the use of [k] or [g] align with the speaker’s home town?* To answer these questions, we talk to more people from each community and we collect more words, marking down the pronunciation of each word and where the speaker is from.

In this case, we do find more words with this pattern. Some of them are listed in (2). The sounds we are focusing on, [k] and [g], are in normal black text, while other symbols are gray.

- (2) Are there more examples? Yes — here are two more

	Lower Aguaytía		San Alejandro		gloss
a.	<i>chúkuma</i>	[tʃukúma]	<i>chúkuma</i>	[tʃugúma]	‘small’
b.	<i>nipákēte</i>	[nipákiti]	<i>nipákēte</i>	[nipágiti]	‘fall, drop’

We keep track in our list of the home town of the speakers, and we find that speakers from Lower Aguaytía use [k] where speakers from San Alejandro use [g]. We conclude that the sound difference and the societal difference align.

From this preliminary work, we see that the use of [k] or [g] appears to be a dialectal difference, and shows which community a speaker is from. We make a preliminary conclusion; two versions are given in (3). The first, in (3a), is stated in terms of the segments involved, [k] and [g]. The second, in (3b), is stated in terms of the features involved, voiceless and voiced dorsal stops. We return to this difference shortly.

- (3) First statement of variation for Lower Aguaytía and San Alejandro plain dorsal stops
- In terms of segments*
San Alejandro [g] corresponds to Lower Aguaytía [k].
 - In terms of features*
Voiced dorsal stops in San Alejandro correspond to voiceless ones in Lower Aguaytía.

4.2.3.3 What is the dialect difference with dorsal stops?

Our next step is to identify the phonetic properties that mark the difference between the dialectal variants. In this case, we ask *What property distinguishes [k] from [g]?*

Reviewing Chapter 2 again, we see that [k] and [g] are both dorsal stops, and that [k] is voiceless while [g] is voiced: The two sounds differ only in voicing.

At this point, we no longer simply suspect that there is a dialectal pattern. We have found more instances of the same pattern and have analyzed the sounds to know that the pattern involves voicing of dorsal stops. We modify our preliminary conclusion: Lower Aguaytía voiceless dorsal stop [k] and San Alejandro voiced dorsal stop [g] correspond to each other. (We could also refer to voiceless and voiced dorsals, instead of voiceless and voiced dorsal stops. Since all dorsals are stops in Kashibo-Kakataibo, both ways of referring to the relevant sounds are correct.)

Our next question is whether [k] and [g] are the only sounds that differ between the two dialects, or whether they are examples of a larger pattern.

4.2.3.4 Is there a larger pattern?

Now that we are pretty sure that we've found a dialectal difference, our next step is to check whether that difference is simply between [k] and [g], or whether the [k]~[g] difference is part of a larger general pattern. Linguistic tools give us a rational way to make this determination.

The goal is to find out whether sounds that are similar to [k] and [g] have the same pattern. To do this, we look at the phonetic properties of [k] and [g]. (This is why Chapter 2 is included.) The back (dorsum) of the tongue makes contact with the velum for [k] and [g], so they are dorsal sounds, due to dorsal constriction at the velum. Complete closure is made with the tongue, stopping the air, so they are stops. Putting both facts together, we can say that [k] and [g] are dorsal stops, characterized by the features [dorsal, -continuant].

These two properties are independent: There can be other dorsal sounds and there can be other stop sounds. To determine whether the dialectal pattern extends beyond [k] and [g], we consider whether it extends to all stops, and whether it extends to all dorsals. Let's look at stops first.

4.2.3.4.1 What about other stops? When collecting words to create lists like (2), speakers are likely to produce words with the other stops in Kashibo-Kakataibo — labial and coronal stops. The words in (4) both have a **labial** stop. Labial stops are made with lip closure (Chapter 2, Section 2.3.2.1); in general, a labial stop can be either voiceless [p] or voiced [b] — is this the case in Kashibo-Kakataibo? Read the words in (4) out loud. Do you hear [p]? Do you hear [b]?

(4) Labial stops: We only hear [p]

		Lower	San	gloss
		Aguatía	Alejandro	
a.	initial	<i>pěchi</i>	[p̠it̪i]	‘wing’
b.	between vowels	<i>chimápu</i>	[t̪imápu]	‘ashes’

We hope you noticed that we only find [p] in these words, whether the sound is at the beginning of the word or in the middle of the word. This remains true even as more words in Kashibo-Kakataibo are collected.

The next set, in (5), illustrates words with coronal stops. These are sounds made with the front (corona) of the tongue with closure near the alveolar ridge

(Chapter 2 Section 2.3.2.2). In general, a coronal stop can be either voiceless [t] or voiced [d].

Try this. Read the words in (5) aloud, listening for both [t] and [d]. Do you hear them both?

(5) Coronal stops: We only hear [t]

		Lower	San	gloss
		Aguaytía	Alejandro	
a.	initial	táë [tái]	[tái]	‘foot’
b.	between vowels	nēte [níti]	[níti]	‘day, daytime’

We hope you noticed that speakers of these two dialects only use the voiceless [t] in these words; they do not use the voiced [d]. As more words in the language are collected, this general statement holds true: Speakers use [t] and don’t use [d].

When we compare the same word as said by a speaker from Lower Aguaytía with the pronunciation by a speaker from San Alejandro, we find that [p] corresponds to [p] and [t] corresponds to [t]. The words are chosen to show that this is true whether the stop is (a) at the beginning of a word or (b) in the middle of a word. We conclude that it doesn’t matter where a speaker of Kashibo-Kakataibo lives: Speakers use [p] and [t], not [b] and [d].

4.2.3.4.2 What about other dorsals? Many languages have only one set of dorsals, the dorsal stops [k], [g]. Some also have dorsal fricatives, [x], [ɣ]. And some have dorsal stops with secondary articulation, such as [k^w] with labial articulation, [kʲ] with palatal articulation, or [kʔ] with glottal articulation. (In Chapter 2 Section 2.3.2.5, you can find more explanation of secondary articulation.)

The Kashibo-Kakataibo language has dorsal stops with a secondary labial articulation — the lips are rounded as the stop is pronounced. Do we find a dialectal difference in voicing, [k^w]~[g^w]?

Again to answer this question, we must find more words. As we can see from the examples in (6), there is a [k^w]~[g^w] difference, just like the [k]~[g] difference that we started with. The words in (6) show that speakers from Lower Aguaytía use [k], [k^w] while speakers from San Alejandro use [g], [g^w] respectively.

- (6) Labialized dorsal stops: [k^w] vs. [g^w]

	Lower	San	gloss
	Aguaytía	Alejandro	
<i>tákwa</i>	[ták ^w a]	[tág ^w a]	‘liver’
<i>rakwēti</i>	[rak ^w iti]	[rag ^w iti]	‘fear (verb)’

Do we need to improve our preliminary conclusion from (3)? Interestingly, if we stated the correspondences in terms of segments (3a), we must change our statement as shown in (7). The original statement listed only [k] and [g]; we need to revise it to include [k^w] and [g^w]; the addition is italicized in (7).

- (7) Second statement of variation for Lower Aquaytía and San Alejandro, in terms of segments

San Alejandro [g] corresponds to Lower Aguaytía [k] *and San Alejandro [g^w] corresponds to Lower Aguaytía [k^w].*

However, if we chose to make our statement in terms of features as in (3b), no change is required. Dorsal stops include the voiceless [k], [k^w] and the voiced [g], [g^w]. We repeat (3b) in (8).

- (8) Second statement of variation for Lower Aquaytía and San Alejandro, in terms of features

Voiced dorsal stops in San Alejandro correspond to voiceless ones in Lower Aguaytía.

This is a major advantage of working with features: Patterns are often expressed more simply than if we list all the segments involved, so no change is necessary when we discover the pattern extends to other segments of the same class.

With all of this discussion, you might think we have finished our analysis but in fact there is one last point to check before deciding that the pattern is fully understood: Does it matter where the dorsal is located in a word?

4.2.3.5 Does the location in the word matter?

The examples showing the [k]~[g] pattern in (2) and showing the [k^w]~[g^w] pattern in (6) only show dorsals in the middle of words (**word-medial** position). Does the same pattern occur at the beginning of words (**word-initial** position)?

We give two figures, one with words that begin with a plain dorsal [k] or [g], (9), and the other with words that begin with a labialized dorsal [k^w] or [g^w], (10).

Try this. The words with initial plain dorsal stops are in (9). Read these aloud, listening for [k] and for [g].

When you read the words aloud, you probably will notice that all words begin with a dorsal stop. Do these these dorsal stops follow our general pattern, voiceless for speakers in Lower Aguaytía and voiced for speakers in San Alejandro?

(9) Word-initial [k] in both dialects

	Lower	San	gloss
	Aguaytía	Alejandro	
<i>káti</i>	[káti]	[káti]	‘say’
(none given)	[kúrunsa]	[kúrunza]	‘yellow’

We hope that you realized that at the beginning of words, you only hear voiceless [k], not voiced [g], regardless of the dialect.

Try this. We have words with initial labialized dorsals in (10). Read the words in (10) out loud, listening for whether the initial sound is voiceless [kʷ] or voiced [gʷ] in each dialect. Do you hear a difference?

(10) Word-initial [kʷ] in both dialects

	Lower	San	gloss
	Aguaytía	Alejandro	
<i>kwáiti</i>	[kʷáiti]	[kʷáiti]	‘play, laugh’
<i>kwěbí</i>	[kʷiβi]	[kʷiwí]	‘mouth’

We hope that you noticed that the words in (10) all begin with voiceless [kʷ]: There is no difference. If you didn’t notice this, read them aloud again.

We can now make our final statement about the correspondence between Lower Aguaytía voiceless dorsals and San Alejandro voiced dorsals — that this correspondence only occurs in the middle of a word, not at the beginning of a word. At the beginning of a word, dorsal stops are voiceless in both dialects. In (11), our final statement of the pattern, we qualify (8) by adding a phrase about word-medial position.

(11) Final statement of variation for Lower Aquaytía and San Alejandro dorsal stops

When word-medial, voiced dorsal stops in San Alejandro correspond to voiceless ones in Lower Aguaytía. (Otherwise, stops are voiceless.)

Notice that this statement tells three things: (i) which sounds are in variation (dorsal stops), (ii) the property that varies (voiceless or voiced), and (iii) the **location** of variation (in the middle of words, not at the beginning of words). These are the three properties of explicit statements about phonological relations (the sounds, the relating property, and the location) – whether the statement is about some kind of variation or about the distribution of sounds within the words of a single dialect (explored in Chapters 6–9).

Interestingly, this is a one-way correspondence, between voiced dorsals in one dialect and voiceless ones in the other. We say this because voiceless dorsals in Lower Aguaytía do not necessarily correspond to voiced dorsals in San Alejandro, for example, *títikati* ‘push’ is pronounced [títikati] in both dialects.



Comprehension Check 2

The first step in identifying a dialectal pattern is noticing a pair of words that mean the same thing but that are pronounced differently in the dialects being studied. Here are some other words from these two dialects.

	Lower	San	gloss
	Aguaytía	Alejandro	
a.	<i>kwěbí</i>	[kʷiβí]	[kʷiwi] ‘mouth’
b.	<i>kamabi</i>	[kámaβi]	[kámawi] ‘all’

There is a difference between these words that does not involve dorsal stops. Can you find it?

Say the words aloud as they are said in each dialect. Be sure you are looking at the IPA transcription (in square brackets), and not the orthography (or writing system, in italics).

What is the difference? Identify the different sounds. Can you also identify the features that distinguish the two sounds? Might there be a special location where this difference is found?

To challenge yourself further, *do you think this difference is likely to be a dialectal sound difference?* Explain your answer.



Here's our answer

The different sounds: Lower Aguaytía [β] corresponds to San Alejandro [w]: [kámaβi] vs. [kámawi] ‘all’ and [k^wiβí] vs. [k^wiwí] ‘mouth’.

The distinguishing property: The sounds [β] and [w] are both labial sounds that are [+continuant] — the air continues to flow through the mouth the entire time. There is more closure with [β] than with [w], so [β] is a [–sonorant] obstruent while [w] is a [+sonorant] sonorant.

Possible special context: The [w]~[β] difference may require a special context: All examples occur in the middle of a word. There are no examples showing the pattern at the beginning of a word. This is something to check as more words are found.

Could this be dialectal: This could be a dialectal difference because the choice of sound, [β] or [w], depends on where the speaker lives. To test this, we would want to find more word pairs with the labial continuants from the two villages, to see whether this is a general pattern or whether these two words are the only ones that show the pattern.

A note of warning. You may have noticed that the practical orthography uses the letter *b* where speakers say either [w] or [β]. The unaware person using only the practical orthography might never realize how these words really sound, using the sound [b] for words in both dialects instead of the correct [w] or [β].



Comprehension Check 3

Bonus question! You now know that velar stops are voiceless at the beginnings of words in both dialects, and usually voiced in the middle of words in the San Alejandro dialect. You also know that the two dialects differ in the pronunciation of the sound that is written *b* — it is [β] in the Lower Aguaytía dialect and [w] for speakers in San Alejandro. How do you imagine speakers of the two dialects might pronounce the name of the language, written *Kashibo-Kakataibo*?



Here's our answer

This question is tricky but it is also kind of fun. The language name *Kashibo-Kakataibo* has a medial *k* – *Kashibo-Kakataibo* – and two medial sounds that are written *b* – *Kashibo-Kakataibo*. The table below shows how these sounds might be pronounced by speakers of the two dialects, resulting in the pronunciation *Kashi[β]o-Ka[k]atai[β]o* in Lower Aguaytía and the pronunciation *Kashi[w]o-Ka[g]atai[w]o* in San Aguaytía.

Two ways to say *Kashibo-Kakataibo*

	Lower Aguaytía	San Alejandro
[k] vs. [g]	Kashibo-Ka[k]ataibo	Kashibo-Ka[g]ataibo
[β] vs. [w]	Kashi[β]o-Kakatai[β]o	Kashi[w]o-Kakatai[w]o
result	Kashi[β]o-Ka[k]atai[β]o	Kashi[w]o-Ka[g]atai[w]o

4.2.4 Doing research vs. presenting research

Before going on to our next example, we will make a comment about the relation between how a pattern is presented in writing and how a pattern is discovered when studying a language.

We've organized our discussion of the two *Kashibo-Kakataibo* dialects to emphasize the individual analytic steps that we use to uncover a pattern. In this case, we established the dialectal correspondence between voiceless [k] and voiced [g] before considering (and rejecting) the possibility of correspondences between all voiceless and voiced stops, testing whether there were [p]~[b] or [t]~[d] correspondences. We then tested whether there was a correspondence with other dorsals, [k^w] and [g^w]. Finally, we looked at the location in the word where the correspondence is found. This led us to find examples with corresponding voiceless dorsals at the beginnings of words, thereby identifying a special word-internal context for the voiced-voiceless dialectal correspondence.

The goal with this organization was to highlight each of the several steps in identifying a pattern:

1. Identify a pair of sounds in correspondence (here the dorsal stops [k] and [g]).

2. Determine whether that pair is an example of a larger class of correspondences (here (i) other stops [p]~[b] and [t]~[d] show no correspondences but (ii) other dorsals [k^w] and [g^w] do show correspondences).
3. Determine whether the location in the word is critical for the correspondence to occur (here word-medial sounds participate in the pattern).

However, it is likely that if you were collecting words trying to identify the properties of this pattern, the speakers you worked with would come up with words with initial as well as medial dorsals, words with labialized as well as plain dorsals, words with other stops — and even words where medial [k] corresponds to medial [k] as in *títikati* [títikati] ‘push’. As words are being collected, it might start to look like there is no pattern at all.

To avoid confusion in such a case it can be very helpful to make separate lists. Rather than grouping all words in a single list, label each list for the correspondence shown in that list. In this case, there would be both a list for the [k]~[g] correspondences, and a list for the [k]~[k] correspondences. There would also be a [t]~[d] list (with no words on it), a [t]~[t] list, etc. In making such lists, we want to keep track of where the speaker is from (our column headings), the sounds being compared, and even the location in the word. A partial list example is shown in (12), giving [k] correspondences with [k] and with [g], and in (13), giving [t] correspondences with [t] and [d]. (If you feel a little shaky on this idea, test yourself by making the column and row headings for lists for the labialized dorsal stops and for the labial stops.)

(12) List structure for linguistic analysis for plain dorsals

			Lower Aguaytía	San Alejandro	gloss
List 1. [k]~[k]	initial	<i>káti</i>	[káti]	[káti]	‘say’
	medial	<i>títikati</i>	[títikati]	[títikati]	‘push’
List 2. [k]~[g]	initial		—	—	
	medial	<i>púku</i>	[púku]	[púgu]	‘belly’

(13) List structure for linguistic analysis for coronals

			Lower Aguaytía	San Alejandro	gloss
List 3. [t]~[t]	initial	<i>tákwa</i>	[ták ^w a]	[tág ^w a]	‘liver’
	medial	<i>káti</i>	[káti]	[káti]	‘say’
List 4. [t]~[d]	initial		—	—	
	medial		—	—	

As the number of words in a list increases, you can examine the lists and compare the words for clues about whether (or how) the surrounding context matters. You can also use these lists to help organize your thoughts if you need to present your conclusions to someone else.

If you understand the phonological side of dialectal variation and you are feeling very confident about your analytic skills — or it is not important to you to have those skills — you might wish to skip the next section and go on to Section 4.2.6. You can test your skills with the Comprehension Check on p. 210.

4.2.5 Zenzontepec Chatino vowel variation

Our last example in this section is included to show that dialectal patterns can also involve vowels. The language we explore is Zenzontepec Chatino, spoken in Oaxaca, Mexico; our discussion relies on Campbell (2014). The way words are pronounced varies depending on the speaker's community.

There are several differences with vowels in this language. We focus on the curious patterns involving the vowels [a] and [e] in the San Pedro del Río and La Aurora regions.³

Try this. Read aloud the words in (14), focusing on these two vowels.

To help you with reading aloud, the accent mark over a vowel indicates high tone, as with *á* and *é*. The macron over a vowel indicates a mid tone, as with *ā* and *ē*. Vowels with no mark are described as having no tone. Vowels with a tilde, such as *ã*, *ẽ*, are nasal vowels.

(14) [a]~[e] — no variation

	San Pedro del Río	La Aurora	gloss
<i>tsashī</i>	[t͡saʃi̯]	[t͡saʃi̯]	'little bit'
<i>skuelā</i>	[skʷelā]	[skʷelā]	'school'

As we see with the highlighted vowels in (14), [a] in one dialect corresponds to [a] in the other dialect and, similarly, [e] in one dialect corresponds to [e] in the other dialect. We find these correspondences in the middle of words and at the ends of words.

There are also some words where San Pedro del Río [a] corresponds to La Aurora [e], highlighted in (15).

³Eric Campbell uses the local orthography instead of IPA transcriptions; we give both the orthographic representation and an IPA representation (in square brackets); in consultation with Eric, we have converted the orthography to IPA to show how words are pronounced.

Try *this*. Read the words aloud. Note that The consonant [g^j] is a palatalized dorsal. Campbell discusses whether [g^j] is better thought of as a sequence [gj], but rejects the idea. This discussion suggests that [g^j] sounds a bit like a dorsal [g] followed by a palatal glide [j].

(15) [a]~[e] – variation

San Pedro del Río	La Aurora	gloss		
<i>ngyati</i> '	[ŋg ^j atiʔ]	<i>ngyeti</i> '	[ŋg ^j etiʔ]	'got untied'
<i>ngyánō</i>	[ŋg ^j ánō]	<i>ngyéno</i>	[ŋg ^j énō]	'remained'

We ask *Is the difference between [a] and [e] a case of dialectal variation?*

We have already identified properties of dialectal variation. First, dialectal variation aligns with a social difference of some sort. Indeed, the sound difference (between [a] and [e]) aligns with a social difference (the speaker's hometown): Where [a] and [e] correspond, speakers from San Pedro del Río use [a] and speakers from La Aurora use [e].

Second, sounds in dialectal variation are similar. We identify the phonetic properties of the vowels. (See Chapter 2 Section 2.4.2 on these phonetic properties.)

- What is the same?

Both [a] and [e] are [–high]; both [a] and [e] are [–round].

- What is different?

[a] is [+low], [e] is [–low]; [a] is [+back], [e] is [–back].

We conclude that the non-high, non-round vowels participate in the variation. No other vowels participate, so we conclude that this difference is limited to [a]~[e] variation.

Third, the difference between sounds in dialectal variation can be described easily in terms of sound features. The differences between [a] and [e] have to do with tongue body height ([–low] for [e] and [+low] for [a]) and tongue position in the mouth ([–back] for [e] and [+back] for [a]).

Finally, there is a consistent context where the variation occurs. Let's look at the reasons behind this statement. First, every vowel in the Zenzontepec Chatino words appears after a consonant, and some occur before a consonant as well. Both invariant and varying vowels are found between consonants. This means that the variation could be driven by the preceding consonant, by the following consonant, or by the combination of them both. We have to look at the consonants.

To do this, we list the consonants that come before and after the vowels, dividing the consonantal contexts between those where we find the [a]~[e] variation and those where the vowels do not vary. The table in (16) shows the consonant frames that we've seen so far. The contexts in (16) show the consonants that precede and follow the vowels [a] and [e] in different words. Gray text shows consonants that are different in the various contexts while black text shows consonants that are the same. In (16a), we see that when the vowel follows [gʲ], it shows variation between [a] and [e], regardless of the following consonant. As seen in (16a), other consonants are not compatible with the [e]~[a] variation.

(16) Contexts and the Zenzontepec Chatino [a]~[e] variation

- a. San Pedro del Río [a]~La Aurora [e]

[gʲ]__[t]

[gʲ]__[n]

- b. No variation: Words with [a] or with [e] in both dialects

[ts]__[ʃ]

[kʷ]__[l]

These steps lead us to the conclusion that in Zenzontepec Chatino, there is a dialectal difference between the language as spoken in San Pedro del Río and the language as spoken in La Aurora: Immediately following [gʲ], in San Pedro del Río speakers use [a] while in La Aurora speakers use [e] in this context.

Understanding the phonetic properties of features may help explain the nature of particular variations. In this example, a palatalized dorsal [gʲ] involves a very high tongue body, for dorsal closure, that is fronted, for palatalization. The vowel [a] has a low and central tongue position. In contrast, the vowel [e] has a higher and more fronted tongue body. These facts suggest that we find [gʲe] instead of [gʲa] because there is co-articulation from the consonant to the vowel, raising the vowel and making it more fronted.



Comprehension Check 4

Several Ayutla Mixe speakers are having a discussion. The older speakers use the words [tɻ:] 'rusty' and [akʰɻn] 'very' in their discussion. (The vowel [ɻ] is a mid, back, non-round vowel.) The younger speakers respond

with different forms of those words: [to:] ‘rusty’ and [akʰson] ‘very’. Using your newly honed skills in linguistic analysis, how would you explain this situation? (Our source for this question is Romero-Méndez 2008.

Ayutla Mixe is a Mixe-Zoquean language that is spoken in Southern Mexico.) We thank Rodrigo Romero-Méndez for sharing his knowledge of the language with us. He tells us that the speakers call their language *Ayuujk*, meaning both ‘word’ and ‘language’.



Here's our answer

Among speakers of Ayutla Mixe, we see two groups of people divided by age who use different forms of the “same” words, a case of an age-based dialect. The variants are [ɣ] for older speakers and [o] for younger speakers. These are both non-high back vowels; [ɣ] is not round, while [o] is round. The younger generation is using rounding while the older generation is not.

4.2.6 Summary

Dialects vary from each other in myriad ways. In Yauyos Quechua, one dialectal variation is between [l] and [ɾ]. Lower Aguaytía and San Alejandro are differentiated in part by whether dorsals are voiceless or voiced in the middle of a word. And in Zenzontotec Chatino, we have seen a dialectal difference in whether a speaker uses [a] or [e] after [gʲ]. These variants contribute to distinctions between dialects, giving each dialect its unique phonology.

To find a pattern of variation, we started with finding a pair of words that people pronounce differently. To determine whether this is dialectal variation, we checked whether the differences correlated with some social difference between the speakers. To determine whether it is a pattern, we checked whether other words showed the same sound difference. When everything checks out, we're pretty sure we have a pattern; the next step is to figure out the properties that define the pattern.



Exploring my language

Think about the ways that people who speak your language pronounce different words. Try to find an example of dialectal variation in the sounds of your language. Answer these questions.

- *What social difference aligns with the pronunciation difference?*
- *What sounds differ between the two dialects?*
- *Can you come up with three words that show the difference?*

In order to figure out the phonological side of the pattern, we worked through an analytic strategy with three basic steps, (17).

(17) The three basic analytic steps

- a. *What is the difference between the two sounds?* Look at the two sounds that differ and determine what features distinguish the two from each other.
- b. *Does the pattern extend to other sounds in the language?* To answer this, look at the features that are the same between the two sounds. Use these features to determine whether other sounds might participate in the same pattern because they share some of these features. Then check other words to see whether the variation is found with any of the additional sounds.
- c. *Is the pattern limited to a particular place in the word?* Look at words which show the two sounds so you can see whether the pattern occurs only in certain places in a word.

Once the analysis is complete, we write a statement defining the nature of the variation pattern — what the social communities are, what the sound differences are, where the difference is found in words.

You may be happy to know that this analytic strategy is the same one that we use with other types of sound patterns, in Chapters 6–9. If you keep reading, you get more practice with them — and there are no other basic strategies to learn.



Exploring my language

If you want to explore your language further, try answering the three questions in (17), using the words you identified in the previous exploration. If you need some review, the questions are repeated below with sections in the chapter where you can re-read about how we went about answering each question.

- a. *What is the difference between the two sounds?*
→ See sections 4.2.2.1, 4.2.3.1, and 4.2.5.
- b. *Does the pattern extend to other sounds in the language?*
→ See Section 4.2.3.4
- c. *Is the pattern limited to a particular place in the word?*
→ See sections 4.2.3.5 and 4.2.5.



Variation in Highland Mayan Sign Language

Sybil Vachaudéz and Samantha Prins

All languages show **variation** along the lines of region, gender, age, and other factors. This variation can be lexical (completely different words or signs) or phonological (different variants of the same word or sign). An example of the latter is the way two different signers of Highland Mayan Sign Language (HMSL, used by Maya communities in the Guatemalan highlands) produce the sign CHILD.

Variants of ‘child’ in HSML

Kaqchikel signers recorded by Samantha Prins and Sybil Vachandez

a. CHILD

Female signer from Patzicía



b. CHILD

Male signer from Sololá



In this case, the same basic sign for CHILD—a semi-open handshape with five fingers outstretched—is produced with varying orientation: The woman on the left uses palm-down orientation, while the man on the right uses forward-facing palm orientation. (You might also notice that the woman produces CHILD with her left hand, while the man produces it with his right. While signing with the left or right hand can sometimes be meaningful, for example to indicate location in space, often it’s simply due to the signer being left- or right-handed. This would be an example of individual habit or preference rather than linguistic variation.)

Just by observing the way someone produces a given sign, you might learn something about who they are and where they’re from. If the signs used by signers from different regions, gender or age consistently vary in the same way, you can learn something about what the source of the variation might be. Understanding variation of this kind is important for accurately using, documenting, teaching, or learning a particular dialect of a language.

4.3 Idiolectal variation

Think about two people who know the “same” language — and even the same dialect of that language. There’s a lot of overlap in the words each person knows, but there are also some words that one person knows that the other does not, and vice versa. Confusions may arise because the two people have slightly different meanings for a pair of words. Some words will be pronounced differently by the two people. These observations show that each person internalizes their own grammar, their own *idiolect*, based on their own personal language journey.

Since humans are always changing, learning new words and adapting speech patterns in light of our experiences, our idiolects are constantly in flux as well. Every person speaks their own idiolect. No two idiolects can be exactly the same, and no single individual’s idiolect remains unchanged over time.

In this section, we review two examples of *idiolectal variation* in the way words are pronounced, using the same analytic steps that we used for dialectal variation. Our first example is from a Mixe language.

4.3.1 Ayutla Mixe vowels

Several dialects of Mixe are spoken in Oaxaca, Mexico. Our interest is in the variety spoken in the community of San Pedro y San Pablo Ayutla, Oaxaca, Mexico; our source, Romero-Méndez (2008), calls it Ayutla Mixe.

Ayutla Mixe has several vowels, including two central vowels, low [a] and high [i]. When these vowels precede a stressed vowel, some speakers use low [a] while others use high [i]. Since the last vowel of the word is typically stressed, this variation is found with any vowel that is not the last vowel in the word. There is no clear connection to some social group — it is simply the case that some people use the low [a] and others use the high [i], so we conclude it is idiolectal variation. The vowels in variation are black in (18); the other sounds are in gray text. (The sound [ʂ] is a retroflex, with the tip of the tongue curled back. Retroflexes are discussed in Chapter 2 Section 2.3.2.2.)

(18) Ayutla Mixe idiolectal difference: Vowels

	idiolect 1	idiolect 2	gloss
a. <i>atya’aky</i>	[atʰa’akʰ]	[itʰa’akʰ]	‘slowly’
b. <i>kaxë’ëk</i>	[kaʂi’ik]	[kiʂi’ik]	‘to appear’
c. <i>tsapajkx</i>	[tsapaʰkʂ]	[tsipaʰkʂ]	‘peach’

We answer the questions in (17).

What is the difference between the two sounds? We've seen that the dimension of variation is vowel height, [+low] [a] vs. [+high] [i].

Does the pattern extend to other sounds in the language? We identify the common properties shared by the vowels in idiolectal variation: [a] and [i] are both central vowels. They are the only central vowels, but there are other vowels. Do any of the other vowels share the high-low variation? According to Romero-Méndez (2008: 129), only [a] and [i] have this pattern. (A full study of the language would examine many words to check on this assertion.)

Finally, *is the pattern limited to a particular place in the word?* Yes: Variation is restricted to non-final, unstressed vowels.

We make our statement of idiolectal variation in Ayutla Mixe: When unstressed, central vowel height is in idiolectal variation between low and high: Some speakers use [+low] [a] and others use [+high] [i].

4.3.2 Expressive vowel length in Cupeño

Our next example raises an interesting question about working with a very small number of speakers, however that comes about. When there aren't many speakers, the researcher is always asking the question *Are the differences that I am hearing idiolectal or dialectal?* We illustrate with an example from Cupeño, a Uto-Aztecan language spoken in California, USA; our source is Hill (2005). Jane Hill carried out the original research in the 1960s for her doctoral dissertation; Hill worked primarily with Roscinda Nolasquez, a "terminal-generation" speaker, that is she was a speaker from the last generation to learn the language from infancy. Leroy Miranda, Roscinda Nolasquez's great-grandson and for several years director of the Cupa Cultural Center, urged Hill to publish a revised version of her work, resulting in Hill (2005).

In discussing pronunciation, Hill (2005: 24) observes that "Roscinda Nolasquez... frequently used lengthening of stressed syllables for expressive purposes, so that words like *met'ish* 'many', *peta'ama* 'all', *mangin* 'slowly', or *akulyi* 'little' often appear as *me:::t'ish*, *peta:::ama*, *ma:::ngin*, or *aku:::lyi* respectively." Hill uses three colons in a row to show that these vowels are extremely long. (See Chapter 2, Section 2.4.3.4 on the various ways people use to show vowel length.) (Hill says *stressed syllables*; she could also say *stressed vowels* since it is the stressed vowel that is long. See Chapter 2 Section 2.4.3.6 on stress and Chapter 6 on syllables.)

No other speakers are reported using expressive length but that may simply be because Hill worked primarily with Roscinda Nolasquez. Perhaps all speakers used this same type of expressive length, but on the other hand, this use of vowel

length may have occurred in the speech of only some individual speakers of Cupeño.

There is often no way to distinguish idiolectal usage from language or dialectal usage when the number of speakers is extremely small.



Comprehension Check 5

In writing about Choctaw, George Broadwell, a scholar at the University of Florida, says “In my own fieldwork in Mississippi, I spent some time trying to identify words that varied according to community, and often encountered situations like the following. I asked a speaker of Choctaw who lives in Pearl River the word for ‘one’, and she replied *achaffah*. When asked about the pronunciation *chaffah*, she said that this is what people say in Conehatta. The next day in Conehatta, I asked the word for ‘one’, and my consultant said *achaffah*. The pronunciation *chaffah*, he said, is what people say in Pearl River. Clearly both speakers could not be right.

After asking people in several communities, I found that the pronunciations *chaffah* and *achaffah* are not correlated with community of residence at all... This sort of situation turned out to be extremely common.” Broadwell (2006: 11). (Choctaw is a Muskogean language.)

Is the use of *chaffah* and *achaffah* dialectal or idiolectal? Give yourself bonus points if you can explain why.



Here's our answer

The use of *chaffah* and *achaffah* is likely to be idiolectal, because the use varies by individual, not by community. And we know that a dialect is the ways of speech that identify a community while an idiolect is the ways of speech that identify an individual.

4.3.3 Summary

Idiolect refers to the ways of speech unique to an individual. The examples in this section focused on productions that vary between speakers in a single speech community.

Idiolectal variation is found with all types of sounds. As with dialectal variation, the classes of sounds that are found in idiolectal variation can typically be identified by sound properties, for example the central vowels in Ayutla Mixe. The difference between idiolectal variants can also be described in terms of sound properties. The variation in Ayutla Mixe is between low and high vowels, while in Cupeño the variation is between short and very long vowels. Variation may be found wherever a particular sound occurs; it may also be limited to particular contexts, like the unstressed vowels in Ayutla Mixe and the stressed vowels in Cupeño.

We see that the phonetic properties of idiolects and of dialects follow the same general structure — a class of sounds varies along some dimension, possibly in a restricted context. These three elements (sound class, dimension of difference, and location in word) determine the phonology of dialects, of idiolects, and, as the next section shows, of free variation.



The idiolect/dialect continuum

Robert Autrey, Mosiah Bluecloud, Kickapoo, and Corey Roberts, Occaneechi Saponi

Idiolects and dialects can sometimes be a matter of scale. Dialects are often thought of as varieties of a language spoken in different geographical regions or by people at different levels of social status in a community. However, in some of the smallest speech communities (often having 10 speakers or less) geographical distance and social status are not necessarily the most important factors to consider. In these cases, an idiolect can easily be interpreted to be a dialect because of the small number of people involved. Without evidence coming from many speakers to prove otherwise, it can be easy to assume that the speech pattern of one individual applies to everyone in a small language community.

Idiolect and dialect are also classifications that can overlap. The Kickapoo communities of Oklahoma, Texas, Kansas, and Mexico provide an example of how this could take place. Two generations ago, it was common to hear the sound [ʃ] in use in the Kansas community, whereas the Kickapoo speakers in Oklahoma, Texas, and Mexico would use the sound [s]. One elder spent the first 40 years of his life in Kansas before marrying into the Mexico community. He spent the remainder of his life between the Mexico, Texas, and Oklahoma communities, where he adopted the dialectal use of [s]. Even though he became the last speaker of the Kansas dialect in his 70s, his idiolect contained a mixture of both [s] and [ʃ]. Using each sound half of the time he spoke made his idiolect neither Kansas Kickapoo, nor the language exactly as it was spoken in the other communities. This last speaker of the Kansas dialect had a habit of speech that was entirely unique to him — an idiolect.



Exploring my language

Think about your own speech and the speech of someone else who speaks your language. Find a difference in the way the two of you pronounce words.

Is this difference idiolectal or dialectal? Explain your answer.

Can you identify the *class of sounds*, the *dimension of variation*, and any *special context for variation* for the difference you found?

4.4 Free variation

Free variation is the ability for speakers to use distinct sounds interchangeably in the same word, pronouncing each sound in different ways at different times without changing the meaning of the word and without being viewed as having incorrect pronunciation. Two (or more) different sounds — vowels or consonants — can be in free variation with each other. Meaning remains the same, regardless

of which sound is used. Free variation may be dialectal (a property of a speech community) or it may be idiolectal (found in the speech of an individual). Some people think that there is no true free variation, that all variation is determined by some social or stylistic factors. However, it is not uncommon to find reference to “free variation” in language descriptions, so it is useful to know what the term refers to.

We illustrate free variation with two examples, one involving consonants and one involving vowels. We use familiar categories in our descriptions – the sound class that varies, the dimensions of variation, and a special context, if any.

4.4.1 Mushuau Innu labial consonants

Mushuau Innu is an Algonquian language from northeastern Labrador.⁴ There is a large amount of free variation in the Mushuau Innu consonantal system. We focus here on labial consonants.

Scott notes that there are multiple ways to say a voiceless labial consonant: “In Mushuau Innu, [hp] alternates with [p], [ɸ], [hɸ] and [f]... occurring in free variation.” (Scott 2000: 62). (The sound [ɸ] is a voiceless labial continuant; see Chapter 2, Section 2.3.2.1.) These variants are shown in (19), with the word meaning ‘coat’. Note that the first three words differ only in the last consonant of the word (in black text); despite this difference, all three words mean ‘coats’. For the other two words, there are additional *morphemes* at the start or end of the word, resulting in some meaning differences, but we can still find the sequence *ʌku*: followed by a labial consonant, and we see that the consonant has yet other forms, a stop [p] or a continuant [ɸ].

The practical orthography, shown in italics, comes from Scott (2000). Note that in each word, the spelling sequence (*h*)*p* corresponds to the various labial consonants that we are examining. Scott notes that communities had yet to come to consensus on how to write the language in 2000. The orthographic symbols used include symbols that connect to historical forms or forms from other dialects of the language. Letters in parentheses indicate sounds in Mushuau Innu that are absent or have a different form in related dialects.

⁴We consulted Scott (2000), a work based on primary research by José Mailhot, and we thank Mark Scott for reviewing and improving our discussion of the labial consonants in Mushuau Innu. In 1963, Mailhot began a lifelong working relationship with the community.

(19) Different ways to say ‘coat’

- | | | | | |
|----|------|---------------|---------------------------|------------------------|
| a. | [hp] | [ʌku:hpʌ] | <i>akû(h)pa</i> | ‘coats’ |
| b. | [f] | [ʌku:fa] | <i>akû(h)pa</i> | ‘coats’ |
| c. | [hɸ] | [ʌku:hɸʌ] | <i>akû(h)pa</i> | ‘coats’ |
| d. | [p] | [utʌku:pihts] | <i>tshitakû(h)p</i> | ‘on his coat’ |
| e. | [ɸ] | [stuku:ɸ] | <i>utakû(h)pi(h)t(sh)</i> | ‘(take off) your coat’ |

Although the last consonant in ‘coat’ is said in five different ways, there is always a labial component. The consonant can be pronounced as a labial fricative [f] or [ɸ], as a stop [p], or as a sequence of [h] followed by a labial, [hp] or [hɸ]. It does not matter which pronunciation is used; the speaker is understood and is not viewed as mispronouncing the word. These sounds are used interchangeably.

Once again, we can identify the class of sounds that are in free variation, the voiceless labials [–voice, labial]. We can identify the range of variation by their articulation: A voiceless labial is [labial, –voice]; the fricatives are [+continuant], there may be a preceding [h]. We can make a statement, (20).

(20) Mushuau Innu free variation of labial consonants

Some labial consonants are in free variation with each other, between [+continuant, labial] [f] and [ɸ] (fricatives), labials preceded by aspiration, both the stop [hp] and the fricative [hɸ], and the stop [p].



The curious status of [p] in Mushuau Innu

If a speaker hears any of [f], [ɸ], [hp] or [hɸ], they know that they can also hear the other sounds in the same word, or they can hear a [p]. But the case is different with the sound [p].

If a speaker hears the sound [p] in a word, the speaker does not know whether only the sound [p] is acceptable in this word, or whether any of these other voiceless labial sounds – [f], [ɸ], [hp], or [hɸ] – are acceptable as well. An example is the final sound in *shîshîp* [ʃi:ʃip] ‘bird (duck)’ – the other four sounds are not acceptable in this word.

With [f], [ɸ], [hp], or [hɸ], the free variation is complete. With [p], we only know that it *might* be in free variation with these other sounds.

With an understanding of this pattern of free variation, Mushuau Innu learners will be better at understanding and producing fluent speech. Those documenting the language will be able to class words as “the same” even though they sound different.

4.4.2 Nahuat Pipil [u] and [o]

Nahuat Pipil is a Uto-Aztecan language spoken in some areas of western and central El Salvador. Our data comes from Lemus (1996). We thank Jorge Lemus for reviewing this section for accuracy in our description of Nahuat Pipil, including phonetic transcriptions that differ from those in Lemus (1996).

Listeners can hear Nahuat Pipil speakers use five vowels, [i], [e], [a], [o], and [u] in their words. However, [o] is not very common. At the ends of words, Nahuat Pipil vowels are glottalized (shown by [ʔ]); glottal vowels are not found anywhere else except at the ends of words. For the free variation, speakers use mid [oʔ] as an alternative to high [uʔ] at the ends of words. There is virtually no other use of [o] or [oʔ].

Both [o] and [u] are [+back] and [+round]. Lemus (pc, 2022-8-15) observes “These vowels are in free variation for all speakers, though [u] is preferred in slow speech”. (Lemus does not include a practical orthography, so we only have IPA symbols for the Nahuat Pipil words.)

(21) Free variation: [o] and [u]

- | | | | |
|----|--------------|--------------|-------------------------------|
| a. | [péluʔ] | [péloʔ] | ‘dog’ (borrowed from Spanish) |
| b. | [metsʰkújuʔ] | [metsʰkújoʔ] | ‘leg’ |

These word pairs mean exactly the same thing, whether the speaker uses [oʔ] or [uʔ] as the final vowel.

To test whether the **word-final** restriction is correct, we find words where [u] is not final. Examples are given in (22). When the back, round vowel is not final in the word, only [+high] [u] is used. There is no variation with [–high] [o]. The symbol [ɣw] refers to a labial-velar fricative.

(22) No variation; only [u]

<i>non-final [u]</i>		<i>[u] before final consonant</i>	
[ʰuɣiʔ]	‘male’	[ʃultukʔ]	‘soul’
[guʰtuna]	‘to cut’	[ɣwaktukʔ]	‘dry’

Knowing when to use [u] and when to use [o] is part of knowing Nahuat Pipil. We would be surprised to hear [ʰoɣiʔ] for ‘male’ and [ʃoltokʔ] for ‘soul’; hearing [ʰuɣiʔ] and [ʃultukʔ] is expected.

For our analysis, we once again looked for the class of sounds involved, the properties that vary, and the required location, if any. We use this knowledge to formulate a statement of free variation, (23).

(23) Nahuat Pipil free variation of round vowels

In Nahuat Pipil, [+round] vowels are in free variation with each other, between [+high, +round] [uʔ] and [−high, +round] [oʔ] at the very end of words.



Words borrowed at different times

Jorge Ernesto Lemus, Universidad Don Bosco

Some Spanish loans into Nahuat Pipil have only [o], with no [u] option: [ombrón] ‘big’, [kotón] ‘shirt’, [tʃortʃah] ‘rooster’s crest’. Other Spanish **loanwords** containing [o] have changed it to [u]. I think it has to do with the time when the word was incorporated into Nahuat Pipil. Older loans (16th to 18th centuries) changed [o] to [u], e.g. Spanish *tomín*, Nahuat Pipil [tumin] ‘money’. Newer loans (after Independence, 19th century) retained [o], e.g. Spanish *hombrón*, Nahuat Pipil [ombrón] ‘big’.

4.4.3 Summary

Free variation is the term used when the same speaker has two or more different pronunciations of some word which can be used interchangeably and are still perceived of as fluent by other speakers. Reflecting on the examples in this section, we can identify a variety of properties found with free variation, properties that are familiar from our discussion of dialects and idiolects.

We saw that free variation is found with both consonants (Mushuau Innu) and vowels (Nahuat Pipil), the two major classes of sounds. We found that the sound class that varies in a free variation pattern can be described by features ([labial] consonants in Mushuau Innu, [+round] vowels in Nahuat Pipil). We also found that we can describe what varies by using features ([+continuant] or [−continuant] in Mushuau Innu; [+high] or [−high] in Nahuat Pipil vowels). We learned that sometimes free variation occurs with a particular sound regardless

of where it is found in a word, as with the consonantal variation in Mushuau Innu, and sometimes it is found only in certain locations in a word, as with the word-final Nahuat Pipil pattern.

Whether the usage is marking dialects or idiolects, or is a pattern of free variation, we find the same three general properties of phonological patterns — a class of sounds, properties by which they vary, and a context for where the variation is found. The way that we define a sound pattern is independent of whether the pattern is dialectal, idiolectal, or free variation — or even a pattern used in some other way. As we will see in the remainder of this volume, these properties not only define variation, but they contribute in other ways to the unique phonology of each language.



Exploring my language

Think about the way you or a speaker familiar to you pronounces the words of your language. Pay close attention — *can you find examples of sounds that are in free variation with each other?*

Once you find an example, *identify the sounds, the phonetic dimensions of variation, and the context for that variation.*

4.5 Discussion

We looked at three different types of variation in this chapter, dialectal variation, idiolectal variation, and free variation, summarized in Figure 1.

In identifying sounds that are in variation with each other, we saw that there are three basic components — the class of sounds that are engaged, the features that vary, and the location for variation. In the examples that we examined, we saw that variation occurs with vowels and with consonants and we saw that some instances occur anywhere in a word while other instances occur in restricted locations.

How can we put this knowledge to good use? Before answering that important question, we turn briefly to two other issues — resolving our initial puzzle, and understanding why language variation exists.

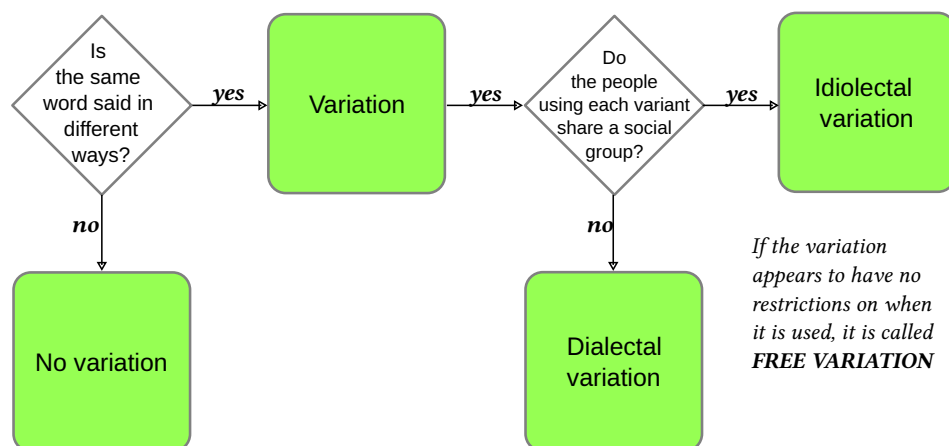


Figure 1: Types of variation

4.5.1 Resolving the puzzle

This chapter focused on strategies to use in answering our initial puzzle.

How does a single word tell you about home town or gender?

We focused on the properties of sounds in variation. We examined the properties of a word that are clues to social connections — the different sounds, the way they are different, and the locations where the difference is relevant.

These properties can serve to identify how a speaker is socially situated, in dialects. The same type of patterns show up in the speech of individuals, in their idiolects. And the same type of patterns define free variation within a speech community. We looked at a set of strategies, summarized in Figure 2, to determine the dimensions of a variation pattern: The class of varying sounds, the varying features, and any relevant contexts. The figure is useful: The same strategies help to understand the sound patterns of your language.

4.5.2 Why are there so many varieties?

Languages would be so much easier to figure out, teach, or learn if there was only one version that everyone used. Why is there so much variety within a single language?

Language variation is a result of language users being human. As humans, we are social creatures (showing our affinity for our social networks) and we are

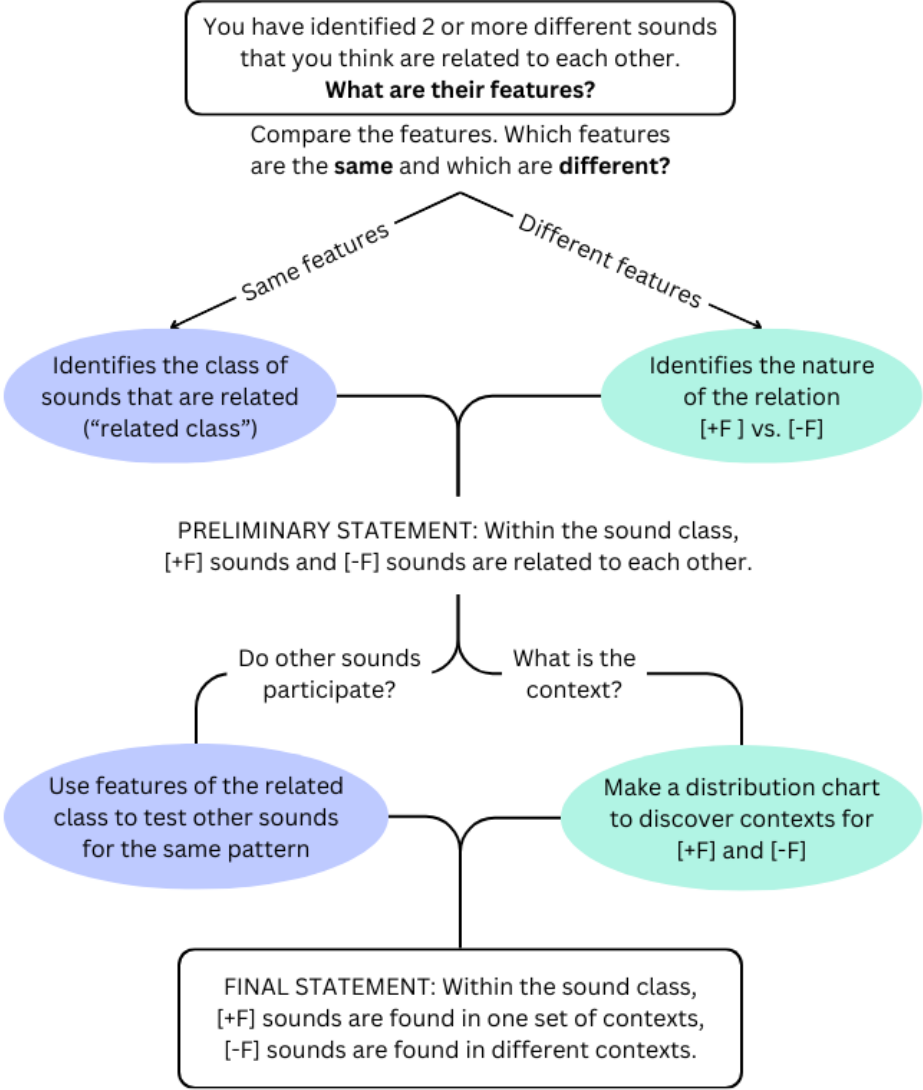


Figure 2: Determining the dimensions of variation

life-long learners (including language learning). As the group around us changes, our language may change too. The result is that, in general, we tend to sound more like the people around us, particularly if we want to belong to that social group: Language is a means of establishing social ties and participating in social activities. Conversely, we may choose to distinguish ourselves from others by speaking differently.

Dialectal variation can lead to significant language change over time, resulting in versions of the same language that are no longer mutually intelligible. Language change over time is called *diachronic variation*. One method of understanding diachronic change is to compare similarities and differences in two or more related languages: Similarities are thought to be unchanged from an earlier time, while differences suggest changes from that earlier time in one or both languages. To make an educated guess about the earlier version, we can use the same strategies identified here, identifying the class of sounds that are different, what the difference is, and any restrictions on the location of the differences. The additional piece that diachronic change introduces is figuring out what that earlier version could have been.

Another way to analyze diachronic variation is to use the same approach that we used to analyze dialectal variation, although this is only possible if we have evidence of the older version of a language as well as the current version. In cases where resources on a language come from very different points in time, the “same” word may be represented in different ways in the different resources. A challenge is to sort out whether the differences are due to (i) different transcription systems, (ii) differences in the perception of the person reporting on the language, or (iii) changes in the form of the word.

4.5.3 Using my hard-earned knowledge

From reviewing this chapter, you know that variation is a natural fact of language, and you can explain to someone else why people who “speak the same language” may sound different from each other. Furthermore, you have the skills to identify variation of different types, and to analyze the variation you’ve found to determine whether there is a pattern and, if so, what the phonetic properties of that pattern are. Why would you want to do this?

Language learning. Variation can be very confusing to students, because students work hard to learn some aspect of language only to hear people saying something else. The teacher who understands about variation can both explain the phenomenon to students (and so take away some of the frustration) and can

also make use of the details of variation to improve and enliven the learning experience.

Variation is interesting. It is fun for students to learn about different dialects. The more the teacher understands the details that make up different dialects, the easier it is for the teacher to accurately explain the dialectal differences and how a student can adjust their learning to speak a particular dialect of their choosing.

Variation improves learning. The more rich the learning environment, the better the learning experience. This is a challenge in a classroom or a language mentor setting because these situations typically have one fluent speaker only. A teacher who is aware of variation in their language — not just of its existence, but of the properties themselves — can use different varieties of the language to consciously enrich the classroom, whether the teacher is the one speaking the different varieties at different times, or the students are the ones challenged to stick to a particular dialect.

Variation is used consciously in some Indigenous languages. People are aware of idiolectal variation, even if they do not know the term. Culturally, storytellers rely on their audience's intuitive understanding of idiolectal variation, so the audience can identify characters by their idiolects. In some communities, there are stories about animals who speak, and those speaking animals may have their own idiolects. For example, in the Sahaptian language Nez Perce, Coyote uses [l] instead of [n] and [ʃ] instead of [s] while Grizzly Bear uses [ɬ] instead of all stops and fricatives. A teacher can use this concept to make classes more engaging, even if their own community does not have talking animal stories: A student needs to know the correct form in order to modify it “so the animal can understand”, or to “translate” the animal's speech for the rest of the class — or even to talk as the animal. An animal hand puppet might add to the fun.

Language resources — dictionaries, grammars, orthographies. Once a person making a dictionary or a grammar recognizes that there is variation within their language, new questions come up. Decisions must be made about which variety or varieties to include, and how to show the differences. There is a trade-off between a complete description and a useful resource. Knowledge of the different types of variation can factor into decisions about how or whether to include certain patterns. Understanding a pattern of variation helps answer these questions. If you can concisely describe a variation pattern, then you're able to complete a more accurate description without necessarily listing every variant directly: The resource could include a short section concisely describing alternative dialects and alternative idiolects — concise because you now have the tools to understand and describe the patterns concretely.

Orthography development is an interesting puzzle when multiple mutually intelligible dialects are involved, even though they don't sound the same. It is much easier to learn to read and write in a way that directly reflects how we speak. On the other hand, it takes time, effort, and money to produce materials, making it easier to have a single orthographic system that adequately represents multiple dialects where the sounds are mostly similar. Using knowledge of the phonological properties of patterns of variation can help communities in determining what symbols to use to represent sounds, to maximize ease of learning and use vs. production costs.

Language documentation and description. A key goal with language documentation is to create an accurate and complete representation of the language as used in a particular community. The more speakers there are in the community, the more likely there is to be idiolectal and dialectal variants. Once variation is suspected, the strategies introduced in this chapter can be used to help identify words where that variation is likely, thereby determining the complete nature and scope of the variation. The consequence is a more accurate record of the language as it is used.

Sometimes there are already materials to work with when documenting a particular language, and in some cases, variation may already be noted. You may know something or see something that leads you to wonder whether the reported variation is described accurately — and now you know how to examine the language for yourself to make the determination. As you work through materials, you may notice something that looks like variation that has not been commented on. The analysis methods introduced here give a means of checking whether your hunch is on the right track.

An additional challenge faced in documentation is transcription, faithfully representing the sounds of a language in written form. As with orthography, there is value to having a transcription system that is not too detailed. Variation is one of the details that might be lost, especially idiolectal variation. Being aware of the phenomenon and how to analyze it means that the researcher is positioned to evaluate transcription choices, and to make those choices in an informed manner.

Language revitalization. Efforts to gain even some speaking ability in an Indigenous language can be challenging when the language is sleeping or the documentation is sparse. One strategy that some communities adopt is to create words to fill gaps in the dictionary. This strategy can be enhanced by a good understanding of dialectal variation, where there is documentation of a related dialect or even a related language. When there is a gap in our own language but not in the

related dialect or language, we can take the related form, then modify the way it sounds in line with the patterns of variation between the two dialects/languages. In this way, the created words fill gaps in the collective knowledge with a greater degree of connection to the language itself.

Reading linguists' resources about your language. The terms introduced in this chapter are often used in linguistic writing, in particular **variation**, **variety**, and **variant** as well as the more specific terms **dialect**, **idiolect**, and **free variation**. Often, authors assume the reader is familiar with the concept and give no explanation. Some authors state that a pattern is dialectal, or idiolectal, or in free variation, without giving their reasons for the claim. Armed with your knowledge from this chapter, you now know the type of evidence they should have to support their claims about variation.

4.6 Conclusion

Linguistic descriptions of languages often mention patterns of variation; having worked through this chapter, you now are ready to understand what this means.

You know that to establish a pattern of variation we need to see that there are many forms that show the variation (not just one or two word pairs). We also need some discussion of the use of the different forms, to establish the social aspects of the variation (dialectal, idiolectal, and/or free).

You also know how to determine the phonological nature of the pattern by identifying

- the class of sounds that vary
- the dimension(s) of variation
- the location in which the variation occurs

This knowledge prepares you to understand linguistic literature involving variation, to analyze variation in your language, and to explain the patterns to others in a clear and succinct way.

Finally, with the ideas in Section 4.5.3 as inspiration, you are ready to use this knowledge as you work with your language.

Chapter 5

Contrast: Where sounds meet language

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Here's a puzzle. We know that the better we know a language, the better we are at recognizing when someone is speaking that language. Suppose your language teacher gives your class the challenge of identifying which words might belong to your language, just based on the sounds. The first word is [p^hekənə], the second is [bənɛ:], the third is usually said [plɛku] but is sometimes pronounced [pələgu]. (Remember that symbols inside square brackets are from the [International Phonetic Alphabet](#) (IPA), to show the pronunciation of the word. See both Chapter 2 and Appendix A for more information on these symbols.)

If your language is the Algonquian language Kickapoo, you might even recognize the first word as the word meaning 'nut'.¹ The second word, [bənɛ:], you would reject outright as not Kickapoo. The third word, [plɛku], might give some confusion, until the alternative pronunciation, [pələgu], is given — this can't be a Kickapoo word either, regardless of what it might mean.²

What is it about language sounds that tells you that they can or cannot be used in the words of a particular language?

¹Kickapoo is now spoken in Oklahoma and Mexico. Speakers of the language refer to it as Metotheeenihaatoweeweni. We thank Mosiah Bluecloud for his careful checking of our use of Kickapoo in this volume.

²The word [bənɛ:] is from Potawatomi, a language that is discussed in this chapter in sections 5.2.2 and 5.4.2.1. The third word, [plɛku]~[pələgu], is from Newfoundland Mi'kmaq, another Algonquian language, spoken in much of Eastern Canada, including in the Atlantic provinces, and possibly in the northeastern United States (Steeves 2022). We thank Sydney Steeves for checking that our discussion of Newfoundland Mi'kmaq is appropriate. Steeves told us that the speakers of Mi'kmaq refer to the language as Lnu'k.

There are a lot of different sounds found in languages, but not all sounds are found in every language. Each language has its own **inventory**, or collection, of sounds. The simple answer to the Kickapoo word puzzle is that the sounds [b] and [g] are not in the sound inventory of Kickapoo.

Knowing the sounds that are in a language is part of knowing that language. Sounds that are “part of a language” are used in making words of that language; sounds that are not in the language are not used in making words of that language. But how do you know what the sounds of a language are? This chapter gives tools to figure out the answer to that question — for any language.

We can be confident that any two sounds are part of the sounds of a language if we find that they can make a difference in what a word means. In this case, we say that the sounds **contrast** with each other.

Contrast is the second major component of a language’s unique **phonology**, along with variation. In this chapter, we show how to find the sounds that contrast in a language (also known as the **contrastive** sounds) so we know what the sounds of the language are. The list of sounds for a language is called its sound inventory (or its **phonemic inventory**). You may also see mention of sub-inventories of a language, such as the consonant inventory, the vowel inventory, the stop inventories, and so on. In this section, we also explore some common properties of inventories and the variety found in different inventories.

Here’s our promise. After you’ve read this chapter, you will know how to identify the sounds that make up the inventory of your language.

Identifying the inventory introduces a new puzzle: How do we convey those segments over time and space? This chapter also looks at some different ways of representing a language on paper, through both **transcription** (**symbols** that directly represent sounds to know how a language is pronounced) and **orthography** (letters, also called **graphemes**, used for writing and reading a language).

Here’s our plan. To fulfill our promise, this chapter does three things: First, we explain what contrast is, why it matters, and how you can be sure you have found examples of contrastive sounds. Second, we show how to determine the precise nature of each contrast and why that is important. Finally, we work through different ways of arranging the consonants and vowels of a language on a page to highlight the contrasts.

The list below tells what is in each section. It also gives guidance on whether you might want to skip a particular section.

1. Section 5.1 expands on the concept of contrast in language. We also present the basic tool for identifying contrast, the **minimal pair**.

2. Section 5.2 explores how to use minimal pairs to identify contrastive sounds in a language.

If your language has shorter words, you may find a lot of minimal pairs and you won't need to read the next section — in that case you might want to skip ahead to Section 5.4.

3. Section 5.3 introduces the close cousin of minimal pairs, called **near-minimal pairs**, and a more distant cousin, which we call **context pairs**. Sometimes it is hard or impossible to find true minimal pairs; in these cases we can use near-minimal pairs, or even context pairs, to identify contrasts.
4. Section 5.4 reviews how to identify the precise nature of the contrast between two sounds, as well as why this is useful to do.

If you've read all the sections up to this point, you may have a solid understanding of how to figure out the sounds that are in your language. At this point, you may realize that you want to have a good way to present the sounds to other people (including your future self). The next section helps with that task.

5. Section 5.5 discusses the conventions used in presenting the consonant and vowel inventories of a language, conventions that make use of and highlight the contrasts. This section may be a good reference for you if you are figuring out a visual arrangement of the consonants and vowels of your language. On the other hand, you may have seen these in your other readings and can safely skip this section.
6. Section 5.6 shows how the material in this chapter answers the puzzle we started with. We revisit the important question of using both orthography and transcription to represent the words of a language. Finally, there is a discussion of ways that a community language worker might use the knowledge gained in this chapter. You may choose to jump ahead to that section first, so that you are primed to think about how to make use of the knowledge as you read the chapter.

5.1 Introduction: What is contrast?

When we think of things that contrast, we think of things that are different in some way. When people speak, we recognize some of the words that they say as the same and others as different. To be the same, there must be no contrast. Yet most often there are differences: Differences in words as well as in voice. Perhaps people are saying the same word, so the voices are different. Or the same person

is speaking, but says the word softly and then more loudly — or is trying to help a child sleep, and the word is said slowly as well as softly.

These kinds of differences are true differences in the utterances, but they don't make one word into another word. For a language, whose voice it is, or how loudly or quickly a word is said does not change the word's meaning. It's when the sound differences correspond to meaning differences that we know we have different words. It is this type of sound difference that is meant by the term *contrast*: Two sounds contrast in a language when the sound difference is connected with a meaning difference — or has the potential of creating a meaning difference. If this doesn't make sense yet, we show some examples momentarily.

We can examine the sounds we hear in the words of a language to determine which ones contrast. The contrastive sounds are sometimes called the *phonemes* of the language. The term *inventory* is used for the set of sounds in a language. Often, an inventory (or *sound inventory* or *phonemic inventory*) is restricted to the contrastive sounds of a language. (These terms are explained later in the chapter, but if you can't wait, you can find them in the glossary.)

Before we continue, we mention two points that you might be wondering about. First, the existence of contrastive sounds in a language implies the existence of *non-contrastive* sounds as well. Understanding patterns of non-contrastive sounds in a language is the topic of Chapter 8. Second, there can be meaning differences with no sound differences. Linguists distinguish two types of cases — *homophones*, where there are two distinct words that sound the same but have unrelated meanings, and a *polysemous word*, a single word which has multiple layers of meaning differences. Cheyenne (an Algonquian language of the Algic family, spoken in Montana) has examples which show how hard it can be to tell the difference between homophonous words and a polysemous word.



Turtles and fog

Rosalia Badhorse, Northern Cheyenne

In the Cheyenne worldview, everything on this earth is interrelated or connected somehow. Whether you are searching for homophones or polysemous words, Cheyennes would just understand the entirety of this as a series of relationships. Therefore, it is not surprising that there are words

in the Cheyenne language that can have more than one meaning, and that each is connected. Much of my ancestors' philosophy comes from observing the natural world around them. This involves noticing and understanding all the special characteristics of animals, the land, the water, the weather, and connecting the dots, or seeing the similarities between concepts. This is how the Cheyenne perspective operates, and it is revealed in the language. See examples below.

Ma'ēno [ma'ēno] is used to refer to 'turtle' but also to 'fog', 'copycat', and 'someone who doesn't listen'. Close observers of the natural world can find the connections.

When it is foggy in our part of the country, the fog often hides all the hills and trees just as a turtle hides inside its shell. Young turtles have also been observed to follow a leader, mimicking everything they do, creating the meaning of a 'copycat'. In addition, turtles do not have ears that physically protrude from their heads; hence, creating the idiom for 'no ears', or 'someone who doesn't listen'.

The explanations above are the things my Cheyenne-speaking grandparents used to tell me while I was growing up, which are also verified in the online Cheyenne Dictionary at the link below.

Cheyenne Dictionary is online at <http://cdkc.edu/cheyennedictionary/index-en.htm>.

5.1.1 Identifying contrast

The goal of this section is to learn how to identify sounds that contrast in a language. We already know that when we say "sounds contrast" they can make a difference in meaning (also known as a *meaningful difference*).

There are many different layers of meaning when a person says something. Some of these include the speaker's mood ("I'm angry", "I'm confused", "I respect you", etc.). Other meanings include information about the speaker — where the speaker comes from, what their gender is, something about their age or their socio-economic class. (See Chapter 4 for discussion and examples of this type of meaning.)

Other meaning has to do with the ideas being communicated, carried by the words themselves. We call this last kind of meaning the *semantic interpretation*, based on the word *semantics*. In this chapter, when we refer to meaning, we refer to this last type of meaning, the semantic interpretation.

5.1.2 Introducing minimal pairs

The clearest method for showing that two sounds contrast is a pair of words that differ by only one sound yet have different meanings; we call this type of word-pair a *minimal pair*. Let's look at an example.

The language Kumiai (also Kumeyaay, formerly known as Diegueño) is a Cochimi-Yuman language spoken across the Mexico-California (United States) border; Mai et al. (2019) describe the sounds of the Ja'a Kumiai dialect. The language has a large number of consonants.³

Listening to a speaker, you might hear what sounds like a plain coronal stop [t] as well as something that sounds like a palatalized stop [tʰ]. (Palatalized stops like [tʰ] are introduced in Section 2.3.2.5 in Chapter 2.) Are you hearing different ways of saying the same sound, or are these two different sounds? When you find the words in (1), you can tell that these are two different sounds. We downplay the identical sounds in these words by using gray text; the sounds that contrast are shown with black text. (*SUBJECT* refers to the actor a sentence, the one performing the verb. If you are confused by the dot or the small caps in 'you.SUBJECT', please see Chapter 3 Section 3.2.1 and Section 3.2.3 respectively for explanations.)

- (1) Ja'a Kumiai [t] and [tʰ]
- a. *matt* [mat] 'earth'
 - b. *math* [matʰ] 'you.SUBJECT'

This pair has different meanings. When we look at the sounds, we see that everything is exactly the same except for the last sound. In the word meaning 'earth', the last sound is a plain [t], while in the word meaning 'you. SUBJECT', the last sound is a palatal [tʰ].

A word-pair with these properties — different meanings, and all sounds the same except for one (in the same place) — is called a *minimal pair*.

5.1.3 Which sound pairs should we test?

When we are trying to find the contrastive sounds in a language, the question comes up of whether we need to test every pair of sounds for contrast. In general, the answer is no; we only test for contrast in questionable pairs of sounds.

³We thank Anna Mai for checking our use of the Kumiai language. She told us that "Kumiai and Kumeyaay are both endonym and exonym for the language in Mexico and the US, respectively. Diegueño is a former exonym, and like many former exonyms can be considered pejorative, though you can still find it mentioned (as... on Wikipedia, on some tribal websites, etc.) since it makes it easier for new scholars to find earlier literature..."

What makes a pair of sounds questionable? It is similarity. The more *similar* a pair of sounds is, the greater the likelihood is that there is no contrast between the two. When sounds are quite dissimilar, they are likely to be contrastive sounds, so we don't need to check them unless we find something telling us that in a particular case, our rule of thumb isn't working. Consonants and vowels are quite dissimilar, both in where they occur in *syllables* (see Chapter 6) and in their feature make up. We very rarely test for contrast between a consonant and a vowel. We do want to be sure that vowels contrast with each other — especially pairs of vowels that share most features, for example [i] and [e] (differing only by the feature [high]) or [æ] and [a] (differing only by the feature [back]). We want to check whether there is contrast between consonants with the same place features but different manner features or laryngeal features; a few examples are given in (2). In each case, the example sounds are only examples. Use information from your language to figure out sounds to check to determine the contrasts in your language. (We review how to determine whether sounds are *similar* in Chapter 8 Section 8.1.1, if you want more practice with the concept now.)

- (2) Examples of consonants to check for contrast
- a. Same place and voicing, different manner features.
example: voiced labial consonants [b], [β], [v], [m]
 - b. Same place and manner but different voicing.
examples: [labial] stops [p] and [b]; [labial] fricatives [ɸ] and [β]
 - c. Same manner and voicing but different place.
example: voiceless fricatives [θ], [s], [ʃ]

Try this. Consider the pair of sounds [k], [x]. If your language had both, would it be reasonable to check them to determine whether they are contrastive? What about [k] and [ð]?

The first pair, [k] and [x], are both voiceless [dorsal] consonants, differing by manner of articulation ([k] is a [–continuant] stop and [x] is a [+continuant] fricative). This is an example of (2a), a pair worth checking for contrast. On the other hand, [k] is a voiceless [dorsal] stop while [ð] is a voiced [coronal] fricative — there are too many differences between the two so we assume that they contrast with each other without checking.



Exploring my language

Go back to the list of sounds that you created through the Language Explorations in Chapter 2. Starting with consonants, try picking pairs that are reasonable to check for contrast. Write down your reasons for choosing each pair. It may help to also write down pairs you don't need to check, and the reasons.

This list of pairs to check (and to not check) put you in position for finding evidence of contrastive sounds in your language.

5.1.4 Summary

Minimal pairs are words that are almost identical. There are two differences: There is a meaning difference and there is a single sound difference. When a pair of sounds is quite similar to each other, it is most important to check whether there is a contrast — sounds that are quite different from each other typically contrast with each other, so we start by assuming that they contrast.

Minimal pairs are the best test to determine whether two sounds contrast in a language. The next section shows some examples of minimal pairs in different languages, introducing the term *frame* for the identical part of the words being tested.

5.2 Determining contrast

How do you find contrast in the sounds of your language? Sometimes it is easy; sometimes it can be challenging. In this section, we explain the easy cases, [minimal pairs](#) and [minimal sets](#).

5.2.1 Testing for minimal pairs

The Ja'a Kumiai words in (1), *matt* [mat] 'earth' and *math* [matʰ] 'you.SUBJECT', both begin with the sequence [ma__] and have one other sound. We use the term *frame* for sequences like this: As with a picture, the frame surrounds the item of interest with the underline showing where the framed sound goes. In our example, [ma__] frames the sounds [t] and [tʰ] in the two words. In discussing

minimal pairs, we typically show the frame segments in gray text to focus attention on the potentially contrastive sounds: *matt* [mat] ‘earth’ and *math* [matʰ] ‘you.SUBJECT’.

To determine whether we have found a minimal pair, we make a table showing the frames which surround the sounds being tested, as in Table 1. We have columns for the test sounds, [t] and [tʰ] in Table 1. We fill in the cells with the words that we have. The column labeled “frames” shows the remainder of the word.

Table 1: Ja’a Kumiai [t] and [tʰ]

frames	[t]	[tʰ]
[ma_]	<i>matt</i> [mat] ‘earth’	<i>math</i> [matʰ] ‘you (SUBJECT)’

We see in Table 1 that the [ma_] row has a word in each column, showing us that there are words that begin with [ma] and have one more sound, either [t] or [tʰ].

We check meanings and see that they are different: [mat], with final plain [t], means ‘earth’; [matʰ], with final [tʰ], means ‘you (SUBJECT)’. The meanings are different and the words are identical except for the very last sound in each – [t] or [tʰ]. It is this different sound that tells the listener which word is being said. We conclude *The sounds [t] and [tʰ] contrast in Ja’a Kumiai*.

A pair of words with these two properties – a meaning difference and a difference in only one sound – is called a **minimal pair**. Minimal pairs are the easiest way to show that two sounds are contrastive. A summary of steps is in Figure 1.

As indicated by the oval surrounded by the dashed ring in the bottom right of Figure 1, you may not find a minimal pair but you may find another type of pair that is useful in identifying contrast. This is explained in the discussion of Figure 2 in Section 5.3.

5.2.2 Initial contrast in Potawatomi

Contrastive sounds can occur anywhere in a word. The example in (1) shows contrastive sounds at the end of words. Our next example shows a minimal pair differing at the start of words.

The words in (3) are from Potawatomi, a language of the Algonquian branch of the Algic family, spoken in Illinois, USA; our source is Lockwood (2017).⁴

⁴The **practical orthography** used in Lockwood (2017) is the Wisconsin Native American Languages Project writing system. Lockwood chooses to use this system for easy cross-referencing with the dictionary (Forest County Potawatomi 2014). Lockwood notes that there are other orthographies in use, and provides a comparison.

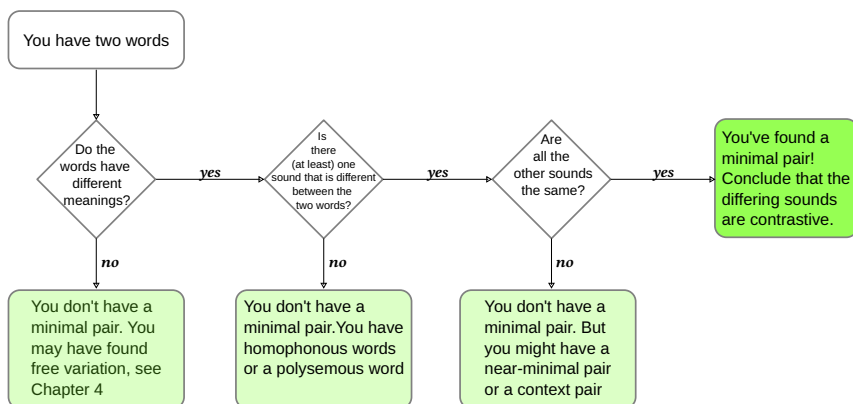


Figure 1: Minimal pairs and contrast

(3) Potawatomi [z] and [ʒ]

- a. *zadi* [zɑ:di:] ‘poplar, white or yellow; aspen, quaking aspen’
- b. *zhadi* [ʒɑ:di:] ‘arrowhead; bullet (or ball) for a muzzle-loader’

In looking at (3), we see that the frame is [__ɑ:di:]. The sound difference in (3) is found at the beginning of these words, shown by the underline in the frame. The word in (3a), meaning ‘poplar, aspen’, starts with a coronal [z] while the word in (3b), meaning ‘arrowhead’, starts with a post-alveolar [ʒ]. Everything else in the two words — [__ɑ:di:] — is the same.

The clue to which meaning is intended is found in the difference between [z] and [ʒ]. Because there is a meaning difference and only one sound difference, we conclude that coronal [z] contrasts with post-alveolar [ʒ] in Potawatomi.

It is also possible to find minimal pairs where the sound is in the middle of a word. And it is possible to have more than two words in a set that illustrates contrast. Our next example illustrates both of these points, with a minimal set (not pair) showing that four vowels contrast in Shipibo.



Comprehension Check 1

We did not give you the table for the Potawatomi words from (3) because we think you can create the table, based on what you saw in Table 1.

Try making a table. Identify the frame to label the row; identify the sounds being tested to label the columns. That gives the basic table. Now fill in the words in the appropriate cells. For bonus points, make a statement about whether this table shows that [z] and [ʒ] are contrastive and how you came to your conclusion.



Here's our answer

Here's the way we filled made our table for Potawatomi.

frames	[z]	[ʒ]
[_ɑ:di:]	<i>zadi</i> [zɑ:di:] 'poplar, aspen'	<i>zhadi</i> [ʒɑ:di:] 'arrowhead'

From this, we can see that [z] and [ʒ] occur in exactly the same frame, and have different meanings. The two sounds contrast with each other.

Our Potawatomi statement follows: The two sounds, [z] and [ʒ], contrast with each other because they are found in a minimal pair: *zadi* [zɑ:di:] 'poplar, aspen' and *zhadi* [ʒɑ:di:] 'arrowhead', have different meanings and the only sound difference is whether the initial consonant is [z] or [ʒ].

5.2.3 Minimal sets in Shipibo

We turn now to an example of a *minimal set* — a group of three or more words, each of which differs from all of the others by a single sound. In this example, the contrastive sounds are in the middle of the word.

An example of a four-member minimal set is given in (4). In this case, we have four words that differ by only one sound, the second sound in each word (which is a vowel in each of these words). These words are from Shipibo, a Pano-Tacanan language spoken in eastern Peru, in the Upper Amazon watershed. Our source is Valenzuela et al. (2001); the second author is a member of the Shipibo-speaking Dos Unidos de Pachitea community. (The function morpheme glossed as *REMOTE.PAST* means that the activity took place quite some time ago.)

- (4) Shipibo vowels
- a. *kini* [kini] ‘hole’
 - b. *kani* [kani] ‘went.REMOTE.PAST’
 - c. *koni* [koni] ‘eel’
 - d. *keni* [kuni] ‘whisker, beard’

When we examine several words together, it can help to write down the frame for each word, as we do in (5). (Doing this makes it easy to check that you really do have the same frame for each word.)

- (5) Shipibo vowels
- | frame | | word | |
|---------|-------------|--------|--------------------|
| [k__ni] | <i>kini</i> | [kini] | ‘hole’ |
| [k__ni] | <i>kani</i> | [kani] | ‘went.REMOTE.PAST’ |
| [k__ni] | <i>koni</i> | [koni] | ‘eel’ |
| [k__ni] | <i>keni</i> | [kuni] | ‘whisker, beard’ |

We can use the same method for testing for contrast that we used in Section 5.2.1. When we look closely at the frames in the leftmost column in (5), we see that each word has the frame [k__ni], so there will be only one row in our table. We have four vowels to test, so we need a table with four columns (a column for each vowel). The filled-in table is shown in Table 2.

Table 2: Shipibo vowels

frame	[i]	[a]	[o]	[u]
k__ni	[kini] ‘hole’	[kani] ‘went...’	[koni] ‘eel’	[kuni] ‘whisker,...’

We fill in the table with the words that we have, and see easily that each has a different first vowel filling the blank in the frame. Each word has a unique meaning. We know that a sound difference coupled with a meaning difference is the clue that shows contrastive sounds. Our set of four Shipibo words shows

that the four vowels have a four-way contrast with each other in Shipibo: [i] contrasts with [a], [o], and [u]; [a] contrasts with [i], [o], and [u], and so on.

Because there are more than two words in this list, we call it a **minimal set** instead of a minimal pair. The minimal set gives more information about sound contrasts than does a minimal pair, so they are a very compact way of getting information. It is usually easier to find minimal sets for vowels than for consonants.

Finally, notice that the blank in the frame is in the middle of the word, not at the start nor at the end. In minimal pairs and minimal sets, the sounds being contrasted can be anywhere in a word, as long as the frame surrounding the sounds stays the same.

Before we move on to a how-to on finding minimal pairs and minimal sets, we want to emphasize that contrast is not necessary for all sounds in a language. In some cases, sounds are non-contrastive, meaning that they do *not* contrast, even though they are different from each other.

5.2.4 Finding minimal pairs or minimal sets

You may be wondering how to find minimal pairs or minimal sets in your own language. It is not necessarily easy, but it can be fun. As you do this, keep track of your pairs if there is any chance that you will want to share your knowledge of contrasting sounds in your language with other people. The pairs are evidence of the contrastive sounds in your language.

Following up with the point of Section 5.1.3, start with a pair of **similar** consonants or vowels — you can choose from your list of potential contrasts that you made for the Language Exploration on p. 238. The challenge is to find matching words with those sounds at the beginning of the word (initial position), in the middle of the word (medial position), and at the end of the word (final position).

Here are tips on identifying frames for sounds in initial, medial, and final position. As you work with your language, you may want to jot down the frames you try with a pencil and paper so you don't repeat yourself.

We saw an example of contrast in word-initial position with the Potawatomi pair in (3), *zadi* [zɑːdiː] 'poplar, aspen' vs. *zhadi* [ʒɑːdiː] 'arrowhead': Two words with different meanings that differ only in the first sound. The identical part, the frame [__ɑːdiː], makes the words rhyme with each other. You can use rhyming words to identify minimal pairs.

Try this. For initial sounds, if you know your language well, you can try to find words that rhyme, differing only in their first sounds, to show that the initial sounds contrast.

Medial sounds — sounds somewhere in the middle of a word — can be tricky. You find these pairs by taking a word, choosing a medial sound, and running through other sounds in the language to put in place of your chosen sound to see whether it is a word. In some languages, there are few, if any, vowel-initial words, so looking at medial sounds may be the best way to test whether vowels are contrastive with each other. Especially with vowels, it is sometimes possible to find a minimal set that shows the contrast with all vowels in the language, as was done with Shipibo in (4).

Try this. Start with one word from your language that has one of the test sounds in it, then change that sound to the other test sound, without changing any other sounds. Does it make another word when you switch the sounds?

Third, even if you know your language well, it can help to have a dictionary to work with, provided the dictionary gives a good guide telling how to pronounce words. With dictionaries, you can look at entries that are close to each other, or even adjacent — because they will be identical at the start of the word. You may even find words that are identical except for the sound at the end of the word — true minimal pairs.

Try this. If you have a dictionary for your language, open it to a page with several entries. Scan through the entries to see whether you can find a pair that differs only by the last sound in the word.

In general, having a list of words is very helpful in looking for minimal pairs and minimal sets. If you've been doing the Language Explorations, you already have a long list of words. These words may include some minimal pairs already (accidentally or intentionally); at the very least they give you a starting point for finding minimal pairs and/or minimal sets.

The next Comprehension Check gives you a chance to try making a table for minimal pairs from a list of words, rather than simply a pair of words. This practice may be useful to you before you work on minimal pairs in your language.



Comprehension Check 2

Mundurukú is a Tupian language spoken in the very center of Brazil. Our source is Picanço (2005).⁵

Mundurukú is described as having both plain and creaky vowels. (See (21) in Chapter 2 for discussion of creaky vowels.)

Here are six words from Mundurukú; each has either the plain low vowel [a] or the creaky low vowel [ǎ] (creak is indicated by the wavy line diacritic beneath the vowel).

- | | | | |
|-----------|------------------|------------|------------|
| a. [dat] | ‘scorpion’ | d. [daǎa] | ‘firewood’ |
| b. [áj] | ‘rodent species’ | e. [dǎt] | ‘vomit’ |
| c. [wǐdǎ] | ‘jaguar’ | f. [wadʒe] | ‘cocoa’ |

Your job is to determine whether any of these words show that the two sounds [a] and [ǎ] are contrastive in Mundurukú. To do this, put what you’ve read about into practice: Make a table and use it to make a statement about whether or not [a] and [ǎ] are contrastive in Mundurukú, giving your reasons. The key questions to answer are:

1. Are [a] and [ǎ] found in identical frames?
2. If yes, do the words have different meanings?

If you answered yes to the two questions, you have evidence that [a] and [ǎ] contrast. Otherwise, you do not have the necessary evidence.

⁵We thank Gessiane Picanço for improving the discussion of Mundurukú; she told us that the Mundurukú refer to themselves as *Wǎyǎyǎ* ‘our people’ while outsiders use the term *Mundurukú*.



Here's our answer

The table below shows the frames for the two vowels. Words are placed in the appropriate cells, with a column for [a] and another for [ə]. (You may have words in a different order in each column; this does not matter. If you have words in the wrong column, it may help to review Section 5.2 from the beginning.)

frames	[a]	[ə]
d_fa	[daʃa] 'firewood'	
daʃ_	[daʃa] 'firewood'	
_j	[áj] 'rodent species'	
w_je	[waje] 'cocoa'	
d_t	[dat] 'scorpion'	[dət] 'vomit'
wid_		[widə] 'jaguar'

Are [a] and [ə] found in identical frames? Yes: The frame *d_t* has a word in each column. This shows that both [a] and [ə] are found in this frame. *If yes, do the words have different meanings?* Yes: [dat], with a plain [a], means 'scorpion' while [dət], with a creaky [ə], means 'vomit'.

Since the answers to both questions are yes, we know that [a] and [ə] contrast — the minimal sound difference makes a meaningful difference.

Notice that most of the words do not illustrate contrast. This may be more similar to the situation you find in working with your language. Also notice that [daʃa] 'firewood' appears twice in the [a] column, once for the first [a] and once for the second [a].

5.2.5 Summary

The point of knowing what minimal pairs and minimal sets are, and how to find them, is to determine which sounds are contrastive in a language. Once we know which sounds are contrastive, we have the core of the sound inventory of the language.

The words in minimal pairs and minimal sets differ from each other by one sound; every other sound in the word is identical. We call the identical part the frame for the contrastive sounds; we find minimal pairs and minimal sets by

finding a frame and determining whether more than one word uses the same frame.

The concept is simple and in many languages it is relatively easy to find minimal pairs and minimal sets. However, in some languages, it can be hard to find them. In such cases, we need to consult relatives of minimal pairs — [near-minimal pairs](#) and [context pairs](#).

We turn to these other types of word-pairs that show contrast in the next section. Be sure to try the Language Exploration first. If you have no trouble identifying minimal pairs in your language, you might decide to skip Section 5.3 and go right to Section 5.4, to learn how to precisely define the contrastive sounds in your language.



Exploring my language

Try finding a minimal pair in your language. Choose one of the pairs of sounds that you identified in the Language Exploration on p. 238 as being very [similar](#), so a good pair to check for contrast.

If you find at least one minimal pair for these sounds, you will show that these sounds are contrastive in your language.

There is no easy quick way to find minimal pairs, though with practice it becomes easier — and some people have a knack for it. If you need ideas on how to find potential minimal pairs, review Section 5.2.4.

If you don't find minimal pairs after you've given it a good try, read on. There are other ways to establish that sounds are contrastive.

(We talk about non-contrastive sounds in Chapter 8; if you find sounds that show no contrast, keep track of them to examine them further when you read Chapter 8.)

5.3 Near-minimal pairs and context pairs

Some languages do not have many minimal pairs. This may just be the way that the language is. This is common if most words are very long — the longer words are, the harder it is to find identical words that vary by only one sound. It can also happen when you don't have a lot of words in your language. And some

languages simply don't have many minimal pairs. In such cases, there are two other ways that we can establish that there are contrastive sounds, by using near-minimal pairs and by using context pairs. (If you think you've identified all the contrastive sounds in your language using minimal pairs and minimal sets, you might decide to skip this section and go on to Section 5.4 – you can always come back to this section if you discover you need the extra knowledge.)

5.3.1 Near-minimal pairs

Minimal pairs are pairs of words that differ from each other in two ways – one sound (and one sound only) is different, and the meaning is different. *Near-minimal pairs* differ from each other in three ways:

- the meanings are different.
- the sounds being tested are different.
- there is one other sound difference, a difference that is not right next to the sounds being tested.

Everything else in the words is identical.

If we left out the third point (that there is one other sound difference), we have the definition of minimal pairs. It is this third point that makes this kind of pair a near-minimal pair. We stipulate that the “other difference” between the two words is not right next to the sounds being tested for contrast because adjacent sounds sometimes influence each other, and we don't want those influences to affect our decisions about contrast. To be safe, we keep any differences as far from the test sounds as possible. (The impact of adjacent sounds is explored in Chapters 7 and 8.)

5.3.1.1 Near-minimal pairs with consonants

We find an example of near-minimal pairs in the Yakama dialect of Northwest Sahaptin, a Sahaptian language spoken in Washington, USA; our source is Hargus & Beavert (2014). Virginia Beavert was a Yakama Nation Elder who wrote about Yakama culture, for example, Beavert (2017) and Beavert et al. (2021). She was also a co-author of a dictionary of Yakama Sahaptin, Beavert & Hargus (2009).⁶

The words *ilúx* [ʔilux] ‘willing’ and *isúx* [ʔisux] ‘roe’ are given in Hargus & Beavert (2014) to show that the velar and uvular fricatives, [x] and [χ] are

⁶We thank Sharon Hargus for checking the discussion of Yakama, including verifying the transcription and orthography used here. She also told us that speakers of the language refer to it as *Ichishkiin*, which literally translates as ‘in this language’.

contrastive. If you look carefully, you see that these words are not a minimal pair. (The diacritic on the second vowel, *ú*, is an accent mark.)

The situation is illustrated in Table 3. Gray text in the frames shows the identical sounds; the sound differences are in black text. At the ends of the words we see the sounds we are testing for contrast, velar [x] and uvular [χ], following the vowel [u]. Both words begin with the sequence [ʔi]. However, between [ʔi] and [u], there is another sound difference: We see [l] in Table 3a and [s] in Table 3b. That second sound difference means that these words are not a true minimal pair. They form a near-minimal pair.

Table 3: Yakama near-minimal pairs for [x] and [χ]

frames	[x]	[χ]
a. ʔilu__	<i>ilúx</i> [ʔilux] ‘willing’	
b. ʔisu__		<i>isúχ</i> [ʔisuχ] ‘roe’

In Table 3, the velar and uvular fricatives are found in almost the same context. Both are at the end of a word, and both are immediately preceded by [u]. In both words, the first part begins [ʔi]. The only difference is that one word has [l] between the [ʔi]-part and the [u]-part while the other has [s] in that location. It is this second sound difference, [l] vs. [s], that makes these words a near-minimal pair instead of a minimal pair.

Near-minimal pairs are a strong indicator of contrast, but they aren’t a sure thing. We conclude that [x] and [χ] are very likely contrastive. Because of the “very likely” part, anyone continuing to work with Yakama would want to keep an eye out for further evidence that these are contrastive sounds.

5.3.1.2 Near-minimal pairs with vowels

Examples of a minimal pair, a minimal set, and a near-minimal pair are found with Seri vowels. The language Seri is spoken in Sonora, Mexico; it is thought to be a language isolate. Our source is Marlett et al. (2005); a co-author of this article, F. Xavier Moreno Herrera, is a speaker of Seri.⁷

In Seri, some vowels are held for a longer time than other vowels; these are called long and short vowels respectively. The IPA convention to show vowel length is to give a vowel symbol followed by [:], so [a:] is a long [a], [i:] is a long [i], and so on. (See also Chapter 2 Section 2.4.3.4.) The minimal pairs in Table 4 illustrate that some long vowels and short vowels contrast in Seri.

⁷We thank Steve Marlett for checking that our use of Seri is appropriate.

Table 4: Seri minimal pairs with short and long vowels

frame	[i], [o], [ɑ]	[i:], [o:], [ɑ:]
[k__s]	[kis] ‘that which is raw’	[ki:s] ‘s/he who misses’
[k__s]	[kos] ‘species of plant’	[ko:s] ‘s/he who sings’
[ʔ__p]	[ʔap] ‘mule deer’	[ʔɑ:p] ‘tepany bean’

The long and short vowels shown in Table 4 are six of the eight vowels in Seri. There are also words with the vowels [ɛ] and [ɛ:], such as [moxɛt] ‘bighorn sheep’ and [pɛ:m] ‘carrying pole’, but no minimal pairs for [ɛ] and [ɛ:] are given in Marlett et al. (2005). Does the lack of minimal pairs with [ɛ] and [ɛ:] mean that these vowels are not contrastive? Looking more closely at Marlett et al. (2005), we find that there are near-minimal pairs involving [ɛ] and [ɛ:], as in Table 5.

Table 5: Seri near-minimal pairs for [ɛ] and [ɛ:]

	frame	word
a.	ʃ__p	[ʃɛp] ‘golden eagle’
	ʔ__p	[ʔɛ:p] ‘mesquite sap’
b.	k__mɛ	[kɛmɛ] ‘that which is used up’
	k__mɛx	[kɛ:mɛx] ‘that which moves slowly’

Again, there are two sound differences in these words — the underscore shows the position of the vowels [ɛ] and [ɛ:], while the black text shows the second sound difference. In Table 5a, the difference is in the initial sound, [ʃ] or [ʔ], while in Table 5b the difference is in whether the word has no final consonant or ends with [x]. The meanings are all different and there is only one sound difference in addition to the sounds being checked for contrast. We conclude that [ɛ] and [ɛ:] very likely contrast with each other. This conclusion is supported by the evidence that the other vowels all show a long/short contrast.



Comprehension Check 3

We introduced this section saying that we would show a minimal pair, a minimal set, and a near-minimal pair for Seri vowels. The words in

Table 5 show near-minimal pairs. Review the words in Table 4 and find the minimal pair and the minimal set. If you need a reminder, review the concepts of minimal pair and minimal set in Section 5.2.



Here's our answer

Minimal pairs show contrast with a pair of words; we see this with [ʔa:p] 'mule deer' and [ʔa:p] 'tepary bean', where the two words have the frame [ʔ_p]. Minimal sets show contrast in three or more words; we see this with the four words [kis] 'that which is raw', [ki:s] 's/he who misses', [kos] 'species of plant' and [ko:s] 's/he who sings', where each one has the frame [k_s].

5.3.1.3 Summary

Near-minimal pairs show one sound difference in addition to the sound being tested. Ideally this second sound difference is not adjacent to the sounds being tested, in case there is an influence of adjacent sounds. The question in Figure 2 tests for near-minimal pairs, following from the oval with the dashed ring in Figure 1.

5.3.2 Context pairs

It is not always possible to find even near-minimal pairs for sounds that we strongly suspect are contrastive. This may be because we don't have many words to look at, it may be because the language simply doesn't have many minimal or near-minimal pairs, it may be a random fact about those particular sounds in the language.

When this happens, we look for *context pairs*, words that show the test sounds in the same "local context" — that is, words that have identical sounds right next to the sounds being checked for contrast, but different sounds elsewhere. (To the best of our knowledge, there is no widely-used term for the concept of *context pair*. We find it useful to have a term for this concept, but you may not find the

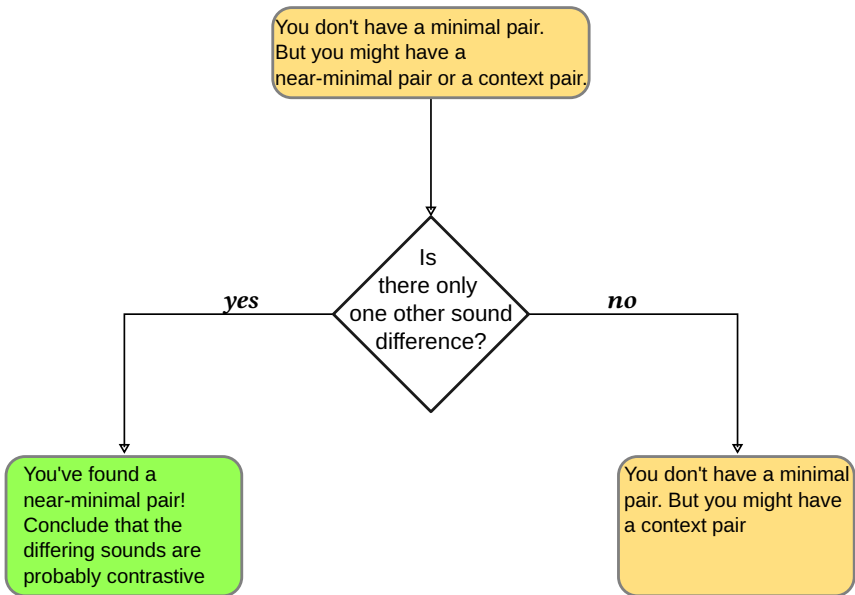


Figure 2: Near-minimal pairs and probable contrast

term used in other sources, even though the concept is used. These words may be called “near-minimal pairs”, but the concept of “near” is getting stretched quite a bit.)

Let’s look at a couple of examples from Salasaca Quichua, a Quechua language spoken in Ecuador whose sounds are described in Masaquiza & Marlett (2008). Fanny Chango Masaquiza was a resident of Salasaca at the time of publication; Quichua is her mother tongue and principal language.

There are very few minimal pairs or even near-minimal pairs included in the description in Masaquiza & Marlett (2008). The language is described as contrasting plain and aspirated stops, such as [t] and [tʰ]. While there are no minimal pairs nor near-minimal pairs listed for these sounds, there is a context pair, where the immediately adjacent contexts are identical. The context pair is shown in Table 6.

Table 6: Salasaca Quichua context pairs for [t] and [tʰ]

frame	local context	words	
<u> </u> uta	[<u> </u> u...]	[tuta]	‘night’
<u> </u> uktu	[<u> </u> u...]	[tʰuktu]	‘cornflower’

In the pair in Table 6, the sounds being tested for contrast are at the beginning of a word, and the immediately following vowel is identical, [u] for both words. The identical local contexts – [#_u...] for both sounds – suggest that [t] and [tʰ] are contrastive sounds.

A similar situation is found for the contrast between coronal [ts̺] and post-alveolar [tʃ̺] – the immediately surrounding context is identical but the rest of the word is different. This is shown with the words in Table 7.

Table 7: Salasaca Quichua context pairs for [ts̺] and [tʃ̺]

frame	local context	words
_ala	[_a...]	[ts̺ala] ‘skinny’
_aki	[_a...]	[tʃ̺aki] ‘foot’

In the words in Table 7, the sounds being tested for contrast are at the beginning of a word, and the immediately following vowel is identical, [a] for both words. The existence of the identical local contexts suggests the likelihood of contrast between [ts̺] and [tʃ̺].

In some cases, you will be successful finding words with the sounds in identical local contexts rather than finding true minimal pairs or near-minimal pairs.

The question to ask for context pairs is shown in Figure 3, which follows from Figure 2.

5.3.3 Summary

Whether we are looking at true minimal pairs, near-minimal pairs, or context pairs, the method is the same – we check to make sure the meanings are different and we check the context frame for the sounds in question. If the sounds are found in identical contexts, we have identified a likely contrast.

We say “likely” because it is only with true minimal pairs that we can be absolutely assured of contrast. With near-minimal pairs, contrast is highly likely, but it isn’t certain. This is because the second sound difference might in some way influence the sounds we are testing. With context pairs, there are even more sound differences between the two test words, so while contrast is likely, it is neither certain nor almost certain. It is nice to have more than one pair of words and more than one local context in these cases, to add support for the claim that the two sounds are probably contrastive.

These tests can show more than which sounds are contrastive – we can also identify specific properties of sounds that are contrastive – or distinctive. We

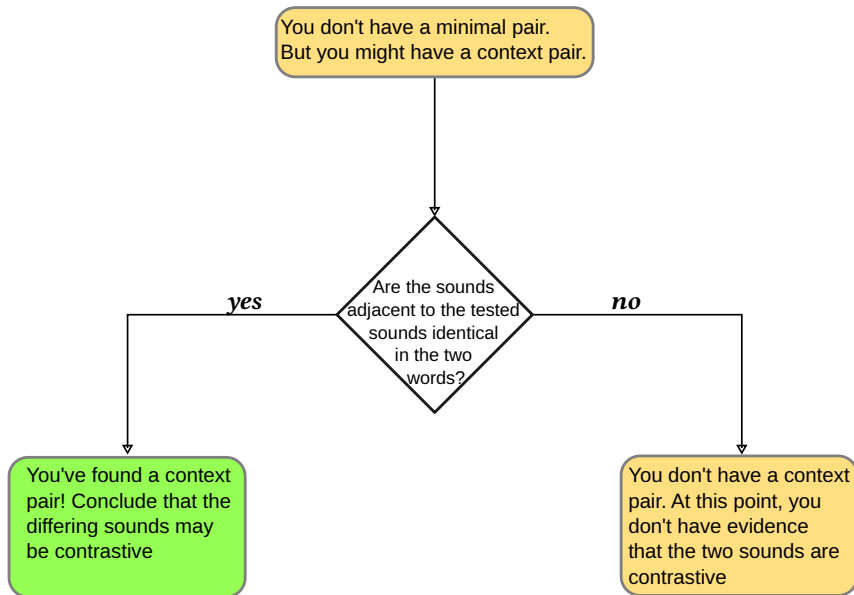


Figure 3: Context pairs and likely contrast

talk about these **distinctive features** in Section 5.4 of this chapter. Knowing the precise features that distinguish two contrastive sounds is useful in finding more contrastive pairs — a feature that is contrastive for one pair of sounds may well be contrastive for another pair. Additionally, when we look deeper into the sound patterns of a language, we often see that the distinctive features in a language are central in defining its sound patterns, a topic taken up in Chapters 7–9.



Contrast in Mũtẽã

Oliver Barraza

Barraza (2022) gives the following minimal pairs and near minimal pairs from Mũtẽã.

Labial	[p]	[pue]	‘verano’	[b]	[bue]	‘arco iris’
Alveolar	[t]	[tutu]	‘tronco’	[d]	[duturu]	‘doctor’
Velar	[k]	[niko]	‘raíz’	[g]	[migo]	‘tía’

“The labial sounds show a set of words where if one sound differs, the meaning of the word is different. This is how we know they are separate phonemes. This becomes less obvious with our alveolar and velar set. Instead of finding a set of exact minimal pairs, presented are the two sounds occurring in the same environment. In the alveolar example, both sounds start the word and are followed by the vowel /u/. We are able to determine that these are separate phonemes because the sounds can occur in the same environment, showing that there is not a predictable environment in which only one occurs. Similarly, our velar sounds both are fronted with the vowel /i/ and followed by the vowel /o/. Because they occur in the exact same environment, they are separate phonemes.” Barraza (2022: 16–17)

The glosses for these words are given in Spanish because, as Barraza explains, the work “is part of a continuous documentation project directly requested by a Tukanoan language community called the Mütëã, located in the Vaupés region of Colombia. The purpose of this paper is to create lessons that can be translated and brought back to the community. The use of these lessons will be to explain the linguistic background of Mütëã in a way that allows teachers and students to generate their own examples...”

Barraza (2022: 8)

Before moving on, you might like to check your understanding of the degree of certainty about contrasting sounds from different pairs of words. Check out the words from Nuu-chah-nulth in the Comprehension Check.⁸

⁸Our source is Carlson et al. (2001). Katie Fraser, a co-author, is a speaker of Nuu-chah-nulth and was part of the team that developed the *Nuu-chah-nulth Language Kit*, which includes a phone app that enables the user to hear spoken Nuu-chah-nulth words (Titian 2017). We thank John Esling, another co-author, for his review of our discussion of Nuu-chah-nulth. He tells us that the term Nuu-chah-nulth refers to a group of languages, and that Katie Fraser would refer to her own speech as Ahousaht.



Comprehension Check 4

Nuu-chah-nulth is a Wakashan language of Vancouver Island, British Columbia.

Your task is to determine how likely it is that the initial consonants are contrastive in each pair (the consonant in black text), and why. Each pair involves a plain consonant and one with secondary articulation, either a labialized consonant or an ejective consonant. (See Chapter 2, Section 2.3.2.5 for a reminder about secondary articulation.)

Here are guidelines to likelihood; you can also refer back to Figure 1, Figure 2, and Figure 3 for visualizations of the decision process.

- *certain* because the words make a minimal pair
- *highly likely* because the words make a near-minimal pair
- *likely* because the words make a context pair
- *can't tell* because the words don't have the same sounds next to the initial sound

contrast	reason?	test the initial consonants in the pair
		[ka:ʔa] 'give it to me' [k'a:ʔin] 'crow'
		[qawi:] 'berry, salmonberry' [q^wini:] 'seagull'
		[t̪aʔak] 'water' [t̪aʔak] 'island'
		[k^wa:t̪it̪] 'motion backwards' [k^wat̪it̪] 'to break'



Here's our answer

Here are our decisions about contrast and our reasons for these decisions, briefly. How do our answers compare to your answers? (We elaborate on our decisions after the table.)

contrast/reason		
contrast likely	[kaʔa]	'give it to me'
context pair	[k'aʔin]	'crow'
no evidence about contrast	[qawi:]	'berry'
no contrast shown	[q ^w ini:]	'seagull'
contrast certain	[tʃaʔak]	'island'
minimal pair	[tʃ ^h aʔak]	'water'
contrast probable	[k ^w a:tʃitɪ]	'motion backwards'
near-minimal pair	[k ^w atʃitɪ]	'to break'

- The words [kaʔa] and [k'aʔin] are context pairs because they have the same context immediately surrounding the initial consonants, the frame [__aʔ...]. The remainders of the two words are different however, [k'aʔa] and [k'aʔin]. Identity of only the immediate context is the hallmark of context pairs. Context pairs indicate that it is likely that the sounds contrast, but not certain.
- The words [qawi:] 'berry' and [q^wini:] 'seagull' have almost nothing in common — only the final vowel, and the final vowel is not adjacent to the initial consonant in these words. When words have (almost) nothing in common, there is no evidence one way or the other about contrast. Other words may show that [q] and [q^w] contrast with each other, but this pair is no help.
- The words [tʃaʔak] 'island' and [tʃ^haʔak] 'water' make a true minimal pair, showing that [tʃ] and [tʃ^h] are contrastive sounds in the language. These words make a true minimal pair because the meanings are different and the words are identical except for the initial sound: Both [tʃ] and [tʃ^h] are found in the frame [__aʔak].

- The words [k^wa:tʃɪtɪ] ‘motion backwards’ and [k^watʃɪtɪ] ‘to break’ make a near-minimal pair for their initial sounds [k^w] and [k^w]. The second sound difference, which is what makes this a near-minimal pair, is the length of the first vowel in each word. In [k^wa:tʃɪtɪ] and [k^watʃɪtɪ], the two sound differences are shown with black text, while the sounds that are identical are in gray text.



Exploring my language

Choose two pairs of sounds that you believe are contrastive sounds in your language. For one, try to find a near-minimal pair, and for the other, try to find a context pair.

If you can find a near-minimal pair or a context pair for your sounds, you will show that these sounds are likely to be contrastive.

Refer to the suggestions in Section 5.2.4 for ideas about how to find near-minimal pairs and context pairs.

We hope you now understand how to determine whether two sounds are contrastive in a language, using the different types of pairs discussed in this chapter. We also hope you can make wise decisions about which pairs of sounds to test for contrast.

Through your efforts in the previous chapters, you now have a long list of consonants and vowels for your language. The next step with that list is to identify pairs to check for contrast and to test those pairs. In language descriptions, it is typical to list the inventory of sounds for the language and to support those sounds by showing the minimal pairs, near-minimal pairs, or context pairs.

You are now prepared to determine the contrastive sounds for your language.

5.4 Dimensions of contrast

Up to this point, our discussion of contrast has addressed the question of whether sounds contrast with each other. In this section, we take a closer look at the

sounds that do contrast with each other, to identify very precisely the dimension(s) of contrast between the sounds. This is useful in finding other contrastive pairs because it is often the case that the same dimension of contrast is used with multiple sound pairs.

The concepts introduced in this section build on the point made in Section 5.1.3, namely that the best sounds to test for contrast are highly similar sounds, and that we can measure *similarity* between sounds by the number of features that are the same — with only one feature difference between two sounds being the best indicator that this is a pair to test for contrast.

We define the dimensions of contrast in terms of the features introduced in Chapter 2. Features that differentiate a pair of contrastive sounds are *distinctive features*. Distinctive features are very useful in identifying phonological patterns in your language. If you have identified the contrastive sounds and want to know more about the distribution of sounds in your language, you will appreciate having this knowledge.

Importantly, the sounds we check for contrast are quite similar to each other — consonants [t] and [tʰ], [z] and [ʒ], [x] and [χ], [t] and [tʰ], [ts] and [tʃ], vowels [ɛ] and [ɛː], and so on.

Not all pairs of sounds are similar to each other — some sounds are very different. For example, consider the Nuuchah-nulth sounds [h], as in [haː] ‘yes’, and [n], as in [naqʃitʰ] ‘to drink’. The sound [h] is a voiceless glottal fricative while [n] is a voiced nasal stop. The words [haː] and [naqʃitʰ] are context pairs for [h] and [n] (because the initial consonants are both followed by [a]), but finding this pair of words does not contribute a lot to understanding the language. The sounds are simply so different from each other that there is little likelihood that they are alternative pronunciations of the same sound. We suspect they contrast with each other even if we don’t find any words that support the decision.

In this section, we continue to focus on sounds that contrast minimally — examples where contrastive sounds are differentiated by only one feature, or by very few features.

5.4.1 Distinctive features

Let’s start with the likely contrast we found in Salasaca Quichua, shown in Table 6. The sounds [t] and [tʰ] were found in the context pair [tuta] ‘night’ and [tʰuktu] ‘cornflower’.

Let’s look at [t] and [tʰ]. You may recall that [t] and [tʰ] are both voiceless coronal stops. How are these sounds different from each other? The sound [tʰ] is an aspirated stop — the raised [ʰ] tells us that the sound is made with extra

air coming out when it is released. The sound [t] is called a stop or a plain stop because it has no special additional property, such as aspiration. (If you'd like more review on any of the features of [t] or [t^h], go back to Chapter 2 where Section 2.2.1 introduces voicing and voicelessness, Section 2.3.1.1 explains stops, Section 2.3.2.2 discusses coronals, and aspiration is presented in Section 2.3.2.5.)

We can visually display the key properties for these two sounds, with the features labeling rows and the appropriate feature values filling in the cells for the two sounds [t] and [t^h]. This is done in Table 8. The boxed row shows the feature difference between the two sounds — there is a difference of only one feature, [±spread glottis].

Table 8: Distinctive feature for [t] and [t^h]

plain [t]	aspirated [t ^h]
[−voice]	[−voice]
[−continuant]	[−continuant]
[coronal]	[coronal]
[−spread glottis]	[+spread glottis]

The Salasaca Quichua pair [t] and [t^h] is found in a context pair, suggesting the sounds are contrastive. With our feature investigation, we can say that the contrast is made by the feature [±spread glottis], with [t] being plain and [t^h] being aspirated. Features that define a contrast, like the contrast between plain and aspirated stops, are known as **distinctive features** within a language: [±spread glottis] is a distinctive feature in Salasaca Quichua.



Comprehension Check 5

What is the distinctive feature demonstrated by Potawatomi [z] and [ʒ]? Examine the table below to figure it out.

[z]	[ʒ]
[+voice]	[+voice]
[+continuant]	[+continuant]
[coronal]	[coronal]
[+anterior]	[−anterior]



Here's our answer

The table shows us that [z] and [ʒ] are distinguished by the place feature [$\pm$ anterior], showing that this is a distinctive feature in Potawatomi. The difference is shown by the boxed features in the table below.

[z]	[ʒ]
[+voice]	[+voice]
[+continuant]	[+continuant]
[coronal]	[coronal]
[+anterior]	[−anterior]



Comprehension Check 6

What is the distinctive feature that distinguishes Yakama [x] and [χ]? Create a feature table to help figure it out.



Here's our answer

The table below shows us that [x] and [χ] are distinguished by place of articulation. Velar [x] and uvular [χ] are both [dorsal] and [+back], but velars are [+high] and uvulars are [−high]. Thus, [$\pm$ high] is a distinctive feature in Yakama. The difference is shown by the box in the table below; we only show place feature here (manner and laryngeal features are the same for these two sounds, [+continuant, −voice]).

[x]	[χ]
[dorsal]	[dorsal]
[+back]	[+back]
[+high]	[−high]

In many languages, we can find a single feature difference between pairs of contrastive sounds. With multiple contrasts (like we find with minimal sets), there will be multiple features to distinguish all the contrastive sounds in the set, but if we look at the sounds pairwise, individual contrastive sounds are likely to be distinguished from some other sound by a single feature. This is the usual case; in some languages, though, a contrast cannot be defined by a single feature. If you need to know about how multiple features can define a contrast, Section 5.4.2 gives some examples. Otherwise, you might want to skip ahead to Section 5.4.3.



Exploring my language

Choose some of the contrasting sounds that you’ve found in your language — include both a consonant pair and a vowel pair. Can you identify a single distinctive feature that distinguishes one sound from the other for each pair?

5.4.2 More than one feature making up a contrast

In some instances of contrast, there is a contrast that cannot be categorized neatly as the result of a single distinctive feature. Instead, two or more properties combine to make the difference. We look briefly at two examples, one from Potawatomi and one from maqlaqsyals, also known as Klamath-Modoc. According to Dupris 2015: 10, “the capitalization of letters in the language represents a quality of sound represented. As a result, if words in maqlaqsyals are the initial word of a clause, they may not be capitalized per the revised orthography.”⁹

⁹We thank Joseph Dupris for checking our discussion of the maqlaqsyals forms and helping improve the discussion.

5.4.2.1 The lenis/fortis distinction in Potawatomi

We return to Potawatomi (see also Section 5.2.2) to look at a contrast that some researchers on the language have called a *fortis/lenis* distinction.

The terms *fortis* and *lenis* are used to describe sounds with stronger and weaker articulation. The strength dimension involves tendencies towards a variety of factors — length, likelihood of voicing, likelihood of frication, whether or not the sound is aspirated. Fortis sounds are typically longer and tend towards voiceless, aspirated stops. Lenis sounds are typically shorter and tend toward voiced, unaspirated fricatives. The fortis/lenis distinction is one of tendencies rather than absolutes — when pronouncing fortis sounds, speakers tend to use the stronger properties and when pronouncing lenis sounds, speakers tend to use the weaker properties. This is what we find in Potawatomi.

In Potawatomi, there are two series of stops, those written *p, t, k* and those written *b, d, g*. When words are recorded and sounds are measured for length, the sounds in the *p-t-k* series are around twice as long as the sounds in the *b-d-g* series. (This measurement is based on the length of time that the stop closure is held before opening the lips or releasing the tongue.) Sounds in the *p-t-k* series can also be aspirated and typically they sound voiceless. The *b-d-g* series is typically held for a short time, is never aspirated and may sound voiced. These two sets of properties are summarized in Table 9 (from Lockwood 2017: 18).

Table 9: Potawatomi fortis and lenis consonants

	fortis [p], [t], [k]	lenis [b], [d], [g]
length	longer	shorter
aspiration	sometimes	no
sounds voiceless	usually	occasionally
sounds voiced	occasionally	usually

We can see from these properties that, while length divides the two series consistently, there are also a variety of articulatory tendencies in each category. We often find sounds described as fortis and lenis when there is a constellation of tendencies towards a more precisely produced sound (fortis) or a less precisely produced sound (lenis), rather than a single property distinguishing the two sets of consonants from each other.

Fortis and lenis consonants are represented in different ways using IPA symbols. The Potawatomi system is common: In Potawatomi, the symbols used are those that correspond to voicing, not to length — fortis [p], [t], [k] and lenis [b],

[d], [g]. These representations follow conventions established by the Wisconsin Native American Languages Project (Nichols et al. 1981). Contrasts are shown in (6); a minimal pair in (6a) shows that [p] and [b] contrast while, in (6b,c), context pairs show the likelihood of contrast between [t] and [d], and between [k] and [g]. (Sometimes it is hard to find good illustrations that two sounds contrast: Lockwood (2017) gives the pair in (6b) to illustrate the contrast between fortis [t] and lenis [d], even though the following vowel is not the same in this pair of words.)

(6) Potawatomi fortis/lenis contrasts

- | | | | |
|----|-------------------|-----------------|-------------------------------|
| a. | <i>pené</i> | [pəne:] | ‘always’ |
| | <i>bené</i> | [bəne:] | ‘stop it’ |
| b. | <i>tebyankwé</i> | [təbjɑ: nkwe:] | ‘s/he has curled up hair’ |
| | <i>débabjegen</i> | [de:ba:bdʒəgən] | ‘watchtower’ |
| c. | <i>kadénesh</i> | [ka:de:nəʃ] | ‘braid him/her/it (animate)!’ |
| | <i>gadén</i> | [ga:dʔen] | ‘push him/her/it (animate)!’ |



Another way to represent fortis and lenis

Early researchers used a length distinction to show whether a sound was fortis or lenis, representing fortis consonants as [pp], [tt], and [kk] and lenis consonants as [p], [t], and [k]. This led to ambiguities: Is [ppp] a sequence [pp] followed by [p], or is it [p] followed by [pp]? Using the Wisconsin Native Languages Project system removes the ambiguity: [pb] and [bp] are obviously distinct from each other.

5.4.2.2 Interacting length and quality in maqlaqsyals

The maqlaqsyals language provides another example where contrastive sounds are distinguished from each other by a complex of properties. (The language, also called Klamath-Modoc, is an isolate from southern Oregon and northern California, USA; our source is Dupris 2015.) With maqlaqsyals, we are looking at vowels, not consonants. As with fortis and lenis consonants, in maqlaqsyals, there is an interaction between quality and length in vowels.

As explained in Section 2.4.1.5 of Chapter 2, **quality** refers to the sound of a vowel independent from its length. Quality properties include height ([±high], [±low]), tongue body position [±back], lip position ([±round]), and tongue root position ([±atr]). Thus, [i] and [i:] have the vowel quality of [+high, -back, -round]; they differ in **length**.

There are eight vowels in maqlaqsyals, four short vowels and four long vowels, written *i* and *ii*, *e* and *ee*, *a* and *aa*, and *u* and *uu*. These symbols suggest that there are four vowel qualities cross-cut by length since each vowel sound can be either long or short. However, a closer check on the way these vowels are produced shows that the length difference aligns with a quality difference for each pair.

The high front vowels written *i* and *ii* differ not only by length but also by quality, a distinction shown by the feature [±advanced tongue root] (abbreviated [±atr]). Both of these vowels are [-high, -back, -round], but the short *i* [ɪ] is [-atr], having retracted tongue root, while long *ii* [i:] is [+atr], due to its advanced tongue root, shown in (7).

As you look at these examples, you may note an interesting stylistic convention in the example titles involving the maqlaqsyals language. The writing conventions of maqlaqsyals do not capitalize the language name. However, this book has adopted a convention of capitalizing the first word of examples. Out of respect for the people whose language this is, we let their convention dominate our convention, and so do not capitalize the language name maqlaqsyals in these example titles.

(7) maqlaqsyals *i* and *ii*

shared	short [ɪ]	long [i:]
[+high, -back, -round]	[-atr]	[+atr]

The [-high, -back] vowels written *e* and *ee* differ not only in length but also by height and tongue root position. Both vowels are front but neither is high nor round; short *e* [ɛ] is [-low, -atr] while long *ee* [æ:] is [+low, +atr], shown in (8).

(8) maqlaqsyals *e* and *ee*

shared	short [ɛ]	long [æ:]
[-high, -back, -round]	[-low, -atr]	[+low, +atr]

The remaining non-round vowels are written *a* and *aa*, yet only one of the vowels is low. The vowels are neither high nor front nor round: [-high, +back, -round]. The short vowel *a* is produced as [-low] [ə] while the long vowel *aa* is produced as [+low] [ɑ:], (9).

- (9) maqlaqsyals *a* and *ɑ*

shared	short [ə]	[long [ɑ:]]
[-high, -round, +back]	[-low]	[+low]

Finally, the two [+round, +back, -low] vowels are written *o* and *oo*, suggesting that they are mid vowels. This is true of short *o* [o]. However, long *oo* is pronounced as [u:] – a [+high] vowel, not mid at all despite the spelling, (10).

- (10) maqlaqsyals *o* and *oo*

shared	short [o]	long [u:]
[-low, +back, +round]	[-high]	[+high]

When we look at these differences as a whole, we see that long vowels are more extreme – they are high or low while most short vowels are mid; the front long vowels are advanced where the front short vowels are retracted. There is no single quality dimension to characterize the differences between long and short vowels in maqlaqsyals. The strategy made when teaching the language is to use the symbols orthography, or writing system – *i/ii*, *e/ee*, *a/aa*, and *o/oo* – and to explain to students how each vowel is pronounced. In some transcriptions, you might see [i]/[i:], [e]/[e:], [a]/[a:], and [o]/[o:], where the author either ignores the different details, or explains them but chooses to use a more systematic transcription system anyway.

5.4.3 Summary: The value of finding the precise contrast

When we identify contrast that is based on a single feature, we have the strongest possible evidence that that distinctive feature is relevant in our language. This is important because the distinctive features of a language provide tools for better understanding other aspects of the sound pattern of the language. We list some of the advantages here.

1. A feature that is contrastive for one pair of sounds is often contrastive for other, **similar** sounds so can lead to discoveries about those other pairs of sounds. For instance, knowing that aspiration is distinctive in Salasaca Quichua coronal stops [t] and [t^h] leads us to check for aspiration contrasts with other sound pairs: The stops [p]/[p^h] and [k]/[k^h], and the affricates [tʃ]/[tʃ^h]. Testing these reveals that these pairs, too, are contrastive.
2. The distinctive features in a language are typically the features that play an important role in the sound patterns of the language – governing distribution and alternations. If we know the distinctive features for a language,

we are better prepared when we approach distribution and alternations. We will see distinctive features playing important roles in Chapters 7–9.

3. Distinctive features provide a very handy way to visually organize the sounds of a language so that it is easy to see what the sounds are in a language. This is the topic of our next section.



Exploring my language

Revisit the contrastive pairs of sounds that you identified in the previous explorations in this chapter.

What are the features that distinguish the contrastive pairs from each other? Make a table for each pair of sounds to help you identify the distinctive features for these sounds in your language. Use Table 8 as a model. Decide whether your pair illustrates a distinctive feature.

If you found a pair illustrating a distinctive feature, write out your statement of that the evidence is and what the distinction is.

This exercise supports claims about what the contrastive sounds are in your language. It also helps you to think about the distinctive features of your language, preparing yourself to develop an inventory for the sounds of your language.

5.5 Inventories

We have said that the term *inventory* refers to the sounds used in a language. Exactly what “the sounds used in a language” means is tricky because precise physical sound varies so much from utterance to utterance, let alone from speaker to speaker. Typically, *inventory* refers to one of two things. The most common use is in reference to only the sounds that contrast. This kind of inventory is sometimes called a *phonemic inventory*, because linguists use the term *phoneme* to refer to contrastive sounds. Another use refers to both the sounds that contrast and the sounds whose properties, while not contrastive, are systematic. This kind of inventory may be referred to as *phonetic*, meaning it includes some sounds that are not contrastive, or as *allophonic*, meaning it includes *allophones* as well as

phonemes. The concepts of *phonemes* and *allophones*, which give rise to the terms *phonemic* and *allophonic*, are discussed in Chapter 8 Section 8.7.2.

Because you know how to test whether sounds contrast, you know how to determine many, if not all, of the contrasting sounds in your language. These are the sounds that belong in your language's inventory.

In this section, we talk about the most common way to display the sounds of an inventory — the first step being to separate all the sounds into consonants and vowels, to create both a [consonant inventory](#) and a [vowel inventory](#). Knowing how consonant and vowel inventories are represented is useful in reading discussions of your language. It is also useful when presenting information about your language to others (including to your future self).

5.5.1 Consonant inventories

The [consonant inventory](#) is the collection of contrastive consonants used in the words of a language. As such, it is a critical element when describing or documenting a language.

Typically, each consonant inventory is presented in a table with a very specific arrangement. The arrangement makes use of the distinctive features that are used to differentiate the language's contrastive consonants from each other. In this section, we show how these tables are constructed. Consonant inventories are often accompanied by a list of the contrastive consonants, often with minimal pairs demonstrating the contrasts. You're prepared to do this, if you've been keeping up with the Language Explorations.

Once you have started to identify the distinctive features for the contrastive consonants in your language, you are in position to begin to build a consonant inventory for your language. This can be done using distinctive features or descriptive terms. We illustrate each with the Seneca consonant inventory, starting with the distinctive features.

Seneca is an Iroquoian language spoken in New York, USA; we follow Dean-John (1983). Hazel Dean-John not only earned a doctoral degree, she also was a Faith Keeper of the Seneca Nation.



Language and spirit of an Indigenous community

Hazel Dean-John, Seneca

“It is important for the reader to understand that as a Faith Keeper of the extant oral traditional native way of life (spiritual) of the Seneca Nation Longhouse of Steamburg, New York, I could not do an empirical study or conduct a random sampling of taped or recorded speech of the Seneca people from my reservation or any of the other Seneca reservations in New York State and Canada. Consequently, this analysis will be an analysis of my personal speech...

Not only would it have been immoral, but unethical, for me to exploit the people to whom my first responsibility is the maintenance of, and adherence to, the principles and teachings of the religious code of life. Neither [sic] may I benefit personally or financially from the knowledge I possess of the language, the culture, and traditions of the Senecas and the other Iroquois Nations. Moreover, I may not publish any material without the approval of the Head Faith Keepers or the Wampum Keepers of the Seneca Nation Longhouse.

The language of the Senecas and the Iroquois Nations was developed for a culture which evolves around a spiritual core in unity and oneness with the Creator, the universe and all it contains. Because of the emotion and feeling in Seneca utterances, one sentence can reduce the hearer to tears because of its content. As youngsters, we are taught to think before we speak and speak only when the heart and mind are in unison.”

Dean-John (1983: v–vi)

An inventory showing distinctive feature contrasts is given in Table 10 for Seneca. We label the rows with sounds and the columns with the distinctive features. Cells are filled with feature values for the sound-feature pair. Gaps show features that are not contrastive for the segment. (Please refer back to the discussion of figures like (1), (2), and (5) in Chapter 2 if you need a refresher.)

A distinctive feature layout is helpful as a reference tool when examining the phonological patterns of a language. Customarily, place of articulation features are grouped together and manner of articulation features are grouped together; we have put place of articulation features at the left edge of our table in Table 10.

Table 10: Seneca consonant inventory — distinctive features

	[cor]	[ant]	[dor]	[cont]	[del rel]	[nas]	[vce]
[t]	[cor]	+		—			—
[ts]	[cor]	+		—	+		—
[s]	[cor]	+		+			—
[d]	[cor]	+		—			+
[dz]	[cor]	+		—	+		+
[n]	[cor]	+		—		+	+
[tʃ]	[cor]	—		—	+		—
[ʃ]	[cor]	—		+			—
[ɲ]	[cor]	—		—		+	+
[j]	[cor]	—		+			+
[k]			[dor]	—			—
[g]			[dor]	—			+
[ŋ]			[dor]	—		+	+
[w]			[dor]	+			+
[ʔ]				—			—
[h]				+			—

Another way to present the consonant inventory of a language is to use descriptive terms, as in Table 11. We usually see rows labeled with manner and laryngeal descriptors, and columns labeled with place descriptors. Which place descriptors are included? Those that are distinctive in the language. In Seneca, [labial] is omitted entirely since there are no labial consonants. Similarly, we include only the distinctive manner and laryngeal descriptors.

Table 11: Seneca consonant inventory — descriptive terms

	coronal		post-alveolar		velar		glottal
stops	t	d			k	g	ʔ
affricates	ts	dz	tʃ	dʒ			
fricatives	s		ʃ				h
nasals		n		ɲ			
approximants				j		w	

The Seneca inventory in Table 11 is arranged in one of the most typical ways we find for inventories. Let’s look at the details, starting with column labels.

Column labels. Columns show place of articulation. The place descriptors are ordered in a specific way, from the front of the mouth to the back of the mouth.

Typically, the only column labels that are used are the ones that are contrastive in the language. For Seneca, as with many languages, this leaves some gaps – such as labial (which in another language would be the leftmost column), dental (between labial and coronal), alveo-palatal (between coronal and post-alveolar), and uvular (between velar and glottal).

Another column convention is used in Table 11, and highlighted with column shading in Table 12. The left side of each column shows only voiceless sounds. The unshaded right side of each column shows only voiced sounds. In Table 12, we use gray shading to show the leftside of the column space that is reserved for voiceless consonants. Voiced consonants belong on the right side, in the white space.

Table 12: Seneca consonants: Voicing shown within columns

	coronal		post-alveolar		velar		glottal
stops	t	d			k	g	ʔ
affricates	ts	dz	tʃ	dʒ			
fricatives	s		ʃ				h
nasals		n		ɲ		ŋ	
approximants				j		w	

An alternative way to represent laryngeal properties is to add additional rows. This is shown in Table 13, where there is a row for voiceless stops and a row just below it for voiced stops, and so on.

Table 13: Seneca consonant: Voicing shown in separate rows

	coronal	post-alveolar	velar	glottal
voiceless stops	t		k	ʔ
voiced stops	d		g	
voiceless affricates	ts	tʃ		
voiced affricates	dz	dʒ		
fricatives	s	ʃ		h
nasals	n	ɲ	ŋ	
approximants		j	w	

The same information is presented in all three tables, Table 11, Table 12, and Table 13. The visual layout on the page differs as does the amount of space each requires, but the information has not changed.

Row labels. Each row in a consonant inventory shows a manner of articulation. This can be coupled with laryngeal features (like voice, as we saw in Table 13). The typical arrangement for a consonant inventory follows a pattern of increasing openness, from the most consonantal (the consonants with the greatest constriction of the airflow, the stops) and ending with the least consonantal of the consonants (the consonants with the least constriction of the airflow, the approximants).

As you read different descriptions, you may come across any or all of the conventions introduced here. Some writers assume that their readers understand the conventions and do not explain them, so please refer back to this section if you get puzzled.



Comprehension Check 7

Samantha Prins

Niisti'powahsin (Blackfoot) is a Plains Algonquian language spoken in the northwestern plains of North America, including in Montana, United States, and Alberta, Canada. Niisti'powahsin is written with a [practical orthography](#) (Frantz 1978).

orthography	IPA	orthography	IPA	orthography	IPA
<i>p</i>	[p]	<i>n</i>	[n]	<i>h</i>	[x]
<i>t</i>	[t]	<i>ts</i>	[ts̥]	<i>'</i>	[ʔ]
<i>k</i>	[k]	<i>ks</i>	[ks̥]	<i>w</i>	[w]
<i>m</i>	[m]	<i>s</i>	[s]	<i>y</i>	[j]

Some Niisti'powahsin sounds can be either long and short. Long consonants are shown by double letters in the [practical orthography](#), such as *nitákkawa* [niták:awa] 'my friend'. Here are some words that illustrate the long and short consonants in Niisti'powahsin.

<i>poos</i>	‘cat’	<i>nínaawa</i>	‘man’
<i>kiipíppo</i>	‘one hundred’	<i>nínna</i>	‘my father’
<i>takáa</i>	‘who’	<i>tsimá</i>	‘where’
<i>áipottawa</i>	‘he’s flying’	<i>ksisóyi</i>	‘teakettle’
<i>kitsími</i>	‘door’	<i>ísska</i>	‘pail’
<i>nitákkawa</i>	‘my friend’	<i>ihkítsika</i>	‘seven’
<i>mamííwa</i>	‘fish’		
<i>iyímmiit</i>	‘laugh!’	<i>otahkóínattsi</i>	‘yellow’

Your job is to fill in the Niisti’powahsin inventory chart below, putting the sounds in the correct places. *C* indicates a short consonant and *C:* indicates a long consonant.

	labial		coronal		post-alveolar		velar		glottal	
	C	C:	C	C:	C	C:	C	C:	C	C:
stops										
affricates										
fricatives										
nasals										
liquids										
glides										



Here’s our answer

Here’s how we filled in the table for Niisti’powahsin, using the IPA symbols, with the Frantz symbols in parentheses when they differ from the IPA. (Using parentheses in an inventory is a common way of showing that a symbol is of a different status than the symbols that are not in parentheses. In this case it shows the Frantz symbols. In another case it might show sounds used only in borrowed words, or only in some dialects, or something else.)

	labial		coronal		post- alveolar	velar		glottal
	C	C:	C	C:	C	C	C:	C C:
stops	p	p: (pp)	t	t: (tt)		k	k: (kk)	ʔ (ʔ)
affricates			ṭs (ts)	ṭs: (tts)		ḵs (ks)		
fricatives			s	s: (ss)				h (x)
nasals	m	m: (mm)	n	n: (nn)				
glides	w				j (y)			

Notice that we have interpreted the sequence *tts* as a long affricate, [ṭs]. An alternative interpretation would be as a sequence of *t* followed by *ts*, in which case there would be no long counterpart to [ṭs] in the inventory chart. Looking more closely at the language could provide clues telling us which interpretation is better.



Exploring my language

Try making a consonant inventory chart for your language. Do you need more column or row labels than are needed for Seneca or Niisti'powahsin?

If you are having trouble figuring out where to put the additional columns and/or rows, please review the above two sections about column and row labels.

5.5.2 Vowel inventories

In this section, we turn to *vowel inventories*. Once you have started to identify the contrastive vowels in your language, you can start constructing a vowel inven-

tory chart. As you identify more contrastive vowels, you can add them to your chart, adjusting the chart as needed.

As with consonants, vowel inventories can be presented in terms of distinctive features, or using descriptive terms. An example is given in Table 14, showing the vowels of Ja’a Kumiai, a Cochimi-Yuman language spoken near the border between Mexico and the United States (California).¹⁰ Our source for Ja’a Kumiai is Mai et al. (2019), who checked our use of the language.

Table 14: Ja’a Kumiai vowel inventory

	[i]	[e]	[a]	[o]	[u]
[±high]	+	–	–	–	+
[±low]	–	–	+	–	–
[±back]	–	–	+	+	+
[±round]	–	–	–	+	+

As with consonants, each cell shows how a specific feature classifies a specific sound. Again, this kind of array is very useful when you are working through a phonological analysis of a language.

You may want a more descriptive tool for other purposes. We turn now to inventory layouts using descriptive features.

Column labels. The most common chart structure uses the front/back dimension to label columns, sometimes augmenting this with round and non-round as needed. Front is at the left and back is at the right. (This corresponds to the consonant chart, keeping the front of the mouth at the left side.)

Row labels. Height descriptors are used to label rows. The descriptor for the highest tongue position is at the top, going down to the lowest tongue position at the bottom. In many systems, the labels will be high, mid, and low (you may also see close, mid, and open).

Some languages only make use of two height dimensions while others have more than three dimensions, adding labels like advanced and retracted tongue root, or upper mid (for mid advanced vowels) and lower mid (for mid retracted vowels). (See Chapter 2, Section 2.4.3.1 on advanced and retracted tongue root features).

An example is given in Table 15 for the five Ja’a Kumiai vowels.

In Table 15, we see the most common way that these five vowels are represented. The columns show only the back/central/front dimension, ignoring

¹⁰Arrays of vowels using distinctive features are also found in Section 2.4 of Chapter 2.

Table 15: Ja'a Kumiai vowel inventory

	front	central	back
high	i		u
mid	e		o
low		a	

rounding. In Table 16, detail about rounding is added. The system shown in Table 15 is used more commonly, but both are correct. Labels for roundness are safely omitted in Table 15 because the other properties listed are able to distinguish all five vowels. It does not mean that there are no round vowels — or no non-round vowels — in the language, just that rounding isn't critical for distinguishing the five vowels. (You might also have thought about representing the vowels using the non-round/round dimension rather than front/back to label rows. While correct, it is less commonly used.)

Table 16: Ja'a Kumiai vowel inventory — rounding added

	non-round		round
	front	central	back
high	i		u
mid	e		o
low		a	

While front is typically at the left edge of a vowel chart and back at the right edge, when rounding is added, it is typically done within the front and back categories. The standard is to put non-round on the left side of each column and round on the right side. (This is similar to the use of columns-within-columns to show voicing, as in Table 11.) We illustrate with the vowel inventory of Chuxnabán Mixe (of the Mixe-Zoquean family, spoken in Oaxaca, Mexico; Jany 2011).¹¹

Chuxnabán Mixe has six modal vowels, including front round vowels along with the more common front non-round vowels. These vowels are given in Table 17. See Chapter 2 Section 2.4.2.5 on modal vowels and Section 2.4.3 on non-modal vowels. (The vowels [ø], [y], and [æ] that are heard in Chuxnabán Mixe speech are sometimes called “marginal vowels” because they have limited distribution. On distribution, see chapters 7 and 8.)

¹¹We thank Carmen Dagostino (formerly Jany) for her help improving our discussion of Chuxnabán Mixe. She tells us that, in general, Mixe people call their language *ayuuk* [aju:k] and call themselves *ayuukjä'äy* [aju:khi?ij].

Table 17: Chuxnabán Mixe modal vowel inventory

	front		central		back	
	non-round	round	non-round		round	
high	i	ɤ	ɨ		u	
mid	e	ø			o	
low	æ		a			

Modal vowels are “plain” vowels. When linguists talk about the number of vowels that are found in a language, they often count only the modal vowels. Thus, Chuxnabán Mixe is said to have nine vowels, despite the fact that Chuxnabán Mixe also has four distinct cross-cutting features that are used with all of the modal vowels, giving 36 distinct vowels — length ([a] vs. [a:]), nasality ([a] vs. [ã]), creak ([a] vs. [a̰], symbolized as [aʔ] and [aʔa] in Janý 2011), and aspiration ([a:] vs. [aʰ]).

When languages have both modal and non-modal vowels, there is a question of how to present the vowel inventory. Rather than using Chuxnabán Mixe and its four non-modal categories as our example, we switch to the Siouan language Hiráaca, which has only one non-modal category, length. However, Hiráaca also has diphthongs, so we get to see how those are represented as well.

Hiráaca (spoken in North Dakota, United States, Park 2012) is described as having five vowel qualities, or five modal vowels. Some of these vowels occur both long and short while others have only a long version. The Hiráaca vowel inventory is given in Table 18.

Table 18: Hiráaca vowel inventory

	front	central	back
high	i, i:		u, u:
mid	e:		o:
low		a, a:	
diphthongs	ia		ua

Non-modal vowels in vowel inventories. Non-modal vowels can be presented alongside modal vowels in a single chart, as in Table 18. This way of presentation makes it a little tricky to see which vowels do not have both counterparts. Because of this, some researchers will present two inventories for a language that uses a non-modal feature with vowels, as in Table 19.

Table 19: Hiráaca vowel inventory

a. short vowels			b. long vowels				
	front	central	back		front	central	back
high	i		u	high	i:		u:
mid				mid	e:		o:
low		a		low		a:	
c. diphthongs:			ia	ua			

The two-chart method shown in Table 19a, b can be used in many different ways. For example, it is sometimes used in languages with plain and nasal vowels, or plain and creaky vowels. It can also be expanded to more than two charts if a language has multiple cross-cutting features, as in Chuxnabán Mixe.

Diphthongs in vowel inventories. **Diphthongs** are sequences of two vowels that belong together. They usually act rather like long vowels, except that there are two different vowel qualities in a diphthong, not just one. Typically, the two vowels in each diphthong are vowels from the vowel inventory of the language. For this reason, diphthongs may be counted as composites of two other vowels, not independent vowels of a language. (See Chapter 6 Section 6.1.3.1 for more information about diphthongs.)

For this reason, diphthongs are sometimes not listed with a vowel inventory chart at all. If they are included, they are listed separately from the inventory itself, as we’ve done in both Table 18 and Table 19.



Comprehension Check 8

Kickapoo is an Algie language spoken in Oklahoma. Our source, Blue-cloud (2020), gives eight Kickapoo vowels: [i], [ɪ], [o], [ɔ], [ə], [æ], [ɑ]. Try making a vowel inventory chart displaying these eight vowels. You can make one with distinctive features or with descriptive terms or, for bonus points, make both.



Here's our answer

Here's the distinctive feature chart that we came up with for the eight Kickapoo vowels, [i], [ɪ], [o], [ɔ], [ə], [æ], [ɑ]. The vowels [ə], [æ], [ɑ] are not clearly advanced or retracted from the descriptions so we do not give them values for [±advanced tongue root] ([±atr]).

	[i]	[ɪ]	[e]	[æ]	[ɑ]	[ə]	[ɔ]	[o]
[±high]	+	+	–	–	–	–	–	–
[±low]	–	–	–	+	+	–	–	–
[±atr]	+	–	+				–	+
[±back]	–	–	–	–	+	+	+	+
[±round]	–	–	–	–	–	–	+	+

Here's the descriptive chart we came up with for the eight Kickapoo vowels.

	front	central	back
high advanced/upper high	i		
high retracted/lower high	ɪ		
mid advanced/upper mid	e		o
mid retracted/lower mid		ə	ɔ
low	æ	ɑ	

How did you do with these? Notice that [ə] is placed slightly below the mid advanced/upper mid row and slightly above the mid retracted/lower mid row because we don't know which category it belongs to.



Exploring my language

Try making a vowel inventory chart for your language. Do you need more column or row labels than are needed for Ja'a Kumiai, Chuxnabán Mixe, or Hiráaca?

Do you have a cross-cutting feature like nasality or length or creak, that might be best represented by having two (or more) charts?

If you are having trouble figuring out where to put additional columns and/or rows, it might help to review the guidelines in this section.

5.6 Discussion

This chapter has introduced the concept of contrastive sounds in a language, the skills needed to identify contrast, and the techniques for presenting contrastive sounds visually.

5.6.1 Resolving the puzzle

We started this chapter with the puzzle of how we can know that a word can or cannot be part of a language just by hearing someone say the word.

What is it about language sounds that tells you that they can or cannot be part of words in a particular language?

The short answer is that each language uses only some of the sounds that humans can make. A word with sounds that are not part of your language is not a word in your language. The longer answer shows how to use minimal pairs and minimal sets to identify [segments](#) that are in a particular language. We extended that answer to look at which [features](#) are critical to define the sounds of a language, including the features that distinguish sounds from each other. Those features are the distinctive features in the language.

Identifying the sounds of a language gives the tools necessary to transcribe words of the language into the IPA. Finding the distinctive features of the language gives tools that are necessary to precisely characterize the sound patterns found in a language, a topic explored next, in Chapters 7, 8, and 9.

5.6.2 Sounds and writing systems

Throughout our discussion, we have presented words in three ways when possible — using the community’s writing system ([orthography](#)), using transcription in the [IPA](#), and using the word’s meaning. For example, in our discussion of the

Potawatomi fortis/lenis contrast, we gave a minimal pair in Table 9a showing the community writing system, IPA, and meaning, repeated in Table 20.

Table 20: Potawatomi, repeated from Table 9a

orthography	IPA transcription	meaning
<i>pené</i>	[pənɛː]	‘always’
<i>bené</i>	[bənɛː]	‘stop it’

To review, the orthographic presentation of a word and the transcription of the same word are similar in that both are visual presentations of the word. They are different because an orthography creates a word to be read, while a [transcription](#) shows how the spoken word sounds. In this work, we include IPA transcriptions because our focus is on the sound patterns of languages. The IPA transcriptions are meant to be consistent across languages, so we can use them to get a rough approximation of how each word sounds.

We also include the orthographic representations where possible. These are useful if you happen to know one of the languages we are discussing, and are accustomed to the community writing system, but not accustomed to seeing words transcribed in the IPA. It is also useful even if you don’t know the language, to see how different communities have dealt with problems they face in converting the sounds of a word to an alphabetic system, for writing words to be read.

As an example, a challenge for the Potawatomi community is how to represent different vowels in their written language. The example in Table 20 has two different vowels, [ə], which is written *e*, and [ɛː], which is written *é*. The orthographic system uses the same symbol, *e*, for both vowels, except that one of the *es* has a diacritic above it, *é*. (A [diacritic](#) is an extra mark on a letter.) The two letters, *e* and *é*, indicate that the two vowels sound different from each other, which helps the writing system show that words are distinct from each other.

With no further information, however, we might imagine that the vowel written *e* is an advanced mid front vowel [e] and that the vowel written *é* is the same vowel, but it is also high-toned [é] or stressed, [eː]. As the transcriptions show, neither of these corresponds to the pronunciation of either vowel. Rather, *e* is the way to write the central mid vowel [ə] while *é* is the way to write a long, front, retracted mid [ɛː]. The orthographic system helps you know how to pronounce the words only if you know the correspondences between the letters and the sounds.

We see from this example how important it is to explain the use of symbols even for a phonetic writing system, where each letter corresponds fairly directly

to a single contrastive sound. Without knowing the correspondences between orthography and sound, Potawatomi words with *e* and *é* are very likely to be mispronounced.

Explaining the pronunciation of each letter used in a community writing system is critical in dictionaries and in language classes in order for others to know how to pronounce words that are not transcribed with IPA symbols.

5.6.3 Using my hard-earned knowledge

Having worked through this chapter, you now understand how to determine the sounds that are critical in a language, the contrastive sounds. These are the sounds that allow us to perceive words as different from each other simply by the way they sound. You have developed skills to figure out the specific features that result in sounds being contrastive. And you know how to use those features in visually presenting the sounds of your language. How does this kind of knowledge serve a language worker?

Language learning. An early lesson for students is to learn the sounds of a language. By knowing which sounds are contrastive, a teacher knows which sounds the students must learn. Sometimes the sounds to be learned are not contrastive in the language of instruction. When such sounds are quite similar to each other, they may require special attention in learning both how to perceive the sounds and how to produce them. Knowing the distinctive feature that distinguishes two such sounds gives the teacher clues about how to help students, especially with production. Some of the pronunciation teaching tips from Chapter 2 may be useful if you are faced with this situation.

Language resources — dictionaries, grammars, orthographies. Some communities want a writing system for their language, to create books, grammars, and dictionaries. Writing systems use letters that generally correspond to the contrastive sounds in the language, so knowing those sounds is a critical step.

Sometimes there are sounds in a language for which there is no obvious symbol on a keyboard. Communities make different decisions in these cases. For example, speakers of a language with contrastive velar [x] and uvular [χ] may choose to use the letter *x* for both sounds because *x* is easy to type. However, this can lead to confusion — if you don't already know a word with *x* in it, you won't know whether to use velar [x] or uvular [χ] if you read *x*. There are different options available, that avoid the confusion. One is to use a diacritic to indicate one of the sounds. For instance, an underlined *x* — x — could be used for the uvular fricative [χ], saving *x* for the velar fricative. Another option is to create a

digraph — a sequence of two letters to indicate one sound. For example, *xh* might be used to indicate the uvular fricative.

There is a concern with digraphs. Suppose the language uses the letter *h* to represent [h], *x* to represent [x], and *xh* to represent [χ]. In this case, a person seeing a word written with *xh* might think it means the sound [χ], but the correct pronunciation is [xh], or vice versa. Depending on the number of words of each type, this sort of ambiguity might be very frustrating for people trying to use a writing system to correctly pronounce words.

A writing system specific to a language is quite useful in creating a dictionary: if the entire language can be written using letters from the *ABC* alphabet, then there is an order already established for dictionary entries. However, some communities choose to introduce special symbols for specific sounds, as in the examples above for [χ]. In this case, decisions must also be made about the order of those letters along with the other letters used, to standardize the order of entries in the dictionary.

Language documentation and description. A description of the sounds of a language is the foundation of a general description of a language. Without understanding the sounds, it is hard to represent words; without words, we can't make sentences. The sounds are the starting point.

In a description, the consonant and vowel inventories provide a snapshot of the sounds, a very handy reference tool. They also provide structure for a follow-up discussion that explains how the sounds are made. Many descriptions augment those explanations with a list of words illustrating the sounds in a variety of contexts, and showing minimal pairs, near-minimal pairs, or context pairs to establish contrast. Such word lists are useful for learners because they give examples to practice with, to learn to pronounce and perceive the contrasts.

Inventories are also useful to help when creating language documentation. The consonant and vowel inventories give a guide to the sounds that will ideally be represented in recordings of the spoken language.

Language revitalization. Efforts to re-awaken a language include increasing the resources written in the language in an effort to make it easier to use the language on a daily basis. To do this, a first step is having a writing system to use when creating materials. A writing system also facilitates active engagement with the language, through texting or emailing, list-making, and other daily casual uses — including labeling objects in our homes to make it easier to practice and remember words. There are also passive ways that written language supports a community revitalization effort, through road signs, building signs, product names, and so on.

Depending on the resources available about your language, revitalization may involve a fair amount of “behind the scenes” detective work, especially when most (if not all) resources about the language are written works. This situation can be especially challenging when the written resources span centuries, are written with varying degrees of attention to sounds, and from different points of view (including the primary language spoken by the recorder). It may be necessary in such a case to reconstruct the consonant and vowel inventories of a language from piecemeal records. Understanding that an inventory shows the contrastive sounds in a language gives an important tool for interpreting the resources available to you. (It’s important to remember that you may be reading an author who had no understanding of the concept of contrastive sounds or inventories.)

A reasonable strategy when dealing with multiple resources separated by time and language of researcher is to create the inventory for each resource and then compare those inventories with each other. The goal is to be able to appropriately pair distinct symbols to understand how the representations of sounds differ from each other. Doing so empowers you to cross-reference between sources, essentially creating a single unified source from many separate writings.

Sometimes, when the gaps in the record are quite large, it can be helpful to compare the inventories of related languages to the inventory you have reconstructed for your own language, provided inventories are available for the related languages. Finding contrasts of a particular type in related languages gives an idea of types of contrasts to look for in your own language. This does not mean you will find evidence of those contrasts — it may be that the record is too sparse; we also know that related languages are not identical to each other (though they do bear similarities).

Once you are fairly confident about the sounds of your language, you know the sounds to use in creating a new word for your language. Typically, this is done when a community discovers the need for a word that is not in the historical record (perhaps because no one recorded it; perhaps because the concept came into the community after the language was first recorded). Another source for adding words is to search related languages, and then adapt the sounds of the word from the related language to the sounds and sound patterns of your own language.

There is more to creating words than simply using sounds from your language; in subsequent chapters we look at the way that sounds are distributed in a language so that you know not only what sounds are appropriate to use but also where in a word those sounds are most appropriate.

Reading linguist’s resources about your language. Linguists often write for other linguists, rather than for speakers of a language. This means that there

are many specialized terms in linguistic writings — even in descriptions of languages. In this chapter, we have made use of the terms — and concepts — of contrast, minimal pairs, distinctive features, and inventories. These are all terms that you are likely to find in a linguist's description of the sounds of a language.

One resource that may be of interest to you is a series of articles called *Illustrations of the IPA*, in the *Journal of the International Phonetic Association*. These are known more casually as *JIPA Illustrations*. *JIPA Illustrations* are short articles that blend description and documentation, listing and describing the contrastive sounds of a language, with example words that are linked to recordings if you read the articles online. You can find the *JIPA Illustrations* here: <https://www.cambridge.org/core/journals/journal-of-the-international-phonetic-association/illustrations-of-the-ipa>. The material in Chapter 2 and this chapter has prepared you for reading these articles. (You might get bogged down in some of those articles by reference to *allophones* and *phonemes*; if so, Chapter 8 will be helpful.)

5.7 Conclusion

Through reading this chapter, you have learned what is meant when we talk about the contrastive sounds of a language. You have also learned how to determine whether there is evidence that sounds are contrastive, by using minimal pairs, near-minimal pairs, and context pairs. With these skills, you can identify the sounds that contrast in your language.

You have also learned how to use those contrastive sounds to identify the distinctive features for your language. We do this by calling on the features introduced in Chapter 2 and isolating the one feature that is different in the descriptions of a pair of contrastive sounds.

You have learned how to use those distinctive features to create charts showing the contrastive segments of your language, both the consonants and the vowels. We reviewed the conventions about column and row labels, and when it is appropriate to put two symbols in one column.

Knowing the contrastive sounds of your language is the beginning of understanding the system that gives your language its unique phonology. One of the first things we notice — and you may have already noticed this — is that even though we have found the consonants and the vowels of a language, we still cannot be confident that any sequence of those vowels and consonants makes a viable word in the language — in fact, most sequences do not make viable words. We also need an understanding of the distribution of the sound segments of the

language — where each type of sound can be found in a word. The remaining chapters of this book examine the distribution of consonants and vowels, and the role of distribution in creating the music of each language.

Chapter 6

Syllables: Putting segments into words

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Here's a puzzle. Imagine a language worker using with archival resources on Djeoromitxí, a language spoken in the state of Rondônia, Northwest Brazil. One day, she finds a source containing a list of words mixed in with other materials about Djeoromitxí. This list, however, has no information about the language that the words are from. The worker wants to know whether these could be words from Djeoromitxí that she might add to the master list she is compiling.

In the unlabeled word list, she finds the word [kanta]. Could this be a word from Djeoromitxí? She knows that Djeoromitxí has the four sounds, [k], [n], [t], and [a], so the sounds are consistent with the Djeoromitxí language. Nonetheless, her reaction is “Aha, this is probably not a Djeoromitxí word”. (Recall that the symbols inside square brackets are written in the [International Phonetic Alphabet](#) (also referred to as the IPA), showing how the sounds are pronounced. See Chapter 2 and Appendix A for more on the IPA.)

Why can a word made up of sounds of a language not be possible as a word of that language?

The answer depends on the concept of *syllable*. A syllable is a unit made up of one or more sounds, so a syllable is typically bigger than a single sound. Syllables combine together to make words, so a syllable is typically smaller than a word. Each language has a different way of combining sounds into syllables, so in some cases, it is possible to figure out if a word belongs to a language or not by identifying the syllables in the word. Djeoromitxí syllables show why [kanta] cannot be a well-formed Djeoromitxí word.

The Djeoromitxí language worker understands that syllables follow one of two patterns in the language: (i) a consonant and a vowel combine to form a syllable, or (ii) a vowel can stand alone as a syllable. With this knowledge, she would have tried analyzing the word [kanta] into syllables that match the patterns. [k] and the first [a] can be combined into a syllable [kanta] and [t] and the second [a] can be combined into another syllable [kanta]. The sequences [ka] and [ta] are both made up of a consonant followed by a vowel, the first pattern given above. However, the sound [n] in [kanta] is left over. It doesn't make a Djeoromitxí syllable for two reasons. First, there is no vowel immediately following [n] (pattern 1) and, second, [n] cannot stand alone as a syllable since it is a consonant, not a vowel (pattern 2). Therefore, the language worker concludes that the word [kanta] does not belong to Djeoromitxí.¹

The **syllable** is the third major component of a language's unique **phonology**, along with variation and contrast. Generally speaking, syllables consist of a vowel and optional consonants before and/or after the vowel. We have to figure out the syllable types for each language. We saw that Djeoromitxí has only two syllable types, a consonant plus a vowel (CV) or just a vowel (V), but there are languages that show far more variety in their syllables. In this chapter we will explore what a syllable is and how syllable types can differ by languages. We will also learn how to analyze words into syllables.

Here's our promise. After reading this chapter, you will understand the concept of **syllable**. You will have the tools to analyze a language's syllables by looking at words of the language. This will enable you to determine if a word can possibly belong to your language — by analyzing the syllables in the word, as illustrated in the puzzle above. You can use your knowledge and analytic skills to enhance language teaching/learning experience, develop an orthography system that better represents your language's syllables, be more effective in documenting your language, create new words that follow your language's syllable system, and read linguistic resources about your language's syllables.

These applications are discussed at the end of the chapter. But first let's look at our plan and then get on to understanding syllables.

Here's our plan. To fulfill our promise, this chapter does four things. It explains what syllables are (Section 6.1). It shows how to find syllables in the words of a language (sections 6.2 and 6.2.4). We show how linguists represent syllables visually and explain why different representations are used (Section 6.4). Finally,

¹Our information about Djeoromitxí comes from De Castro (2012). Djeoromitxí is a Jabutí language spoken in the Brazilian state Rondônia.

we explore a few ways that syllables help us understand phonological patterns in a language (Section 6.4.3).

More details about each section, including why you might want to read particular sections, are in the list below. It also gives guidance on why you might want to read — or skip — a particular section.

1. Section 6.1 explains what syllables are and presents various syllables types with examples showing their diversity. We also introduce terms linguists use related to syllables here. This section is useful for talking or reading about syllables and for understanding the rest of the chapter. If the terms are already familiar, you can skip the section. Read the first few paragraphs of the section to learn what the terms are.
2. Section 6.2 shows how to use word edges to figure out syllable types in a language, probably the most useful method for determining the syllables of your language. This section also shows how to use the sequences of sounds inside the word to confirm the results from word edges. Read this section if the idea is new to you and you want to know your syllables.
3. These two sections together may be sufficient for your needs. However, sometimes even the strategies in Section 6.2 aren't enough to know how a particular word is syllabified. Section 6.3 has useful tips for challenging situations.

These first three sections are designed to explain what syllables are and to help you find them in your language. This may be sufficient for your needs. If this is the case, try the Language Explorations in the rest of the chapter, but skip reading Section 6.4 and jump straight to 6.5.

On the other hand, you may find that you need to know more about syllables. It may be that syllables seem to play different types of roles in your language and just knowing where the syllables are is not sufficient. It may be that you are reading linguistic literature about syllables in your language, and the papers are difficult to understand. The next section addresses these issues.

4. Section 6.4 shows different ways to represent syllables. It also addresses reasons to use the different representations. This section is useful in helping you think about different aspects of the syllables in your language, to understand them better. It can also be very helpful in understanding work done by linguists, to make sense of the figures showing syllable structure.
5. The chapter ends with Section 6.5, answering the puzzle, explaining how loanwords can be adapted to match the target language's syllable types, and exploring ways to use the knowledge of syllables in language work.

As always with this book, read what matters most to you at this time, and determine when it is better for you to skip ahead to another section. We encourage you to keep up with the Language Explorations even when you skip a section, so that you have a more complete a description of the phonology of your language when you finish the book.

You can always go back at another time to read a section you skipped, if it becomes relevant to you.

6.1 What is a syllable?

A *syllable* is a linguistic unit that is bigger than a sound but smaller than a phonological word. A syllable usually consists of a vowel and one or more consonants, but there is a lot more to syllables than that statement. The concept of syllable may already be familiar to you; to determine if you need to read this section or can safely skip ahead to Section 6.4, read through the following list of terms to see whether you already know them: Syllable type, onset, nucleus, coda, rime, open syllable, closed syllable, light syllable, heavy syllable. You can check yourself with (10) in the summary discussion of this section. If these terms aren't familiar to you now, they will be by the time you finish this section. If you decide to skip this section, we urge you to do the Language Explorations anyway, to have a preliminary list of syllables in your language for use in later Explorations.

6.1.1 Basic terms used in talking about syllables

We use the symbol *V* for a vowel. The symbol *C* stands for a consonant. We use the symbols *V* and *C* as a shorthand when we talk about syllables. For example, we can say that in the Siouan language Nakoda, word [hi] 'tooth, teeth' consists of a CV syllable since [h] is a consonant (C) and [i] is a vowel (V). (Chapter 2 Section 2.4 explains that vowels are made with the vocal tract open so that airflow is uninterrupted; from Section 2.3, you may recall that consonants are sounds made with complete or nearly complete closure somewhere between the lips and the glottis (that is, somewhere in the vocal tract).)

Consonants are found at the beginning of a syllable before the vowel: CV, CCV, etc. Consonants can also be found at the end of a syllable after the vowel: VC or CVC, VCC or CVCC, etc. We don't find a single syllable with a consonant between two vowels. A sequence VCV is made up of two syllables, either V-CV or VC-V.

If a syllable ends in a vowel, it is called an *open syllable*. On the other hand, if a syllable ends in a consonant, it is called a *closed syllable*. The sequence V-CV is

two open syllables. What is the sequence VC-V? (We hope you said it is a closed syllable followed by an open syllable.)

If a syllable has one short vowel with no consonant after the vowel, it is a **light syllable**. A **heavy syllable** has either a long vowel, a diphthong, or a consonant after the vowel.

What kind of syllable is a V? A V syllable is an open, light syllable. It is open because it ends in a vowel, and it is light because it has only one short vowel without any consonant after it. The most common kind of syllable is a light open syllable that begins with a consonant, the CV syllable. The consonant that starts a syllable is called an onset. Every language has CV syllables.

There are other kinds of syllables besides CV syllables. Looking closely at different languages, we see that in some languages, syllables can be as small as just a single sound: V. We also see that in some languages syllables can be as large as four or even more sounds: CCVC, CVCC, CCVCC, etc. While there is the core CV-syllable similarity, there can be a lot of variation between the syllables in different languages. In general, you have to figure out what the syllables are for every language.

One place to start is by figuring out the number of syllables in a word. You can do this by counting the vowel sounds in the word. This is because each syllable typically contains a vowel.

Try this. How many syllables are in the Sierra Popoluca word [put^hpaʔ] ‘he is leaving’? Are there more syllables or fewer syllables in the Nuu-chah-nulth word [tʃ^haʔak] ‘water’? How about the Potawatomi word *tebyankwé* [təbjɑːnkwɛː] ‘s/he has curled up hair’?

Both [put^hpaʔ] and [tʃ^haʔak] have two syllables because each has two vowels (In [tʃ^haʔak], there is one vowel quality, [a], but there are two instances of [a] in the word, so there are two syllables.) There are three vowels in the third word, *tebyankwé* [təbjɑːnkwɛː] ‘s/he has curled up hair’, so it has three syllables.

Knowing the number of syllables in a word is a good start, but it doesn’t tell us the types of syllables in each language, and each language has its own set of possible syllables. As you saw in the puzzle, Djeoromitxi has only CV and V syllables — it does not have any syllables ending with a consonant. However, many languages do allow syllables ending in a consonant. In this section, we introduce a variety of syllable types found in different languages. By looking at the examples, we learn more about syllables.

6.1.2 Light syllables

Recall that a light syllable has a short vowel and no following consonant. Light syllables can have onsets, but they are not required. We look at the smallest possible syllable first, a syllable made up of only a vowel.

6.1.2.1 Syllables made up of only a vowel: V

Syllables can consist of a single vowel, so the smallest possible syllable is simply V. We see examples in (1), from Nakoda (also called Assiniboine), a Siouan language spoken in southern Saskatchewan and northeastern Montana, and from Djeoromitxí.²

In (1), we introduce a common convention — using a dot “.” to separate syllables. We use the syllable-separating dot convention a lot in this chapter; you may also see it if you read works written by linguists about your language. (We also continue to use our convention of italics to indicate how the word is written and square brackets for the IPA symbols showing how the word sounds.)



The ambiguous dot conventions

You may remember another dot convention from Chapter 3, where the dot “.” is used to separate words in a gloss, as in the Zenzontepec Chatino example *ndetiyaa* [ndetija:] ‘PRG.arrive’ (‘is arriving’). You might think that it will be very confusing, since these dots have very different meanings. However, the dot “.” separating syllables is found only in the transcription and the dot “.” showing complex glosses is found only in the gloss.

- (1) Words containing V syllables
- | | | | | |
|----|-------------|-----------|-------|---------|
| a. | Nakoda | <i>í</i> | [í] | ‘mouth’ |
| b. | Djeoromitxí | <i>oo</i> | [o.o] | ‘wind’ |

In both Nakoda and Djeoromitxí we can find syllables consisting of a single vowel. The Nakoda word *í* consists of one syllable that is just a vowel [í], while the Djeoromitxí word *oo* is composed of two syllables, each consisting of a single vowel [o].

²Our source for Nakoda is Cumberland (2005).

In most cases, a vowel is a required part of a syllable. The required part of a syllable is also called the *nucleus* of a syllable. In the Nakoda syllable [í], [í] is the nucleus. Similarly, in the Djeoromitxí syllable [o], [o] is the nucleus.

A nucleus is usually a single vowel, but there are cases where a combination of vowels takes a nucleus position. Theoretically speaking, a nucleus is the most *sonorant* part of a syllable. In some languages, it is possible that a consonant (mostly sonorant ones such as [r], [l], [m], [n]) might fill the nucleus position. Sonorants are sounds made without any impediment to the airflow. To review (or learn) about sonorants, see Chapter 2 Section 2.3.1.4.

6.1.2.2 CV syllables, the core syllable

Next let's have a look at CV syllables, the one syllable type that is found in every language. Because of this, it is sometimes called the *core syllable*. In a CV syllable, there is one consonant, followed by one vowel. Below are some examples of core syllables (CV syllables).

(2) Words containing CV syllables

- a. Nakoda *hí* [hí] 'tooth, teeth'
- b. Djeoromitxí *djiri* [dʒi.ri] 'child'

In the Nakoda word, there is one CV syllable, [hí]: A consonant [h] precedes a vowel [i]. In the Djeoromitxí word, there are two CV syllables: [dʒi] and [ri]. In both syllables, a consonant, either [dʒ] or [r], precedes a vowel, [i]. The sound [dʒ] is an affricate; affricates are discussed in Chapter 2 Section 2.3.1.3. The sound [r] is a tap, discussed in Chapter 2, Section 2.3.1.4.

Recall that the vowel is the nucleus of the syllable. The consonant preceding the nucleus is called the *onset*, because it is the start of the syllable. In the examples above, [h] is the onset of the Nakoda syllable [hí], and [dʒ] and [r] are onsets of the Djeoromitxí syllables [dʒi] and [ri], respectively. We can say that these are onset-nucleus syllables. They are also called CV syllables and core syllables.

In many languages, the onset is an optional part of a syllable. When a language has optional onsets, we find both CV syllables (syllables with onsets) and V syllables (syllables without onsets). Onsetless V syllables are also sometimes called core syllables.

Try this. Are Nakoda and Djeoromitxí languages that require onsets, or are onsets optional in those two languages?

We hope you said that onsets are optional in Nakoda and Djeoromitxí. You've seen that both languages have syllables made up only of a vowel, in (1) — these are syllables using the “no-onset option”. You've also seen that both languages have syllables made up of a consonant and a vowel, in (2) — these are syllables using the “onset option”. There are also languages where the onset is a required part of a syllable. We discuss this further in later in this chapter, in section (6.2.1.1).

Try this. Are CV syllables open or closed? Are they heavy or light? Why do you think so?

You may have noticed this question reviews concepts introduced earlier. Review Section 6.1 if you didn't answer that CV syllables are open, light syllables. (They are open because they end in a vowel, and they are light because they have only one vowel without any consonant after it.)



Comprehension Check 1

Lisa Casarez, Mandan, Hiraaca, Arikara Nation (MHA Nation), and Corey Roberts, Occaneechi Saponi

Can you identify CV syllables in words? Test yourself.

Here are five words from Hiráaca, a Siouan language spoken in North Dakota, in the United States of America. The words are provided by Lisa Casarez.

- | | | | |
|----|-----------------|---------------------------------------|-----------------------------|
| a. | <i>gagúwi</i> | [kaguwi] | ‘squash’ |
| b. | <i>nakcí</i> | [nakt̪sí] | ‘to quill something’ |
| c. | <i>abátupe</i> | [abát ^h up ^h e] | ‘vest’ |
| d. | <i>mirúxi</i> | [mirúxi] | ‘ice’ |
| e. | <i>beericga</i> | [pe:rit̪ská] | ‘sleet’ (lit. ‘raven poop’) |

Your job with these words is to sort them into two lists. One list is the words that have only CV syllables and the other list is the words that have some other syllables besides CV syllables.

A few helpful tips follow.

1. In doing this task, be sure you check the column that shows the pronunciation of these words (in square brackets), not the orthog-

- raphy (in italics), since orthography cannot always be trusted to match letters and segments in a one-to-one fashion.
2. The IPA symbol $[\text{ts}]$ represents the voiceless alveolar affricate. The sound $[\text{ts}]$ is spelled *c* in the Hiráaca writing system. See Chapter 2 Section 2.3.1.3 for more on affricates.
 3. We introduced the length symbol $[:]$ in Chapter 2 Section 2.4.3.4 – there is a long vowel in (e). Hiráaca orthography uses a double letter to indicate length: *ee* corresponds to $[\text{e}:]$.



Here's our answer

Here are our two lists of the Hiráaca words, those with only CV syllables and those with some other types of syllables.

- i. These two words have only CV syllables: Their structure is CV.CV.CV.
 - a. *gagúwi* $[\text{ka.gu.wi}]$ ‘squash’
 - d. *mirúxi* $[\text{mi.rú.xi}]$ ‘ice’

The remaining three words have other syllable types. In case you included one of the other words in your list of words with CV syllables, we explain why the consonants and vowels of these other words cannot be interpreted as a sequence of CV syllables.

2. The sequence $[\text{kts}]$ contains two consonants, $[\text{k}]$ and $[\text{ts}]$.
 - b. *nakcí* $[\text{nak.t̥sí}]$ or $[\text{na.kt̥sí}]$ ‘to quill something’
Perhaps $[\text{k}]$ is part of the preceding syllable (CVC.CV, as in $[\text{nak.t̥sí}]$); perhaps it is a part of the following syllable (CV.CCV, as in $[\text{na.kt̥sí}]$). In either case, we have something besides a CV syllable.
3. The vowel $[\text{a}]$ at the beginning is not preceded by a consonant, so the first syllable of the word is a V syllable, not a CV syllable: V.CV.CV.CV.
 - c. *abátupe* $[\text{a.bá.t̥u.p̥e}]$ ‘vest’

4. In the last word, there is a long vowel [e:] at the beginning of the word, so there is a CVV syllable, not a CV syllable.

e. *beericga* [pe:.ri.t̪ská] ‘sleet’ (lit. ‘raven poop’)

Additionally, there is a CC sequence, [t̪sk], just like we saw in [nak-t̪sí] in (b). These two issues are highlighted in CV:CVCCV.



Exploring my language

Some languages have only words with CV syllables. Is yours one of them?

To figure this out, find at least six words in your language. Divide your words into two lists, headed *CV syllables only* and *Other syllable type(s)*. In the first list, add words that have only CV syllables; put all other words in the other list. Try to get words in both lists. (You won’t be able to have words in both lists if your language has only CV syllables.)

Look at your *Other syllable type(s)* list. Do any of those words have V syllables in them? If so, it looks like your language has both CV and V syllables. You could add them to the first column, and relabel it *Light syllables only: CV or V*.

These lists give you a starting point for further explorations into the syllable types in your language. They also give you practice in thinking about the patterns of Cs and Vs in words.

6.1.3 Heavy syllables

Heavy syllables are syllables with a long vowel, or with a diphthong, or with a consonant after the vowel. Syllables can be heavy even if there is no onset. We start our discussion of with syllables with long vowels or diphthongs.

6.1.3.1 Syllables with long vowels or diphthongs (VV)

A long vowel is exactly that — a vowel that takes longer to say. It doesn’t change quality as you say the long vowel, it just gets dragged out longer. (Length is

also discussed in Chapter 2, Section 2.4.3.4.) A diphthong is different: The key property of a diphthong is that the vowel quality changes as you go from the start to the end of the diphthong, so it sounds like a sequence of two vowels smooshed together. Diphthongs also typically take longer to say than short vowels do.

In this section, we look at syllables where the nuclei are long vowels or diphthongs. If your language seems to have long vowels and/or diphthongs, you are likely to find this section useful.



What about “vowel clusters”?

You may see reference to “consonant clusters” in your reading, to talk about more than one consonant in a row. From that, you might think that linguists would use the term “vowel cluster” in discussing sequences of vowels like the sequences that make up diphthongs. However, the term “vowel cluster” is not in common usage.

We give examples of syllables which have long nuclei — either a **long** vowel or a **diphthong** — in (3). Arapaho is an Algonquian language spoken on and around the Wind River Reservation in Wyoming and in and around western Oklahoma (Cowell & Moss Sr 2008).³

Paresi-Haliti is a southern Arawakan language spoken in the state of Mato Grosso, Brazil (Brandão 2014).⁴ In these figures, we show the transcription in IPA both without the dots separating syllables as in [né:tʃɛ:] ‘chief’ in (3a), and with the dots to separate the syllables as in [né:.tʃɛ:] ‘chief’.

(3) Syllables with a long vowel or a diphthong

- | | | | | | | |
|----|------------|---------------|---------------|-----------|------------|---------|
| a. | long vowel | Arapaho | <i>néécee</i> | [né:tʃɛ:] | [né:.tʃɛ:] | ‘chief’ |
| b. | diphthong | Paresi-Haliti | <i>kamae</i> | [kamai] | [ka.mai] | ‘sun’ |

In the Arapaho syllables [né:] and [tʃɛ:], the nuclei are long vowels [é:] and [ɛ:], respectively. In the Paresi-Haliti syllable [mai], the nucleus is a diphthong [ai].

³We thank James Cowell for checking our use of the Arapaho language here and elsewhere in this volume, to make sure forms are correct.

⁴We have tried to connect with Ana Brandão to check our use of the Paresi-Haliti language, but have yet to succeed in that effort.

You may already be thinking that a syllable with a long vowel or a diphthong (CVV) is an open, heavy syllable. It is open because it ends in a vowel, and it is heavy because it has either a long vowel or a diphthong as the nucleus.

When we hear a vowel that is long, or see something written with two vowel symbols (like *oo*), we want to know whether we are dealing with a long vowel, or with a sequence of two short vowels. It is sometimes hard to figure this out. See the examples of *oo* in (4).

- (4) Two ways of syllabifying [oo]
- | | | | | | | |
|----|--------------|-------------|--------------|----------|-----------|---------|
| a. | long vowel | Arapaho | <i>hócoo</i> | [hótʃo:] | [hó.tʃo:] | ‘steak’ |
| b. | short vowels | Djeoromitxí | <i>oo</i> | [oo] | [o.o] | ‘wind’ |

In the Arapaho word *hócoo* and Djeoromitxí word *oo*, we see two [o]s adjacent to each other. In Arapaho, the two [o]s represent a single long vowel [o:], the nucleus in the syllable [tʃo:]. On the other hand, in Djeoromitxí, the two [o]s are separate sounds belonging to two distinct syllables. Once you figure out whether you are hearing a long vowel or two short vowels, you can use IPA conventions to distinguish them: [o:] is a long vowel while [oo] or [o.o] is a sequence of two short vowels. (Be warned: There are many writers who represent [o:] as [oo]. Read carefully.)

It is also challenging to figure out if two different vowels in a row constitute a diphthong or are two adjacent vowels belonging to two different syllables. The examples in (5) from Paresi-Haliti and Djeoromitxí, illustrate the puzzle.

- (5) A vowel sequence as two separate short vowels or as one diphthong
- | | | | | | | |
|----|--------------|---------------|--------------|---------|-----------|----------|
| a. | short vowels | Djeoromitxí | <i>baodü</i> | [βaodɯ] | [βa.o.dɯ] | ‘sloth’ |
| b. | diphthong | Paresi-Haliti | <i>kaoka</i> | [kaokɐ] | [kao.kɐ] | ‘arrive’ |

We can see that there are [ao] sequences in both words, [βaodɯ] and [kaokɐ]. Even though these are the same sounds and both are transcribed as [ao], the way they are syllabified is different. In Djeoromitxí, [a] and [o] are two separate vowels belonging to distinct syllables, [βa] and [o], [βa.o]. On the other hand, in Paresi-Haliti, [ao] is a diphthong and it is a nucleus of a single syllable [kao].

The same vowel sequences can be syllabified differently in different languages, depending on whether or not the language has long vowels or diphthongs as syllable nuclei. How do you figure out whether two vowels are in one syllable or two? Hold on to this question — we get to it in Section 6.3.



Exploring my language

Look at the words that you found for the Language Exploration on p. 296. With this exercise, you will add a third column titled *Words with CVV-syllables*. You now have columns labeled *Light syllables only: CV or V*, *Words with CVV-syllables*, and *Other syllable type(s)*.

To start, look at the *Other syllable type(s)* list, the words that you think have at least one syllable that is not a CV syllable. Do any of these words have a long vowel or a diphthong? If so, put them in the *Words with CVV-syllables* column. If you don't have any such words in your list, do you think there are such words in your language? If so, try to come up with a few to add to your *Words with CVV-syllables* list.

If you can't find any CVV syllables in your language, put *none?* in the *Words with CVV-syllables*. We use a question mark because you are in the early stages of exploring the syllables in your language. Later you may find CVV syllables, and then you will change your list.

6.1.3.2 CVC syllables

In addition to syllables with consonants at the beginning, there are syllables with consonants at the end, VC and CVC syllables. In these syllables, the vowel is followed by a consonant. If you think your language has some syllables with consonants at the end, this is a good section for you to read.

The examples below show CVC syllables from two languages, Nakoda and Zihuateutla Totonac.⁵

⁵Our source for Zihuateutla Totonac is Garcia-Vega (2022). Zihuateutla Totonac is a member of the Northern branch of the Totonacan language family spoken in the town of Zihuateutla in northern Puebla State, Mexico. Garcia-Vega (2022: 2) notes: "The name Zihuateutla is of Nahuatl origin (from *cihuatl* 'woman' and *teuctli* 'lord, knight'..., the meaning of which is said to mean *mujer que gobierna* 'the woman that governs.' The name, according to my consultants, is based on a legend that the mountain on which Zihuateutla sits is so high that it watches over and protects all the other villages in the region." We thank Michelle Garcia-Vega for checking and improving our discussion of Zihuateutla Totonac.

(6) Words containing CVC syllables

- a. Nakoda *hík* [hík] ‘and’
- b. Zihuateutla Totonac *tej* [teχ] ‘road, path’

In the Nakoda CVC syllable [hík], the vowel [í] is preceded by a consonant [h] and followed by another consonant [k]. In the Zihuateutla Totonac syllable [teχ], the vowel [e] is preceded by the consonant [t] and followed by another consonant, [χ]. Consonants at the end of a syllable are called the *coda*, the concluding part of a syllable.

There are some symbols in these examples that you may want to review. In the Nakoda word *hík* [hík] ‘and’, [í] is a nasal vowel as indicated by the symbol [̃] above the vowel [i]. Nasal vowels are discussed in Chapter 2, Section 2.4.3.2. In the Zihuateutla Totonac word *tej* [teχ], [χ] is a voiceless uvular fricative. Voiceless, uvular, and fricative sounds are all discussed in Chapter 2: Voiceless sounds are discussed in Section 2.2.1, uvular sounds are discussed in Section 2.3.2.3, and fricative sounds are discussed in Section 2.3.1.2.

Try this. In (6), what is the coda in the Nakoda syllable [hík]? What is the coda in the Zihuateutla Totonac syllable [teχ]?

We hope you said that [k] is the coda in [hík] and [χ] is the coda in [teχ]. The sounds [k] and [χ] end these two syllables, so they are the codas. CVC syllables are onset-nucleus-coda syllables.

In many languages, the coda is a possible part of a syllable — some syllables have codas and others do not. However, in some languages, there are simply no codas — there are no syllables with a consonant after the nucleus. For example, in Djeoromitxí, syllables do not have codas.

There is another term that is used when talking about syllables, the term *rime*. A rime is made up of a nucleus and a coda. (You might see *rime* spelled as *rhyme* sometimes. When referring to a part of a syllable, it is more common to spell it as *rime*.)

Try this. Using this definition, can you find the rimes in the Nakoda syllable [hík] and the Zihuateutla Totonac syllable [teχ]?

We hope you identified the sequences [ík] and [eχ] as the rimes in these syllables — the rime is the nucleus and coda (if any) of a syllable.

Try this. Are CVC syllables open or closed? Are they heavy or light? What are your reasons?

CVC syllables are closed, heavy syllables. They are **closed syllables** because they end in a consonant, and they are **heavy syllables** because they have a consonant after a vowel.



Comprehension Check 2

Below we have three **monosyllabic** words, from Nakoda, Chol, and Kubeo. What is the syllable structure of each of these words: V, CV, or CVC? For bonus points, identify the onsets, nuclei, codas, and rimes in each syllable.⁶

- | | | | | |
|----|--------|------------|-------|---------------|
| a. | Nakoda | <i>hí</i> | [hí] | ‘tooth/teeth’ |
| b. | Chol | <i>pul</i> | [pul] | ‘to burn’ |
| c. | Kubeo | <i>o</i> | [‘o] | ‘wild banana’ |



Here’s our answer

We’ve rearranged the words from the problem sets into syllable types:

- | | | |
|-----------|----|-------|
| Kubeo V | c. | [ú] |
| Nakoda CV | a. | [hí] |
| Chol CVC | b. | [pul] |

The table below shows the onsets, nuclei, codas, and rimes.

⁶Our source for Chol is Vázquez Álvarez (2011). Chol is a Mayan language from the greater Tzeltalan branch mostly spoken in the state of Chiapas, Mexico. Our source for Kubeo is Chacon (2012). Kubeo is an Eastern Tukanoan language spoken in the Northwest Amazon. We thank Thiago Chacon for help with our discussion of Kubeo (here and elsewhere in this chapter).

	syllable	onset	rime	
			nucleus	coda
Kubeo V	[ú]		ú	
Nakoda CV	[hí]	h	í	
Chol CVC	[pu]	p	u	ɿ



Exploring my language

In this exploration, you will add a column *Words with CVC syllables* to the others, giving four lists: *Light syllables only: CV or V*, *Words with CVV-syllables*, *Words with CVC syllables* and *Other syllable type(s)*.

Your job is to look at your words for CVC syllables. There shouldn't be any in the *Light syllables only* list, but there might be some in either of the other two lists, *Words with CVC syllables* and *Other syllable type(s)*.

Do any of these words have CVC syllables? If so, add the words to the *Words with CVC syllables* column. If you don't find any in your lists, do you think that your language has codas? If so, try to come up with a few examples to add to the *Words with CVC syllables* column.

6.1.4 Other ways that consonants show up in syllables

You now know that consonants can be onsets, beginning a syllable, and they can be codas, ending a syllable. Consonants are found in syllables in other ways too. A cluster (a group of two or more consonants) can begin or end a syllable. A nucleus can be made up of a single consonant, called a syllabic consonant. If you suspect your language has consonant clusters or syllabic consonants, this section may be very useful to you.

6.1.4.1 Syllables with consonant clusters (CC)

A sequence of two or more consonants is called a *consonant cluster*. There are syllables that contain more than one consonant in the onset or the coda — that

is, the onset or the coda contains a consonant cluster. You may also see these referred to as a *complex onset* and a *complex coda*. Below are examples of such syllables from two languages, Nakoda (Cumberland 2005) and Chol (Vázquez Álvarez 2011).

(7) Words containing syllables with consonant clusters

- a. Nakoda *štén* [ʃtén] ‘when’
- b. Chol *mujk* [muxk^h] ‘navel’

The Nakoda syllable [ʃtén] contains two consonants, [ʃ] and [t], in the onset. Chol syllable [muxk^h] contains two consonants, [x] and [k^h], in the coda. (The sound [ʃ] is a voiceless postalveolar fricative, a sound that is fricative is discussed in the *Try this* box on p. 73. [x] is voiceless velar fricative, discussed in Chapter 2, Section 2.3.2.3.) These examples show that it is possible for a syllable to have more than one consonant in the onset or coda position.

How many consonants can be found in an onset or in a coda differs in different languages. We’ve seen that Djeoromitxi has only CV and V syllables — at most, only one consonant in the onset, and no coda consonants at all. Compare this to Chol, where we find up to two consonants in onsets and up to two in codas, shown by the examples in (8).

(8) Chol syllables containing two consonants in the onset or coda

- a. CCVC *kpam* [kpam] ‘my forehead’
- b. CVCC *mujk* [muxk^h] ‘navel’

There are even languages that allow more than two consonants in the onset or coda. In Sierra Popoluca, up to three consonants are found in the coda as demonstrated by the example below. A note of explanation is in order about this example. Our source for Sierra Popoluca, de Jong Boudreault (2009), includes the “+” in the orthographic representation of *?a+so?pspa*, and we have faithfully done so as well. This symbol indicates that [ʔa] is a particular type of morpheme that is not included in the stress pattern of the word. (A stressed syllable has some kind of prominence to it — length, loudness, pitch, or a combination of properties.) Park calls such morphemes *clitics*, a term that is used to refer to many different kinds of morphemes in linguistic writings.

(9) Sierra Popoluca syllable containing three consonants in the coda

- CVCCC *?a+so?pspa* [ʔa.səʔps.paʔ] ‘It tires me.’

Try this. Are CCV syllables open or closed? Heavy or light? What about CVCC syllables? Think about the reasons for your answers.

We hope you are now comfortable with the terms *open* and *closed*, *heavy* and *light*. If you are, you probably said that a CCV syllable is both an open syllable and a light syllable because it ends in a vowel and has only one short vowel. The consonant cluster in the onset has no effect on syllable weight or type.

You probably also said that a CVCC syllable is both a closed syllable and a heavy syllable, closed because it ends in a consonant (in this case a consonant cluster), and heavy because it has a consonant (or, in this case, two consonants) after the vowel. Whether there is one or more than one consonant in a coda, the effect is that the syllable is closed and heavy.

As you saw from this section, the number of consonants found in the onset or coda varies when we look at different languages. As you work on your language, you may want to figure out the maximum number of consonants found in the onset or coda to start to understand your language's syllables. You might be wondering how you can do that. Please be patient a little longer — we discuss that in Section 6.3. First, however, we consider a couple other properties of syllables in general.



CV syllables or consonant clusters in O'odham?

Cheyenne Wing

You may find statements about languages that don't make sense in terms of what you observe when people speak the language. For example, Wing (2020: 25) tells us that "O'odham consonants are always followed by a vowel, however consonant clusters occur...", giving the examples here:

cluster of two	cipkan	'working'
cluster of three	maistla	'teacher'
cluster of four	kotst'abe	'on a cross'

Many readers find it confusing to say that "O'odham consonants are always followed by a vowel" and then say that "consonant clusters oc-

cur”. Wing goes on to give her linguistic analysis, that the consonant clusters arise through vowel deletion. The vowel-deletion interpretation of O’odham words says that even words like *kotst’abe* ‘on a cross’ are represented with vowels between all consonants — *kotVsVt’abe*. The vowels shown as *V* are always deleted when the word is pronounced.

You may find a similarly confusing description of sounds in your language that does not match your observations. Usually, this means there is some linguistic analysis that is not being explained fully; sometimes it is simply an error.

O’odham is a Uto-Aztecan language.

6.1.5 Syllables without a vowel

This is just a reminder, because it was mentioned already, in Section 6.1.2.1. It is not very common, but you can find syllables without a vowel in some languages.



Exploring my language

Are there any words left in your *Other syllable type(s)* list? If so, check if any of them have a consonant cluster inside of a syllable, giving either a complex onset or a complex coda. Add columns with appropriate headings: *Syllables with complex onsets*, *Syllables with complex codas*, *Syllables with syllabic consonants*.

If you don’t find words with these kinds of syllables, you may want to make a note of them anyway. This would act as a reminder of these possibilities in case you come up with additional syllable types later as you explore your language further.

6.1.6 Summary

In this section we learned what a **syllable** is and saw the diversity in syllable types in different languages.

A syllable is a linguistic unit that is bigger than a sound and smaller than a word. In general, it consists of a vowel and some number of consonants before and/or after the vowel. Throughout the section, we've seen many types of syllables: V syllables that consist of a single vowel, CV syllables that have a consonant and a vowel, CVC syllables that have a consonant both before and after the vowel, CCVC and CVCC syllables with consonant clusters, and VV syllables with long vowels or diphthongs. As we looked at these diverse syllable types, we also saw that there is a lot of variation in which types of syllables are found in different languages. We also learned that the same sound sequence can be syllabified differently in different languages.

We also covered many of the terms used when discussing syllable types: *onset*, *nucleus*, *coda*, *rime*, *open syllable*, *closed syllable*, *light syllable*, and *heavy syllable*. These are quickly summarized in (10).

(10) Syllable terms

<i>syllable type</i>	The basic structure of a syllable, ignoring the specific segments that make up the syllable.
<i>onset</i>	One or more consonants that precede the nucleus.
<i>nucleus</i>	The required part of a syllable, usually a vowel or diphthong.
<i>coda</i>	One or more consonants that follow the nucleus.
<i>rime</i>	A unit made up of the nucleus and coda (if there is a coda).
<i>open syllable</i>	A syllable that ends in a vowel.
<i>closed syllable</i>	A syllable that ends in a consonant.
<i>light syllable</i>	A syllable with a single short vowel and no coda.
<i>heavy syllable</i>	A syllable with a rime holding more than a single short vowel — a coda and/or a long vowel or diphthong.

Linguists use the term *template* to refer to a generic syllable type without specifying the segments. You might see references to a *CV template* for a language with open light syllables or to a *CVC template* for a language that allows codas (and so has closed syllables). To emphasize that onsets are optional, you can put the onset C in parentheses: The (C)V template includes both CV and V syllables. However, this emphasis is not always included, and some writers use “CV template” to refer to a language with both CV and V syllables. With codas, because we know that not every syllable has to have a coda, it is common to see CVC instead of CV(C): A CVC template includes both CV and CVC syllables.



Comprehension Check 3

Test your skill in understanding the condensed syllable structure schema. Suppose a language is described as having (C)V(C) syllables. Which of the following are syllables in that language?

- | | |
|-------|--------|
| a. CV | c. CCV |
| b. VC | d. CVC |

For bonus points, (C)V(C) expands into four syllable structures. What is missing (if anything) from the set of syllables given in this Comprehension Check?



Here's our answer

In the figure below, we show (using gray and black text) the expansion that results in a syllable structure.

- | | |
|-----------------|---|
| a. CV = (C)V(C) | c. CCV = impossible from a (C)V(C) template |
| b. VC = (C)V(C) | d. CVC = (C)V(C) |

The fourth syllable type that is missing uses neither the onset option nor the coda option, (C)V(C), resulting in a single V syllable.



Exploring my language

In the discussion above, we looked at words of only one syllable — monosyllabic words. Does your language have monosyllabic words? (Not every language does.) If your language has monosyllabic words, try to find three

such words in your language. (There may already be some on your lists.)

Examine your monosyllabic words. Characterize the syllables in terms of the symbols *C* and *V* so that you show the type of each syllable — *V*, *CV*, *CVC*, *CCVC*, *CVV*, or something else.

If you found the above to be fairly straightforward, you can go further by identifying the onset, nucleus, coda, and rime in each syllable. (Remember that the only required part of a syllable is the nucleus, so some syllables may not have a coda, and some syllables may not have an onset.)

6.2 Identifying the syllable template

At this point, you know that syllables typically have at least one vowel in the nucleus. You also know that every language has *CV* syllables — syllables made up of an onset and a short vowel.

You also know that there is variation in syllable types in different languages. Does your language have long vowels or diphthongs? Does your language allow complex onsets? Does it allow codas? How about complex codas? To answer these questions, it is important to know how to figure out possible syllable types in a language. In this section, we figure out possible syllable templates in a language, by finding out what syllables are possible. There are three general questions to ask.

1. What kind of **onsets** does my language have? There are two parts to answering the onset question: Are onsets required on all syllables, or are there also syllables without onsets? What is the maximum number of consonants found in onsets in your language?
2. What kind of **codas** does my language have? There are two parts to answering this question: Are there any codas in my language? (Not all languages have codas, right?) What is the maximum number of consonants found in codas in your language?
3. What kind of **nuclei** does my language have? You know that vowels generally count as a syllable nucleus, but you also know there are other types of nuclei — long vowels, diphthongs, or consonants. Each of these needs to be examined for your language: Does your language have long vowels as well as short ones? Does it have diphthongs? Does it have syllabic consonants?

The answer to these questions for a particular language will lead to a statement

about the syllable **template** for the language. The template is given in terms of C and V. The goal is to find the sequence of Cs and Vs, the template, that fits all syllables in the language.

This section introduces methods you can use to answer these questions and create a syllable template for your language. Answering these questions and creating a template is an important part of syllable analysis.

6.2.1 Word-initial onset tests

Every word starts with a syllable, so the start of a word matches the start of some syllable. This means we can examine the beginnings of words to see what is possible at the beginning of a syllable. When a sound is at the beginning of a word, we say that sound is **word-initial**, or in *word-initial position*. We use word-initial onset tests to learn about onsets.

There are two word-initial onset tests, *Are onsets necessary?* and *How big can an onset be?*

6.2.1.1 Are onsets necessary?

Look at the beginning of the words in a language to determine whether **onsets** are required. If there are words that begin with a vowel, it means that the language does not require syllable onsets. If you cannot find any word that begins with a vowel, it is likely that all syllables in the language begin with an onset.

Let's try the word-initial onset test with a couple of languages.

We start with the Djeoromitxí words in (11). We want to find out whether onsets are necessary or optional. To do this, we focus on the beginning of words, asking *do any words begin with a vowel?* If the answer is “yes”, then onsets are not required.

Word-initial sounds that are in black text in (11), making them easier to spot. Are there any vowel-initial words in (11)? (To test yourself, look at the forms and come to your own answer before reading further.)

(11) Djeoromitxí word list

- | | | | |
|----|---------------|----------|-----------------|
| a. | <i>hau</i> | [hau] | ‘lizard’ |
| b. | <i>oo</i> | [oo] | ‘wind’ |
| c. | <i>uhiri</i> | [uhiri] | ‘alligator’ |
| d. | <i>kōaka</i> | [kōaka] | ‘my head’ |
| e. | <i>takahō</i> | [takahō] | ‘stone, gravel’ |
| f. | <i>hotō</i> | [hotō] | ‘grandfather’ |

We find two words that begin with a vowel in (11), [oo] ‘wind’ and [uhiri] ‘alligator’. Since there are words that begin with a vowel, we can safely conclude that Djeoromitxi has syllables without onsets as well as syllables with onsets, summarized in (12).

- (12) Djeoromitxi syllable template (preliminary)
(C)V...

The notation in (12) is very cryptic. Let’s unpack it. You may remember that C means *consonant* and V means *vowel*. By putting the C inside parentheses, (C), we are saying that a consonant is possible in this position but is not required: (C)V is a shorthand for *both CV and V*. So we are saying that some Djeoromitxi syllables have onsets and some do not. By following this with ellipses (ellipses are the three dots, “...”) we are saying that we aren’t done yet — which is true, since we haven’t checked for codas yet.

Now let’s try the word-initial onset test with words from Chol. (Chol, a Mayan language, was introduced in (7).) Again, only the first sound is black, to make them easy to identify.

- (13) Chol word list
- | | | | |
|----|--------------|------------------------|-------------------|
| a. | <i>wiñik</i> | [wip̃ik ^h] | ‘man’ |
| b. | <i>kpaṃ</i> | [kp̃aṃ] | ‘my forehead’ |
| c. | <i>mujk</i> | [muxk ^h] | ‘navel’ |
| d. | <i>ax</i> | [ʔax] | ‘a type of fruit’ |
| e. | <i>ixik</i> | [ʔiʃik ^h] | ‘woman’ |
| f. | <i>ok</i> | [ʔok ^h] | ‘feet’ |

We ask the word-initial onset test question again: *Do any words begin with a vowel?* Looking at the transcription, we see that every word in this list begins with a consonant. We begin to suspect that every Chol syllable has an onset. We show this in (14) by not including parentheses around the C — we say that Chol appears to have CV... syllables, not (C)V... syllables.

- (14) Chol syllable template (preliminary)
Chol syllable template: CV...

There are three important points illustrated by the Chol example. First, we concluded with a supposition, not a definitive statement (our supposition is that *We begin to suspect that every Chol syllable has an onset.*) We can’t be absolutely sure about onsets from a list of six random words. Maybe the next word we find will begin with a vowel. We have a hypothesis that we can test every time we

find a new Chol word. Many of our answers about language properties end this way, with a hypothesis to test as we move forward.

Second, it is important to look at the transcription, not the way words are spelled, because it is the sounds that matter, not the way words are written. Someone who looked at the spelling might be tempted to say that (13d, e, f) begin with vowels — these words are spelled *ax*, *ixik*, and *ok*. However, the transcriptions tell us how the words sound. Each one of these words begins with a glottal stop. That’s why we say that none of these words begins with a vowel and it is likely that onsets are required in Chol syllables.

Third, we asked the question *do any words begin with a vowel?*. The answer “yes” tells us that onsets are not required. We could phrase the question a different way and get to the same answers. We could ask *do all words begin with a consonant?* If all words in our list start with a consonant, we answer “yes”, and understand that onsets may be required. If some words start with a vowel, we answer “no”, and understand that onsets are optional because there are both vowel-initial words and consonant-initial words. You can get to the same answers whether you focus on vowel-initial or consonant-initial words.



Comprehension Check 4

Identify word-initial sounds for yourself. Here are some words from two languages: Sierra Popoluca and Kubeo. Your job is to identify the initial sound in each word. Be sure to focus on the transcription, and not the spelling. (We do this because you need to know the first sound of each word in order to answer the key question *do any words begin with a vowel?*)

Before you start, here’s some help with the transcription. The words in this Comprehension Check indicate which syllable has **stress**: The syllable following the symbol ‘ is the syllable with stress. For example, in the Sierra Popoluca word *nuuma* [‘nu:maʔ] ‘certainly’, the syllable [‘nu:] is stressed, and in *tzabatz* [‘tsabats] ‘red’, the syllable [‘tsa] is stressed. (In [‘tsa], the ‘ may be a little tricky to see because of the tiebar in [ts].) A syllable with stress is more prominent than other syllables in some way — it is longer, or louder, or has a different pitch, etc.

Sierra Popoluca	<i>xix</i>	[ʃi]	‘cow’
	<i>wiʔk</i>	[wiʔk]	‘he ate’
	<i>nuuma</i>	[nu:ma ^ʔ]	‘certainly’
	<i>tzabatz</i>	[tsabats]	‘red’
Kubeo	<i>bia</i>	[bia]	‘chili pepper’
	<i>jo</i>	[dʒo]	‘this count thing, here’
	<i>abi</i>	[aʔbi]	‘he said’
	<i>jābī</i>	[jã'mi]	‘night’

We will build on your answers about Sierra Popoluca and Kubeo in other Comprehension Checks in this chapter, so please keep track of your work.



Here's our answer

We've marked the initial sounds with our usual convention of black and gray text — the initial sounds are in black text. (As a reminder, gray text is used for parts of words that are not the focus of our discussion. Normal black text is used for the sounds we are most concerned about.)

Sierra Popoluca	<i>xix</i>	[ʃi]	‘cow’
	<i>wiʔk</i>	[wiʔk]	‘he ate’
	<i>nuuma</i>	[nu:ma ^ʔ]	‘certainly’
	<i>tzabatz</i>	[tsabats]	‘red’
Kubeo	<i>bia</i>	[bia]	‘chili pepper’
	<i>jo</i>	[dʒo]	‘this count thing, here’
	<i>abi</i>	[aʔbi]	‘he said’
	<i>jābī</i>	[jã'mi]	‘night’



Comprehension Check 5

Try the word-initial onset test yourself. Use the words in the previous Comprehension Check to determine whether onsets are optional in either Sierra Popoluca or Kubeo. Explain the reasons for your answer. Again, keep track of your answer to this Comprehension Check.



Here's our answer

We think that onsets are optional in Kubeo but may be required in Sierra Popoluca. Do you agree? Here are our reasons.

In Kubeo, we found a word that begins with a vowel, *abi* [a'bi] 'he said'. This means that the first syllable in the word has no onset. This is possible only if onsets are optional, not required.

In Sierra Popoluca, every word starts with a consonant, so we suspect that onsets may be required in Sierra Popoluca. Anyone working on this language would keep an eye out as more words are encountered, to determine whether every word does start with a consonant.



Exploring my language

Now, it's time to apply the method to your language and make a syllable template for your language. We know that all languages have CV syllables. We are trying to find out whether there is anything more to syllables in your language.

Take your word list from the previous Language Exploration. Do any of these words begin with a vowel? (Pay attention to sounds, not spelling.)

If every word begins with a consonant, think a little bit more: Do you know any words that start with a vowel? (If your language has a dictionary, you can try finding any word starting with a vowel from there. You can also ask other people, if that helps.) With any of these methods, can you find words that begin with a vowel?

If you succeeded in finding a word starting with a vowel, it means that the syllables in your language don't need onsets. On the other hand, if you couldn't find any word that starts with a vowel even if you tried your best, then it is likely that the syllables in your language need onsets. You have a hypothesis to test as you learn additional words.

6.2.1.2 How big can an onset be?

Many languages have only one consonant in the onset of a syllable, but some languages also have syllables with two or even more consonants in them. (Review Section 6.1.4.1 if you need to review this point.)

Again, since words are made of syllables, the beginnings of words and the beginnings of syllables coincide with each other. This means that word-initial consonant sequences are onsets.

We can use this to figure out how many consonants are possible in an onset: Examine the beginnings of words and count how many consonants you find in the longest sequence at the beginning of a word. This is likely to be the largest possible onset in the language.

Let's try this test with Djeoromitxí words. Below we repeat the word list from (15). This time, we count the number of consonants at the beginning of each consonant-initial word. (All consonants before the first vowel are black in these words; all other sounds are gray.)

(15) Djeoromitxí word list, repeated from (11)

- | | | | |
|----|---------------|-------------------|-----------------|
| a. | <i>hau</i> | [hau] | 'lizard' |
| b. | <i>oo</i> | [oo] | 'wind' |
| c. | <i>uhiri</i> | [uhiri] | 'alligator' |
| d. | <i>kōaka</i> | [kōaka] | 'head' |
| e. | <i>takahō</i> | [takahō] | 'stone, gravel' |
| f. | <i>hotō</i> | [hotō] | 'grandfather' |

What do you see when you look at the beginnings of words? You probably noticed right away that a couple of the words, (15b, c) begin with vowels, not

consonants. We can set them aside and look at the other four words, (15a, d, e, f). What is the longest onset among these words?

As we scan the words with initial consonants, we see that each one of them has only one consonant at the beginning. We can conclude that it is likely that the biggest onset in Djeoromitxí words is only one consonant long. If we were continuing to work with Djeoromitxí, we would expect to find no more than one consonant in an onset — but we would also keep an eye out for longer onsets, just in case we missed something. At this point, however, our preliminary syllable template statement for Djeoromitxí is still appropriate, given again in (16).

- (16) Djeoromitxí syllable template (preliminary, repeated from (12))
(C)V...

Let's try the word-initial onset test with the Chol words. The words from (13) are repeated in (17), with the consonants before the first vowel of each word highlighted by black text. Examine these words: Does Chol have only one consonant in its onsets, like Djeoromitxí, or do we find longer onsets in Chol?

- (17) Chol words, repeated from (13)
- | | | | |
|----|--------------|-----------------------|-------------------|
| a. | <i>wiñik</i> | [wɪñɪk ^h] | 'man' |
| b. | <i>kɔpam</i> | [kɔpam] | 'my forehead' |
| c. | <i>mujk</i> | [muxk ^h] | 'navel' |
| d. | <i>ax</i> | [ʔax] | 'a type of fruit' |
| e. | <i>ixik</i> | [ʔiɪk ^h] | 'woman' |
| f. | <i>ok</i> | [ʔok ^h] | 'feet' |

Most Chol words start with a single consonant. However, we hope you noticed that in (17) there is one Chol word that begins with two consonants, [kɔpam] 'my forehead' in (17b). This is the longest word-initial sequence in our words. Word-initial sequences of consonants are onsets, so this is the longest onset we have found for Chol: Chol syllables may have onsets at least as long as two consonants.

We again have a hypothesis to test if we continue to work with the Chol language: We expect onsets of one or two consonants as we encounter other words. We revise our statement from (14) to include the optional second consonant in an onset, as in (18).

- (18) Chol syllable template (preliminary)
(C)CV...

The statement in (18) says that Chol syllables start with one or two consonants. We do not expect to find longer onsets (onsets with three or more consonants).

But we must recognize that we are looking at a very small number of words. If we look at a lot more Chol words, we may find longer onsets in some of them.



Comprehension Check 6

Here we have word lists of Sierra Popoluca and Kubeo. Your job is to count the number of consonants in the word-initial syllable onset for each word.

Sierra Popoluca	<i>wejpa</i>	['wehpa [?]]	'he cries'
	<i>p̄iwi</i>	['ni:wi [?]]	'chili'
	<i>kiʔps</i>	['kiʔps]	'measure'
	<i>kruuna</i>	['kru:na [?]]	'crown'
Kubeo	<i>hia</i>	['hia]	'river'
	<i>ō</i>	['ō]	'she'
	<i>dā</i>	['nā]	'they'
	<i>dībĩ</i>	[nĩmĩ]	'season, epoch, time'

There will be more questions building on your answers to this one, so keep track of the answers.



Here's our answer

We've marked the initial sounds with our usual convention of black and gray text – the initial sounds are in black text. The number of consonants in the onset position is given in the last column.

Sierra Popoluca

<i>wejpa</i>	['wehpa [?]]	'he cries'	1
<i>p̄iwi</i>	['ni:wi [?]]	'chili'	1
<i>kiʔps</i>	['kiʔps]	'measure'	1
<i>kruuna</i>	['kru:na [?]]	'crown'	2

Kubeo

<i>hia</i>	['hia]	'river'	1
<i>ō</i>	['ō]	'she'	0
<i>dā</i>	['nā]	'they'	1
<i>dībĩ</i>	[nĩmĩ]	'season, epoch, time'	1



Comprehension Check 7

Try the second part of the word-initial onset test yourself. Use the words in the previous Comprehension Check to determine the maximum onset size in either Sierra Popoluca or Kubeo. Explain the reasons for your answer. Again, there will be more questions about these two languages.



Here's our answer

We think that the maximum onset size for Sierra Popoluca is two while for Kubeo it is one. Here are our reasons.

In Sierra Popoluca, we found a word that has two consonants in the onset, *kruuna* ['kru:na⁷] 'crown'. This means that the syllable onset can have two consonants in the onset position. Until we find a word that has more than two consonants in the onset position, we can assume that the maximum onset size for Sierra Popoluca is two.

In Kubeo, we only found words with at most one consonant in the onset (i.e., *hia* ['hia] ('river'); *dā* ['nā] ('they'); *dībĩ* [nĩmĩ] ('season, epoch, time')). Until we find a word that has more than one consonant in the onset position, we can assume that the maximum onset size for Kubeo is one.



Exploring my language

Now let's apply the word-initial onset test to your language. Take your word list from the previous Language Exploration again. This time, we'll try finding a word that starts with the largest number of consonants. (Again, pay attention to sounds, not spelling.)

Can you find any word that starts with one consonant? If so, what about a word starting with two or three consonants? Is there any word that starts with more than three consonants?

What is the largest number of consonants that you found from looking at word-initial consonants? It is highly likely that your answer to the previous question is the same number as the longest syllable onset for your language.

Add onset consonants to your syllable template, if appropriate.

6.2.2 Word-final coda tests

We again start with the observation that words are made of syllables. This means that the ends of words and the ends of syllables coincide. **Codas** are at the end of a syllable. Just as we used the beginnings of words to learn about possible onsets, we can use the end of the word to learn about possible codas.

When a sound is at the end of a word, we say that the sound is **word-final**, or in *word-final position*. We use word-final coda tests to learn about codas in a language. We again have two questions: *Are codas possible? How big can a coda be?*

6.2.2.1 Are codas possible?

One way to learn whether your language might have codas is to look at the ends of the words in your language. If you find words that end in a consonant, it means that the syllables can have codas in the language. If not, it is likely that syllables cannot have codas in the language.

Let's try this with our words from Djeoromitxi. The word-final sound in each word is in black text in (19). (These are the same words you saw in (11) and (15).)

The word-final coda question is *do any words end with a consonant?* Look at these words and see whether any of them end in a consonant. If you find a word-final consonant, you know that Djeoromitxí is a language that has codas in some of its syllables.

(19) Djeoromitxí word list, repeated from (11), (15)

- | | | | |
|----|---------------|----------|-----------------|
| a. | <i>hau</i> | [hau] | ‘lizard’ |
| b. | <i>oo</i> | [oo] | ‘wind’ |
| c. | <i>uhiri</i> | [uhiri] | ‘alligator’ |
| d. | <i>kōaka</i> | [kōaka] | ‘head’ |
| e. | <i>takahō</i> | [takahō] | ‘stone, gravel’ |
| f. | <i>hotō</i> | [hotō] | ‘grandfather’ |

Did you find any word that ends in a consonant? We hope your answer is *no, I did not*. All the words (19) end in a vowel. That means that it is very likely that Djeoromitxí syllables cannot have codas. We summarize the template of Djeoromitxí syllables in (20): Syllables in Djeoromitxí consist of a vowel, which may be preceded by an onset.

(20) Djeoromitxí syllable template (final)

(C)V

Another way this can be shown is to list the various syllable structures that are condensed into the template in (20). The list of syllable structures is shown in (21). We call this an expansion because it expands (20) to show all the syllable types that are condensed into the succinct (C)V template.

(21) Djeoromitxí syllable template expansion (final)

(C)V expands into V and CV

The V syllable is the result of not using the onset option in (C)V, resulting in a V syllable, not a CV syllable. The CV syllable is the result of using the onset option, (C)V. We could just say that Djeoromitxí has two syllable types, V and CV, instead of condensing the two into (C)V. We take the time to explain how to expand the (C)V schema for two reasons. First, if you are reading something written by a linguist, you may well see the CV — or (C)V — template notation instead of a list of syllable types. If you do, you now know what it means (or, if you forget, we hope you remember that we talked about it here and can come back and refresh your memory). Second, the (C)V notation is a very quick way to encode a lot of information, so it may prove useful to you in your own research notes as well.

We mark both (20) and (21) as final statements because at this point we think we have discovered what there is to know about Djeoromitxí syllables. Since there are no codas in this language, we don't need to check the size of coda. Of course, we can't be absolutely sure that this is the final word on Djeoromitxí syllables because we haven't checked very many words in the language. But we have a complete and rational hypothesis, that Djeoromitxí words do not have codas and have optional single consonant onsets. Someone working with the Djeoromitxí language would test this hypothesis as they went forward in their explorations.

Now let's look at the Chol words from (13) and (17). Again, we've focused attention on the final sound of each word with black text. Examine those word-final sounds as you ask the word-final coda question: *do any words end with a consonant?*

(22) Chol word list, repeated from (13), (17)

- | | | | |
|----|--------------|-----------------------|-------------------|
| a. | <i>wiñik</i> | [wiɲik ^h] | 'man' |
| b. | <i>kpam</i> | [kpam] | 'my forehead' |
| c. | <i>mujk</i> | [muxk ^h] | 'navel' |
| d. | <i>ax</i> | [ʔax] | 'a type of fruit' |
| e. | <i>ixik</i> | [ʔiɟik ^h] | 'woman' |
| f. | <i>ok</i> | [ʔok ^h] | 'feet' |

Did you find any word that ends in a consonant this time? The answer this time is *yes, there are words that end in a consonant*. This means that Chol syllables can have codas, summarized in (23).

(23) Chol syllable template (still preliminary)
(C)CV(C)...

We aren't done yet with Chol syllables because we haven't checked how big codas can be, but we do know that codas are possible in the language.

You may have noticed that every word in the list ends in a consonant, but that we marked the coda as optional. We are supposing that Chol follows the pattern of virtually every language and has vowel-final syllables, syllables without codas, and we just haven't seen the vowel-final words yet. Another possibility is that every word in Chol ends with a consonant. These are hypotheses to keep track of as we go forward with Chol. We talk about how to figure out more about codas in Section 6.2.4.



Comprehension Check 8

Identify word-final sounds for yourself. Here we again have word lists from Sierra Popoluca and Kubeo. To get ready to answer the question *do any words end with a consonant?* identify the final sound in each word. (Again, be sure to focus on the transcription, and not the spelling. And again, be sure to keep track of your answers.)

Sierra Popoluca	<i>xix</i>	[ʃiʃ]	‘cow’
	<i>wiʔk</i>	[ʔwiʔk]	‘he ate’
	<i>kiʔps</i>	[ʔkiʔps]	‘measure’
	<i>tzabatz</i>	[ʔtsabats]	‘red’
Kubeo	<i>wako</i>	[waʔko]	‘cacao’
	<i>idã</i>	[iʔnã]	‘these ones’ (animate)
	<i>bia</i>	[ʔbia]	‘chili pepper’
	<i>jo</i>	[ʔdʒo]	‘this count thing, here’



Here’s our answer

We’ve marked the final sounds with our usual convention of black and gray text - the final sounds are in black text.

Sierra Popoluca	<i>xix</i>	[ʃiʃ]	‘cow’
	<i>wiʔk</i>	[ʔwiʔk]	‘he ate’
	<i>kiʔps</i>	[ʔkiʔps]	‘measure’
	<i>tzabatz</i>	[ʔtsabats]	‘red’
Kubeo	<i>wako</i>	[waʔko]	‘cacao’
	<i>idã</i>	[iʔnã]	‘these ones’ (animate)
	<i>bia</i>	[ʔbia]	‘chili pepper’
	<i>jo</i>	[ʔdʒo]	‘this count thing, here’



Comprehension Check 9

Try the word-final coda test yourself. Use the words in the previous Comprehension Check to determine whether codas are possible in either Sierra Popoluca or Kubeo. Explain the reasons for your answer. And make a note of your answers to have ready for the next Comprehension Check.



Here's our answer

We think that codas are possible in Sierra Popoluca but may be impossible in Kubeo. Do you agree? Here are our reasons.

In Sierra Popoluca, every word ends with a consonant. This means that the last syllable in each word has coda. This can only happen if codas are possible.

In Kubeo, every word ends with a vowel, so we suspect that codas may be impossible in Kubeo. Anyone working on this language would keep an eye out as more words are encountered, to determine whether every word does end with a vowel.



Exploring my language

Now let's apply the method to your language. Take your word list from the previous Language Exploration again. Do any of these words end in a consonant? (Pay attention to sounds, not spelling.)

If every word ends with a vowel, think a little bit more: Do you know any words that end with a consonant? Can you find any by looking in a dictionary or asking other speakers?

If you succeeded in finding a word that ends in a consonant, it means that the syllables in your language can have codas. On the other hand, if you couldn't find any word ending in a consonant even if you tried your best, then it is likely that your syllables don't have codas. You have a hypothesis to test as you learn additional words.

6.2.2.2 How big can a coda be?

Another thing you can check by looking at the end of the words is the maximum number of consonants found in the coda. You can simply count the number of consonants at the end of words. It is likely that the largest number of consonants at the end of a word is the largest number of consonants to be found in the syllable coda.

We already suspect that Djeoromitxí syllables do not have codas, so we can go right to the Chol word-final consonant sequences. The word list is repeated below with any word-final consonant strings left in black text. This time, we count the number of consonants in the word-final position.

(24) Chol word list, repeated from (13), (17), (22)

- | | | | |
|----|--------------|-----------------------|-------------------|
| a. | <i>wiñik</i> | [wiɲik ^h] | 'man' |
| b. | <i>kpam</i> | [kpaɱ] | 'my forehead' |
| c. | <i>mujk</i> | [muxk ^h] | 'navel' |
| d. | <i>ax</i> | [ʔax] | 'a type of fruit' |
| e. | <i>ixik</i> | [ʔiʃik ^h] | 'woman' |
| f. | <i>ok</i> | [ʔok ^h] | 'feet' |

What is the longest string of consonants at the end of a word? This number answers the question of the size of the longest possible coda (as far as the words in our list are concerned – more on this momentarily).

We hope you noticed that there are two consonants at the end of [muxk^h] (24c) while the rest of the words have only one consonant at the word-final position. This means that the words we have collected so far show that the maximum number of consonants found in the syllable coda is likely to be two. We put this together with the CV-schema from (23) to come up with a statement about our thoughts on Chol syllable template, given in (25). To understand (25), remember that C stands for *consonant*, V for *vowel*, and that parentheses tell us when a C or V is optional. You can test your understanding of this way of showing the syllable

template by trying to interpret (25) before reading our explanation (which comes right after (25)).

- (25) Chol syllable template (final)
(C)CV(C)(C)

The syllable pattern in (25) is very dense; we unpack it in (26), showing how the various parenthesized Cs can be used to expand the syllable.

As for onsets, it is possible to have two consonants before the vowel, shown by (C)CV(C)(C). As for codas, it is possible to have one or two consonants after the vowel, (C)CV(C)(C) and (C)CV(C)(C) respectively.

The expansion is shown in (26). We've made two rows in (26), one to show syllables with single C onsets and the other to show syllables that have CC-onsets. There are three columns: The first column shows syllables without codas, the second shows syllables with a single C coda, and the final column shows syllables with two-consonant codas. Count them up: You will see that succinct schema (C)CV(C)(C) condenses six different syllables into one statement.

- (26) Chol (C)CV(C)(C) syllable template expansion (final)

	no coda	C-coda	CC-coda
C onset	CV	CVC	CVCC
CC onset	CCV	CCVC	CCVCC

You may have looked at the expansions in (26) and said to yourself: “Chol syllables always have a CV sequence in them.” This is true; that is why there is a CV sequence that is not in parentheses in the schema of (25). This means that a CV sequence is required in every syllable: (C)CV(C)(C). This statement is generally true in languages that require onsets in all syllables. And it makes sense: Speech sounds are created by closing and opening our mouths while we send air out of our mouths; closing creates consonants and opening creates vowels, the CV syllable.

As noted in our discussion of (21), it is very common to find syllable representations like (C)CV(C)(C) in works by linguists, especially when there are several syllable structures in a language, like the six in Chol.



Comprehension Check 10

Here we have the same word lists that we saw in the previous Comprehension Checks. This time, can you find the number of consonants in the word-final coda for each?

Sierra Popoluca	<i>xix</i>	[ʃiʃ]	‘cow’
	<i>wiʔk</i>	[ˈwiʔk]	‘he ate’
	<i>kiʔps</i>	[ˈkiʔps]	‘measure’
	<i>tzabatz</i>	[ˈtsabats̩]	‘red’
Kubeo	<i>wako</i>	[waˈko]	‘cacao’
	<i>idã</i>	[iˈnã]	‘these ones’ (animate)
	<i>bia</i>	[ˈbia]	‘chili pepper’
	<i>jo</i>	[ˈdʒo]	‘this count thing, here’



Here’s our answer

We’ve marked the final sounds with our usual convention of black and gray text – the final consonant or consonant sequence is in black text. The number of consonants in coda position is given in the last column.

Sierra Popoluca			
<i>xix</i>	[ʃiʃ]	‘cow’	1
<i>wiʔk</i>	[ˈwiʔk]	‘he ate’	2
<i>kiʔps</i>	[ˈkiʔps]	‘measure’	3
<i>tzabatz</i>	[ˈtsabats̩]	‘red’	1
Kubeo			
<i>wako</i>	[waˈko]	‘cacao’	0
<i>idã</i>	[iˈnã]	‘these ones’ (animate)	0
<i>bia</i>	[ˈbia]	‘chili pepper’	0
<i>jo</i>	[ˈdʒo]	‘this count thing, here’	0



Comprehension Check 11

Try the second part of the word-final coda test yourself. Use the words in the previous Comprehension Check to determine the maximum coda size for Sierra Popoluca and for Kubeo. Explain the reasons for your answer. For bonus points, try writing down the “CV” template for each language. And do keep track of your answers, there is more to come with these languages.



Here's our answer

We think that the maximum coda size for Sierra Popoluca is three while for Kubeo it is zero (meaning that there are no codas in Kubeo syllables). Here are our reasons.

In Sierra Popoluca, we found a word that has three consonants in the coda, *kiʔps* [ˈkiʔps] ‘measure’. This means that the syllable coda can have three consonants in coda position. Until we find a word that has more than three consonants in coda position, we can assume that the maximum coda size for Sierra Popoluca is three.

In Kubeo, we did not find any word ending in a consonant. Until we find a word that ends in a consonant, we can assume that Kubeo syllables do not have codas.

As for the CV templates, we found that Sierra Popoluca had one or two consonants in the onsets, and up to three in the coda, giving a (C)CV(C)(C)(C) template (maximum syllable is CCVCCC, and onsets are required). Kubeo did not require onsets and had at most one onset consonant and no coda consonants, giving a (C)V template, just like Djéromitxí.



Exploring my language

This time we'll apply the word-final coda test to your language. Take your word list from the previous Language Exploration. This time, look for words that end with a sequence of consonants. (Again, pay attention to sounds, not spelling.)

Can you find any words that end with one consonant? If so, what about a word ending with two or three consonants? Are there any words that end with more than three consonants?

What is the largest number of consonants that you found in word-final position? It is highly likely that your answer to the previous question is the same number as the longest syllable coda for your language.

Add coda consonants to your syllable template if appropriate.

6.2.3 Summary/conclusion

In this section, we discussed what we can learn about syllables by looking at word edges. By examining the beginnings of words, you can learn about syllable onsets. You can find out whether an onset is necessary in each syllable by looking at the word-initial sounds. If you find a word that begins with a vowel, it means that onsets are not necessary for the language's syllables. If you don't find any words beginning with vowels, you may suspect that onsets are necessary in the language. You can also figure out the syllable onset size by looking at word-initial position. It is likely that the longest sequence of consonants in word-initial position shows the maximum size for an onset.

You can learn about syllable codas by looking at the end of words. You can tell whether syllables can have codas or not by looking at the word-final position. If you find any word that ends in a consonant, you can tell that the language's syllables can have codas. If you don't find any, we suspect that the language's syllables cannot have codas. Another thing you can figure out from word-final position is the size of codas. We expect that the longest sequence of consonants at the end of the word is the same as the largest coda found in the language.

There are two additional things to note here. First, when applying these methods, it is important to keep in mind that we need to focus on the transcription, not the orthography. Second, it is not always possible to get the answer with

100% certainty with these methods. This is especially true if you are working with a limited set of words. However, even with a limited number of words, you can apply the methods introduced in this section and make hypotheses about the language. You can test your hypotheses further if you get more words later.

We also introduced a notation for syllable templates. In this notation, consonants are represented by C and vowels are represented by V. Optional components are marked with parentheses. This notation is used to represent various syllable types in a condensed way. For example, you can represent two syllable types, V and CV, with one template, (C)V. The CV notation is frequently used in linguistics publications when discussing syllables, so it is worth keeping in mind. This notation will be used throughout this chapter as well.

6.2.4 What we can learn about syllables from the middle of words

In this section, we'll have a look at what we can learn from syllables in the middle of words. Word-internal consonant sequences and vowel sequences give us a way of checking the templates we created by looking at word edges. We first look at *word-internal consonant clusters*, or sequences of consonants inside of words.

6.2.4.1 Word-internal consonant clusters

Word-internal consonant clusters can be longer than the longest clusters found at the beginning or end of a word. At the beginning of a word, we have onsets. At the end of a word, we have codas. In the middle of a word, we can have a coda followed by an onset — a sequence longer than is found either at the start or end of a word.

Our word-initial onset tests and word-final coda tests tell us how long the longest onset and the longest coda might be. We can test our proposed syllable template for a language by seeing whether it correctly predicts the length of the longest word-internal consonant clusters. If there is a discrepancy, we know that we have more work to do on figuring out the syllables.

Again, we rely on the idea that words are made up of syllables. When you get two or more syllables in a word, you may have a coda followed by an onset, creating a consonant cluster. How big can these clusters be? We expect that the longest ones will be as long as the longest coda and the longest onset added together.

As an example, let's consider the Nakoda case. In sections 6.1.3.2 and 6.1.4.1, we used word-initial and word-final tests to find that Nakoda syllables are at most CCVC — up to two consonants in the onset and one consonant in the coda, illustrated in (27).

- (27) Nakoda word with the maximum number of consonants in both onset and coda positions

CCVC [štén] ‘when’

What is the longest sequence of consonants that we expect to find in the middle of a Nakoda word? Think about this before reading on.

The longest sequence of consonants that we expect to find in a Nakoda word is made up of the longest coda (in Nakoda, that is a single consonant, C) followed by the longest onset (in Nakoda, that is two consonants, CC). Together, these make a sequence of three consonants. This is illustrated in Figure 1.

$$\begin{array}{cc} \text{Coda} & \text{Onset} \\ \boxed{\text{C}} + \boxed{\text{CC}} & = \boxed{\text{CCC}} \end{array}$$

Figure 1: Nakoda coda + onset = word-internal consonant sequence

We see an example in the Nakoda word in (28), where the dot (.) shows the syllable boundary.

- (28) Nakoda word with the word-internal CC

p^het-mná [p^hen.mná] ‘be the odor of smoke’

Notice that the word-internal consonant cluster in [p^hen.mná] is three consonants long — longer than either an onset or a coda. We emphasize this because it can be easy to forget that inside of words, we may see longer sequences of consonants than at either the beginning or the end of a word.

We can use this to double-check the results of our word-edge tests: We expect no more than three consonants in a word-internal cluster in Nakoda. If we found Nakoda words with four (or more) consonants in the middle of a word, we would start to wonder whether we had missed a word-initial onset with three consonants or a word-final coda with two consonants.

Before we move on to vowel sequences, let’s look at another example of consonants. We revisit Chol; in (26), we summarized our study of the Chol syllable template as (C)CV(C)(C): The biggest possible syllable then is CCVCC. How big is the longest sequence of consonants we might find in the middle of a Chol word?

Because onsets and codas can both be two consonants, we might find some Chol words with four consonants together in the middle of a word, as shown in Figure 2.

$$\begin{array}{c} \text{Coda} \quad \text{Onset} \\ \boxed{\text{CC}} + \boxed{\text{CC}} = \boxed{\text{CCCC}} \end{array}$$

Figure 2: Chol coda + onset = word-internal consonant sequence

Interestingly, in Chol it is much more common to find two consonants and sometimes three, but not four in the middle of Chol words. We see examples in (29).

- (29) Word-internal consonants in Chol
- a. *jiñku* [xiŋku] ‘yes, it is’
 - b. *majchki* [maxtʃki] ‘who’

The Chol words in (29) contain word-internal consonant clusters with two or three consonants, like [ŋk] and [xtʃk]. We don’t find the maximum length of four consonant clusters inside of words, the maximum we expect by adding together the longest coda and the longest onset. Importantly, though, the two and three consonants we find do not exceed the possible four consonants. What we find is consistent with our understanding of Chol syllables, because internal clusters are not longer than the expected maximum length.

Is it a problem that the internal clusters are shorter than the expected maximum length? Not really. Many languages have more variety at word edges than in the middle of words, and Chol is one of them.



Comprehension Check 12

By now you have a hypothesis that Sierra Popoluca syllables can have up to two consonants in an onset and up to three consonants in a coda. Below you have a list of Sierra Popoluca words which contain word-internal consonant clusters.

Given the number of consonants in the word-internal consonant clusters, do you think the words in the list support your hypothesis? Give your reasons.

- | | | | |
|----|-------------------|----------------------------|------------------|
| a. | <i>jimpa</i> | [himpa [?]] | ‘it is spicy’ |
| b. | <i>kaʔnpu</i> | [ʔkaʔnpu [?]] | ‘egg’ |
| c. | <i>ʔa+soʔpspa</i> | [ʔa'soʔpspa [?]] | ‘I’m very tired’ |
| d. | <i>wiʔkkuj</i> | [ʔwiʔkkuj] | ‘food’ |



Here’s our answer

The words in the list support our hypothesis about the Sierra Popoluca syllable template. According to our hypothesis, Sierra Popoluca words can have up to three consonants in a coda and two in an onset. If a syllable with a three-consonant coda is followed by a syllable with a two-consonants onset, we get a sequence of five consonants. In order for our hypothesis to be true, there should be no word that contains more than five consonants in a row. The word-internal consonant clusters in our words all contain four or fewer consonants. Therefore, we can conclude that the given word list aligns with our hypothesis.

- | | | |
|----|----------------------------|------------------|
| a. | [himpa [?]] | two consonants |
| b. | [ʔkaʔnpu [?]] | three consonants |
| c. | [ʔa'soʔpspa [?]] | four consonants |
| d. | [ʔwiʔkkuj] | three consonants |



Exploring my language

Given the onsets and codas you identified in your language from the word-edge tests, how long is the longest word-internal consonant sequence that you expect?

If you expect clusters of two or more consonants inside words, try to find five example words to illustrate these clusters. As always, feel free to use a dictionary or consult with other people.

Are the word-internal clusters in your list in accord with the expectations from your word-edge tests? Do you think the word list supports your hypothesis? Why or why not?

6.2.4.2 Word-internal vowel sequences

There is another question we can ask about word-internal sounds based on syllable templates: Do we expect to see vowel sequences in the middle of a word? We can verify our findings from word-edge tests with this question. If it's a language where every syllable begins with a consonant and every syllable nucleus is a short, single vowel, we wouldn't expect to see vowel sequences in the middle of a word. Figure 3 shows why this is the case.

$$\boxed{\text{CV}} + \boxed{\text{CV}} = \boxed{\text{CV.CV}}$$

Figure 3: CV syllables in sequence

In a language where every syllable begins with a consonant, there should always be at least one consonant between vowels inside of words. Chol is such a language. We decided in our earlier discussion of Chol syllables that every syllable begins with a consonant (see discussion of (26)). This means that there should always be at least one consonant between vowels inside of Chol words. Therefore, we wouldn't see any vowel sequences in Chol words. A couple of examples are given in (30).

(30) A consonant in between vowels in Chol

- a. *ja'al* [xaʔa_i] 'rain'
- b. *chikiñ* [tʃiki_n] 'ear'

We can see that there is a consonant in between two vowels in the words given. These words are consistent with the requirement that every Chol syllable begins with a consonant.

When onsets are not required in syllables, we expect to find sequences of vowels in the middle of words, where the vowels are in different syllables. The

Djeoromitxí syllable template, summarized as (C)V in (20), allows just such sequences. We’ve seen onsetless syllables in Djeoromitxí words like [oo] ‘wind’ and [hau] ‘lizard’.

We also expect to see word-internal vowel sequences in a language of which syllable nucleus can have two vowels (i.e., long vowels, diphthongs). Arapaho is such a case. As discussed in 6.1.3.1, Arapaho syllables can have two vowels as their nuclei. We can easily find vowel sequences in Arapaho words as expected.

- (31) Arapaho word list
- a. *néécee* [né:tʃɛ:] ‘chief’
 - c. *hou* [hou] ‘blanket’



Comprehension Check 13

In the previous Comprehension Checks, we’ve explored Sierra Popoluca and Kubeo syllable structure. The questions here ask you to think about expected word-internal vowel sequences in Sierra Popoluca and Kubeo using what you learned about these languages in earlier Comprehension Checks.

1. Do you expect to see any vowel sequences made up of vowels in two different syllables inside Sierra Popoluca words?
2. Do you expect to see any vowel sequences made up of vowels in two different syllables inside Kubeo words?
3. Do the words below support your answers to these two questions? Why or why not?

Sierra Popoluca			Kubeo		
<i>xix</i>	[ʃiʃ]	‘cow’	<i>bia</i>	[ˈbia]	‘chili pepper’
<i>nuuma</i>	[ˈnu:maʔ]	‘certainly’	<i>abi</i>	[aˈbi]	‘he said’
<i>wejpa</i>	[ˈwehpaʔ]	‘he cries’	<i>hia</i>	[ˈhia]	‘river’

By the way, this is the last Comprehension Check in the series building on these two languages.



Here's our answer

1. No, we don't expect to see any vowel sequences made up of vowels in two different syllables in Sierra Popoluca words. Sierra Popoluca syllables always begin with an onset. Therefore, there will always be at least one consonant — the onset — between two syllables, and so between two vowels, as in CV.CV (CVCV).
2. Yes, we expect to see vowel sequences made up of vowels in two different syllables inside of Kubeo words. Kubeo syllables do not require an onset. Therefore, two syllables can meet without a consonant in between, as in CV.V. (While the forms we have seen in Kubeo suggest that CV.V sequences are possible in Kubeo, Chacon (2012) concludes that Kubeo words do not have CV.V sequences. What looks like it might be a CV.V sequence actually is a single CVV syllable. The reasons take us afield from our primary purpose.)
3. Yes, the given word lists support the earlier answers. In the Sierra Popoluca word list, there are no vowel sequences made up of vowels in two different syllables. There is a long vowel in ['nu:ma²], but it is within a single syllable, syllabified ['nu:.ma²]. In the Kubeo word list, we see vowel sequences that are likely to be a result of two syllables meeting each other (e.g., ['bia], ['hia], likely to be ['bi.a], ['hi.a]).



Exploring my language

In the previous Language Explorations, you were asked to figure out your language's possible syllable template. (If you haven't tried the previous Language Explorations yet, please try them first before attempting this Language Exploration.) Based on your proposed template, answer the questions below about what you expect inside of words.

1. Do you expect to see any vowel sequences made up of vowels in two different syllables inside of your language's words?
2. Look at the words from your earlier explorations. Do they support your answer to question 1? What is your reasoning? If there are discrepancies, what do you need to do?

6.2.5 Summary

In this section, we learned how we can relate word-edge information and word-internal information. By looking at the consonant sequences in the middle of words, we can check if our findings from the word-edge tests are correct. We also learned how a vowel sequence can (or cannot) occur when two syllables meet each other inside of a word. Our findings from the word-edge tests tell us whether to expect to find vowel sequences made up of vowels in two different syllables inside of words in the language. If the syllables of a language always begin with an onset, we will not find any vowel sequences where the vowels are in two different syllables. On the other hand, if a language has onsetless syllables, we expect to see vowel sequences split across two (or even more) syllables.

A question that we have yet to answer is *where is the boundary between syllables inside of a word?* In the next section, we see why figuring out syllable boundaries can be tricky in some cases. When we know where the syllables are in a word, we know the [syllabification](#) of a word. We discuss some challenges for syllabification in the next section.

6.3 Syllabification

While learning about syllables, we saw many words divided into syllables, shown by dots somewhere inside the words. Our next question is *how do we know where to put the syllable boundaries inside words?* The act of dividing words into syllables is called [syllabification](#). Syllabification is needed for various reasons – to teach the pronunciation, to find out if a word follows the syllable pattern of the language, to analyze prosodic patterns such as stress, etc. It is not very difficult to syllabify a word in many cases. In this section, we show you the steps to follow to syllabify words, and what to do if you get stuck with a challenging case.

6.3.1 The basics of syllabification

There are three things to know when you start to syllabify a word in a language: *What are the vowels? what are the consonants? What is the syllable template for the language?*

If you know what the vowels and consonants are in a word and you know the template, then there are only three steps in syllabification.

1. Assign each vowel to a **nucleus**.
2. Assign each consonant to an **onset**, up to the maximum defined by the template.
3. Assign remaining consonants to **coda**, up to the maximum defined by the template.

You can see that the three steps match up with the three parts of a syllable, nucleus, onset, and coda. The critical piece is to do the three steps in this order: nucleus, onset, coda.

Let's see how it works. For our starting example, we use the Djeoromitxí word [takahõ] 'stone, gravel'. Recall from (20) that the Djeoromitxí template is (C)V.

6.3.1.1 Nuclei

The first step is: *Assign each vowel to a nucleus*, since vowels make the nuclei of syllables. What are the vowels in the word [takahõ]? (We hope that you know they are [a], [a], and [õ].) We can show that each vowel is the nucleus of a syllable in a figure by drawing a line from each vowel to a symbol designating syllable, as in Figure 4. We follow linguistic convention and use the symbol σ (the Greek letter sigma).

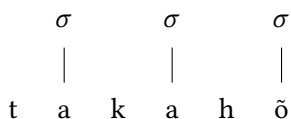


Figure 4: Identifying syllable nuclei

6.3.1.2 Onsets

Once all the vowels are assigned to syllables, we are ready to move on to the second step: *Assign each consonant to an onset, up to the maximum defined by the template*. An onset is at the beginning of a syllable, and is followed by a vowel.

So we check our word to see whether there is a consonant in front of any of the vowels.

In [takahō], we find a consonant before each vowel. To show that these consonants are onsets, we use a line to connect each consonant to the syllable of the following vowel, resulting in the syllables [ta], [ka], and [hō], as in Figure 5. Since the template is (C)V, only one onset can be put into each onset.

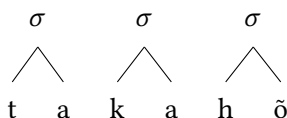


Figure 5: Identifying onsets

After connecting the onset consonants, we check whether there are any sounds left to be syllabified. There are not: Every sound in the word [takahō] is assigned to a syllable, shown by the connecting lines. We don't have consonants left over. Once the syllabification is done, we know where to put the dots if we want to use that representation: Dots go between syllables, giving [ta.ka.hō].

Before we move on to an example with codas, let's pause a moment to think about what we just did. By assigning onsets in step 2, right after assigning the nucleus vowels to syllables, we make sure that a single consonant between two vowels is the onset of the second syllable, not the coda of the first syllable. There is a very strong general tendency that onsets are preferred over codas. This is such a strong tendency that linguists call it the *Onset First Principle*.

6.3.1.3 Codas

We now work through an example involving codas, the Ashéninka Perené word [tʰoŋka:ntsʰi] 'to bathe in steam'.⁷ The Ashéninka Perené syllable template is CV(V)(C): onsets are required, vowels can be short, long, or diphthongs, and up to one coda consonant is permitted. (In Ashéninka Perené, the coda is always a nasal consonant.)

We start our syllabification by reviewing the first two steps: Assigning vowels (the nuclei), to syllables, and then adding onsets.

⁷Ashéninka Perené is an Arawakan language spoken in Southeastern Peru. Our source for Ashéninka Perené is Mihás (2010).

Try this. Try carrying out the first two steps in syllabification, assigning vowels to syllables then adding onsets. You can check yourself against Figure 6 and Figure 7 to see whether you've got it right.

First step is *assign each vowel to a nucleus*. We identify the vowels in the word [tʰoŋka:ntsʰi]: [o], [a:], and [i]. These vowels are the nuclei of their syllables, and so we assign each to its own syllable, as shown in Figure 6.

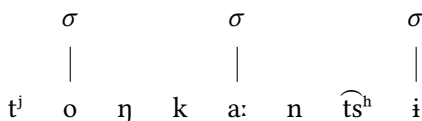


Figure 6: Identifying syllable nuclei

The second step, *assign each consonant to an onset, up to the maximum defined by the template*, is to connect the consonants before vowels as onsets. The consonants that are right in front of the syllable nuclei are [tʰ], [k], and [tsʰ]. We connect these consonants with each syllable nucleus, as shown in Figure 7.

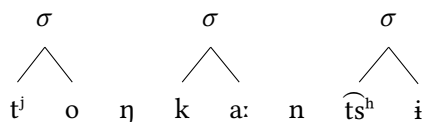


Figure 7: Identifying onsets

At this point when we inspect Figure 7, we can see that two consonants are not yet in a syllable, [ŋ] and [n]. (We don't put them into onsets because the syllable template, CV(V)(C), does not allow more than one consonant in an onset.)

We are ready for step three: *Assign remaining consonants to codas*. These consonants get drawn into the preceding syllable, to show that they are the codas of those syllables, shown in Figure 8.

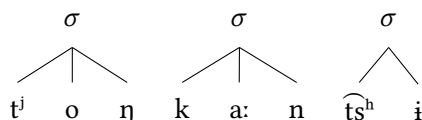


Figure 8: Adding codas

Since there are no more sounds left to be syllabified, the syllabification is done. We now know where to place the syllabification dots in [tʰoŋka:ntsʰi] since they go between the syllables: The result is [tʰoŋ.ka:n.tsʰi].

6.3.1.4 Summary

In this section, we learned the three steps in syllabifying a word, starting with vowels to find nuclei, then turning to consonants to add onsets, then codas. Once all the consonants are in syllables, syllabification is done. In many cases, syllabification can be done by following the three steps.

There is a very strong general tendency that a single consonant in between two vowels is an onset rather than a coda. That is why adding onsets precedes adding codas in our three steps. Linguists call this the **Onset First Principle** because it is such a strong tendency.

For some words in some languages, the three steps are too simple, or they create syllables that “feel wrong” to speakers. Section 6.3.2 will introduce some cases where syllabification can become more challenging, and some strategies to try if you encounter a challenge.



Exploring my language

Find a word from your language that you wish to syllabify. Following the steps explained earlier, try syllabifying the word. Draw a figure as you proceed.

Did you succeed in syllabifying all the sounds in the word? If you encountered a challenge, make a note and see if the next section gives you tools to resolve the challenge.

6.3.2 Syllabification challenges

How can the three steps (find nuclei, add onsets, add codas) fail? Problems might come up with syllabifying consonants (Section 6.3.2.1) or with syllabifying vowels (Section 6.3.2.2). Choose the sections that matter to you.

6.3.2.1 The puzzle of word-internal consonant clusters

In languages that have CV(C) syllables (at most one onset consonant and one coda consonant), it is clear how word-internal consonant clusters, ...VCCV..., are syllabified – the boundary comes between the two consonants: VC.CV. We can't syllabify before the two consonants, V.CCV, because that makes an onset cluster and the template only allows one onset consonant. We can't syllabify after the two consonants, VCC.V, because that makes a coda cluster which again is bigger than the single coda consonant that the template allows. (It also would leave no onset for the second syllable.) A CV(C) language is unambiguous in syllabifying a VCCV sequence as VC.CV.

What happens if you have a (C)CV(C) language or a (C)CV(C)(C) language? In these cases, there are questions about how to syllabify word-internal consonant clusters, both ...VCCV... and ...VCCCV.... Let's look at the two-consonant cluster first, ...VCCV....

In a language with (C)CV(C) syllables, a word-internal CC sequence could not be syllabified as VCC.V (with a complex coda and no onset), for two reasons. First, the template allows only one coda consonant so a complex coda is impossible, and the template requires an onset but the second syllable has no onset. Still, there are two possibilities. The syllabification could be either a two-consonant onset, V.CCV, or a coda-onset sequence, VC.CV. Both of these are possible in a (C)CV(C) language. How do we resolve the puzzle?

To make this concrete, we look at an example from Chol. Chol is a language where syllables can have multiple consonants in both the onset and the coda (review Section 6.2 if you need to). Chol has (C)CV(C)(C) syllables – up to two consonants in both onset and coda. This means there is ambiguity when we try to syllabify a word like *jiñku* [xiŋku] 'yes, it is'. Is it syllabified [xi.ŋku] (CV.CCV) or [xiŋ.ku] (CVC.CV)? Here are some ways to approach the problem.

6.3.2.1.1 Ask speakers to speak very slowly

One way of figuring out syllable boundaries is to use speaker intuition. Speakers have internal knowledge about where syllables are, so you can ask for their help when figuring out syllable boundaries.

You might ask speakers to pronounce a word, phrase, or sentence very slowly and see whether they separate sounds into syllables when they do this task.

Remember our Chol word *jiñku* [xiŋku] 'yes, it is'. Speaking very slowly, speakers might say [xi...ŋku], suggesting the syllabification [xi.ŋku]. They might say [xiŋ...ku], suggesting the syllabification [xiŋ.ku]. (Some speakers are confused

by the idea of trying to speak slowly, so may not really be able to help with this strategy.)

6.3.2.1.2 Learn about the distribution of CC onsets

Sometimes there are restrictions on where CC onsets can occur. The Chol situation is resolved when more is understood about Chol syllables. In fact, the complex onsets are only found at the beginnings of words when a single consonant prefix is added to a word: [pam̩] ‘forehead’, [kpam̩] ‘my forehead’. In this case, understanding the word and morphological structure explains how to syllabify word-internal sequences: CC is not a good onset in the middle of a word, only at the beginning of words. (If you need a refresher on how to find morphological structure, you could look at Chapter 3.)

There can also be restrictions on which consonants are possible in an onset or in a coda. Looking at word edges can give clues about which sequences make a good onset or a good coda. If your language has complex onsets or codas, there may be limits on exactly what sounds can occur together at syllable edges.

6.3.2.1.3 Look at the sonority of the consonants

Ayutla Mixe (Mixe-Zoquean family, spoken in Southern Mexico, Romero-Méndez 2008) has syllables with up to two onset consonants and four coda consonants. We have possibilities, then, for really long word-internal consonant clusters. There is a three-consonant cluster in *teety ja’a* [teːtʰ.ɟaːa] ‘father.this’; there is also one in [θkwentʰ.pikj] *tkentpiky* ‘(s/he).pays.attention’. The first is syllabified with a complex onset (...C.CC...) and the second with a complex coda (...CC.C...). Why does Romero Mendez propose different syllabifications?

The answer is found when we look at the sonority of the three consonants in each cluster. Remember sonority, from Chapter 2? (See Chapter 2, Section 2.3.1 if you want to review sonorants).

Sonority has to do with how open your mouth is when you make a sound. CVC syllables start with the mouth fairly closed (the onset C), then have the mouth fairly open (the V), then the mouth closes again (the coda C).

This syllable sonority pattern of closed-open-closed is also found when the syllable has a consonant cluster in the onset or the coda — the consonant closest to the nucleus is more sonorous than the one that is further away, so that the segments of the syllable continue to rise in sonority as they get closer to the nucleus and fall in sonority as they get further away.

We can use sonority to create a *sonority hierarchy* of sounds — vowels are the most sonorous because the mouth is the most open, and obstruent stops are

the least sonorous because the mouth is completely shut. In between we find sonorous consonants, which are more sonorous than obstruent continuants. The sonority hierarchy is shown in (32). Some languages even make divisions within these categories: Within vowels, the lower a vowel is the more sonorous it is (because the mouth is more open); glides are more sonorous than liquids (again due to how open the mouth is); nasals are the least sonorous of the sonorants (because the mouth is closed even though air flows through the nose).

(32) Sonority hierarchy

vowels > sonorants > obstruents

Now that we know the relative degrees of sonority, we can use the sonority contour within a syllable to resolve the Ayutla Mixe puzzle. We start by figuring out the sonority of the six consonants; our results are shown in Table 1. Recall that our two words are *teety ja'a* [te:tʰ.ɕja²a] 'father.this' and [θkwent.pikj] *tkentpiky* '(s/he) pays attention'. In this figure, we put a dashed line where the syllable boundary is.

Table 1: Sonority in [...tʰ.ɕj...] and [...nt.p...]

	[tʰ]		[c]	[j]		[n]	[t]		[p]
sonorants				x		x			
obstruents	x		x				x		x

When we lay the information out like this, we can easily see that [tʰ] and [c] are both obstruents and so have equal sonority; the same holds of [t] and [p]. The sonorants [j] and [n] have greater sonority. And we see that the syllable boundary is placed to create an onset of increasing sonority in [...tʰ.ɕj...] and a coda of decreasing sonority in [...nt.p...].

The Ayutla Mixe syllable onsets and codas follow the general tendency for more sonorant consonants to be closer to the nucleus, and less sonorant consonants to be further away from the nucleus. If your language has complex onsets and/or codas, using sonority may help you figure out where the syllable boundaries are.

6.3.2.1.4 Summary

Determining syllable boundaries within a word can be tricky if there are word-internal consonant clusters. In most languages, there are some guidelines for figuring out where the syllable boundaries are within words: Asking a speaker

of the language to say the word very slowly, examining the distribution of consonant clusters in the language, and looking at sonority are all useful. In some cases, it might not be possible to figure out exactly where the syllable edges are. And in some cases, it may not matter exactly where they are.

6.3.2.2 The puzzle of word-internal vowel sequences

Similar to the consonant cluster case, when there is a vowel sequence, it can be challenging to determine the location of syllable boundaries. When we are dealing with a language that can have a long vowel or a diphthong as its syllable's nucleus, we have a question: Are we looking at two short vowels (V.V) or either a long vowel (.VV., also .V:) or a diphthong (.VV.)? Here are some strategies to follow in trying to sort out the different types of syllable nuclei.

6.3.2.2.1 Tapping or clapping in time with syllables

If you are uncertain about the number of syllables in a word, saying a word and clapping or tapping once for each syllable can help. You can do this with your own language or with other speakers of a language.

For example, suppose you are working with Ashéninka Perené native speakers and you find the word *intsoNpoi* [ɪntsʰompɔi] 'inside'. You are not sure if the two different vowels in sequence – [...poi] – make one syllable or two. Is it syllabified [ɪn.tsʰom.pɔi] or [ɪn.tsʰom.po.i]? If you can, ask a speaker to tap for each syllable while saying the word. If the speaker gives three taps, then it is likely that [poi] is a single syllable – [ɪn] [tsʰom] [poi]. If the speaker gives four taps, then it is more likely that [poi] is two syllables – [ɪn] [tsʰom] [po] [i].

Similarly, Ashéninka Perené has words with long vowels, like *aa*. This raises the question: Is a word like *naaka* a two syllable word ([na:ga]) or a three syllable word ([na.a.ga])? You can ask the speakers to tap or clap while pronouncing each syllable in the word. If the speakers clap twice while they pronounce the word, it suggests that the word consists of two syllables ([na:] and [ga]). On the other hand, if the speakers clap three times while pronouncing the word, it suggests that the word consists of three syllables ([na], [a], and [ga]).

Note: Not all speakers are adept at tapping or clapping with syllables. Some people need coaching to understand the task, and some people never figure out what to do.

6.3.2.2.2 Restrictions on sequences, or phonotactics

Every language has some restrictions on sounds that can occur in a sequence. The term linguists use for these restrictions is *phonotactics*. A statement like “only voiced consonants are found between vowels” is a phonotactic because it tells us about the kinds of sequences we find in a language. (You can learn more about phonotactics in Chapters 7 and 8.)

Knowing about phonotactics sometimes helps when determining syllable boundaries, because syllabification will conform to the phonotactics. This means that you can rule out syllabifications that violate the phonotactics of the language.

Let’s have a look at an example with a phonotactic constraint on vowels. Rarámuri (also known as Tarahumara) is a Uto-Aztecan language spoken in the southwestern part of the Mexican State of Chihuahua. The example we will see is from Choguita Rarámuri. Choguita Rarámuri is the variety of Rarámuri spoken in the community of Choguita (also known as Choguita de Guachochi or Choguita de Norogachi), located in the municipality of Guachochi. Our source is Caballero (2008).

Choguita Rarámuri has five vowels, [i], [ɛ], [a], [ɔ], and [u]. It also has diphthongs. And it has onsetless syllables inside of words. This raises a question: When you encounter a sequence of two vowels, is it one syllable or two? For example, how do we syllabify the sequence [éa] in the Choguita Rarámuri word *čokéari* [tʃókéari] ‘mountain dove’.

Does the vowel sequence [éa] belong to a single syllable or two different syllables? These two syllabifications are shown in (33).

- (33) Two different ways to syllabify a word [tʃókéari]
- a. one syllable [tʃo.kéa.ri] CV.CVV.CV
 - b. two syllables [tʃo.ké.a.ri] CV.CV.V.CV

To answer this question, we look at Choguita Rarámuri phonotactics, in particular the phonotactics about diphthongs. When we examine diphthongs, we find that every diphthong has the vowel [i] in it: Five have [i] following one of the four other vowels, [ei], [oi], [ui], [ai], and two have [i] preceding a non-high, non-front vowel, [ia] and [io]. These six are the only vowel sequences that can occur within a same syllable.

Vowel sequences without the vowel [i], or with the vowel [i] before [e] or [u], may occur in Choguita Rarámuri words, but they cannot occur within a same syllable, because that would violate a phonotactic on diphthongs in the language.

This means that if we see such vowel sequences in Choguita Rarámuri words, we can be certain that the two vowels belong to two different syllables.

When we look at the word *čokéari* [tʃokéari] ‘mountain dove’, we see that neither vowel in the sequence [éa] is the vowel [i], so this cannot be a diphthong. The only possible syllabification that is consistent with the diphthong phonotactics is to put the two vowels into separate syllables, [tʃo.ké.a.ri], shown in (33b).

In the above discussion, we learned that the vowel sequence [éa] is not a possible diphthong in Choguita Rarámuri and so cannot belong to a single syllable. With this phonotactics, we can exclude (33b) from the possibilities.

As we saw from the two examples, knowing a language’s phonotactics can be useful when determining syllable boundaries.

6.3.2.2.3 A look at stress patterns

In some languages, when there are both heavy syllables (CVV and CVC) and light syllables (CV), **stress** is typically found on the heavy syllables, not the light syllables (unless there are no heavy syllables in the word). When you are questioning whether a VV sequence is in one heavy syllable, .CVV., or two light syllables, CV.V, stress might help.

Ashéninka Perené has a stress system of this sort, stressing a final light syllable only if there are no heavy syllables in the word. For example, in the word *aka* [a.ˈka] ‘water’, both syllables are light, and stress is on the final syllable (shown by the mark ˈ before the last syllable, [ka]).

Can this tell us about how “VV” is syllabified? If we see ...VˈV, then we know it is two light syllables because if there are only light syllables, the last syllable is stressed. If we see ...ˈVV, then we know it is one heavy syllable (and it is stressed). What do we see?

In (34) we see a Ashéninka Perené word: What does the stress tell you about the syllables? Is it [...pˈja.a] or [...pˈjaa]? Try to answer before reading on!

- (34) Ashéninka Perené word with [aa] or [a:]
taNˈpyaa [tamˈpˈjaa] or [tamˈpˈja:] ‘wind’

We hope you said the long vowel is in one syllable: *taNˈpyaa* [tamˈpˈja:] ‘wind’ (recall that you might see [tamˈpˈjaa] in some writings, if the author does not use the “:” symbol). The stress is not on the final vowel (we do not hear [tamˈpˈja.a]), so the sequence can’t be two light syllables.

6.3.2.2.4 Recap of strategies for syllabifying vowel sequences

We presented three strategies for syllabifying word-internal vowel sequences: The clapping strategy (which requires a fluent speaker) and two analytic strategies. One analytic strategy looks at phonotactics, the restrictions on sequences that can occur inside a syllable or word. The other looks at a phonological pattern — in our example, stress — as a means for determining whether you are looking at two short vowels or one long vowel.



Exploring my language

In the previous Language Exploration, you were asked to write down any challenges you encountered while syllabifying a word from your language. (If you haven't tried the previous Language Exploration, please attempt it before trying this one.) This time, try syllabifying the word again and see if you can overcome the challenges with the tips introduced in this section. Were you able to syllabify the word this time?

6.3.3 Summary

Because it can be useful to know where syllables are in a word, in this section we addressed the question of how to put the sounds of a word into syllables.

In Section 6.3.1, we dealt with the basic three steps in syllabification — finding nuclei, adding onsets, adding codas. An important point is to do the three steps in this order, so that consonants are syllabified as onsets rather than codas (in languages where having a coda is an option).

In Section 6.3.2, we looked at challenges that come up when you are syllabifying sequences of consonants inside a word (Section 6.3.2.1) and those that come up when you are syllabifying a sequence of vowels inside a word (Section 6.3.2.2).

You now have the tools to figure out the syllables of the words of your language. Some words may be harder to syllabify than others; we recommend that, if some word presents a challenge to you, you try working with other words that are more straightforward first, until you are really confident in your syllabification skills.

This may give you as much as you want or need to know about syllabification. If so, jump to Section 6.5 to read about how to make use of this knowledge in language work. On the other hand, you may want to know a little more.

The next section, Section 6.4, presents four different ways to visually show the syllables in a word. Each of the four highlights different properties of syllables. Some people are visual learners, so reading this section may help solidify your understanding, and/or make it easier for you to explain syllables to others. As a bonus, you already are familiar with the first two representations.

6.4 The visual representation of syllables

Visual representations of syllables can help make syllabification concepts more clear. Also, it is worth learning about the conventions before reading a linguistic discussion about syllables in your language, because the conventions discussed here are used in linguistic literature.

On the other hand, if you aren't reading linguistic literature, or you've been finding our discussion clear, you may want to skip this section and go on to Section 6.4.3, where we look at how you find syllables in a word.

We look at four types of representations: the dot convention (already introduced), flat structure (already introduced), internal structure, and moraic representation. To check whether this section will cover new material for you, look at flat structure in figure Figure 9, internal structure in Figure 13, and moraic representation in Figures 22, 24, and 25, and especially Figure 26. If these figures all make sense — or if they are not relevant for your interests and needs — skip this section and go on to Section 6.5.

We first review the dot convention and flat structure, then go on to the other two representations.

6.4.1 The dot convention

The dot convention shows the syllables in a word by putting “.”, a dot or period, between syllables in an IPA transcription; we started using this convention in figure (1). The dot convention shows where the syllable edges are. For example, the Paresi-Haliti word [kamai] ‘sun’ can be represented as [ka.mai], with the dot showing that the first syllable is [ka] and the second syllable is [mai]. The dot convention does not show the internal structure of a syllable — the onset, rime, etc.

Try this. Show the syllables in the Karajá word [kɔwɔrɔ] ‘tree’, using the dot convention. If you are uncertain about the syllables in the word, they are [kɔ], [wɔ], and [rɔ]. (Karajá is a Macro-Je language, spoken in Brazil. We discuss variation in Karajá in Chapter 4.)

We hope that you put two dots in the word as follows, [kɔ.wɔ.rɔ], showing that it has three CV syllables.

Typically, dots are not put at the edges of words, so we rarely see representations like [.kɔ.wɔ.rɔ.]. Putting in the extra dots is not wrong, and it can emphasize that the edges of the word are also edges of syllables. However, the extra dots look a little cluttered, so are not often used. Since we rarely put dots at the edges of words, monosyllabic words (those with only one syllable) usually don’t have dots on each side of them.

6.4.2 Syllable trees

Syllable trees represent the segments and structure of a syllable. Following linguistic conventions, we use the symbol σ and the C, V shorthand when drawing syllable trees of any type. The lower case sigma symbol, σ , is used to represent a syllable. The capital C and V represent consonants and vowels respectively.

There are two types of syllable trees: one with a flat structure and one with an internal structure. We will cover both types of trees in this section.

6.4.2.1 Syllable trees with flat structure

You are already familiar with flat structure — it is the kind of syllables that we created in Section 6.3.1. We review this structure briefly here. A syllable template with flat structure is illustrated in Figure 9a. The lines connecting the Cs and V to the σ show the types of segments found in syllables that fit this template. (You may occasionally see a similar figure that is inverted — having the Cs and Vs at the top and the σ at the bottom. The meaning is the same.)

If you are syllabifying a particular word, rather than showing a syllable template, you add the segments instead of the C and V symbols, as shown in Figure 9b, with the Zihuateutla Totonac syllable *tej* [tɛχ] ‘road, path’ (from (6)).

The flat syllable template shows that the syllable contains a CVC sequence. It does not label the onset, nucleus, coda, or rime, so it shows no internal structure. If you are interested in showing only which segments are in a syllable, this type of syllable tree can be useful.

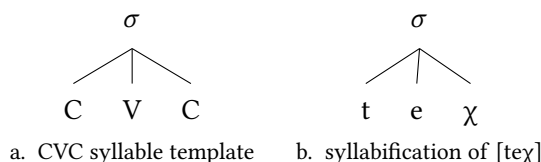


Figure 9: Flat structure: CVC – syllable with coda

Another use of the flat structure is to show the difference between a long vowel and a diphthong. To do this, the CV layer is essential. Long vowels and diphthongs are illustrated in Figure 10. The syllable tree in Figure 10a shows the visual representation of the syllable [nɛ:] from the Arapaho word [né:.cɛ:] ‘chief’; [nɛ:] has a long vowel. The syllable tree in Figure 10b shows how to represent a diphthong using the syllable [kao] from the Paresi-Haliti word [ˈkao.kɐ] ‘arrive’. (These words were first introduced in (3); they may feel familiar.)

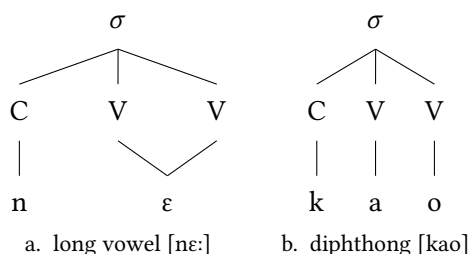


Figure 10: Flat structure: CVV – long vowels and diphthongs

The long vowel in Figure 10a is shown by linking the vowel segment [ɛ] to two Vs. The two Vs show that the [ɛ] is long – [ɛ:]. The diphthong in Figure 10b also has two Vs. One is linked to [a] and the other to [ɪ], showing that this syllable has two vowels in it, the essence of a diphthong.

Leaving the CV layer in flat structure representations makes it easy to check your syllable template: The CV layer in each syllable should be a possible syllable in your language.



Exploring my language

Choose two of the words you identified in the previous Language Exploration. Do you think they have different syllable structure?

Check your answer by drawing a syllable tree for each word using Figure 9b as your model. If you are having trouble figuring out whether the structure of the two syllables is the same, try substituting in the more general C and V labels (instead of specific sounds, as in Figure 9a).

6.4.2.1.1 The visual representation of a syllabified word

Let's see how a word with multiple syllables can be visually represented. We demonstrate this with the disyllabic Paresi-Haliti word [ka.mai] 'sun'. The first step is to create the tree for each syllable.

Try this. Make the visual representations of the syllables [ka] and [mai] using the flat structure. Since one of these syllables has a diphthong, keep the CV layer in your representation.

We hope that the syllable trees you drew are the same as the ones in Figure 11.

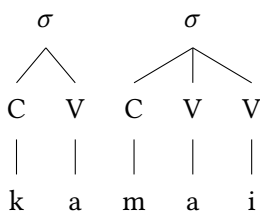


Figure 11: Syllable trees of [ka] and [mai]

Now that we have the syllable trees for [ka] and [mai], all we have to do is assemble them into a word. We do this by grouping the two syllables with additional tree structure, as in Figure 12. We use “Wd” to stand for “word”. The structure shows that the syllables [ka] and [mai] together form a single word, [kamai].

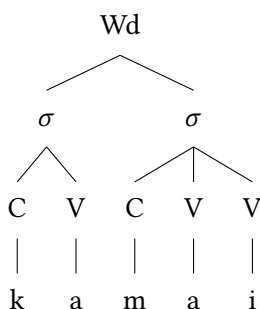


Figure 12: Word and syllable structure of the word [kamai]

Given the word and syllable structure in Figure 12, we know where to put a dot if we want to use that representation — the dot goes between the syllables and the tree structure shows us what the syllables are — [ka] and [mai]. So the dot goes between them: [ka.mai].

If you are using syllable trees, think about what you want to show your readers. Depending on your purpose, you can choose to show the CV layer or to leave it out. Throughout the rest of this section, you will see both types of trees. Our choice depends on the point under discussion.

Both the dot representation and the flat structure fail to show the onset and rime inside of each syllable. If that is important to you, you need a syllable template with internal structure. Read on.

6.4.2.2 Syllable templates with internal structure

We can represent the internal structure of a syllable by including onset, rime, nucleus, and coda, as shown in Figure 13.

Try this. Can you remember or figure out what the symbols σ , O, R, N, C, and V refer to in Figure 13? (Warning: The symbol C is tricky because it has two different meanings.)

Check whether you sorted out what the letters mean. Starting with the σ (which stands for syllable), we see that it branches into two parts, O and R. O stands for Onset and R stands for Rime. So this shows that the syllable has both an onset and a rime. The onset O is connected to a C, showing that a single consonant makes up the onset of this syllable.

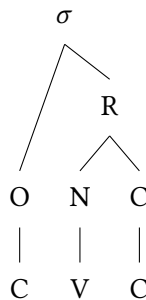


Figure 13: Syllable template with internal structure

The rime also branches, into the nucleus N and the coda C. The nucleus N is connected to one V, showing that there is a single vowel in the nucleus. The coda C is connected to one C — this shows that there is a single consonant in the coda.

Did you notice that we use the same symbol, C, for both a consonant and for a coda? Usually the context will help make it clear which is being referred to (both in this text and in other readings you might do). Being aware of this ambiguity may help you avoid any confusion.

Try this. Use the template in Figure 13 to draw a syllable tree for the monosyllabic Nakoda word [hík] ‘and’. Use these steps:

1. Label the vowel as nucleus (N).
 2. Label the nucleus as a rime (R).
 3. Label any consonant preceding a rime as an onset (O).
 4. Group onset O and rime R together as a syllable (σ).
 5. An unsyllabified consonant following a nucleus is labeled as a coda (C).
 6. The nucleus N and a following coda C group together in the rime R.
- (Notice that the onset step precedes the coda step, ensuring syllables have onsets if possible, even if they don’t have a coda. This is how syllable representations showing internal structure encode the Onset First Principle.)

Is one of the trees in Figure 14 the tree that you came up with? If not, see if you can determine where your picture and ours differ.

Replacing the Cs and V with symbols for the specific sounds, [h], [í], and [k] gives the structure in Figure 14a. Alternatively, you can put the sounds underneath the CV layer as in Figure 14b. However, if you are representing a long

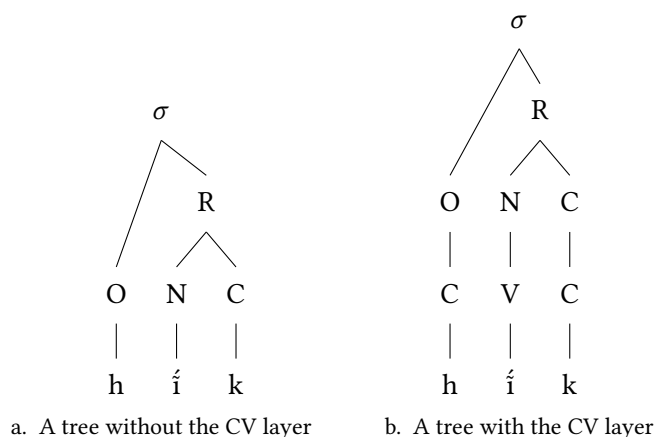


Figure 14: Syllable trees for Nakoda [hík] ‘and’

vowel or a diphthong, you again need the two Vs, as in Figure 12. The Vs always link to the nucleus.

The advantage of showing internal structure comes if knowing what the rime is matters to you. This representation also makes the onset and coda clearer, because there is a label for each. Which type of visual you use in your work depends on what you want to show your audience. Think about your purpose and what picture will be most helpful. As before, if displaying the CV layer better serves your purpose, you can choose to do so. If not, you can omit it.

Every type of syllable can be represented with a syllable tree showing internal structure.

Try this. Try making the syllable tree for the Nakoda word [i] ‘mouth’ showing the internal structure of the syllable, before looking at the figure in Figure 15. Use the steps laid out in the *Try this* box on p. 352. Syllable structure shows only the structure that is actually included in the syllable — think about whether this syllable has an onset, a rime, a nucleus, and/or a coda as you make your representation.

Figure 15a shows the syllable tree for a generic syllable made up of only one vowel. The figure in Figure 15b shows the syllable tree for the Nakoda word [i] ‘mouth’.

The syllable template and the specific syllable tree in Figure 15 both show a V syllable structure. There is no onset or coda. (You might see this referred to

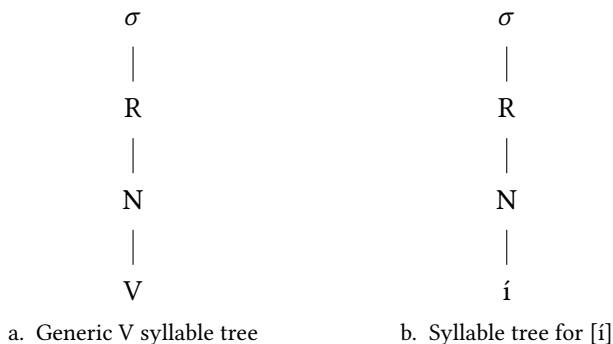


Figure 15: Syllable trees for V syllables

as a non-*branching* structure, because there is only one unit under each label. There is a single V under the nucleus N so the nucleus does not branch; there is a nucleus N but no coda in the rime R, so the rime does not branch; there is only a rime R in the syllable but no onset, so the syllable does not branch.)

Try this. You've seen the tree templates for CVC and V syllables, and example syllabified words. Now try making a syllable tree template for a CV syllable, and a syllable tree for the Nakoda word [hí] 'tooth', again using the steps laid out in the *Try this* box on p. 352. Draw it on a piece of paper before you look at our figures, so you can see how you did.

You can see our structures for CV syllables in Figure 16. These syllable trees branch into an onset and a rime, but the rime does not branch: The rime has only a nucleus, N. There is no coda in a CV syllable, so there is no coda under the rime.

We've seen *branching* rimes in Figure 13 and branching syllables in Figure 16. What about syllables with branching nuclei? That's how we represent both a long vowel and a diphthong. The internal structure of both long vowels and diphthongs is given in Figure 17. A branching nucleus represents the weight or *quantity* of the syllable while the segment represents the *quality* of the long vowel.

The syllable tree in Figure 18a shows the visual representation of the syllable [nɛ:] from the Arapaho word [né:.ce:] 'chief'; [nɛ:] has a long vowel. The syllable tree in Figure 18b shows how to represent a diphthong, using the syllable [kao] from the Paresi-Haliti word ['kao.kɛ] 'arrive', the same words we used in Figure 10 to illustrate long vowels and diphthongs using flat structure. As with the flat

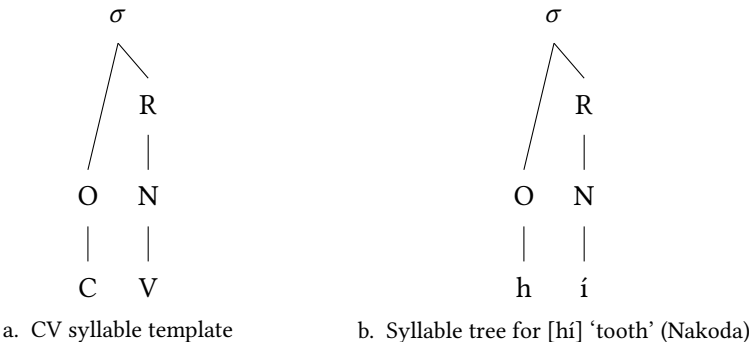


Figure 16: Syllable template and tree for CV syllables

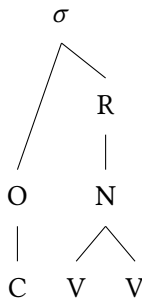


Figure 17: Syllable template for a CVV syllable (long vowel or diphthong)

structure representations of long vowels and diphthongs, the internal-structure representation also includes two Vs in the CV layer. The nucleus of the syllable tree branches to accommodate two Vs. As with the flat structure, a long vowel has a single vowel segment linked to both of the Vs, shown with the long vowel [ɛ:] in Figure 18a, while a diphthong has two different vowel segments linked to one V each, shown with the diphthong [ao] in Figure 18b. The CV layer is helpful in showing this, while the syllable-internal structure is shown by the onset, nucleus, and rime nodes in the trees.

As with the flat structure representations in Figure 10, it is important how segments link to the CV layer. With long vowels, there is one vowel segment linked to two Vs in the nucleus. Figure 18a shows how a long vowel maintains a steady vowel quality within the syllable. Figure 18 shows how with diphthongs there are two vowel segments within a single nucleus. This represents how a

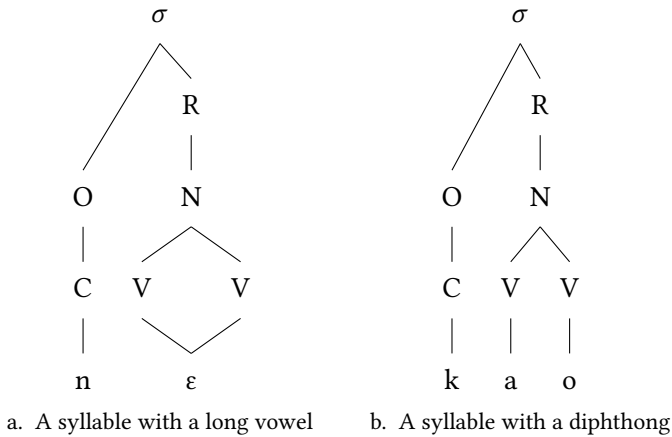


Figure 18: Syllable internal structure for a long vowel and for a diphthong

diphthong slides from one vowel to the next with no syllable break in between.

We’ve seen that nuclei can branch; onsets and codas can also branch. In Figure 19 we show how a CCVCC syllable, with both an onset cluster and a coda cluster, can be represented with a syllable tree, using the syllable [kruʔj] from the Sierra Popoluca word [ˈkruʔj.tʰiː] ‘quail’.

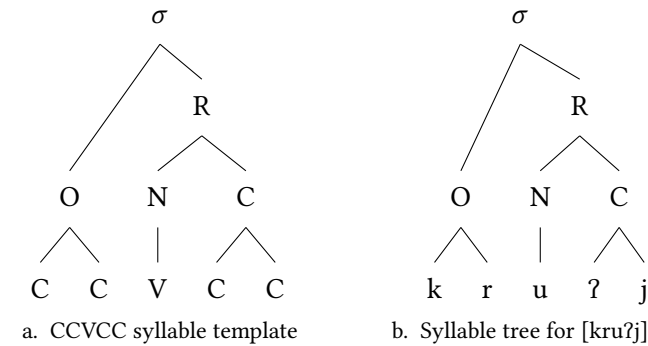


Figure 19: Syllable with internal structure for CCVCC syllables

The syllable template in Figure 19a shows how onset and coda consonant clusters are represented in syllable trees. A consonant cluster is shown by a branching unit — a branching onset for the syllable-initial cluster, CCVCC, and a branch-

ing coda for the syllable-final cluster, **CCVCC**. Remember that the label *C* means “coda” in a syllable tree and “consonant” in the CV layer.



Comprehension Check 14

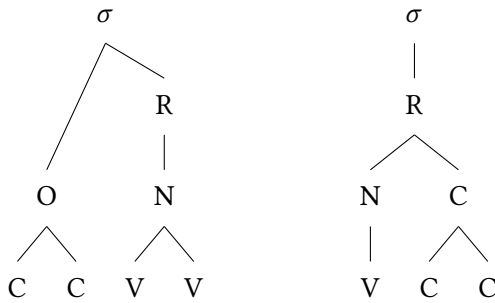
Can you draw syllable templates showing the internal structure for the syllables below? Try drawing the trees without looking back at the examples given earlier, but do remember to start with the nucleus. Refer to the example cases only after you’ve tried on your own.

- a. CCVV b. VCC

For bonus points, is either of these syllables a branching syllable?



Here’s our answer



a. CCVV syllable template b. VCC syllable template

The first syllable tree, in (a), represents the CCVV syllable structure. The onset branches to accommodate a consonant cluster (CC), and the nucleus branches to accommodate two Vs (VV). The second syllable tree,

in (b), shows the VCC syllable structure. As there is no onset, the syllable does not branch into onset and rime, but rather goes straight to the rime. The rime branches as there is a coda, and the coda also branches to accommodate a consonant cluster (CC).

Both syllables are branching. In the CCVV syllable, branching is found in the syllable (branching into onset and rime), in the onset (branching into two Cs) and in the nucleus (branching into two Vs). In the VCC syllable, branching is found in the rime (branching into nucleus and coda), and in the coda (branching into two Cs).

6.4.2.2.1 Representing a word with syllable trees that show internal structure

As noted in Section 6.4.2.1.1, sometimes the author wants to show that certain syllables belong together in a single word. Whether syllables show internal structure or flat structure, the means is the same: We connect the syllables of the word to a tree node labeled Wd, for Word.

Try this. Create a word tree for the Paresi-Haliti word [kamai] ‘sun’, this time showing the internal structure of the two syllables. We have already identified syllables [ka] and [mai]. Draw the syllable template for each syllable, then group the syllables into one word.

After you draw your word tree for [kamai], jump forward to figure (21). If your syllable structures for this word match the ones in our figure, you probably don’t need to read this section, and might want to go straight to the Language Exploration on p. 361, then on to Section 6.4.3. On the other hand, if you didn’t get it quite right or don’t feel confident, read on.

The first step is to create the tree structure for the syllables [ka] and [mai]. We start the syllables with a nucleus for the vowels and a rime for the nucleus, then go on to add the onsets. You can check your structure for [ka] with Figure 16, and your structure for [mai] with Figure 18b. The result is the two syllables in Figure 20.

Now that we have the syllable trees for the two syllables [ka] and [mai], all we have to do is to assemble them into a word. This is shown in Figure 21.

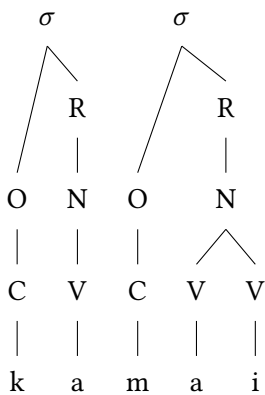


Figure 20: Syllable trees of [ka] and [mai] with their internal structure

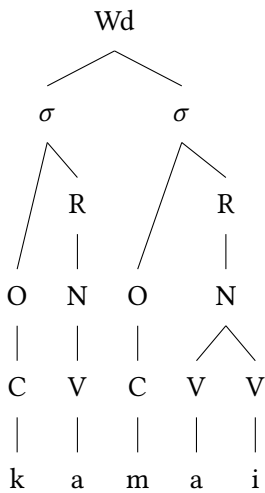


Figure 21: Word and syllable internal structure of the word [kamai]



Comprehension Check 15

Here is a trisyllabic Sierra Popoluca word: *kwidaadoj* [kwi'da:doh] 'be careful'. (The symbol ' indicates that the next vowel is stressed.)

Can you create the word tree for this word, using syllable trees that show the internal structure of the syllables?

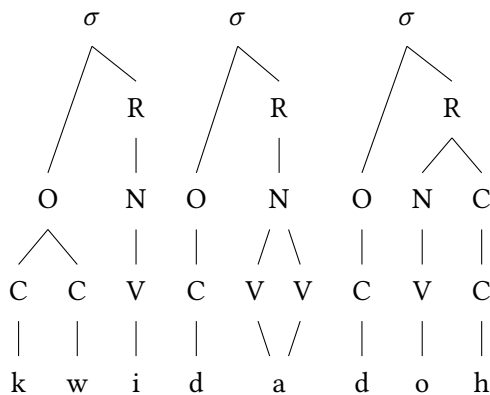
If this feels daunting, here are the steps to use.

1. Identify the syllables (there are three), starting with nucleus, then rime, onset, and coda.
2. Create the syllable tree for each syllable. Hint: Each syllable has an onset, so all three syllables branch into onset O and rime R.
3. Group the three syllables into a word, labeled Wd.

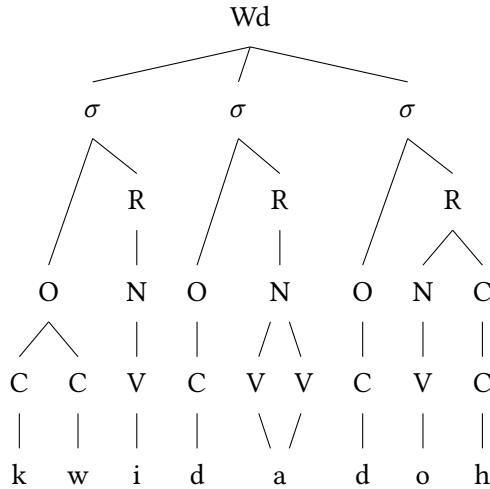


Here's our answer

For [kwi.'da:doh], it helps to draw the trees for [kwi], [da:], and [doh] first. Below are the trees for the syllables.



To make the word tree, assemble the syllable trees by putting “Wd” on top of them and connecting them to it, as shown below.



Exploring my language

Can you think of two words from your language that have at least two syllables in each? Try to make the word trees showing internal syllable structure. Use the steps laid out in the previous Comprehension Check.

You will learn the most if your words have different kinds of syllables in them — so try to avoid words that only have CV syllables.

6.4.3 Syllable trees that show syllable weight

Throughout this chapter, we’ve occasionally referred to syllables as open or closed, heavy or light. These terms were introduced in Section 6.1. In this section, we introduce a representation type that focuses on syllable weight.

6.4.3.1 What is syllable weight?

Syllable weight refers to how “heavy” or “light” a syllable is. Weight, or quantity, is determined by the nature of a syllable’s rime. In general, a syllable whose rime has only one short vowel is a *light syllable*. Any other kind of rime is a *heavy syllable*: A long vowel, a diphthong, a vowel (long or short) followed by a coda. These points are summarized in (35). (Occasionally, syllables with short vowels followed by a coda count as light in a language, not as heavy.)

(35) Syllable weight

light syllable A light syllable has only a short vowel in the rime.

heavy syllable A heavy syllable has a bigger rime — a long vowel, a diphthong, a vowel and a coda.

Note: Onsets do not count for syllable weight.

Linguists use the concept of the *mora* (symbolized as μ) to represent syllable weight. A syllable with more morae is heavier than a syllable with fewer morae. A light syllable has only one mora. In Figure 22, we show the moraic template for a CV syllable. It has one mora; it is a light syllable.

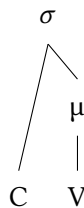


Figure 22: Moraic template for a light syllable

6.4.3.1.1 Representing light syllables with morae

In a moraic representation, as in Figure 22, the syllable, σ , connects directly to the onset consonant C and to one mora, μ . Onset consonants are not represented with morae, since they have no weight. (If there were no onset, there would be no consonant in the structure.) The mora is connected to a V, showing it connects to vowel features. Because there is only one mora, the vowel in this syllable is short and the syllable is light — it has only one weight unit.

Look at Figure 23 to see a specific CV syllable, the Nakoda word [hi] ‘tooth’, represented using the moraic template.

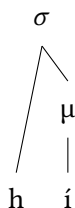


Figure 23: Moraic representation of the Nakoda word [hí] (‘tooth’)

There are three major differences between the moraic representations and the representations presented in sections 6.4.2.1 and 6.4.2.2. First, the weight of a syllable is shown by the number of morae. A syllable with two morae is heavy while a syllable with one mora is light. It doesn’t matter whether the morae are linked to the same vowel, two different vowels, or a vowel and a consonant — as long as there are two morae in the syllable, it is a heavy syllable. This point is explored more fully in Section 6.4.3.1.2.

Second, with morae, segments attach directly to a mora or to the syllable. There is no place for a “CV layer” between segments and syllable structure — if you see a representation with morae and with “C” and “V”, the C and V labels stand for any consonant and vowel features, not a separate layer labeled C and V.

Third, there is internal structure as shown by the morae. However, there are no labels identifying the onset, rime, nucleus, or coda. The only structure in a moraic template is the syllable (σ) and the mora (μ).

6.4.3.1.2 Representing long vowels and diphthongs with morae

Here we look at syllables with two morae, starting with representations for a **long** vowel and for a **diphthong**. Both a long vowel, like [ɛ:], and a diphthong, like [ai], have two morae linked to the syllable, as seen in Figure 24. The moraic tree structures are identical. However, the linkage to vowel features differ. Really look at Figure 24 to see what the difference is and think about what it represents. We use the two syllables that we’ve used before to illustrate long vowels and diphthongs, [nɛ:] from the Arapaho word [nɛ:.cɛ:] ‘chief’ and [kao] from the Paresi-Haliti word [‘kao.kɛ] arrive’.

Structurally, these syllables are identical: Both have two morae in one syllable, and so are heavy syllables. The two representations differ in the segments linked to the morae: There is only one vowel symbol in Figure 24a, so it shows a long vowel, while in Figure 24b there are two vowel symbols, so it shows a diphthong.

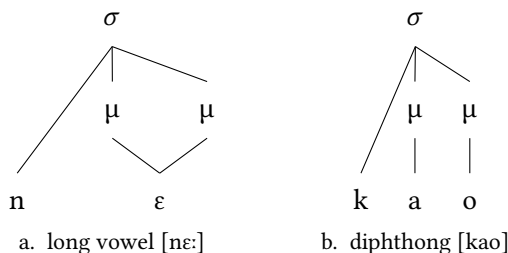


Figure 24: Moraic representation of a long vowel and of a diphthong



Light diphthongs

You might be able now to figure out how to represent a *light diphthongs*. Because it is a diphthong, there must be two different vowel qualities. And because it is light, there can be only one mora. Putting those two together gives a moraic representation showing two different vowels linked to a single mora — the single mora shows that the syllable is light while the two different vowels show that it is a diphthong.

6.4.3.1.3 Representing codas with morae

Syllables with codas are typically also heavy syllables. By now you know that “heavy syllable” means that the syllable has two morae. There is one mora for the coda itself, and one for the vowel.

That being said, occasionally we find a language where codas do not contribute to syllable weight, so even syllables with codas are light. The moraic representations differ then, because heavy syllables have at least two morae while light syllables have only one mora.

Try this. Before looking at Figure 25, try drawing the generic moraic syllable template for syllables with codas, one where the coda adds to syllable weight and one where it does not count for syllable weight. (A generic syllable structure uses “V” for vowels and “C” for consonants instead of segments or features.)

If a CVC syllable is considered heavy, coda consonants have moraic structure. This means that there are two morae in the syllable – one for the vowel, one for the coda consonant, Figure 25a. Light syllables have only one mora for the vowel. This means that if a CVC syllable is considered light, a coda consonant does not connect to a mora, but rather connects directly to the syllable, as shown in Figure 25b.

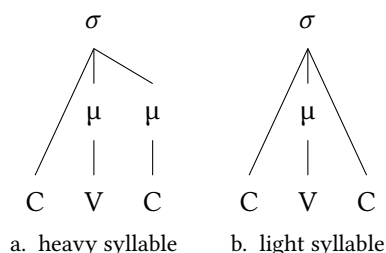


Figure 25: Moraic templates for syllables with codas

6.4.3.1.4 Representing a word with moraic syllables

You can represent a word using moraic structure. You may even have already figured out how to do this. It is essentially the same as was seen with flat structure in Section 6.4.2.1.1 and with internal structure in Section 6.4.2.2.1. Simply group the syllables of a word together under a node labeled Wd for word.

Try this. Draw the moraic syllable structure for the Sierra Popoluca word [kwida:doh] ‘be careful’. First, draw moraic structure for each syllable. Once you have the syllables, try grouping them into a word. For this, consider CVC as a heavy syllable.

Jump forward to Figure 27: If your moraic representation for this word matches our figure, go on to Section 6.4.3.2. If you didn’t get it quite right, note the differences then read on.

The Sierra Popoluca word [kwida:doh] ‘be careful’ consists of three syllables: [kwi], [da:] and [doh]. The first syllable [kwi] is a CCV syllable, which is a light syllable with a complex onset. We haven’t seen templates for complex onsets in moraic structure, but we know that onset consonants connect directly to the syllable so that’s where we put them, as shown in Figure 26a.

The next syllable [da:] is a familiar type: A CVV syllable, a heavy syllable with a long vowel, Figure 26b. The long vowel requires two morae, so we look at

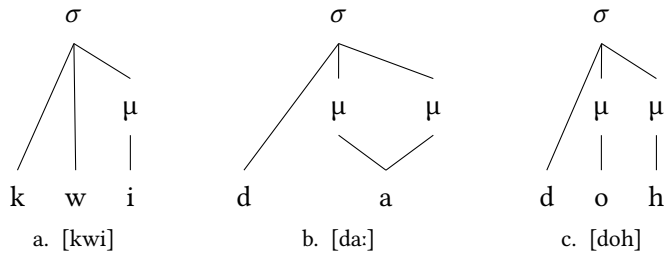


Figure 26: Moraic representations of the syllables in the Sierra Popoluca word [kwida:doh]

Figure 24a for the template. The last syllable [doh] is a CVC syllable, Figure 26c. We consider it to be a heavy syllable, so we use Figure 25a as our model.

Once you get the moraic representations of the syllables, all you have to do is to group them into a word. See Figure 27 for the final moraic representation of the word [kwida:doh].

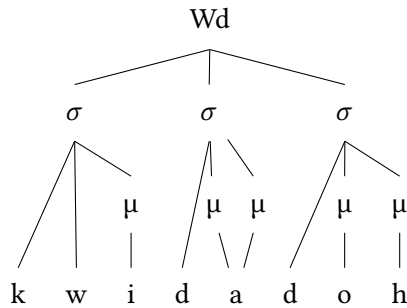


Figure 27: Word and syllables with moraic representation for the Sierra Popoluca word [kwida:doh]

6.4.3.2 Open syllables and closed syllables

So far we’ve explored the difference between light and heavy syllables, and how to represent them using moraic trees. A related but distinct concept is that of open and closed syllables. Any syllable that ends in a vowel is an *open syllable*. Any syllable that ends in a consonant is a *closed syllable*.

(36) Open vs. closed syllables

open syllable A syllable that ends with a vowel.*closed syllable* A syllable that ends with a consonant (a coda).

This distinction is not the same as syllable weight because there are both light and heavy open syllables, and there can be both light and heavy closed syllables.

Try this. Identify the template for each of the four types of syllables, light open, heavy open, light closed, and heavy closed. Tip: It may help to think about the light/heavy distinction in terms of moraic structures.

An open syllable that has a single vowel because its nucleus is light. An open syllable that has a long vowel or a diphthong as its nucleus (CVV) is a heavy syllable. A closed syllable (e.g. CVC) is typically a heavy syllable. Some languages class weight by vowels alone, and codas do not contribute weight. In the moraic structure, these codas would connect directly to the syllable, not to a mora.



Comprehension Check 16

Below are the words from various languages that we've seen from the previous section.

- | | | | |
|----|---------------|----------------------|---------|
| a. | Chol | [muxk ^h] | 'navel' |
| b. | Tutelo-Saponi | [mɤs] | 'but' |
| c. | Paresi-Haliti | [ka.mai] | 'sun' |
| d. | Arapaho | [hó.tʃo:] | 'steak' |

There are six syllables in the word list: [mɤs], [ka], [mai], [hó], [tʃo:], and [muxk^h]. Can you classify these syllables into open syllables and closed syllables?

For bonus points, can you try classifying them into heavy syllables and light syllables? For this exercise, consider any syllable that has a coda, a long vowel, or a diphthong to be a heavy syllable.



Here's our answer

An open syllable is any syllable that ends in a vowel, and a closed syllable is any syllable that ends in a consonant.

A heavy syllable has two morae, regardless of how they are linked to segments. This means that syllables containing a long vowel, diphthong, or a short vowel with a coda are all heavy.

We can put this information into a single table in order to see how the heavy/light and open/closed distinctions are different from each other. Note the absence of light, closed syllables. This is because we set this exercise up saying that syllables with codas are heavy.

	light	heavy
open	[ka], [hó]	[mai], [tʃo:]
closed	—	[maʃ], [muxk ^h]

6.4.3.3 The importance of syllable weight

We have just noted two ways of thinking about the non-onset part of a syllable, syllable weight and whether a syllable is open or closed. In this section we look at why syllable weight matters.

6.4.3.3.1 Syllable weight and word size

In some languages, there are phonological patterns that attend to syllable weight. For example, in San Martín Peras Mixtec, words are fairly short, having just one or two syllables. But not any monosyllabic sequence is a possible word — every word has to have two morae, a minimum weight requirement. Also, not every sequence of two syllables is a word: Disyllabic words are made up of exactly two light syllables, so no word has more than two morae.⁸ We show the syllabification of the two morae in each word at the far right in (37).

⁸San Martín Peras Mixtec is an Otomanguan language spoken in Oaxaca. Our source is Eischens (2022). Ben Eischens kindly checked the forms for us, and told us that speakers of San Martín Peras Mixtec use different ways to refer to their language — Tu'un Nta'vi ([tù'ũ ˈná'βi]) or Tu'un Savi ([tù'ũ sàʰβi]), and as 'mixteco' when speaking Spanish.

(37) Bimoraic words in San Martín Peras Mixtec

						mora
a.	Two syllables	[léló]	‘skunk’	[ámá]	‘when?’	μ.μ
b.	One syllable	[lěè]	‘baby’	[ĩ]	‘one’	.μμ.

6.4.3.3.2 Syllable weight and stress

Some languages have a stress system that takes account of syllable weight. **Stress** (see Chapter 2 Section 2.4.3.6) marks one or more syllables in a word as more prominent in some way – it may be a bit louder perhaps, or a bit longer, or a bit higher pitched than the rest of the word – or be made prominent by some combination of these properties. If your language has stress and you want to figure out the pattern, you may need to know about syllable weight.

Stress in Zihuateutla Totonac nouns pays attention to syllable weight. (Zihuateutla Totonac is a Totonacan language spoken in Mexico; our source for Zihuateutla Totonac is Garcia-Vega (2022).) Stress in Zihuateutla Totonac is on one of the last two syllables in the word – but which one?

Here are some Zihuateutla Totonac words, with stress indicated. We put the unstressed syllables in gray text, to make it very clear which syllable is stressed in each word. We also use the IPA symbol for a stressed syllable – a small raised vertical line (ˈ) right before the syllable that is stressed. In the leftmost column, the same symbol (ˈ) is used in the orthography to indicate a laryngeal vowel – there are laryngeal vowels in (38a, c, e).

(38) Primary stress assignment in Zihuateutla Totonac nouns

Stress on next-to-last syllable

- | | | | |
|----|-----------------|--------------|---------------|
| a. | <i>pó:ti</i> | [ˈpo:ti] | ‘boat’ |
| b. | <i>kílhniˈ</i> | [ˈkił.ni] | ‘mouth’ |
| c. | <i>ka:tániˈ</i> | [kə :ˈta.ni] | ‘celebration’ |

Stress on final syllable

- | | | | |
|----|----------------|------------|--------|
| d. | <i>tintzé:</i> | [tin.tseː] | ‘boss’ |
| e. | <i>qe:láˈt</i> | [qe:ˈlæt] | ‘wasp’ |

Can you see a pattern that tells you which of the last two syllables is stressed? Notice that in (38a, b, c), the final syllable is light – so those syllables each have one mora. And they are not stressed. The final syllable in (38d) is a CVV syllable, [tse:], and the final syllable in (38e) is a CVC syllable, [læt]. We know that both

CVV and CVC syllables are represented with two morae, so structurally they are identical. Final syllables with two morae are stressed.

Putting these two observations together, we can say: “Stress a final heavy syllable, otherwise stress the penultimate (next-to-last) syllable”.

6.4.3.3.3 Summary

We looked briefly at a couple of ways that languages make use of syllable weight — the restrictions on word size in San Martín Peras Mixtec and restrictions on the location of stress in Zihuateutla Totonac nouns. Importantly, weight pays attention to the mora count of a syllable, not to the type of segments filling the morae. If you suspect a syllable-weight pattern in your language, pay close attention to what counts as heavy and what counts as light.

6.5 Discussion

We hope you have gained an understanding of what a syllable is, including the variety of syllables we find. We also hope you can find syllables in your language with confidence. You may also understand visual representations of syllables.

These tools lead to solving the puzzle that we started the chapter with.

6.5.1 Resolving the puzzle

We started the chapter by asking how it is that the sounds of a language can be put together in ways that sound like words, but that do not sound like words of the language itself.

Why can a word made up of sounds of a language not be possible as a word of that language?

We can use our knowledge of syllable structure to answer this question. We looked at the sound sequence [kanta] and asked whether it was possibly a word in Djeoromitxí. Once we take into consideration Djeoromitxí syllable structure, it is clear that [kanta] cannot be a word in the language, even though each sound in the sequence is a sound in Djeoromitxí. The explanation starts with the Djeoromitxí syllable template (C)V (20): There are no codas and no consonant clusters. The sequence [nt] in [kanta] does not conform — either the [n] must be a coda, [kan.ta], or [nt] must be a complex onset, [ka.nta]. Neither syllabification conforms to Djeoromitxí’s (C)V template, so we conclude that this sound combination cannot be a Djeoromitxí word.

Every sound combination must conform to a language's syllable structure in order to be considered a legitimate word in the language. If it doesn't, the sound combination cannot be a word of the language.

6.5.2 Syllable structure and loanwords

It is common for speakers of two different languages to interact, to the point that they adopt words from each other's language. These words are called **loanwords**. What happens when the syllable structures in the two languages are different? Typically, the loanwords are adapted to fit the syllable structure of the adopting language. The most common ways to adapt the syllable structure of a loanword are (i) to add a sound, (ii) to delete a sound, (iii) to change a sound. Let's look at a couple of examples.

6.5.2.1 Changing and adding sounds in Ashéninka Perené

Ashéninka Perené has a number of loanwords from Spanish. The two languages differ in a variety of ways. For example, Spanish uses both [l] and [r], while Ashéninka Perené uses neither, but does have the sound [r]. Ashéninka Perené allows only nasal consonants as codas, while Spanish allows a variety of different coda consonants. How do Ashéninka Perené speakers resolve the differences when borrowing words from Spanish?

Look at the words in (39) and see if you can figure out what is happening. (Parts of the words that remain unchanged are in gray text.)

(39) Ashéninka Perené loanwords from Spanish

	Ashéninka Perené			Spanish original	
a.	<i>borisa</i>	[bo.ri.sa]	'bag'	<i>bolsa</i>	[bol.sa]
b.	<i>paperi</i>	[pa.pe.ri]	'paper'	<i>papel</i>	[pa.pel]
c.	<i>aroso</i>	[a.ro.so]	'rice'	<i>arroz</i>	[a.ros]

The examples in (39) show sound changes and syllable changes. First, in the three words, [l] and [r] from Spanish are replaced with [r] when borrowing words into Ashéninka Perené. Why speakers use [r] for both [l] and [r] is explained by the similarity of their phonetic properties. (See Chapter 2 Section 2.3.1.4 if you want a review.)

The second change is syllabic. Each of the Spanish words has a non-nasal coda, and the Ashéninka Perené syllable template allows only nasal codas. To solve this problem, Ashéninka Perené speakers add a vowel after the Spanish coda consonants, turning those consonants into onsets. With these two changes,

the Spanish words now fit the Ashéninka Perené sound **inventory** and syllable template.

6.5.2.2 Deleting sounds in Chol

Chol has also borrowed words from Spanish. Chol syllables require onsets, (25), while Spanish syllables do not. Spanish allows the voiced velar fricative [ɣ] while Chol has only the voiceless velar stop [k]. Both of these differences came up when Chol speakers borrowed the Spanish word *iglesia* [i.ɣle.sja] ‘church’, as shown in (40).

(40) A Chol loanword from Spanish

	Chol		Spanish original
a.	<i>klesya</i>	[kle.sja] ‘church’	<i>iglesia</i> [i.ɣle.sja]

There are two changes to make [i.ɣle.sja] sound like Chol. First, the Chol sound [k] is substituted for the Spanish sound [ɣ] so that the sounds of the word are all standard Chol sounds. Second, the Chol word begins with [k] while the Spanish word begins with [i] — Chol speakers drop the word-initial [i]. Dropping the [i] gets rid of the onsetless syllable, so that all remaining syllables fit the Chol syllable template.

Adapting loanwords to fit the syllable structure of the target language not only makes them sound more natural, but also easier to pronounce, making them more accessible and easier to integrate into everyday use.



Exploring my language

Choose a foreign word that you’d like to adapt into your language. First, if there’s any sound that is not in your language, replace it with the most similar one that is in your language. Next, analyze the syllable structure of the foreign word and figure out if it matches your language’s syllable structure. If it does not, try adapting it to match your language’s syllable structure by inserting and/or deleting sounds.

6.5.3 Using my hard-earned knowledge

By working through this chapter, you have gained a comprehensive understanding of syllables. You understand about different syllable types and how each language has its own syllable template. You learned how to use word edges to figure out onsets and codas in a language. You learned how to use a template and the syllabification steps to syllabify words in a language. Finally, you may also have learned how to represent syllables visually. What more can you do with your knowledge of syllables?

Language learning. Understanding syllables can enhance language learning, particularly in pronunciation.

Long words can be hard for learners to pronounce, and some languages have a lot of long words. You can use syllabic “backward build-up” to help in pronouncing long words. Here’s how it works. Consider this Hiráaca word: *beer-icgisdabeedhé* [peeritskiʃtabeet^hé] ‘sleet’ (lit. ‘raven’s eye secretion’).⁹ It is quite long, and might be challenging for a learner, especially if they are learning using an oral method. Using an analysis of the syllables, a teacher can identify the six syllables in the word — [pee.ri.tski.ʃta.bee.t^hé] — and use backward build-up. You start with the last syllable of the word: [t^hé]. Now add the next to last syllable: [bee.t^hé]. Then add the syllable before that: [ʃta.bee.t^hé]. Continue until you have the whole word: [pee.ri.tski.ʃta.bee.t^hé].

The advantage of backward build-up over the more standard system of starting from the front of the word is that as you go through the word, the word becomes increasingly familiar and so easier and more fluent — you’re done with the new part after the first syllable. (You can also use this method with phrases and sentences to increase fluent speaking.)

Syllables act as a useful unit for reading, especially when dealing with long words or sentences. Suppose you are a Sierra Popoluca teacher and you are dealing with a sentence *?a+so?pspa* [ʔa’so?pspa[?]] (‘I’m very tired’). The sentence might be challenging for students to read due to the complex consonant cluster ([ʔpsp]). But if you syllabify the sentence and present it in three distinct syllables ([ʔa.’so?ps.pa[?]], students can try reading one syllable at a time — maybe even coupling this with backward build up. After students get used to reading each syllable, they can try reading the whole word or sentence at once.

Language resources — dictionaries, grammars, orthographies. If you are developing an orthography for a language, knowing about the language’s syllable

⁹Hiráaca is a Siouan language spoken in North Dakota in the United States. Our source is Park (2012). Park says that there are no codas in Hiráaca — every syllable is open, ending with a vowel.

structure can help you make better choices. A major choice in writing is whether writing symbols will represent sounds — an *alphabet* — or syllables — a *syllabary*. Cherokee, an Iroquoian language spoken in Oklahoma and North Carolina in the United States, uses a syllabary for writing. Each symbol represents a syllable rather than a single sound. Syllabaries are most effective for languages that do not have many possible syllables, typically for CV or CVC languages. (If the syllable structure of your language is too complicated, you might want to consider an alphabetic system instead.)



The Cherokee syllabary

The Cherokee syllabary system was invented in the 1820s by an Alabama Cherokee named Sequoyah (*sikhowya*, also known as George Gist or George Guess). According to Mooney (1995), Sequoyah first attempted to create a writing system in which each symbol represent a different word of the language (a *logographic system*). Eventually, he realized that there were simply too many words to represent each one with its own symbol. However, he listened closely to how Cherokee is spoken and realized that the words in his language can be analyzed into syllables. With this realization he formed a new set of symbols, the Cherokee syllabary. It has 85 symbols; once those symbols are mastered, a person can fluently read Cherokee texts.



The Rocky Boy Cree syllabary

Kaylene Big Knife, Rocky Boy Cree

Big Knife observes that, while there are only seventeen phonetic sounds in the Rocky Boy Cree dialect of the Algonquian family, the syllabary has 46 symbols, shown in the image below.

AH: SOUNDS AS IN FATHER	E: SOUNDS AS IN GET OR ATE	I: SOUNDS AS IN EAT OR INDIAN	O: SOUNDS AS IN OPEN OR MOON	ENDINGS
◁ ah	▽ e	△ i	▷ o	◌ wa ◌ w
< pah	∇ pe	△ pi	> po	◌ p
◡ tah	∪ te	∩ ti	◡ to	◌ t
◡ kah	◡ ke	◡ ki	◡ ko	◌ k
◡ chah	◡ che	◡ chi	◡ cho	◌ ch
◡ mah	◡ me	◡ mi	◡ mo	◌ m
◡ nah	◡ ne	◡ ni	◡ no	◌ n
◡ sah	◡ se	◡ si	◡ so	◌ s
◡ yah	◡ ye	◡ yi	◡ yo	◌ y

“Observing the Rocky Boy Cree syllabary from a phonetician standpoint, the syllabary contains a total of seventeen (17) phonetic sounds... Using the International Phonetic Alphabet (IPA), there are nine consonants found in the Rocky Boy Cree syllabary in the column “Endings.” The consonants are [p], [t], [k], [m], [n], [s], [j], [tʃ], and [w]. The vowels are depicted in the syllabary’s top row, and the eight vowels are [i], [ɪ], [e], [ɛ], [u], [o], [ɑ], and a single diphthong [oʊ]. This equals the seventeen phonetic sounds of the syllabary.”

Big Knife (2019: 17)

Language documentation and description. The syllable structure of a language gives another tool in determining whether you are documenting all types of phonological forms in the language. Once you have the syllable template of a language, you can determine whether your examples show all consonants that can be onsets in that position, to be sure your documentation covers all possibilities. Similarly, you can check on codas, long vowels, and diphthongs.

Being aware of syllables is also helpful if you are dealing with a stress language, to be able to check whether syllable weight affects the location of the stressed syllables in words.

Language revitalization. Expanding a language's vocabulary is an important part of language revitalization. Once you are aware of the syllable structure of your language, you will be able to make sure that the **neologisms** you create are consistent with the syllable template.

Some communities decide that they want to remove borrowed words from the vocabulary of their language, but it is sometimes hard to figure out which ones are borrowed. Sometimes the sounds suggest that a word is borrowed, but perhaps there are too many or too few sounds in the word you are looking at. This might be explained by syllable structure: If the syllables of your language are different from the syllables of a source language, the word may have changed by adding or removing segments to make the syllables fit the template.

On the other hand, you may want to use borrowed words in your language as a quick way to increase the vocabulary, but you want them to sound like your language, not like the source language. Looking at syllable structure gives clues about whether adding or removing segments will help in improving the sound of the borrowed words.

Reading linguists' resources about your language. Linguists will rarely explain syllable terms. After reading this chapter, you are familiar with onset, coda, nucleus, and rime, and with open syllables, closed syllables, heavy syllables, and light syllables. If you read Section 6.4, you also know how to interpret the different visual representations of syllable structure. With this acquired knowledge, you are better equipped to understand linguistic resources about your language. You may even be able to evaluate claims that are made about your language's syllable structure because of your knowledge and skills.

6.6 Conclusion

By reading this chapter, you have learned what a syllable is and how its structure varies by languages. You know the terms used to talk about the parts of syllables.

You know that you can figure out a language's syllable structure by observing words. You can focus on word edges to find out about syllable onsets and codas, and you know how to use information in the middle of a word to confirm your findings from word edges. You can use this information to find the syllable template of your language, and you have learned how to syllabify words in your language. You've also learned some methods that can be useful if you meet with syllabification challenges.

You may also know the conventions that linguists use to visually represent them syllables, and have some ideas about how syllables interact with language patterns.

Overall, syllables help us to understand how consonants and vowels are sequenced in words. However, there are other types of restrictions on where particular segments are found in the words of a language. The next two chapters look closely at these other types of sound distribution.

Chapter 7

Distribution: Where sounds are found in words

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Here's a puzzle. You are reading about Djeoromitxí, a Jabutí language of the Brazilian state Rondônia. You find the statement that $[p\phi]$, $[ps]$, $[bz]$, $[tj]$, $[dʒ]$ “are considered affricates, so they each constitute one consonantal segment” (De Castro 2012: 33). (If you are starting the book with this chapter, this may be the first time you’ve seen the **tiebar** — the $\wedge$ above two symbols to indicate that together they represent a single consonant. The tiebar is the [International Phonetic Alphabet \(IPA\) diacritic](#) used to show that two symbols make a single sound. On the IPA, see Chapter 2, Section 2.5.2.)

You’re curious. You might have encountered affricates like $[tj]$ and $[dʒ]$ before (see Chapter 2 Section 2.3.1.3 if you need a review). The other three sounds may be new for you, especially $[ps]$ and $[bz]$. (Affricates generally involve a very similar place of articulation, like the two labial sounds in $[p\phi]$, but $[ps]$ and $[bz]$ are labial-coronal affricates.) Why does De Castro call these sounds “one consonantal segment” rather than a sequence of segments — $[p]$ and $[\phi]$ instead of $[p\phi]$, $[p]$ and $[s]$ instead of $[ps]$, and $[b]$ and $[z]$ instead of $[bz]$?

You search through De Castro (2012), and find no explanation of the claim that the five sounds, $[p\phi]$, $[ps]$, $[bz]$, $[tj]$, $[dʒ]$, are affricates. Are they really acting like one consonant in the language? Or are they all two consonants? When you look at the way Djeoromitxí is written, you see that the writing system uses two letters to represent certain affricates — *ps* for $[ps]$, *bz* for $[bz]$, and *tx* for $[tj]$ — but uses one letter to represent other affricates — *p* for $[p\phi]$ and *c* for $[tʃ]$. Could

the writing system reflect speakers' knowledge, and some of these be a single consonant and others be two consonants?

How can you determine whether sounds that are spelled with two letters (in writing for regular reading) count as one consonant or two in the way your language uses consonants in words?

The short answer is that each of these affricates appears in the same places that we see single consonants, not where we see two consonants. In other words, the **distribution** of these sounds is the distribution of single segments, not of segment sequences. We can use distribution to help determine properties of the segment **inventory** of a language, among other things. (See Chapter 5, Section 5.5 on inventories.)

Distribution is the fourth major component of a language's unique **phonology**, along with variation, contrast, and syllables.

Usually when we start to think about the sounds of a language, our focus is on what the sounds are and how to pronounce them alone and in words. But simply knowing the sounds of a language and being able to pronounce them does not mean that any sequence of those sounds can be used as a word in that language. Rather, there are restrictions on where specific segments can appear in a word — some segments are found only at the start of a word, for example, or only between vowels. We use the term **distribution** to refer to where a sound can occur in a word of a specific language. This may sound familiar to you from our discussion of syllables in Chapter 6. Here we look at sequential restrictions on where sounds occur in words, which is similar to, yet different from, syllabic restrictions on where sounds are found.

In this chapter, we explain the concept of distribution and show how to determine whether the distribution of some sound is limited in some way. This discussion is preparation for Chapter 8, where we explore **complementary distribution**, a very important type of distribution in understanding how sounds are put together in making the words of a language.

Here's our promise. After reading this chapter, you will be able to construct a distribution table for the segments of your language and you will know how to use that table to determine whether any specific segment has regular or limited distribution. This information can be useful in determining the inventory of segments in your language, a necessary precursor to determining whether a word belongs to a particular language, to creating new words, and to setting up a writing system. Knowledge of distribution also helps in understanding claims you might read or hear about the sounds of your language.

We discuss these applications and more at the end of the chapter. But now, let's look at our plan, and then get on to distribution.

Here's our plan. We recommend that you start with sections 7.1 and 7.2, to get an introduction to the concept of limited distribution and to the strategies for identifying distribution patterns. For some readers, this will be enough and they will jump right to the discussion in Section 7.8. Other readers may choose to read more about the strategies for identifying patterns of distribution, for consonants in Section 7.3 and for vowels in 7.4. Some readers may want to see examples putting the analytic pieces together — this is done in sections 7.5–7.7. These sections are particularly helpful if you want to do some analysis yourself. Read what matters most to you, and determine when it is best for you to skip ahead to the discussion of how to use knowledge about distribution, in Section 7.8. You know you can always go back at another time to read a section you skipped.

1. Section 7.1 gives useful symbols. You may already be familiar with consonants being represented as C and vowels as V, from Chapter 6, Section 6.1. This section introduces some new convenient symbols and explains how they are used in abbreviating statements about the distribution of particular segments.
2. Section 7.2 introduces the concept of limited distribution and shows the basic strategy for determining whether the distribution of some segment is limited, by examining the distribution of [r] in Djeoromitxí. Some readers may want to skip ahead to the discussion section (Section 7.8) at this time; other readers may want to read more of this chapter, in order to learn the strategies for identifying limited distribution.

The general strategy for determining whether distribution is limited for some segment is to compare its distribution to other segments in its class — to other consonants or to other vowels.

3. Section 7.3 shows how to determine the general distribution of consonants. We do this by looking at sequences, in Section 7.3.1, and by looking at syllables, in Section 7.3.2.
4. Section 7.4 shows how to determine the general distribution of vowels.
5. Section 7.5 takes the next step in figuring out whether distribution is limited for some segment, by showing how to check the distribution of a particular sound against the distribution table for that class of sounds. This section answers the question of whether the sound chunk that is called an affricate is one segment or two in Djeoromitxí.
6. Section 7.6 puts all the pieces together through a case study of the limited distribution of voiced stops in Nakoda. This example shows how to go from

finding limited distribution of one segment to checking limited distribution of a class of segments.

The next section presents another example of distribution, which you might want to read in order to solidify your own understanding. The example also shows how to determine whether some chunk of sound is one segment or two. If you are happy with your understanding after Section 7.6 and you don't need help in understanding how to test whether something is one or two sounds, you might want to skip ahead to the final discussion, in Section 7.8.

7. A second case study is given in Section 7.7, where we show how distribution can be used to determine whether a stop-fricative sound series is a sequence of two consonants or is an affricate.
8. The chapter ends with Section 7.8, answering the puzzle, explaining a bit about why languages have patterns of limited distribution, and exploring ways to use patterns of limited distribution in community language work.

7.1 Introduction

In reading linguistic descriptions of the distribution of segments in a language, it is common to see descriptions given in terms of the six symbols: C, V, #, __, /, and *. This common shorthand is widely used in all sorts of linguistic writings. C (consonant) and V (vowel) were already introduced in Chapter 6 Section 6.1. This short section gives definitions and explains how to use the rest of these symbols.

The symbol # stands for word edge, whether the beginning or the end of the word.

We can use the three symbols C, V, and # to concisely describe the structure of words or parts of words. Instead of writing “a word that is made up of a consonant followed by a vowel”, we can use the shorthand and say “a CV word”. Instead of writing “a cluster of two consonants” or “a sequence of two consonants”, we can write “CC” or “a CC cluster” or “a CC sequence” – they all mean the same thing. Instead of writing “a consonant at the end of a word” or “a word-final consonant”, we can write “C#”.

Try this. How would you write “a consonant at the beginning of a word” or “a word-initial consonant”? (Both mean the same thing so the shorthand is the same.) Write down your answer before reading our answer.

We hope you wrote “#C” for “a consonant at the beginning of a word” or “a word-initial consonant”. If you had something else, try writing “a vowel at the end of a word” (or a “word-final vowel”) in the shorthand.

Did you come up with “V#”? (If you are still having trouble, we suggest you review this section before reading on.)

Continuing with the symbols, the role of “___” is to say “this is the location of the segment we are focusing on”. If we want to say that a certain segment occurs at the beginning of a word, we can write “#___”. To say that a segment is between two vowels, we can write “V___V”, and to indicate that a segment is at the end of a word, we can write “___#”.

Sometimes you will see a “/” used between a sound segment and a context, to tell us that the segment is found in the context that follows the “/”. For example, the shorthand statement “[b] / V___V” is read as “the segment [b] is found between two vowels”. The forward slash “/” corresponds to “is found”.

Try this. Write down the meaning of “[p] / ___#”.

Do you need to review “___” and “/” or did you come up with “[p] is found at the end of a word”?

Can you create a shorthand statement?

Try this. Write down a shorthand version of “the sound [ʃ] is found before the vowel [i]”.

This might have been harder to do. We hope you came up with “[ʃ] / ___[i]”.

We can also use features instead of segments: “[ʃ] / ___[V, +high, –back]” says “[ʃ] is found before high front vowels”. With a statement like this, we expect to see [ʃ] before any high front vowel that the language has, typically [i]. (If you need a list matching features and sounds, see Appendix B; for discussion of features more generally, go to Chapter 2. Specific features, such as [high] and [back] are defined in the glossary of terms linguists use.)

Try this. Write down your interpretation of ‘[ʃ] / ___[V, +high]’. Look at your statement. Do you expect to find [ʃ] before [i]? [u]? [o]?

The statement would be very similar to our first statement about [ʃ], with a slight difference — this time, [ʃ] occurs before *any* high vowel. That means we expect [ʃ] before both [i] and [u] (but not before [o], because [o] is a mid vowel, not a high vowel). (This is dense material; go slowly and review if needed.)

Finally, the asterisk “*” is used before something that does not occur. This is used in two ways. One is to negate a statement: “*[p] / V___V” means “[p] is not found between vowels”. It can also be used to say that a sequence of sounds does

not form a word in a language. In Djeoromitxí, which has only **open syllables**, a description might include “*[par]”, to say “the sequence [par] is not a word in Djeoromitxí”. (Syllables are discussed at length in Chapter 6, if you need to remind yourself.) More generally, we might say “in Djeoromitxí, *C]_σ” or “in Djeoromitxí, *C / ____]_σ” to say ‘in Djeoromitxí, consonants are not allowed at the end of a syllable’. (Note that “]_σ” is a different use of the square bracket, indicating the boundary of a syllable rather than that something is written in IPA symbols.)

Try this. Write down the meaning of “*[–voice] / V__V”.

Here’s our interpretation of “*[–voice] / V__V” – this statement means that “voiceless sounds are not found between vowels”.

Before ending this section, we put these shorthand symbols and their definitions in a single figure in (1) for easy reference.

(1) Shorthand symbols

- | | | |
|----|-------|--|
| a. | C | A consonant |
| b. | V | A vowel |
| c. | # | The edge of a word, either the beginning or the end. |
| d. | _____ | The location where a segment of interest is found. |
| e. | / | “is found”. The symbol / is used between some segment and some context to say “this segment is found in this context”. |
| f. | * | “is not”. The “*” says that whatever follows is not accepted in the language. |

You’ve had some chances to test yourself already. If you need more practice, try the next Comprehension Check. Otherwise, read on.



Comprehension Check 1

If you want to practice a little more, match the items in the two columns.

- | | | | |
|----|----------|------|--|
| a. | / ____ # | i. | a cluster of three consonants is not found |
| b. | ____ C | ii. | is found at the end of a word |
| c. | *CCC | iii. | is found between a vowel and a consonant |
| d. | / V__ C | iv. | before a consonant |

Write down your answers before looking at our explanations. If you get any wrong, please review this section before reading further, since we will use this shorthand a lot in the rest of the book.



Here's our answer

- a. / __# matches ii. *is found at the end of a word* (or *word-final*). The forward slash “/” tells us that we are looking at the context where some sound is found. Since # comes after the underscore, we know that the edge of the word immediately follows the location of the segment we are interested in — for that to be the case, the sound must occur at the end of a word.
- b. __C matches iv. *before a consonant* (or *pre-consonantal*). The underscore is followed by a C, the symbol for a consonant, which means the sound we are interested in comes right before a consonant.
- c. *CCC matches i. *a cluster of three consonants is bad* (or *a tri-consonantal cluster is not used in the language*). C is the symbol for consonants. Three Cs in a row tells us we have three consonants in a row. The * tells us this is not an accepted sequence in some language.
- d. / V__C matches iii. *is found between a vowel and a consonant*. Again, the forward slash “/” tells us that we are looking at the context where some sound is found. V is the symbol for vowels and C is the symbol for consonants. V comes before the context of interest and C comes after that location, as shown by the position of the underscore.

7.2 Limited distribution: [ɾ] in Djeoromitxí

Now that we've introduced the shorthand notation that we use to make statements more concise, we turn to a pattern to illustrate *limited distribution* — when a particular sound or class of sounds is not found in a context where we might expect to find it. For example, in Djeoromitxí, there are words that begin with consonants, for example *paro* [paro] ‘chicken’. There are no surprises here —

we expect to find consonants at the beginning of words in Djeoromitxí and in all languages. Typically languages have consonant-initial words, although there can also be words that start with a vowel in some languages. Chapter 6 explains this: It is because syllables typically have *onsets*, and words begin with a syllable.

However, De Castro (2012: 15) makes the unexpected statement that [r] “does not occur word-initially”. What this means is that, while words like *paro* [paɾo] ‘chicken’ are found in Djeoromitxí, a word like *[raɾo] is not found, because [r] cannot be at the beginning of a word in the language: *[r] / # __.

Saying that [r] “does not occur word-initially” is a statement of limited distribution: Consonants are found at the beginnings of words and [r] is a consonant – we expect to find [r] at the beginnings of words, but we do not.

How can you look at the words of a language and determine whether some sound has limited distribution? A straightforward way is to create a table showing the expected distribution of sounds in your language, and then check whether the sound you are interested in is found in every one of those contexts. For Djeoromitxí, we need to understand the general distribution of consonants. Then we need to check whether [r] is found in all of the contexts that consonants generally occur in.

To determine the general distribution of consonants in a language, we can make a distribution table. Hopefully, this idea is familiar from the tables we made in Chapter 5.

7.2.1 A distribution table for Djeoromitxí consonants

We know that in general consonants and vowels alternate in words (CVCVCV...). Sometimes there are consonant clusters (...VCCV...), some words start with a vowel or end with a consonant and sometimes a word is made up of just a vowel. However, in general, most words alternate between consonants and vowels.

We can use this observation to make a table showing where we expect to find consonants in the language – at the start of a word, between vowels, and at the end of a word. We try to fill in each cell with words from Djeoromitxí to determine whether consonants are found in all of these contexts. The distribution table (Table 1) shows what this looks like for Djeoromitxí. The row labels give the context in words; the cells show where consonants are found. The first row is for word-initial consonants (# __), the second row is for word-final consonants (__ #), and the third row is for intervocalic consonants (V __ V). The final row, labeled *other*, is a catch-all category in case the language has consonant clusters. The column headed C means that we are looking at the general distribution of consonants in Djeoromitxí, not the distribution of a specific consonant.

Table 1: Djeoromitxí consonant distribution

	C
# ___ V	[ha'u] 'lizard' [paɾo] 'chicken'
V ___ #	
V ___ V	[paɾo] 'chicken'
<i>other</i>	

In Table 1 we have also filled in Djeoromitxí words in each cell where we found appropriate examples, using black text to show the consonant that fulfills the requirement – word-initial [h] in [ha'u] and intervocalic [ɾ] in [paɾo]. We see words in only two of the cells.

We found no words with a consonant at the very end, so that cell is empty: We use gray shading in that row to indicate that we are pretty sure there are no consonant-final words in Djeoromitxí. We also see no words in the row labeled *other*, where we would put words with consonants in locations other than the three specific contexts we have identified (such as between a vowel and a consonant). The *other* row is empty and so it, too, is shaded gray.

We conclude that the general distribution of consonants in Djeoromitxí is that they are found in word-initial position and they are found between vowels, but they are not found at the end of a word, nor anywhere else (that is, consonants are not next to another consonant). (By the way, if you are thinking that it looks like Djeoromitxí has only CV or V syllables, you are right.)

Now that we know where we expect to find consonants in Djeoromitxí, we can check the distribution of individual consonants to determine whether a consonant is found in all of the expected contexts. Since we are focusing on [ɾ], we make a table showing the distribution of [ɾ] in Djeoromitxí. We do this by using the consonant distribution table template, but filling each cell with words showing [ɾ] in the appropriate context.

7.2.2 Distribution table for Djeoromitxí [ɾ]

The distribution in Table 2 shows where we find [ɾ] in Djeoromitxí words – the only row with words in it is Table 2b, the row for intervocalic consonants. We've left out the two rows corresponding to general restrictions on consonants in Djeoromitxí (no word-final consonants, no consonant clusters) since these are not restrictions specific to [ɾ].

A surprise shows up in the row for word-initial consonants, row Table 2a. There are no words in this row, showing that we found no words beginning with

Table 2: Djeoromitxí [r] distribution

		[r]		
		#		
a.				
b.	V__V		<i>paro</i>	[paɾo] ‘chicken’
			<i>piro</i>	[piɾo] ‘language, talk’

[r]. We used light gray shading for this row to show that we expected to find words that begin with [r], but we didn’t find any such words.

Since we expected word-initial [r] but could not find it, we conclude that the distribution of [r] is limited in Djeoromitxí – the consonant [r] is not found in word-initial position.

(2) Distribution of [r] in Djeoromitxí

In Djeoromitxí, [r] appears only between vowels. [r] is not found at the beginning of words.

In shorthand: *[r] / # ____.

We could also include the statement “[r] is also not found at the end of a word or next to another consonant” in (2), but it is not necessary, and it distracts from the unexpected pattern. Both the end-of-the-word part and the next-to-another-consonant part are general distributional properties of all consonants in Djeoromitxí, not special properties of [r]: It is better to really focus on the unusual property of [r]. When you are making distribution statements about sounds in your language and encounter a similar situation, think about what will be most helpful to your audience – to include the extra information or to leave it out.

7.2.3 The take-away from the investigation of Djeoromitxí [r]

You may read a description that says some sound or group of sounds has limited distribution in some way, and you may want to check whether that statement is correct for your language. Or you may suspect limited distribution in your language for some particular sound or class of sounds. You can use the method presented here to check the distribution: you find the general distribution for the class of sounds, then you determine whether each sound in the class has regular distribution, or whether there are unexpected gaps. The unexpected gaps are the signal that distribution is limited for those sounds.

The next three sections give a how-to on determining the general distribution of consonants (Section 7.3) and vowels (Section 7.4), and the distribution of spe-

cific sounds (Section 7.5). If you are feeling confident with those determinations, try the Language Explorations in those sections to test yourself. If you are still confident, you may want to jump to more illustrative examples, starting with Section 7.6.



Comprehension Check 2

Soyaltepec Mazatec is an Otomanguean language spoken in the state of Oaxaca, Mexico. The Mazatec of the Soyaltepec Municipio, described in Beal (2012), uses a labial fricative [β], as in these words. (Please note that no practical orthography is given in Beal 2012.) Not all Soyaltepec Mazatec consonants are shown in these words, but we do show all contexts where [β] is found. (We are grateful to Heather Beal for checking the use of the Soyaltepec Mazatec examples and for reminding us about the consonant clusters.)

Your task is to determine whether [β] has limited distribution.

- a. [βasɛ] b. [βɛ] c. [βifu]
 ‘half’ ‘know’ ‘they fell’

Since all the words in the list have a CVCV... structure, you can use the consonant distribution table below, which is based on the one for Djeoromitxí in Table 1. We omit rows for word-final consonants and for consonants next to other consonants since Soyaltepec Mazatec, like Djeoromitxí, has no consonants in either of these contexts. This table shows where we expect to find consonants in Soyaltepec Mazatec.

		C
#	V	
V	V	

If you find that [β] has limited distribution, make a statement about its distribution, modeled after (2). (Soyaltepec Mazatec has contrastive tones on vowels. Since tone does not make a difference in the distribution of [β], we have omitted the tone markings.)

If you know something about Soyaltepec Mazatec, you probably know that there are consonant clusters in the language. Since this is not a complete study of the language, we are ignoring consonant clusters for this Comprehension Check.



Here's our answer

We put the three words into the Soyaltepec Mazatec distribution table.

	C	
# ____	[βase]	'half'
	[βε]	'know'
	[βifu]	'they fell'
V __ V		

We see that the only the cell for consonants at the beginning of a word is filled in in this table. There are no examples with [β] between vowels. This shows limited distribution for [β]. Our statement is given below.

[β] appears only word-initially in Soyaltepec Mazatec: *[β]/V __ V.

7.3 The general distribution of consonants

From Section 7.2 in this chapter, you have learned that determining limited distribution of a sound requires knowing the general distribution of that class of sounds. You also have learned about the two major classes of sounds, consonants and vowels in Chapter 2. And you have learned how consonants and vowels can be organized into syllables within words, in Chapter 6. We can put all of this knowledge together to determine the general distribution of consonants and of vowels in any language. These general distributions can then be refined to determine whether a specific sound has limited distribution.

When we think about the distribution of consonants in words, we can think about a sequence of sounds, CVCVCV... We can also divide the sequence into syllables, as in CV.CV.CV... We can use either sequences or syllables to make a distribution table. We talk about each; you may prefer to read only about sequences, in Section 7.3.1, or only about syllables, in Section 7.3.2. You can always come back to the other section if you discover you need the additional information for understanding your language.

7.3.1 Consonant distribution based on sequences

Consonants can begin or end words, and they can come between vowels. We considered these locations in our discussion of Djeoromitxí in Section 7.7. (Linguists also use the words *context* and *environment* for “location”; we use the three terms interchangeably.) However, Djeoromitxí had no sequences of consonants; some languages do have them. In this section, we expand our discussion of the distribution of consonants to include languages with CC sequences.

By looking at several words in a language, we can usually determine whether the language allows sequences of consonants. If there are no sequences of consonants, then the generic distribution table for consonants has exactly the structure of the one we used in Table 1 for determining the distribution of [r] in Djeoromitxí.

However, if the language rules allow consonant sequences, there are more contexts. Suppose we can have two consonants between vowels: VCCV. What are the contexts for each consonant? The first consonant (VCCV) follows a vowel and precedes a consonant, V__C. The second consonant (VCCV) follows a consonant and precedes a vowel, C__V. This adds two contexts to our table, in the last two rows of Table 3.

Table 3: Adding contexts to C distribution

		C
a.	#__V	
b.	V__#	
c.	V__V	
d.	V__C	
e.	C__V	

We can continue in this fashion if we find additional CC sequences. Word-initial and word-final CC clusters add more rows to the table, as do clusters of three or more consonants, like #CCCV..., ...VCCCV..., ...VCCCCV..., ...VCCC#. The additional contexts are added in Table Table 4.

Try this. Compare Table 3 and Table 4. Can you figure out the addition due to word-initial consonants clusters? The addition due to word-final consonant clusters? The addition due to sequences of three or more consonants? (The last one may take a bit of thinking.)

Look at Table 3a and Table 4a, b. All three refer to word-initial consonants. If consonant sequences are found in a language, then there may be some word-

Table 4: Adding more contexts to C distribution

	C
a. # ___ V	
b. # ___ C	
c. V ___ #	
d. C ___ #	
e. V ___ V	
f. V ___ C	
g. C ___ V	
h. C ___ C	

initial consonants that precede another consonant, # ___ C in Table 4b, rather than a vowel, # ___ V in both Table 3a and Table 4a. If the language has word-final consonant clusters, we can use the same logic to relate rows Table 4c, d to Table 3b.

What about sequences of three (or more) consonants within words — #CCCV..., ...VCCCV..., ...VCCCCV..., ...VCCC#, and so on? Only one additional context is added, C ___ C, despite the variety of sequences. In each case, the CCC (or longer) sequence results because one consonant has a consonant on each side of it, whether word-initial #CCCV..., word-final ...VCCC#, or in the middle of a word, ...VCCCV..., ...VCCCCV..., and ...VCCCCV.... Thus, we require only one additional context, C ___ C, shown in Table 4h.



Exploring my language

Can you create a consonant distribution table for your language, based on sequences of consonants? Your table should look a lot like either Table 3 or Table 4.

7.3.2 Consonant distribution based on syllables

In some cases, whether a consonant is next to a vowel or next to another consonant may not be the right kind of information that you need. You may want to know whether a consonant is part of an **onset** or part of a **coda**. In that case,

you will need to think about contexts in terms of syllable structure instead of consonant sequences.

As an example, consider aspirated voiceless consonants in Sierra Popoluca, a Mixe-Zoquean language from the Mexican state of Veracruz. Words showing examples of all the locations where [tʰ] is found are given in (3); there are no words with [tʰ] in different location in the word. (Our source is de Jong Boudreault (2009); the language is also called Soteapanec.)

From these examples, we can tell that Sierra Popoluca has no words that start with [tʰ] although there are word-initial consonants. We can also tell that Sierra Popoluca has consonant clusters and word-final consonants.

- (3) Aspirated [tʰ] in Sierra Popoluca
- | | |
|----------------------|------------------------------|
| word edge | middle of word |
| [petʰ] ‘he swept’ | [putʰpaʔ] ‘he is leaving’ |

When we look closely at (3), we see that [tʰ] is not found at the beginning of a word, showing that the distribution of aspirated [tʰ] is restricted. More examination of (3) shows that [tʰ] is found only at the end of a word (in [petʰ]) or before a consonant (in [putʰpaʔ]). We see this in Table 5. End-of-a-word and before-a-consonant are two separate, quite different, contexts.

Table 5: Distribution of [tʰ] in Sierra Popoluca in terms of sequences

		[tʰ]
initial	# __ V	
final	V __ #	[petʰ] ‘he swept’
medial	V __ V	
medial	V __ C	[putʰpaʔ] ‘he is leaving’
medial	C __ V	

Let’s consider syllables now. With syllables, there are two contexts for consonants, in the onset σ [__ and in the coda __] σ . (As you learned in Chapter 6, some languages have complex onsets and/or complex codas, allowing more than one consonant in those positions. In such cases, there are more contexts to consider, within the onset and within the coda.) We can construct a distribution table showing these two contexts.

When we look at the locations for aspirated [tʰ] in terms of this table, we see that [tʰ] has an asymmetrical distribution. We use light gray shading to highlight the absence of [tʰ] in onsets.

Table 6: C distribution table based on syllables

		C
onset	$\sigma[\rule{1cm}{0.4pt}]$	
coda	$\rule{1cm}{0.4pt}]\sigma$	

Table 7: Distribution of [t^h] in Sierra Popoluca in terms of syllables

		[t ^h]	
onset	$\sigma[\rule{1cm}{0.4pt}]$		
coda	$\rule{1cm}{0.4pt}]\sigma$	[pet ^h]	‘he swept’
		[put ^h paʔ]	‘he is leaving’

From Table 7, we construct a distribution statement for [t^h] in Sierra Popoluca: [t^h] is found in syllable codas. Using syllables, the context for [t^h] is unified; using sequences, we have two separate statements so we do not see the unity.

From examples like this one, we know that some contexts are better described in terms of syllable structure. Naturally, this relies on the concept [syllable](#), introduced in Chapter 6.



Exploring my language

Can you create a consonant distribution table for your language, based on syllables?

Your table may look exactly like Table 6. However, in some languages, either (or both) onsets and codas may have more than one consonant in them. If your language has complex onsets and/or complex codas, your distribution table will have more rows in it, to accommodate whether the targeted consonant is at the beginning, middle, or end of the onset or the coda. It combines elements of Table 4 with the basic structure of Table 6.

7.3.3 Summary

In this section, we have shown two ways of determining the general distribution of consonants, looking at consonants in sequences of sounds and looking at consonants in terms of their position in a syllable.

We have seen that in Djeoromitxí, the word-initial context is appropriate for identifying the distribution of [r]. Onsets are inappropriate because [r] is found in word-medial onsets.

We saw from Sierra Popoluca that coda position is appropriate for identifying the distribution of [tʰ]. If we were to describe this in terms of CV sequences, we would have two contexts where [tʰ] is found, before another consonant and at the end of a word. Invoking coda position unites these distinct contexts.

In each case, the system used makes the pattern clear and simple. As you work on your language, if you are trying to express the limited distribution pattern of some sound or class of sounds, pay attention to how simple your statement is. If your statement is complex, look at whether you are expressing it in terms of sequences or syllables. You may find simplicity by looking at the pattern from another angle.

7.4 The general distribution of vowels

Determining the general distribution of vowels in a language is easier than figuring out the general distribution of consonants, because vowels are always in the nucleus of a syllable. This means we only need to consider sequences of sounds in working out the general distribution of vowels: The syllabic position shows little variation. Because we only need to think about sequences of vowels, we can use the same procedure that we used with consonants: Finding the row labels by considering the possible locations for vowels. Vowels can come at the start of a word, at the end of a word, and between consonants, so these are the labels for the rows. We can add a “V:” column if the language has long vowels and a V_iV_j column if the language has diphthongs. (The subscripted “i” and “j” are to show that the two vowels in the diphthong are different from each other. There are often limits on which vowels are found in diphthongs and on which vowels can come first and which come second.

Table 8: Generic vowel distribution table

	V	V:	V _i V _j
# _____			
_____ #			
C _____ C			

For Djeoromitxí, the table would have V and V_iV_j columns. In the sample Djeoromitxí vowel distribution table (in Table 9), we see that words can be found to fill every cell.

Table 9: Djeoromitxí vowel distribution table

	V	V _i V _j
#___	[oreka] 'Brazil nut burr'	[oitʃi] 'one'
___#	[pako] 'woman'	[hau] 'lizard'
C___C	[kəβi] 'dance'	[βaudɨ] 'sloth'

When working with a particular language, we can change the column headings to match the types of vowel sequences that we find in words; we then fill in the distribution table with words fitting the criteria defining each cell and use that information to create a statement of vowel distribution for the language. The method is exactly the same as that seen for consonants in Section 7.3, except that the focus is on vowels, not consonants.

Try this. Look at Table 8 and think about your language. Would you use both a V column and a V_iV_j column in your language? Would you use a V: column? Explain your answers.

The discussions in this section and in Section 7.3 addresses the general distribution of consonants and of vowels in languages. Determining whether specific sounds have limited distribution begins with this general distribution. We looked at two examples of this, the distribution of Djeoromitxí [r] and the distribution of aspirated stops in Sierra Popoluca, illustrated with [tʰ].

To determine whether any vowel or diphthong has limited distribution in Djeoromitxí, we would compare the distribution of a specific vowel (or diphthong) against the generic pattern established in Table 9. How this is done is the focus of the next section.

7.5 Determining the distribution of specific sounds

Knowing the contexts where we can find the consonants and the vowels in the words of a language gives us a new way of looking at specific sounds. We can now answer the question of whether a specific segment actually occurs in all of those contexts. For example, we have seen that in Djeoromitxí, consonants appear before vowels, either at the beginning of a word or inside a word, but never

appear next to another consonant. Because we know these contexts, we can examine words for specific segments to determine whether individual consonants appear in each of those contexts.

A good strategy for doing this is to use the distribution table that we have developed for a language and to change the general “C” headings to specific consonants. For example, if we wanted to determine whether [t] occurs in all contexts available for consonants in Djeoromitxí, we can use the distribution table shown in Table 1, but change the column heading to [t], instead of the generic “C”. Looking through Djeoromitxí word lists shows words with [t], as in *tee* [tɛɛ] ‘nephew/niece’, *kuti* [kuti] ‘smooth’, and even *teteno* [tɛtɛno] ‘pepper’. We can put these words in the appropriate places in the Djeoromitxí distribution table.

Try this. Take the Djeoromitxí consonant distribution table from Table 1 and put the three words with [t] into the correct cells. (Our answer is in Table 10.)

The Djeoromitxí distribution table illustrated in Table 10 shows us that [t] is found in every place that we expect to find it because we find words illustrating [t] both at the beginning of a word and between vowels, the two contexts for consonants in this language.

Table 10: Djeoromitxí [t] distribution

	[t]	
# ____	[tɛɛ]	‘nephew/niece’
	[tɛtɛno]	‘pepper’
V ____ V	[kuti]	‘smooth’
	[tɛtɛno]	‘pepper’

This is the pattern that we expect to find with all the consonants in Djeoromitxí. In most languages it will be true that individual consonants appear in any of the contexts that consonants can appear in generally. Similarly, if a vowel is in a language, we expect to find that it can appear in any context where a vowel is allowed in that language.

The strategy to check whether a given vowel or consonant has the expected general distribution involves two steps. We first create a distribution table showing the contexts for consonants (beginning and ends of words and between two vowels, or the syllabic positions of onset and coda) and for vowels (beginning and ends of words and between two consonants).

We then examine words in the language to be sure that there are words with the target sound in each of the contexts that the general class of sounds is allowed in the language. We did this with Djeoromitxí [t], showing that there are words that start with [t] and words that have [t] between vowels, the two locations for consonants in Djeoromitxí: [t] has regular distribution in the language.

When you check individual segments, you may find limited distribution, as we found with [r] in Djeoromitxí. When you check classes of segments, you may find that an entire class of sounds has limited distribution. This is the case with the class of voiceless stops in Sierra Popoluca. We demonstrated that [t^h] is found only in coda position, but the same holds true of [p^h] and [k^h], as shown by words like [ʔan̥ɲip^h] ‘my mouth’ for [p^h] and both [pak^h] ‘bone’ and [nik^hpam] ‘he’s going already’ for [k^h].

Our next section works through another example showing the limited distribution of a class of sounds, this time the voiced stops in Nakoda. If you are feeling confident about how to determine general and limited distribution, you might decide to skip this section and go on to the next section.

Section 7.7 discusses how distribution can help determine whether a sound is an affricate or a sequence of two segments. If that doesn’t interest you, then skip ahead to Section 7.8 which discusses how to use distribution in language community work.



Exploring my language

Choose a consonant from your language and try to fill in your consonant distribution table with example words from your language.

If you come across a word that doesn’t match any of the cells you have in your table, write it down. Keeping track of “misfits” provides information that may prove useful, for example by alerting you to having omitted a critical context for your language.

7.6 The limited distribution of voiced stops in Nakoda

Nakoda is a Siouan language (also called Assiniboiné) which is spoken in southern Saskatchewan and northeastern Montana. Our sources for Nakoda are Cum-

berland (2005) and an online dictionary hosted at the American Indian Studies Research Institute at Indiana University Bloomington.¹ The first step is working out the general distribution of consonants in Nakoda. Since there are words like [wɪtkó] ‘be crazy’, we know that there are consonant sequences (the [kt] sequence). This means that the generic Table 3 is missing a few contexts – we also need the context V__C (from Table 4f) to target [t] in [wɪtkó] and the context C__V (from Table 4g) to target [k] in [wɪtkó].

7.6.1 The generic distribution table for Nakoda consonants and what to do if you miss something along the way

We take a little bit of time in working out the distribution table for Nakoda consonants, to show what to do if you find a word in your language that doesn’t fit into the distribution table that you started with.

Suppose we are part of a team working on the Nakoda language, and we’ve filled in the preliminary generic distribution table Table 3 as best we can, using the words in (4).

(4) Nakoda words for filling in table

<i>wɪtkó</i>	[wɪtkó]	‘be crazy’
<i>kapéya</i>	[kapéja]	‘to surpass, to do more’
<i>tuwénuk</i>	[tuwénuk]	‘there’s no one’

Our preliminary filled-in table is given in Table 11. Notice that the words appear more than once. Each word starts with a consonant, so each word is in the top left cell. The initial consonants are in black to make them easier to spot. One word ends with a consonant, *tuwénuk* [tuwénuk] ‘there’s no one’, so it is in the cell for a consonant at the end of a word, with the final [k] in black text. Two words have single intervocalic consonants, while *wɪtkó* [wɪtkó] ‘be crazy’ has a two-consonant sequence between vowels – again, the relevant symbols are in black text to make them stand out.

From our filled-in Table 11, we make a preliminary statement about the distribution of consonants in Nakoda – that words can begin and end with a single consonant, and there can be one or two consonants between vowels. However, there are a couple problems with Table 11: Inspection of more Nakoda words shows that we have missed two important properties of consonant distribution in the language because we didn’t look at enough words. These missing properties

¹The online dictionary is available at <http://zia.aisri.indiana.edu/~dictsearch/>; we consulted it between November 22, 2022 and June 7, 2023.

Table 11: Filling in the Nakoda distribution table, preliminary

	C	
#__V	[wɪtkó]	‘be crazy’
	[kapéja]	‘to surpass, to do more’
	[tuwénuk]	‘there’s no one’
V__#	[tuwénuk]	‘there’s no one’
V__V	[kapéja]	‘to surpass, to do more’
	[tuwénuk]	‘there’s no one’
V__C	[wɪtkó]	‘be crazy’
C__V	[wɪtkó]	‘be crazy’

mean that our general statement about the distribution of Nakoda consonants is not quite right.

It’s useful to look at both of these missed properties, in case you encounter something similar when you are working on your language. We discuss one of them in Section 7.6.1.1 and the other in Section 7.6.1.2.

7.6.1.1 What do we do when we encounter *šmá* [ʃma] ‘deep’?

We keep collecting words in Nakoda because we want to make a complete description of the language. Eventually, we start finding words like *šmá* [ʃma] ‘deep’ and *tká* [tká] ‘be heavy’. These words surprise us because they begin with two consonants, and they are the first words we have encountered that begin this way. At the same time, they are easy to add to our table because we already know that CC-initial words are a possibility in some languages. So we add the row for an initial consonant that is followed by another consonant, #__C, giving rise to Table 12 – and modify our summary statement of consonant distribution to include the possibility of two consonants together at the start of a word.

With this new information, we can say that Nakoda words start with one or two consonants, have at most one consonant at the end, and can have one or two consonants between vowels. (If you are excited about syllables, you will be thinking that Nakoda allows complex onsets as well as allowing single consonant codas, and you are right.)

This scenario – of having filled in a table and then later finding words that go into cells that you hadn’t included (or had shaded gray because you thought they were empty) – sometimes happens. It is an easy one to recover from – add the row (or take away the gray shading showing the cell is empty) and put the words in the cell instead, then modify your summary statement of the distribution of sounds in your language.

Table 12: Filling in the Nakoda distribution table, preliminary

	C	
#__V	[wɪtkó]	‘be crazy’
	[kapéja]	‘to surpass, to do more’
	[tuwénuk]	‘there’s no one’
#__C	[ʃmá]	‘deep’
	[tká]	‘be heavy’
__#	[tuwénuk]	‘there’s no one’
V__V	[kapéja]	‘to surpass, to do more’
	[tuwénuk]	‘there’s no one’
V__C	[wɪtkó]	‘be crazy’
C__V	[wɪtkó]	‘be crazy’
	[ʃmá]	‘deep’
	[tká]	‘be heavy’

Try this. What would you do if you were working on Nakoda and came across the word *penmná* [p^henmná] ‘be the odor of smoke’? Try to come up with your own answer before reading the next section.

7.6.1.2 What do we do when we encounter *penmná* [p^henmná] ‘be the odor of smoke’?

Someone who continues to work on Nakoda may come across one of the few words in the language that has three consonants in a row, for example the word *penmná* [p^henmná] ‘be the odor of smoke’ which has a sequence of [nmn] right in the middle.

We are caught off guard, but we recover quickly. All that is necessary is to admit we had missed the CCC sequences and add the relevant row to our distribution table – the C / C__C. The adjusted distribution table is shown in Table 13. (Again, if you are excited about syllables, you will see view this as a combination of a coda followed by a complex onset, *penmná* [p^hen.mná] ‘be the odor of smoke’.)

It can be helpful in identifying parts of the table to group the rows into those looking at initial consonants, those looking at final consonants, and those looking at medial consonants. We repeat the table from Table 13 in Table 14 to show what we mean. We also shorten the table by including only one example of each type.

We can say that Nakoda words start with one or two consonants, have at most one consonant at the end, and can have between one and three consonants

Table 13: Filling in the Nakoda distribution table, final

	C	
# __ V	[wɪtkó]	‘be crazy’
	[kapéja]	‘to surpass, to do more’
	[tuwénuk]	‘there’s no one’
# __ C	[ʃmá]	‘deep’
	[tká]	‘be heavy’
__ #	[tuwénuk]	‘there’s no one’
V __ V	[kapéja]	‘to surpass, to do more’
	[tuwénuk]	‘there’s no one’
V __ C	[wɪtkó]	‘be crazy’
	[p ^h enmná]	‘be the odor of smoke’
C __ V	[wɪtkó]	‘be crazy’
	[ʃmá]	‘deep’
	[tká]	‘be heavy’
	[p ^h enmná]	‘be the odor of smoke’
C __ C	[p ^h enmná]	‘be the odor of smoke’

Table 14: The final Nakoda distribution table, rearranged

		C	
initial	# __ V	[wɪtkó]	‘be crazy’
	# __ C	[ʃmá]	‘deep’
final	__ #	[tuwénuk]	‘there’s no one’
medial	V __ V	[kapéja]	‘to surpass, to do more’
	V __ C	[p ^h enmná]	‘be the odor of smoke’
	C __ V	[wɪtkó]	‘be crazy’
	C __ C	[p ^h enmná]	‘be the odor of smoke’

between vowels. (If you are keeping track of syllables, you expected this — a coda followed by a CC onset results in three consonants between vowels.)

This scenario, of suddenly finding words that show that the original distribution table needs additional rows, may happen as you work on your language. Incorporating the new information is straightforward: We simply add a row with the appropriate label to our distribution table, and adjust any statements we’ve made based on the original formulation of the distribution table.

7.6.1.3 The take-away

New discoveries that we might make about a language — such as discovering that we missed words that start with a CC sequence — are frequent occurrences. In many cases, these oversights are easily remedied when we come across new information, simply by adding the relevant row and filling the new cells. Perhaps the hardest part with a discovery of this sort is reviewing the summary statements you have made about the distribution of those sounds and adjusting them appropriately.



Exploring my language

Back on p. 398, you were asked to keep track of words that didn't seem to fit into a cell in the consonant distribution table for your language.

Examine those words and modify your consonant distribution table, if necessary, so that the table accommodates all types of words in your language. Put those words in the new distribution table, to make sure you've covered all possibilities.

7.6.1.4 Isn't there an easier way to do this?

Wouldn't it be easier just to go through a lot of Nakoda words and write down where we find the consonants in them, without making all these tables? For example, we could look at the word *én* [én] 'at, to, there, in', and see that it has the consonant [n] at the end of the word following a vowel. This shows us that consonants are found following a vowel at the end of a word, so we would put the "end of the word" context on our list, C/V___# *a consonant can be found after a vowel at the end of a word*.

Using this intuitive method, you would continue like this until we thought we had identified all the contexts for Nakoda consonants.

There are a few of problems with the intuitive method.

First, it isn't clear when you are done. It isn't a matter of looking at "enough" words: Being done depends on what the C and V patterns are in the words you check. If all your words had the VC pattern of *én* [én] 'at, to, there, in', for example,

you would never know that you could also find words with consonants at the beginning of a word or between vowels.

Second, every time you find a new word, you need to run through your entire list to make sure that all the consonant contexts in the word are on your list. This makes it hard to realize whether you have discovered a new context — you may miss something in your list, especially when it starts to get long.

Third, it really helps to have an organized list when you start looking at the distribution of particular consonants. With the intuitive method, you need to inspect all the contexts on your list and organize them yourself.

These problems don't occur if you use the systematic approach presented in this chapter, so you may find the systematic method more effective. Because it is systematic, it is easy to tell whether you've considered all the relevant contexts. It has an internal organization so it is easy to see whether there is a pattern. It is easy to see if you have found a consonant in a new context, and if so, it is easy to expand the system to check on that additional context. And it extends to all languages, not just Nakoda.

7.6.2 Limited distribution of the voiced labial stop [b] in Nakoda

Now that we have finalized the general consonant distribution table for Nakoda, we can start looking at the distribution of individual consonants. With the general distribution table, it is easy to check for limited distribution. We choose the consonants we want to look at, and test whether they occur in all of the Nakoda consonant contexts. If there are empty cells, we have found limited distribution.

We start with the voiced labial stop consonant, [b], and show that it has limited distribution in Nakoda. This is straightforward now that we have identified the contexts where consonants can occur in Nakoda. All we need to do is check whether there are words with [b] in each of the contexts listed in Table 14. Since we are looking at the distribution of [b], we use [b] to head our column. Each cell can be filled with words that show [b] in the relevant context.

7.6.2.1 Filling in the distribution table for [b]

We give a filled table showing the distribution of [b] in Table 15. What do you notice about the distribution of [b] — is it limited or not? (As before, we use black type to emphasize the [b] while other segments are in gray type. We use gray shading in the cells where we found no examples to fill the cell.)

In our distribution Table 15, just one cell is filled. Every other cell is empty. The only context for voiced labial [b] in Nakoda is between vowels — a very limited distribution.

Table 15: Nakoda [b] distribution

		[b]
final	V __ #	
initial	# __ V	
	# __ C	
medial	V __ V	sába [sába] ‘black’
	V __ C	
	C __ V	
	C __ C	

7.6.2.2 Distribution statement for the voiced labial stop [b]

When we examine Table 15, we see that the voiced labial stop [b] appears in only one row and so in only one context – between two vowels. This makes it easy to characterize the distribution of [b], (5).

(5) Nakoda [b] limited distribution statement

In Nakoda, the voiced labial stop [b] is in limited distribution, found only between vowels. In shorthand with features:

[labial, +voice, –continuant] / V __ V

The labial voiced stop ([b]) is found between two vowels.



Exploring my language

Do you suspect that there is limited distribution of some consonant or class of consonants in your language? Use the consonant distribution table for your language to determine whether your suspicions are well-founded.

End with a statement summarizing your results, stating either that the distribution is regular or that it is limited. If you find limited distribution, model your statement on our statement in (5).

7.6.2.3 The take-away from making an explicit statement

Making an explicit statement about limited distribution can be useful for a variety of reasons. It gives something to share with others to help clarify the rules of pronunciation, whether in a classroom, a grammar, or a dictionary.

An explicit statement also might encourage us to look more deeply at the language. For example, the limited distribution of [b] in Nakoda refers to a voiced stop. This might make you wonder — is distribution limited only for [b], or do other voiced stops behave the same way? How would you go about answering this question? Think about how you would answer this question before you continue reading — in the next section, we show one way to determine whether other Nakoda voiced stops have the same distribution.

7.6.3 What about other voiced stops in Nakoda?

We sometimes see [similar](#) segments follow the same pattern. Since [b] is a voiced stop, it makes sense to check whether the other voiced stops in Nakoda, [d] and [g], have the same limited distribution.

In doing this, we are using a general analysis strategy of starting with one segment, and then moving on to other, similar, segments — rather than trying to tackle all segments in a language at the same time. We start small, then build to a larger picture. Starting big can get confusing very quickly, which can be frustrating — and can make it hard to see a pattern. You may find the small-to-big strategy useful in your own analytic work or when explaining patterns to others.

7.6.3.1 A distribution table for three consonants, [b], [d] and [g]

We use the same strategy we used with [b] for [d] and [g] — checking whether each consonant is found in all the contexts that consonants are found in generally. We extend our distribution table by adding columns for the other voiced stops, [d] and [g], so we have a table with three columns. The extended distribution table is given in Table 16, with words filled in. Examine this distribution table — do [d] and [g] have the same limited distribution as [b]?

We hope that you said that the three voiced stops have the same pattern of limited distribution in Nakoda. Only one row is filled in each of the columns in Table 16, the row where the voiced stops [b], [d], and [g] are between vowels.

Table 16: Nakoda voiced stop distribution

		[b]	[d]	[g]
final	# _ V			
initial	# _ C			
medial	V _ V	sába [sába] ‘black’	nedám [nedám] ‘over.here’	wagú [wagú] ‘covet’
	V _ C			
	C _ V			

7.6.3.2 Describing the distribution of the voiced stops in Nakoda

What remains is to modify our statement of the limited distribution of [b] in Nakoda, (5), to include all voiced stops. The original, with the references to labials in gray text, said: *In Nakoda, the voiced labial stop [b] is in limited distribution, found only between vowels. In shorthand with features, [labial, +voice, –continuant] is found only between vowels.* Instead of referring to the labial [b], we want to refer to all stops, an easy change to make. We simply refer to “voiced stops” instead of “the voiced labial stop”. The revised statement is given in (6).

- (6) Nakoda limited distribution of voiced stops
- In Nakoda, voiced stops [b], [d], and [g] are in limited distribution, found only between vowels. In shorthand with features:

[+voice, –continuant] / V _ V

Linguists prefer to use features to describe such statements because features unify the sounds that are found in this context — it is not a random list of any group of sounds, but rather it is a set of sounds that is uniquely classified as [+voice, –continuant]. Using features in this way also suggests what we should find if there are other stops in the language besides the ones we’ve considered — we expect to find voiced stops only between vowels.

The statement (6) tells how to pronounce Nakoda stops — they are voiced only when they are found between vowels.

If you look at a lot of languages, you will find that it is not unusual to find voiced consonants between vowels, just like we see in Nakoda. This type of pattern is called voicing *assimilation*. Assimilation occurs when neighboring sounds are more like each other. We know that voicing occurs when the vocal folds continue to vibrate while a sound is being pronounced. Vowels are almost always voiced, so a stop with a vowel on each side is surrounded by voicing. A voiced

stop between the two vowels is more similar to the vowels than a voiceless stop would be. We can also find assimilation of nasality, roundness, and so on.



Comprehension Check 3

Suppose you continued to work with Nakoda, and realized that the language also has a voiced affricate [dʒ].

Do words like *ecé* [edʒé] ‘only’ make you curious whether [dʒ] follows the same pattern of limited distribution as we found with the voiced stops [b], [d], and [g]? Explain your answer.



Here's our answer

We hope that these words do make you curious about whether the limited distribution of voiced stops in Nakoda includes the voiced affricate [dʒ] as well. The voiced [dʒ] is between vowels in *ece* [edʒé] ‘only’ — exactly where we saw voiced stops. In fact, further exploration of Nakoda words leads to the conclusion that the limited distribution pattern extends to the voiced affricate as well as the voiced stops.

7.6.3.3 The take-away from the distribution of [b], [d], and [g]

Our exploration of the distribution of [b], [d], and [g] shows us that similar consonants sometimes have the same distribution. The consonants [b], [d], and [g] are similar because each is voiced and each is a stop; they differ in having different places of articulation. They have the same distribution, appearing only between vowels. We often find a class of similar sounds behaving the same way in a language.

We used a single member of the class, in this case [b], as an entry to the pattern. Once we found limited distribution, we checked to see if other members of the same class had the same pattern. We could have begun with any of the voiced

stops, [b], [d], or [g], as our entry point into the pattern. If a class of sounds has the same pattern, you can start with any member of the class to find the pattern, then extend the distribution table to check other members of the class.

It is also possible to begin with the extended table, jumping right to Table 16, rather than using a single member of a class as an entry point. As you become more confident, you may decide you prefer testing all members of a class at the same time.

7.7 One or two consonants? Djeoromitxí affricates and stops

In this section, we use the concept of the general distribution of consonants to determine whether a particular sound we hear in Djeoromitxí is a single segment, one consonant, or a sequence of two segments, two consonants. Specifically, we are trying to determine whether what we hear is an affricate or a stop followed by a fricative. This issue arises in many languages, and is important in determining the sounds of the language. If you don't face this question in your language, and you don't need another example showing how to determine limited distribution, you might jump ahead to the closing discussion, in Section 7.8.

7.7.1 Affricates or stop-fricative sequences?

De Castro (2012) states that Djeoromitxí has stops, fricatives, and affricates: Stops include voiceless [p], [t], [k], and voiced [b], [d]; fricatives are voiced labial [β] and voiceless laryngeal [h]. Our focus here is on the third class of segments, the affricates.

Affricates are segments that begin with a stop closure but end with a fricative. Two of the Djeoromitxí affricates are fairly common, post-alveolar voiceless [tʃ] and voiced [dʒ]. The other Djeoromitxí affricates are relatively uncommon. One has labial closure and labial release, written [p̪̚]. Others have labial closure and alveolar release, written [ps̪̚] and [bz̪̚].

This raises a question: Why are these sounds treated as single consonants, the one-consonant view? Why not simply say they are all sequences of two consonants — [p] and [ɸ], [p] and [s], [b] and [z], [t] and [ʃ], [d] and [ʒ] — the two-consonant view? One could even imagine a hybrid view — with [tʃ] and [dʒ] alongside [p] and [ɸ], [s], [b] and [z].

The answer is clear when we consider distribution. The one-consonant view says each of these sounds is one segment, an affricate, and each should pattern

just like all the other single consonants. If we adopt the two-consonant view, then each is a sequence of a stop and a fricative, and the distribution of certain stops and the fricatives is quite different from the distribution of any other consonant in the language — as well as being different from each other. Distribution shows why it makes sense to adopt the one-consonant view, to think of these sounds as affricates and not as a CC sequence.

In the next two sections, we look at distribution, first in terms of the one-consonant view (thinking of the sounds as affricates), and then in terms of the two-consonant view (thinking of the sounds as sequences of stop and fricative). If your language has sounds that might be affricates, but that might also be stop-fricative sequences, this section shows you how to determine which analysis makes most sense for your language.

7.7.2 One-consonant view: Affricates [p̪ɸ], [ps̪], [bz̪], [tʃ̪], [dʒ̪]

In this section, we see that there are no surprises in the distribution of consonants if we take the one-consonant view and treat these sounds as affricates.

Given the general distribution of consonants in Djeoromitxi, we expect to find each consonant at the beginning of some words as well as between vowels in some words. So we expect to see affricates in exactly those two positions. To determine whether this is the case, we make a distribution table similar to the one made for [t] in Table 10. We modify the distribution table by having a column for each of the affricates, [p̪ɸ], [ps̪], [bz̪], [tʃ̪], [dʒ̪]. As Table 17 shows, each affricate is found at the beginning of a word and each is found between vowels, the standard distribution for consonants in Djeoromitxi.

Table 17: Djeoromitxi [p̪ɸ], [ps̪], [bz̪], [tʃ̪], and [dʒ̪] distribution

	[p̪ɸ]	[ps̪]	[bz̪]	[tʃ̪]	[dʒ̪]
#_	<i>pō</i> [p̪ɸō] 'grandson'	<i>psihe</i> [ps̪ihe] 'to wed'	<i>bziru</i> [bz̪iru] 'water'	<i>txori</i> [tʃ̪ori] 'fish eagle'	<i>djiri</i> [dʒ̪iri] 'child'
V_V	<i>kupū</i> [kup̪ɸū] 'forest'	<i>nitopsi</i> [nitops̪i] 'to fart'	<i>djebzi</i> [dʒ̪ebzi] 'full, flood'	<i>txitxika</i> [tʃ̪itʃ̪ika] 'tick' (insect)	<i>todjo</i> [todʒ̪o] 'crooked'

All the appropriate cells are filled in Table 17. If we adopt the one-consonant view and think of these sounds as affricates, we see that their distribution is exactly as expected given what we know of the distribution of consonants in Djeoromitxi.

As the next section shows, distribution is quite different if we adopt the two-consonant view, and think of these sounds as sequences of two consonants.

7.7.3 Two-consonant view: CC sequences [p][ɸ], [p][s], [b][z], [t][ʃ], and [d][ʒ]

Under the two-consonant view, each of these sounds is composed of two consonants, a stop and a fricative. This means that there are nine sounds whose distribution we need to check — four stops, [p], [b], [t], [d], and five fricatives, [ɸ], [s], [z], [ʃ], [ʒ]. To check distribution, we again follow use our consonant distribution table, first filling it in with words that show the distribution of all four stops, in Table 18. We start with the stops, because De Castro says that these stops are part of the consonant inventory of Djeoromitxí.

Table 18: Djeoromitxí [p], [b], [t], [d] distribution

	[p]	[b]	[t]	[d]
#__	<i>piti</i> [piti] <i>cicada species</i>	<i>berece</i> [bɛɾɛtʃɛ] ‘bat’	<i>tee</i> [tɛɛ] ‘nephew/niece’	<i>dudu</i> [dudu] ‘to walk’
V__V	<i>kupa</i> [kupa] ‘moon’	<i>kuti</i> [hibo] <i>an animal</i>	<i>hibo</i> [kuti] ‘smooth’	<i>kudi</i> [kudi] ‘deer’

But this doesn’t capture the full distribution of stops and fricatives under the two-consonant view because if the sounds we hear are a sequence of two consonants, then we must admit that there are CC sequences in Djeoromitxí — and now we need to check V__C and C__V contexts.

Let’s look at stops first. Table 19, pairing stops (heading the columns) with fricatives (labeling the rows), makes this highly skewed distribution visually clear. Before reading the next paragraphs, it might help to flip the page and look at Table 19 to see how the table reveals the skewed distribution.

If we insist that these sounds are each a sequence of two consonants, we must say that stops have a very unusual distribution for consonants in Djeoromitxí, in four ways. First, we must say that stops — and only stops — can appear before another consonant. Second, these sounds cannot occur before just any other consonant. They can only occur before a fricative. Third, each of them occurs before a very specific fricative, not just any fricative. Finally, the fifth stop in Djeoromitxí, the velar [k], does not appear before any fricatives.

As shown by the empty rightmost column, [k] has the expected distribution for consonants. This same distribution holds for other consonants in Djeoromitxí, except for the ones in this table. As shown by the diagonal, [p] is found only before [ɸ] or [s], while the others are found before exactly one consonant each – [b] is found only before [z], [t] only before [ʃ], and [d] only before [ʒ]. Both the stops and the fricatives have very limited distribution.

As Table 19 shows, the two-consonant view requires a special and very complex restriction on the distribution of both stops and fricatives. They do not occur freely in combination with any consonant. Stops only precede fricatives, fricatives only follow stops, and each occurs only in combination with one specific consonant, except [p] which occurs before two different fricatives.

Table 19: Djeoromitxí [p], [b], [t], [d], [k] before specific consonants

	[p]	[b]	[t]	[d]	[k]
___[s]	<i>psihe</i> [psihe] 'to wed'				
___[ɸ]	<i>põ</i> [pɸõ] 'grandson'				
___[z]		<i>bziru</i> [bziru] 'water'			
___[ʃ]			<i>txori</i> [tʃori] 'fish eagle'		
___[ʒ]				<i>djiri</i> [dʒiri] 'child'	

7.7.4 Comparison of the one-consonant and two-consonant results

When we compare distribution under the one-consonant and two-consonant views of Djeoromitxí sounds, we see that, with the one-consonant view, these sounds have completely normal distribution for consonants. On the other hand, with the two-consonant view, the distribution of both the first consonant and the second consonant is unlike that of any other consonant in the language. Furthermore, the distribution is very difficult to state.

Our general rule of thumb — if a distribution statement is very complex, try looking from a different perspective — is relevant. That other perspective is the one-consonant view. It is much simpler to say that in Djeoromitxí, every consonant in a Djeoromitxí word appears directly before a vowel. This is possible only if we accept that the language has five affricates, [p̪t̪], [ps̪], [bʒ̪], [tʃ̪], [dʒ̪]. (Again, if you are keeping track of syllables, you would say that the language has only open syllables with at most one consonant in the onset, (C)V syllables. Such syllables require that these five sounds are affricates, not clusters.)

A further point in support of the one-consonant analysis is that while Djeoromitxí has words with these five affricates, [p̪t̪], [ps̪], [bʒ̪], [tʃ̪], [dʒ̪], the corresponding fricatives are not found independently in the language. There are no words with initial [ɸ], [s̪], [z̪], [ʃ̪], or [ʒ̪] or with any of these sounds between vowels. They do not behave like independent consonants.

7.7.5 The take-away from the investigation of Djeoromitxí affricates

You may read a description of a language that says the language has affricates. In this case, if the source is a good source, the statement about affricates is based on the type of investigation we just saw for Djeoromitxí. Alternatively, a source may say that a sequence like [ts] is two consonants, not an affricate. In this case, if the source is reliable, there was an investigation of the same type, but the results were different. Sources will not always discuss the investigation itself. You now have the tools to check whether a language has affricates, consonant sequences, or both: You use the consonant distribution table for the language and see where consonants are found.



Exploring my language

Does your language have affricates or sequences of two consonants? Choose one (affricates or two-consonant sequences) and use a consonant distribution table to determine whether the analysis of “affricate” or of “two-consonant sequence” makes more sense. Follow the steps in Section 7.7 as you make your determination.

7.8 Discussion

This chapter focused on both general and limited distribution of sounds in a language, with examples from the consonant systems of Djeoromitxí, Sierra Popoluca, and Nakoda.

7.8.1 Resolving the puzzle

We began the chapter with a puzzle — the question of whether what is written with two letters counts as one sound or as two sounds in a language.

How can you determine whether sounds that are spelled with two letters (in writing for regular reading) count as one consonant or two in the way your language uses consonants in words?

In this chapter we gave a method for solving this kind of puzzle — building a distribution table for the sounds in question. We showed how to make a distribution table very specific in order to test whether affricates or CC sequences make more sense in the language. In our example from Djeoromitxí, we saw that the language has affricates, not CC sequences (Section 7.7.1).

We used distribution tables to confirm that an individual sound has limited distribution. We saw that [r] is restricted to word-medial position in Djeoromitxí, (Section 7.2) and that the aspirated stop [t^h] is found only in syllable-final position in Sierra Popoluca (Section 7.3.2). Distribution tables showed that a class of sounds, the voiced stops, have limited distribution in Nakoda, restricted to intervocalic position (Section 7.6).

We used the distribution tables to systematically answer questions about these languages. In each case, we started with a table showing the general distribution of consonants or of vowels in the language. Then we followed a five-step procedure.

1. *Modify the generic distribution table* by making a column (or columns) for the sound (or sounds) that you are interested in.
2. *Fill in the cells* in the distribution table as best you can.
3. *Check the distribution pattern* by looking at the distribution table; remember that empty cells suggests limited distribution.
4. *Adjust the table* (if needed) by making row and/or column headings more specific.
5. *Create a concise statement* of the distribution pattern that you find.

Understanding patterns of limited distribution help us understand how sounds are arranged in the words of a particular language. This knowledge can be used in a variety of ways by language workers, as we discuss in Section 7.8.3. First, we address the question of why languages have limited distribution.

7.8.2 Why do languages have limited distribution?

Limited distribution is a common property of languages but can be tough to figure out. Where does it come from?

Many instances of limited distribution involve sounds that are easier to say in a particular context. One way that this can be done is by *assimilation*, making a sound more like its neighbor. Examples of assimilation include the limited distribution of voiced stops in Nakoda. Another example of assimilation is the limited distribution of voiced stops in Baure — voiced stops in Baure are only found following a voiced nasal consonant, as in [kəʈəmbə] ‘new (e.g. house)’ (*[...mp...] in Baure).²

Another common way to make it easier to say a consonant is to use a fricative instead of a stop; this is an example of *lenition*. In general, lenition means that the sound used requires a minimum of effort — the articulators can relax a bit. An example of lenition occurs in Tutelo-Saponi, a Siouan language spoken in Ohio, Virginia, and North Carolina. Tutelo-Saponi consonants include a voiceless velar stop [k]. This sound can occur by itself, illustrated in (7a), and in clusters, illustrated in (7b). The words in (7) are from Hale (1883).

- (7) Tutelo-Saponi [k]
- | | | | |
|----|---|---------|-------------------------------|
| a. | # | [kite:] | ‘kill’ |
| | V | # | [hu:k] ‘all’ |
| | V | V | [wahtaka] ‘man’ |
| b. | C | # | [wisʊtk] ‘my younger brother’ |
| | C | V | [tapnɪskai] ‘star’ |

When [k] is the first consonant in a cluster, there are several ways that the word can be pronounced — with the voiceless velar stop [k] or with a voiceless fricative, either velar [x] or laryngeal [h], as shown in (8). These words are from Hale (1883) and Oliverio (1997).

²Bauré is an Arawakan language spoken in Bolivia, Danielsen (2007). We thank Swintha Danielsen for checking our Bauré example. She told us that speakers refer to their language as Bauré (though the accent on the final *e* — Bauré — was introduced by French people to show that the vowel is pronounced).

(8) Tutelo-Saponi lenition: [k]~[x]~[h]

___[t] [waktɑːkɑ] ‘man’
[waxtɑːkɑ]
[wahtɑːkɑ]

Producing a consonant cluster is more challenging for the articulators than producing a single consonant; lenition is one way to make it easier. Making a fricative [x] or [h] requires less effort than moving the tongue to make complete closure for the stop [k].

To the extent that hearers can still understand the speaker, strategies like assimilation and lenition can help the speaker communicate more easily. One generation’s “sloppy speech” may become the next generation’s “limited distribution” as the pronunciations get integrated into the rules a learner acquires about their language.

The converse of lenition is fortition. Fortition involves making sounds more clear — by making them longer, by devoicing, and so on.

7.8.3 Using my hard-earned knowledge

Having worked through this chapter, you now understand that some segments have limited distribution, appearing in only some of the contexts where we expect to find that class of sounds. You have developed skills for identifying the general distribution of both consonants and vowels in a particular language, and to use that general distribution to test whether specific segments have limited or unrestricted distribution. Why is it useful to have this kind of detailed understanding of a language?

Language learning. A teacher who understands which sounds (or classes of sounds) have limited distribution can use that knowledge to identify sound sequences that will be challenging to the students, and design pronunciation exercises to target the potential trouble spots. A teacher can also use the knowledge to design drills that highlight the limited distribution so that these sequences become automatic for the learners.

A teacher might also explain the generalizations that produce gaps in the distribution of certain sounds so that the learners gain a more complete understanding. With an understanding of distribution, learners gain the ability to apply the knowledge in different situations. Pronunciation becomes more fluent when learners begin using their knowledge of limited distribution to pronounce new words.

Language documentation and description. Including which segments have limited distribution and which have general distribution is part of a complete language description. The completeness of these descriptions is one metric to use when evaluating the quality of a resource on your language.

Some communities are fortunate enough to have digital resources available already, others may develop digital materials in the future. A critical design element is how to structure the search functions for digital resources. Knowledge of segments with limited distribution can be used in the design of the resources, including the letters used to represent segments, as well as the design of the search feature.

If the digital resources include automated speech recognition that recognition — even of less clearly produced speech — is improved when the distribution of sounds in the language is encoded in the programming. The converse is also true with text-to-speech capabilities.

Language revitalization. There are many challenges in understanding and using resources on a language that has been sleeping. Resources may come from different times and different regions; descriptions may be fragmented. How can these varied resources be used to revitalize the sound system of a language?

As noted in Chapter 5 Section 5.6.3, a great place to start is to create inventories of the segments included in each resource, and compare the different ways they are represented. What this chapter brings out is that we also need to consider the distribution of the segments as found in the different resources. If there are discrepancies, then there are puzzles to figure out — is this due to inadequate coverage in some of the resources? is it because the resources are from different points in time or place, and so representing different varieties of the language?

Some efforts at revitalizing a language include efforts to remove borrowed words. One way to identify a borrowed word is that borrowed words may not follow patterns of limited distribution in a language. For example, we saw in Section 7.2 that Djeoromitxí has no words beginning with the segment [ɾ], so finding a word in that language that begins with [ɾ] would indicate that it is a borrowed word.

Dealing with inadequate records and eliminating borrowed words can lead to a new problem for revitalization — concepts for which there is no word in the language. Understanding the distribution of the segments in a language's inventory, whether limited or general, enables workers to create words that feel more natural to the language to fill these gaps, by making sure segments with limited distribution only appear in legitimate contexts, following their pattern of limited distribution. For example, a Djeoromitxí language worker who knows

about the limited distribution of [r] would be unlikely to propose a sequence beginning with [r] for a new word.

Reading linguist's resources about your language. Basic descriptions of a language typically include some discussion of the distribution of the segments of the language. You now know what this is referring to, and you know what a researcher does to establish distribution patterns.

As we see in the ensuing chapters, the concept of distribution is central to deeper levels of language analysis. For example, Chapter 8 addresses a special type of distribution called *complementary distribution*. Understanding distribution and limited distribution gets us ready to explore complementary distribution.

7.9 Conclusion

Distribution of the language's segments in its words is at the core of the sound pattern of any language: You now have an understanding of the concept of distribution, both general and limited, and can make use of distribution when talking about your language.

You also know how to carry out an analysis of distribution, to establish generally where consonants and vowels occur in your language and then to determine whether any of the segments of your language have limited distribution. You've learned the skill and value of making and filling in tables showing the possible locations of consonants (or vowels) in your language.

Inspired by the ideas in Section 7.8.3, you are prepared to use your knowledge of distribution as you work with your language.

Chapter 8

Complementary distribution: The same but different

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Here's a puzzle. Suppose you are learning Nakoda, a Siouan language spoken in northeastern Montana and southern Saskatchewan, and you are becoming familiar with the sounds of the language. Being an eager student, you look at an online Nakoda dictionary and find several words containing [sapsaba], including *pahásapsaba* 'to have black hair' and *ptewánuwq-sapsababi* 'Black Angus'. In these examples, we use text color to focus attention on the black letters, not the gray ones. Text in italics shows the way words are written for reading; symbols in square brackets show how words are pronounced, by using the [International Phonetic Alphabet](#), or IPA. For more on IPA symbols, you can review Chapter 2, as well as the summary lists in Appendix A.

From these words, you might notice that Nakoda words use both the voiceless sound [p] and the voiced sound [b]: [sapsaba]. Then you find a full sentence in a doctoral dissertation — *ma-p^háha sap-sapa* 'I have black hair' — the word for 'black' is written with two instances of the letter *p*, and no *b*: *sapsapa*. But you know it is pronounced [sapsába].¹

¹Nakoda, a Siouan language, is also called Assiniboiné. The online dictionary that we consulted is found at <https://indigenouknowledge.indiana.edu/resources/index.html>, using the the Resources tab to get to the IIK Dictionary Portal; we consulted the Assiniboiné dictionary on November 22, 2022. The dissertation we consulted is Cumberland (2005), a dissertation written for linguists. The hyphens in *ma-p^háha sap-sapa* were used in the source to indicate subparts of the words (called *morphemes*, a concept we address in Chapter 3).

Why would someone write different sounds with the same letter?

Typically, we use different letters to indicate different sounds. For example, when writing Nakoda, people use the letters *s* and *š* for the sounds [s] and [ʃ] — using different letters is important because words that start with these sounds have different meanings, like *sápa* [sába] ‘black’ and *šápa* [ʃába] ‘soiled’. In this case, the writing system used in the dissertation uses distinct symbols only for sounds that **contrast** with each other — that is, only for sounds that can make a meaningful difference between words as we see with *sápa* [sába] ‘black’ and *šápa* [ʃába] ‘soiled’. (Chapter 5 introduces the concept of contrast.)

Sometimes however, a single symbol is used to write two different sounds when the sounds do not contrast, when they make no meaningful difference. By using *p* when writing both [b] and [p], the creators of this Nakoda writing system are implying that [b] and [p] are **non-contrastive** sounds. We know sounds are non-contrastive when they are in **complementary distribution** with each other, the topic of this chapter.

Complementary distribution is a particular type of distribution, the fifth major component of a language’s unique **phonology** (along with variation, contrast, syllables, and distribution). In this chapter, we explore examples of complementary distribution, where two sounds or sound classes never appear in the same contexts in words.

Here’s our promise. After reading this chapter, you will understand the concept of complementary distribution and be able to use the concept in talking and thinking about your language. Additionally, you will be able to determine whether a pair of sounds is or is not in complementary distribution. Importantly, sounds in complementary distribution are non-contrastive, so they can’t create a meaningful difference between two words. (Non-contrastive sounds can also be due to **free variation**, discussed in Chapter 4, Section 4.4.)

Knowledge about which sounds are in complementary distribution in your language is particularly useful in helping students produce the words of a language more fluently. It can be useful in making decisions about writing systems. Additionally, the material in this chapter can be useful in understanding linguistic writings about the sounds of your language: For example, descriptions written for linguists and/or by linguists may use terms like **phoneme** and **allophone** when talking about sounds that are in complementary distribution; we show the connection in this chapter.

Here's our plan. To make good on our promise, this chapter gives a more complete explanation of the elements involved in complementary distribution as well as a strategy for determining whether a pair of sounds is in complementary distribution. Examples illustrate several variations on the core concept of complementary distribution and show how to analyze each. Here's what the different sections include, so you can decide which are the most important for you at this time.

1. Section 8.1 introduces the concept of similarity between sound segments and the concept of complementary contexts, the two key elements in complementary distribution.
2. Section 8.2 shows how to identify sounds that might be in complementary distribution with each other, building on the concept of similarity. This section also shows the basic analytic strategy for determining whether there is a pattern of complementary distribution, including its properties.

These two sections may be enough for your purposes. If so, when you finish them, we suggest that you skip ahead to the discussion in Section 8.7.

The next four sections show some of the variety we find with complementary distribution, and how the analysis works. If you want to know how to analyze complementary distribution, we recommend reading sections 8.3 and 8.4. If your language has more complex complementary distribution, you may find it helpful to go through either or both of sections 8.5 and 8.6.

3. Section 8.3 extends the Nakoda example from the introduction, showing that the pattern of complementary distribution affects a class of sounds, not just a pair of sounds.
4. The example in Section 8.4 is from Djeoromitxí. This example puts features into the distribution table: Djeoromitxí has a pattern of complementary distribution that depends on the features of the following vowel.
5. Complementary distribution sometimes involves multiple segments in a single complementary relation, as the example in Section 8.5 illustrates. This will be particularly useful if your language has this type of pattern and you are trying to figure out how to understand it.
6. In some cases, there are instances of words that do not follow a fairly general pattern of complementary distribution. This situation of exceptions is discussed in Section 8.6. If you suspect that your language has a general pattern of complementary distribution but also has exceptions to the pattern, you may find this section extra helpful.

7. Complementary distribution is at the heart of the linguistic concepts of **phoneme** and **allophone**, as well as **surface representation** and **underlying representation**. These concepts are explained in Section 8.7, along with resolving the chapter puzzle and presenting ways that language workers can make use of complementary distribution.

8.1 Introduction: What is complementary distribution?

From Chapter 7, you may recall that we use the term **distribution** to refer to the location of sounds in words. If this makes you think that **complementary distribution** has to do with the location of sounds in words, you are right. Complementary distribution is a very specific type of distribution that we find in languages, that opens a new way of understanding the sound patterns in a language. Sounds in complementary distribution do not **contrast** with each other. If you can't find **minimal pairs** for a particular pair of similar sounds, it may be a case of complementary distribution. (See Chapter 5 if you need a review of contrast and minimal pairs.)

Chapter 7 introduced the concept of distribution, and showed how to determine the distribution of consonants and vowels in a language, using distribution tables. For analyzing complementary distribution, it is important to be able to construct and use a distribution table. (If you need a review, please see Section 7.3 in Chapter 7.) In this chapter, we focus directly on complementary distribution.

We say that **similar** sounds are in **complementary distribution** with each other when they are found in complementary contexts in a language. From this definition, it is important to understand when sounds are similar to each other; it is also important to understand what complementary contexts are. Let's start with similarity.

8.1.1 Similarity between sounds

If we are asked to say a pair of sounds, and then are asked whether the sounds are **similar** to each other, we may have a very strong intuition about which sounds are similar and which are not similar.

Try this. Say the sounds [m] and [n]. They are different, but are they also similar? Now say the pair [m] and [a]. Do you find similarities as well as differences with this pair? Is one of the pairs more similar than the other?

We hope your intuitions led you to say that [m] and [n] are fairly similar to each other, and definitely more similar to each other than are [m] and [a], which we hope felt quite different to you. With a linguistic understanding of these sounds, it is easy to talk about the similarities and the differences, and make those intuitions more concrete.

The way we do this is by comparing the features of the sounds. We compare the features of the sounds [m], [n], and [a], in Table 1. (You can go to Chapter 2 if you need to review these features. The nasal/oral contrast is found in Section 2.2.2, manner of articulation is found in Section 2.3.1, and place of articulation in Section 2.3.2. If you just want a list matching segments and their features, please check Appendix B, and go to the glossary of terms linguists use for definitions of specific features such as [nasal].)

Table 1: Comparing features of [m], [n], and [a]

	[m]	[n]	[a]
place of articulation	[labial]	[coronal]	—
tongue position	[−low]		[+low]
C or V?	C		V
nasal/oral	[+nasal]		[−nasal]
stop?	[−continuant]		[+continuant]

When we inspect the features of these sounds, we see that [m] and [n] are the same with respect to four features, boxed in Table 1 — both are consonants, both are [+nasal], both are [−continuant] stops, the tongue body is in a neutral position, [−low]. The difference comes with the place of articulation features: [m] is a [labial] sound, while [n] is a [coronal] sound. The third sound differs on all five properties — it is an oral vowel, not a stop, made with a low tongue position and with neither labial nor coronal place of articulation — this is a [+low, -nasal] vowel. When we look at the properties closely like this, we can see reasons for saying that [m] and [n] are similar while [a] is not similar to [m] nor to [n].



Comprehension Check 1

Consider the three sounds [β], [w], and [s] found in the Mazatec language from Soyaltepec Municipio (an Otomanguean language spoken in the state of Oaxaca, Mexico; it is described in Beal 2012).

Which sounds are more similar to each other and which sound is different, and why?

Since all three sounds are consonants, you need other properties to distinguish them. It helps to isolate the features of the individual sounds, to find ones where there are differences. You might think about place and manner of articulation, and whether sounds are voiced or not.



Here's our answer

We found that [β] and [w] are quite similar, and [s] is fairly different. The three sounds are similar in that all are continuant consonants and none is a stop. When we look at other properties, though, finer points of difference come to light. For place, both [β] and [w] have lip constriction making them [labial] sounds; with [s] the tongue position is raised in the front, making it [coronal]. While both [β] and [w] are [+voice]; [s] is [−voice]. On the other hand, [s] is a “hissy” [+strident] sound while both [β] and [w] are quiet sounds. Our evaluation table lays out these points below.

	[β]	[w]	[s]
stop?	[+continuant]		
voicing	[+voice]		[−voice]
place	[labial]		[coronal]
strident?	[−strident]		[+strident]

8.1.2 Complementary contexts

When we talk about the **contexts** for some sound, we are talking about where that type of sound can be found in words of the language. This is the **distribution** of Chapter 7. There are many different ways to describe the contexts for a single class of sounds in a language. For example, in Chapter 7, we saw that consonants in Djeoromitxí are found immediately preceding a vowel — there are no **word-final** consonants and there are no consonant sequences inside words. We can describe this as one context, before a vowel (or ____V). We could also say that consonants are only found in **onset** position in Djeoromitxí. (See Chapter 6 if you need a review of onsets.) We can also describe the distribution of Djeoromitxí consonants using two contexts — Djeoromitxí consonants are found either at the beginning of a word before a vowel or between two vowels. It is when we start using multiple contexts to describe the distribution of sounds that the concept of complementary contexts becomes relevant.

Complementary contexts for a class of sounds must meet two criteria — no gaps in coverage and no overlap between contexts. Let's look at both.

No gaps. The contexts taken together must include all possible contexts for that class of sounds. Nothing can be left out. Our contexts for Djeoromitxí — either at the beginning of a word before a vowel or between two vowels — meet this criterion. There are no other locations where consonants are found in Djeoromitxí.

Compare this to Nakoda, where these two contexts are inadequate in describing consonant distribution. Consider again the word *sapsapa* [sapsaba] 'black'. We find [s] at the beginning of the word, [sapsaba], and [b] between vowels, [sapsaba], but there are two consonants whose location is not described by these contexts, the two consonants in the sequence [ps], as in [sapsaba]. Because we do not include these contexts for consonants, we have a gap, and that gap tells us that we have not yet identified complementary contexts for Nakoda.

Try this. See if you can figure out the missing context(s) for Nakoda consonant distribution.

To remedy the gap, we need to add that consonants in Nakoda are found after a consonant and before a vowel, C____V, as well as after a vowel and before a consonant, V____C. If you are fond of syllables, you may have noticed that we can reduce the four contexts down to two contexts using **syllable** structure: Consonants are found in **onset** position and in **coda** position. Reducing the number of contexts is one advantage of the concept of *syllable*.

No overlap. Each context defines a unique location. Consider the Djeoromitxí word [paro] ‘chicken’, remembering that Djeoromitxí consonants are found at the beginning of a word before a vowel or between vowels. This is true of [paro] ‘chicken’ — the [p] is at the beginning of a word before a vowel in [paro] and the [r] is between vowels in [paro]. The consonant [p] is not also between vowels in [paro], and the [r] is not also at the beginning of the word [paro].

What would overlapping contexts look like? Suppose we proposed the two contexts *before a vowel* and *between vowels* for consonants in Djeoromitxí. This pair of contexts is true of consonants in the language. For example, [p] and [r] are both before vowels in [paro] ‘chicken’, and [r] is between vowels. Notice, though, that we have defined the location of [r] *in two ways* — both before a vowel and between vowels. This means there is overlap in the contexts, so the contexts are not complementary. With complementary contexts, there is no overlap.

Try this. Is there overlap in these four contexts for Nakoda consonants? (i) before a consonant, (ii) after a consonant, (iii) before a vowel, (iv) after a vowel?

We hope that you figured out that there is a lot of overlap in the four Nakoda contexts we gave you — in the word *sapsapa* [sapsaba] ‘black’, the initial [s] is only before a vowel, but every other consonant is put into two categories. The [p] is both after a vowel and before a consonant — [sapsaba]. The second [s] is both after a consonant and before a vowel — [sapsaba]. Finally, the [b] is both after a vowel — [sapsaba] — and before a vowel — [sapsaba].

8.1.3 Complementary distribution

We turn now to *complementary distribution*. Complementary distribution puts the similar sounds of Section 8.1.1 into the complementary contexts of Section 8.1.2 so that each sound (or sound class) has its own, unique set of contexts. These sounds do not contrast with each other; there are no minimal pairs, no near minimal pairs, and no context pairs.

For example, in Soyaltepec Mazatec, the sound [β] is found at the beginning of words and [w] is found between vowels: [βase] ‘half’, [βifu] ‘they fell’, etc. vs. [tiwa] ‘plate’, [tʃiwu] ‘goat’. There are no words with an initial [w], and no words with [β] between vowels. Here, a pair of similar sounds, [β] and [w], is distributed among complementary contexts (*word-initial* and *between vowels*) so that each sound has its own unique context. These sounds are in complementary distribution with each other.

The focus of this chapter is on learning how to determine whether sounds are in complementary distribution with each other in some language. Before beginning to explore the analytic strategies, we take a moment to consider clues that might lead you to suspect that certain sounds are in complementary distribution with each other.

8.2 Strategies for identifying sounds that might be in complementary distribution with each other

This section addresses the question *what are the clues we look for, that lead us to suspect complementary distribution?*

The first step is to identify similar sounds in the language, since sounds that are in complementary distribution are typically very similar to each other. How to tell whether sounds are similar is discussed in Section 8.1.1 in this chapter. Without similar sounds, we don't expect to find complementary distribution.

Similarity alone, however, is not enough to identify likely cases of complementary distribution. Ideally, you will have similarity coupled with another indicator. Several possible other indicators are discussed in the following subsections.

8.2.1 Are there minimal pairs?

As shown in Chapter 5, minimal pairs, near-minimal pairs, and context pairs provide evidence that sounds contrast with each other. If you have a pair of sounds (or sound classes) in your language and can find no evidence of contrast, you are encouraged to investigate whether the sounds are in complementary distribution (and so are non-contrastive).

The absence of contrast for similar sounds, shown by the absence of minimal pairs for those sounds, is the most common clue for checking for complementary distribution. You can remind yourself about contrast by reviewing Chapter 5, if you need a reminder.

8.2.2 Clues from the sound inventory

A language's sound **inventory** is the collection of sounds that are used in the language (review Chapter 5 Section 5.5 on inventories if needed). Examining the sound inventory is a rich source of clues that might suggest complementary distribution.

A clue from inventories comes when you are working to identify the contrastive sounds for the language. If you find similar sounds, but cannot find evidence that those sounds contrast with each other, it is a good idea to check for complementary distribution. For example, in setting up the Djeoromitxí inventory, there would be no words showing a contrast between [p] and [p̠]. We look at this pair in Section 8.4 of this chapter.

Another inventory clue relies on the expectation that sound inventories have a tendency towards symmetry — for example, we often find voiced and voiceless pairs of consonants, or stop and fricative pairs; with vowels we might find high and mid pairs, or round and non-round pairs, and so on. When an inventory has a general symmetry except for one pairing with either too few or too many sounds in one part, we may have a clue to complementary distribution. For example, suppose you have a language with a paired series, but there is one sound in the series without a counterpart — a case of too few sounds. In such cases, the unpaired sound might be in complementary distribution with some other, similar, sound. In Section 8.4 (this chapter), we explore a too-few-sounds example in Djeoromitxí, a language rich in affricates with voiced/voiceless pairs — except for the voiceless affricate [p̠].

Alternatively, suppose that you have an unpaired series for the most part, except that one sound (or a few sounds) in the series is paired — a case of too many sounds. In such cases, the extra sound(s) may be a clue to complementary distribution. We examine a too-many-sounds example in Section 8.5 (this chapter): Nahuatl Pipil stops form a single plain voiceless series — except that dorsals are pronounced in four different ways.



Comprehension Check 2

Kickapoo is an Algonquian language, spoken in the United States in Oklahoma and Texas, and in Mexico in the state of Coahuila. Our source is Bluecloud (2020).

The Kickapoo consonant inventory is shown here. Is there either a too-many-sounds or too-few-sounds situation among the fricatives, suggesting the possibility of complementary distribution?

	labial	dental	coronal	palatal	dorsal	glottal
stop	[p]		[t]		[k]	
nasal	[m]		[n]			
fricative		[θ] [ð]	[s]			[h]
affricate				[tʃ]		
approximant	[w]			[j]		



Here's our answer

The pair [θ] and [ð] is a reasonable choice to explore for complementary distribution. The Kickapoo fricatives have a [−voice] series, [θ], [s], [h]. In addition, there is one voiced fricative, [ð], which has the same phonetic properties as [θ], except [ð] is [+voice] while [θ] is [−voice].

8.2.3 Clues from speaker intuitions

Another general source of clues to possible complementary distribution is speakers with intuitions about their own language. The type of intuition that suggests complementary distribution is when a speaker says that a certain sound cannot occur in certain places, but another, similar, sound can occur in exactly those places. When you find this kind of intuition, it is a smart to check for complementary distribution.

8.2.4 Clues from orthography

Another tip that can be gathered from speakers is the choice of symbols used to write a language (for signs, books, texts, etc.). If there are two distinct sounds that are written with the same letter, it may be because the sounds are in complementary distribution.

8.2.5 Clues from existing resources

Writings about a language may also be a guide to complementary distribution. You might read something about your language that claims certain sounds are in complementary distribution. (Testing such claims yourself is one way to determine how reliable the resource is.)

You might also have access to a dictionary for your language, or you may be creating a dictionary for your language. In creating comprehensive word lists, we sometimes find gaps that we simply cannot fill (such as the lack of [ɾ]-initial words in a Djeoromitxí dictionary). Gaps can indicate complementary distribution.

8.2.6 Clues from limited distribution

Another clue to complementary distribution is limited distribution, addressed in Chapter 7. When a sound or sound class is found in one set of contexts but not in another, we say it has limited distribution. When we find a pair of similar sounds each with limited distribution, we want to know whether the contexts are complementary, making the pair a case of complementary distribution, not two separate cases of limited distribution.

Whenever we see limited distribution it is smart to ask *is there a similar sound or class of sounds that also has limited distribution, in exactly the complementary contexts?* In Chapter 7 Section 7.6.2, we discovered that voiced stops are found only between vowels. Nakoda also has voiceless stops, sounds that differ from voiced stops only in terms of voicing. And since the voiced stops have limited distribution (only between vowels), we wonder whether voiceless stops are found only in the complementary contexts — everywhere in words *except* between vowels. We show how to determine whether sounds are in complementary distribution in Section 8.3.

8.2.7 Conclusion

Once you suspect complementary distribution, only one thing remains — we need the analytic strategies to determine whether sounds are in complementary distribution or not, the topic of the remainder of this chapter. We suspect complementary distribution between voiced and voiceless stops in Nakoda, so we begin there.



Exploring my language

This is the first of a connected sequence of explorations — the next ones make use of the results you come up with here.

The first task is to find sounds that you think might be in complementary distribution in your language, trying out some of the suggestions in this section. See if you can find two different sets of sounds that you might test for complementary distribution. For each set of sounds you identify, check whether you are focusing on consonants or on vowels.

8.3 Complementary distribution in Nakoda

We dig deeper into the distribution of Nakoda stops in this section, continuing from where we left off at the end of Section 7.6 in Chapter 7. As before, we begin with a pair of sounds, [b] and [p], before expanding our search to include all pairs of voiced and voiceless stops.

To recap from Chapter 7, Section 7.6, the voiced labial stop [b] is found only between vowels; this is illustrated in the distribution table in (2). (If you need a refresher, look especially at Section 7.6.2 in Chapter 7.) (We continue to use our conventions. We use italics to show the *orthography* (when we can find one). Pronunciation is shown with *transcription* using IPA symbols placed between square brackets. When looking at an example like this, be sure to attend to the way words are pronounced, not the way they are spelled — phonological analysis has to do with pronunciation, not with writing systems.)

This is our starting point. It shows us that Nakoda [b] has limited distribution. Words like *sapsapa* [sapsaba] ‘black’ show us that Nakoda has both voiced [b] and voiceless [p], sounds that are identical except for voicing. Similar sounds, one with limited distribution? We are smart: We ask the question *are [b] and [p] in complementary distribution in Nakoda?*

8.3.1 The distribution of [b] and [p] Nakoda

We have seen that we can determine whether a sound has limited or regular distribution by using a distribution table to find the contexts where that sound is

Table 2: Nakoda [b] distribution (repeated from Table 15 in Chapter 7)

		[b]		
final	V _ #			
initial	# _ V			
	# _ C			
medial	V _ V	sába	[sába]	‘black’
	V _ C			
	C _ V			
	C _ C			

found. We start with a distribution table that shows the complementary contexts for consonants in great detail – with no gaps and no overlap. We test the distribution of our two consonants, [b] and [p], by having a column for each in our distribution table. The distribution table to test whether Nakoda [b] and [p] are in complementary distribution is given in Table 3. The acronym RDP in *sapsába* [sapsába] ‘black’ (RDP) stands for *REDUPLICATION*, a morphological pattern which includes repetition of some part of the base word.

Table 3: Nakoda [b]/[p] distribution

	[b]			[p]		
V _ #						
# _ V				bahá	[pahá]	‘hill’
# _ C				ptá	[ptá]	‘otter’
V _ V	sába	[sába]	‘black’			
V _ C				sapsába	[sapsába]	‘black’ (RDP)
C _ V				nqbíkpa	[nabíkpa]	‘mittens’
C _ C				kpamní	[kpámni]	‘serve’

We again use gray shading to show empty cells, highlighting the contexts where we found no words with the sound in that question.

Try this. Look closely at Table 3 to determine whether there are any contexts where neither [p] nor [b] is found.

In examining Table 3, we see that there are two contexts where neither [p] nor [b] is found – the top row and the bottom row. Every cell in these two rows has gray shading.

The top row is shaded gray because we found no words with a final labial stop, whether [b] or [p]. This makes sense in the larger picture of Nakoda. There is limited distribution of stops in word-final position — the only word-final Nakoda stop is dorsal [k] (Cumberland 2005: 70–71). So there can be neither [b] nor [p] at the end of a word.

The bottom row is shaded gray because we found no words with a labial stop anywhere in a three-consonant sequence. This too makes sense in the larger picture of Nakoda. CCC sequences are very rare in Nakoda. While it may be possible to have a CCC sequence with a [p] or a [b] in Nakoda, it is not surprising that we haven't found any words of that type (though examining even more words in the language may show that there are a few such words).

We repeat Table 3 in Table 4, omitting the first and last rows because neither [p] nor [b] appears in those contexts.

Table 4: Nakoda [b]/[p] distribution (final)

	[b]		[p]	
# _ V			<i>bahá</i>	[pahá] 'hill'
# _ C			<i>ptá</i>	[ptá] 'otter'
V _ V	<i>sába</i>	[sába] 'black'		
V _ C			<i>sapsába</i>	[sapsába] 'black' (RDP)
C _ V			<i>nabí̃kpa</i>	[nabí̃kpa] 'mittens'
			<i>kpámni</i>	[kpámni] 'serve'

At this point, we have completed the preparation necessary to understand the complementary distribution of [b] and [p] in Nakoda.



Exploring my language

Select one of the pairs of consonants or vowels that you identified as possibly being in complementary distribution in your language, so that you are focusing on two sounds that you think might be in complementary distribution with each other.

In order to determine whether sounds are in complementary distribution, you need a distribution table — for consonants if you are looking at

consonants and for vowels if you are looking for vowels. (If you are looking at consonants, find the distribution table you created when exploring your language in Chapter 7 (see p. 392 and p. 394). If you are looking at vowels, you will need to create a distribution table for vowels now.)

Once you've constructed your distribution table, find words that have the sounds you are targeting, and put them in the correct cells in your distribution table. When you have completed this step, you will be in position to test whether your sounds are in complementary distribution.

8.3.2 Complementary distribution of [b] and [p] in Nakoda

We are now going to focus on Table 4 to find whether there is a pattern of complementary distribution.

Try this. Look at Table 4 to see whether there are any contexts where both [b] and [p] are found. (Hint: A row with an example for [b] and an example for [p] shows a context where both [b] and [p] are found.)

A filled cell means there is at least one word with the targeted sound in the context shown by the row. A row with words in both columns indicates that the sound can occur in both contexts, so the distribution is not complementary. Whenever you construct a distribution table to check for complementary distribution, the key question is *are there any rows where there are words in both columns?*

Did you find any such rows in the Nakoda [b]/[p] table in Table 4? We hope you did not, because the answer is *no* — there are no rows with words in both columns. The absence of such rows shows that the contexts divide into two sets — those where we find [b] and those where we find [p]. For each context, we find one sound but not the other sound — we do not find both sounds in the same context. This is the essence of complementary distribution.

Along with inspecting rows, we can also look at the pattern of empty gray cells and filled white cells — do so now with Table 4. See how the pattern of gray and white cells makes a checkerboard? Each row has both white or gray cells, and each column has both white and gray cells. The exact checkerboard pattern showing complementary distribution will vary depending on how the rows are labeled, but you will see a checkerboard if there is complementary distribution. In Tables 5 and 6 we show two other ways of ordering the row labels that result

in different checkerboard patterns, to illustrate that the pattern can differ but the information is the same.

Table 5: Final Nakoda [b]/[p] distribution, rearranged

	[b]		[p]	
#_V			<i>bahá</i>	[pahá] ‘hill’
#_C			<i>ptá</i>	[ptá] ‘otter’
V_C			<i>sapsába</i>	[sapsába] ‘black’ (RDP)
C_V			<i>nabíkpa</i>	[nabíkpa] ‘mittens’
			<i>kpámni</i>	[kpámni] ‘serve’
V_V	<i>sába</i>	[sába] ‘black’		

Table 6: Simplified final Nakoda [b]/[p] distribution

	[b]		[p]	
V_V	<i>sába</i>	[sába] ‘black’		
#_V			<i>bahá</i>	[pahá] ‘hill’
#_C			<i>ptá</i>	[ptá] ‘otter’
V_C			<i>sapsába</i>	[sapsába] ‘black’ (RDP)
C_V			<i>nabíkpa</i>	[nabíkpa] ‘mittens’
			<i>kpámni</i>	[kpámni] ‘serve’

Visually, Table 4, Table 5, and Table 6 are quite different from each other. The cause not because of a conceptual difference — it is due solely to the order of the rows in each table, in particular where the row labeled V_V is located.

Through inspection of Table 4 (or Table 5 or Table 6), we have established that [b] and [p] are in complementary distribution in Nakoda. However, we have not made the pattern of distribution explicit yet. We need a complementary distribution statement.

When we examine Table 4, we see that [b] appears in only one row and so in only one context, between two vowels. This makes it easy to characterize the distribution of [b] and [p], because once we establish the location of [b] (between vowels), we can say that [p] occurs in all other contexts.

(1) Nakoda [b]/[p] complementary distribution statement

In Nakoda, labial stops [b] and [p] are in complementary distribution; the voiced stop [b] is found between vowels and the voiceless stop [p] is found in all other contexts.

In features: [labial, –continuant, +voice] / V ___ V, else [–voice]

This type of statement is sometimes written in shorthand, for example [b] / V ___ V, else [p]. This shorthand can be interpreted as *Use [b] when a labial stop is found between two vowels, otherwise use [p]*.

Making an explicit statement about complementary distribution can be useful for a variety of reasons. It gives something to share with others – in a classroom, grammar, or dictionary for example – to help clarify the rules of pronunciation.

It also gives us a statement that might encourage us to look more deeply at the language. For example, the Nakoda [b]/[p] complementary distribution statement refers to (labial) voiced stops and (labial) voiceless stops. This might make you wonder – is complementary distribution restricted to [b] ~ [p], or do other stops have the same pattern? We already know that [d] and [g] have the same distribution as [b] (see Chapter 7, Section 7.6). Now we know that [b] and [p] are in complementary distribution – we are on high alert for complementary distribution with [t] and [d], and with [k] and [g]. We look at this next. (If you are feeling very confident about what you would look for in Nakoda to test for complementary distribution of the other voiced/voiceless stop pairs, you might want to jump ahead to Section 8.4.)



Comprehension Check 3

Here are three schematic tables. Which one(s) – if any – show a pattern of complementary distribution. Give one sentence for each to explain why. Because these are schematic, we use *A* and *B* for our two sounds in each case, and we use *C1* and *C2* for the two contexts. Gray cells show that no words were found meeting the sound-context pair while white cells show that a word was found.

a.	A	B	b.	A	B	c.	A	B
C1			C1			C1		
C2			C2			C2		



Here's our answer

Only table (c) shows complementary distribution. Our explanations follow.

In (a), the column for sound A has no words — this shows that the sound A is not found in the language. There are words with B in the language. We conclude that B is in the [inventory](#), but A is not.

In (b), every cell is white, telling us that both sounds A and B are found in both contexts. In this case, A and B [contrast](#); we should be able to find [minimal pairs](#), [near-minimal pairs](#), or at least [context pairs](#).

In (c), we find sound A in context C1 but not in context C2. Conversely, we find B in context C2 but not in context C1. Visually we see a checkerboard of gray and white cells. This is the pattern that shows us that the sounds A and B are in [complementary distribution](#).



Exploring my language

Examine the table you created in response to the Language Exploration on p. 433. As you look at it, determine whether you have found a pattern of complementary distribution. (If you need help to understand your pattern, refer back to the explanations in the previous Comprehension Check.)

If you find complementary distribution, make an explicit statement, modeling your statement after (1).

If you have not found complementary distribution, state that the two sounds are not in complementary distribution and give the forms that show how you determined that they are likely to be contrastive sounds.

8.3.3 What about other stops in Nakoda?

Most examples of complementary distribution involve very similar sounds. The stops [p] and [b] are labial, as are aspirated [p^h] and glottalized [pʰ]. Using the strategies presented in this section and more words from Nakoda, you can show that these stops do not participate in the voiced/voiceless pattern of complementary distribution. Nor do the other aspirated and glottalized stops, [t^h], [k^h], [tʰ], and [kʰ]. That still leaves four stops to consider, [t] and [d], [k] and [g].

The sounds [b] and [p] are similar in that both are oral labial stops; they differ only in that [b] is voiced and [p] is voiceless. This observation leads very naturally to the question of other voiced/voiceless stop pairs in Nakoda, [t] and [d], [k] and [g]. Do other voiced and voiceless stops follow the same pattern?

In this section, we use distribution tables to explore whether there is complementary distribution with other, similar, pairs of sounds. We are using the general strategy of starting with one pair — [b] and [p] — and then moving on to other pairs — [d] and [t], [g] and [k] — rather than trying to tackle all stops in a language at the same time. You may find the small-to-large strategy useful in your own work or for explaining patterns to others.

What we need to do is find the other voiced/voiceless stop pairs and put them into a distribution table like the one in Table 4, to see whether the other pairs follow the same pattern as [b]~[p]. To do this, we use two copies of our distribution table for consonants in Nakoda, one to compare the distribution of [d] and [t], and the other to compare [g] and [k].

Try this. If you want to reassure yourself about your understanding, stop here and try constructing one or both of these tables yourself before reading further.

Our filled-in distribution tables follow, showing the distribution of [d] and [t] in Table 7, and of the [g] and [k] in (8). We include the word-final context for [k] in (8) because [k] is the only stop found in that position (Cumberland 2005: 70–71) — and we leave this row out in the other distribution tables. Look at Tables 7 and 8 carefully. Pay special attention to the contents of each row. Do you think

that voiced and voiceless stops are in complementary distribution in Nakoda? Why or why not?

Table 7: Nakoda [d]/[t] stop distribution

	[d]	[t]
#_V		<i>tópa</i> [tóba] ‘four’
#_C		<i>tká</i> [tká] ‘heavy’
V_V	<i>netám</i> [nedám] ‘over.here’	
V_C		<i>atkúku</i> [atkúgu] ‘her fathers’
C_V		<i>záptq</i> [záptq] ‘five’
		<i>ptq</i> [ptq] ‘otter’

Table 8: Nakoda [g]/[k] stop distribution

	[g]	[k]
#		<i>hík</i> [hík] ‘and’
#_V		<i>kíza</i> [kíza] ‘squeak’ (verb)
#_C		<i>kpámni</i> [kpámni] ‘serve’
V_V	<i>wakú</i> [wagú] ‘I covet it’	
V_C		<i>pakcá</i> [paktʃá] ‘comb’
C_V		<i>yatká</i> [jatká] ‘drank’
		<i>ská</i> [ská] ‘melt’

We hope that you said that voiced and voiceless stops are in complementary distribution in Nakoda. Looking closely at each row in Tables 7 and 8 shows that there are no rows with a word in each column. Instead, we see a checkerboard of shaded and white cells.

When we compare Tables 7 and 8 with Table 4, we see that in all three distribution tables, the voiced stop is found only in the row labeled V_V – the between-vowels context – while voiceless stops are found in all contexts except the V_V context. Voiced stops [g] and [d] are found only between vowels; voiceless stops [k] and [t] are found everywhere else. This is complementary distribution.

What remains is to modify our statement of complementarity in Nakoda that we gave in (1), to include all stops. This is an easy change to make – rather than referring to labials, we refer to all stops. The following is the text from (1) with

the references to labials struck out and in gray text. In Nakoda, *labial stops* ~~[b]~~ and ~~[p]~~ are in complementary distribution; the voiced stop ~~[b]~~ is found between vowels and the voiceless stop ~~[p]~~ is found in all other contexts. We can do the same thing with the shorthand statement from (1): ~~[labial, -continuant, +voice]~~ / V ____ V, else ~~[-voice]~~. We put these together in our revised statement in (2).

(2) Nakoda complementary distribution

In Nakoda, stops are in complementary distribution; the voiced stops are found between vowels and the voiceless stops are found in all other contexts.

In features: [-continuant, +voice] / V ____ V, else [-voice]

Notice that using natural classes ([–continuant, +voice], [–voice]) means that we don't have to list the six stops [b], [p], [d], [t], [g], and [k]. The features create coherent groups and help us see how there is a system to the way sounds are used in the language. It is not some random set of sounds that behaves differently between vowels; rather, it is the [–continuant] stops, and they are [+voice] when between vowels — a single group of sounds with a shared pattern.

Nakoda stops are voiced between vowels. If you look at a lot of languages, you will find that it is not unusual to find intervocalic voiced consonants. Voicing of the consonant occurs when the vocal folds continue to vibrate while the consonant is being pronounced. Since the vowels are voiced and vowels flank the consonant, the consonant is surrounded by voicing: It is not unusual to find that voicing continues from vowel to consonant to vowel without interruption. This kind of phonetic effect is known as [assimilation](#). (We also saw examples of assimilation in Chapter 7 Section 7.6.)



Comprehension Check 4

Suppose you have done some other work with Nakoda, and you remember that the language also has affricates, both [tʃ] and [dʒ]. Do words like *ece* [edʒé] 'only' and *wikcémní* [wiktʃémna] 'ten' make you suspect that [tʃ] and [dʒ] follow the same pattern of complementary distribution as we found with the stops? Explain your answer.



Here's our answer

We hope that these words do make you curious about whether the Nakoda voicing complementary distribution includes affricates as well as stops. The voiced [d͡ʒ] is between vowels in *ece* [ed͡ʒé] ‘only’ – exactly where we saw voiced stops. The voiceless [t͡ʃ] is not between vowels in *wikcémn̩i* [wik͡t͡ʃém̩na] ‘ten’, since there is a [k] next to it. Also, you may remember from Chapter 2 that affricates are classed as [–continuant], and so are very similar to stops (differing by the feature [±delayed release]).

In fact, further exploration of Nakoda words leads to the conclusion that the complementary distribution pattern extends to affricates as well as stops.

8.3.4 Summary

We started this section with two observations about Nakoda: (i) voiced stops have limited distribution, and (ii) the language has both voiced and voiceless stops. This raised a question – could voiced and voiceless stops be in complementary distribution?

To answer this question, we constructed distribution tables to compare the distribution of pairs of sounds, first with [b] and [p]. When we found that [b]~[p] are in complementary distribution, we went on to check [d] and [t], and [g] and [k]. Examining the words of the language laid out in distribution tables showed three things: (i) each row had a word in only one column, not both columns, showing us that each pair was in complementary distribution, (ii) the voiced member of each pair is used only between vowels, and (iii) the voiceless member of each pair is used in all other contexts. We concluded by making an explicit statement of the complementary distribution about voicing in the class [–continuant] stops in Nakoda.

Notice that we use the symbol ~ to connect two sounds in complementary distribution. This is another type of shorthand that some linguists use; we like to use ~ because you can tell at a glance which sounds are independent in a language and which are related to each other by complementary distribution: [b]~[p] means that [b] and [p] are in complementary distribution; [b] and [p] means simply that [b] and [p] are two sounds.



Minimal pairs in Highland Mayan Sign Language

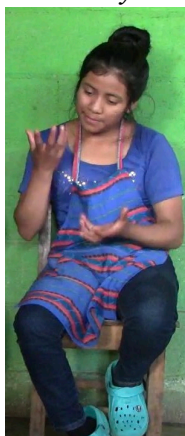
Sybil Vachaudéz and Samantha Prins

Minimal pairs allow us to identify the meaningful, contrastive units of a language. For spoken languages, these units are sounds. For Sign languages, these units are phonological parameters like **handshape**, **orientation**, **movement**, **location**, and **non-manual markers**. When two signs with different meanings share all but one parameter, those two signs form a **minimal pair**. We can then infer that the distinguishing parameter is contrastive in that language.

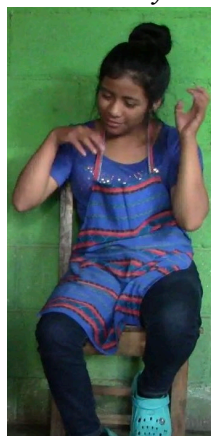
Let's look at some examples of minimal pairs in Highland Mayan Sign Language (HMSL, used by Maya communities in the Guatemalan highlands).

'Fire' and 'rain' in HMSL

Kaqchikel signer recorded by Samantha Prins and Sybil Vachaudéz



FIRE



RAIN

Comparing FIRE and RAIN, we can see that the two signs share the same handshape (curved, open palm), movement (the hands alternate up and down), and location (neutral space). Importantly, the two signs vary in orientation (palm-up vs. palm-down). This tells us that these orientations are meaningfully contrastive in HMSL.

For BICYCLE and MOTORCYCLE, we can see that the signs have the same handshape (O-handshape), location (neutral space), and orientation (downwards), but vary in movement (both hands spinning vs. both hands trembling and moving side-to-side). This tells us that these movements are also meaningfully contrastive in HMSL.

‘Bicycle’ and ‘motorcycle’ in HMSL

Kaqchikel signer recorded by Samantha Prins and Sybil Vachaudéz



BICYCLE



MOTORCYCLE

Identifying minimal pairs is a good first step towards understanding the phonemic inventory — the building blocks — of any language, including Sign languages. Understanding what these building blocks are (and how to combine them) is an essential part of language teaching, learning, and documentation.

8.4 Complementary distribution in Djeoromitxí

We return now to Djeoromitxí, a Jabutí language of the Brazilian state Rondônia, to examine the distribution of one of the several affricates in the language, the [affricate](#) $[\text{p}\phi]$. We again rely on De Castro (2012).

We begin this discussion with suspicions about the affricate $[\text{p}\phi]$: It is an unusual affricate in that it has no voiced counterpart, unlike the other affricates in Djeoromitxí — $[\text{ps}]$, $[\text{bz}]$, $[\text{tj}]$, $[\text{dʒ}]$. This is an example of having too few sounds.

Also, it is the only non-strident affricate — the only affricate with a soft release rather than a hissy **strident** release (as is found with the other Djeoromitxí affricates, [ps̠], [bz̠], [tʃ̠], [dʒ̠]). We consider [p̠] in opposition to [p] to test whether the source is correct in saying these two sounds are in complementary distribution. (For review on affricates see Chapter 2 Section 2.3.1.3. Chapter 7, Section 7.7, presents reasons to view Djeoromitxí affricates as single segments rather than as a sequence of two consonants. If you need a refresher on stridents, see Section 2.3.1.2 in the same chapter.)

This example raises an important point: Sometimes, we have to look at the phonetic properties of the sounds that make up the context in order to determine whether there is complementary distribution. The C or V classification is too broad. In this example, if we characterize the context simply with Cs and Vs (as we have done up to this point), we do not see a pattern of complementary distribution. However, if we characterize the context in terms of whether the following vowel is round or not, the complementary distribution pattern is revealed — [p̠] is used before round vowels and [p] is used everywhere else.

8.4.1 Construct a distribution table.

You may remember from Chapter 7 that consonants in Djeoromitxí can be word-initial and they can be between vowels. We begin with a distribution table with these two contexts, and fill in words as we find them. The result is in Table 9.

Table 9: Djeoromitxí [p]/[p̠] distribution table (problematic)

	[p]			[p̠]		
# ____	<i>paro</i>	[paro]	‘chicken’	<i>põ</i>	[p̠p̠õ]	‘grandson’
V ____ V	<i>kupa</i>	[kupa]	‘moon’	<i>kupü</i>	[kup̠p̠u]	‘woodland’

We see that with both contexts, there is a word in each column — this shows that both sounds are found in each context. We also see that for a given sound, there is a word in each row — this shows that each context allows a given sound. We conclude that these contexts do not differentiate between sounds. Either of these situations is sufficient to say that the sounds are not in complementary distribution. Our suspicions may be correct, but we don’t have the evidence yet.

Our findings suggest that the sounds [p] and [p̠] are contrastive after all. We have two options here — to look for **minimal pairs** and the like to show that these sounds are **contrastive**, or to dig deeper. A quick search of a Djeoromitxí word list would show no evidence of contrast, so we need to look further into

complementary contexts — so far, we have only looked at one way of defining the complementary contexts. It may be too early to conclude that the sounds must be contrastive.

The problem may be that we have configured the contexts in a way that does not reveal the complementary distribution of this pair of sounds. Our challenge is to discover other ways to configure the contexts to determine whether the two sounds are in complementary distribution.

At this point, it helps to look closely at the distribution we have seen for the two sounds. Importantly, both sounds can occur at the beginning of a word — so if there is a difference in context, it must be in the sound *following* the consonant. We get evidence to support this observation with the two consonants in the middle of a word. For example, the pair *kupa* [kupa] ‘moon’ and *kupü* [kup̥u] ‘woodland’ both begin with the sequence [ku], but in one word we find [p] and in the other we find [p̥]. In short, there is nothing to be gained by examining the preceding context because it can be the same for both [p] and [p̥]. We decide to focus on the following context.

8.4.2 Construct another distribution table!

Every consonant is followed by a vowel in Djeoromitxi. (We established this point in Chapter 7. See the discussion leading up to (20) in that chapter if you want to review.) However, we can’t create complementary contexts using just V for the following context because that context is the same for all consonants. To differentiate the following context, we need to differentiate the vowels from each other. The way to do this is by the phonological features of vowels. So our next step is to look at the vowels in Djeoromitxi and their features.

The Djeoromitxi vowels and their features are given in (3). We abbreviate the feature [±back] as [±bk] and the feature [±round] as [±rd] to make the table fit the page.



High and low, close and open

For consistency within this volume, we use the terms high, mid advanced, mid retracted, and low for different degrees of vowel height. De Castro uses different terms — close, close-mid, open-mid, and open. We point this

out in case you encounter a resource that uses different terms to describe the features of vowels. **Open** vowels have the tongue far from the roof of the mouth — in our terms, they are [+low]. **Close** vowels have the tongue close to the roof of the mouth — in our terms, they are [+high]. The **IPA** descriptions tend to use “close” and “open” terms; we use “high” and “low” because they are more common in the **Americanist transcription** tradition of language description.

(3) Djeoromitxi vowels

		[−nasal]			[+nasal]	
		[−bk]	central	[+bk]	[−bk]	[+bk]
		[−rd]	[−rd] [+rd]	[+rd]	[−rd]	[+rd]
[+high]	i		ʊ	u	ĩ	
mid [+ATR]			ə	o	ẽ	õ
mid [−ATR]	ɛ					
[+low]			a			

Examining (3) shows that vowels differ by height, nasality, the front-central-back dimension, and by rounding; the wavy line above a vowel, as in [õ] means that the vowel is nasal so air comes through the nose as well as through the vocal tract. The symbol [ʊ] indicates a high, round, central vowels. Any of these dimensions — or even a combination of these — might be used to create complementary classes of vowels. Once we have complementary classes of vowels, we can create complementary contexts to test for complementary distribution. How do we choose among the many possibilities?

Here’s where it can really be helpful to have a word list like those in (4) and (5), to read the words aloud as best you can. Focus on how your mouth feels as you read the words — focus especially on the consonants [p] and [p̥] and the vowel that immediately follows.

Try this. Read aloud the [p] words in (4) and the words with [p̥] in (5). The labials ([p] and [p̥]) and the immediately following vowel are in black text to help draw them to your attention. Focus on those sounds. What do you notice?

(4) Words with [p] in Djeoromitxí

- | | | | | |
|----|----------------|-------------|--------|-----------------------|
| a. | word-initial | <i>paro</i> | [paɾo] | ‘chicken’ |
| | | <i>piti</i> | [piti] | <i>cicada species</i> |
| b. | between vowels | <i>bupe</i> | [bupe] | ‘that’ |
| | | <i>kupa</i> | [kupa] | ‘moon’ |

(5) Djeoromitxí words with [p̞] (rounded)

- | | | | | |
|----|----------------|-------------|----------|---------------------|
| a. | word-initial | <i>pō</i> | [p̞ō] | ‘grandson’ |
| | | <i>pupu</i> | [p̞up̞u] | <i>frog species</i> |
| b. | between vowels | <i>kupü</i> | [kup̞u] | ‘woodland; forest’ |
| | | <i>hipo</i> | [hip̞o] | ‘animal’ |

What did you feel? You may have noticed that with [p̞], you rounded your lips with the following vowel, but this never happened with [p]. This observation would lead you to test whether a pattern can be discovered by using a distribution table using the complementary contexts ___[round] and ___[non-round].

Maybe you didn’t feel this; maybe you are still unsure enough of how to wrap your tongue around the sounds of some other language so you don’t trust your own observations. Don’t give up — there is another way to identify contexts to test — it takes more thinking, but is very logical.

Now that we have figured out that the following sound is critical, we can make a very detailed distribution table, with a column for [p] and a column for [p̞]. Each row shows the following vowel. We include a row for every vowel in Djeoromitxí (see (3)), even though we haven’t found words to fit every case yet.

If you squint at Table 10 so you don’t see the words, but just see the pattern of shaded and white cells, you will see the familiar checkerboard pattern that indicates complementary distribution. You will also see the section at the bottom of the table where we don’t have words; we set these aside momentarily but will return to them. Our next step is to look at the groups of vowels that follow each consonant to determine whether there is a pattern.

Let’s compare the vowels that follow [p] — [a], [i], and [ε] — with the vowels that follow [p̞] — [ō], [u], [ʌ], and [o]. In making our comparison, we use the dimensions identified in (3). Let’s go through these one by one — vowel height, backness, rounding, and nasality.

Could vowel height be a decider? No. Both sounds [p] and [p̞] are found in both height contexts, so vowel height does not decide between the two consonants. The evidence? [p] is followed by [+high] [i] (*piti* [piti] *cicada species*) and

Table 10: Distribution of [p] and [p̥] in Djeoromitxi

	[p]	[p̥]
__[a]	<i>paro</i> [paɾo] ‘chicken’	
__[i]	<i>piti</i> [piti] <i>cicada</i> <i>species</i>	
__[ɛ]	<i>bupe</i> [bupe] ‘that’	
__[õ]		<i>põ</i> [p̥õ] ‘grandson’
__[u]		<i>pupu</i> [p̥up̥u] <i>frog species</i>
__[ʊ]		<i>kupü</i> [kup̥ʊ] ‘woodland; forest’
__[o]		<i>hipo</i> [hip̥o] ‘animal’
__[ə]		
__[ĩ]		
__[ẽ]		

[p̥] is followed by [+high] [u] (*pupu* [p̥up̥u] *frog species*). Additionally, both are followed by mid vowels, [p] by mid [ɛ] in *bupe* [bupe] ‘that’ and [p̥] by mid [o] in *hipo* [hip̥o] ‘animal’. The vowel height distinction does not align with the [p] and [p̥] pronunciations.

Could the front/central/back dimension make a difference? No. Both sounds [p] and [p̥] are found preceding central vowels. The evidence? In *kupa* [kupa] ‘moon’, [p] precedes central [a] while [p̥] precedes central [ʊ] in *kupü* [kup̥ʊ] ‘woodland; forest’. The front/central/back distinction does not align with the [p] and [p̥] pronunciations.

Could nasality differentiate? No. The same sound [p̥] is found before both a nasal vowel and an oral vowel. The evidence? In the word *põ* [p̥õ] ‘grandson’, [p̥] is followed by both a nasal vowel [õ] while in *hipo* [hip̥o] ‘animal’, [p̥] is followed by an oral vowel [o]. The oral/nasal dimension does not align with the [p] and [p̥] pronunciations.

We are left with [+round] vs. [−round] — the same property that you may have noticed when reading the words out loud. In Table 10, the vowels that follow [p] are [a], [i], and [ɛ], all of which are [−round]. The vowels that follow [p̥] are [õ], [u], [ʊ], and [o], all of which are [+round]. We have found a dimension, [±round], which aligns with the [p] and [p̥] pronunciations.

Try this. Try making a distribution statement for these two Djeoromitxí labials, [p] and [p̥], before looking at our statement in (6).

Our statement of the [p]~[p̥] complementary distribution is found in (6).

(6) Djeoromitxí [p]~[p̥] complementary distribution (tentative)

In Djeoromitxí, stop [p] and affricate [p̥] are in complementary distribution; the stop [p] is found only before [−round] vowels and the affricate [p̥] is found only before [+round] vowels.

We call our statement “tentative” because of the final three rows in Table 10, where we have yet to find words to fill in the cells. Given our tentative statement in (6), we expect to find [p] before each of these vowels, because each is a [−round] vowel.



Comprehension Check 5

We have concluded that the consonants [p] and [p̥] are in complementary distribution in Djeoromitxí words as follows.

- i. [p̥] is found before [+round] vowels [o], [õ], [u], [ʊ]
- ii. [p] is found before [−round] vowels [i], [ε], [a], [ə], [ε], [ĩ])

Test your understanding by choosing [p] or [p̥] in these words.

- a. _____ [ihõ__ĩ] ‘warrior’
- b. _____ [__ɛdʒʊ] ‘elder’
- c. _____ [__əri] ‘flea’
- d. _____ [ha__õ] ‘left’ (opposite of right)



Here's our answer

Here are the labial sounds we chose for each word, along with a short explanation.

- a. [p] is found in [ihõ__ĩ] [ihõpĩ] 'warrior'
- b. [p] is found in [__ɛdʒu] [pɛdʒu] 'elder'
- c. [p] is found in [__əri] [pəri] 'flea'
- d. [p̸] is found in [ha__õ] [hap̸õ] 'left (opposite to right)'

- a. The following vowel is [ĩ], a [–round] vowel. The sound [p] is used before [–round] vowels (rule ii).
- b. The following vowel is [ɛ] a [–round] vowel. The sound [p] is used before [–round] vowels (rule ii).
- c. The following vowel is [ə] a [–round] vowel. The sound [p] is used before [–round] vowels (rule ii).
- d. The following vowel is [õ] a [+round] vowel. The sound [p̸] is used before [+round] vowels (rule i).

8.4.3 Conclusion

The Djeoromitxí example illustrates that the context for complementary distribution may require an understanding of the distinctive features of the surrounding sounds. We then needed a strategy for identifying what features to examine for complementary distribution. One strategy is to simply say words aloud, paying close attention to the sound and its surrounding sounds. The second strategy is to logically identify specific contexts for each sound and test each one. Sometimes fine-tuning the contexts for our sounds reveals complementary distribution, as it did with Djeoromitxí.

In the remainder of this chapter, we use these strategies to explore some variations on complementary distribution that you might encounter in your language. You may wish to jump ahead to the final discussion in Section 8.7, and return to these examples later.

8.5 Complementary distribution is not necessarily pairwise: Nahuat Pipil

We turn now to Nahuat Pipil, already discussed in Chapter 4, Section 4.4.2. We again use Lemus (1996) for information about this Uto-Aztecan language, spoken in some parts of El Salvador. We do not include an orthography in this section because Lemus did not include one in his dissertation.

8.5.1 The four dorsal obstruents consonants in Nahuat Pipil

Nahuat Pipil has voiceless stops at three places of articulation, labial [p], coronal [t], dorsal [k]. With the labial [p] and coronal [t] stops, the voiced counterparts [b] and [d] are found only in borrowings from Spanish (such as [hwabor] ‘favor’ from Spanish *favor*); there are no other counterparts for [p] and [t]: There are no glottalized stops like [pʰ] or [tʰ] and there are no voiced fricatives, no [β] and no [ð].

With dorsals the story is different: Dorsals show a surprising range of variety – plain voiceless [k], plain voiced [g], glottalized [kʰ], and voiced fricative [ɣ], all found in native Nahuat Pipil words. This is an asymmetric system, illustrated in the partial consonant inventory in (7), where the parenthesized sounds [b] and [d] are restricted to words borrowed from Spanish.

(7) Nahuat Pipil partial consonant inventory

	labial	coronal	dorsal
voiceless stop	[p]	[t]	[k]
voiced stop	([b])	([d])	[g]
voiced fricative			[ɣ]
glottalized			[kʰ]

This asymmetry – with too many dorsal sounds – leads us to ask *are all of the dorsals contrastive, or is there some complementary distribution?* (Remember that asymmetry in the sound inventory is often a clue for complementary distribution. See Section 8.2 for other clues.)

To determine whether any or all of these dorsal consonants are in complementary distribution, we need to know the general distribution of consonants in Nahuat Pipil.

At the beginning of a word, we find at most one consonant, #___V, illustrated in [gal] ‘house’ (CVC) with an initial [g]. At the end of a word we find at most one consonant, V___#, seen in [jekʰ] ‘fine’ (CVC), with a final [kʰ]. (If you are

thinking that Nahuat Pipil has CV(C) syllables, you are right – but this won’t be important in our discussion.)

Between vowels, we find one or two consonants. One consonant between vowels is found in [náɣatʔ] ‘meat’ (CVCVC), with [ɣ] in the context V__V. Two consonants are found between vowels in [máktukʔ] ‘dry’ (CVCCVC), with [k] in the context V__C, and in [iɰkaʔ] ‘to roast’ (VCCV), with [k] in the context C__V. Another example of this last type is [tsúŋgal] ‘hair’ (CVCCVC), with [g] in the context C__V. There are no medial sequences of three or more consonants.

Our next step is to make a distribution table for Nahuat Pipil consonants.

Try this. If you want to test yourself, try constructing a generic distribution table for Nahuat Pipil consonants, using the five contexts we’ve just identified to label the rows. The column will be headed with C. We give our empty distribution table for Nahuat Pipil consonants in Table 11, so finish yours before you continue reading.

Let’s now use the basic distribution table in Table 11 to construct a distribution table for the four dorsal consonants in Nahuat Pipil.

Table 11: Empty generic distribution table for Nahuat Pipil consonants

	C
V__#	
#__V	
V__V	
C__V	
V__C	

To make our distribution table, we keep the same row labels, but we construct columns for each of the dorsal consonants, plain voiceless [k], plain voiced [g], glottalized [kʔ], and voiced fricative [ɣ], as is done in Table 12.

You may have noticed that the words used to illustrate the different contexts for consonants (leading up to Table 11) each have a dorsal consonant in them. We use those words to fill in our distribution table for Nahuat Pipil dorsals, again using gray shading to show where cells are empty because no relevant words could be found. (We are showing only one word in each cell. Searching through Lemus 1996 shows many words of each type.)

This distribution table is a little more interesting than the tables that we have seen earlier. Let’s unpack it to form an understanding of the distribution of dorsals in Nahuat Pipil.

Table 12: Empty distribution table for Nahuat Pipil dorsal consonants

	[k]	[g]	[kʔ]	[ɣ]
V ___ #				
# ___ V				
V ___ V				
C ___ V				
V ___ C				

Table 13: Nahuat Pipil dorsal distribution table

	[k]	[g]	[kʔ]	[ɣ]
a. V ___ #			[jekʔ]	
b. # ___ V		[gal]		
c. V ___ V				[náɣatʔ]
d. V ___ C	[máktukʔ]			
e. C ___ V	[ijkaʔ]	[tsúŋgal]		

8.5.2 Interpreting the Nahuat Pipil distribution table

There are several interesting properties of the distribution table in Table 13.

- Overall, there are lots of cells shaded gray. In fact, each row except the last one has only one word in it. Each of these contexts is paired with exactly one dorsal consonant.
- As for rows, most rows have only one word, because we found only one dorsal consonant in the context shown by the row.
- The last row has two words in it, in separate columns. (That tells us that two different dorsals are possible following a consonant and before a vowel, C ___ V.)
- As for columns, two columns have only one word in them, showing that there is only one context for the two sounds ([kʔ] and [ɣ]).
- On the other hand, two columns have two words in them, showing that the sounds [k] and [g] are each found in two contexts.

It looks like we have a combination of complementary and contrastive distribution. Let's understand this, starting with the easy cases, where the distribution is clearly complementary.

The top row, Table 13a, shows dorsals in word-final position. Only the glottalized dorsal [kʔ] is found in this position and the column shows that this is the only place the glottalized dorsal is found. This is generally true in Nahuat Pipil; more such words are shown in (8a).

The second row, Table 13b, shows that only [g] is possible at the beginning of a word, and this is also generally true. Examples are in (8b).

(8) Nahuatl Pipil [kʔ] and [g]

- | | | | | | |
|----|-----------|---------------|----|-----------|------------|
| a. | [kʔ]/# | | b. | [g]/# | |
| | [tikʔ] | ‘in’ | | [ginéyiʔ] | ‘he wants’ |
| | [súhjakʔ] | ‘burnt smell’ | | [gal] | ‘house’ |

The third row Table 13c shows that only the voiced dorsal fricative [ɣ] is found between vowels, and the column shows that this is the only location for [ɣ]. More examples are in (9a).

The fourth row, Table 13d, shows us that only [k] can appear before another consonant. We see a couple examples in (9b).

(9) Nahuatl Pipil [ɣ] and [k]

- | | | | | | |
|----|-----------|------------|----|------------|-----------------|
| a. | [ɣ]/V_V | | b. | [k]/V_C | |
| | [páyɪʔ] | ‘to laugh’ | | [íkʃiʔ] | ‘foot’ |
| | [ginéyiʔ] | ‘he wants’ | | [niktʃiaʔ] | ‘I wait for it’ |

Let’s turn now to the last row, Table 13e. Here, we find both [g] and [k] can follow another consonant. This does not look like complementary distribution. Did we make a mistake? Do [k] and [g] contrast in this context? Have we missed a nuance of the contexts? With the high degree of complementary distribution among the dorsals in Nahuatl Pipil, we are suspicious of the apparently contrastive distribution of [k] and [g]. This observation may get you thinking that we ought to check the types of consonants that precede [g] and [k].

This might be a good time to try reading the words aloud, so see whether there is something about the consonant sequences that feels different in your mouth.

Try this. Try saying the words aloud again — notice that [ŋg] feels very different in the mouth than [ŋk].



An analytic question

Have you figured out why we won't be looking at the sound following the velar to determine whether the context affects whether we hear [g] or [k]? Think about it for a minute before reading our explanation, then read on to see whether you figured it out.

It's because we know that both [g] or [k] are found before the vowel [a], from Table 9. This shows that the following context can be the same. The only thing left to look at is the preceding consonant.

It may be that phonological features of the preceding sound are relevant. We look at a few more words, those with [k] in (10a) and those with [g] in (10b). The entire CC sequence is shown in black text, to draw attention to it.

(10) Nahuatl Pipil [k] and [g] after another consonant

a.	[k]/C_V	b.	[g]/C_V
	[iʃkaʔ] 'to roast'		[tsúŋgal] 'hair'
	[metskújuʔ] 'leg'		[tsingíftiʔ] 'to get near'

When we examine these words, where the second consonant in a CC sequence is a dorsal, we see that [–voice] [k] follows a [–voice, –nasal], consonant [ʃ] and [ts] in these examples. In contrast, [+voice] [g] follows a [+voice, +nasal] consonant, [ŋ]. A complementary distribution table for the post-consonantal dorsals shows that these are complementary contexts.

Try this. Try making the detailed distribution table for Nahuatl Pipil [k] and [g]. What will you use for the contexts? (If you need a hint, look at (10) again and re-read the two preceding paragraphs.)

In Table 14, we show how we modified the post-consonantal row of our distribution table to show the two types of consonants that precede the dorsal – [+voice, +nasal] and [–voice, –nasal]. Filling in the distribution table cells shows the familiar complementary distribution checkerboard.

Let's make a statement of the complementary distribution of dorsals in Nahuatl Pipil. It is fairly complex. Try yours before reading on; ours is given in (11).²

²We propose here that [g] follows voiced consonants. We could also say that it follows nasals.

Table 14: Post-consonantal dorsals in Nahuat Pipil

	[k]	[g]
[voiceless oral] __	[iʃka ^ʔ] 'to roast' [metskúju ^ʔ] 'leg'	
[voiced nasal] __		[tsingíʃti ^ʔ] 'to get near' [tsúŋgal] 'hair'

- (11) Complementary distribution of dorsals in Nahuat Pipil
- Between vowels, we find [ɣ].
 - At the start of a word, we find [g].
 - At the end of a word, we find [k^ʔ].
 - After voiced consonants, we find [g].
 - Everywhere else, we find [k]. (This amounts to before another consonant and after a voiceless consonant.)



Comprehension Check 6

Look at the vowels in these Nahuat Pipil words.

[páɣi ^ʔ]	'to laugh'
[ginéɣi ^ʔ]	'he wants'
[metskúju ^ʔ]	'leg'
[niktʃíia ^ʔ]	'I wait for it'
[gal]	'house'
[súhjak ^ʔ]	'burnt smell'

Do you think glottalized vowels could be in complementary distribution with plain vowels in Nahuat Pipil? Why or why not?

By saying it follows voiced consonants, we imply that this is a case of voice assimilation, since [g] is a voiced consonant.



Here's our answer

We notice that in these Nahuat Pipil words, every word-final vowel is glottalized and only plain vowels are found in the middle of words. This definitely suggests complementary distribution. If you are puzzled about this, make a distribution table and check it out.

The Nahuat Pipil example shows a new side to complementary distribution — four distinct sounds in complementary distribution with each other. We see that complementarity is not necessarily pairwise. This example also emphasizes that sometimes we need to extend the basic contexts (which involve consonants, vowels, and word edges) with more detailed contexts involving specific phonological features of adjacent sounds.

Despite the different types of complementarity, our analytic strategies remain the same, systematically examining the distribution of sounds in the language to be sure that all contexts are considered, testing those contexts logically using distribution tables, and refining the distribution tables when necessary to determine the full range of complementarity — or of contrast.



Exploring my language

Continue to use the results from your previous exploration into your language.

Examine your filled-in distribution table. Do you have complementary distribution, contrastive distribution, or some of each? If the distribution is not obviously complementary, are there other contexts to try to make sure you aren't missing something?

Make a note of what you find, so it is ready for the last of this series of questions.

8.6 Languages are not perfect: Partial complementarity

A further type of complementarity is where sounds or sound classes are almost in complementary distribution, but there are a few words where one of the sounds appears in the context where the other is expected: There is only partial complementarity. The analytic methodology used in this section is familiar, but the results are a little ambiguous because the complementary distribution, though general, is not absolute. It would be nice if languages were neat and tidy all the time, but they are not. We do our best to recognize and respect this.

8.6.1 [tʃ] and [k] are almost complementary in Tapiete

Tapiete is a Tupian language spoken in parts of Argentina, Bolivia, and Paraguay; González (2005) worked with a community in Northern Argentina.³ Part of González' work was to create a lexicon, a word list with definitions in English and Spanish of Tapiete words. In this lexicon, there are words that begin with [tʃ] and others that begin with [k], the two [dorsal] consonants – [+back] [k] is further back than [–back] [tʃ].

Interestingly, in the [tʃ] category, almost every word listed begins with one of only two sequences, [tʃe] or [tʃi]. Conversely, in the [k] category, there are a lot of words, but very very few of the words begin with [ki] or [ke]. This is suggestive. It may have inspired González to check the distribution of the two consonants for general complementary distribution.

González found that in Tapiete the [–back] [tʃ] has a reduced distribution, appearing primarily before the vowels [i] and [e] – the [–back] vowels. Conversely, except for a few words, [k] is found everywhere *except* before [i] and [e] – in other words, the [+back] [k] is found before [+back] vowels. The consonant [tʃ] is more front in the mouth and appears before front vowels; [k] is more back and appears before back vowels. This looks like assimilation between the consonant and the following vowel.

González (2005: 58–59) discusses the complementary distribution of [k] and [tʃ], and states that “...the palatal voiceless affricate [tʃ]... is never found before the non-front vowels [i], [o] and [u].” However, “never” is too strong – González points out one minimal pair with [k] and [tʃ], showing that the two sounds make a meaningful contrast: *yocha* [ʒotʃa] ‘to stop’ and *yoka* [ʒoka] ‘to break’.

When we check González' lexicon for words with [tʃ] and with [k], we find that both consonants are found before any of the vowels, although the numbers

³We thank Hebe González for communicating with us about Tapiete.

of words in each cell is quite different. In Table 15, gray shading is used to indicate that there are very few words for the cell.

We have split this distribution table so it fits better on the page, putting words with [tʃ] first, then words with [k]. Note that the top two rows in the [tʃ] section have lots of words, and the last four rows of the [k] section have lots of words – these are complementary contexts. (The word [tʃinini] ‘quiet’ is an alternative pronunciation for [[tʃinini]. In the source, *apitʃuri man’s name* is not given in IPA; the description implies that this is an appropriate phonetic rendering of the name.)

González notes that many of the words that do not follow the general [tʃ]~[k] pattern are borrowed from Spanish, such as *chocolate* [tʃokoláte] ‘chocolate’ and *kancha* [kántʃa] ‘playground’ (with [tʃ] before a non-front vowel, [o] and [a]); *shikireta* [ʃikireta] ‘bicycle’ and *kasike* [kasike] ‘cacique’ (with [k] before the front vowels [i] and [e]). However, these contradictory sequences are not limited to borrowed words: *kina* [kína] *kind of plant* and *Kiñi* [kiɲi] *female name* have [k] before front [i], and *ichomí* [itʃomí] *kind of fish* and *yochapiwa* [dʒotʃapɪa] ‘mosquito’ have [tʃ] before non-front vowels, [o] and [a] – these are not Spanish borrowings. (González also suggests that [k] is used before front vowels if there is a preceding back vowel, as in *sapuke* [sapuke] ‘to shout’, with back [u] preceding the [ke] sequence. The González lexicon has 12 words with [k] or [tʃ] between a back vowel and a front vowel, six with [k] (like [sapuke]) and six with [tʃ] (for example *oche* [otʃe] ‘s/he sleeps’), so this doesn’t seem to be a general pattern.)

We see that [k] and [tʃ] are not completely complementary, but they are not completely contrastive either. We say that there is *partial complementary distribution* of [k] and [tʃ] in Tapiete.

The Tapiete example highlights another very important point. When you read a description of a language, it is important to pay attention not just to the summaries (like González’ statement that [tʃ] and [k] are in complementary distribution), but also to read the descriptions and to attend to what the language itself is telling you.

And now you’re ready for the next Comprehension Check, to create a statement about the distribution [k] and [tʃ] in Tapiete.

Table 15: [k] and [tʃ] in Tapiete

		[tʃ]		
front vowels	___[i]	<i>chinchí</i>	[tʃintʃi]	'(s)he cuts'
	___[e]	<i>chense</i>	[tʃense]	'(s)he closes'
non-front vowels	___[ɨ]	<i>chinini</i>	[tʃinini]	'quiet'
	___[a]	<i>Chakara</i>	[tʃakára]	<i>man's name</i>
	___[u]	<i>Apichuri</i>	[apitʃuri]	<i>man's name</i>
	___[o]	<i>chokolate</i>	[tʃokoláte]	'chocolate' (Sp.)
		[k]		
front vowels	___[i]	<i>kina</i>	[kína]	<i>kind of plant</i>
	___[e]	<i>sapuke</i>	[sapuke]	'to shout'
non-front vowels	___[ɨ]	<i>kí</i>	[kix]	'louse'
	___[a]	<i>kawi</i>	[kawi]	'well'
	___[u]	<i>kutu</i>	[kutu]	'(s)he stabs'
	___[o]	<i>hekowe</i>	[hekoʔe]	'his/her life'



Comprehension Check 7

Try making the statement of the partial complementary distribution for Tapiete [tʃ] and [k].



Here's our answer

In Tapiete, [tʃ] and [k] are in partial complementary distribution, with [tʃ] typically appearing before front vowels and [k] typically appearing before back vowels. There are a few exceptions, both in original words and in borrowed words.

8.6.2 Near complementarity in Filomeno Mata Totonac

Our last example illustrates just-about-completely-complementary sounds, from a Totonacan language Filomeno Mata Totonac, spoken in the state of Veracruz in Eastern Mexico. There is a general pattern in the language with a small number of exceptions according to McFarland (2009).⁴ The pattern is found with high vowels [i], [u] and mid vowels [e], [o]. We do not go through the full analysis of Filomeno Mata Totonac vowel distribution, but instead jump to McFarland's conclusion: In general, there is always a uvular consonant on one side or the other of a mid vowel. For example, [e] precedes the uvular [g] in [tʃeqɬ] 'puddle', and [o] follows the uvular [q] in [sqon] 'it glows'.

Finding non-high vowels next to uvular consonants is fairly common in the world's languages. It is a type of assimilation of phonetic properties. Uvular consonants are further back in the mouth than dorsals, with the back of the tongue pulled to the uvula. Because the tongue is rather like a water balloon – expansion in one part is coupled with compression in another part – pulling the tongue far back in the mouth makes it hard to also raise the tongue. A consequence is that it is very common to find non-high vowels next to uvular sounds.

In Filomeno Mata Totonac, if there is no uvular consonant in sight, there are no mid vowels either – with very few exceptions. In the word [toxóx] 'backpack', mid [o] is found next to dorsal [x], not next to some uvular. McFarland points out that high vowels are typically found next to [x], as we see in words such as [spuxúx] 'straight' and [yux] 'it falls from a height'. In the word [ɬe:n] (the common greeting), mid [e], not the expected high vowel [i], is found next to two coronals, lateral fricative [ɬ] and nasal [n]. The typical pattern is found in [taqaɬín] 'domestic animal', where the vowel [i] follows [ɬ] and precedes [n], exactly the location of [e:] in [ɬe:n].⁵

We state the complementarity of non-low vowels, noting the exceptions.

- (12) Filomeno Mata Totonac complementary distribution of non-low vowels
 In Filomeno Mata Totonac, mid vowels are found next to uvular consonants and high vowels are found when there is no adjacent uvular consonant. (*Exceptions*: [toxóx] 'backpack', [ɬe:n] (common greeting), ...)

⁴We thank Teresa McFarland for checking our discussion of Filomena Mata Totonac.

⁵We thank Teresa McFarland for bringing the second example, [ɬe:n], to our attention. She suspects there are "a handful more of such words".



Comprehension Check 8

Try making the distribution table for Filomeno Mata Totonac mid and high vowels.



Here's our answer

	uvular	uvular	other consonants
high vowels			[spuxúx]
mid vowels	[sqon]	[tʃeqɬ]	

8.6.3 Summary

Near-complementarity is not rare. There is a continuum between contrastive distribution and complementary distribution, rather than just the two extremes. Contrastive distribution is illustrated in Chapter 5; complementary distribution is the focus of this chapter. The Tapiete and Filomeno Mata Totonac examples show partial complementarity, illustrating points along the continuum between fully contrastive and fully complementary.



Comprehension Check 9

Chuxnabán Mixe is a Mixe-Zoquean language with labial, coronal, and dorsal nasals. Some of these are voiced and some are voiceless. (Recall that a little circle beneath an IPA symbol indicates voicelessness, so [ᵐ̥] is a voiceless labial nasal.) Below we show words with labial nasals, [ᵐ] and [ᵐ̥]. All items are from Jany (2011).

Chuxnabán Mixe nasals

voiced

[mits]

'cat'

[paʔam]

'illness'

[ma'mokʂ]

'fifteen'

voiceless

[poʂm̥]

'spider'

[poʔt̪m̥]

'summer'

[at̪m̥]

'custard apple'

Are the voiced and voiceless [m] and [m̥] in complementary distribution?

In answering this question, make a consonant distribution table given the location of consonants in these words. Fill in the cells and look for patterns.

Tip: Many people are unsure about how to say a voiceless nasal (it's like whispering the sound), which makes saying these Chuxnabán Mixe words aloud difficult. It's a good idea to try it anyway — when a sound is hard to say, the effort helps draw our attention to these sounds.



Here's our answer

In Chuxnabán Mixe, the difficulty in saying voiceless nasals helps us identify two things about the voiceless [m̥]: (i) it only occurs towards the very end of a word and (ii) it always follows a voiceless consonant — either at the end of the word or before another voiceless consonant. When next to a vowel, we find only voiced [m], which is very easy to say.

Here's a distribution table that demonstrates the complementary distribution of [m] and [m̥] in Chuxnabán Mixe — note the gaps. (We abbreviate [±voice] as [±vce] in the distribution table.)

	<u> </u> V	V <u> </u>	[-vce]__#	[-vce]__[-vce]
[+vce]	[mits]	[paʔam]		
	[ma'mokʂ]	[ma'mokʂ]		
[-vce]			[poʂm̥]	[at̪m̥] [poʔt̪m̥]

The gaps show the complementarity of voiced [m] and voiceless [m̥]. Voiced [m] is found adjacent to a vowel; voiceless [m̥] is found at the end of a word after a voiceless consonant or it is found between voiceless con-

sonants. If you were to find words with the other nasals, [ɲ], [ŋ], [ɳ], [ɰ], you would find that all voiced and voiceless nasals follow this pattern. We can make a concluding statement: In Chuxnabán Mixe, nasal consonants are voiced when next to vowels and voiceless when there is no adjacent vowel.

This example gives more practice with different environments for sounds that are in complementary distribution. It also presents an example of voiced/voiceless sounds that are not oral stops. You may recall from Chapter 2 that nasals are of the class of sounds called **sonorants**, a class which also includes vowels, glides, and liquids. Sonorants are usually voiced; when voiceless sonorants are identified, look for complementary distribution.



Comprehension Check 10

If you want a little more practice analyzing complementary distribution, try this exercise.

The words in the distribution table below are from Zenzontepec Chatino, an Otomanguean language that we explored in Chapter 3. This distribution table shows the distribution of the plain and palatal coronal oral stops [t] and [tʰ], nasal stops [n] and [nʲ], and laterals [l] and [lʲ].

Decide whether the distribution table shows that these sound pairs are in full complementary distribution, partial complementary distribution, or that they occur freely.

	initial		after [i]		after other vowels	
[t]	[tunũ]	‘large’			[káti]	‘seven’
[n]	[nĩ]	‘now’			[tunũ]	‘large’
[l]	[leta]	‘path’			[tēló]	‘skirt’
[tʰ]	[tʰēēʔ]	‘root of’	[kitʰē]	‘pine’	[ŋgatʰélē]	‘yanked’
[nʲ]	[nʲāfēʔ]	‘angry’	[tʰinʲi]	‘smoke’	[punʲō]	‘handful’
[lʲ]	[lʲoʔō]	‘chicken’	[liʲa]	‘straight’	[ndelʲā]	‘is watering’



Here's our answer

The quadrant with gray shading shows that the plain [t], [n], and [l] do not follow the vowel [i]. Otherwise, all six consonants can occur after the other vowels and all six are found at the beginning of a word. We conclude that these pairs are in *partial complementary distribution*, with plain coronal stops not found following the high front vowel [i].

Hang on to your answer to this Comprehension Check – we return to it in Chapter 9, in a Comprehension Check on p. 489.



Exploring my language

This question ends the series of connected questions about complementary distribution in your language.

Write up a clear statement about your sounds, whether you found complementary distribution, contrastive distribution, or some of each.

8.7 Discussion

By working through Chapter 7 and this chapter, you have learned what is meant by “complementary distribution” and you understand why two distinct sounds might not be perceived as distinct by speakers of a language.

8.7.1 Resolving the puzzle

You now know how to answer the puzzle posed at the beginning of the chapter.

Why would someone write different sounds with the same letter?

Sounds that are in complementary distribution with each other are sometimes not perceived as distinct from each other by speakers of that language. In this

case, it is common to find a writing system that uses the same letter to represent both of the sounds that are in complementary distribution with each other.

This is not always the case. In some instances, even though sounds are in complementary distribution, a community may choose to use symbols that are more representative of the way a word sounds. We saw this in Nakoda, which has at least two writing systems — some writers use the letter *p* for the sound [p] and *b* for the sound [b] as in *sapsaba* [sapsaba] ‘black’ while others use the letter *p* regardless of whether they hear [p] or [b] as in *sapsapa* [sapsaba] ‘black’.

This example highlights choices that communities must make when devising a writing system; it also highlights the importance of understanding what those decisions were — it would be easy to read *sapsapa* as *[sapsapa] rather than the correct [sapsaba].

8.7.2 Complementary distribution and reading linguistic writings

The concept of complementary distribution is integral to the way linguists approach the sound system of any language. Now that you have an understanding of this foundational concept, you are prepared to make sense of several concepts that are so basic in linguistics that many works that use these concepts don’t define them — they assume the reader knows what the terms mean. Now that you understand complementary distribution, you can understand these terms too. We address them before turning to other ways that knowledge of complementary distribution supports efforts by language workers.

Published work on a language can be an essential part of the resources available to many language workers. However, published works on languages are often written by linguists for other linguists rather than for language workers, and so are written with no effort to explain technical terms like *complementary distribution* (which you now understand.) or related concepts like *phone*, *phoneme*, *allophone*, *phonemic representation*, *surface representation*, and *underlying representation*. These concepts are so deeply entrenched in linguistic thought that many works never explain what is meant when these terms are used. Having a grasp of complementary distribution puts you in position to really understand these other concepts.

8.7.2.1 Phone

Phones are sounds used in language. Phones encompass both the **contrastive** sounds and the **non-contrastive** sounds in a language, as well as language sounds found in other languages. Ideally, in written work, square brackets are used with

phones: Labial [m], coronal [n], [–back] dorsal [ɲ], dorsal [ŋ], and uvular [ɴ] are some nasal phones; voiced stop [b], voiced fricative [β], nasal [m], voiceless stop [p] and voiceless affricate [pʰ] are all labial phones, and so on. Chapter 2 focused on phones, the sounds used in languages.⁶

8.7.2.2 Phonetic representation

A *phonetic representation* is a written form of a language meant to represent the sounds (or signs) that were used in some utterance. We use the *IPA* for phonetic representations.

A phonetic representation is written in phones, and typically appears in square brackets. However, not all sources adhere to the square bracket convention. Sometimes it is very difficult to figure out whether a form you see is phonetic or not.

Phonetic representations are not exact replications of utterances. Many details are omitted. Researchers presenting words in a phonetic representation choose which details to include. A reliable source will indicate which details are omitted.

8.7.2.3 Allophones

Allophones are the sounds of a specific language, not just a generic set of language sounds. (Phones are generic, for any language.) Like phones, allophones are written in square brackets: [b] and [p] are the plain labial allophones in Nakoda; Nahuat Pipil has two oral labial allophones, voiceless stop [p] and, in borrowed words, voiced stop [b]; Nahuat Pipil also has four oral dorsal allophones, voiceless stop [k], voiced stop [g], voiced fricative [ɣ], and glottalized [kʔ].

But this is not the end of the story: Individual or groups of allophones may correspond to individual *phonemes*. Read on.

8.7.2.4 Phoneme

Linguists use the term *phoneme* to refer to a somewhat abstract concept — the term phoneme is used for the *contrastive* phonological units in a language. Some of these contrastive units correspond to a single allophone — there are no alternative pronunciations. In Djeoromitxí, there are four contrasts among the five labial allophones: Nasal [m], voiced stop [b], voiced fricative [β] all contrast with each

⁶People who work with Sign languages have the same concept for the gestures used in Sign language.

other. Each of these three allophones corresponds to a single phoneme – the allophone [b] corresponds to the phoneme /b/, the allophone [m] corresponds to the phoneme /m/, the allophone [β] corresponds to the phoneme /β/. (Notice that phonemes are conventionally written between forward slashes: /b/ is a phoneme while [b] is an allophone or phone.) But what about the remaining two allophones, [p] and [p̸]?

We already know that [p] and [p̥] are in complementary distribution with each other; together they form a unit, a phoneme. It is the phoneme, the unit composed of [p] and [p̥], that contrasts with all of the other labials. We say, then, that [p] and [p̥] are *allophones of the same phoneme* – together they make up a contrastive unit that contrasts with other sounds in the language, and contrastive units are called phonemes.

Typically, a single symbol is chosen to represent the phoneme. Often, the more common allophone is chosen: Since [p] appears before non-round vowels in Djéromitxí and there are more non-round vowels than round vowels, it is likely that [p] is more common. Using this logic leads us to select /p/ as the phoneme that corresponds to [p] and [p̠].

These two types of representation, the abstract phoneme /p/ and the concrete allophones [p] and [p̞], leads to statements like */p/ is realized [p̞] before round vowels and is realized as [p] elsewhere*. A shorthand for this statement uses a rightward arrow → in place of “is realized as”:

$/p/ \rightarrow [p\phi] / ______ [V, +round], \text{ else } [p].$

Another alternative is to use a symbol that is different from both allophones. For the allophones [p] and [p̠], we might choose /P/. The /P/ is similar to both [p] and [p̠] (which helps with memory), but it is different from both. An advantage of this solution is that when you see a representation with /P/ in it, you are reminded that this is a sound with allophones, and so are less likely to pronounce it as [p] before a round vowel.

8.7.2.5 Phonemic representation

Using a single symbol for each phoneme means that forms can be written without all phonetic detail. For example, in Nahuatl Pipil, Lemus uses /K/ to represent the dorsal phoneme corresponding to the four allophones [k], [g], [ɣ], or [kʰ]. (Another researcher might choose /k/ or /g/ — usually a symbol is chosen that is mnemonically helpful.) If you know the Nahuatl Pipil pattern of dorsal complementary distribution and see a word with /K/ in it, you know whether to pronounce it as [k], [g], [ɣ], or [kʰ], simply by where it appears in the word —

because you know the pattern of complementary distribution. If you don't know the Nahuat Pipil pattern, the symbol /K/ might trigger you to look for a description of how to interpret the /K/.

A form that is written in terms of phonemes is in a *phonemic representation*. Phonemic representations, standardly written between forward slashes, show only the phonemes, that is, the contrastive sounds. A phonemic representation does not show the allophones of those sounds. It does not show the real pronunciation — the phonetic details of complementary distribution are missing in a phonemic representation, and other details may be missing too.

Recall the four dorsal allophones in Nahuat Pipil, [k], [g], [ɣ], or [kʰ]. The phonetic representation [tikʰ] (meaning 'in') includes an allophone of the phonemic /K/. The phonemic representation is /tiK/ if we use Lemus' convention. The glotalization of word-final /K/ is phonetic but not contrastive — that means the glotalization is not phonemic, so it is not included in the phonemic representation.

Try this. What would the phonemic representation be for Nahuat Pipil [gal] 'house'?

Don't read any further until you try to figure it out. Be sure you put your answer between forward slashes — /.../ — to show it is a phonemic representation.

The square brackets tell us that [gal] is the actual pronunciation for the word meaning 'house'. We know that dorsals are in complementary distribution, so the initial [g] can be represented phonemically as /K/, using Lemus' convention. This gives /Ka/ as the phonemic representation for [gal].

One way that phonetic representations and phonemic representations are used in language work is in developing a writing system. Some communities prefer a writing system which focus on a letter for each of the sounds (or phones) of the language; these are called phonetic writing systems. Another approach is to have letters for each of the phonemes, a phonemic writing system.

Try this. You may recall that [p] and [b] are allophones of /p/ in Nakoda, with [b] appearing between vowels and [p] being used elsewhere. You may also recall that there are two different writing systems that are used with Nakoda. In one, the word [sapsaba] 'black' is written *sapsapa* and in the other 'black' is written *sapsaba*. Which of these writing systems is phonemic and which is phonetic?

The phonemic writing system corresponds to phonemes, not allophones. Since the phoneme /p/ corresponds to the allophones [p] and [b], the writing system

that uses only *p* is the phonemic one: *sapsapa*. A writing system whose symbols correspond to the allophones, as happens in *sapsaba*, is a phonetic writing system.



Comprehension Check 11

In Chuxnabán Mixe, voiced and voiceless nasals are in complementary distribution. Which of the two practical orthographies below is phonetic (allophonic) and which is phonemic, and why?

	word	meaning	spelling
Orthography 1	[mits]	‘cat’	<i>mits</i>
	[paʔam]	‘illness’	<i>pa’am</i>
	[poʃm]	‘spider’	<i>poʃam</i>
	[poʔtmp]	‘summer’	<i>po’tmp</i>
Orthography 2	[mits]	‘cat’	<i>mits</i>
	[paʔam]	‘illness’	<i>pa’am</i>
	[poʃm]	‘spider’	<i>poʃaM</i>
	[poʔtmp]	‘summer’	<i>po’tMP</i>



Here’s our answer

The spelling system illustrated in Orthography 1 is phonemic and the one in Orthography 2 is phonetic. Orthography 1 uses *m* for labial nasal consonants, regardless of whether they are voiced or voiceless, recognizing their similarity and ignoring their differences. Orthography 2 uses *m* for the voiced labial nasal [m] and it uses *M* for the voiceless labial [ɱ], recognizing the differences as well as the similarities. Note that each system has value; it depends on each community to determine their own preferences. (The Chuxnabán Mixe community opted for a phonemic system in this regard, Janý 2010.)

8.7.2.6 Surface representation

The *surface representation* is another way of showing how a word is pronounced. In this way, the surface representation is similar to the phonetic representation. The difference comes in how a researcher determines which details to include in representations. With phonetic representations, a researcher can decide which details to include based on what the researcher thinks is important.

The information in a surface representation depends on the analysis that a researcher is using (or proposing). The analysis consists of underlying representations (see Section 8.7.2.7) and rules that add or remove properties from the underlying representation: The result of applying all the relevant rules is the surface representation. If there is no rule to provide some property, then that property is not shown in the surface representation.

8.7.2.7 Underlying representation

The concept of an *underlying representation* was introduced in the 1960s, when Generative Phonology was proposed (Chomsky & Halle 1968). Practically speaking, an underlying representation is a phonemic representation. Both are abstract representations, missing a lot of the phonetic detail of the language. The phonetic detail can be filled in if you know the rules of allophony — as well as the phonological patterns that accompany morphological interactions, the topic of Chapter 9. If you don't know the rules of allophony, then you don't know how to pronounce words just from the underlying (or phonemic) representation.

One challenge with published material is to determine whether representations in a source are written phonemically or phonetically. If forms are given without brackets or forward slashes, it can be very hard to tell what is going on unless you read very very closely — and even then it can be tricky.

If representations are phonemic, it means that complementary distribution patterns have been stripped away from the representations. It simplifies the writing of the language, but it does not give an accurate picture of the way that the language sounds because it is missing the allophonic variation introduced by complementary distribution. In general, the most useful sources give both phonemic and phonetic (allophonic) representations; ideally, each is indicated very clearly. Practically speaking, many authors use phonetic transcription only when directly discussing the phonology of a language, and revert to phonemic transcription when discussing *morphology* (word-building) and *syntax* (sentence structure).

We discuss underlying representations further in Chapter 9.

8.7.3 Using my hard-earned knowledge

By working through this chapter, you have acquired the skills needed to identify whether sounds are in complementary distribution — complete or partial — with each other. You have also learned how to describe the characteristics of the pattern with a fair degree of precision. You might even be able to explain the general concept of complementary distribution to another person, and maybe even the concepts of partial complementary distribution or of complementary distribution among several sounds. What else can you do with your new skills and knowledge?

Language learning. While the concept of complementary distribution may be more than a teacher wants to explain in the language classroom, and may not be something students want to learn about as a concept, a practical understanding of complementary distribution in a language has great benefits to teachers and learners alike.

Pronunciation tips and drills based on complementary distribution can help students learn words. Students are struggling to learn vocabulary items — what they mean, how to say them, how to use them in sentences. The job of learning pronunciation is simplified for students once patterns of complementary distribution are understood — what is learned can be applied to many words, not just one vocabulary item. Understanding the pattern speeds learning.

Pronunciation fluency is greater when learners internalize complementary distribution patterns of the target language. It is also increased when learners reduce interference of complementary distribution patterns in a language that they already know well. When preparing lesson plans, a teacher can drill students both on the complementary distributions of the target language and also on ways that the target language does not follow the familiar language's patterns. For example, English speakers aspirate [p], [t], and [k] before stressed vowels and at the beginnings of words; many languages do not do this. A learner whose most familiar language is English would benefit from drills involving unaspirated [p], [t], and [k] in those two contexts. Another example is that in Spanish, [β], [ð], and [ɣ] are used between vowels and [b], [d], and [g] are used elsewhere. A Spanish speaker learning a language with intervocalic voiced stops can benefit from practicing [b], [d], and [g] between different vowels.

Communication with fluent speakers is improved when the learner is able to use complementary distribution patterns appropriately. It can be difficult for fluent speakers to adjust to pronunciations that contradict the general patterns of their language. Improving fluency with complementary distribution patterns can improve comprehension and so make communication more effective.

Complementary sounds are *non-contrastive* with each other, so using one instead of the other (or one of the others) does not produce a different word, but you might sound a little funny or be confusing if you don't use the two sounds in their proper locations. For instance, if a Nahuat Pipil learner used [k] in the word [gal] 'house', a more fluent speaker might be confused — [g] should be used at the beginning of a word, not [k].

Language resources — dictionaries, grammars, orthographies. It is possible to create language resources with little or no recognition of patterns of complementary distribution. However, that can make resources difficult to create and unhelpful to users. One of the biggest questions is to determine how to represent sounds that are in regular complementary distribution. The choice of representation affects the way words are written, which affects the creation of dictionaries, teaching materials, and reading materials.

The essence of the choice that must be made when two sounds are in complementary distribution is whether to use one symbol or two to represent the sounds, essentially the puzzle for this chapter.

Developing a writing system involves numerous different considerations, especially whether the speakers of a language will be comfortable with using the proposed writing system (orthography).

There are a lot of decisions to make when designing a writing system for a language. Some people like to have smaller alphabets. When they find sounds in complementary distribution, they use a single symbol to represent all the allophones. For example, when devising a writing system for Nahuat Pipil (see Section 8.5), we might choose to represent the dorsal allophones with a single symbol, such as *k*, regardless of its actual pronunciation. That keeps the alphabet smaller. On the other hand, the community may prefer a more transparent relation between the writing system and the pronunciation and so will have a different symbol for each type of dorsal, using a phonetic writing system. The phonetic system is more transparent but has a much larger alphabet. Another consideration is typographical ease: Standard keyboards typically do not have a symbol for [ɣ]; trying to represent all four dorsal consonants in writing for Nahuat Pipil may tax creativity for the creators and typographical ease for users.

There are advantages and disadvantages to both approaches. Phonetic writing systems are generally easiest for novices to learn, but this is countered by the weight of learning and using novel symbols, especially when a pattern of complementary distribution is well understood. On the other hand, the “small alphabet” approach can make it more difficult for a novice to learn correct pronunciations because of the strong expectation that words “sound like they are

spelled”. A phonemic writing system may also make it easier to produce and share materials between dialects with different patterns of complementary distribution. In this case, the phonemic representations may be identical, but the allophonic (phonetic) representations are different. If a phonemic writing system is used, it may work well for both dialects.



Karapanã allophones and orthography

Christian DiRado-Owens

With the Karapanã practical orthography, allophones are represented in two different ways. (Karapanã is a Tukanoan language spoken in Columbia.)

Examples in DiRado-Owens (2020) show that the nasal allophones of oral consonants are represented with symbols that reflect their pronunciation (as in a. below) while only *h* is used to represent the voiceless glottal fricative, whether oral or nasal (as in b. below).

- a. *m* not *b* *kũkãmaĩnĩmĩ* ‘loves he who’
- b. *h* not *h̃* *mãhã* ‘beings/spirits/people’

“All oral consonants have nasal allophones when they occur in nasal morphemes... The voiceless glottal fricative /h/ is the only voiceless segment that has a nasal allophone [h̃]... I will use the graphemes of the practical orthography for the Karapanã examples instead of the phonetic transcription. Table 2.3 [not reproduced here — DA] shows the Karapanã phonemes and graphemes used by some community members to write the language. These phonemes and graphemes cannot yet be considered a part of a concerted community developed orthography, but rather serve the community as part of an informal practical orthography with similarities to Spanish for the time being.”

DiRado-Owens (2020: 25-28)

Dictionary organization relies on how a language is written; once decisions are made about orthography, alphabetizing is relatively straightforward. However, when novel sounds are created, the dictionary maker must decide where those sounds fit in the alphabetic sequence of the language. This is often done by or-

dering “modified letters” after the unmodified letter — giving sequences like *m*, *n*, *ñ* and *a*, *ä*, etc.

If a phonemic writing system is used, the most helpful dictionaries include discussion of how to pronounce the letters in different contexts.

Language documentation and description. Grammars and language descriptions usually include a section on the sounds of the language, including both contrastive and non-contrastive (complementary) sounds. Contrastive and complementary sounds form the basis of how all words in a language are pronounced. The best basic descriptions include phonemic inventories and minimal pairs to support the contrasts; they also include rules of allophony and examples of words with each allophone as evidence to support those rules. If you worked through this book up to this point (rather than jumping in at the middle), you have learned the tools needed to create this kind of basic description of the sounds of your language.

There is still more to the phonology of a language. Up to this point we have looked primarily at words, though in Chapter 3 we looked at **morphemes**, the subparts that combine together to make some words. When words are made this way, there is sometimes an interaction between the morphology and the phonology. The next chapter enables you to improve on your basic description by coupling phonological patterns and morphological patterns.

Language revitalization. In working to create a robust vocabulary for a language that has been sleeping or nearly asleep, language workers may choose to create new words, or **neologisms**, to fill lexical gaps (whether due to new inventions, like cell phones, or due to incomplete records of more common vocabulary). Neologisms that follow complementary distribution patterns in the language sound more like words of the language than do those that are created without regard to these patterns.

Reading linguist’s resources about your language. As discussed in Section 8.7.2 of this chapter, complementary distribution is the principle behind the concepts of phoneme and allophone as well as underlying representation and surface representation, concepts that are used in many linguistic analyses. When linguists write for other linguists, they often do not explain the reasoning that leads to statements about representations or phonemes. With your knowledge from this chapter, you are prepared to understand the hidden assumptions.

8.8 Conclusion

By making a statement of complementary distribution we clarify our thinking about the possible sound patterns in a language. If you are trying to figure out complementary distribution in your own language, you may be thinking “what if I’m wrong?”, a very natural response especially when you are in the early stages of figuring out the sound patterns of a language. With every new word that we encounter in the language, we can check whether the word is consistent with the pattern we think we have identified. As more and more words are encountered, we are increasingly confident in our statements — or we persuade ourselves that (some part of) a statement is wrong and needs to be revised or even discarded as a principle of our language.

Descriptions of the sounds of languages typically address the contrastive and non-contrastive sounds of a language. You now know that complementary distribution is at the heart of demonstrating that sounds are non-contrastive with respect to each other. You know that, to show a pattern of complementary distribution among a set of allophones, we need to see that each allophone appears in a unique contexts.

You are now familiar with the methods used to determine the phonological properties of the pattern — both the similar and the different phonological features of the sounds themselves as well as the contexts for each. You may remember these skills from earlier chapters; they are used in Chapter 9 as well.

Familiarity with complementary distribution and contrast means that the linguist terms **phoneme** and **allophone** make sense — and you have an understanding of why these concepts are critical for anyone using published linguistic descriptions to better understand their own language.

You are ready to use these concepts and skills as you work with your own language.

Chapter 9

Alternations: Tuning into a language

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Here's a puzzle. Suppose you are in the early stages of creating a dictionary for your language, collecting words and their meanings. As you progress, you find words with similar sounds and related meanings — for instance, if you were working with Hopi, a Uto-Aztecan language spoken in Northern Arizona in the United States, you might find the following words that all share a *REPETITIVE* (REP) meaning: *asisita* [asisita] 'sneeze.REP', *eykita* [ejkita] 'be groaning REP', *hiqta* [hiqta] 'have a crush on REP', and *warta* [waʃta] 'run REP'. Our sources for Hopi are Jeanne (1978, 1982), Hill et al. (1998). (Following our conventions in this book, italics show the practical orthography used in Hill et al. (1998); IPA is given in square brackets [...].)

Your morphological senses tingle. You carry out a subtractive analysis, identifying [-ta] as a morpheme meaning REP. You're now on high alert, watching out for roots [asisi-] 'sneeze', [ejki-] 'be groaning', [hiq-] 'have a crush on', and [waʃ-] 'run'.

As you keep learning more about Hopi, you come across a word *wari* [wazi] 'run'. Since the meaning is 'run', you expected *[waʃi] and you're surprised to hear [wazi]. Did you hear incorrectly? You double-check and sure enough, people say [wazi], not *[waʃi]. It looks like there are two almost-identical morphemes meaning 'run', [waz...] and [waʃ...].

Is it normal for languages to have two almost-identical morphemes meaning exactly the same thing?

The short answer is yes — this is quite common. In fact, in some cases there are multiple almost-identical forms of a single morpheme. The distinct forms

of a morpheme mean that words of the language respect the sound patterns of the language. In Hopi, [z] and [ʂ] are in **complementary distribution**: We find [+voice] [z] at the beginning of a **syllable** — as in *wari* [wazi] ‘run’ — and we find [–voice] [ʂ] at the end of a syllable — as in *warta* [waʂta] ‘run.REP’. The morpheme form used in each word means that the word as a whole conforms to the complementary distribution pattern for [z] and [ʂ] in Hopi. (See Chapter 6 on syllables and Chapter 8 on complementary distribution.)

We call this **alternation** — a morpheme is pronounced in different ways depending on the context. We examine alternations occurring in both phonological and morphological contexts in this chapter.

Alternation is the sixth and final major component of a language’s unique **phonology**, along with variation, contrast, syllables, distribution, and complementary distribution.



Apico-alveolar sounds

The Hopi sound represented with [ʂ] is a “voiceless apico-alveolar fricative (somewhat as *rsh* in *harsh*)” Hill et al. (1998: 863), and the sound [z] is a “voiced apico-alveolar fricative (something like a cross between Spanish *rr* and the sound in the middle of English *vision*” Hill et al. (1998: 863). The difference is voicing, but what is an apico-alveolar sound?

The feature [±distributed] distinguishes [coronal] sounds, those involving the tongue tip. Apical sounds involve a pointed tongue tip, signified by the feature [–distributed]. Laminal sounds, with a flattened tip, are [+distributed, coronal].

Sounds made with the tongue forward in the mouth are [+anterior] (see Chapter 2 Section 2.3.2.2). Thus, apico-alveolar sounds are defined by the features [coronal, –distributed, +anterior].

(Features are introduced in Chapter 2; definitions for features, like [distributed] and [coronal], are found in the glossary of terms linguists use; there is also a list of sound-feature pairings in Appendix B.)

Here’s our promise. Analysis of alternation calls on all the phonological skills that you’ve learned in working through the previous chapters. You have acquired analytic skills to identify classes of sounds and the properties by which they vary

through the characterization of language sounds (Chapter 2). You have learned to identify patterns of difference and the contexts for those patterns when you learned about variation (Chapter 4), contrast (Chapter 5), distribution (Chapter 7), and complementary distribution (Chapter 8). You may even find reason to call on syllables from Chapter 6, just as we did with our brief introduction to Hopi in this section. In this chapter, we put these phonological analysis skills together with the morphological analysis skill of subtractive analysis that you acquired in Chapter 3.

Here's our plan. When phonological analysis intersects with morphological analysis, we get a *morpho-phonological analysis*, the final skill necessary for untangling the sound patterns of a language.

This chapter introduces you to morpho-phonological analysis by building on the skills from previous chapters. After working through this chapter, you will understand what a morpho-phonological analysis is and you will have solidified the skills necessary to discuss a morpho-phonological analysis of your own language. You may even have the confidence to try some morpho-phonological analysis on your own. Understanding the alternations in a language provides the finishing touch to sounding natural when speaking a language, creating *neologisms*, and making decisions about dictionary entries, among other goals.

1. Section 9.1 introduces the basic concept of alternation, where a particular morpheme has various related phonological forms.
2. Section 9.2 works through an example of a common type of alternation, one that results when putting morphemes together creates the context for a complementary distribution pattern. This puts us in position to address the linguistic concepts of *underlying representation* and *derivation* (in Section 9.2.3).

Section 9.2 shows a basic approach to understanding alternations. Depending on your situation, after reading Section 9.2 you may want to jump right to Section 9.7. On the other hand, you may want to continue reading. There is a continuum in alternations, from phonological alternations driven by complementary distribution, to morpho-phonological alternations that show phonologically-conditioned differences in some morphemes but not in others, to morphological alternations where there is no phonological reason for the alternation, yet a morpheme has more than one phonological form. Examples of each are found in the next few sections.

3. Section 9.3 illustrates an alternation which is not related to complementary distribution, but rather is found only with some morphemes (even though

there are others that have the phonologically-relevant parts in them). If you felt a bit shaky about how to work through an alternation after reading Section 9.2, you may feel more solid if you work through this section too.

4. Section 9.4 brings up some polymorphemic words that are extra fun because they involve multiple alternations in the same word. This kind of interaction can be confusing at first, but the overall logic is revealed when you see it as the interaction of separate patterns.
5. Section 9.5 introduces various forms of morphemes which are phonologically unrelated. This is known as **suppletion**: Semantically related morphemes which have distinct phonological forms.
6. Section 9.6 offers a sense of the richness of interactions between phonology and morphology. We discuss a few of these: *Reduplication*, an *apparent alternation* (something that looks like an affix, but has no semantic contribution), an *alternation variation* (the intersection of variation and alternations), and a *metrical alternation* (one that connects to where the stress is placed in a word).

If you are looking for practical analytic knowledge, you may want to skip Section 9.6 and go straight to the discussion, Section 9.7. You can always return to read Section 9.6 if you need inspiration about understanding an unusual pattern.

Alternations connect phonological forms and the morphemes that make up words. Analytic decisions about alternations are reviewed in Section 9.7; we also revisit the puzzle and sketch other types of alternations you might come across. We close the chapter with a discussion of why knowing about alternations can be very useful to language workers.

9.1 Introduction: What is alternation?

When morphemes are put together to make a word, sometimes all that is needed is to put an affix and a root together and the result is a well-formed word. We saw this in the examples in Chapter 3. An example from Hopi is the word *eykita* [ejkita] ‘groan REP’, made up of the suffix *ta* [-ta] ‘REP’ and the root *eyki* [ejki] ‘groan’. Hopi may be somewhat familiar to you from the Comprehension Checks in Chapter 3; in this chapter we focus on Hopi so that you can see how morphology and phonology work together in a single language.

Sometimes, doing subtractive analysis and then recombining the morphemes does not result in a word of the language. For example, subtractive analysis with *warta* [waʃta] ‘run.REP’ gives a morpheme [waʃ] ‘run’, leading us to expect *wari* *[waʃi], but we observe *wari* [wazi] ‘run’.

Conversely, adding *-ta* to the Hopi morpheme *wari* [wazi] ‘run’ suggests *warita* *[wazita], not the observed *warta* [waʂta] ‘run.REP’. The two different forms of the Hopi morpheme for ‘run’ indicate that there is an *alternation*: A morpheme has different phonological forms; typically each form is used in a particular context or set of contexts.

As we have seen, Hopi has complementary distribution of the apico-alveolar fricatives, with [+voice] [z] found only at the beginning of a syllable and [–voice] [ʂ] found only at the end of a syllable. When the Hopi morphology conspires to put a [coronal, –distributed, +anterior] fricative at the beginning of a syllable, as in *wari*, the fricative is voiced [z], [wazi] (not *[waʂi]). On the other hand, when the result of morphology is that the fricative is at the end of a syllable, as in *warta*, we find voiceless [ʂ], [waʂta] (not *[wazta]).

In short, the morpheme meaning ‘run, race’ has two forms, [wazi] and [waʂ] (we’ll get to the “missing final vowel” later). When there are two (or more) forms of a morpheme, such as [wazi] and [waʂ], we say that the morpheme alternates. The different forms of each morpheme are called *allomorphs* or *alternants*. Alternations are sometimes labeled by the sounds that alternate with each other. This example from Hopi could be referred to as a [z]~[ʂ] alternation.

We have just identified the existence of an alternation by using our subtractive analysis skills from Chapter 3. To understand the phonological properties of the analysis, we turn to our phonological analysis skills from Chapters 4, 6, and 8; this is the primary focus of the discussion in this chapter. Importantly, no new skills are introduced in this chapter. Rather, we combine familiar skills in novel ways.



Comprehension Check 1

Can you identify allomorphs? The following words from Zenzontepec Chatino, an Otomanguean language spoken in Oaxaca (Campbell 2014), show three prefixes, POTENTIAL [ki-], PROGRESSIVE [nde-], and COMPLETIVE [ŋgu-]. The verb stems are in black typeface — notice that there is an alternation with some verb stems. The verb in (a) did not have a completive form in Campbell (2014). (You may remember these Zenzontepec Chatino prefixes from Chapter 3.)

	POTENTIAL	PROGRESSIVE	COMPLETIVE	
a.	[kit ^h ěě]	[ndetěě]		‘carry’
b.	[kin ^h ǎǎ]	[ndēnǎǎ]	[ŋgūnǎǎ]	‘get cleared (field)’
c.	[kil ^h ák ^w i]	[ndelák ^w i]	[ŋgūlák ^w i]	‘boil (intransitive)’

For each verb, list its allomorphs. Hint: You should find two allomorphs for each verb.



Here’s our answer

Here are the forms (allomorphs) of each of the four verbs in the question. We highlight any differences using black text.

a.	[t ^h ěě]	[těě]	‘carry’
b.	[n ^h ǎǎ]	[nǎǎ]	‘get cleared (field)’
c.	[l ^h ák ^w i]	[lák ^w i]	‘boil (intransitive)’



Comprehension Check 2

Let’s find the general patterns with the allomorph pairs. We start with the pair [t^hěě] and [těě], meaning ‘carry’.

1. *What is the same?* (When you remove the part that is the same, the part that is left shows you the difference — subtractive analysis.)
2. *What is different?*

We hope you noticed that the vowel is the same, [ẽẽ] in each allomorph. What is left is the initial sounds, [tʰ] in one case and [t] in the other.

Now do the same for the next two pairs — what is the same and what is different with [nʲãã] and [nãã], meaning ‘get cleared (field)’, and with [lʲákʷi] and [lákʷi], meaning ‘boil (intransitive)’?



Here's our answer

What is the same? With both pairs, everything except the initial consonant is the same.

What is different? The initial consonant, [nʲ] vs. [n] and [lʲ] vs. [l].

To summarize, each verb has two allomorphs, one that begins with a plain coronal t, [n], or [l], and the other that begins with a palatalized coronal, [tʰ], [nʲ], and [lʲ] respectively.

At this point, you might want to review the Comprehension Check about the complementary distribution of plain and palatal coronals in Zenzontepec Chatino, from Chapter 8, on p. 464.

By the way, hang on to your answer. You will use it again with a Comprehension Check on p. 489.

9.2 Alternations

We'll start by exploring the puzzle a little more. We will first show the complementary distribution of [z] and [ʂ] in Hopi, and then show how alternations connect complementary distribution and morphology. Our sources for Hopi include Jeanne (1978, 1982).



LaVerne Jeanne

In 1978, Dr. LaVerne Jeanne, Hopi, and Dr. Paul Platero, Navajo, became the first two Native Americans to earn PhDs in Linguistics, both at the Massachusetts Institute of Technology and both working with Professor Ken Hale. Here is more information about Dr. LaVerne Jeanne.

“LaVerne Jeanne has a PhD. in linguistics and is a retired professor from the University of Nevada... She grew up on the Hopi Reservation in Hotevilla at a time when Hopi was the primary language; everyone spoke Hopi. Dr. Jeanne developed an interest in the Hopi Language after taking an anthropology class and that interest intensified when she was invited to participate in weekend seminars on Arizona Native languages spoken in Arizona, including Hopi, Navajo, Hualapai, Apache, Tohono O’odham, Yavapai, and Supai. Dr. Jeanne has experience working with tribal groups engaged in language revival programs including:

- Consulting with the Washiw Language Immersion School (1997- 2002)
- Directing the Hotevilla-Bacavi Community School Bilingual Program, Hopi Reservation, Hotevilla (1981–1983)
- Consulting with the Hualapai Bilingual and Bicultural Program, Peach Springs; ideas for Hualapai Language Teaching (1978)
- Conducting an Orthography workshop for the Tohono O’odham Nation (1978)”

Jeanne (2013: 2)

9.2.1 Complementary distribution in Hopi

A quick review of Chapter 8 reminds us that complementary distribution refers to similar sounds that are not contrastive with each other — each appears in its own unique context. They are not contrastive, so they do not appear in the same context.

Jeanne (1978, 1982) and Hill et al. (1998) say that [z] and [ʒ] are in **complementary distribution**. This claim is based on words like those in (1) and (2). In (1), we

see [z] in each word, occurring before a vowel — that is, at the beginning of a **syllable**. Dots are included to separate syllables from each other and black font is used for the sounds we are focusing on. (See Chapter 6 if you need to review syllables.)

- (1) Examples of [z] before a vowel

kura' [ku.zaʔ] 'so it seems'
toriho [to.zi.ho] 'crooked arrow'

In (2), we see [s] but not [z]. Note that each [s] occurs before a consonant — at the end of a syllable.

- (2) Examples of [s] before a consonant

kurpi [kuʃ.pi] 'darn it!'
tarpihma [taʃ.pik.ma] 'ricochet out of sight'

A complementary distribution table shows that the two sounds [z] and [s] occur in complementary contexts.

Table 1: Complementary distribution table for Hopi [z] and [s]

		[z]	[s]
syllable-initial	$\sigma[_]$	[kuzaʔ]	
syllable-final	$_]\sigma$		[kuʃpi]

In Table 1, we see the classic complementary distribution checkerboard. The apico-alveolar fricatives [z] and [s] are in complementary distribution, with voiced [z] at the beginning of syllables and voiceless [s] at the end of syllables. Recalling that, while [z] is [+voice] and [s] is [−voice], both sounds are [+continuant, coronal, −distributed, +anterior], we create a precise statement about the pattern. (We abbreviate some of the feature names in (3): [±cont] for [±continuant], [cor] for [coronal], [±dist] for [±distributed], and [±ant] for [±anterior].)

- (3) Hopi complementary distribution of [z] and [s]

[z] and [s], the [+continuant, coronal, −distributed, +anterior] sounds, are in complementary distribution, with the [+voice] [z] at the beginning of syllables and [−voice] [s] at the end of syllables.

Alternatively in shorthand,

[+cont, cor, −dist, +ant] → [+voice]/ $\sigma[_]$, [−voice] / $_]\sigma$



Alternative analyses

You may have noticed that the Hopi fricative is voiceless before consonants and voiceless elsewhere. This observation suggests that there is no need to refer to syllables in the analysis.

In fact, this situation is not uncommon when trying to understand a pattern. It is often true that there is more than one way to characterize a pattern. This means that two people can talk about a particular pattern in different ways and both people can have accurate descriptions of the pattern. It is important to remember this, and to keep an open mind when examining alternative analyses.

Having said this, here we continue to use the syllable-based analysis to help familiarize readers with how syllables can be used to help understand the patterns of a language. If you want, you can try reframing the Hopi discussion without using syllables, to determine for yourself whether the two approaches are equally sufficient given the forms presented here.

9.2.2 When complementary distribution meets morphology

In Hopi, the complementary distribution of [z] and [ʃ] determines which allomorph to use, giving syllable-initial [z] in *wari* [wazi] ‘run’ and syllable-final [ʃ] in *warta* [waʃta] ‘run.REP’. We give a couple more examples in (4), again separating syllables with dots.

(4) Examples of the Hopi [z] and [ʃ] alternation

- | | | | |
|----|---------------|-----------|-------------------|
| a. | <i>yori</i> | [jo.zi] | ‘see’ |
| | <i>yorta</i> | [joʃ.ta] | ‘keep looking at’ |
| b. | <i>tsuru</i> | [tsu.zu] | ‘get stuck in’ |
| | <i>tsurta</i> | [tsuʃ.ta] | ‘be squeezing in’ |

The [z]~[ʃ] alternation connects complementary distribution with morphology: The morphology puts morphemes together, and the phonology determines what is a well-formed sequence. This is a morpho-phonological pattern.

9.2.2.1 One place to look for alternations in roots and/or affixes

It is not uncommon to find a morpheme-final consonant that is sometimes at the start of a syllable and sometimes at the end of a syllable. Consider a morpheme of the shape CVC, such as *som* [som] ‘tie’. If a consonant-initial suffix is added, the final [m] will be at the end of the syllable, *somni* [som.ni] ‘tie.FUT’. If a vowel-initial suffix is added, the final [m] will be at the beginning of a syllable, *soma* [so.ma] ‘tie.NONFUTURE’. This context is a source of alternation in languages with syllable-dependent complementary distribution. Hopi is such a language, since the choice between [z] and [ʃ] is dependent on syllable position.

9.2.2.2 A note on Hopi orthography

You may have noticed that in the written Hopi forms (those in italics), the letter *r* is used for both [ʃ] and [z], as in *yor*i [jozi] and *yort*a [joʃta]. Those who created the Hopi [writing system \(orthography\)](#) chose to use a single symbol rather than distinct symbols despite the difference in pronunciation. The result is an orthography that shows the morphological connection between [z] and [ʃ] by using the same letter *r*, to represent the two related sounds.

Another result is that if a person were trying to read Hopi, they might mispronounce the words spelled with the letter *r*, using either an *r*-like sound (rather than a fricative, see footnote 9), or using the voiced fricative [z] in all contexts, or using the voiceless fricative [ʃ] in all contexts. Any one of these is incorrect. To use an orthographic system in order to know how to pronounce words requires understand the code relating the writing system to sounds.



Comprehension Check 3

Consider your responses to the previous Comprehension Checks about allomorphs on p. 481 and on p. 483.

Look again at the verbs with two allomorphs, repeated here. The left-hand column shows the allomorphs which begin with a palatalized coronal while in the other columns the verb allomorphs begin with plain coronals.

What is the sound that immediately precedes the allomorphs in each column?

	POTENTIAL	PROGRESSIVE	COMPLETIVE	
a.	[kitʰēē]	[ndetēē]		‘carry’
b.	[kīnʰāā]	[ndēnāā]	[ŋgūnāā]	‘get cleared (field)’
c.	[kilʰákʷi]	[ndelákʷi]	[ŋgūlákʷi]	‘boil (intransitive)’

What sound precedes palatal coronals? Does that sound precede plain coronals? What does come before plain coronals?



Here are our answers

What is the sound that immediately precedes the allomorphs in each column?

- The leftmost column, with the palatalized coronals at the start of each verb allomorph, has the vowel [i] before the verb: [kitʰēē], [kīnʰāā], [kilʰákʷi].
- The verbs are preceded by the vowel [e] in the middle column: [ndetēē], [ndēnāā], [ndelákʷi].
- The verbs are preceded by the vowel [u] in the rightmost column: [ŋgūnāā], [ŋgūlákʷi].

What sound precedes palatal coronals? The vowel [i] is found right before every palatal coronal.

Does that sound precede plain coronals? No, the vowel [i] is not found before plain coronals.

What does come before plain coronals? The vowels [e] and [u] are found before plain coronals.



Comprehension Check 4

Is there an alternation corresponding to complementary distribution in Zenzontepec Chatino?

To answer this question, first review the Comprehension Check about complementary distribution in Zenzontepec Chatino on p. 464.

Also review the distribution of plain and palatalized coronals in Zenzontepec Chatino from the Comprehension Check on p. 487.



Here's our answer

Yes, we find an alternation corresponding to complementary distribution in Zenzontepec Chatino.

Zenzontepec Chatino has complementary distribution of plain and palatalized coronals, with palatalized coronals only following [i].

Zenzontepec Chatino has alternations between plain and palatalized root initial coronals, [t] and [tʲ], [n] and [nʲ], [l] and [lʲ].

The distribution of the plain and palatalized coronals in verb roots is exactly what we expect given the pattern of complementary distribution in Zenzontepec Chatino: After a prefix ending with [i], we find the allomorph with a palatalized coronal and elsewhere we find the allomorph with a plain coronal.



Exploring my language

In Chapter 8, you identified patterns of complementary distribution in your language, and in Chapter 3, you identified morphemes in your language based on word-internal structure.

The challenge now is to put the two together and determine whether any of the complementary distribution patterns you identified contribute to alternations when morphemes are combined.

1. What is the context for the complementary distribution pattern?
2. Can that context be created by putting morphemes together to make a word?
3. If your answer to 2 is “yes”, do you find alternations? List words to show whether you do or do not find alternations in this context.

You can use this information in teaching language students how to expand their vocabulary (through adding morphemes) while still speaking fluently, once the student understands how to make morphemes alternate to satisfy complementary distribution and syllabification.

9.2.3 *Underlying representations and derivations*

It is extremely common for linguists to talk about the **underlying representation** of a morpheme, represented with sound symbols between slashes.

The basic idea of underlying representations is that each morpheme has exactly one phonological form, and that all the different pronunciations of that morpheme are related to that one phonological form by rules. The rules convert the underlying representation into a **surface representation**. The surface representation shows how the word is pronounced, based on the underlying representation and the rules.

For example, the Hopi word for ‘see’ has two pronunciations, [joʔi] in *yori* [joʔi] ‘see’ and [joʃ] in *yorta* [joʃta] ‘keep looking at’. A linguistic analysis typically adopts a single representation for the root ‘see’ — in this case, /joʔi/ might be chosen as the underlying representation of ‘see’ in Hopi. This directly relates to the pronunciation in [joʔi] — no change is needed. But how do you get from /joʔi/ to [joʃ]?

We know that there is a regular phonological relation between [z̥] and [s], depending on the syllable position. The other necessary piece is to delete the /i/ at the end of *yorɪ* [jozi] ‘see’. With underlying representations, this relation is expressed as a rule that deletes the final vowel in certain morphological contexts. This means you start with the underlying representation /jozi/, add any relevant suffixes, apply the two rules, and the resulting form is what gets said.

This series of steps is called a *derivation*. We show two derivations in (5), for [jozi] ‘see’ and [joʂta] ‘keep looking at’ in Hopi.

A derivation starts with a meaning and an underlying representation (5a, b) respectively. The first action is to add the morphemes together into a single string, (5c). Now we can apply the rules in turn. Here we start with vowel deletion, in (5d), followed by the [z̥]~[s] rule of complementary distribution, (5e). The last step is to convert the form to IPA symbols. This is seen in (5f), the *surface representation*. The surface representation is a representation of how the word is pronounced (and so it is put in square brackets).

(5) Example derivation

a.	meaning	‘see’	‘keep looking at’
b.	underlying representation	/jozi/	/jozi/ + /ta/
c.	combine morphemes	—	jozi-ta
d.	apply vowel deletion rule	—	joʂta
e.	apply [z̥]~[s] rule	—	joʂta
f.	surface representation	[jozi]	[joʂta]

The last step of a derivation is the surface representation, telling how the word is pronounced. This means that underlying representations are not intended to be pronounced. Trying to pronounce an underlying representation without checking the rules of the language (as proposed by the linguist), means you will be saying something that speakers may never ever say. Underlying representations relate to pronunciations once you know the rules.

Unfortunately, many works describing a language introduce complementary distribution and alternations in the language, and then go on to represent examples in terms of underlying representations rather than pronunciations. It is a shorthand of convenience for the author but can be misleading and frustrating for the language worker who needs to know how the words are to be said. Since both underlying and surface representations use IPA symbols, the unsuspecting reader may be confused, thinking that the underlying representation shows the pronunciation directly, not indirectly.



Generative phonology and optimality theory

You may read something that talks about *generative phonology*. This is a model of how phonologies are structured and it makes use of underlying and surface representations related to each other by rules. The basic idea is that the rules *generate* the surface representation from the underlying representation.

You may also come across reference to *optimality theory*. This is another model of how phonologies are structured. Optimality theory uses the terms *input* and *output* instead of underlying representation and surface representation. Inputs and outputs are related to each other by general principles which result in multiple contending outputs; constraints stating preferred sound sequences in the language determine the best of the outputs for a given input.

9.3 An alternation that is not based on complementary distribution

Alternations that result from complementary distribution are explained in terms of a phonological pattern, ensuring that words are phonologically well-formed when morphemes are put together. Not all alternations involve straightforward phonological well-formedness. In our examples showing [z]~[s] alternants, we also saw that one form of the root ended with a vowel, *wari* [wazi] ‘run’. The other form ended with a consonant ([waʂ-] in *warta* [waʂta] ‘run fast’). We can now understand this as yet another alternation, one with a vowel and the other without the vowel.

This kind of alternation is called a V~0 alternation (‘V-zero alternation’). The expression V~0 alternation refers to an alternation where one form of a morpheme has a vowel (‘V’) which is missing in the other form of the morpheme (hence “zero”). In our example, we see the “V” alternant in [wazi] and the “0” alternant in [waʂ] (where the entire form is in gray text because the vowel is missing).

9.3.1 The Hopi alternation between V and nothing

There are several Hopi verbs that end in a short vowel, like *wari* [wazi] ‘run’; two more are shown in the NONFUT column in (6). We see a difference when a suffix with the shape CV is added, such as the suffix *-ni* [-ni] FUT. In this case, the root is consonant-final, not vowel-final. In (6), the final sounds in each root are highlighted by keeping them in black text. (Recall that a hyphen separates morphemes, as in [som-ni] ‘tie-FUT’.)

(6) Verb roots that alternate

	NONFUT	FUT	
a.	<i>soma</i> [soma]	<i>somni</i> [som-ni]	‘tie’
b.	<i>so’a</i> [soʔa]	<i>so’ni</i> [soʔ-ni]	‘die.PL’

We see that these verbs have both a vowel-final allomorph and a consonant-final allomorph — [soma] and [som] ‘tie’. The vowel-final allomorph is used at the end of a word while the consonant-final allomorph is used before the FUT suffix.

If this alternation is a general phonological pattern, we expect that every verb would follow this pattern. But this is not the case. Some verbs only have vowel-final forms, highlighted in (7).

(7) Verb roots that don’t alternate

	NONFUT	FUT	
a.	<i>maqa</i> [maqa]	<i>maqani</i> [maqa-ni]	‘give’
b.	<i>öki</i> [ʔœki]	<i>ökini</i> [ʔœki-ni]	‘arrive.PL’

We have an interesting situation: Some CVCV verbs show the V~0 alternation, but others do not. The only way to tell whether a verb has the V~0 alternation is to hear the two forms and remember that the verb alternates.

Whether a verb alternates or not cannot be determined by any phonological property of the verb. In cases like this, linguists will talk about **lexical classes** — classes of words with a phonological behavior that must be memorized for that class. Fluent speakers of Hopi have learned which verbs show the alternation and which ones do not. When learning a language with lexical classes, students must simply memorize which class a word belongs to. Fortunately, it is often the case that there is one lexical class that is larger than any of the other classes. Once learners know the common pattern, the focus shifts to remembering which words are different, learning which words are in which of the small classes.

The advantage of lexical classes is that once you know what the alternations are that define a class, when you learn a new word, you may also have a clue

about its class because of its behavior. And we know the different forms to use in different contexts. For instance, in Hopi, once a verb root is identified as an alternator, we know to use a form without the final vowel when the FUT suffix is attached. The learning challenge is to remember which items belong to the alternating lexical class.



Comprehension Check 5

In Chapter 3 Section 3.6.1, we saw that Zenzontepec Chatino has at least three different prefixes to mark the COMPLETIVE:

[ŋgu-]	in	[ŋgu-tõõ]	‘COMPLETIVE-stand’
[ŋg ^w i-]	in	[ŋg ^w i-tʰẽẽ]	‘COMPLETIVE-carry’
[ŋga-]	in	[ŋgatāá]	‘COMPLETIVE-give’

Does Zenzontepec Chatino have lexical classes to show COMPLETIVE?



Here's our answer

It looks like Zenzontepec Chatino has lexical classes, where the classes each have a different COMPLETIVE prefix. The alternative would be that there is some phonological property which determines which prefix is used. But when we look at the different verbs to which the prefixes are attached (including those in Chapter 3 Section 3.4), we see that there are verbs with the same initial consonant in the different classes, and verbs with the same vowel. So the sounds don't appear to provide clues for which prefix to use. Prefix choice must be memorized for each verb, the hallmark of lexical classes.

9.3.2 Phonological vs. morpho-phonological alternations

This section extends the concept of alternations — morphemes with more than one allomorph — to alternations occurring within a lexical class.

Alternations driven by complementary distribution are *phonological alternations* because the alternations make a word conform to the general phonological properties of a language, properties that we've established in our understanding of the distribution of sounds in a language.

Regular alternations restricted to a lexical class are *morpho-phonological alternations*. The *morpho-* part refers to the alternation being restricted to a lexical class; the *phonological* part tells us that the alternation can be characterized phonologically (such as the relation between the presence and absence of a vowel). So morpho-phonological alternations are phonological alternations that take place in a lexical class, a learned subset of the words of the language. In Hopi, the V~0 alternation is a morpho-phonological alternation.



Exploring my language

The task here is to examine some affixes in your language to search for alternations in the affixes or in the roots to which the affixes attach.

Start with your answer to the Language Exploration on p. 169, showing a paradigm in your language. There are three steps in this task.

1. Add three more roots to the paradigm, making sure that each begins or ends with a different sound. (Tip: If your paradigm uses prefixes, you want different sounds at the start of the roots; if your paradigm uses suffixes, you want different sounds at the end of the roots.)
2. Add the correct form of the affix(es) in your paradigm to each of those roots.
3. Do any roots alternate to accommodate the affix? Or do any affixes alternate to accommodate the roots? List the allomorphs.

If you have found allomorphs, you have found an alternation. If not, there may be no alternation. On the other hand, there may be an alternation with a sound or sound class you haven't considered yet.

9.3.3 Analyzing alternations

In our discussion to this point, you may have identified some clues to finding an alternation.

You can start with known complementary distribution and distributional patterns because distribution often results in alternations when morphemes are combined into words. Knowing the patterns of your language means you can test what happens when the morphology could create a form that doesn't meet the phonological requirements of the language.

You can also observe your language, to spot a pair of words that seem like they are morphologically related, yet subtractive analysis results in more than one allomorph for a particular morpheme.

If you find either of these in your language, you may have found an alternation to explore.

You may have recognized the steps to figuring out the dimensions to an alternation — what sounds alternate, what features stay the same, whether there is a critical context. These steps may be familiar to you from earlier chapters — they are essentially the same steps used in identifying the dimensions of variation in Chapter 4, of distribution in Chapter 7, and of complementary distribution in Chapter 8. These steps are summarized in Figure 1.

When you work through Figure 1, it should feel fairly familiar. The step of identifying classes of sounds by similar features and determining which features are different was introduced with our discussion of variation in Chapter 4 and has been used throughout the subsequent chapters. Figuring out the context for a pattern is familiar not only from looking at variation but also from finding complementary distribution in Chapter 8. Using features to identify classes began in Chapter 2 and is used in every kind of phonological analysis. The challenge with alternation is integrating these several different types of analysis, but the types themselves are familiar.

9.4 Putting morpho-phonological alternations together

Hopi has another morpho-phonological alternation, an alternation between long and short vowels — a VV~V alternation. In this section, we see how the VV~V alternation interacts with the morpho-phonological V~0 alternation and with the phonological [z]~[s] alternation — three alternations that interact with each other.

9.4.1 The Hopi alternation between VV and V

Hopi vowels can be either long or short. Interestingly, verb roots with long vowels can also be members of the V~0 lexical class.

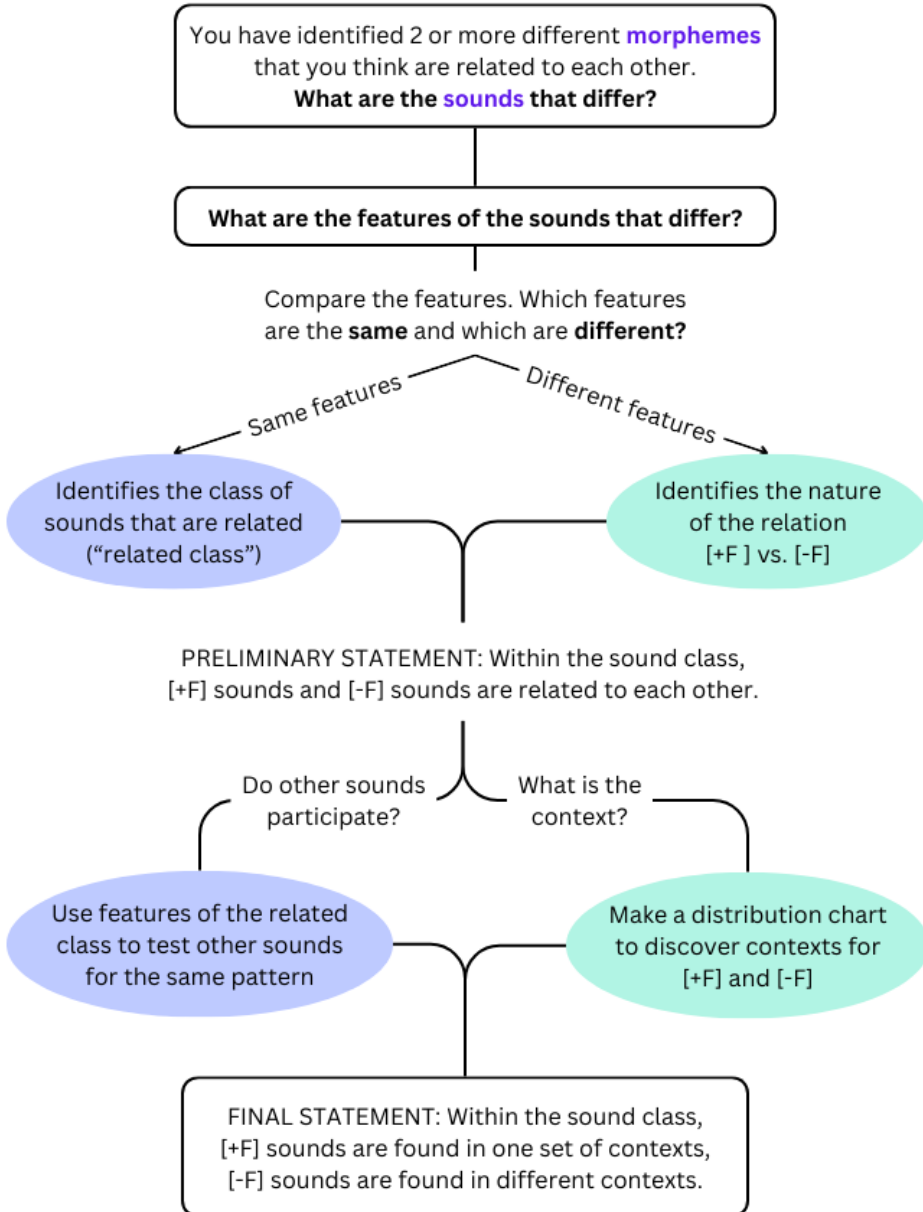


Figure 1: Determining the dimensions of alternation

That said, we might expect a verb like *aasi* [ʔa:si] ‘wash hair’ to show up in the future as [ʔa:s-ni]. But no. What we see is [ʔas-ni] — not only is the final vowel missing, but the long vowel is missing as well, and there is a short vowel in its place. The allomorphs are [ʔa:si] and [ʔas] ‘wash hair’. See the examples in (8).

(8) Vowel length alternations

	NONFUT		FUT	
a.	<i>aasi</i>	[ʔa:si]	<i>asni</i>	[ʔas-ni] ‘wash hair’
b.	<i>mooki</i>	[mo:ki]	<i>mokni</i>	[mok-ni] ‘die’

If we simply add the FUTURE suffix *-ni* [-ni] to [ʔa:si], we expect *[ʔa:si-ni], a CVVCVCV sequence. That is incorrect for [ʔa:si], but it is exactly the right thing with some verbs, as seen in (9).

(9) Suffixes and verb roots with long vowels, no alternations

	NONFUT		FUT	
a.	<i>tuuva</i>	[tu:va]	<i>tuuvani</i>	[tu:va-ni] ‘throw away’
b.	<i>peena</i>	[pe:na]	<i>peenani</i>	[pe:na-ni] ‘write’

We conclude that the VV~V alternation is morpho-phonological, just like the V~0 alternation. And we also see that a single word can participate in both morpho-phonological alternations at the same time. (You might have noticed that the long vowel appears in an open syllable and the short vowel appears in a closed syllable. This is often the case with VV~V alternations, but it is not the case in Hopi, since Hopi has long vowels in closed syllables.)

Just for fun, we next take a quick look at forms that participate in *three* alternations at once — both of the morpho-phonological alternations, VV~V and V~0, as well as the familiar phonological alternation, [z]~[s] illustrated in example (3).

Try this. Before reading further, what do you think will happen with such words? (Our answer is in the next section. But think about it first.)

9.4.2 Three alternations in the same word!

In this section, we show that all three alternations, VV~V and V~0, and [z]~[s], can be found in the same word. To do this, we introduce another way morphology works, *compounds*.

In addition to adding affixes to roots, Hopi words can be made by combining two nouns into a single word. The resulting word type is called a compound (in this case, a compound noun). The Hopi word for ‘crow’ is *angwusi* [ʔaŋwu:si], and the word for ‘feather’ is *masa* [masa]. In *angwusmasa* [ʔaŋwu:s-masa] ‘crowfeather’ we see compound nouns where the first noun is a bird name — *angwusi* [ʔaŋwu:si] ‘crow’ — and the second noun is *masa* [masa] ‘wing, feather’.

Sometimes when two nouns are compounded in Hopi, we see a V~0 alternation: There is a final vowel when the noun stands alone, but that vowel is missing in the middle of the compound. We see this with the final vowel of *angwusi* [ʔaŋwu:si] ‘crow’, which is missing in *angwusmasa* [ʔaŋwu:s-masa] ‘crowfeather’. The same pattern is also seen with the final [o] in *koyongo* [kojoŋo] ‘turkey’, missing in the middle of *koyongmasa* [kojoŋ-masa] ‘turkey feather’.

With bird names that have a long vowel, we see both the V~0 alternation at the end of the noun and the VV~V alternation in the first syllable, (10). Consider the word *kyaaro* ‘parrot’. The word has a long vowel by itself, but when it appears in a compound, *kyarmasa* ‘parrot wing’, the final [o] is missing (the V~0 alternation) and the first vowel is short [a], not long [a:] (the VV~V alternation).

Now, if you haven’t picked up on it already, notice the last consonant in *kyaaro* ‘parrot’. It is spelled *r*. Remember from our earlier discussion that in Hopi writing, *r* corresponds to both sounds [z] ~ [ʂ], sounds that are in complementary distribution with each other (Section 9.2.1). As shown by the words (10), we find the [z] ~ [ʂ] alternation in words that have both the V~0 alternation and the VV~V alternation, three alternations in one word: [z] is found at the beginning of syllables and [ʂ] at the end of syllables. The alternating consonant is exactly what the complementary distribution pattern leads us to expect.

(10) V~0, VV~V, and [z]~[ʂ] alternations in one word!

- | | | | |
|----|-----------------|------------|-----------------|
| a. | <i>kyaaro</i> | [kʲa:zɔ] | ‘parrot’ |
| | <i>kyarmasa</i> | [kʲaʂmasa] | ‘parrot wing’ |
| b. | <i>tsooro</i> | [tso:zɔ] | ‘bluebird’ |
| | <i>tsormasa</i> | [tsoʂmasa] | ‘bluebird wing’ |

The words in (10) suggest that there is no limit on how many alternations can show up in a single word. Alternations are observed if the context is right, and some words contain the contexts for more than one alternation.



Exploring my language

Does your language have compound nouns or compound verbs? Try to find two.

Note: Compounds are very useful if your language needs a word for some novel meaning (a **neologism**). By putting two existing words together, the result fits in with the language's history. Watch out for alternations, so you can be sure to make the appropriate adjustments.

9.4.3 Dictionaries and lexical classes

When you encounter lexical classes in your language and are putting together a dictionary, it is very helpful to the user if you figure out a way to consistently show the lexical classes in dictionary entries. Ken Hill and colleagues faced this problem when putting together the entries for Hill et al. (1998). Before we continue to morphological alternations (*suppletion*), we digress to briefly explore how the Hopi Dictionary Project handled the lexical class challenge.

Non-alternating roots are listed with their single form, such as non-alternating **maqa** 'give', with a short vowel, and non-alternating **tuuva** 'throw', with a long vowel. Alternating verbs are shown with the symbol | (called a "pipe") to indicate that the final vowel can be omitted in certain contexts: Contrast alternating **som** | a 'tie' with non-alternating **maqa** 'give'. Additionally, a separate entry gives "**som-** *comb. of soma*".

Alternating roots with long vowels are treated somewhat differently. Since the dictionary alphabetizes long vowels differently from short vowels, a root that has the VV~V alternation has two separate entries in the dictionary. Helpfully, each refers to the other, the entry "**ööyi** *comb. öy-*" 'get full' and "**öy-** *comb. of ööyi*" are each found, alphabetized appropriately.

This convention encodes information about the V~0 and VV~V alternations, so that dictionary users can create appropriate words when combining entries with affixes. The dictionary uses other such conventions as well, explained carefully in Hill et al. (1998: xvii-xviii).



Exploring my language

Does your language have lexical classes determined by whether sounds alternate? If you have found lexical classes, try one of these.

1. Work on your language description: Make a table for each lexical class, showing how the classes differ, like the two tables in (8) and (9). Add a statement describing the pattern for each class. Finally, list the roots you know that belong to each class.
2. Work on a dictionary entry: Think about ways to show the lexical class alternation in a dictionary entry, and write down entries for a member of each class. What works well? What might be confusing?
3. Work on a lesson plan: How would you teach the lexical class differences to students? Would you start with one pattern, then on another day show them the other pattern? Or would you do both patterns on the same day? Why do you think your method would be effective?

9.4.4 Summary

An important subtext of this discussion of alternations is that each alternation is fairly straightforward by itself.

With the $V \sim \emptyset$ alternation, there is a lexical class whose final vowel is missing when there is a following morpheme in the same word. We saw this with verbs and suffixes and we saw it with nouns and compounding.

With the $VV \sim V$ alternation, in that same lexical class, VV in an **open syllable** alternates with V in a **closed syllable**.

Finally, with the $[z] \sim [s]$ alternation, we find allomorph pairs where syllable-initial $[z]$ corresponds to syllable-final $[s]$.

Individually, each pattern is relatively easy to describe and understand. It becomes complicated when we look at word pairs like *tsooro* [tso:zo] ‘bluebird’ and *tsormasa* [tsoʃmasa] ‘bluebird wing’, where all three alternations together distinguish the allomorph [tso:zo] from [tsoʃ]. Understanding the three separate alternations shows that the quite disparate allomorphs are related in a regular way.

9.5 Completing the continuum: Related forms without phonological similarity

We have looked at phonological alternations, where a phonological pattern is driven by the phonological form. We have also looked at morpho-phonological alternations, where a phonological pattern is driven by a particular morphological context. We now turn to cases where there is no phonological pattern — everything is driven by the morphology.

In Hopi, if the subject of the verb is either one or two **actors** (singular or dual), there is usually no marker added to the verb. When there are three or more actors, a plural suffix *-ya* [-ja] ‘PLURAL.SUBJECT’ is added. Examples are given in (11): Subtractive analysis shows the final [-ja] in the plural column.

- (11) Subject number is marked on verbs if PLURAL

	SINGULAR/DUAL		PLURAL	
a.	<i>aniwna</i> [ʔaniwna]	<i>aniwnaya</i>	[ʔaniwnaja]	‘raise, perfect’
b.	<i>wuuwa</i> [wu:wa]	<i>wuuwaya</i>	[wu:waja]	‘think, consider’
c.	<i>ipwa</i> [ʔipwa]	<i>ipwaya</i>	[ʔipwaja]	‘take out (plural object)’

Once you’ve figured out the *-ya* [-ja] suffix to mark plural actors, if you hear the verb *pitu* [pitw] ‘arrive’, you might reasonably use *pituya* [pitwja], intending to mean ‘they arrive’. You would be wrong. The verb *pitu* [pitw] ‘arrive’ is unusual in that there is a completely different, unrelated verb form used when the subject is plural. We call this **suppletion**. A few examples of suppletive verbs from Hopi are given in (12).

- (12) Subject number is marked on verbs

	SINGULAR/DUAL		PLURAL	
a.	<i>pitu</i> [pitw]	<i>öki</i>	[ʔœki]	‘arrive’
b.	<i>wunima</i> [wunima]	<i>tiiva</i>	[ti:va]	‘be dancing’
c.	<i>qatu</i> [qatu]	<i>yeese</i>	[je:se]	‘sit, dwell’

The [-ja] suffix leads us to expect forms like **pituya* *[pitwja] when multiple people are arriving; instead Hopi speakers use the word *öki* [ʔœki]. (You may recall from Chapter 7 Section 7.1 that a * in front of a form means it is not a form used in the language by fluent speakers.) There is no phonological connection between the two forms — [pitw] and [ʔœki] are unrelated as far as sounds are concerned. We can say the same thing about *wunima* [wunima] and *tiiva* [ti:va]

‘be dancing’ and about *qatu* [qatu] and *yeese* [je:se] ‘sit, dwell’. There is no phonological relation between the singular/dual and plural forms of these verbs.

Reflecting on our discussion of Hopi up to this point, we can see that there is a continuum in the phonology-morphology interface. At one end, the role of morphology is to put morphemes together and the role of phonology, through distribution patterns, is to ensure that the combination is phonologically well-formed. In the middle of the interface between morphology and phonology are the morpho-phonological patterns. Morphemes are put together and phonology determines well-formedness within each lexical class. At the other end of the spectrum we find *suppletion*, where the role of morphology is to put morphemes together and there is no role for phonology.

In a sense, suppletion carries lexical classes to the morphological edge. When we encounter suppletive elements, the morphemes have to be memorized individually to know what sounds to use to express a particular meaning. Both lexical classes and suppletive pairings have to be memorized. With lexical classes, a group of morphemes shares a pattern while with suppletion, there is no pattern.



Exploring my language

List examples of suppletion in your language that you know of. Keep adding to the list as you learn more about your language.



The *domain* of a phonological pattern

So far, we have looked at phonological patterns in words. Another way of saying this is that the *domain* for the pattern is the word — we find the pattern inside a word. There are other domain types, and each of these categories can be limited to some lexical class.

- *Domain = word*. The pattern holds within words, regardless of internal morphological structure — that is, you see the pattern whether

the context is contained wholly within one morpheme or the context is created when two morphemes come together.

- *Domain = morpheme*. The pattern holds within individual morphemes but does not result in alternations. You may see this kind of pattern referred to as a *morpheme structure condition*. In these cases, you can get sequences that violate the pattern if the sequence is created by two morphemes coming together.
- *Domain = derived forms*. A *derived form* is a form composed of two or more morphemes. When the domain of a pattern is derived forms, the pattern holds only when the context is created by two morphemes coming in contact. It does not hold within morphemes themselves. You may see this kind of pattern referred to as a *derived environment condition*. You don't see this pattern within morphemes themselves, but when two morphemes come together, they may create the appropriate context for the pattern.

9.6 Other kinds of alternations you might find

There is a lot of exuberance in the way sounds behave when morphemes are combined to make words, sometimes in surprising ways. This section sketches a few of the permutations that languages make use of. You might want to skim the first paragraph of each section, to determine whether the type of pattern discussed is relevant in your language.

9.6.1 Let's do it again! Reduplication

Some morphemes don't have a particular shape of their own that is independent of the stem they are attached to. Instead, they take their phonological form from the stem itself. This general class of "self-affixation" is called *REDUPLICATION*. Examples of one kind of Hopi reduplication are shown in (13). The base word has a long vowel with a low tone in the first syllable, as in [kì:si] 'things that make shade'. (Vowels marked with a *grave accent*, like ì in [kì:si] have a falling tone in the Third Mesa dialect, going from a higher pitch to a lower pitch; other dialects treat these vowels differently (Hill et al. 1998: 864).)

The DISTRIBUTIVE PLURAL form of this word is *kiikiisi* [ki:kì:si]. Subtractive analysis shows us that the prefix is [ki:-], since that is what differentiates [kì:si]

and [ki:kì:si]. When we look at another word of this class, we might expect the DISTRIBUTIVE PLURAL to add the prefix [ki:-], but this is not what happens. Instead, with *ùutsi* [ʔù:tsi] ‘partition’, the DISTRIBUTIVE PLURAL is *uu’ùutsi* [ʔu:ʔù:tsi], not *[ki:ʔù:tsi]. Subtractive analysis says the prefix is [ʔu:-], not [ki:-].

(13) One way of making plurals in Hopi

	singular		plural/distributed	
a.	<i>kìisi</i>	[kì:si]	<i>kiikìisi</i>	[ki:kì:si] ‘things that make shade’
b.	<i>ùutsi</i>	[ʔù:tsi]	<i>uu’ùutsi</i>	[ʔu:ʔù:tsi] ‘partition’

We see that the form of the plural is dependent on the form of the stem — it is a copy of the first syllable (but without the tone). Subtractive analysis shows us the syllabic shape of the prefixes; the stem shows us what the segments of each prefix will be. This class of affixation is called *reduplication*.

Reduplication can take many forms. In Hopi alone, there are several other ways to reduplicate words. For instance, some words that are like *kìisi* and *ùutsi* in having a long initial syllable have the same kind of reduplicated prefix but when that syllable is added, there is a short vowel in the base word: *leena* [le:na] ‘flute’ has a long vowel in the prefix — *leelena* [le:le:na] — but a short vowel in the stem — [le:le:na]. Yet another option is to simply add a short syllable, as in *kikiipo* [kiki:po] ‘PLURAL.attack’. (We hope you are thinking “more lexical classes in Hopi”.) Hopi also has a number of ways to reduplicate stems that have an initial short vowel. In some languages, reduplication involves suffixation instead of prefixation. (Reduplicated forms can also have many different meanings. **PLURAL** and **DISTRIBUTIVE** are common meanings. In Hopi, **PLURAL** means 3 or more; **DISTRIBUTIVE** means something like ‘in multiple places’.)



Exploring my language

Does your language make use of reduplication? If you answer “yes”, list several forms that show the pattern — or patterns, putting each different pattern type in its own list.

9.6.2 Are all phonological word components morphological?

Hopi has several short words that end with the syllable [hu] which is not present when the word is suffixed: *soohu* [so:hu] ‘starlight’ but *sootu* [so:tu] (not *[so:hutu]) ‘starlight.PLURAL’. Subtractive analysis might suggest suffixes [-hu] and [-tu]. The meanings indicate that [-tu] means ‘PLURAL’ — there is both a change in meaning and in form. But what about [-hu]? There is no meaning attached to this syllable.

(14) Meaningless syllables

	orthography	IPA	gloss	
a.	<i>kiihu</i>	[ki:hu]	‘house’	
	<i>kii’at</i>	[ki:-ʔat]	‘his/her house’	(not [ki:hu-ʔat])
b.	<i>pöhu</i>	[pœhu]	‘road’	
	<i>pö’at</i>	[pœ-ʔat]	‘his/her road’	(not [pœhu-ʔat])

Some languages, and Hopi appears to be one of them, avoid words that are “too short”. Languages differ in what “too short” can mean: In Hopi it appears to be a single syllable (whether a **heavy syllable** or a **light syllable**). Languages also deal with this limitation in different ways. Some languages simply do not have monosyllabic words. It looks like Hopi speakers attach a phonological syllable to monosyllabic nouns. When a suffix is added, the word is disyllabic without the extra syllable, so the additional syllable need not be added.

Stems with two or more syllables are long enough that there is no need to augment their length.

(15) Longer roots don’t need -hu

<i>tsoongo</i>	[tso:ŋo]	‘smoking pipe’	(not [tsoŋohu])
<i>tsoongo’at</i>	[tso:ŋo-ʔat]	‘his/her smoking pipe’	

These added syllables look like morphemes, but don’t have any meaning attached to them. They fulfill a phonological well-formedness condition, not for morphological purposes. So even though they look like morphemes, they are not. If you read about “dummy syllables” or “augments”, it might be referring to this kind of pattern.

9.6.3 Is there variation with alternation?

Speakers in different social groups sometimes have their own dialect (recall **variation**, discussed in Chapter 4). One kind of variation is how a particular group

uses an alternation. In Hopi, there is a [p]~[v] alternation that is used differently by speakers depending on their age.

In Hopi, the sound [p] is found after consonants and the sound [v] is found between vowels. This is a morpho-phonological alternation, not the result of complementary distribution. According to Jeanne (1982), the [p]~[v] alternation is found in the speech of all Third Mesa speakers when either V~0 (and VV~V) or reduplication results in the consonant of some morpheme being either intervocalic [v] or post-consonantal [p].

With the possessive prefixes, however, we see a generational difference. The prefix meaning ‘my’ is vowel-final: Adding *i-* [ʔi-] ‘my’ to *sipmasmi* [sipmasmi] ‘bracelet’ gives us *isipmasmi* [ʔi-sipmasmi] ‘my bracelet’.

The prefix *i-* [ʔi-] ‘my’ ends with a vowel. What happens when it is added to a word that generally starts with [p]? Examples are given in (16) and (17) to show what older and younger speakers do.

Try this. Can you find the [p]~[v] difference between the speech of older and younger people?

(16) Older speakers

- | | | | |
|----|------------|--------------|----------|
| a. | ‘eye’ | <i>poosi</i> | [po:si] |
| | ‘my eye’ | <i>ivosi</i> | [ʔivosi] |
| b. | ‘knife’ | <i>poyo</i> | [pojo] |
| | ‘my knife’ | <i>ivoyo</i> | [ʔivojo] |

(17) Younger speakers

- | | | | |
|----|------------|--------------|----------|
| a. | ‘eye’ | <i>poosi</i> | [po:si] |
| | ‘my eye’ | <i>iposi</i> | [ʔiposi] |
| b. | ‘knife’ | <i>poyo</i> | [pojo] |
| | ‘my knife’ | <i>ipoyo</i> | [ʔipojo] |

Comparing (16) and (17) shows us that younger speakers use [p] in all these words. It is only the older speakers who show the alternation. For older speakers, the Hopi morpho-phonological [p]~[v] alternation is relevant whenever morphemes are put together to make words. For younger speakers, the alternation is not relevant for the possessive prefixes.

There are different lessons from this story. One lesson is that different social communities can use alternations in different ways, contributing to the systematic differences in the way speakers of each community sound — defining part of their phonology by the alternations they use.



A Mesa dialect difference

Jeanne (1982) and Hill et al. (1998) focus on Third Mesa Hopi, with brief mention of dialects spoken on the other mesas. Jeanne notes that Second Mesa Hopi speech is marked by the lack of the [p]~[v] alternation. Where Third Mesa speech has *hepni* [hepni] ‘will seek’, Second Mesa speakers have *hevni* [hevni]. This is another example of variation.

9.6.4 Are alternations limited to segments?

While most of our attention has been on segmental patterns, it is possible for [suprasegmental](#) properties to be involved in alternations as well. Hopi has a regular stress pattern within words. What this means is that the location of a word’s stress is calculated over the word, not over specific morphemes within the word. Consequently, if affixes change the length of the word, the location of the stress may shift with respect to the stem or root on which the word is built.

The essence of the Hopi stress pattern ensures that each word has one stressed syllable. Generally, stress falls on one of the first two syllables of the word. If the first syllable is a [heavy syllable](#) — CVV or CVC (remember Chapter 6?) — it is stressed, as in *Weenima* [‘we:.ni.ma] (place name) and *pàngqawu’u* [‘pàŋ.qa.wu.ʔu] ‘say it like that’. (The stressed syllable is marked by a preceding ‘. Syllables are separated by dots.) Otherwise, stress is on the second syllable, as in *nonani* [ɡno.‘na.ni] ‘badger’ and *kawayom* [ka.‘wa.jom] ‘horses’. Our focus is on regular stress in words with three or more syllables. Stress is rarely found on the final syllable. This means that in words of two syllables, the first syllable is stressed regardless of the syllable type, as in *sino* [‘si.no] ‘person’ and *Hopiit* [‘ho.pi:t] ‘Hopis’.

Since stress falls on one of the first two syllables of a word, adding a prefix can change stress in the stem. Stress is on the second syllable, [sa:], of the stem *pösaala’at* [pœ:‘sa:.la.ʔat] ‘his/her blanket’.

- (18) No prefix, stress on second syllable of stem
pösaala’at [pœ:‘sa:.la.ʔat] ‘his/her blanket’

When a CV- prefix is added as in (19), stress shows up on the second syllable of the word, as per usual in Hopi. The word’s second syllable is the first syllable

of the stem, so it looks like the stress shifts. In a way it does, if you focus on the stem alone. But if you look at the whole word, which is the domain of stress, we see that the location of stress is found exactly where we expect it — on the second syllable when the first syllable is light, as with the prefix [ʔi-] ‘my’. (Notice we also see the [p]/[v] alternation here. This was discussed in Section 9.6.3.)

- (19) CV prefix, stress on first syllable of stem
ivösaala [ʔi.vœ.sa:.la] ‘my blanket’

We know that stress falls on the initial syllable of words which begin with a heavy syllable. So we expect that if a prefix has a heavy syllable, that prefix will be stressed and all syllables in the stem will be unstressed. This is exactly what happens in Hopi, which has CVV prefixes, like [ʔúr:-] ‘your (sg.)’, shown in (20).

- (20) CV prefix, stress on first syllable of stem
úupösaala [ʔúr:.pœ.sa:.la] ‘your (sg.) blanket’

The Hopi example shows one way that stress patterns and morphological combinations interact. There are many other ways. We included this example as a reminder that there can be interaction and to give some ideas about how to sort out the patterns.



Exploring my language

Do you suspect that your language has an alternation involving stress or tone? If you answer “yes”, list several forms that lead you to this suspicion. These forms can be the beginning of an analysis of a suprasegmental alternation in your language.

9.6.5 Summary

This section looked at a few of the surprising ways that alternations show up in languages:

- reduplication in Section 9.6.1
- augments (or “dummy syllables”) in Section 9.6.2
- variation between older and younger speakers in Section 9.6.3
- a suprasegmental alternation in Section 9.6.4

Alternations show up in a lot of different ways in languages. The sketches here cover only a small fraction of them.

9.7 Discussion

We looked at **alternations** in this chapter, where a morpheme identified by subtractive analysis has two (or more) different forms, or **allomorphs**. Sometimes the different allomorphs are the result of a **phonological alternation**, due to following the phonological dictates of the language — **complementary distribution**, **syllable** structure, and other distributional patterns. In other cases, the different allomorphs are due to a sound difference that occurs in a context that is, at least in part, morphologically defined, the **morpho-phonological alternations**. In the case of **suppletion**, the different allomorphs are not related phonologically; we simply memorize which allomorph to use in which context — a purely morphological alternation.

With your understanding of this chapter, you are equipped to tackle alternations — the phonological consequences of morphological analysis. You have learned how all the skills developed in this book work together to bring you to this point. You know how to identify and understand the sounds of your language, their overall distribution in terms of inventories, syllables, and complementary distribution, and how they get put together to make words, through morphemic composition, sometimes with accompanying alternation. By stepping through these different aspects of phonological systems, you have learned the analytic skills necessary for each, and you’ve seen in this chapter how those same skills are used in understanding alternation. The questions we ask to determine which type of alternation we are looking at are summarized in Table 2.

9.7.1 Resolving the puzzle

We can now see how to resolve the puzzle from the introduction.

Is it normal for languages to have two almost-identical morphemes meaning exactly the same thing?

We now know that this is a completely normal situation. The two “almost-identical morphemes meaning exactly the same thing” are allomorphs, forms which differ from each other phonologically but with no meaning difference. Allomorphs arise due to both phonological considerations (like complementary

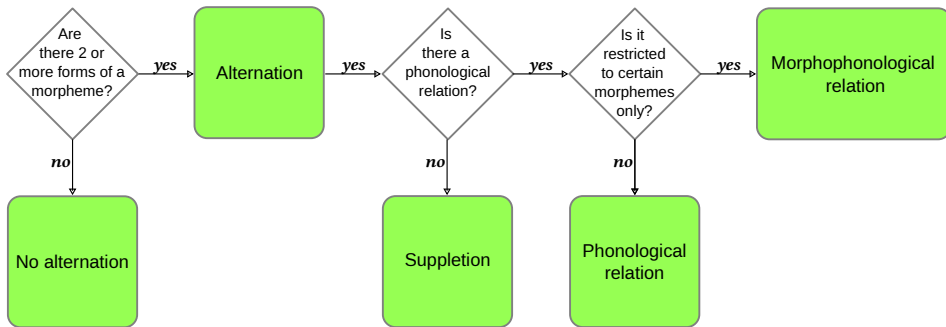


Figure 2: Types of alternation

distribution, syllable structure, other distributional effects) and morphological considerations (where a lexical class of morphemes follow some phonological pattern that is not general in the language).

In addition to knowing that this is completely normal, you also know how to carry out an analysis so that you can precisely describe each pattern of allomorphy.

9.7.2 Welcome to the world of alternations!

We chose to stick with one language in this chapter, to provide a sense of how a whole language fits its sounds and morphemes together. This necessarily limited the types of patterns to those that are found in Hopi. But even if we had used a different language in each section, we could not do justice to the range found in the world's languages.

For segmental alternations, ours in this chapter involved the voiced/voiceless pair [s]~[z] and the oral labial consonants, [p]~[v]. Reflecting back on Chapter 2, we know that languages have far more than one or two pairs of consonants; Chapter 7 showed us that not only pairs of sounds (vowels as well as consonants) but pairs of *sets* of sounds can appear in patterns (see, for example, Nakoda voiced and voiceless stops in Section 7.6.3). Any of these can be involved in an alternation – vowels, consonants, pairs, sets. And, as seen with complementary distribution in Chapter 8, those sounds or sets can define a context for a pattern as well as define the pattern itself (see, for example, the role of [+round] vowels in determining the complementary distribution of labial affricates in Djeoromitxi, in Section 8.4).

For suprasegmental alternations, some involve syllable structure like the V~0 and VV~V alternations in Hopi. Syllable-based alternations can affect consonants (for example, a C~0 alternation) or both vowels and consonants (for instance, CC can alternate with CVC). Some involve stress, as we saw with Hopi. Still others involve tone. Essentially, any sounds organized in some way in a language are fair game for alternation, either for the context or for the alternation itself.

With this much variety, identifying and analyzing alternation can feel daunting. But we've worked through the necessary steps already. For phonological background, it helps to know the sounds of our language, the syllables, and complementary distribution. It also helps to understand the morphemes in words, which we get by subtractive analysis. If subtractive analysis shows you that you have an alternating morpheme, the first thing to do is to try to find the same type of affixation or compounding in a word pair *without* alternation, if possible. When there is no alternation, you have a baseline for understanding the relation between roots and affixes. With that in hand, you can then turn to pairs differing by one alternation, to determine properties of that single alternation. Where more than one alternation occurs in a word, it is easiest to sort out the individual pieces before tackling the complex form. (A word of warning: Languages can be tricky, and it may be difficult or even impossible to find relevant forms without any alternations.)

9.7.3 Using my hard-earned knowledge

Let's now consider some of the ways that understanding alternations can be used in community language work.

Language learning. Each time a general pattern is learned, there is a leap in what a student can do. If words are taught one at a time, every word is an individual object to be learned, independent of all other words in the language. When a teacher shows how relations within words create relations between words — the common morphological structures — then students see the flow of their language fitting together. We know now that morphological structure can be obscured by alternations. With this chapter, a teacher can unravel the sound patterns for their students. Teachers have the skills to differentiate the one-off suppletive forms from the general patterns of alternation. Teachers are able to identify the in-between lexical class patterns as well — those that appear in one set of words, but not in all words. And so the teacher is better prepared to design class materials.

Where there is an alternation, a teacher might choose to begin with forms which to do not show the alternation, so that students are comfortable with the

morphological pattern. Once the general morphology is understood, forms with alternations can be introduced — a teacher might even present the alternation as a puzzle for their class to figure out. These patterns can be drilled, so that students become automatic in using the patterns. Where appropriate, a teacher might then bring up more affixes that follow the same alternation to make it clear that what the student has just learned will extend beyond one affix or affix class. With lexical classes, the challenge for students will be to remember which class a word belongs to; once class membership is known, the phonological part is an easy pattern.

A teacher can play word games with their class in different ways. Students can fill in a word missing from a sentence, where the missing word involves an alternation. Students can be challenged to change a word or sentence by one morpheme; the teacher can choose the morpheme so that an alternation is introduced. Students can be challenged to create new words from familiar morphemes — and get the sounds right because they know the alternation patterns. The imagination is the only limit on word play of this sort that can enliven the classroom while challenging the students.

Language resources — dictionaries, grammars, orthographies. There is an incredible number of decisions to be made when developing language resources; in this chapter we have introduced the concepts of lexical classes and alternations, two more decision-types, and we have shown how these concepts relate to complementary distribution from Chapter 8. With these tools, the language worker can make informed decisions about how — or whether — to show each in their dictionaries, grammars, and orthographies.

It can be very helpful to have a grammar sketch as part of a dictionary. Dictionary makers rely on an orthography — either one that they create or one that already exists. That orthography will treat complementary distribution either phonetically or phonemically.

A grammar sketch can explain what the pattern is and how it is being represented in the orthography and in dictionary entries. The grammar sketch can include a discussion of word formation — how affixes are put together with roots to make words. This explanation is useful in explaining how polymorphemic words are represented in the dictionary. Alternations can also be included in a grammar sketch, with examples to illustrate the patterns. (It works best to find forms illustrating only one alternation if possible.) The discussion of alternations is a background for explaining how alternations are represented in the dictionary.

Be aware that not everyone will read the grammar sketch or read the explanation of how entries are presented. This creates a risk if the orthographic system

is phonemic rather than phonetic, because users will assume that the phonemic orthography represents how words sound. If your language uses a phonemic writing system, you may wish to include reminders about the pronunciation of words throughout the resource rather than just at the beginning.

Language documentation and description. If alternations are left out of the description of a language, then the description is difficult to use as a guide to the language. For example, if you had a “description” of Hopi that made no mention of the V~0 alternation, you would be studying about something that wasn’t quite Hopi. Your speech would be quite puzzling to fluent speakers.



Who was it written for?

The issue of studying a source that does not directly show how words are pronounced may arise when using materials created by linguists for linguists, not for communities. Many linguistic descriptions, especially from the 1960s onwards, simply assume the concept of **underlying representation**. But you now know that underlying representations are not spoken forms — they are proposed as a common skeleton from which all spoken forms of a word can be determined. The underlying representations have to be translated back into **surface representations** to show how they are pronounced — by adding in sound properties determined by complementary distribution and by alternations.

Language revitalization. Revitalizing a language may involve nurturing a language based on very few written records — records that don’t do justice to the richness the language enjoyed. When the materials available do not include words for actions or objects that the community wishes to be able to talk about, the community may choose to create new words, i.e. **neologisms**. Knowing how to put words together by combining morphemes is an important step that was introduced in Chapter 3; doing so while using the sounds of the language in their correct distribution was discussed in Chapters 5 — 8. This chapter brings in the last piece: Incorporating any relevant alternations into **polymorphemic** neologisms, so the words sound right. This means that a community can experience rapid vocabulary expansion while sounding more like fluent speakers. Where words are completely missing from the vocabulary, it is possible to propose new

morpheme combinations (or even new morphemes) and combine them in ways that sound natural in the language because alternations are incorporated.

Reading linguist’s resources about your language. Linguists write a lot about alternations, because it is fascinating to see the many different ways that sounds are related to each other in different languages. The concepts of [underlying representation](#), [surface representation](#), and [derivation](#) figure prominently in writings by linguists (or the concepts of [input](#) and [output](#); see Section 9.2.3 for a review of the four terms). The discussion in this chapter gives you some tools for understanding technical writings that assume familiarity with these concepts.

As we have noted, there are some works that present most words in terms of underlying representations (without surface representations), and it is important to understand the allomorphy patterns if you wish to pronounce the words fluently. And you have now learned what is needed in order to do this.

9.8 Conclusion

Descriptions of languages may refer to alternations; by working through this chapter, you have an understanding of what an alternation is in general. You also understand the details of several specific alternations in Hopi.

Through this, you now have different tools to use in understanding the specifics of alternation in your language. You know that you start with morphemic analysis and that you need to identify patterns of complementary distribution. With that background, you are in position to determine whether a pattern of complementary distribution affects the pronunciation of individual morphemes, leading to allomorphs. You know that this kind of pattern is called a [phonological alternation](#).

You know that some alternations between sounds are not purely phonological, but occur only in certain morphological contexts. So if you discover a morpheme with different pronunciations, and see that the differences are not due to complementary distribution, you do not give up in despair. Rather, you look systematically at the morphology of your language to determine the contexts where the pattern occurs and does not occur. Sometimes these are morphological, sometimes phonological, and sometimes both: You know that a [morpho-phonological alternation](#) includes elements of both.

And you know that there are suppletive forms, where the alternants are lexical and there is no (morpho-)phonological alternation involved at all.

That is the new information in this chapter. But you also now know that you can bring the skills and strategies that you honed in the previous chapters to the

analysis of alternations. You are able to identify the sounds of your language, its syllable structure, the patterns of complementary distribution, and its morphological structure, using the same analytic steps introduced in Chapters 3 – 8, and then, with this background, you are able to turn those same skills to understanding alternation in your language.

Chapter 10

Conclusion: Looking back and going forward

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We began the book with two puzzles.

How can I get answers to questions about my language when I don't even know what questions to ask?

How do I create a description of the phonology — the sounds and sound patterns — of my language?

We hope that, having read this book, these two questions no longer challenge you. If that is the case, we have met our four goals in writing this book.

One goal of the book is to give you the knowledge to frame your questions and to understand the answers. Each chapter addresses a different aspect of phonological systems, providing the basic knowledge so that you are equipped to continue learning and understanding. We review key points in Section 10.1 of this chapter.

Another goal of the book is to give you the tools and skills needed to understand or create a sketch of the **phonology** of your language. This is done especially through the Language Explorations. The results of your explorations into your own language create a preliminary phonological sketch of your language. Additionally, those same skills equip you to extend the phonological sketch as much as you like. We go through a standard organization of a phonological sketch in Section 10.2 of this chapter.

The third goal of this book is to provide a reference to return to when needed, to review terms, concepts, or skills, or expand your understanding.

A final goal of the book is to inspire you to use the knowledge you have acquired in your language work. Each chapter gave several suggestions on ways the new skills might be used in a variety of different situations. We hope that as you read each chapter, you thought of other ways you could use the material, and even tried some of the applications yourself. In this chapter, we go a step further with using your hard-earned knowledge, addressing ways to make use of your phonological sketch of your language in Section 10.3.

Here's our plan. We review the analytic skills you have acquired first, in Section 10.1, and then lay out the structure of a phonological sketch, in Section 10.2. We then summarize the many ways that this knowledge can be used in community language efforts, in Section 10.3.

10.1 What you have learned to do

In this section, we review the analytic skills you have acquired, repeating some of the key summary figures from the book along the way.

10.1.1 Background skills, Chapters 2 – 4

We began this book with background information and context for understanding the rest of the book.

10.1.1.1 Sounds, Chapter 2

Chapter 2 addresses sound [segments](#) — how they are made, how they are described, how they are represented using phonological features, and the [International Phonetic Alphabet](#) (IPA) symbols used to indicate each sound. In considering each different property of sounds, we saw that features divide sounds into two classes, those with a particular property (e.g. [+voice] or [+nasal]) and those without that property (e.g. [–voice] or [–nasal]).

We began with consonants, looking first at [manner of articulation](#). Manner of articulation begins with the division between the [+sonorant] [sonorants](#) and the [–sonorant] [obstruents](#), based on whether airflow was regular ([+sonorant]) or turbulent ([–sonorant]).

Sonorant consonants are [+nasal] [nasals](#) (with air coming through the nose), [liquids](#) (the 'l' and 'r' sounds), and [glides](#) ([w] and [j], often spelled y). Sonorants are discussed in Section 2.3.1.4 of Chapter 2.

The [–sonorant] obstruents are divided by whether airflow is completely blocked, in the [–continuant] [stops](#), or whether airflow is released through a

narrow opening, in the [+continuant] **fricatives**. Fricatives themselves can be noisy [+strident] sounds or soft [–strident] sounds. Stops with a slow release have the feature [+delayed release]; these sounds are called **affricates**. Affricates sound like a stop and fricative smooshed together.

These binary options are shown in Figure 1. (As a reminder about the formatting in this figure, italics are used to emphasize the descriptive terms. By convention, phonological features and IPA symbols are both presented in square brackets.)

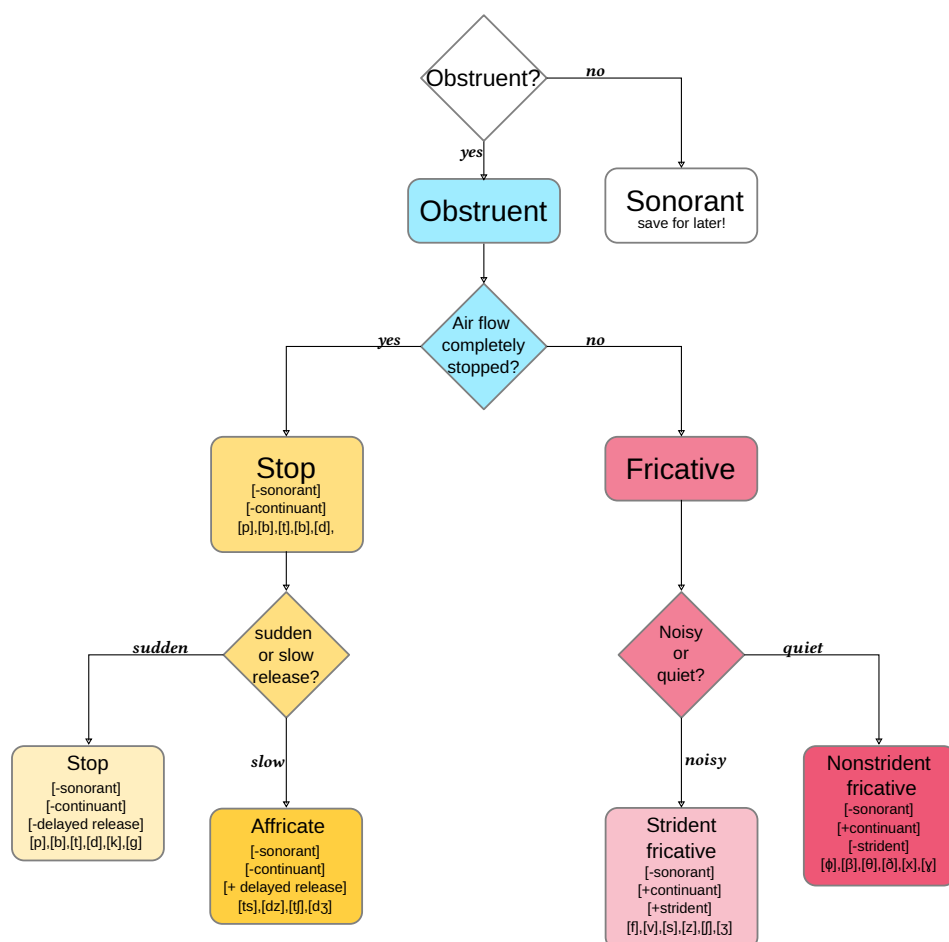


Figure 1: Identifying types of obstruents, identical to Figure 9 in Chapter 2

Consonants are also distinguished from each other by place of articulation — where in the mouth the greatest constriction is made, the topic of Section 2.3.2 in Chapter 2.

Constriction can be made with the lips, creating [labial] sounds; it can also be made with the tongue. If the tip of the tongue is involved, the sounds are [coronal] while if the back of the tongue is used, the sounds are [dorsal].

When we combine the place of articulation and the manner of articulation, there are a lot of different possibilities. Some of these are shown in Figure 2.

To help in identifying the different types of sounds, we use red for sounds involving the lips, yellow for sounds involving the tip of the tongue, and blue for sounds involving the back of the tongue.

The other major class of sounds is vowels, the topic of Section 2.4 in Chapter 2. The basic divisions for vowels come from tongue height ([±high], [±low]), tongue backness ([±back]), and lip position ([±round]). These four features serve to differentiate 12 different vowels (three heights, since [+high, +low] is not possible, and four combinations of [back] and [round]). In some languages, speakers manipulate the position of the root of the tongue, with advanced tongue root ([+advanced tongue root], or [+ATR]) or retracted tongue root ([−advanced tongue root], or [−ATR]). Some languages make use of a nasal/oral contrast in vowels ([±nasal]). We also find contrasts between long and short vowels, between vowels with and without glottalization of some sort, and between vowels with different pitch levels (called **tone** in linguistic writing).

10.1.1.2 Morphemes, Chapter 3

Chapter 3 examines how words can be created from smaller parts (**morphemes**), and how you can figure out the components in complex words, using subtractive analysis. The decisions to make are shown in Figure 3.

The first step is to determine whether the words you are looking at might be complex. This is done by comparing both the sounds and the meanings of two words, to see whether there is similarity in sound and similarity in meaning. We suspect a morphological relation when there is both a similar (or identical) sound sequence and a similar (or identical) meaning.

Once it appears that two words are morphologically related, you might want to figure out the component morphemes. In Section 3.3.1, we introduced **subtractive analysis** as a means for determining the morphemes within words. The basic idea is that you identify the similar parts of sounds and meanings that led you to believe the words were related. These are likely to be one morpheme. When

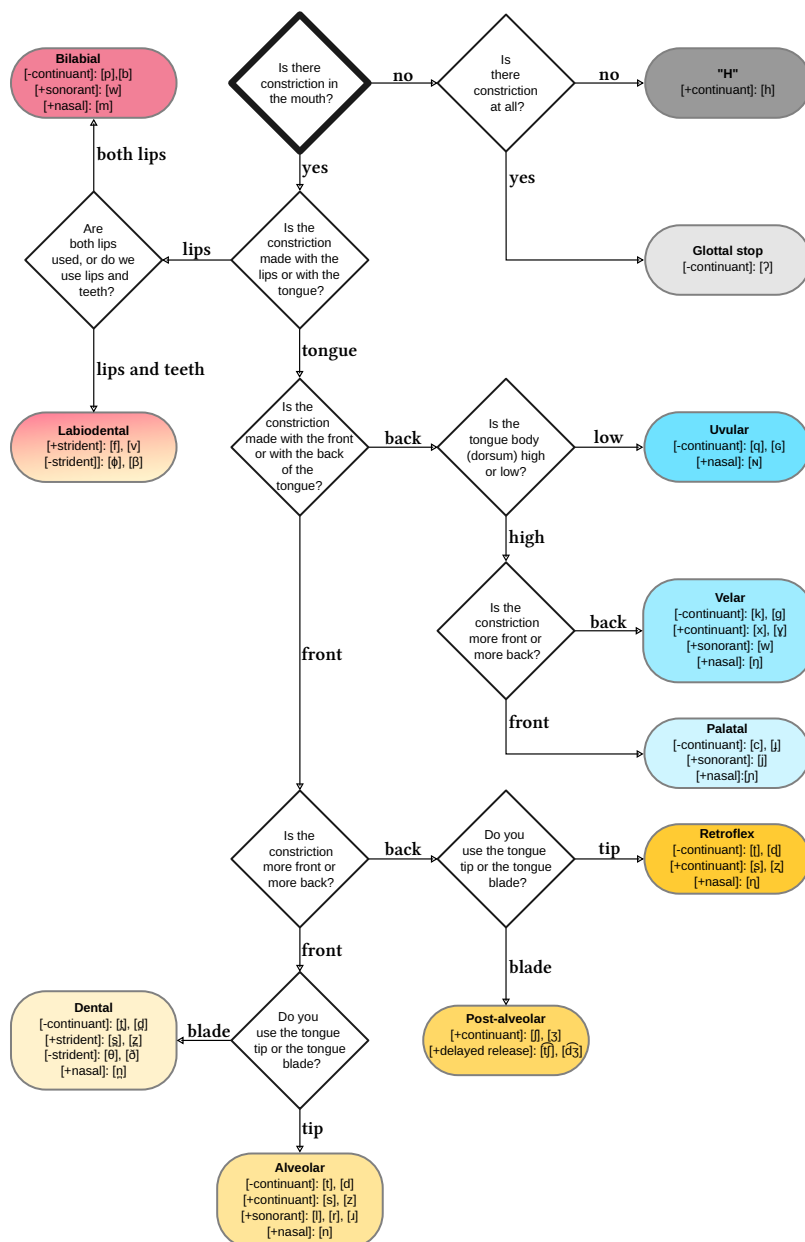


Figure 2: Identifying the place of articulation, identical to Figure 18 in Chapter 2

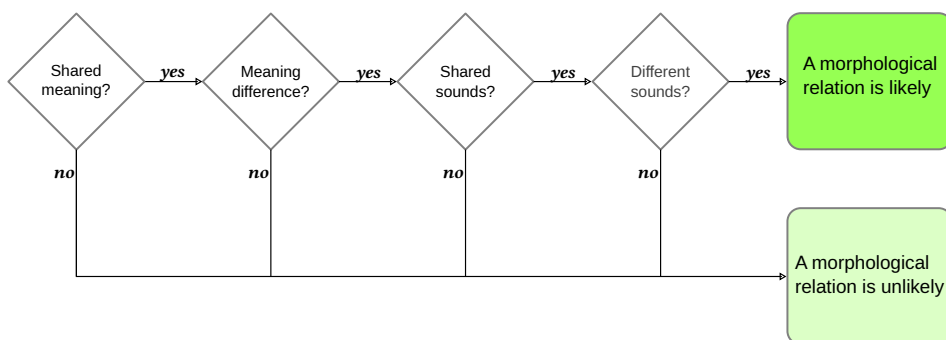


Figure 3: Are words morphologically related? identical to p. 149

you subtract those parts (both sound and meaning) from each word, what is left in each is the sounds and meaning of another possible morpheme.

You can test the analysis by seeking more words with the same morphemes — the same sound-meaning pieces that you just found. As explained in Section 3.4 from Chapter 3, it can help to think about the morphemes as either **roots** or **affixes**, so you know where to look for the identical parts in each new word you consider.

Once you have identified a few morphemes, you can use your analysis to find more affixes (by subtracting roots or stems) and more roots/stems (by subtracting affixes), as shown in Section 3.5 of Chapter 3. You can also create new words by combining roots/stems with affixes in new ways. This is particularly helpful when learning a language, to build the vocabulary. It can also prove useful when the language records are somewhat sparse, and you want to be able to say something for which there is no record.

Chapter 3 also explained standard conventions for representing morphemes in linguistic works, the **Leipzig Conventions** (Comrie et al. 2015). The Leipzig Conventions use three-line glosses, with small capital letters for **function** morphemes and hyphens to separate known morphemes. See Section 3.2 in Chapter 3 to review these conventions.

10.1.1.3 Variation, Chapter 4

Chapter 4 provides context about how the same word may sound different depending on the speaker, a phenomenon that may be important in your community.

Variation may be the result of different social groups speaking differently from each other, **dialectal variation**. (How these social groups are defined is very interesting, but our focus is on the way words sound, not on who uses different

sounds.) Another type of variation is **idiolectal variation**, differences between the speech of individuals in the same dialect community. Another way that variation works is sporadic, with variant pronunciations used sometimes — but not always — when people speak, **free variation**. The questions to ask in determining the type of variation are given in Figure 4.

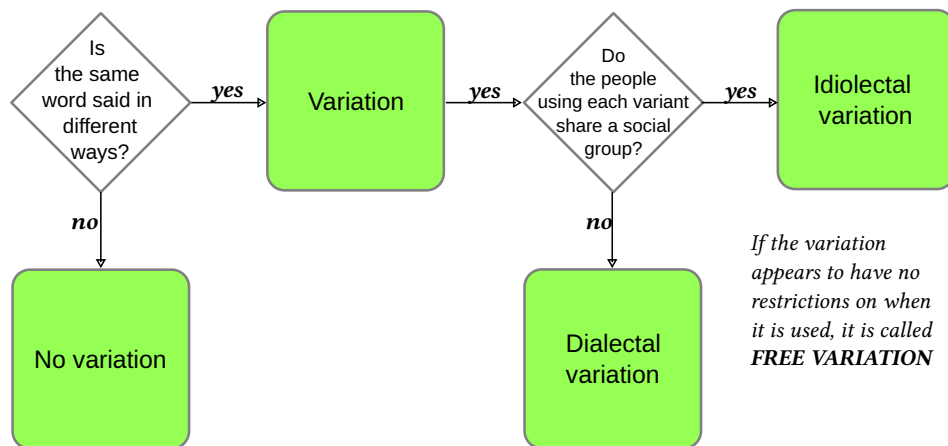


Figure 4: Types of variation, identical to Figure 1 in Chapter 4

It can be useful to know the dimensions of a pattern of variation, so that you know which words are likely to show variation. Chapter 4 lays out steps for determining these dimensions, once you have identified the sounds in the different **variants** of a word. These steps are summarized in Figure 5 and reviewed here.

To determine the precise dimensions of variation, it helps to start with the features of the variant sounds, to identify what is the same between them and what is different. The identical features identify the class of sounds that vary, the related class; the differences show the nature of the variation, or nature of the relation.

Once you know the class of sounds that vary, you can experiment to determine whether the class you have identified is the correct class, or whether it is too narrowly defined. You do this by removing features. For example, if you find that [labial, –continuant] sounds (labial stops) vary, you can remove the place feature to determine whether other [–continuant] sounds (other stops) also vary along the same dimension. And you can remove [–continuant] to determine whether other labials vary along the same dimension.

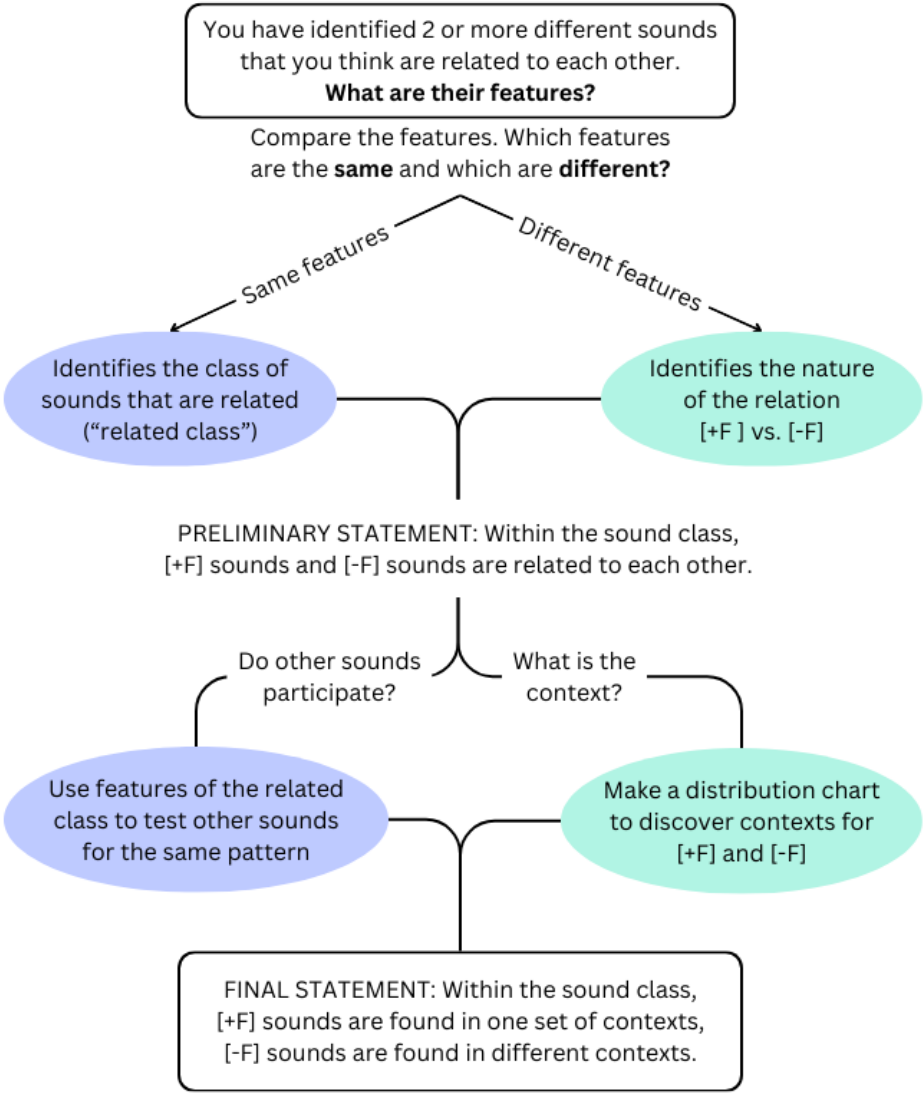


Figure 5: Determining the dimensions of variation, identical to Figure 2 in Chapter 4

Another refinement is to determine the location in the word where the variation occurs. In some cases, one dialect (or idiolect) will use a particular sound in all contexts while another dialect (or idiolect) uses a different sound. In other cases, this will happen only in a particular part of the word. Checking whether there is a location restriction is important.

The last piece of an analysis of variation is to make a clear, concise statement of the pattern you have found. This gives you a reminder of what you have done (it can be hard to remember the next day) and it gives you something to share with other people.

10.1.2 The sounds of a particular language, Chapters 5 – 7

The chapters in this part of the book address the sounds of a particular language, making use of the classifications of sounds introduced in Chapter 2. We go beyond what the sounds of a language are, and also look at their distribution in syllables and words.

10.1.2.1 Contrast, Chapter 5

The concept of **contrast** is central to language analysis. We have already seen contrast in the way sounds are classified – those that are [+voice] and those that are [–voice] contrast in voicing, for example. In Chapter 5, we use the concept of contrast to identify the sounds that are important in a language, namely those that contrast with each other.

The tool used to identify contrastive sounds is the **minimal pair** or **minimal set** – words that differ from each other by one sound and, ideally, the differing sounds differ from each other by only one feature. This allows us to identify not only the contrastive sounds of the language, but also the features that are critical to identifying those sounds, the **distinctive features** for the language.

Not all languages have minimal pairs or minimal sets to illustrate each contrastive sound pair. In these cases, we try to find **near-minimal pairs** or **context pairs** for sounds in the language. Chapter 5 explains how to find these different types of examples to illustrate the contrastive sounds in a language (Section 5.2.4) and how to find the critical distinctive features (Section 5.4.1).

Finding contrasting sounds in a language means that we have identified the **sound inventory** of the language. Chapter 5 Section 5.5 shows ways to represent the **consonant inventory** and the **vowel inventory** of a language. There are clear conventions for labeling rows and columns in inventory tables.

10.1.2.2 Syllables, Chapter 6

Syllables are a way of organizing consonants and vowels within the words of a language, into a rhythm alternating between less sonorant sounds and more sonorant sounds — roughly consonants and vowels, as shown in Chapter 6.

Each syllable has at its core a vowel, short or long, with the possibility of preceding and/or following consonants. The vowel is called the nucleus, the preceding consonants are called the onset, and the following consonants are called the coda.

All languages have syllables with onsets of at least one consonant and nuclei of at least one short vowel. Some languages also have syllables with onsets made up of more than one consonant and/or nuclei with more than one vowel (diphthongs). Codas are not found in every language; when they are found, they may be restricted to certain types of consonants (so not every consonant can be a coda). There are also typically limits on how many consonants can appear in a coda.

We find out about onsets and codas by looking at the edges of words (Section 6.2 of Chapter 6). Since each word begins with a syllable, then each word also may begin with an onset. (Words that start with vowels allow syllables without onsets.) Since each word ends with a syllable, then some words may end with a coda — if the language allows them. Languages whose words have no final consonants typically do not have any codas.

We can think about syllables in different ways, as shown by the different graphic representations, introduced in Section 6.4 of Chapter 6. People use different visual representations to highlight different aspects of the structure of their language.

10.1.2.3 Distribution, Chapter 7

The sounds of a language do not occur anywhere in a word. In general, sounds are found in particular places in words and not in other places, a phenomenon called the **distribution** of a sound. As shown in Chapter 6, consonants and vowels are strictly organized by syllables, a particular type of distribution. Distribution can also be determined by locations in a word that are not defined by syllables. In Chapter 7, we present ways to look at words so that we can find the distribution of different sounds.

To do this, we create a distribution table. Because of the contrast between consonants and vowels, it is relatively straightforward to create a distribution table that shows the various contexts where consonants are found (see Section 7.3

in Chapter 7), as well as one that shows the contexts for vowels (Section 7.4 in the same chapter).

We find words to fill in the cells in these tables; where there are gaps, we see that the distribution is limited in that context. The final step in an analysis showing limited distribution is a clear, concise statement of the pattern found.

In creating such a statement, it may be useful to return to the steps laid out in Figure 4 to find the features that define the class of sounds with limited distribution (the features that are the same) and the features that differ between the sounds found in different contexts (to identify the nature of the variation).

10.1.3 Sounds in action, Chapters 8 & 9

The final chapters look at the way sounds pattern within a language, both patterns that can be found in any word and patterns that are found in morphologically complex words.

10.1.3.1 Complementary distribution, Chapter 8

Complementary distribution is a particular type of distribution in which a pair of similar sounds is found in distinct contexts. When you build a distribution table, there is no overlap in the distribution of the sounds if they are completely complementary. The distribution table will look like the one in Table 1, where ‘Context A’ and ‘Context $\neg A$ ’ contain no overlapping contexts and no gaps in coverage of possible contexts: They are complementary. Sound class 1 occurs only in ‘Context A’, not in ‘Context $\neg A$ ’. Conversely, Sound class 2 occurs only in ‘Context $\neg A$ ’, and not in ‘Context A’.

Table 1: Schematic showing complementary distribution

	‘Context A’	‘Context $\neg A$ ’
Sound class 1	yes	no
Sound class 2	no	yes

The complementary contexts may be defined in terms of position in the word (word-final vs. not word-final, for example). They may also be defined in terms of features (before a [+round] vowels vs. before a [–round] vowel, for example). In figuring out a pattern of complementary distribution, we use the basic analytic strategies given in Figure 5. (These strategies were introduced in Chapter 4 and repeated in Chapter 7.) We identify the sounds we suspect to be in complementary distribution with each other, we examine their features to find what

is identical to determine the class of sounds in complementary distribution. We figure out the features that are different to determine the nature of the complementary distribution. Finally, we use distribution tables, introduced in Chapter 7 to figure out where each of the sounds are found.

It can be tricky to figure out which sounds to examine for complementary distribution. We discuss several strategies in Section 8.2 of Chapter 8 — clues from the sound inventory itself, from speaker intuitions or from existing resources, and clues from limited distribution that you may have identified.

10.1.3.2 Alternation, Chapter 9

When a morpheme has two or more phonological forms, we say that there is an **alternation**, or that the morpheme alternates. There are various types of alternation.

Phonological alternations result from syllable structure, complementary distribution, or some other distributional requirement that makes no reference to morpheme structure.

Morpho-phonological alternations arise when there is a phonological context for the alternation, but there is also some kind of morphological limitation — the pattern is found with some morphemes but not with all morphemes (even when they have the same phonological properties).

The third type of case is found when there are two phonological forms of a morpheme that bear no phonological connection to each other at all — there is no regular pattern. This type of morphological alternation is called **suppletion**. These categories are summarized in Figure 6.

When you identify an alternating morpheme, the next step is to figure out the dimensions of the alternation. This process is a familiar one, with one difference: We start with potential **allomorphs**, morphemes with the same meaning and with very similar sounds that we suspect are related to each other. Once we identify the morphemes, we can find the sounds that differ by comparing the allomorphs with each other. At that point, the familiar strategy takes over: The shared features define the class of alternating sounds while the differing features define the type of alternation. Further exploration shows whether the pattern extends to a larger class of sounds than the class originally identified, and determines any critical context for the pattern.

We used this same strategy with variation in Chapter 4, with distribution in Chapter 7, and with complementary distribution in Chapter 8. The only difference is that with alternation, we start with a pair of morphemes, not a pair of sounds. These steps are summarized in Figure 7. The purple text in the top boxes

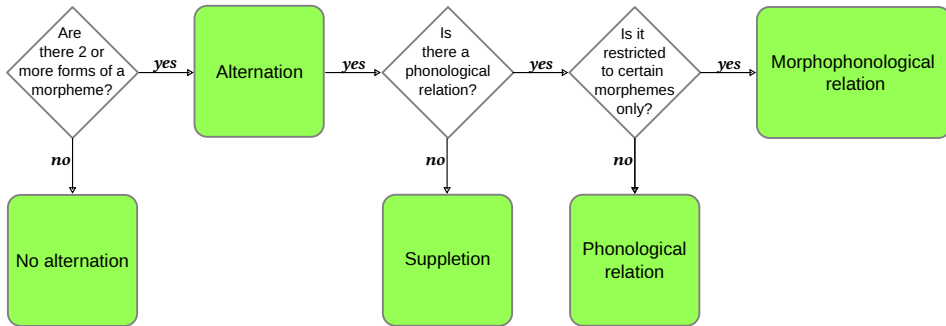


Figure 6: Types of alternation, identical to Figure 2 in Chapter 9

highlight the differences — we start with *morphemes* that we believe are related and identify the alternating *sounds* within those morphemes.

10.1.4 Summary

Phonological analysis requires knowing the sounds of a language, the features of those sounds, and the ways that sounds are related to each other. The analytic skills emphasized in this book repeat the same four steps, whether the analysis is addressing variation, distribution, complementary distribution, or alternation.

1. Identify the features that are the same between the sounds in order to define the class of sounds in the relation
2. Identify the features that are different between the sounds in order to characterize the sound relation
3. Determine whether the class of sounds is larger than initially thought
4. Determine whether there is a special context for the pattern

The phonological analysis of a language can be presented in a descriptive phonological sketch. A phonological sketch gives both the sounds of a language — the inventories — and the sound patterns of a language — syllable structure, the distribution of sounds, the alternations. By completing the Language Explorations in this book, you have created a basic phonological sketch of your language. We turn to phonological sketches in the next section.

10.2 What you have created

If you have kept up with most of the Language Explorations, you now have the basis for a phonological sketch of your language. A standard way of organizing

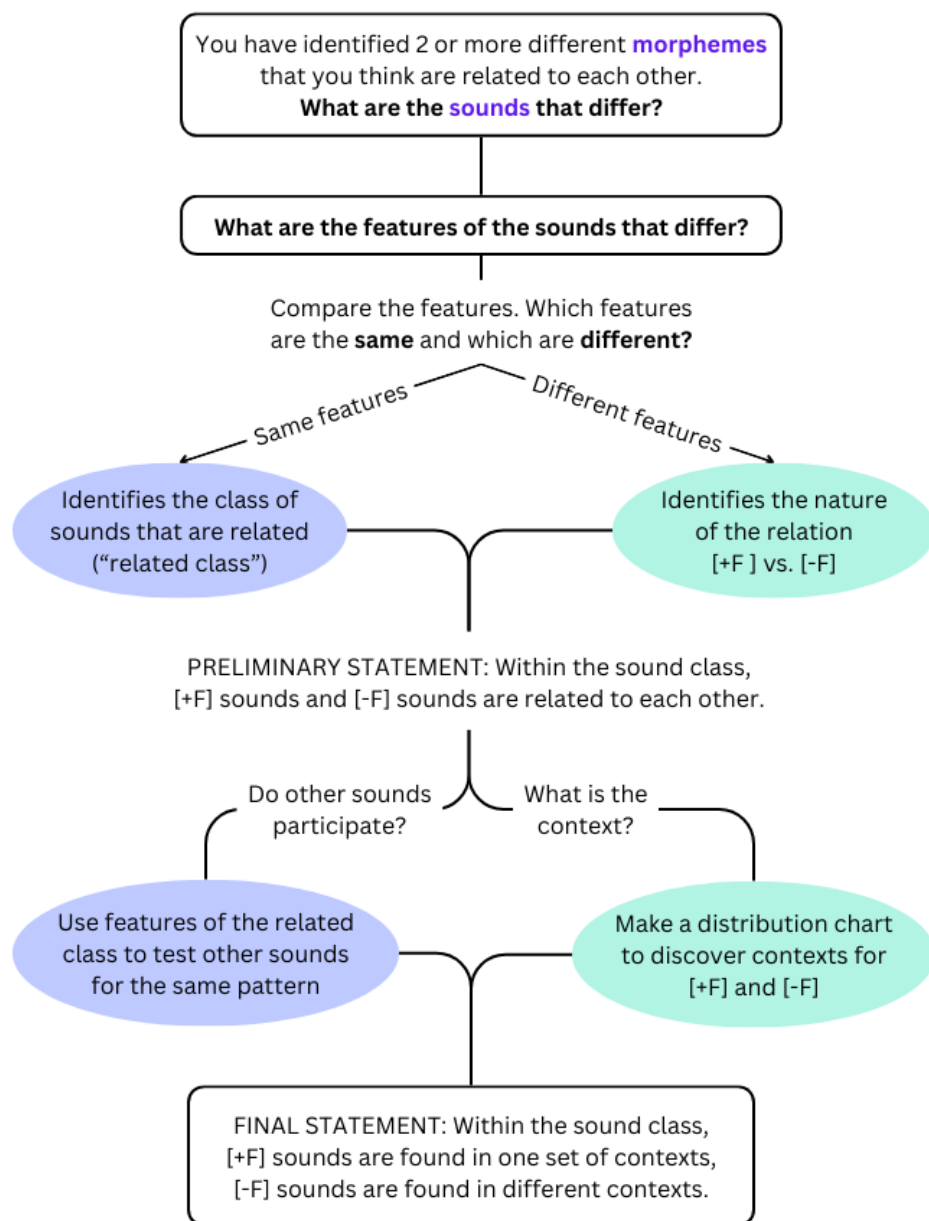


Figure 7: Determining the dimensions of alternation, identical to Figure 1 in Chapter 9

a phonological sketch is to first identify and describe the sounds of a language, then discuss the distribution of those sounds, and finally to show the relations between sounds (both complementary distribution and alternations).

A phonological sketch typically begins with a section (or a chapter, depending on the scope of your project) that introduces the language and its historical and social context. This introductory chapter may also explain how the information on the language was collected and other points of interest. It usually will also outline the rest of the sketch, so that your readers know what to expect.

After the introduction, the phonological sketch typically turns to the contrastive sounds of the language (in either a section or a chapter). In your case, this will be the sounds that you identified in Chapter 5, when you looked for minimal pairs to demonstrate the contrastive sounds of your language. You can present the sounds in tables showing the inventories of consonants and of vowels with columns and rows labeled to show features. It is most helpful to readers if you also include words that show the contrast — the minimal pairs, near-minimal pairs, and the context pairs that you found. Your exposition of the sounds of the language can be augmented by material you put together in the Language Explorations of Chapter 2, telling how the sounds are made, showing their symbols, and including example words. If there are any unusual sounds in your language, you may want to think about ways to help your readers understand how to produce them.

The next part (one or two sections or chapters) of a phonological sketch usually talks about the distribution of sounds in words. The Explorations from Chapter 6 show the distribution of consonants and vowels in syllables while the ones from Chapter 7 look at where specific sounds are found, when their distribution is restricted.

The last major part of a phonological sketch shows how different sounds relate to each other, either through variation, complementary distribution, or alternation. You have explored your language in each of these areas, in Chapters 4, 8, and 9 respectively, so you are well-prepared for this section as well. As you discuss alternations, you may need to explain some of the basic morphological patterns in your language, relying on the Language Explorations from Chapter 3. (Again, this can be put together in chapters or in sections, depending on the overall scope of your project.)

Customarily, a phonological sketch ends with a concluding section or chapter. The conclusion often summarizes material that is presented. It may also highlight some of the more interesting points that you found as you explored the phonology of your language. It also is very helpful to include a discussion of parts of the phonological system that you haven't covered yet.

This skeleton is re-apped in (1).

- (1) The parts of a phonological sketch
 - a. Introduction
 - b. The consonant and vowel inventories, Chapters 2, 5
 - c. The distribution of sounds within words, Chapters 6 and 7
 - d. Relations between sounds, Chapters 8, 9, and possibly 3 and 4
 - e. Conclusion

10.3 Using my hard-earned knowledge

Once you have created a phonological sketch of your language, what can you do with it? Here are some ideas; we hope they inspire you to more ways to use your sketch.

Language learning. Whether you are a language teacher or a language learner, there is a role for the phonological sketch. The overview of sound patterns that the sketch provides shows teachers the scope of the challenges that face their students in mastering the sound patterns. This enables the teacher to make intentional decisions about the sequence used in teaching different aspects of the sound system, in particular to go from easier and clearer patterns to more complex ones. A few examples follow to give you a concrete idea of what we mean:

- Introduce familiar sounds before unfamiliar ones (for example, back round vowels before front round vowels)
- Make students familiar with an allophonic pattern before introducing an affix which alternates based on that allophonic pattern
- Combine a root and a suffix where there are no alternations before introducing words with the alternations.

Students who have access to a phonological sketch of their language have a tool for seeing how the sounds fit together to make words. A good description will include a discussion of how sounds are pronounced, which a student can use to improve their own pronunciation. It will show how sounds are sequenced in words, so a student can identify any difficult sequences as ones to practice a lot to improve fluency. It will include explanations of allophones and alternations so a student is better prepared to pronounce new words, even **polymorphemic** words.

Language resources — dictionaries, grammars, orthographies. A phonological sketch contributes to dictionaries in two ways. Many dictionaries include

a phonological sketch to give a general picture of how sounds are used in the language. This helps users pronounce words more correctly. For example, if a writing system is based on the [phonemic representation](#), the sketch can explain the use of the different [allophones](#) so that users of the dictionary are more able to pronounce words correctly.

A sketch can guide the construction of dictionary entries, to make them more informative without making them excessively long. For example, when roots or stems alternate when certain affixes are added, the sketch can include a discussion of the way the alternate forms are used, so that this discussion does not get repeated in every dictionary entry. As an example, recall that in Hopi (Chapter 9) some stems are always vowel-final, like [maqa] ‘give’, while others alternate between a consonant-final and a vowel-final form, like [som]~[soma] ‘tie’. In such a case, dictionary entries can include either one form or two while the sketch tells when to use each form if there are two forms, rather than repeating that information in each entry.

A second example is to do the same with alternating affixes. The affix alternations can be explained in the sketch. Dictionary entries can show roots and stems; users can add affixes according to the discussion in the sketch. The entries are shorter because there is no need to repeat the addition of each affix with each root or stem entry.

A complete grammar includes a phonological sketch as well as discussion of how words and sentences are put together, so a phonological sketch is important if you are putting together a grammar.

If you need to develop an orthographic system for your language, the sketch can help in knowing which sounds are most suited to having their own symbols – the [contrastive](#) sounds. It can also help in understand which sets of sounds might best be represented with a single symbol – the [allophones](#).

Language documentation and description. When trying to document a language, one challenge is knowing what material you want to be sure to document. The phonological sketch can provide a checklist in this endeavor, giving an overview of the sounds and their use in words. As recordings are made and annotated, the researcher can determine whether some patterns are missing and try to devise means of filling in the missing records.

Conversely, while documenting a language, the annotations made may reveal patterns that had not been noted before. It is easier to identify these discoveries if there is already a sketch in hand, so that you are more aware when something surprising comes up.

Language revitalization. In working with a language with incomplete records, it is tempting to adapt words from another language to fill in gaps in the word-lists that are available. A good phonological sketch provides a means of filling those gaps with new words that have a connection to the language itself.

Some languages are quite free with making new words by combining morphemes to express novel meanings. In such languages, the phonological sketch can show whether the novel words correctly use the alternations found in the language. Sometimes it is more appropriate to make up a word; the sketch gives guidelines for how to put sounds together to make words that sound like words of the language. Another source is to borrow a word from a related language; the phonological sketch can tell how to adapt a word from another language to meet the distributional patterns and allophonic patterns of your language.

Reading linguist's resources about your language. A lot of writing by linguists makes use of concepts like [phonemes](#), [allophones](#), [underlying representation](#), [syllables](#), [complementary distribution](#), and [alternations](#), without explanation. You may find something written about your language which includes statements about underlying representations or syllables or alternations in your language yet it does not include any explanations. Such writings can be very confusing to read. However, if you have a phonological sketch of your language, then you have a resource that explains the phonemes of your language, complementary distribution and allophones, syllables, and alternations. A phonological sketch of your language is a resource that gives examples and reasoning. It gives background that is often omitted in linguistic writings, especially writings that are not descriptive discussions.

10.4 Conclusion

We hope that this book has satisfied goals that you set for yourself when you chose to read it.

You are now prepared to complete the phonological sketch of your language that you have been working on while reading this book. We hope you are excited about that.

You have read about ways to use the material in this book as you expand your work with your language to support your community. We hope you are inspired to try new approaches.

We expect that you have not read the entire book, but have skipped some sections because they did not seem relevant to you at this time. We hope that

you have seen that even the parts of the book that you did not read may serve as a reference for you to enhance your efforts.

If you are now confident about asking questions about the sounds of your language, if you are now confident going forward to complete a sketch of the phonology of your language, if you have found ways to use your knowledge in your role as a language worker, then we, too, have succeeded with our goals for this book.

Appendices

Appendix A

International Phonetic Alphabet (IPA) symbols

The lists here show the [International Phonetic Alphabet](#) (IPA) symbols for all the sounds used in this book, along with some common [Americanist Phonetic Notation](#) (APN) symbols for the same sounds. In the rightmost column, there is a brief description of the sound, and the features that distinguish the sound from others in its class.

In each feature classification, we omit several features. There is an explanation at the beginning of each major section about the features that are omitted in that section. Generally, some features are omitted because they are the same for the entire class of segments in the list. Others are omitted because they can be determined by looking at the features that are included. Still others are omitted because they can vary for the segments. Generally, such enhancements are indicated with diacritics. There are sections diacritics (vowel diacritics in section A.1.4 and consonant diacritics in section A.2.5).

A.1 Vowel symbols

Here is an explanation of the features we omit in our classification of vowels.

Major class features. Vowels are typically both [+sonorant] and [+continuant].

Some sources also mark them as [–consonantal] and/or as [+vocalic].

Predictable features. The high vowels are always non-low, so for [+high] vowels, we leave out [–low]. Similarly, low vowels are always non-high, so for the [+low] vowels, we leave out [–high].

Enhancing features. Vowels are standardly [–nasal] and [+voice], though both nasal vowels ([+nasal]) and voiceless vowels ([–voice]) are found in the world's languages. Other enhancing features are glottal configurations, length, tone, and stress. The diacritics for features that enhance vowels are found in section A.1.4.

A.1.1 High vowels

High vowels are always non-low, so we omit [–low] for these vowels. High vowels usually have advanced tongue root ([+advanced tongue root]). We only show a tongue root contrast on the vowels [i], [u] (which are advanced) and [ɪ], [ʊ] (which are retracted). All other vowels shown here are classed as [+advanced tongue root].

IPA	APN	description & features
[i]	[i]	A high, or close, front non-round vowel. [+high, –back, –round]
[y]	[ü]	A high, or close, front round vowel. [+high, –back, +round]
[ɪ]	[ɪ]	A high, or close, central non-round vowel. [+high, –back, –front, –round]
[ʊ]	[ʊ]	A high, or close, central round vowel. [+high, –back, –front, +round]
[ɯ]	[ɨ]	A high, or close, back non-round vowel. [+high, +back, –round]
[u]	[u]	A high, or close, back round vowel. [+high, +back, +round]
[ɪ]	[ɪ]	A retracted high, or close, front non-round vowel. [+high, –back, –round, –advanced tongue root]
[ɤ]	[ɵ]	A retracted high, or close, front round vowel. [+high, –back, +round, –advanced tongue root]
[ʊ]	[ʊ]	A retracted high, or close, back round vowel. [+high, +back, +round, –advanced tongue root]

A.1.2 Mid vowels

We only show a tongue root contrast on the vowels [e], [o] (which are advanced) and [ɛ], [ɔ] (which are retracted). All other vowels shown here are classed as [+advanced tongue root].

IPA	APN	description & features
[e]	[e]	A mid front non-round vowel. [–high, –low, –back, –round, +advanced tongue root]
[ø]	[ö]	A mid front round vowel. [–high, –low, –back, +round]

[ɤ]	[ë]	A mid back non-round vowel. [-high, -low, +back, -round]
[ə]	[ə]	A mid central non-round vowel. [-high, -low, -back, -front, -round]
[ɵ]	[ð̥], [ɵ]	A mid central round vowel. [-high, -low, -back, -front, +round]
[o]	[o]	A mid back round vowel. [-high, -low, +back, +round, +advanced tongue root]
[ɛ]	[ɛ]	A retracted (or lower) mid front round vowel. [-high, -low, -back, +round, -advanced tongue root]
[ɔ]	[ɔ]	A retracted (or lower) mid back round vowel. [-high, -low, +back, +round, -advanced tongue root]

A.1.3 Low vowels

Low vowels cannot also be high vowels, so [-high] is omitted in our feature classifications of low vowels. Low vowels are generally retracted, so [-advanced tongue root] is also omitted, except in the case where [-advanced tongue root] [a] contrasts with [+advanced tongue root] [æ].

IPA	APN	description & features
[æ]	[æ]	A fairly low, or near-open, front, non-round vowel. [+low, -back, -round, +advanced tongue root]
[œ]	[ɔ̃]	A low, or open, front round vowel. [+low, -back, +round].
[a]	[a]	A low, or open, front, non-round vowel. [+low, -back, -round, -advanced tongue root]
[ɐ]	[a]	A low, or open, central, non-round vowel. [+low, -back, - front, -round]
[ɑ]	[ɑ]	A low, or open, back, non-round vowel. [+low, +back, -round]
[ɒ]	[ɒ]	A low, or open, back, round vowel. [+low, +back, +round]

A.1.4 Diacritics used with vowels, illustrated with [o]

IPA	APN	description & features
[õ]	[ɔ̃]	A nasal vowel; [+nasal]
[ɔ̥]	[ɔ̥]	A voiceless vowel; [-voice]

[oʔ]	[oʔ], [aʔ]	A glottal vowel; [+constricted glottis]
[ɔ]	[ɔ]	A creaky vowel.
[ɔ̃]	[ɔ̃]	A murmured vowel.
[o:]	[o:]	A long vowel.
[ó]	[ó]	A high-toned vowel.
[ò]	[ò]	A low-toned vowel.
[ō]	[ō]	A mid-toned vowel.
[õ]	[õ]	A vowel with rising tone.
[ô]	[ô]	A vowel with falling tone.
[ˈo]	[ó]	A vowel with primary stress.
[ˌo]	[ò]	A vowel with secondary stress.

A.2 Consonant symbols

Here is an explanation of the features we omit in our classification of consonants.

Major class features. Consonants are typically both [+consonantal] and [−vocalic]. Obstruents (the oral stops, fricatives, and affricates) are [−sonorant] and [−nasal]. The sonorants (nasals, liquids, and glides) are [+sonorant] and [+voice]. Only nasal sounds are [+nasal]. We omit [±sonorant] classification in the features, but mention in the description of each sound whether it is an obstruent or a sonorant.

Predictable features. Predictable features vary with the manner of articulation and with the place of articulation. We have organized the sounds here by place of articulation, so discuss the predictable classifications based on the manner of articulation here. We address the predictable classifications based on place of articulation in those sections.

[+sonorant]. Sonorants are [+voice] and [−strident].

[−continuant]. The only stops that might be classed as [+strident] are the affricate stops (segments that are classed as [+delayed release]). We do not give [±strident] values except with fricatives where it is contrastive. Also, we only show [+delayed release]. All other sounds are [−delayed release].

Enhancing features. Laryngeal features (aspiration and glottalization), place features (palatalization and labialization), and length are shown with diacritics in section A.2.5.

A.2.1 Labial consonants

Labial consonants, made with lip constriction of some sort, do not engage the tongue and so features like $[\pm\text{high}]$, $[\pm\text{low}]$, and $[\pm\text{back}]$ are irrelevant and omitted. Labial sounds are necessarily $[\pm\text{anterior}]$, so that feature is also omitted.

IPA	APN	description & features
[p]	[p]	A labial, or bilabial, voiceless obstruent stop. [labial, –continuant, –voice]
[b]	[b]	A labial, or bilabial, voiced obstruent stop. [labial, –continuant, +voice]
[m]	[m]	A labial, or bilabial, nasal sonorant stop. [labial, –continuant, +nasal]
[f]	[f]	A labial, or labio-dental, voiceless obstruent fricative. [labial, +continuant, +strident, –voice]
[v]	[v]	A labial, or labio-dental, voiced obstruent fricative. [labial, +continuant, +strident, +voice]
[ɸ]	[ɸ]	A labial, or bilabial, voiceless obstruent fricative. [labial, +continuant, –strident, –voice]
[β]	[β]	A labial, or bilabial, voiced obstruent fricative. [labial, +continuant, –strident, +voice]
[w]	[w]	A labial, or labio-velar, sonorant glide. [labial, dorsal, +continuant, +sonorant]

A.2.2 Coronal consonants

Because coronal consonants are made with the agile tongue tip, we show values for $[\pm\text{anterior}]$ (whether the tongue is forward or more back in the mouth) and for $[\pm\text{distributed}]$ (whether the tip is pointed or flat). We also show whether a fricative is $[\pm\text{strident}]$, but do not show strident values on stops.

IPA	APN	description & features
[t]	[t̪]	A voiceless dental obstruent stop. [coronal, +anterior, +distributed, –continuant, –voice]
[d]	[d̪]	A voiced dental obstruent stop. [coronal, +anterior, +distributed, –continuant, +voice]
[n]	[n̪]	A voiceless nasal sonorant stop. [coronal, +anterior, +distributed, –continuant, +nasal]
[s]	[s̪]	A voiceless dental obstruent fricative. [coronal, +anterior, +distributed, +continuant, –voice]

[z̥]	[z̥]	A voiceless dental obstruent fricative. [coronal, +anterior, +distributed, +continuant, +voice]
[t̥]	[t̥]	A voiceless alveolar obstruent stop. [coronal, –continuant, –voice]
[d̥]	[d̥]	A voiced alveolar obstruent stop. [coronal, –continuant, +voice]
[n̥]	[n̥]	A nasal alveolar sonorant stop. [coronal, –continuant, +nasal]
[θ̥]	[θ̥]	A voiceless interdental obstruent fricative. [coronal, +anterior, –strident, +continuant, –voice]
[ð̥]	[ð̥]	A voiced interdental obstruent fricative. [coronal, +anterior, –strident, +continuant, +voice]
[s̥]	[s̥]	A voiceless alveolar obstruent fricative. [coronal, +anterior, +strident, +continuant, –voice]
[z̥]	[z̥]	A voiced interdental obstruent fricative. [coronal, +anterior, –strident, +continuant, +voice]
[t̥s̥]	[c], [ts̥]	A voiceless alveolar obstruent affricate. [coronal, –continuant, +delayed release, –voice]
[d̥z̥]	[j̥], [ʒ̥]	A voiced alveolar obstruent affricate. [coronal, –continuant, +delayed release, +voice]
[l̥]	[l̥]	A sonorant lateral. [+lateral, –continuant]
[t̥ɬ̥]	[ɬ̥]	A lateral affricate. [coronal, +lateral, +delayed release]
[ɬ̥]	[ɬ̥]	A lateral fricative. [coronal, +lateral, +continuant]
[ɾ̥]	[ɾ̥]	A sonorant tap (or flap). [coronal, +sonorant, –nasal, –lateral]
[r̥]	[r̥]	A sonorant trill. [coronal, trill]
[t̥̠]	[t̥̠]	A voiceless retroflex obstruent stop. [coronal, –anterior, –distributed, –continuant, –voice]
[d̥̠]	[d̥̠]	A voiced retroflex obstruent stop. [coronal, –anterior, –distributed, –continuant, +voice]
[ɳ̥]	[ɳ̥]	A nasal retroflex sonorant stop. [coronal, –anterior, –distributed, –continuant, +nasal]
[ʂ̥]	[ʂ̥]	A voiceless retroflex obstruent fricative. [coronal, –anterior, –distributed, +continuant, –voice]
[ʐ̥]	[ʐ̥]	A voiced retroflex obstruent fricative. [coronal, –anterior, –distributed, +continuant, +voice]

A.2.3 Dorsal consonants

Dorsal consonants are all made in the back of the mouth. We omit [$\pm$ anterior], [$\pm$ strident], and [$\pm$ distributed] because there are no contrasts characterized by these features when sounds are made with the back of the tongue.

IPA	APN	description & features
[ʃ]	[š]	A palatal voiceless obstruent fricative. [dorsal, +high, –back, +continuant, +strident, –voice]
[ʒ]	[ž]	A palatal voiced obstruent fricative. [dorsal, +high, –back, +continuant, +strident, +voice]
[tʃ]	[č]	A palatal voiceless obstruent affricate. [dorsal, +high, –back, –continuant, +delayed release, –voice]
[dʒ]	[ǰ]	A palatal voiced obstruent affricate. [dorsal, +high, –back, –continuant, +delayed release, +voice]
[c]	[kʷ]	A palatal voiceless obstruent stop. [dorsal, +high, –back, –continuant, –voice]
[j]	[gʷ]	A palatal voiced obstruent stop. [dorsal, +high, –back, –continuant, +voice]
[ɲ]	[ɳ]	A palatal nasal sonorant stop.[dorsal, –back, +high, –continuant, +nasal]]
[j]	[y]	A palatal sonorant glide. [+sonorant, +continuant, dorsal, +high, –back]
[k]	[k]	A dorsal, or velar, voiceless obstruent stop. [dorsal, –continuant, –voice]
[g]	[g]	A dorsal, or velar, voiced obstruent stop. [dorsal, –continuant, +voice]
[ŋ]	[ŋ]	A dorsal, or velar, nasal sonorant stop. [dorsal, –continuant, +nasal]
[x]	[x]	A dorsal, or velar, voiceless obstruent fricative. [dorsal, +continuant, –voice]
[ɣ]	[ɣ]	A dorsal, or velar, voiced obstruent fricative. [dorsal, +continuant, +voice]
[q]	[q]	A uvular voiceless obstruent stop. [dorsal, –high, –continuant, –voice]
[ɢ]	[ɢ]	A uvular voiced obstruent stop. [dorsal, –high, –continuant, +voice]
[ɴ]	[ɳ]	A uvular nasal obstruent stop. [dorsal, –high, –continuant, +nasal]

A.2.4 Glottal consonants

Glottal consonants are often considered [+sonorant], but sometimes they pattern with the [–sonorant] sounds. Because glottal sounds emanate from the position of the glottis, the oral tract is not involved; all oral place features are omitted in these feature classifications.

IPA	APN	description & features
[ʔ]	[ʔ], [ʔʔ]	A glottal stop. [+constricted glottis, =continuant, –voice]
[h]	[h]	A glottal fricative. [+spread glottis, +continuant, –voice]

A.2.5 Diacritics used with consonants, illustrated with [t]

IPA	APN	description & features
[t ^h]	[t ^h]	An aspirated consonant. Add [+spread glottis] to other features.
[tʔ]	[tʔ]	A glottalized consonant. Add [+constricted glottis] to other features.
[tʲ]	[tʲ]	A palatalized consonant. Add [+high, –back] to other features.
[tʷ]	[t]	A labialized consonant. Add [+round] to other features.
[t:]	[t:]	A long consonant. Phonologically shown by syllable structure.

Appendix B

Distinctive features

B.1 Definitions of distinctive features

[±advanced tongue root] Defines two classes of vowels, the [+advanced tongue root] vowels made with the tongue root forward in the mouth and the [–advanced tongue root] vowels made with the tongue root pulled back in the mouth. This feature is often abbreviated [±atr]. See p. 113.

[±anterior] A place of articulation feature that defines the class of sounds made in the front part of the mouth. See p. 73.

[±back] Defines the class of sounds with the tongue body pulled back, the [+back] sounds, or with the tongue body pushed forward, the [–back] sounds. While the feature [±back] is used for some consonants, it is especially important in the classification of vowels. See p. 97.

[±consonantal] Defines the class of consonants, sounds made with complete or almost complete closure in the oral tract. [+consonantal] sounds are almost always found in the onsets and codas of syllables. See p. 28.

[±constricted glottis] A place of articulation feature that defines the class of sounds made by closing the glottis. See p. 84. This place feature can also be used for secondary articulation. See p. 87.

[±continuant] A manner feature that defines whether a sound is in the class with stopped airflow ([–continuant]) or in the class with continuous airflow ([+continuant]). See p. 49.

[coronal] A place of articulation feature that defines the class of sounds made with some part of the tip of the tongue (the tongue corona). See p. 72.

[±delayed release] [+delayed release] defines the class of affricates – stops made with an oral fricative release. All other sounds are [–delayed release]. Often, we see affricates assigned [delayed release], while other sounds do not have a [delayed release] feature. See p. 55.

[±distributed] A place of articulation feature; [+distributed] defines the class of sounds made with a broad tongue blade while [–distributed] sounds are made with the narrow tongue tip. See p. 73.

- [*dorsal*] A place of articulation feature that defines the class of sounds made with the body of the tongue (also known as the tongue dorsum). See p. 78.
- [*±front*] Defines the class of sounds with the tongue body pushed forward, the [+front] sounds, or with the tongue body in neutral position, the [−front] sounds. This feature is sometimes used to classify central vowels. See p. 108.
- [*±high*] Defines the class of sounds with a raised tongue body. While the feature is used for some consonants, it is especially important in the classification of vowels. See p. 96.
- [*labial*] A place of articulation feature that defines the class of sounds made with some kind of lip constriction. See p. 66.
- [*lateral*] Defines the class of *l*-like sounds. The term refers to the position of the tongue, allowing air to escape laterally, along one side or the other of the tongue. See p. 60.
- [*±low*] Defines the class of sounds with a lowered tongue body. While the feature is used for some consonants, it is especially important in the classification of vowels. See p. 96.
- [*±nasal*] Defines the classes of [−nasal] sounds and [+nasal] sounds, oral and nasal sounds respectively. The [+nasal] sounds typically have significant airflow through the nose while the [−nasal] sounds do not. See p. 42. For nasal vowels, see p. 115.
- [*±round*] Defines the [+round] class of sounds (those made with rounded lips) and the [−round] class of sounds (those made with spread lips). The feature [±round] is used primarily to distinguish vowels. See p. 93. [+round] is also used for [w] and for the secondary articulation known as labialization. See p. 88.
- [*±sonorant*] Defines the class of [−sonorant] sounds with turbulent airflow, also called obstruents, all of which are consonants, and the class of [+sonorant] sounds with resonant airflow, also called sonorants. All vowels are sonorants as are some consonants. See p. 48.
- [*±spread glottis*] A place of articulation feature that defines the class of sounds made by opening the glottis wide. See p. 85. This place feature can also be used for secondary articulation. See p. 87.
- [*±strident*] Separates the noisier [+strident] fricatives from the quieter [−strident] fricatives. See p. 52.
- [*trill*] Defines the class of trilled sounds. Usually there is only one such sound in a language, a rhotic. See p. 61.
- [*±vocalic*] Defines the class of vowels, sounds made without closure in the oral tract. Each [+vocalic] sound is typically found in the nucleus of a syllable. See p. 28.

[±voice] Defines the classes of [−voice] sounds and [+voice] sounds. Voiceless sounds behave as though the vocal folds are pulled apart so that the air-flow through the larynx is smooth and quiet; voiced sounds have noisy, turbulent airflow through the larynx. See p. 39.

B.2 Distinctive features for consonants

B.2.1 Place of articulation features

description	example	[labial]	[coronal]	[±anterior]	[±distributed]	[dorsal]	[±high]	[±back]
bilabial	[p]	[lab]						
labial-dental	[f]	[lab]						
labio-velar	[w]	[lab]				[dor]	+	+
dental	[t]		[cor]	+	+			
alveolar	[t]		[cor]	+	−			
retroflex	[ɖ]		[cor]	−	−			
post-alveolar	[ɖ]		[cor]	−	+			
palatal	[c]					[dor]	+	−
velar	[k]					[dor]	+	+
uvular	[q]					[dor]	−	+

B.2.2 Manner of articulation features

description	[±sonorant]	[±nasal]	[±voice]	[±continuant]
nasal stop	+	+	+	−
approximant	+	−	+	+
voiced stop	−	−	+	−
voiced fricative	−	−	+	+
voiceless stop	−	−	−	−
voiceless fricative	−	−	−	+

B.3 Distinctive features for vowels

B.3.1 Modal vowel features

	[i]	[y]	[ɯ]	[u]	[e]	[ø]	[ɤ]	[o]	[æ]	[œ]	[a]	[ɒ]
Hi	+	+	+	+	−	−	−	−	−	−	−	−
Lo	−	−	−	−	−	−	−	−	+	+	+	+
Bk	−	−	+	+	−	−	+	+	−	−	+	+
Rd	−	+	−	+	−	+	−	+	−	+	−	+

B.3.2 Tongue root features

		[+high]			[−high, −low]		
[+advanced tongue root]	[+atr]	i	y	u	e	ø	o
[−advanced tongue root]	[−atr]	ɪ	ʏ	ʊ	ɛ	œ	ɔ

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Terms linguists use

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[back] The feature [back] identifies the class of sounds typically made by constriction with the tongue body in the back of the mouth. 237, 383

[coronal] The feature [coronal] identifies the class of sounds typically made with constriction from the front part of the tongue. 478

[distributed] Distributed sounds use a wide area of the front of the tongue to constrict the airflow through the vocal tract. 478

[high] The feature [high] identifies the class of sounds typically made by raising the tongue body. 237, 383

[nasal] The feature [±nasal] identifies the class of nasal sounds ([+nasal]) and oral sounds ([-nasal]). The [+nasal] sounds are characterized by airflow through the nasal cavity while the [-nasal] sounds do not have significant airflow through the nose. 423

[voice] The feature [±voice] distinguishes between the class of sounds typically made with stiff vocal folds ([-voice], also called *voiceless*) and sounds typically made with relaxed vocal folds ([+voice]). 138, 198

COMPLETIVE A function morpheme that shows that an action has already taken place, typically marking the verb of a clause or a sentence. Acronyms include CPL, COMPL. 144

DISTRIBUTIVE A function morpheme showing that something (an object or an action) is found in multiple places. Acronyms include DIST. 505

DUAL A person-marking function morpheme indicating two people, with meanings like 'we.two' or 'you.two'. Acronyms include DU. 172

- EXCLUSIVE** A person-marking function morpheme that excludes those being spoken to. It would mean something like ‘me and them, but not you’. Acronyms include EXCL. 171
- FUTURE** A function morpheme that shows that an action will take place at a later time, typically marking the verb of a clause or a sentence. Acronyms include FUT. 144
- INCLUSIVE** A person-marking function morpheme that includes those being spoken to. Typically, it would mean something like ‘me and you, but not them’. Acronyms include INCL. 171
- INTRANSITIVE** An intransitive verb has only subject (actor) or only an object (acted-upon), not both. An INTRANSITIVE function morpheme that shows that a verb is intransitive. Acronyms include ITRN, INTRNS. 158
- NONFUTURE** A function morpheme that shows an action either has already taken place or is happening right now, typically marking the verb of a clause or sentence. Acronyms include NONFUT. 487
- PAST** A function morpheme that shows that an action that took place at an earlier time, typically marking the verb of a clause or a sentence. Acronyms include PST. 144
- PLURAL** A person-marking function morpheme that shows more than one is involved. In languages with DUAL morphemes, the PLURAL morphemes typically show that more than two are involved. Acronyms include PL. 171, 505
- POTENTIAL** A function morpheme that shows that an action has yet to take place but that is likely to take place sometime in the future, typically marking the verb of a clause or a sentence. Acronyms include POT. 162
- PREGRESSIVE** A function morpheme that shows that an action is about to take place, typically marking the verb of a clause or a sentence. Acronyms include PREG. 144
- PROGRESSIVE** A function morpheme that shows an on-going action, typically marking the verb of a clause or sentence. Acronyms include PROG, PRG. 144
- REDUPLICATION** A morphological pattern where the form of an affix depends on the root itself, repeating some root consonants and/or vowels to indicate a particular meaning. Reduplication is typically coupled with some meaning change, such as adding PLURAL meaning. Acronyms include RDP, RED, REDUP. 432, 504
- RELATIVIZER** A function morpheme changes an action morpheme into a word meaning one who has done the action. Acronyms include REL. 161

REMOTE.PAST A function morpheme that shows an action took place some time ago, typically marking the verb of a clause or sentence. Also called *DISTANT.PAST*. 242

REPETITIVE A function morpheme that shows that an action is performed multiple times. Acronyms include REP, RPT. 477

SINGULAR A-person marking function morpheme that shows only one is involved. Acronyms include SG. 171

SUBJECT A function morpheme that typically shows the actor for a particular verb. Acronyms include SUBJ. 236

TRANSITIVE A transitive verb has both subject (actor) and object (acted-upon). A *TRANSITIVE* function morpheme that shows that a verb is transitive. Acronyms include TRN, TRANS. 158

accent See stress; some people use the term *accent* for *stress*. 121

active articulator A moving part of the mouth used to make some speech sounds; in sign language, a moving part of the body used to make some speech signs. For example, the tongue tip, body, and back are all active articulators, as is the velum. 34

actor The one that is doing the action conveyed in a phrase or sentence. 143, 170, 502

acute accent A diacritic over a vowel used to show high tone, [á]. 119

affix A morpheme that is added to a root or stem, but cannot stand alone. 137, 153, 522

affricate A consonant that is made with the closure of a stop followed immediately by a fricative release. 55, 63, 443, 519

allomorph One of the forms that we find with an alternating morpheme; same as alternant. 481, 510, 528

allophone Allophones are the sets of sounds (phones) that correspond to each contrastive phoneme of a language. Typically written in square brackets using the International Phonetic Alphabet symbols. 267, 420, 421, 467, 476, 533, 534

alphabet A writing system where each sound is generally represented with a unique symbol, called a letter. Depending on the history of a community, an alphabetic system may have mismatches between the sounds and the symbols. 374

alternant One of the forms that we find with an alternating morpheme; same as allomorph. 481

alternation When a morpheme has different phonological shapes (different sounds or different gestures) depending on which other morphemes (if

- any) it is combined with, we say that there is an alternation in the form of the morpheme. 14, 478, 481, 510, 528, 534
- alveolar* A descriptor for sounds made with constriction at the alveolar ridge. The sounds [t], [d], [n], [s], and [z] are among the alveolars. 73
- alveolar ridge* The bony ridge behind the upper teeth. 53, 71
- Americanist Phonetic Notation* A symbolic system used for representing the sounds of many Indigenous languages in the Americas; unfortunately, there is little consistency across users. Historically, it has largely given way to the International Phonetic Alphabet, but if you are using older texts, you need to pay careful attention to descriptions of what each symbol represents. 23, 124, 539
- Americanist transcription* Alternative name for Americanist Phonetic Notation. 124, 446
- approximant* Glides and liquids – sounds whose active articulators get sort of close to the passive articulators, but do not constrict enough to create turbulent airflow. 59, 64
- articulator* Any part of the mouth and larynx that participates in making speech sounds, such as the lips, teeth, parts of the tongue. 26, 33, 36
- aspirated* Aspirated sounds are produced with a puff of air, a consequence of the vocal folds being pulled far apart. 87
- assimilation* When one sound takes on properties of an adjacent sound, such as a vowel taking on nasality from an adjacent nasal consonant. 408, 415, 439
- bilabial* Any of a class of sounds characterized by the feature [labial]. Also, descriptor for sounds made with constriction using both lips. Examples are [p], [b], [m], [ɸ], and [β]. 66
- borrowing* The process by which speakers of one language adopt a word from another language for regular use. Often when words are borrowed, some sounds are changed to fit the speaker's [phonotactics](#) and syllable [template](#). 193, 567
- bound morpheme* A morpheme which cannot stand on its own as a word. A bound morpheme must be attached (“bound”) to another morpheme in order to form a word. 137
- branching* A syllable tree or a part of a syllable tree that has two or more elements under it. A syllable branches if it has an onset as well as a rime, a rime branches if it has a coda as well as a nucleus, and a nucleus branches if it has a long vowel or diphthong. A single V syllable is completely non-branching because there is no onset, so the syllable doesn't branch, and there is no coda, so the rime doesn't branch. 354

- breathy voice* A voiced sound made with the vocal folds pulled wide apart so a lot of air comes through. Breathy voice is indicated by two dots below the sound: plain [a] and breathy [a̤]. 116
- burst* When the release of a stop is so forceful that it is audible to listeners, the release is called a burst. 51
- C* A shorthand for either *consonant* or *coda*. 290
- central* A descriptive term for vowels where the highest part of the tongue is between the high point for a front vowel and the high point for a back vowel. 108
- circumfix* A type of affix that has two parts, one that preceded the stem and one that follows the stem; neither part can be used without the other part. 179
- circumflex* A diacritic over a vowel used to show falling tone, [â]. 119
- close* A term some sources use for higher vowels, referring to the fact that the mouth is more closed when making a higher vowel. 95, 445
- closed syllable* A syllable that ends with a consonant. 290, 301, 366, 501
- closure* The part of a stop that involves complete closure, stopping the airflow. 50
- coda* The consonant or consonants at the end of a syllable form the coda of the syllable. 13, 300, 308, 318, 336, 393, 425
- complementary distribution* When similar sounds are distributed among complementary contexts so that each sound (or sound class) has its own, unique set of contexts. 13, 380, 420, 422, 426, 436, 478, 484, 510, 527, 534
- complex coda* A coda composed of two or more consonants. 303
- complex onset* An onset composed of two or more consonants. 303
- compound* A compound is a word that is created by putting two independent words or word parts together to make a form whose meaning is related to but different from the two parts. 147, 179, 180, 498
- consonant* A sound made with a partially or completely closed vocal tract. 28
- consonant cluster* A sequence of two or more consonants. 302
- consonant inventory* The collection of consonant sounds used in making the words of a language. Often refers to only the contrastive consonant sounds of a language. 268, 525
- content* A morpheme that contributes basic meaning to a word. 143, 152
- context* Where a sound of interest is located in a word. This may be the sounds preceding and/or following a sound of interest. Context can also include properties like position in a syllable, degree of stress, tone, and so on. Alternative terms: environment, location. 391, 424

- context pair* A pair of words with both a meaning difference and a difference in one sound where the context on each side of the sounds of interest are the same (whether they are other sounds or are word edges). 233, 247, 251, 436, 525
- continuant* A sound made with continuous airflow throughout the production of the sound. Both fricatives (like [s]) and vowels (like [a]) are continuant sounds. 49, 63
- contrast* Two sounds contrast in a language when the sound difference is connected with a meaning difference – or has the potential of creating a meaning difference. 232, 234, 420, 422, 436, 525
- contrastive* Two sounds are contrastive in a language when the sound difference is connected with a meaning difference – or has the potential of creating a meaning difference. 12, 232, 444, 466, 467, 533
- core syllable* A CV syllable – an open light syllable with a single consonant in the onset. This is the most common type of syllable. Every language uses core syllables. 293
- coronal* A member of the class of sounds with the feature [coronal]; a sound made with constriction from the front part of the tongue. 78
- creaky voice* Creaky voice results from significantly slowing down the motion of the vocal folds. The symbol for creak is a tilde under the vowel: plain [a] and creaky [ã]. 116
- dental* A sound made with the tongue touching the teeth. 73
- derivation* A step-by-step demonstration of how a set of rules converts an underlying representation into a phonetic representation. 491, 515
- derived environment condition* A term used in linguistic writing to refer to phonological patterns that are found in poly-morphemic words only when the context for the pattern spans more than one morpheme. 504
- derived form* A form created by putting two or more morphemes together. A word made from a root with one or more affixes is a derived form, as is a compound. 504
- diachronic variation* Differences in earlier and later forms of the same language. 227
- diacritic* An extra mark added to a symbol, that changes the meaning of the symbol slightly. 73, 281, 379
- dialect* A collection of words and grammar rules shared by a particular language community, typically defined by a geographical region or by some social class. 188, 190, 192, 230

dialectal variation Variation that is found in a definable group of speakers; groups are defined by non-linguistic factors such as gender, geographical location, socio-economic class, etc. 188, 192, 522

dictionary An alphabetical listing of words in a language, giving their meanings and usage. Some dictionaries also show pronunciation and/or morphological relations. 168

diphthong A sequence of two different vowels in a single syllable nucleus, whether monomoraic (a light diphthong) or bimoraic (a heavy diphthong). 187, 278, 297, 363

distinctive feature A feature (or single property) that defines a contrast between two sounds of a language. 254, 259, 260, 525

distribution Generically, where a specific sound can occur in the words of a specific language. 380, 422, 424, 526

dorsal A member of the class of sounds with the feature [dorsal]; a sound made with constriction made by the body of the tongue. The term *velar* is also used for dorsal sounds with constriction at the velum. 198

dorsum The large part of the tongue behind the tip and blade of the tongue. Also called the tongue body. 78

ejective A type of secondary articulation involving simultaneous complete closure in the mouth and in the glottis, followed by raising the glottis to create pressure between the glottis and the oral closure; release of the closures creates a sharp kind of explosive noise. Ejectives are symbolized with an apostrophe after the stop: [p'] is an ejective labial. 88

environment The sounds preceding and/or following a sound of interest. The environment of a sound can also include properties like position in a syllable, degree of stress, tone, and so on. Alternative terms: context, location. 391

feature The most basic element of phonological speech sounds. Combinations of features create different sounds. Different sounds can be distinguished from each other by as little as one feature. 22, 26, 34, 280

flap Another name for a tap, an *r*-sound that is made by tapping the tongue very quickly in the vicinity of the alveolar ridge. 61

fortis A sound with stronger articulation in contrast to lenis sounds; there may be no single phonetic characteristic that marks a fortis/lenis contrast. 263

free morpheme A morpheme which can stand on its own as a word. 137

free variation Differences in the production of a form which have no effect on the meaning of the form; free variation can be dialectal (free variation between speakers of the same dialect) or idiolectal (free variation in the speech of a single person). 189, 219, 230, 420, 523

Terms linguists use

- frication* Sounds with an airflow that is so interrupted that there is a hissy or buzzy sound. 48
- fricative* A consonant that is made with a narrow constriction so airflow is turbulent and the sound is hissy or buzzy. 50, 52, 63, 519
- function* A morpheme that contributes grammatical meaning. 143, 144, 152, 522
- generative phonology* A model of how phonology works. A generative analysis posits underlying representations and rules; the rules are applied to the underlying representation and result in a surface representation. The surface representation shows how a word is pronounced. 492
- glide* A sound made with more constriction than used for a vowel, but less constriction than for any other consonant. 59, 64, 518
- gloss* The linguistic meaning of a morpheme or word. 29, 126, 141
- glottal stop* A stop made by closing the vocal folds tightly (constricting the glottis). 84
- glottalized sonorant* Constriction of vocal folds while still allowing voicing (necessary for sonorants) interrupts the airflow through the glottis briefly during production of the sonorant. Glottalized sonorants are symbolized by an added apostrophe: [m'] is a glottalized labial nasal. 88
- grapheme* The letters or other symbols used to represent words in the conventional writing system of a language; graphemes are used for reading and writing and do not necessarily connect directly to how words are pronounced. 125, 232
- grave accent* A diacritic over a vowel used to show low tone, [à]. 119
- hacek* A diacritic over a vowel used to show rising tone, [ǎ]. 120
- handshape* The configuration of hand(s) (fingers and thumb) when making a sign. 36, 181, 441
- heavy syllable* A syllable with more weight units than a light syllable. Typically a heavy syllable has a long vowel, a diphthong, or (in many languages) a coda. 291, 301, 362, 506, 508
- homophone* Distinct words are homophones if they sound the same but have unrelated meanings. 234
- idiolect* The collection of lexical items and grammar principles that a particular speaker can use at a given moment. 189, 190, 215, 230
- idiolectal variation* Variation that marks the way an individual speaks their language; every speaker has their own idiolect. 189, 215, 523
- implosive* A type of secondary articulation involving simultaneous complete closure in the mouth and in the glottis, followed by lowering the glottis to

- create a vacuum between the glottis and the oral closure; release of the closures briefly draws air into the mouth instead of pushing it out. Symbols for implosives are similar to the symbols used for plain stops, except there is a hook at the top, bending to the right: [ɓ] is a labial implosive. 88
- infix* An affix that is placed inside of some other morpheme (an affix that comes *between* other morphemes is not an infix – it is either a prefix or a suffix depending on its relation to the root). 153, 179
- input* This is a term used in optimality theory writings. Roughly speaking, an input is an abstract representation for a morpheme, giving some information about the sounds of the morpheme. It is related to an output by general principles; constraints in a language select the best output for a given input. The input-output relation in Optimality Theory is similar to the underlying-surface relation; the underlying suppositions about how language works are different. 492, 515
- International Phonetic Alphabet* A standard set of symbols for language sounds, based on the roman alphabet, created and maintained by the International Phonetic Association. 22, 123, 133, 188, 231, 287, 379, 419, 518, 539
- inventory* The collection of sounds used in making the words of a language. Often refers to only the contrastive sounds of a language. 232, 234, 267, 372, 380, 427, 436
- labial* A member of the class of sounds with the feature [labial]; a sound made with some degree of lip constriction. 78, 200
- labialization* Labialization is a secondary articulation that adds rounding to a sound, The symbol for this is a raised “w”: [p^w] is a voiceless labialized labial stop. 88
- labio-velar* A sound that is made with constriction both at the lips and at the velum (using the back of the tongue). 81
- labiodental* A descriptor for sounds made with constriction at using lips and teeth, typically the upper lip and the lower teeth. Examples are [f] and [v]. 66
- language variation* See ‘variation’. 190
- larynx* A complex of cartilages, ligaments, and muscles in the neck, with an opening that air passes through between the lungs and the mouth. The larynx is also called the voice box because of its role in making language sounds. 36, 38
- lateral* A class of consonants made with the side of the tongue dropped to allow air to flow freely along the side of the tongue. 60, 64, 195

- lateral affricate* A class of obstruent consonants made with complete closure followed by release with the side of the tongue dropped to allow air to flow freely along the side of the tongue. 61
- lateral fricative* A class of obstruent consonants made with the side of the tongue dropped to allow air to flow freely along the side of the tongue. 60
- Leipzig Conventions* A set of conventions for presenting words to show their phonological and morphological structure and their meanings, proposed in Comrie et al. (2015) and available online at <https://www.eva.mpg.de/lingua/pdf/Glossing-Rules.pdf>. Also called the [Leipzig Glossing Rules](#). 141, 522, 566
- Leipzig Glossing Rules* A set of conventions for presenting words to show their phonological and morphological structure and their meanings, proposed in Comrie et al. (2015) and available online at <https://www.eva.mpg.de/lingua/pdf/Glossing-Rules.pdf>. Also called the [Leipzig Conventions](#). 141, 566
- length* A length contrast occurs when holding a sound (vowel or consonant) for a longer or a shorter time makes a meaningful difference. Also called *quantity*. 118, 265
- lenis* A sound with weaker articulation in contrast to fortis sounds; there may be no single phonetic characteristic that marks a fortis/lenis contrast. 263
- lenition* When a sound is in some way less distinct or less noticeable than one might expect it to be. 415
- lexical* A property of a word or morpheme that is specific to that item and so must be memorized (or stored in the ‘mental lexicon’). 168
- lexical class* A group of morphemes, stems, or words that follows a subpattern in a language, rather than a general pattern. 168, 183, 493
- lexicon* A mental list of information about the words and morphemes in a language. At the minimum, a lexicon includes all information about words and morphemes that has to be memorized because there are no general patterns. 168
- light diphthong* A diphthong that functions like a light syllable in weight-related patterns, not like a heavy syllable. In the moraic representation, a light diphthong shows two different vowels (the diphthong) linked to a single mora (making it light). 364
- light syllable* A syllable with only one weight unit, typically a syllable with a short vowel in an open syllable. Occasionally we find a language with closed light syllables – in these languages, syllables with a short vowel count as light even if there is a coda consonant. 291, 362, 506

- limited distribution* A particular sound or class of sounds is not found in a context where we might expect to find it. 13, 385
- liquid* An *l*-sound or an *r*-sound. 59, 64, 194, 195, 518
- loanword* A word used in one language that originated in another language. Loanwords come into a language through [borrowing](#), so loanwords are also called borrowed words. You may see this written as two words, *loan word*. 193, 223, 371
- location* Where a sound of interest is located in a word. This may be the sounds preceding and/or following a sound of interest. Context can also include properties like position in a syllable, degree of stress, tone, and so on. Alternative terms: context, environment. 36, 204, 441
- logographic* A writing system where each morpheme or word is represented with a unique symbol. 374
- long* A sound that is held for more time is called long. Both consonants and vowels can contrast for length. Here is the IPA symbol to show length differences: short [a] and long [a:]. 118, 297, 363
- manner of articulation* The type of constriction made when producing a sound, for example complete constriction (for stops), narrow constriction (for fricatives). 47, 89, 518
- meaningful difference* When there is a single property that differs between two words that tells you they have different meanings, that property makes a meaningful difference. 235
- melody* A morphological pattern where a single parameter (like tone or nasality) extends over multiple segments to signify some morphological property. 176
- midsagittal* A profile slice through the middle of the head. 33
- minimal pair* A pair of words with both a meaning difference and a difference in only one sound. 12, 138, 232, 236, 238, 239, 422, 436, 441, 444, 525
- minimal set* A set of words where each has a different meaning and all words differ from each other in only one sound. 238, 241, 243, 525
- modal vowel* The oral vowel qualities, made by movements of tongue and lips (that is, qualities that are independent of nasal, tone, stress, length, and laryngealization). 110, 277
- monomorphemic* A stem or word made up of a single morpheme. 152
- monosyllabic* A morphological unit (word, root, affix, stem) that has only one syllable is monosyllabic. 301
- mora* A unit within a syllable used to show syllable weight: heavy syllables have two moras (or morae) while light syllables have only one. Onsets usually

- do not have moraic structure. Typically, people use either moraic structure or rime structure (with nucleus and coda), not the two types of structures together. The mora is traditionally represented with the Greek letter μ . 362
- morpheme* The smallest meaningful unit in a word. 11, 134, 137, 220, 475, 520
- morpheme structure condition* A phonological condition that is true of morphemes, but is not necessarily true when morphemes are combined into words. 504
- morpho-phonological alternation* A regular phonological alternation that is restricted to some lexical class, so it is driven both by phonological and by morphological criteria simultaneously. 495, 510, 515, 528
- morpho-phonological analysis* Analysis of language patterns that requires using both morphological and phonological analytic skills. 479
- morphologically related* In morphologically related words, part of each word is made up of the same morpheme(s) while the rest of the each word has different morphemes. 139
- morphology* The study of how the sounds and meanings of morphemes combine to make words. 11, 134, 471
- movement* How the signing articulators move (if indeed they do move) during the course of making a sign. 36, 181, 441
- nasal* Any sound made with significant airflow through the nose is called a nasal sound. Nasal sounds are characterized by the feature [+nasal]. 42, 59, 63, 66, 518
- nasalization* Nasalization is a secondary articulation added by action of the velum. Both consonants and vowels can be nasalized. 88
- natural class* A group of sounds defined by one or more features is a natural class. 26, 51, 195
- near-minimal pair* A pair of words with both a meaning difference and a difference in two sounds, preferably two sounds which are not close to each other. 233, 247, 248, 436, 525
- neologism* A neologism is a new word in a language. Some neologisms arise spontaneously; others are intentionally created. Consciously creating neologisms is one tool used in language revitalization. 193, 376, 475, 479, 500, 514
- non-contrastive* Two different sounds are non-contrastive in a language when the sound difference makes no meaning difference. This may occur due to free variation or because the sounds are allophones of each other. 234, 420, 466, 472

- non-manual marker* When signing, the components that extend beyond the head, including postures of the mouth, face, head, and body. 36, 441
- non-strident* A continuant sound made gently so it sounds quiet, soft, or murmured. 52, 63
- North American Phonetic Alphabet* An alternative name for the Americanist Phonetic Notation. 23, 124
- nucleus* The nucleus is the most sonorous part of a syllable, typically a vowel though sometimes a consonant. 13, 293, 308, 336
- number* Whether a morpheme refers to one (SINGULAR, abbreviated SG, or more than one entity (PLURAL, abbreviated PL). 171
- obstruent* A sound produced where the airflow is interrupted by some kind of constriction. 48, 63, 89, 518
- ogonek* A diacritic used to show that a sound is nasal or nasalized. The symbol [a] represents an oral vowel; [ã], which has an ogonek, represents a nasal vowel. See *tilde*. Ogonek is often used when a language also has contrastive tone and/or stress, so that diacritics do not get crowded above a single vowel symbol. 115
- onset* The consonant or consonants at the beginning of a syllable form the onset of the syllable. 13, 293, 308, 309, 336, 386, 393, 424, 425
- Onset First Principle* A consonant that precedes a vowel is an onset that makes a syllable together with the following vowel. Even if there is a preceding vowel, that consonant is not a coda. 337, 339
- open* A term some sources use for lower vowels, referring to the fact that the mouth is more open when making a lower vowel. 95, 445
- open syllable* A syllable that ends with a vowel. 290, 366, 384, 501
- optimality theory* A model of how phonology works. An optimality analysis posits input representations which relate to possible output representations by general principles. The work of an individual grammar is to examine the possible outputs and determine which is the best for the relevant input. This is done by filtering the outputs through a set of ranked constraints. The selected output shows how the word is pronounced. 492
- oral* An oral sound is made without airflow through the nasal cavity; the feature for oral sounds is [-nasal]. 42
- orientation* Directional orientation of the palm(s) and/or fingers when making a sign, for example, facing the sky, the ground, towards or away from the speaker, etc. 36, 441
- orthography* The conventional way of writing a language, used for reading and communicating in writing. 10, 87, 123, 133, 232, 280, 431, 487, 576

- output* This is a term used in optimality theory writings. Roughly speaking, the output is a phonetic representation, telling how a word is pronounced. It is related to an input by general principles; constraints in a language select the best output for a given input. The input-output relation in Optimality Theory is similar to the underlying-surface relation; the underlying suppositions about how language works are different. 492, 515
- palatal* A member of the class of sounds with the feature [dorsal], made with the tongue body raised directly toward the palate; a sound made with constriction made by the body of the tongue at the palate. 73, 79
- palatalization* Palatalization is a secondary articulation that adds constriction with the tongue blade toward the palate, adding a [j]-like quality to the basic sound=. The symbol for this is a raised “j”: [p^j] is a palatalized voiceless labial stop. 88
- palate* The roof of the mouth. The front part is hard and immobile; it is a passive articulator. The back part is the velum. 35, 71
- paradigm* A set of affixes that attach to roots of a particular type. 161, 169
- passive articulator* A non-moving part of the mouth used to make some speech sounds; in sign language, a non-moving part of the body used to make some speech signs. For example, the teeth and the hard palate are both passive articulators. 34
- person* Whether a morpheme refers to the speaker (first person, or the number 1), the addressee (second person, or the number 2), or some other party (third person, or the number 3). 171
- pharynx* The tube connecting the larynx and the mouth. 36
- phone* The sounds (or gestures) used in some language. Phones include both contrastive and non-contrastive sounds (gestures). Typically written in square brackets using the International Phonetic Alphabet symbols. 466
- phoneme* A sound that is contrastive in a language. Typically written between forward slashes using the International Phonetic Alphabet symbols. 234, 267, 420, 421, 467, 476, 534
- phonemic inventory* The collection of contrastive sounds used in making the words of a language. 232, 234, 267
- phonemic representation* A phonemic representation includes only the phonemes, or contrastive sounds, of a language. This means it is not a representation of the pronunciation of an utterance – the rules of the language providing allophones and alternations need to be added. Phonemic representation is typically (but not always) written between slashes. 468, 533

- phonetic representation* A way of writing a language meant to represent the sounds (or signs) that were used in an utterance. 467
- phonetic writing system* A writing system that uses symbols that correspond directly to the way a word sounds (the phonetics of the word), so from the spelling you know how to say the word. 196
- phonetics* The study of the physical properties of sound. In language, we hear about (i) articulatory phonetics, the study of how sounds are produced and (ii) acoustic phonetics, the study of the physical properties of the sounds. 10, 36
- phonological alternation* An alternation that is driven by complementary distribution and other phonotactics. 495, 510, 515, 528
- phonology* The sounds and patterns of distribution of those sounds that are found in utterances in a language, making up its unique sound signature. Also, the study of the sounds and sound patterns of a language. 3, 36, 158, 188, 232, 288, 380, 420, 478, 517
- phonotactics* The constraints a language has over sound sequences. Phonotactics may be sequential; they may also be explained by syllable structure. 344, 560
- pitch* Pitch is the physical property of highness or lowness that results from changes in vocal fold vibration rate. Different pitches are measured by the frequency of the sounds; pitch is measurable and not relative. See also tone. 119
- pitch accent* A language that uses tone as the primary marked of stress is called a pitch accent language; see also *tone* and *stress*. 121
- place of articulation* The location in the mouth which has the greatest degree of constriction when making a sound. 47, 64, 65, 90
- plosive* A descriptive term for a stop. 49
- polymorphemic* A stem or word that is made up of more than one morpheme. 134, 152, 514, 532
- polysemous word* A word with multiple nuanced meanings is called a polyseme. 234
- post-alveolar* The region of the palate directly behind the alveolar ridge; also, a sound made with constriction in the post-alveolar region. 73, 78
- practical orthography* A writing system that is used for a language which may not yet be officially accepted by the community of speakers. 123, 136, 142, 239, 272
- prefix* An affix that precedes the root or stem. 153
- quality* The differences in a sound that distinguish one sound from another, re-

sulting from differences in the shape of the vocal tract and the type of airflow, independent of quantity, stress, and tone. 101, 265, 354

quantity A quantity contrast occurs when holding a sound (vowel or consonant) for a longer or a shorter time makes a meaningful difference. Also called *length*. 354

release The part of the stop when the closure is opened, allowing airflow to resume. 50

resonant airflow Airflow that is smooth, so the sounds are not hissy, but they are easy to hum or sing. 48

retroflex Any sound made with the tip of the tongue curling back. 73

rhotic An *r*-sound. 61, 64

rhyme An alternative spelling for *rime*, the unit within a syllable made up of a vowel and any following consonants within the syllable. 300

rime The rime is a unit within a syllable made up of a vowel and any following consonants within the syllable. Another way of saying this is that the rime is made up of the nucleus and the coda. 300, 572

root The morpheme that contributes the basic sound and content to a word. 144, 153, 522

secondary articulation Involving two sets of articulators in producing a single sound; typically, secondary articulation involves an obstruent enhanced by laryngeal or sonorant component. 87

segment A language sound; sometimes *sound segment* is used instead. 121, 129, 280, 518

semantic interpretation The meaning carried by a word, independently of how it is said. 235

semantics The study of meaning in language. 235

sibilant A strident consonant made with the tongue tip or blade so that the consonant has a higher pitch than other fricatives and affricates – [s], [z], [ʃ], [ʒ], and the affricates with these types of release. Labial [f] and [v] have lower pitch so are not considered sibilants. 53

similar Similar sounds differ from each other by a very small number of features, possibly only one feature. For example, voiceless [t] is very similar to voiced [d] because they differ only by their value for the feature [±voice]. 43, 237, 243, 247, 266, 406, 422

similarity The similarity of sounds is determined by the number of features they have in common. 26, 258

sonorant A sound produced with unimpeded airflow. 48, 63, 89, 293, 463, 518, 573

- sonority hierarchy* A phonetic ranking of sounds by their degree of openness (sonority), based on the phonetic properties of the sound. A sound with more openness is more sonorous, or higher on the scale, and a sound with less openness is less sonorous, or lower on the scale. Vowels are higher on the scale, open vowels (like [a]) are at the very top of the scale. Consonants are lower on the scale, with voiceless stops being lowest because they have the least openness. Related to [sonorant](#). 341
- sound inventory* The collection of sounds used in making the words of a language. Often refers to only the contrastive sounds of a language. 234, 525
- spread lips* A descriptor used for an articulatory configuration where the corners of the lips are pulled back, away from each other. 93
- standard language* The dialect of a language that is accepted socially and/or politically as the “correct” variant of the language. 192
- standardization* The effect of deliberate or spontaneous change in a variant of a language to make a standard language. 193
- stem* A morphological unit with a clear meaning, whether monomorphemic or polymorphemic. 144, 153
- stop* A sound made with complete stoppage of airflow. For example, [p] is an oral stop and [m] is a nasal stop; both are made by closing the lips completely to stop the airflow. 49, 63, 66, 198, 518
- stress* The linguistic use of pitch, loudness, and/or length to give prominence to some vowel (or vowels) in a word. 121, 311, 345, 369
- strident* A continuant sound made with extra force so it sounds noisy, hissy, or buzzy. 52, 63, 443
- sub-bridge* A diacritic used to indicate that a sound is dental (made with the tongue blade touching the teeth). 73
- subtractive analysis* The method of comparing two words that are meaningfully related to determine whether there are also sound relations between the two words. When there is both sound and meaning relations between two words, we identify morphemes. We can go on to subtract known morphemes to help identify other morphemes. 134, 148, 150, 520
- suffix* An affix that follows the root or stem. 153
- suppletion* When a morpheme has unrelated allomorphs, and the selection of allomorph is not phonologically systematic. 480, 502, 510, 528
- suprasegmental* Typically, this term is refers to linguistic tone, stress, or length. The basic features of a segment remain the same – a high toned [á] has the features of [a], with a high tone added; similarly a stressed [ˈa] has the features of [a], with stress added, and a long [a:] has the features of [a], with length added. 121, 508

- surface representation* Roughly speaking, a surface representation is a phonetic representation, telling how a word is pronounced. It is created from an underlying representation by applying rules. Abbreviation is SR. 421, 470, 490, 491, 514, 515
- syllabary* A writing system where typically each syllable is represented by a unique symbol. Some systems that are called a syllabary actually have separate symbols for each mora, not each syllable. 374
- syllabification* Syllabification of a word refers to both (i) a word with its syllables, and (ii) a process of assigning syllables to the sounds of a word. 335
- syllable* A unit of sound organization within words. Typically, words are made up of sequences of syllables. The syllable is traditionally represented with the Greek letter σ . 12, 237, 287, 288, 290, 305, 394, 425, 478, 485, 510, 526, 534
- syllable weight* Syllable weight is a relative measure, typically shown by how syllables behave in a stress pattern. Heavy syllables are more likely to be stressed and light syllables are more likely to be unstressed. Weight is sometimes represented by moras (symbolized μ) in linguistic work. Light syllables have one mora which heavy syllables have at least two moras. 362
- symbol* A visual representation corresponding to some sound. 22, 232
- syntax* The study of how words combine to make phrases and sentences. 158, 471
- tap* An *r*-sound that is made by tapping the tongue very quickly in the vicinity of the alveolar ridge. 61, 195
- template* A frame, usually for a syllable, showing the types of segments that are found in that unit. For example, a CV syllable template tells us that the syllable is composed of a consonant followed by a vowel. Occasionally, template have a morphological role as well. For example, words may have a bimoraic template, meaning that each word is made up of two moras. 306, 309, 560
- three-line gloss* A conventionalized way of showing words, morphemes, and meanings. 142
- tiebar* A diacritic that connects two sounds to show they are pronounced together and “count” as a single sound. For example, $[\text{ts}]$ represents a single sound (an affricate) while $[\text{t}][\text{s}]$ represents two sounds, a sequence of $[\text{t}]$ followed by $[\text{s}]$. 55, 379
- tilde* A diacritic used to show that a sound is nasal or nasalized. The symbol $[\text{a}]$ represents an oral vowel; $[\tilde{\text{a}}]$, which has a tilde, represents a nasal vowel. See *ogonek*. 115

tone Tone is the linguistic use of pitch; whether a tone is high or low is relative to the language, the speaker, the words, the context. See also *pitch*. 119, 520

tongue blade The flat part of the tongue right behind the tip of the tongue. 71, 73

tongue body The large part of the tongue behind the tip of the tongue. Also called the *dorsum*. 78

transcribe The act of representing a word with symbols showing the way it is pronounced. 123

transcription A transcription of an utterance is a representation of that utterance in symbols, showing the way it is pronounced. 123, 232, 281, 431

trill An *r*-sound made by relaxing the tongue and letting it vibrate as air passes over it. Also called a *rolled r*. 61

turbulent air flow Airflow that is irregular, so it results in a whooshy or buzzy sound. 48

underlying representation Roughly speaking, an underlying representation is a phonemic representation. Abbreviation is *UR*. 421, 471, 490, 514, 515, 534

unreleased If there is no release phase of a stop, the stop is unreleased. 51

uvula The soft tissue hanging down from the velum in the back of the mouth. 35

uvular A member of the class of sounds with the feature [dorsal], made with the tongue pulled back to the uvula; a sound made with constriction made by the body of the tongue at the uvula. 79

V A shorthand for *vowel*. 290

variant One way to say a given word. Can also refer to one way to pronounce a given language. 190, 230, 523

variation The (normal) situation with language is that different individuals or different groups may express the same meaning differently; this is called *variation*. Groups speaking a particular variant are defined by factors that are not linguistic – factors such as gender, geographical location, socio-economic class, etc. 11, 188, 189, 190, 213, 230, 506

variety A general term for any distinctive form of a language or linguistic expression. 190, 230

velar A member of the class of sounds with the feature [dorsal made with tongue body constriction at the velum]. 78

velum The back part of the roof of the mouth; it is raised for oral sounds and lowered for nasal sounds, and so is an active articulator. The back of the tongue can raise to touch the velum to make dorsal sounds, so it is also an active articulator. xx, 35, 42

Terms linguists use

vocal cords Another term for vocal folds, the soft tissue in the voicebox that is stiffened for voiceless sounds and relaxed for voiced sounds. Other manipulations result in aspirated, creaky, or glottalized sounds. 36

vocal folds Soft tissue in the voicebox that is stiffened for voiceless sounds and relaxed for voiced sounds. Other manipulations result in aspirated, creaky, or glottalized sounds. Vocal folds are also called vocal cords. 36, 38

vocal tract The part of the body that contains the active and passive articulators, from the larynx to the lips and nostrils, including the velum (which can make air go out through the nose). 22, 33

voice box Another term for larynx, a complex of cartilages, ligaments, and muscles in the neck, with an opening that air passes through between the lungs and the mouth. 36

voiced Sounds produced with vocal folds that are relaxed enough to vibrate. There is a buzz-y quality to voiced sounds that you can hear if you put your hands over your ears while you pronounce a voiced sound. 39, 198

voiceless Sounds produced with vocal folds that are too stiff to vibrate. 39, 198

vowel A sound made with an open oral tract. 28

vowel inventory The collection of vowel sounds used in making the words of a language. Often refers to only the contrastive vowel sounds of a language. 268, 274, 525

word-final At the end of a word. We can talk about word-final codas, word-final stress, word-final position, and so on. 222, 318, 424

word-initial At the beginning of a word. We can talk about word-initial onsets, word-initial stress, word-initial position, and so on. 202, 309

word-medial Inside a word, neither at the very beginning nor at the very end. We can talk about word-medial codas, word-medial stress, word-medial position, and so on. 202

writing system A way of writing a language that is intended for reading – books, signs, texting, etc. Also called [orthography](#). 487

zero affixation A type of affix that has no phonological content. 179

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Indigenous languages of the Americas and their structures

What do people mean when they talk about the sound pattern of a language—what linguists call phonology? This book explores that question in a hands-on way, with a focus on applying the knowledge in language revitalization and community language work. The book introduces basic ideas and analytic strategies using examples from Indigenous languages across North and South America. Short comprehension checks invite you to test your understanding, and language explorations encourage you to apply what you've learned to build your own description of the sound pattern of your language. Throughout, the book takes a practical approach to phonological analysis, not a theoretical approach.

The book is designed to be accessible to community members and others without formal linguistic training who are engaged in or preparing for community language work. It is also appropriate for linguistics students and researchers in the early stages of their education about phonology or needing a quick refresher or reference on basic phonological analysis.

Each chapter looks at one key concept, starting with language sounds and how they are made, how words are built, and how communities vary in their use of language. Other topics include the sounds specific to a language and how they are put together into words of that language, both in terms of sequences and in terms of syllables. The culmination is an exploration of how sound sequences and word building interact so that the words follow the general sound patterns of the language. Each chapter ends with some ideas about how to incorporate the topic of the chapter in community language work.

The goal with each discussion is to help you both to understand each idea and to learn how to use it to describe or analyze the way that sounds work in a language. Importantly, the book also includes suggestions on tailoring your reading to your own goals, offering pointers on which sections you might skip and which ones could be especially useful for you.